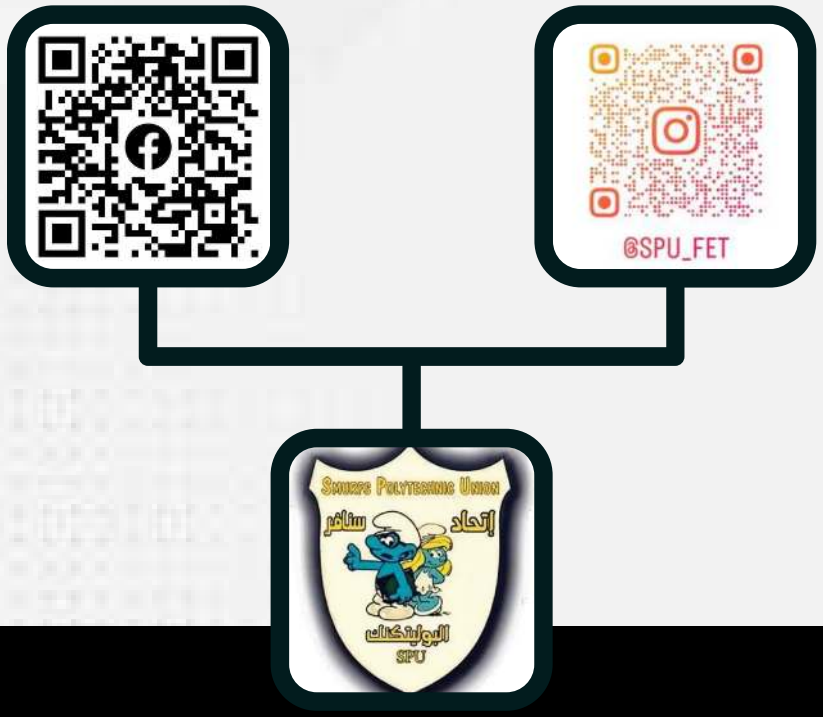
# سلايدات الدكتور علي
# كيمياء 101

▼ من له شغف في طلب العلم، لا يتعبه بحث ولا يرهقه جهد، نحن أمة اقرأ.. أبناء القلم



# Chapter 3. Stoichiometry: Mole-Mass Relationships in Chemical Reactions

## Concept 1. The meaning and usefulness of the mole

- The mole (or mol) represents a certain number of objects.
- SI def.: the amount of a substance that contains the same number of entities as there are atoms in 12 g of carbon-12.
- Exactly 12 g of carbon-12 contains **$6.022 \times 10^{23}$** atoms.
- 1 mole contains **$6.022 \times 10^{23}$** entities (Avogadro's number)
- One mole of $H_2O$ molecules contains $6.022 \times 10^{23}$ molecules.

- One mole of NaCl contains $6.022 \times 10^{23}$ NaCl formula units.
- Use the mole quantity to count formulas by weighing them.

- Mass of a mole of particles = mass of 1 particle **x** $6.022 \times 10^{23}$

Mass of 1 H atom: $1.008 \text{ amu} \times 1.661 \times 10^{-24} \text{ g/amu} = 1.674 \times 10^{-24} \text{ g}$

Mass of 1 mole of H atoms:

$$1.674 \times 10^{-24} \text{g/H atom} \times 6.022 \times 10^{23} \text{H atoms} = 1.008 \text{ g}$$

- The mass of an atom in amu is numerically the same as the mass of one mole of atoms of the element in grams.

- One atom of sulfur has a mass of 32.07 amu; one mole of S atoms has a mass of 32.07 g.

**Concept 2**. The relation between molecular (formula) mass and molar mass

- For compounds, the molecular mass (in amu) is numerically the same as the mass of one mole of the compound in grams.

- **Skill 3-1** Calculate the molecular mass of a compound as the sum of the atomic masses of its elements.

- **Molecular mass $H_2O$ = (2 x atomic mass of H) + atomic mass of O**
  **= 2(1.008 amu) + 16.00 amu = 18.02 amu**

- **So, one mole of water (6.022 x $10^{23}$ molecules) has a mass of 18.02 g.**

- Molar mass of NaCl = atomic mass of Na (22.99 amu) + the atomic mass of Cl (35.45 amu) **22.99** + **35.45** = **58.44** amu

- One mol of NaCl (6.02 x$10^{23}$ formulas) has a mass of 58.44 g.

- To obtain one mole of copper atoms ($6.02 \times 10^{23}$ atoms), weigh out 63.55 g copper.

**Concept 3**. The relations among amount of substance (in moles), mass (in grams), and number of chemical entities

- The molar mass ($M$) of a substance is the mass of one mole of entities (atoms, molecules, or formula units) of the substance.
- Molar mass has units of grams per mole (g/mol).

**TABLE 3.1 Summary of Mass Terminology***

| TERM | DEFINITION | UNIT |
|---|---|---|
| Isotopic mass | Mass of an isotope of an element | amu |
| Atomic mass (also called atomic weight) | Average of the masses of the naturally occurring isotopes of an element weighted according to their abundance | amu |
| Molecular mass (also called molecular weight) | Sum of the atomic masses of the atoms (or ions) in a molecule (or formula unit) | amu |
| Molar mass ($M$) (also called gram-molecular weight) | Mass of one mole of chemical entities (atoms, ions, molecules, formula units) | g/mol |

# Skill 3-2 Mass - Mole Conversions

- **Use the molar mass of an element or compound to convert a given number of moles to mass:**

$$\text{Mass (g)} = \text{no. of moles} \times \frac{\text{no. of grams}}{1 \text{ mole}}$$

- **We can do the reverse with 1/*M*, and convert any mass in grams to the number of moles:**

$$\text{No. of moles} = \text{mass (g)} \times \frac{1 \text{ mole}}{\text{no. of grams}}$$

- **Use Avogadro's number to convert moles of substance to the number of entities:**

$$\text{No. of entities} = \text{no. of moles} \times \frac{6.022 \times 10^{23} \text{ entities}}{1 \text{ mole}}$$

$$\text{No. of moles} = \text{no. of entities} \times \frac{1 \text{ mole}}{6.022 \times 10^{23} \text{ entities}}$$

**Problem**: (a) How many grams of silver, Ag, are in 0.0342 mol Ag? (b) How many atoms of Ag are in 0.0342 mol Ag?

**Plan**: (a) To convert moles Ag to grams Ag, use the molar mass of Ag from the periodic table. (b) To convert moles to number of atoms, use Avogadro's number.

**Solution**: (a) Converting from moles of Ag to mass:

$$\text{Mass (g) of Ag} = 0.0342 \cancel{\text{mol Ag}} \times \frac{107.9 \text{ g Ag}}{1 \cancel{\text{mol Ag}}} = \mathbf{3.69 \text{ g Ag}}$$

(b) Converting from moles of Ag to number of atoms:

$$\text{No. of Ag atoms} = 0.0342 \cancel{\text{mol Ag}} \times \frac{6.022 \times 10^{23} \text{ atoms Ag}}{1 \cancel{\text{mol Ag}}}$$

$$= 2.06 \text{ x } 10^{22} \text{ atoms Ag}$$

**Problem**: Iron is a most important metal in our society. How many iron atoms are present in a piece of iron weighing 95.8 g?

**Plan**:




**Solution**: Converting from mass of Fe to moles:

$$\text{Moles of Fe} = 95.8 \cancel{\text{g Fe}} \times \frac{1 \text{ mol Fe}}{55.85 \cancel{\text{g Fe}}} = 1.72 \text{ mol Fe}$$

Converting from moles of Fe to number of atoms:

$$\text{No. of Fe atoms} = 1.72 \cancel{\text{mol Fe}} \times \frac{6.022 \times 10^{23} \text{ atoms Fe}}{1 \cancel{\text{mol Fe}}}$$

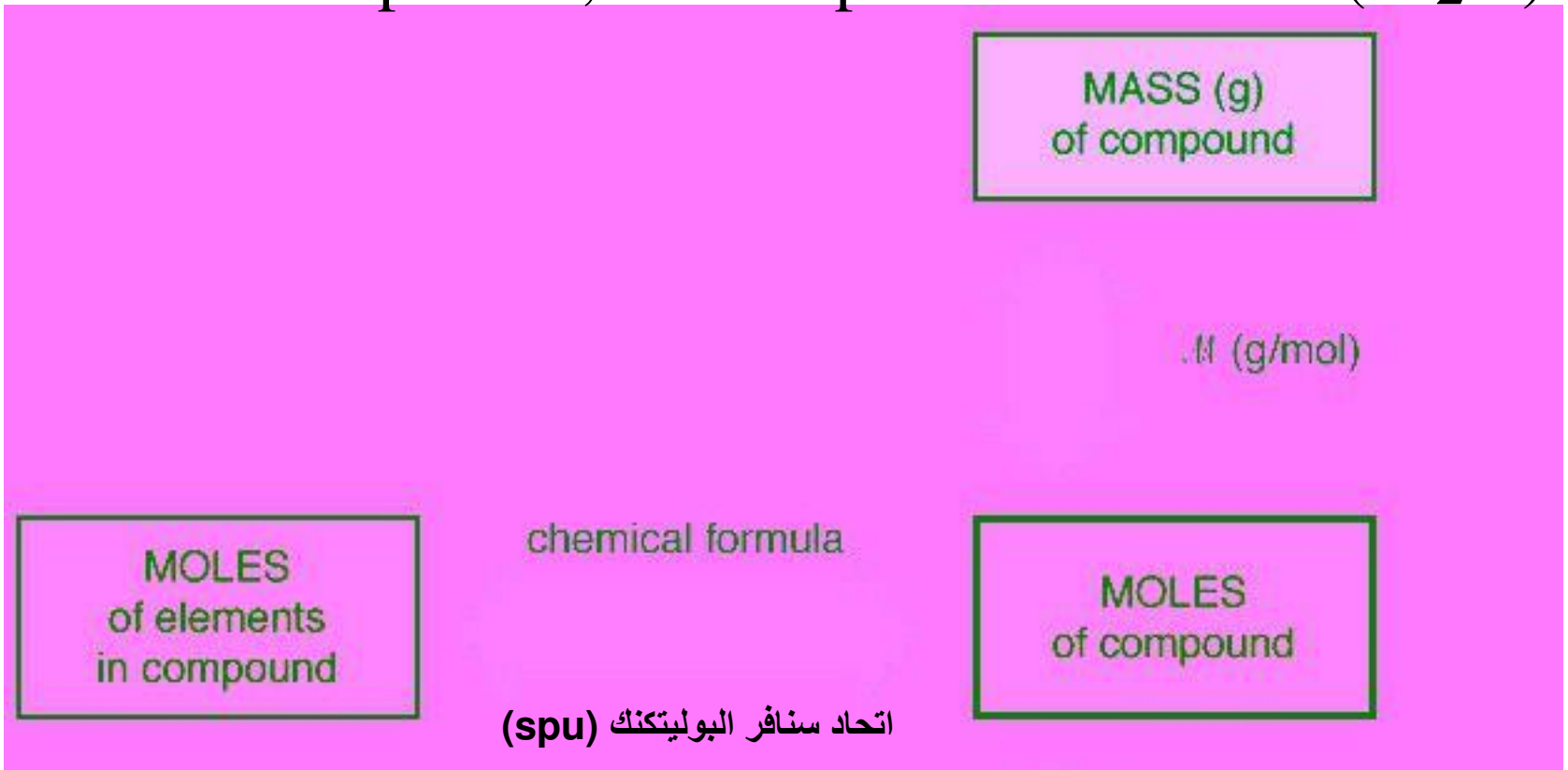
$$= 10.4 \times 10^{23} \text{ atoms Fe} = \mathbf{1.04 \times 10^{24} \text{ atoms Fe}}$$

## Concept 4. The information in a chemical formula

- The sulfur dioxide formula, $SO_2$, shows that its molecules contain one S atom and two O atoms; calculate its molar mass.

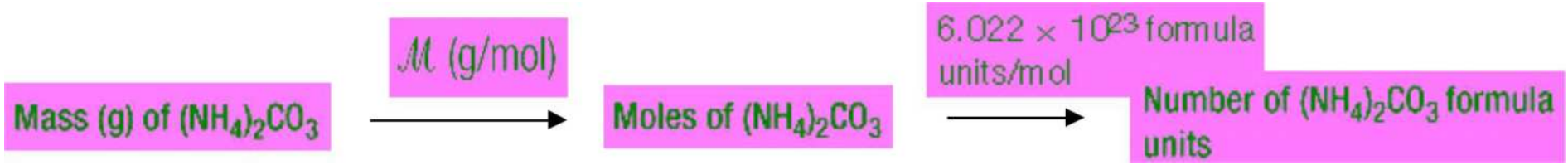
- One mole $SO_2$ contains $6.02 \times 10^{23}$ $SO_2$ molecules, which consist of $6.02 \times 10^{23}$ S atoms and $2(6.02 \times 10^{23})$ O atoms.

- Same for ionic compounds, such as potassium sulfide ($K_2S$):



**Problem**: How many moles and formulas are in 41.6 g ammonium carbonate**?**

**Solution**: Calculating molar mass:




$$\mathcal{M} = (2 \times 14.01\ \text{g/mol}) + (8 \times 1.008\ \text{g/mol}) + 12.01\ \text{g/mol} + (3 \times 16.00\ \text{g/mol})$$
$$= 96.09\ \text{g/mol}$$

Converting from mass to moles:

$$\text{Moles of } (NH_4)_2CO_3 = 41.6\ \cancel{\text{g } (NH_4)_2CO_3} \times \frac{1\ \text{mol } (NH_4)_2CO_3}{96.09\ \cancel{\text{g } (NH_4)_2CO_3}}$$
$$= \mathbf{0.433\ mol\ (NH_4)_2CO_3}$$

Converting from moles to formula units:

Formula units of $(NH_4)_2CO_3$

$$= 0.433\ \cancel{\text{mol } (NH_4)_2CO_3} \times \frac{6.022 \times 10^{23}\ \text{formula units } (NH_4)_2CO_3}{1\ \cancel{\text{mol } (NH_4)_2CO_3}}$$

## Skill 3-3 Mass Percent and the Chemical Formula

The formula shows the number of moles of each element; use it to calculate the mass percent of each element on a mole basis:

$$\text{Mass \% of element X} = \frac{\text{moles of X} \times \text{molar mass of X (g/mol)}}{\text{mass of one mole of compound (g)}} \times 100\ \%$$

**Problem**: The formula of the sugar glucose is $\mathbf{C_6H_{12}O_6}$.
(a) What is the mass percent of each element in glucose?
(b) How many grams of carbon are in 16.55 g glucose?

**Plan**: The formula gives the no. of moles of C, H and O in glucose. Convert from moles of element to grams using the molar mass.

اتحاد سنافر البولبتكنك (spu)

**Dividing by the mass of one mole of glucose gives the element's mass fraction; multiplying this fraction by 100% gives the mass percent.**

$$\text{Mass fraction of C} = \frac{\text{total mass C}}{\text{mass of 1 mol glucose}} = \frac{72.06\ \text{g}}{180.16\ \text{g}} = 0.4000\ \text{g C/g glucose}$$

$$\text{Mass \% of C} = \text{mass fraction of C} \times 100 = 0.4000 \times 100$$

$$= \mathbf{40.00\ mass\ \%\ C}$$

Combining the steps for each of the other two elements:

$$\text{Mass \% of H} = \frac{\text{mol H} \times \mathcal{M}\ \text{of H}}{\text{mass of 1 mol glucose}} \times 100 = \frac{12\ \cancel{\text{mol H}} \times \frac{1.008\ \cancel{\text{g}}\ \text{H}}{1\ \cancel{\text{mol H}}}}{180.16\ \cancel{\text{g}}} \times 100$$

$$= \mathbf{6.714\ mass\ \%\ H}$$

$$\text{Mass \% of O} = \frac{\text{mol O} \times \mathcal{M}\ \text{of O}}{\text{mass of 1 mol glucose}} \times 100 = \frac{6\ \cancel{\text{mol O}} \times \frac{16.00\ \cancel{\text{g}}\ \text{O}}{1\ \cancel{\text{mol O}}}}{180.16\ \cancel{\text{g}}} \times 100$$

$$= \mathbf{53.29\ mass\ \%\ O}$$

(b) Determining the mass of carbon.

**Plan**: To find the mass of C in the glucose sample, multiply the mass of the sample by the mass fraction of C from part (a)

**Solution**:

$$\text{Mass (g) of C} = \text{mass of glucose} \times \text{mass fraction of C}$$

$$= 16.55 \text{ g glucose} \times \frac{0.4000 \text{ g C}}{1 \text{ g glucose}} = \mathbf{6.620 \text{ g C}}$$

## The procedure for finding the empirical and molecular formulas of a compound

- If an unknown pure compound can be **decomposed**, the masses of elements present can be determined.

- Converting the masses to moles of elements leads to the empirical formula, the simplest whole-number ratio of moles of each element in the compound.

- Suppose a sample of unknown compound is found to contain **0.21 mol zinc**, **0.14 mol phosphorus**, and **0.56 mol oxygen**.

- A preliminary formula based on the data is **$Zn_{0.21}P_{0.14}O_{0.56}$**.
- Convert the fractional subscripts to whole numbers. :

1. Divide each subscript by the smallest subscript:

$$Zn_{\frac{0.21}{0.14}}P_{\frac{0.14}{0.14}}O_{\frac{0.56}{0.14}} \rightarrow Zn_{1.5}P_{1.0}O_{4.0}$$

2. If any subscripts are still not integers, multiply through by the smallest integer that will turn all the subscripts into integers

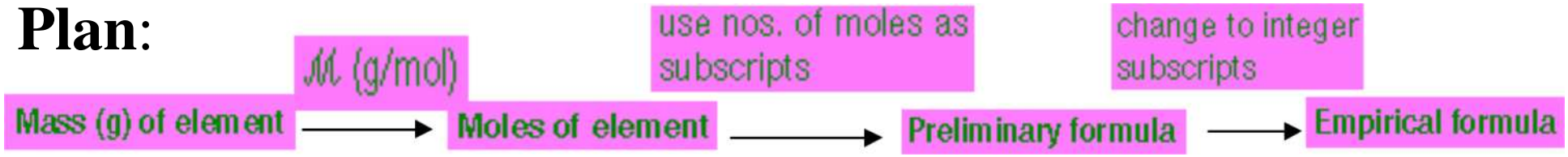
$$Zn_{(1.5\times 2)}P_{(1.0\times 2)}O_{(4.0\times 2)} \rightarrow Zn_{3.0}P_{2.0}O_{8.0}$$

$$\text{or } Zn_3P_2O_8$$

The conventional way to write this formula is **$Zn_3(PO_4)_2$** ; the compound is zinc phosphate, a dental cement.

**Skill 3-4**: Analysis of a sample of ionic compound gave: 2.82 g Na, 4.35 g Cl, and 7.83 g O. What is the empirical formula and name of the compound?

**Plan**:




**Solution**: Find moles of elements:

$$\text{Moles of Na} = 2.82\ \cancel{\text{g Na}} \times \frac{1\ \text{mol Na}}{22.99\ \cancel{\text{g Na}}} = 0.123\ \text{mol Na}$$

$$\text{Moles of Cl} = 4.35\ \cancel{\text{g Cl}} \times \frac{1\ \text{mol Cl}}{35.45\ \cancel{\text{g Cl}}} = 0.123\ \text{mol Cl}$$

$$\text{Moles of O} = 7.83\ \cancel{\text{g O}} \times \frac{1\ \text{mol O}}{16.00\ \cancel{\text{g O}}} = 0.489\ \text{mol O}$$

From the moles of each element, construct a preliminary formula and convert to integer subscripts.

Preliminary formula: $Na_{0.123}Cl_{0.123}O_{0.489}$

Convert to integer subscripts:

$$Na_{\frac{0.123}{0.123}}Cl_{\frac{0.123}{0.123}}O_{\frac{0.489}{0.123}} \rightarrow Na_{1.00}Cl_{1.00}O_{3.98} \approx Na_1Cl_1O_4, \text{ or } NaClO_4$$

- Note that we rounded the subscript of O from 3.98 to 4.

- The empirical formula is $NaClO_4$ ; the name is sodium perchlorate.

# Molecular Formulas

- The actual number of moles of each element in the smallest unit of the compound.
- In water ($H_2O$), ammonia ($NH_3$), methane ($CH_4$), and ionic compounds, the empirical and molecular formulas are identical.
- In some cases the molecular formula is a whole-number multiple of the empirical formula.

- Using the empirical formula to obtain the molecular formula.

Hydrogen peroxide has the empirical formula **HO** (17.01g/mol)

Dividing its molar mass (34.02 g/mol) by the empirical formula mass gives the whole-number multiple:

$$\text{Whole-number multiple} = \frac{\text{molar mass (g/mol)}}{\text{empirical formula mass (g/mol)}}$$

$$= \frac{34.02\ \cancel{\text{g/mol}}}{17.01\ \cancel{\text{g/mol}}} = 2$$

The molecular formula is $H_2O_2$.

19

**Skill 3-5** : Lactic acid ($M$ = 90.08 g/mol) contains 40.0 mass % C, 6.71 mass % H, and 53.3 mass % O.
(a) Determine the empirical formula of lactic acid.
(b) Determine the molecular formula.

**Plan**: Assume 100 g lactic acid to express each mass % as grams. Convert grams to moles and find the empirical formula.

**Solution**: Express mass % as grams; assume 100 g lactic acid:

$$\text{Mass (g) of C} = \frac{\text{40.0 parts C by mass}}{\text{100 parts by mass}} \times 100\ \text{g} = 40.0\ \text{g C}$$

Similarly, there are 6.71 g H and 53.3 g O.

Converting from grams of elements to moles:

$$\text{Moles of C} = \text{mass of C} \times \frac{1}{\mathcal{M}\ \text{of C}} = 40.0\ \text{g C} \times \frac{1\ \text{mol C}}{12.01\ \text{g C}}$$

$$= 3.33\ \text{mol C}$$

Similarly, we have 6.66 mol H and 3.33 mol O. Constructing the preliminary formula: $C_{3.33}H_{6.66}O_{3.33}$ Converting to integer subscripts:

$$C_{\frac{3.33}{3.33}}H_{\frac{6.66}{3.33}}O_{\frac{3.33}{3.33}} \rightarrow C_1H_2O_1;\ \text{the empirical formula is } \mathbf{CH_2O}$$

Determining the molecular formula:

**Plan**: Divide the molar mass by the empirical formula mass.

**Solution**:

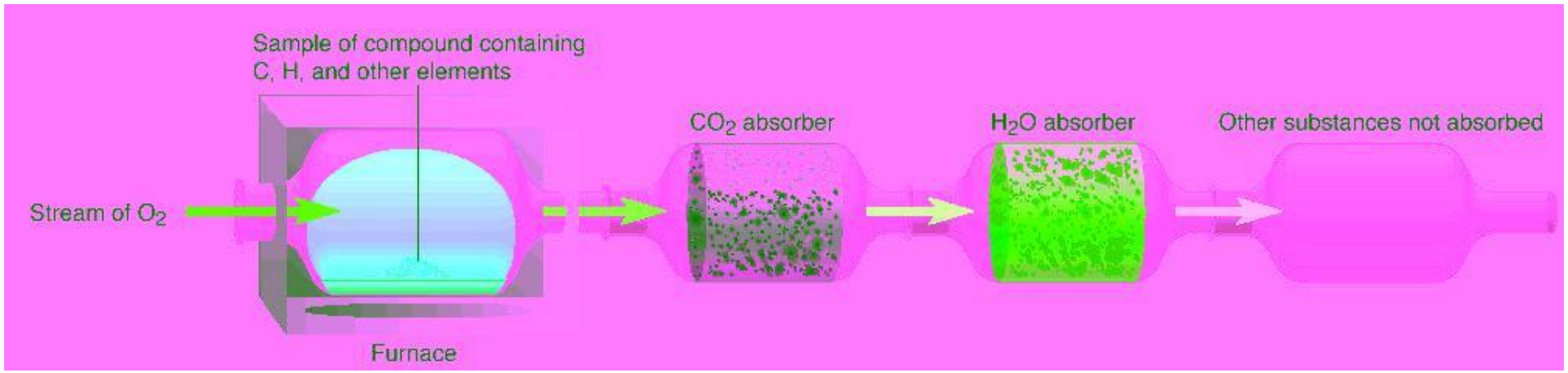
$$\text{Whole-number multiple} = \frac{\mathcal{M} \text{ of lactic acid}}{\text{empirical formula mass}} = \frac{90.08\ \cancel{\text{g/mol}}}{30.03\ \cancel{\text{g/mol}}} = 3.000 = 3$$

Calculate the molecular formula:

$$C_{(1\times 3)}H_{(2\times 3)}O_{(1\times 3)} = \mathbf{C_3H_6O_3}$$

# Combustion Analysis

- Combustion analysis gives the amounts of carbon and hydrogen in a sample of combustible compound.
- The C forms $CO_2$ , which is absorbed in the first chamber.
- The H forms $H_2O$, which is absorbed in the second chamber.




- Weigh the chambers before and after combustion.
- Calculate the masses of $CO_2$ and $H_2O$
- Use them to calculate the masses of C and H in the compound.
- From the masses, calculate the empirical formula.

**Skill 3-5 contd**: Vitamin C (M= 176. g/mol) contains C, H and O. A 1.000 g sample was placed in a combustion apparatus:

Mass of $CO_2$ absorber after combustion = 85.35 g

Mass of $CO_2$ absorber before combustion = 83.85 g

Mass of $H_2O$ absorber after combustion = 37.96 g

Mass of $H_2O$ absorber before combustion = 37.55 g

What is the molecular formula of vitamin C?

**Plan**: -Use changes in mass of $CO_2$ and $H_2O$ absorbers to calculate the moles of C and H present in the sample.

-Find the mass of C, using the mass fraction of C in $CO_2$ . Likewise, find the mass of H from the mass of $H_2O$.

-The mass of vitamin C minus the sum of the C and H masses gives the mass of O.

-Construct the empirical formula and molecular formula.

**Solution**: Finding the masses of combustion products:

$$\text{Mass of } CO_2 = \text{mass of } CO_2 \text{ absorber after} - \text{mass before} = 1.50 \text{ g } CO_2$$

$$\text{Mass of } H_2O = \text{mass of } H_2O \text{ absorber after} - \text{mass before} = 0.41 \text{ g } H_2O$$

Calculating mass fractions of the elements:

$$\text{Mass fraction of C in } CO_2 = \frac{\text{mol C} \times \mathcal{M} \text{ of C}}{\text{mass of 1 mol } CO_2} = \frac{1 \cancel{\text{mol C}} \times \frac{12.01 \text{ g C}}{1 \cancel{\text{mol C}}}}{44.01 \text{ g } CO_2}$$

$$= 0.2729 \text{ g C}/1 \text{ g } CO_2$$

$$\text{Mass fraction of H in } H_2O = \frac{\text{mol H} \times \mathcal{M} \text{ of H}}{\text{mass of 1 mol } H_2O} = \frac{2 \cancel{\text{mol H}} \times \frac{1.008 \text{ g H}}{1 \cancel{\text{mol H}}}}{18.02 \text{ g } H_2O}$$

$$= 0.1119 \text{ g H}/1 \text{ g } H_2O$$

## Calculate masses of C and H:

$$\text{Mass of element} = \text{mass of compound} \times \text{mass fraction of element}$$

$$\text{Mass (g) of C} = 1.50 \text{ g } CO_2 \times \frac{0.2729 \text{ g C}}{1 \text{ g } CO_2} = 0.409 \text{ g C}$$

$$\text{Mass (g) of H} = 0.41 \text{ g } H_2O \times \frac{0.1119 \text{ g H}}{1 \text{ g } H_2O} = 0.046 \text{ g H}$$

Calculate mass of O by difference:

$$\text{Mass (g) of O} = \text{mass of vitamin C sample} - (\text{mass of C} + \text{mass of H})$$
$$= 1.000 \text{ g} - (0.409 \text{ g} + 0.046 \text{ g}) = 0.545 \text{ g O}$$

Find the moles of elements: Divide mass of each by its molar mass gives 0.0341 mol C, 0.046 mol H, and 0.0341 mol O.

Construct the preliminary formula: $C_{0.0341}H_{0.046}O_{0.0341}$

Divide through by the smallest subscript:

$$C_{\frac{0.0341}{0.0341}}H_{\frac{0.046}{0.0341}}O_{\frac{0.0341}{0.0341}} = C_{1.00}H_{1.3}O_{1.00}$$

Determine the empirical formula. By inspection, find that 3 is the smallest multiple that makes all subscripts into integers:

$$C_{(1.00 \times 3)}H_{(1.3 \times 3)}O_{(1.00 \times 3)} = C_{3.00}H_{3.9}O_{3.00} \approx C_3H_4O_3$$

Determine the molecular formula:

$$\text{Whole-number multiple} = \frac{\mathcal{M} \text{ of vitamin C}}{\text{empirical formula mass}} = \frac{176.12 \text{ g/mol}}{88.06 \text{ g/mol}}$$

$$= 2.000 = 2$$

$$C_{(3 \times 2)}H_{(4 \times 2)}O_{(3 \times 2)} = \mathbf{C_6H_8O_6}$$

# Chemical Formulas and the Structures of Molecules

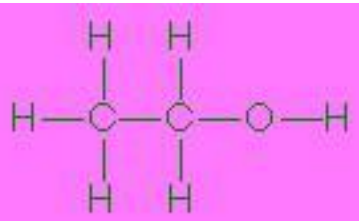
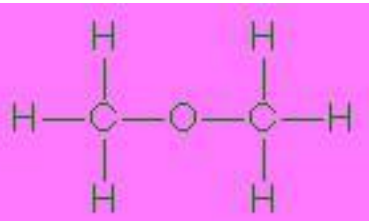
- Chemical formulas represent real three-dimensional objects.
- The molecular formula tells the number of each type of atom.
- Molecular formulas may not be unique; the same types and numbers of atoms can bond to each other in more than one way
- Can be different compounds or structural isomers of the same compound (same molecular formulas).

# Structural Formulas

- Consider $C_2H_6O$; two very different compounds have this molecular formula:

  -ethanol, the intoxicating substance present in wine and beer -
  dimethyl ether, a colorless gas once used in refrigeration

- Their radically different physical and chemical behaviors are the result of different molecular structures.

| | ethanol | dimethyl ether |
|---|---|---|
| Structural formula |  |  |
| Space-filling model |  |  |

**Concept 6**. The importance of balancing equations for the quantitative study of chemical reactions

- For an equation to depict accurately the amounts of chemicals involved in a reaction, it must be balanced.

- The same number of each type of atom must appear on both sides of the equation.

- In a flashbulb Mg wire and oxygen gas react to yield magnesium oxide powder.



A



B



Translate the chemical statement into a "skeleton" equation: chemical formulas arranged in an equation format.

| reactants — "yield"→ products |
|---|

reactants yield product

$$__\ Mg + __\ O_2 \longrightarrow __\ MgO$$

## Balancing the atoms

- Each blank must contain a balancing coefficient, a numerical multiplier of all the atoms in the formula that follows it.
- Stepwise, match the atoms on each side, element by element, beginning with the most complex substance.

$$__\ Mg + __\ O_2 \rightarrow _1MgO$$

Requires one Mg on the left. $_1Mg + __ O_2 \rightarrow _1MgO$

And one oxygen on the left. $_1Mg + _{1/2}O_2 \rightarrow _1MgO$

Adjusting the coefficients.

- The smallest whole-number coefficients are required.
- To remove the 1/2 $O_2$, multiply the whole equation by 2:

$$2Mg + 1O_2 \rightarrow 2MgO$$

- A coefficient of 1 is implied by the presence of the substance and is not written: $2Mg + O_2 \rightarrow 2MgO$
- Check the numbers of atoms:

Reactants (2 Mg, 2 O) → products (2 Mg, 2 O)

Reaction equation is balanced.

Specifying the states of matter.

- A complete equation shows the physical state of each substance or whether it is dissolved in water.

  solid (s)  liquid (l)  gas (g)  aqueous solution (aq)

- From the statement, Mg "wire" is solid, $O_2$ is a gas, and "powdery" MgO is a solid:

$$2Mg(s) + O_2(g) \longrightarrow 2MgO(s)$$

**Skill 3-6**: Octane ($C_8H_{18}$) burns to form carbon dioxide and water vapor. Write a balanced equation for this reaction.

**Solution**: Translate the statement into a skeleton equation (with coefficient blanks). Octane and oxygen are reactants; oxygen is molecular $O_2$. Carbon dioxide and water vapor are products:

$$__\ C_8H_{18} + __\ O_2 \rightarrow __\ CO_2 + __\ H_2O$$

**Balance the atoms**.

Start with the more complex reactant, $C_8H_{18}$, then balance $O_2$:

$$\underline{1}\ C_8H_{18} + __\ O_2 \rightarrow __\ CO_2 + __\ H_2O$$

8 C on left requires 8 $CO_2$ on right:

$$\underline{1}\ C_8H_{18} + __\ O_2 \rightarrow \underline{8}\ CO_2 + __\ H_2O$$

**18** H on left requires **9** $H_2O$.

$$_1C_8H_{18} + __ O_2 \rightarrow _8CO_2 + _9H_2O$$

**25** O on right requires **25/2** $O_2$

$$_1C_8H_{18} + \underline{25/2}\ O_2 \rightarrow _8CO_2 + _9H_2O$$

Adjust coefficients. $2C_8H_{18} + 25O_2 \rightarrow 16CO_2 + 18H_2O$

Check that the equation is balanced:

reactants (16 C, 36 H, 50 O) = products (16 C, 36 H, 50 O)

States of matter: $C_8H_{18}$ is liquid; $O_2$, $CO_2$, and $H_2O$ vapor are gases:

$$2C_8H_{18}(l) + 25O_2(g) \rightarrow 16CO_2(g) + 18H_2O(g)$$

## Concept 7 The mole-mass-number information contained in a balanced equation

- A balanced equation relates quantities of atoms, molecules, formula units, moles of substance, and masses.

$$C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(g)$$

1 molecule $C_3H_8$ + 5 molecules $O_2$ → 3 molecules $CO_2$ + 4 molecules $H_2O$

| | | |
|---|---|---|
| 1 mol $C_3H_8$ + 5 mol $O_2$ | → | 3 mol $CO_2$ + 4 mol $H_2O$ |
| 44.09 amu $C_3H_8$ + 160.00 amu $O_2$ | → | 132.03 amu $CO_2$ + 72.06 amu $H_2O$ |
| 44.09 g $C_3H_8$ + 160.00 g $O_2$ | → | 132.03 g $CO_2$ + 72.06 g $H_2O$ |
| 204.09 g | → | 204.09 g |

## Concept 8 The relation between amounts of reactants and product

- Quantitative relationships in a balanced equation are expressed as stoichiometrically equivalent molar ratios.

- Use stoichiometrically equivalent molar ratios to determine how much of one substance forms from (or reacts with) another.

$$C_3H_8(g) + 5O_2(g) \longrightarrow 3CO_2(g) + 4H_2O(g)$$

- 3 mol $CO_2$ is stoichiometrically equivalent to 4 mol $H_2O$, 5 mol $O_2$ is stoichiometrically equivalent to 3 mol $CO_2$, and so on.
- In the combustion of propane, how many moles of $O_2$ are consumed when 10.0 mol $H_2O$ are produced?

starting amt. molar ratio equiv. amt.

$$\text{Moles of } O_2 \text{ consumed} = 10.0 \text{ mol } H_2O \times \frac{5 \text{ mol } O_2}{4 \text{ mol } H_2O} = 12.5 \text{ mol } O_2$$

## Skill 3-7:

Copper is obtained from sulfide ores by roasting the ore with $O_2$ to form powdered copper(I) oxide and gaseous sulfur dioxide.

(a) How many moles of oxygen are required to roast 10.0 mol copper(I) sulfide?

Write the balanced equation first. The reactants are Cu 2S and O 2 , and the products are Cu 2O and SO 2 :

$$2Cu_2S(s) + 3O_2(g) \rightarrow 2Cu_2O(s) + 2SO_2(g)$$

Given the moles of Cu 2S; need to find the moles of O 2.

$$\text{Moles of } O_2 = 10.0 \text{ mol } Cu_2S \times \frac{3 \text{ mol } O_2}{2 \text{ mol } Cu_2S} = \mathbf{15.0 \text{ mol } O_2}$$

(b) How many grams of sulfur dioxide are formed when 10.0 mol copper(I) sulfide are roasted?

Find mass of $SO_2$ that forms from the given mol of reactant ($Cu_2S$).

$$\text{Mass (g) of } SO_2 = 10.0 \text{ mol } Cu_2S \times \frac{2 \text{ mol } SO_2}{2 \text{ mol } Cu_2S} \times \frac{64.07 \text{ g } SO_2}{1 \text{ mol } SO_2} = \mathbf{641 \text{ g } SO_2}$$

(c) How many kg of oxygen are needed to form 2.86 kg Cu

$_2$O? Convert mass $Cu_2O$ to mol:

$$\text{Moles of } Cu_2O = 2.86 \text{ kg } Cu_2O \times \frac{10^3 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ mol } Cu_2}{143.10 \text{ g Cu}}$$

$$= 20.0 \text{ mol } Cu_2O$$

Mol $Cu_2O$ to mol $O_2$:

$$\text{Moles of } O_2 = 20.0 \text{ mol } Cu_2O \times \frac{3 \text{ mol } O_2}{2 \text{ mol } Cu_2O}$$

$$= 30.0 \text{ mol } O_2$$

Mol $O_2$ to mass $O_2$:

$$\text{Mass (kg) of } O_2 = 30.0 \text{ mol } O_2 \times \frac{32.00 \text{ g } O_2}{1 \text{ mol } O_2} \times$$

$$= \mathbf{0.960 \text{ kg } O_2}$$

## Chemical Reactions That Occur in a Sequence

- A product of one reaction can become a reactant of the next (reaction sequence).
- If a substance forms in one reaction and is used up in the next, write an overall equation that eliminates the substance altogether.

**Skill 3-8**: Roasting chalcocite is the 1st step in extracting copper.

$$2Cu_2S(s) + 3O_2(g) \rightarrow 2Cu_2O(s) + 2SO_2(g)$$ [equation 1]

Next step: copper(I) oxide reacts with powdered carbon to yield copper metal and carbon monoxide gas:

$$Cu_2O(s) + C(s) \rightarrow 2Cu(s) + CO(g)$$ [equation 2]

(a) Write a balanced overall equation for the two-step sequence.

Add the equations together. Only Cu 2O appears as a product in one equation and a reactant in the other, so it is the common substance.

- Two mol $Cu_2O$ forms in eqn. 1; but one mol $Cu_2O$ reacts in eqn. 2.
- Double all the coefficients in eq. 2; the $Cu_2O$ from eqn. 1 is used up

$$2Cu_2S(s) + 3O_2(g) \rightarrow 2Cu_2O(s) + 2SO_2(g) \text{ [equation 1]}$$

$$2Cu_2O(s) + 2C(s) \rightarrow 4Cu(s) + 2CO(g) \text{ [equation 2 doubled]}$$

---

$$2Cu_2S(s) + 3O_2(g) + \cancel{2Cu_2O(s)} + 2C(s) \rightarrow \cancel{2Cu_2O(s)} + 2SO_2(g) + 4Cu(s) + 2CO(g)$$

$$\mathbf{2Cu_2S(s) + 3O_2(g) + 2C(s) \rightarrow 2SO_2(g) + 4Cu(s) + 2CO(g)}$$

Check: Reactants (4Cu, 2S, 6O, 2C) = products (4Cu, 2S, 6O, 2C)

(b) Determine the mass (kg) of copper formed per metric ton of $SO_2$ that is produced.

Convert mass of $SO_2$ to moles, apply the molar ratio from the overall equation to find mol of Cu, and then convert mol to mass (kg) of Cu.

Convert from mass of $SO_2$ to moles of $SO_2$ :

$$\text{Moles of } SO_2 = 1.00 \cancel{\text{ t } SO_2} \times \frac{10^3 \cancel{\text{ kg}}}{1 \cancel{\text{ t}}} \times \frac{10^3 \cancel{\text{ g}}}{1 \cancel{\text{ kg}}} \times \frac{1 \text{ mol } SO_2}{64.07 \cancel{\text{ g } SO_2}}$$

$$= 1.56 \times 10^4 \text{ mol } SO_2$$

Convert from moles of $SO_2$ to moles of Cu:

$$\text{Moles of Cu} = 1.56 \times 10^4 \cancel{\text{ mol } SO_2} \times \frac{4 \text{ mol Cu}}{2 \cancel{\text{ mol } SO_2}} = 3.12 \times 10^4 \text{ mol Cu}$$

Convert from moles of Cu to mass of Cu:

$$\text{Mass (kg) of Cu} = 3.12 \times 10^4 \cancel{\text{ mol Cu}} \times \frac{63.55 \cancel{\text{ g}} \text{ Cu}}{1 \cancel{\text{ mol Cu}}} \times \frac{1 \text{ kg}}{10^3 \cancel{\text{ g}}}$$

$$= \mathbf{1.98 \times 10^3 \text{ kg Cu}}$$

## Concept 9 Why one reactant limits the yield of product

- If amounts of two reactants are not in the stoichiometric ratio of a reaction, one will be in excess; the other is the limiting reactant.

**Problem**: Rocket fuel, [hydrazine ($N_2H_4$) and dinitrogen tetraoxide ($N_2O_4$)], reacts to form $N_2$ gas and water vapor. How many grams of nitrogen gas form from $1.00 \times 10^2$ g $N_2H_4$ and $2.00 \times 10^2$ g $N_2O_4$?

Write the balanced equation:

$$2N_2H_4(l) + N_2O_4(l) \longrightarrow 3N_2(g) + 4H_2O(g)$$

Convert mass of reactants to mol and find the mol of $N_2$ each forms.

$$\text{Moles of } N_2H_4 = 1.00 \times 10^2 \cancel{\text{ g } N_2H_4} \times \frac{1 \text{ mol } N_2H_4}{32.05 \cancel{\text{ g } N_2H_4}} = 3.12 \text{ mol } N_2H_4$$

$$\text{Moles of } N_2 = 3.12 \cancel{\text{ mol } N_2H_4} \times \frac{3 \text{ mol } N_2}{2 \cancel{\text{ mol } N_2H_4}} = 4.68 \text{ mol } N_2$$

- Whichever yields less $N_2$ is the limiting reactant.

$$\text{Moles of } N_2O_4 = 2.00 \times 10^2 \text{ g } N_2O_4 \times \frac{1 \text{ mol } N_2O_4}{92.02 \text{ g } N_2O_4} = 2.17 \text{ mol } N_2O_4$$

$$\text{Moles of } N_2 = 2.17 \text{ mol } N_2O_4 \times \frac{3 \text{ mol } N_2}{1 \text{ mol } N_2O_4} = 6.51 \text{ mol } N_2$$

$N_2H_4$ is the limiting reactant because less N 2 can form (4.68 mol) when all the $N_2H_4$ reacts.

- Convert the lower number of moles of $N_2$ to mass of N 2.

$$\text{Mass (g) of } N_2 = 4.68 \text{ mol } N_2 \times \frac{28.02 \text{ g } N_2}{1 \text{ mol } N_2} = \mathbf{131 \text{ g } N_2}$$

## Concept 10. The causes of lower-than-expected yields and the distinction between theoretical and actual yields

- Ideally 100% of the limiting reactant becomes product.

- This is the theoretical yield, amount of product indicated by the stoichiometrically equivalent molar ratio in balanced equation.

- The theoretical yield is rarely obtained due to several factors.

- The amount of product actually obtained is the actual yield.

- The percent yield (% yield) is the actual yield expressed as a percent of the theoretical yield:

$$\% \text{ yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

**Problem**: Silicon carbide (SiC) is made by heating sand (silicon dioxide) with carbon. Carbon monoxide is also formed. From 100.0 kg sand, 51.4 kg SiC are formed. What is the percent yield of SiC?

**Plan**: Given the actual yield of SiC, need its theoretical yield to be able to calculate its percent yield. **Steps**:

-Write the balanced equation, convert mass of $SiO_2$ to mol, find the mol of SiC from the molar ratio, and convert mol of SiC to mass to obtain the theoretical yield and the percent yield.

Balanced equation. $SiO_2(s) + 3C(s) \rightarrow SiC(s) + 2CO(g)$

Convert the mass of $SiO_2$ to moles:

$$\text{Moles of } SiO_2 = 100.0 \cancel{\text{ kg } SiO_2} \times \frac{1000 \cancel{\text{ g}}}{1 \cancel{\text{ kg}}} \times \frac{1 \text{ mol } SiO_2}{60.09 \cancel{\text{ g } SiO_2}} = 1664 \text{ mol } SiO_2$$

$$\text{Moles of } SiO_2 = \text{moles of SiC} = 1664 \text{ mol SiC}$$

Converting from moles of SiC to mass:

$$\text{Mass (kg) of SiC} = 1664\ \cancel{\text{mol SiC}} \times \frac{40.10\ \text{g SiC}}{1\ \cancel{\text{mol SiC}}} \times \frac{1\ \text{kg}}{1000\ \cancel{\text{g}}} = 66.73\ \text{kg SiC}$$

Calculating the percent yield:

$$\%\ \text{yield SiC} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 = \frac{51.4\ \cancel{\text{kg}}}{66.73\ \cancel{\text{kg}}} \times 100 = \mathbf{77.0\%}$$

- In multistep reaction sequences, the percent yield of each reaction step is multiplied over the whole sequence.
- A six-step sequence has a theoretical yield of 35.0 g of final produc If each step has a 90.0% yield, what would be over-all actual yield?

$$\text{Actual yield} = 35.0\ \text{g} \times 0.900 \times 0.900 \times 0.900 \times 0.900 \times 0.900 \times 0.900 = 18.6\ \text{g}$$

$$\%\ \text{yield} = \frac{18.6\ \text{g}}{35.0\ \text{g}} \times 100 = 53.1\%$$

## Concept 11. The meanings of concentration and molarity

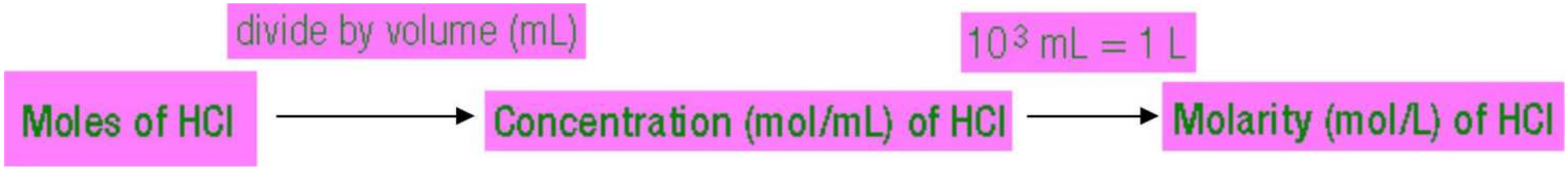
### Solution Concentration and the Calculation of Molarity

- Most reactions occur in solution, so quantitative aspects of solution reactions are important in chemistry and other sciences
- Need to know the concentration of reactants - the number of mol in a given volume - to calculate the volume of solution that contains a given number of mol.
- Solutions consist of a lesser amount the **solute**, dissolved in a larger amount of another substance, the **solvent**.
- Molarity (M) expresses solution concentration in terms of moles of solute per liter of solution:

$$\text{Molarity} = \frac{\text{moles of solute}}{\text{liters of solution}} \quad \text{or} \quad M = \frac{\text{mol solute}}{\text{L soln}}$$

**Problem**: Hydrochloric acid is a solution of HCl gas in water. Calculate the molarity of hydrochloric acid solution if 455 mL contains 1.80 mol hydrogen chloride.

**Plan**:

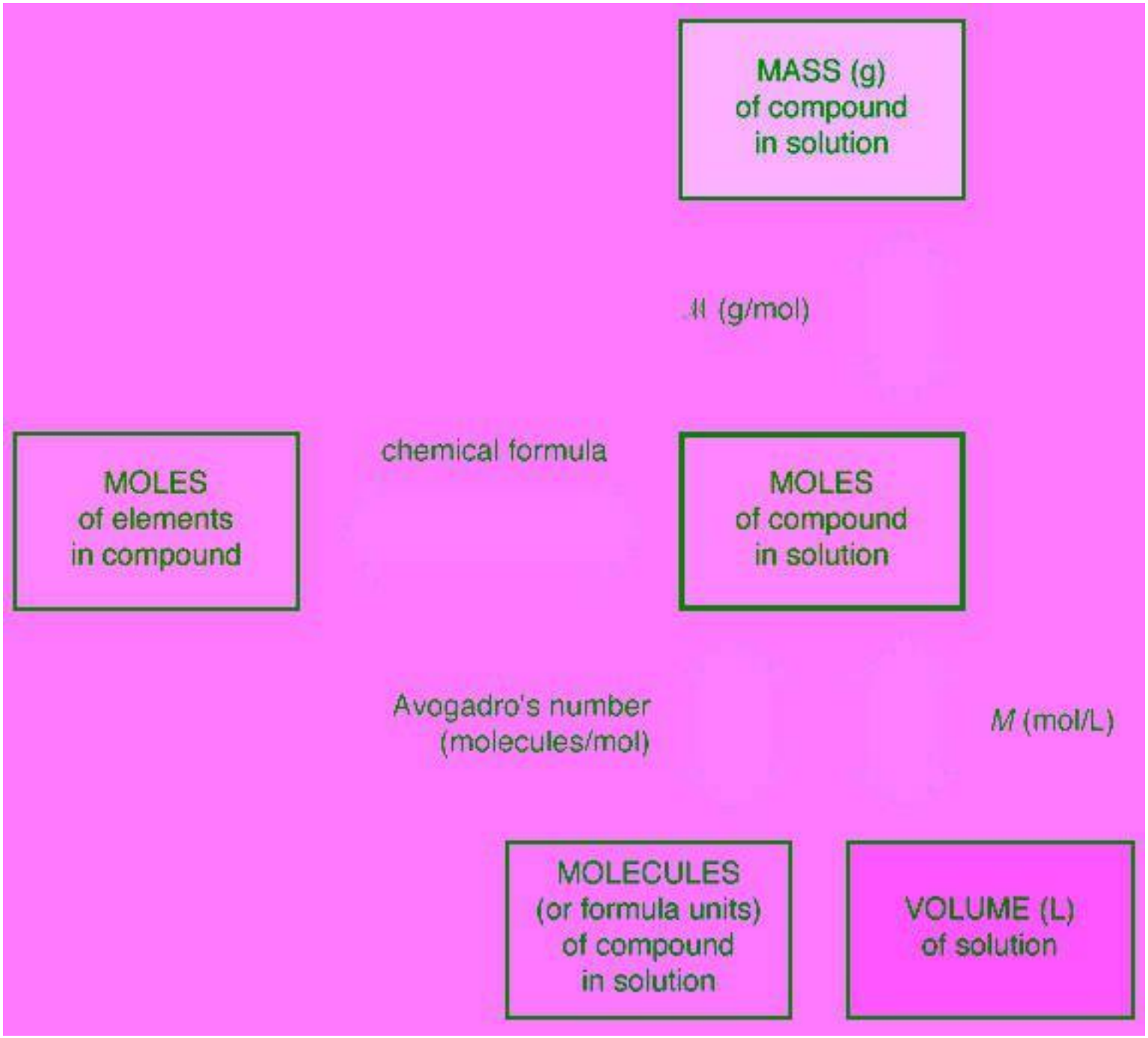



**Solution**:

$$\text{Molarity} = \frac{1.80 \text{ mol HCl}}{455 \text{ mL soln}} \times \frac{1000 \text{ mL}}{1 \text{ L}} = 3.96 \ M \text{ HCl}$$

- Molarity is a conversion factor that extends the stoichiometric relationships among substances.

- It can be used to convert between volume of solution and moles of solute.


MASS (g)
of compound
in solution
M (g/mol)
MOLES
of elements
in compound
chemical formula
MOLES
of compound
in solution
Avogadro's number
(molecules/mol)
M (mol/L)
MOLECULES
(or formula units)
of compound
in solution
VOLUME (L)
of solution

$$\text{Mass (g) of solute} = 0.500 \cancel{\text{ L soln}} \times \frac{0.350 \cancel{\text{ mol Ni(NO}_3)_2\cdot 6\text{H}_2\text{O}}}{1 \cancel{\text{ L soln}}}$$

$$\times \frac{290.83 \text{ g Ni(NO}_3)_2\cdot 6\text{H}_2\text{O}}{1 \cancel{\text{ mol Ni(NO}_3)_2\cdot 6\text{H}_2\text{O}}} = 50.9 \text{ g Ni(NO}_3)_2\cdot 6\text{H}_2\text{O}$$

$$\text{Volume (L) of HCl} = 3.4 \times 10^{-3}\ \cancel{\text{mol HCl}} \times \frac{1\ \text{L}}{0.10\ \cancel{\text{mol HCl}}} = \mathbf{3.4 \times 10^{-2}\ L}$$

# Chapter 3: Stoichiometry

## 3.1 Amount - Mass Relationships in Chemical Systems

### 3.1 The Mole

### 3.2 Determining the Formula of an Unknown Compound

### 3.3 Writing and Balancing Chemical Equations

### 3.4 Calculating Quantities of Reactant and Product

### 3.5 Fundamentals of Solution Stoichiometry

# 3.2 The Mole

**The mole (mol)** is the amount of a substance that contains the same number of entities as there are atoms in exactly 0.012 kg of carbon-12.

The term "***entities***" refers to atoms, ions, molecules, formula units, or electrons – in fact, any type of particle.

**One mole (1 mol) contains $6.022 \times 10^{23}$ entities (to four significant figures).**

This number is called Avogadro's number and is abbreviated as $N_A$.

**Figure 3.1** One mole ($6.022 \times 10^{23}$ entities) of some familiar substances.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



© The McGraw-Hill Companies, Inc./Charles Winters Photographer

# 3.3 Molar Mass

**The molar mass** ($\mathcal{M}$) of a substance is the mass per mole of its entites (atoms, molecules or formula units).

For **monatomic elements**, the molar mass is the same as the atomic mass in grams per mole. The atomic mass is simply read from the Periodic Table.

The molar mass of Ne = 20.18 g/mol.

For **molecular elements** and for **compounds**, the formula is needed to determine the molar mass.

The molar mass of $O_2$ = 2 x $\mathcal{M}$ of O
= 2 x 16.00
= 32.00 g/mol

The molar mass of $SO_2$ = 1 x $\mathcal{M}$ of S + 2 x $\mathcal{M}$ of O
= 32.00 + 2(16.00)
= 64.00 g/mol

3-6

## Table 3.1 Information Contained in the Chemical Formula of Glucose $C_6H_{12}O_6$ ( $\mathcal{M}$ = 180.16 g/mol)

| | Carbon (C) | Hydrogen (H) | Oxygen (O) |
|---|---|---|---|
| Atoms/molecule of compound | 6 atoms | 12 atoms | 6 atoms |
| Moles of atoms/mole of compound | 6 mol of atoms | 12 mol of atoms | 6 mol of atoms |
| Atoms/mole of compound | $6(6.022 \times 10^{23})$ atoms | $12(6.022 \times 10^{23})$ atoms | $6(6.022 \times 10^{23})$ atoms |
| Mass/molecule of compound | 6(12.01 u) = 72.06 u | 12(1.008 u) = 12.10 u | 6(16.00 u) = 96.00 u |
| Mass/mole of compound | 72.06 g | 12.10 g | 96.00 g |

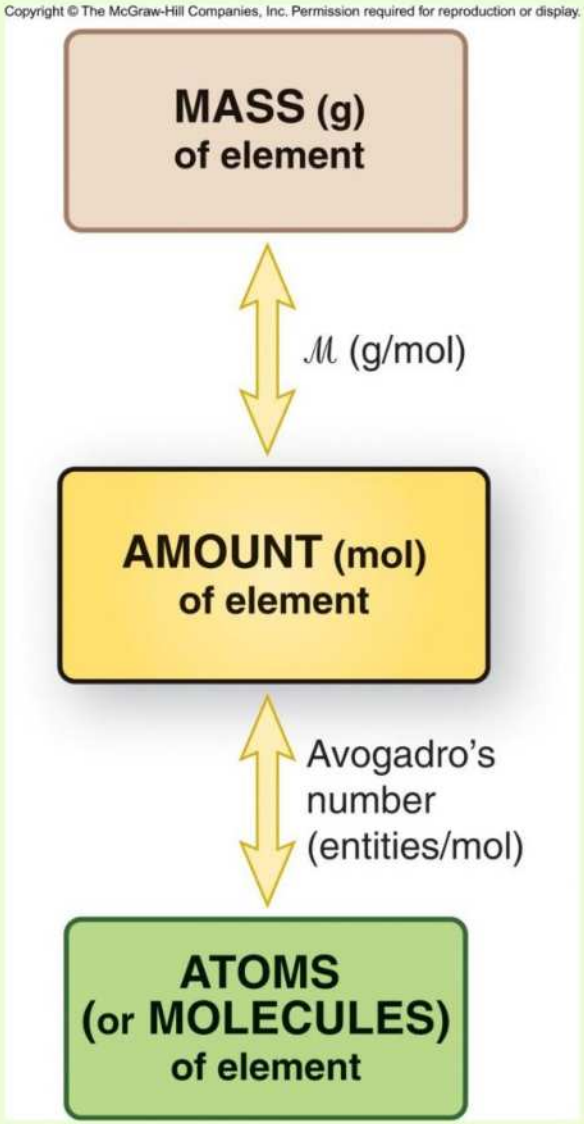
# Interconverting Moles, Mass, and Number of Chemical Entities

$$n\,(mol) = \frac{mass\,(g)}{molar\,mass\,(\,g \cdot mol^{-1})}$$

$$amount\,(mol) = \frac{N\,(entities)}{N_A\,(mol^{-1})}$$

**Figure 3.2 Mass-mole-number relationships for elements.**

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



## Sample Problem 3.1 Calculating the Mass of a Given Amount of an Element

**PROBLEM:** Silver (Ag) is used in jewelry and tableware but no longer in U.S. coins. How many grams of Ag are in 0.0342 mol of Ag?

**PLAN:** To convert mol of Ag to mass of Ag in g we need the molar mass of Ag.

**amount (mol) of Ag**

**multiply by $\mathcal{M}$ of Ag (107.9 g/mol)**

**mass (g) of Ag**

**SOLUTION:**

$$0.0342\,\text{mol Ag} \times \frac{107.9\,\text{g Ag}}{1\,\text{mol Ag}} = \mathbf{3.69\ g\ Ag}$$

3-10

## Sample Problem 3.2 Calculating the Number of Entities in a Given Amount of an Element

**PROBLEM:** Gallium (Ga) is a key element in solar panels, calculators and other light-sensitive electronic devices. How many Ga atoms are in $2.85 \times 10^{-3}$ mol of gallium?

**PLAN:** To convert mol of Ga to number of Ga atoms we need to use Avogadro's number.

**mol of Ga**

**multiply by $6.022 \times 10^{23}$ atoms/mol**

**atoms of Ga**

**Sample Problem 3.2**

**SOLUTION:**

$$2.85 \times 10^{-3} \text{ mol Ga atoms} \times \frac{6.022 \times 10^{23} \text{ Ga atoms}}{1 \text{ mol Ga atoms}}$$

$$= \mathbf{1.72 \times 10^{21} \text{ Ga atoms}}$$

## Sample Problem 3.3 Calculating the Number of Entities in a Given Mass of an Element

**PROBLEM:** Iron (Fe) is the main component of steel and is therefore the most important metal in society; it is also essential in the body. How many Fe atoms are in 95.8 g of Fe?

**PLAN:** The number of atoms cannot be calculated directly from the mass. We must first determine the number of moles of Fe atoms in the sample and then use Avogadro's number.

**mass (g) of Fe**

**divide by $\mathcal{M}$ of Fe (55.85 g/mol)**

**amount (mol) of Fe**

**multiply by $6.022 \times 10^{23}$ atoms/mol**

**atoms of Fe**

**Sample Problem 3.3**

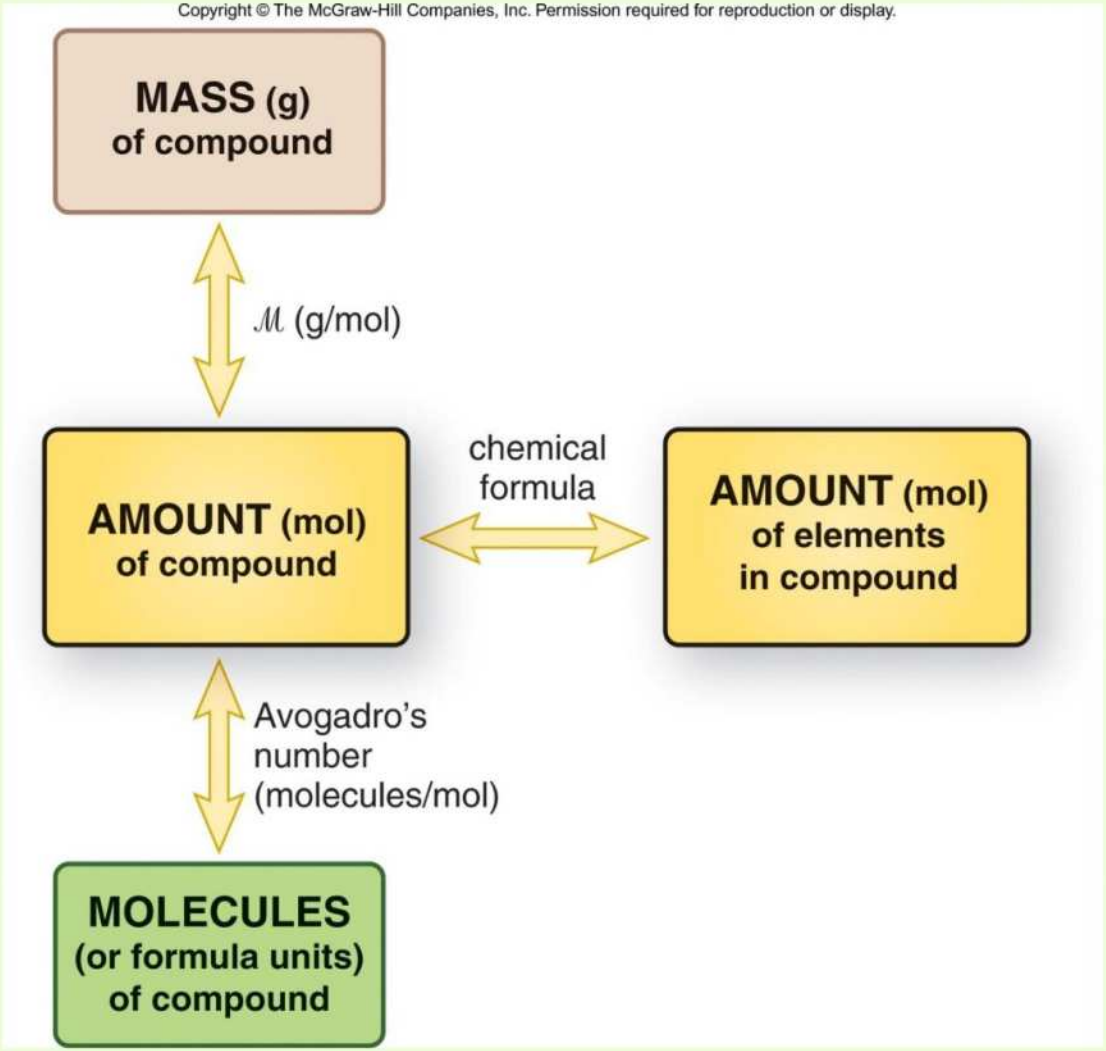
**SOLUTION:**

$$95.8\ \text{g Fe} \times \frac{1\ \text{mol Fe}}{55.85\ \text{g Fe}} = 1.72\ \text{mol Fe}$$

$$1.72\ \text{mol Fe} \times \frac{6.022 \times 10^{23}\ \text{atoms Fe}}{1\ \text{mol Fe}}$$

**$= 1.04 \times 10^{24}$ atoms Fe**

## Figure 3.3 Amount-mass-number relationships for compounds.



## Sample Problem 3.4 Calculating the Number of Chemical Entities in a Given Mass of a Compound I

**PROBLEM:** Nitrogen dioxide is a component of urban smog that forms from the gases in car exhausts. How many molecules are in 8.92 g of nitrogen dioxide?

**PLAN:** Write the formula for the compound and calculate its molar mass. Use the given mass to calculate first the number of moles and then the number of molecules.

**mass (g) of $NO_2$**

**divide by $\mathcal{M}$**

**amount (mol) of $NO_2$**



**multiply by $6.022 \times 10^{23}$ formula units/mol**

**number of $NO_2$ molecules**

**Sample Problem 3.4**

**SOLUTION:** $NO_2$ is the formula for nitrogen dioxide.

$\mathcal{M}$ = (1 x $\mathcal{M}$ of N) + (2 x $\mathcal{M}$ of O)

= 14.01 g/mol + 2(16.00 g/mol)

= 46.01 g/mol

$$8.92 \text{ g } NO_2 \times \frac{1 \text{ mol } NO_2}{46.01 \text{ g } NO_2} = 0.194 \text{ mol } NO_2$$

$$0.194 \text{ mol } NO_2 \times \frac{6.022 \times 10^{23} \text{ molecules } NO_2}{1 \text{ mol } NO_2}$$

**= 1.17 x $10^{23}$ molecules $NO_2$**

3-17

**Sample Problem 3.5** **Calculating the Number of Chemical Entities in a Given Mass of a Compound II**

**PROBLEM:** Ammonium carbonate, a white solid that decomposes on warming, is an component of baking powder.

a) How many formula units are in 41.6 g of ammonium carbonate?

b) How many O atoms are in this sample?

**PLAN:**

Write the formula for the compound and calculate its molar mass. Use the given mass to calculate first the number of moles and then the number of formula units.

The number of O atoms can be determined using the formula and the number of formula units.

**Sample Problem 3.5**

**mass (g) of $(NH_4)_2CO_3$**

**divide by $\mathcal{M}$**

**amount (mol) of $(NH_4)_2CO_3$**

**multiply by $6.022 \times 10^{23}$ formula units/mol**

**number of $(NH_4)_2CO_3$ formula units**

**3 O atoms per formula unit of $(NH_4)_2CO_3$**

**number of O atoms**

**SOLUTION:** $(NH_4)_2CO_3$ is the formula for ammonium carbonate.

$\mathcal{M}$ = (2 x *M* of N) + (8 x $\mathcal{M}$ of H) + (1 x $\mathcal{M}$ of C) + (3 x $\mathcal{M}$ of O)

= (2 x 14.01 g/mol) + (8 x 1.008 g/mol)
+ (12.01 g/mol) + (3 x 16.00 g/mol)

= **96.09 g/mol**

**Sample Problem 3.5**

$$41.6\,\text{g}\ (NH_4)_2CO_3 \times \frac{1\ \text{mol}\ (NH_4)_2CO_3}{96.09\ \text{g}\ (NH_4)_2CO_3} = 0.433\ \text{mol}\ (NH_4)_2CO_3$$

$$0.433\,\text{mol}\ (NH_4)_2CO_3 \times \frac{6.022\times10^{23}\ \text{formula units}\ (NH_4)_2CO_3}{1\ \text{mol}\ (NH_4)_2CO_3}$$

$$\mathbf{= 2.61\times10^{23}\ formula\ units\ (NH_4)_2CO_3}$$

$$2.61\times10^{23}\ \text{formula units}\ (NH_4)_2CO_3 \times \frac{3\ \text{O atoms}}{1\ \text{formula unit of}\ (NH_4)_2CO_3}$$

$$\mathbf{= 7.83\times10^{23}\ O\ atoms}$$

# Mass Percent from the Chemical Formula

Mass % of element X =

$$\frac{\text{atoms of X in formula} \times \text{atomic mass of X (u)}}{\text{molecular (or formula) mass of compound (u)}} \times 100$$

Mass % of element X =

$$\frac{\text{amount of X in formula (mol)} \times \text{molar mass of X (g/mol)}}{\text{mass (g) of 1 mol of compound}} \times 100$$

## Sample Problem 3.6 Calculating the Mass Percent of Each Element in a Compound from the Formula

**PROBLEM:** Glucose ($C_6H_{12}O_6$) is a key nutrient for generating chemical potential energy in biological systems. What is the mass percent of each element in glucose?

**PLAN:** Find the molar mass of glucose, which is the mass of 1 mole of glucose. Find the mass of each element in 1 mole of glucose, using the molecular formula.

The mass % for each element is calculated by dividing the mass of that element in 1 mole of glucose by the total mass of 1 mole of glucose, multiplied by 100.

## Sample Problem 3.6

**PLAN:**

**amount (mol) of element X in 1 mol compound**

**multiply by $\mathcal{M}$ (g/mol) of X**

**mass (g) of X in 1 mol of compound**

**divide by mass (g) of 1 mol of compound**

**mass fraction of X**

**multiply by 100**

**mass % X in compound**

**Sample Problem 3.6**

**SOLUTION:**

In 1 mole of glucose there are **6** moles of C, **12** moles H, and **6** moles O.

$$6 \text{ mol C} \times \frac{12.01 \text{ g C}}{1 \text{ mol C}} = 72.06 \text{ g C} \qquad 12 \text{ mol H} \times \frac{1.008 \text{ g H}}{1 \text{ mol H}} = 12.096 \text{ g H}$$

$$6 \text{ mol O} \times \frac{16.00 \text{ g O}}{1 \text{ mol O}} = 96.00 \text{ g O} \qquad \mathcal{M} = 180.16 \text{ g/mol}$$

$$\text{mass percent of C} = \frac{72.06 \text{ g C}}{180.16 \text{ g glucose}} = 0.3999 \times 100 = 39.99 \text{ mass \%C}$$

$$\text{mass percent of H} = \frac{12.096 \text{ g H}}{180.16 \text{ g glucose}} = 0.06714 \times 100 = 6.714 \text{ mass \%H}$$

$$\text{mass percent of O} = \frac{96.00 \text{ g O}}{} \times 100 = 53.29 \text{ mass \%O}$$

$$\frac{}{180.16\ \text{g glucose}} = 0.5329$$

# Mass Percent and the Mass of an Element

Mass percent can also be used to calculate the mass of a particular element in any mass of a compound.

Mass of element X present in sample =

$$\text{mass of compound} \times \frac{\text{mass of element in 1 mol of compound}}{\text{mass of 1 mol of compound}}$$

**Sample Problem 3.7**

## Calculating the Mass of an Element in a Compound

**PROBLEM:** Use the information from Sample Problem 3.6 to determine the mass (g) of carbon in 16.55 g of glucose.

**PLAN:** The mass percent of carbon in glucose gives us the relative mass of carbon in 1 mole of glucose. We can use this information to find the mass of carbon in any sample of glucose.

**mass of glucose sample**

**multiply by mass percent of C in glucose**

**mass of C in sample**

**Sample Problem 3.7**

**SOLUTION:**

Each mol of glucose contains 6 mol of C, or 72.06 g of C.

$$\text{Mass (g) of C} = \text{mass (g) of glucose} \times \frac{6 \text{ mol} \times \mathcal{M} \text{ of C (g/mol)}}{\text{mass (g) of 1 mol of glucose}}$$

$$= 16.55 \text{ g glucose} \times \frac{72.06 \text{ g C}}{180.16 \text{ g glucose}} = \boxed{\textbf{6.620 g C}}$$

# Empirical and Molecular Formulas

**The empirical formula** is the simplest formula for a compound that agrees with the elemental analysis. It shows the ***lowest*** whole number of moles and gives the ***relative*** number of atoms of each element present.

The empirical formula for hydrogen peroxide is HO.

**The molecular formula** shows the ***actual*** number of atoms of each element in a molecule of the compound.

The molecular formula for hydrogen peroxide is $H_2O_2$.

**Sample Problem 3.8**

## Determining an Empirical Formula from Amounts of Elements

**PROBLEM:** A sample of an unknown compound contains 0.21 mol of zinc, 0.14 mol of phosphorus, and 0.56 mol of oxygen. What is its empirical formula?

**PLAN:** Find the relative number of moles of each element. Divide by the lowest mol amount to find the relative mol ratios (empirical formula).

**amount (mol) of each element**

**use # of moles as subscripts**

**preliminary formula**



**change to integer subscripts**

**empirical formula**

**Sample Problem 3.8**

**SOLUTION:** Using the numbers of moles of each element given, we write the preliminary formula $Zn_{0.21}P_{0.14}O_{0.56}$

Next we divide each fraction by the smallest one; in this case 0.14:

$$\frac{0.21}{0.14} = 1.5 \qquad \frac{0.14}{0.14} = 1.0 \qquad \frac{0.56}{0.14} = 4.0$$

This gives $Zn_{1.5}P_{1.0}O_{4.0}$

We convert to whole numbers by multiplying by the ***smallest integer*** that gives whole numbers; in this case 2:

1.5 x 2 = **3** 1.0 x 2 = **2** 4.0 x 2 = **8**

**This gives us the empirical formula $Zn_3P_2O_8$**

## Sample Problem 3.9 Determining an Empirical Formula from Masses of Elements

**PROBLEM:** Analysis of a sample of an ionic compound yields 2.82 g of Na, 4.35 g of Cl, and 7.83 g of O. What is the empirical formula and the name of the compound?

**PLAN:** Find the relative number of moles of each element. Divide by the lowest mol amount to find the relative mol ratios (empirical formula).

**mass (g) of each element**

**divide by $\mathcal{M}$ (g/mol)**

**amount (mol) of each element**

**use # of moles as subscripts**

**preliminary formula**

**change to integer subscripts**

**empirical formula**

**Sample Problem 3.9**

**SOLUTION:** $2.82 \text{ g Na} \times \frac{1 \text{ mol Na}}{22.99 \text{ g Na}} = 0.123 \text{ mol Na}$

$$4.35 \text{ g Cl} \times \frac{1 \text{ mol Cl}}{35.45 \text{ g Cl}} = 0.123 \text{ mol Cl}$$

$$7.83 \text{ g O} \times \frac{1 \text{ mol O}}{16.00 \text{ g O}} = 0.489 \text{ mol O}$$

$$\text{Na and Cl} = \frac{0.123}{0.123} = 1 \text{ and O} = \frac{0.489}{0.123} = 3.98$$

**The empirical formula is $Na_1Cl_1O_{3.98}$ or $NaClO_4$;**

**this compound is named sodium perchlorate.**

# Determining the Molecular Formula

The molecular formula gives the ***actual*** numbers of moles of each element present in 1 mol of compound.

The molecular formula is a ***whole-number multiple*** of the empirical formula.

$$\frac{\text{molar mass (g/mol)}}{\text{empirical formula mass (g/mol)}} = \text{whole-number multiple}$$

**Sample Problem 3.10**

## Determining a Molecular Formula from Elemental Analysis and Molar Mass

**PROBLEM:** Elemental analysis of lactic acid ($\mathcal{M}$ = 90.08 g/mol) shows it contains 40.0 mass % C, 6.71 mass % H, and 53.3 mass % O. Determine the empirical formula and the molecular formula for lactic acid.

**PLAN:** **assume 100 g lactic acid; then mass % = mass in grams**

**divide each mass by $\mathcal{M}$**

**amount (mol) of each element**

**use # mols as subscripts; convert to integers**

**empirical formula**

**divide $\mathcal{M}$ by the molar mass for the empirical formula; multiply empirical formula by this number**

**molecular formula**

**Sample Problem 3.10**

**SOLUTION:** Assuming there are 100. g of lactic acid;

$$40.0\ \text{g C} \times \frac{1\ \text{mol C}}{12.01\ \text{g C}} = 3.33\ \text{mol C}$$

$$6.71\ \text{g H} \times \frac{1\ \text{mol H}}{1.008\ \text{g H}} = 6.66\ \text{mol H}$$

$$53.3\ \text{g O} \times \frac{1\ \text{mol O}}{16.00\ \text{g O}} = 3.33\ \text{mol O}$$

$$C_{\frac{3.33}{3.33}}\ H_{\frac{6.66}{3.33}}\ O_{\frac{3.33}{3.33}} \longrightarrow CH_2O \quad \text{empirical formula}$$

molar mass of lactate $\dfrac{90.08\ \text{g/mol}}{30.03\ \text{g/mol}} = 3$

mass of $CH_2O$

**$C_3H_6O_3$ is the molecular formula**

## Sample Problem 3.11 Determining a Molecular Formula from Combustion Analysis

**PROBLEM:**

When a 1.000 g sample of vitamin C ($\mathcal{M}$ = 176.12 g/mol) is placed in a combustion chamber and burned, the following data are obtained:

mass of $CO_2$ absorber after combustion = 85.35 g
mass of $CO_2$ absorber before combustion = 83.85 g
mass of $H_2O$ absorber after combustion = 37.96 g
mass of $H_2O$ absorber before combustion = 37.55 g

What is the molecular formula of vitamin C?

**PLAN:** The masses of $CO_2$ and $H_2O$ produced will give us the masses of C and H present in the original sample. From this we can determine the mass of O.

## Sample Problem 3.11

**(mass after combustion – mass before) for each absorber = mass of compound in each absorber**

↓ mass of each compound x mass % of oxidized element

**mass of each oxidized element**

mass of vitamin C – (mass of C + H)

**mass of O**

↓ divide each mass by $\mathcal{M}$

**mol of C, H, and O**

use # mols as subscripts; convert to integers

**empirical formula**

**molecular formula**

## Sample Problem 3.11

**SOLUTION:** For $CO_2$: 85.35 g - 83.85 g = 1.50 g

$$1.50 \text{ g } CO_2 \times \frac{12.01 \text{ g C}}{44.01 \text{ g } CO_2} = 0.409 \text{ g C}$$

For $H_2O$: 37.96 g - 37.55 g = 0.41 g

$$0.41 \text{ g } H_2O \times \frac{2.016 \text{ g H}}{18.02 \text{ g } H_2O} = 0.046 \text{ g H}$$

mass of O = mass of vitamin C – (mass of C + mass of H) = 1.000 g - (0.409 + 0.046) g = 0.545 g O

**Sample Problem 3.11**

Convert mass to moles:

$$\frac{0.409\ \text{g C}}{12.01\ \text{g/mol C}} = 0.0341\ \text{mol C} \qquad \frac{0.046\ \text{g H}}{1.008\ \text{g/mol H}} = 0.0456\ \text{mol H}$$

$$\frac{0.545\ \text{g O}}{16.00\ \text{g/mol O}} = 0.0341\ \text{mol O}$$

Divide by smallest to get the preliminary formula:

$$\text{C}\ \frac{0.0341}{0.0341} = 1 \qquad \text{H}\ \frac{0.0456}{0.0341} = 1.34 \qquad \text{O}\ \frac{0.0341}{0.0341} = 1$$

$$\text{C}_{1}\text{H}_{1.34}\text{O}_{1} = \text{C}_{3}\text{H}_{4.01}\text{O}_{3} \qquad \text{C}_{3}\text{H}_{4}\text{O}_{3}$$

Divide molar mass by mass of empirical formula:

$$\frac{176.12\ \text{g/mol}}{88.06\ \text{g}} = 2.000\ \text{mol}$$

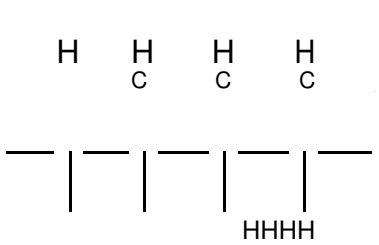
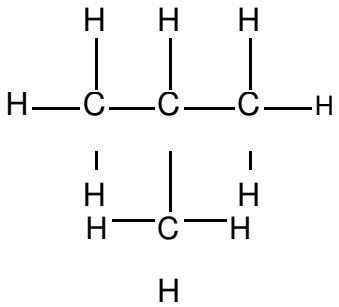
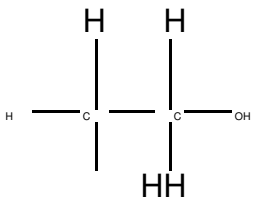
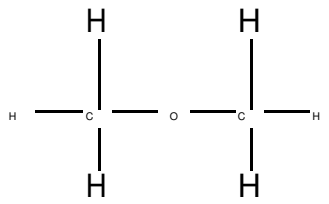
**$C_6H_8O_6$**

**Table 3.2 Some Compounds with Empirical Formula $CH_2O$ (Composition by Mass: 40.0% C, 6.71% H, 53.3% O)**

| Name | Molecular Formula | Whole-Number Multiple | $\mathcal{M}$ (g/mol) | Use or Function |
|---|---|---|---|---|
| formaldehyde | $CH_2O$ | 1 | 30.03 | |
| acetic acid | $C_2H_4O_2$ | 2 | 60.05 | disinfectant; biological preservative |
| lactic acid | $C_3H_6O_3$ | 3 | 90.09 | acetate polymers; vinegar (5% soln) |
| erythrose | $C_4H_8O_4$ | 4 | 120.10 | sour milk; forms in exercising muscle |
| ribose | $C_5H_{10}O_5$ | 5 | 150.13 | part of sugar metabolism |
| glucose | $C_6H_{12}O_6$ | 6 | 180.16 | component of nucleic acids and $B_2$ major energy source of the cell |

$CH_2OC_2H_4O_2$ $C_3H_6O_3$ $C_4H_8O_4$ $C_5H_{10}O_5$ $C_6H_{12}O_6$

**Table 3.3 Two Pairs of Constitutional Isomers**

| Property | $C_4H_{10}$ Butane | $C_4H_{10}$ 2-Methylpropane | $C_2H_6O$ Ethanol | $C_2H_6O$ Dimethyl Ether |
|---|---|---|---|---|
| $\mathcal{M}$ (g/mol) | 58.12 | 58.12 | 46.07 | 46.07 |
| Boiling Point | $-0.5^0$C | $-11.06^0$C | $78.5^0$C | $-25^0$C |
| Density at $20^0$C | 0.579 g/mL (gas) | 0.549 g/mL (gas) | 0.789 g/mL (liquid) | 0.00195 g/mL (gas) |
| Structural formula |  |  |  |  |
| Space-filling model | | | | |

# Chemical Equations

A ***chemical equation*** uses formulas to express the ***identities*** and ***quantities*** of substances involved in a physical or chemical change.

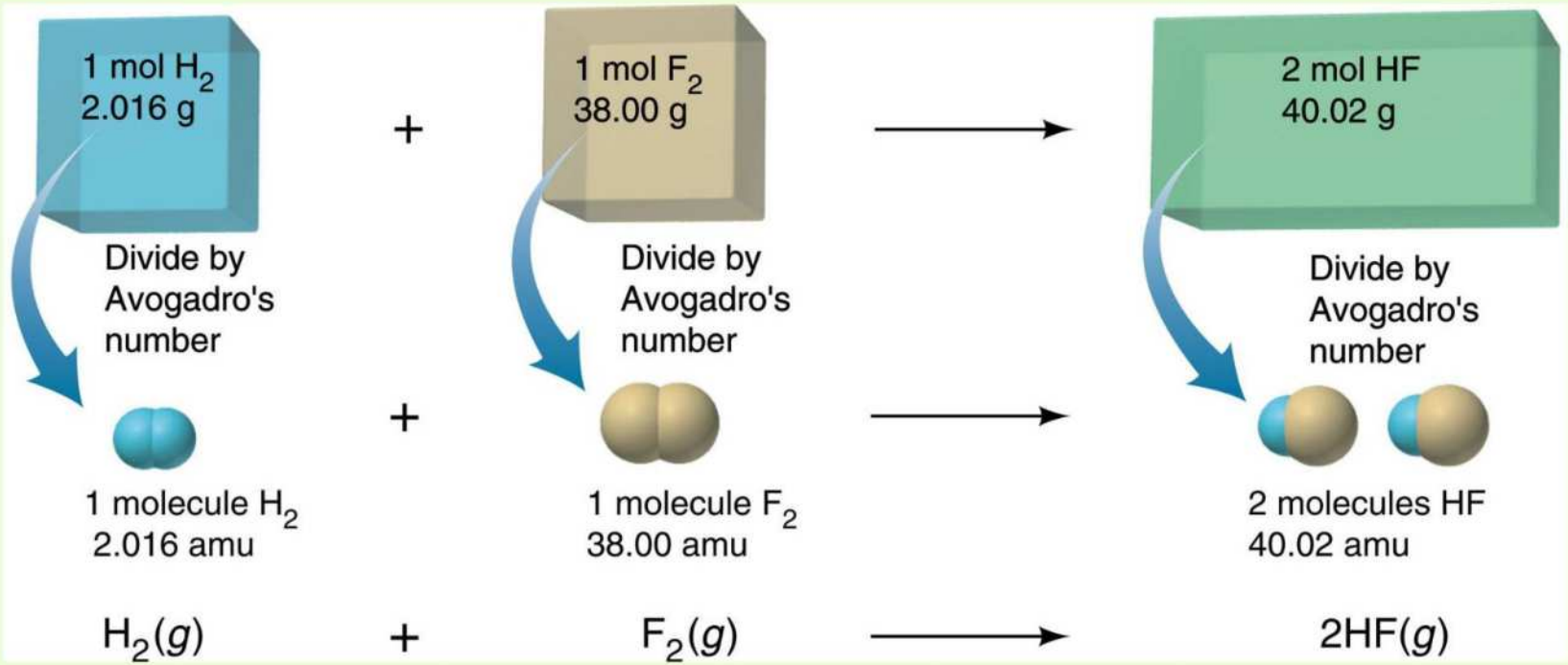
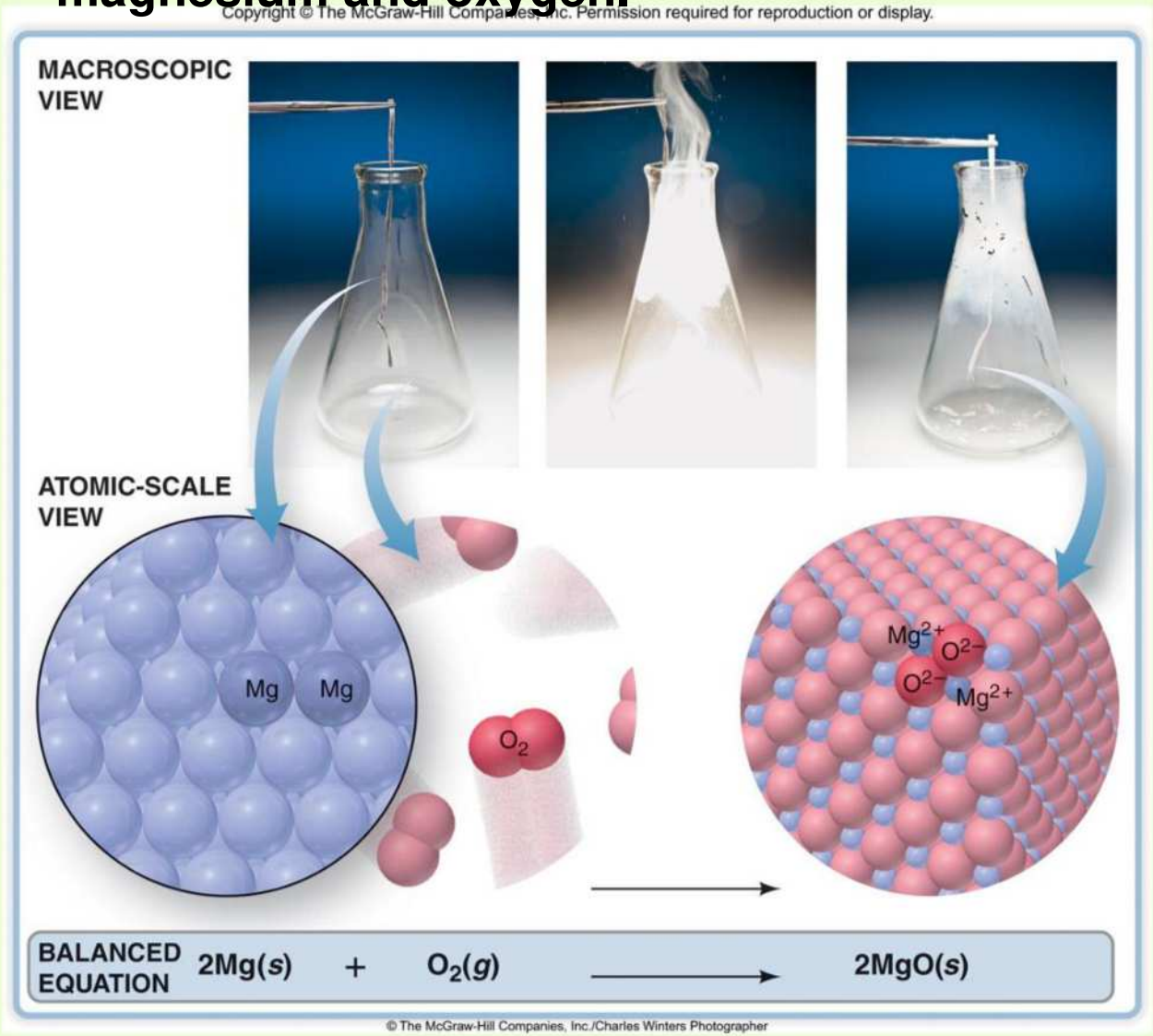



**Figure 3.6**

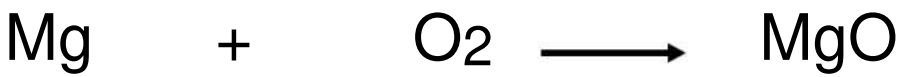
**The formation of HF gas on the macroscopic and molecular levels.**

**Figure 3.7** **A three-level view of the reaction between magnesium and oxygen.**



# Features of Chemical Equations

A **yield arrow** points from reactants to products.




**Reactants** are written on the left.
**Products** are written on the right.

The equation must be **balanced**; the same number and type of each atom must appear on both sides.

# Balancing a Chemical Equation

**translate the statement**

magnesium and oxygen gas react to give magnesium oxide:

$Mg + O_2 \rightarrow MgO$

**balance the atoms using *coefficients*;**
***formulas cannot be changed***

$\mathbf{2}Mg + O_2 \rightarrow \mathbf{2}MgO$

**adjust coefficients if necessary**

**check that all atoms balance**

**specify states of matter**

$\mathbf{2}Mg\ (s) + O_2\ (g) \rightarrow \mathbf{2}MgO\ (s)$

## Sample Problem 3.12 Balancing Chemical Equations

**PROBLEM:** Within the cylinders of a car's engine, the hydrocarbon octane ($C_8H_{18}$), one of many components of gasoline, mixes with oxygen from the air and burns to form carbon dioxide and water vapor. Write a balanced equation for this reaction.

**PLAN:**

**translate the statement**

**balance the atoms**

**adjust the coefficients**

**check the atoms balance**

**specify states of matter**

**SOLUTION:**

$$C_8H_{18} + O_2 \longrightarrow CO_2 + H_2O$$

$$C_8H_{18} + \frac{25}{2}O_2 \longrightarrow 8CO_2 + 9H_2O$$

$$2C_8H_{18} + 25O_2 \longrightarrow 16CO_2 + 18H_2O$$

$$2C_8H_{18} + 25O_2 \longrightarrow 16CO_2 + 18H_2O$$

$$2C_8H_{18}(l) + 25O_2(g) \longrightarrow 16CO_2(g) + 18H_2O(g)$$

# Molecular Scene
# Combustion of Octane

## Sample Problem 3.13 Balancing an Equation from a Molecular Scene

**PROBLEM:** The following molecular scenes depict an important reaction in nitrogen chemistry. The blue spheres represent nitrogen while the red spheres represent oxygen. Write a balanced equation for this reaction.

**PLAN:** Determine the formulas of the reactants and products from their composition. Arrange this information in the correct equation format and balance correctly, including the states of matter.

## Sample Problem 3.13

**SOLUTION:**

The reactant circle shows only one type of molecule, composed of 2 N and 5 O atoms. The formula is thus $N_2O_5$. There are 4 $N_2O_5$ molecules depicted.

The product circle shows two types of molecule; one has 1 N and 2 O atoms while the other has 2 O atoms. The products are $NO_2$ and $O_2$. There are 8 $NO_2$ molecules and 2 $O_2$ molecules shown.

The reaction depicted is **$4\ N_2O_5 \rightarrow 8\ NO_2 + 2\ O_2$**.

Writing the equation with the smallest whole-number coefficients and states of matter included;

**$2\ N_2O_5\ (g) \rightarrow 4\ NO_2\ (g) + O_2\ (g)$**

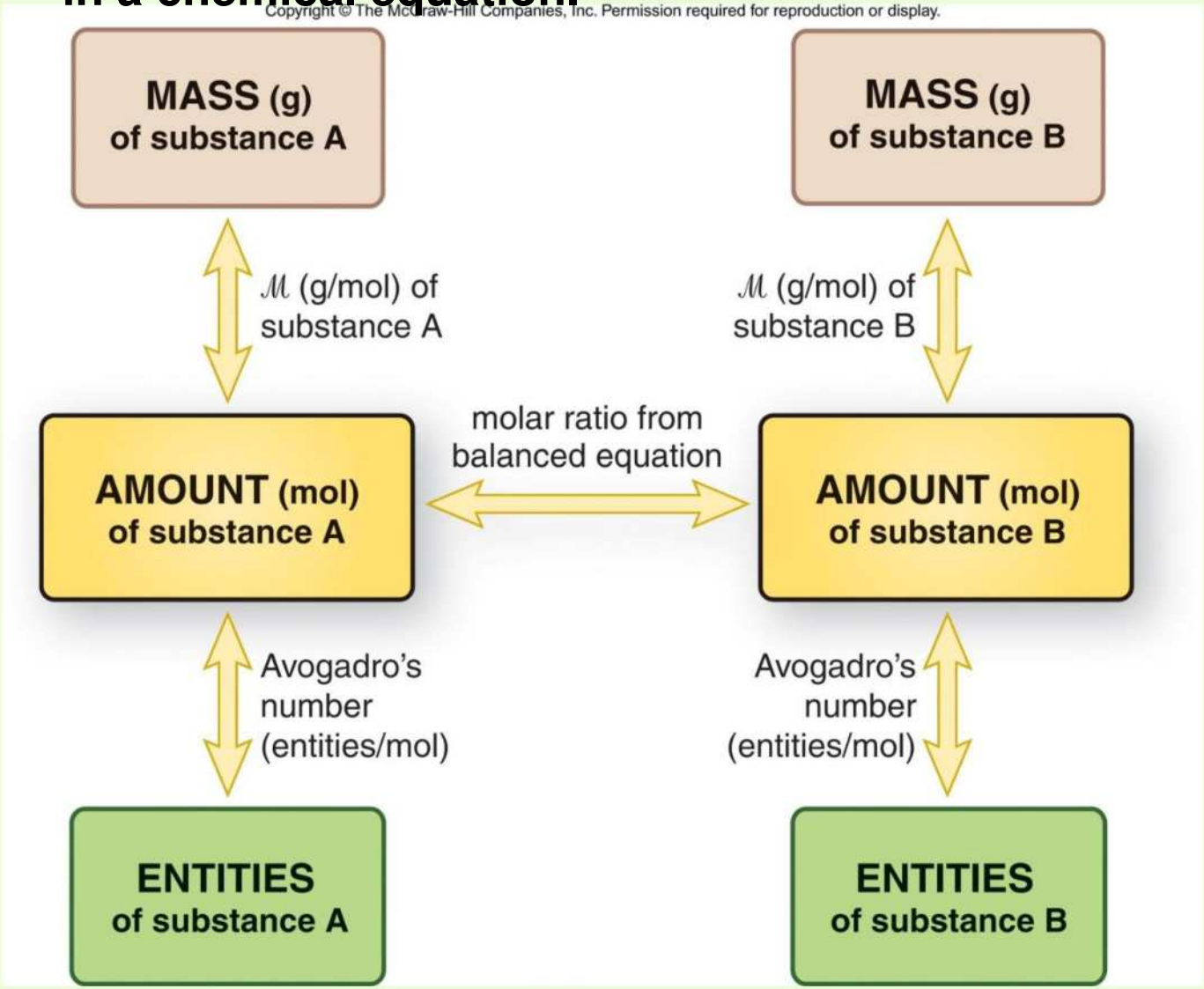
# Stoichiometric Calculations

- The coefficients in a balanced chemical equation
  - represent the relative number of reactant and product particles
  - and the relative number of moles of each.
- Since moles are related to mass
  - the equation can be used to calculate masses of reactants and/or products for a given reaction.
- The mole ratios from the balanced equation are used as conversion factors.

## Table 3.4 Information Contained in a Balanced Equation

| Viewed in Terms of | Reactants $C_3H_8(g) + 5\ O_2(g)$ | Products $3\ CO_2(g) + 4\ H_2O(g)$ |
|---|---|---|
| Molecules | 1 molecule $C_3H_8$ + 5 molecules $O_2$ | 3 molecules $CO_2$ + 4 molecules $H_2O$ |
| (mol) | 1 mol $C_3H_8$ + 5 mol $O_2$ | 3 mol $CO_2$ + 4 mol $H_2O$ |
| Mass (amu) | 44.09 amu $C_3H_8$ + 160.00 amu $O_2$ | 132.03 amu $CO_2$ + 72.06 amu $H_2O$ |
| Mass (g) | 44.09 g $C_3H_8$ + 160.00 g $O_2$ | 132.03 g $CO_2$ + 72.06 g $H_2O$ |
| Total Mass (g) | 204.09 g | 204.09 g |

**Figure 3.8** **Summary of amount-mass-number relationships in a chemical equation.**



**Sample Problem 3.14** **Calculating Quantities of Reactants and Products: Amount (mol) to Amount (mol)**

**PROBLEM:** Copper is obtained from copper(I) sulfide by roasting it in the presence of oxygen gas) to form powdered copper(I) oxide and gaseous sulfur dioxide.

How many moles of oxygen are required to roast 10.0 mol of copper(I) sulfide?

**PLAN:** **write and balance the equation**

**use the mole ratio as a conversion factor**

**moles of oxygen**

**SOLUTION:** $2\ Cu_2S\ (s) + 3\ O_2\ (g) \rightarrow 2\ Cu_2O\ (s) + 2\ SO_2\ (g)$

$$10.0\ \text{mol}\ Cu_2S \times \frac{3\ \text{mol}\ O_2}{} = \mathbf{15.0\ mol\ O_2}$$



$2\,\text{mol}\ Cu_2S$

## Sample Problem 3.15 Calculating Quantities of Reactants and Products: Amount (mol) to Mass (g)

**PROBLEM:** During the process of roasting copper(I) sulfide, how many grams of sulfur dioxide form when 10.0 mol of copper(I) sulfide reacts?

**PLAN:** Using the balanced equation from the previous problem, we again use the mole ratio as a conversion factor.

**mol of copper(I) sulfide**

**use the mole ratio as a conversion factor**

**mol of sulfur dioxide**

**multiply by $\mathcal{M}$ of sulfur dioxide**

**mass of sulfur dioxide**

## Sample Problem 3.15

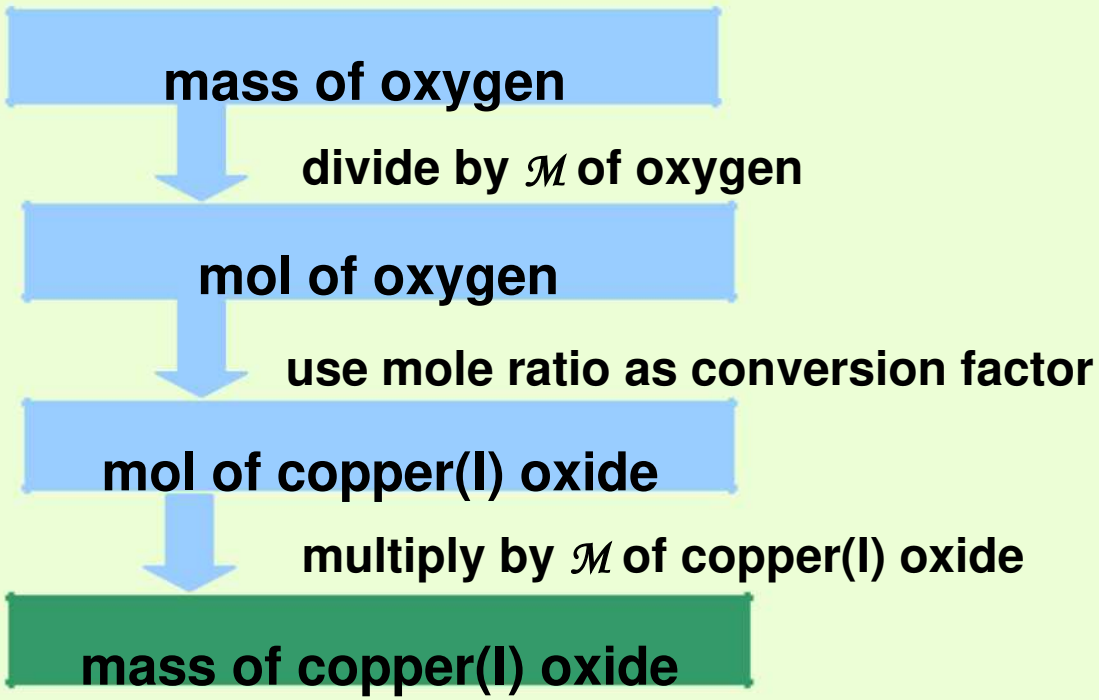
**SOLUTION:** $2\ Cu_2S\ (s) + 3\ O_2\ (g) \rightarrow 2\ Cu_2O\ (s) + 2\ SO_2\ (g)$

$$10.0\ \text{mol } Cu_2S \times \frac{2\ \text{mol } SO_2}{2\ \text{mol } Cu_2S} \times \frac{64.07\ \text{g } SO_2}{1\ \text{mol } SO_2} = \mathbf{641\ g\ SO_2}$$

## Sample Problem 3.16 Calculating Quantities of Reactants and Products: Mass to Mass

**PROBLEM:** During the roasting of copper(I) sulfide, how many kilograms of oxygen are required to form 2.86 kg of copper(I) oxide?

**PLAN:**



## Sample Problem 3.16

**SOLUTION:** $2\ Cu_2S\ (s) + 3\ O_2\ (g) \rightarrow 2\ Cu_2O\ (s) + 2\ SO_2\ (g)$

$$2.86\ kg\ Cu_2O \times \frac{10^3\ g}{1\ kg} \times \frac{1\ mol\ Cu_2O}{143.10\ g\ Cu_2O} = 20.0\ mol\ Cu_2O$$

$$20.0\ mol\ Cu_2O \times \frac{3\ mol\ O_2}{2\ mol\ Cu_2O} \times \frac{32.00\ g\ O_2}{1\ mol\ O_2} \times \frac{1\ kg}{10^3\ g}$$

**= 0.959 kg $O_2$**

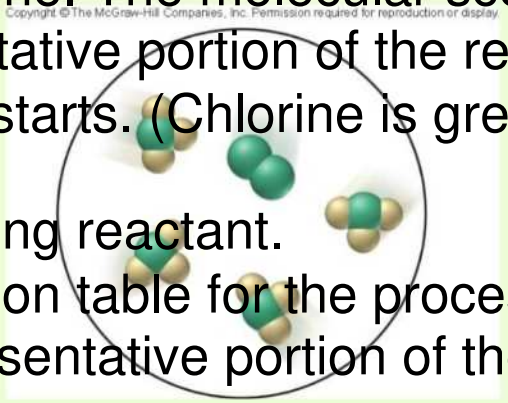
# Limiting Reactants

- So far we have assumed that reactants are present in the correct amounts to react completely.
- In reality, one reactant may ***limit*** the amount of product that can form.
- The ***limiting*** reactant will be completely used up in the reaction.
- The reactant that is *not* limiting is in ***excess*** – some of this reactant will be left over.

## Sample Problem 3.18 Using Molecular Depictions in a Limiting-Reactant Problem

**PROBLEM:** Chlorine trifluoride, an extremely reactive substance, is formed as a gas by the reaction of elemental chlorine and fluorine. The molecular scene shows a representative portion of the reaction mixture before the reaction starts. (Chlorine is green, and fluorine is yellow.)

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



(a) Find the limiting reactant.
(b) Write a reaction table for the process.
(c) Draw a representative portion of the mixture after the reaction is complete. (Hint: The $ClF_3$ molecule has 1 Cl atom bonded to 3 individual F atoms).

## Sample Problem 3.18

**PLAN:** Write a balanced chemical equation. To determine the limiting reactant, find the number of molecules of product that would form from the given numbers of molecules of each reactant. Use these numbers to write a reaction table and use the reaction table to draw the final reaction scene.

**SOLUTION:** The balanced equation is $Cl_2\,(g) + 3F_2\,(g) \rightarrow 2ClF_3\,(g)$

There are 3 molecules of $Cl_2$ and 6 molecules of $F_2$ depicted:

$$3 \text{ molecules } Cl_2 \times \frac{2 \text{ molecules } ClF_3}{1 \text{ molecule } Cl_2} = 6 \text{ molecules } ClF_3$$

$$6 \text{ molecules } F_2 \times \frac{2 \text{ molecules } ClF_3}{3 \text{ molecule } Cl_2} = 4 \text{ molecules } ClF_3$$

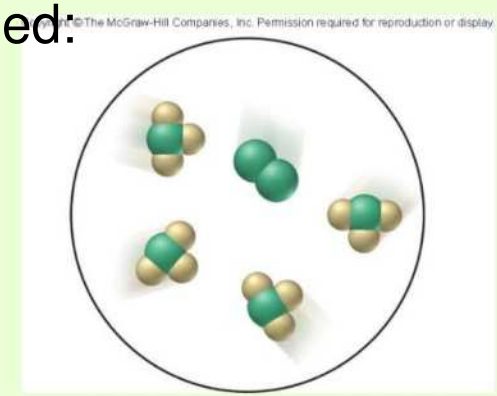
**Since the given amount of $F_2$ can form less product, it is the limiting reactant.**

## Sample Problem 3.18

We use the amount of $F_2$ to determine the "change" in the reaction table, since $F_2$ is the limiting reactant:

| **Molecules** | **$Cl_2$ (*g*) +** | **$3F_2$ (*g*) →** | **$2ClF_3$ (*g*)** |
|---|---|---|---|
| Initial | 3 | 6 | 0 |
| Change | -2 | **-6** | +4 |
| Final | 1 | 0 | 4 |

The final reaction scene shows that all the $F_2$ has reacted and that there is $Cl_2$ left over. 4 molecules of $ClF_2$ have formed:


©The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

**Sample Problem 3.19** **Calculating Quantities in a Limiting-Reactant Problem: Amount to Amount**

**PROBLEM:** In another preparation of $ClF_3$, 0.750 mol of $Cl_2$ reacts with 3.00 mol of $F_2$.

(a) Find the limiting reactant.
(b) Write a reaction table.

**PLAN:** Find the limiting reactant by calculating the amount (mol) of $ClF_3$ that can be formed from each given amount of reactant. Use this information to construct a reaction table.

**SOLUTION:** The balanced equation is $Cl_2\,(g) + 3F_2\,(g) \rightarrow 2ClF_3\,(g)$

$$0.750 \text{ mol } Cl_2 \times \frac{2 \text{ mol } ClF_3}{1 \text{ mol } Cl_2} = 1.50 \text{ mol } ClF_3$$

$$3.00 \text{ mol } F_2 \times \frac{2 \text{ mol } ClF_3}{3 \text{ mol } F_2} = 2.00 \text{ mol } ClF_3$$

**$Cl_2$ is limiting, because it yields less $ClF_3$.**

## Sample Problem 3.19

All the $Cl_2$ reacts since this is the limiting reactant. For every 1 $Cl_2$ that reacts, 3 $F_2$ will react, so 3(0.750) or 2.25 moles of $F_2$ reacts.

| **Moles** | **$Cl_2$ (*g*) +** | **$3F_2$ (*g*) →** | **$2ClF_3$ (*g*)** |
|---|---|---|---|
| Initial | 0.750 | 3.00 | 0 |
| Change | **-0.750** | - 2.25 | +1.50 |
| Final | 0 | 0.75 | 1.50 |

## Sample Problem 3.20

## Calculating Quantities in a Limiting-Reactant Problem: Mass to Mass

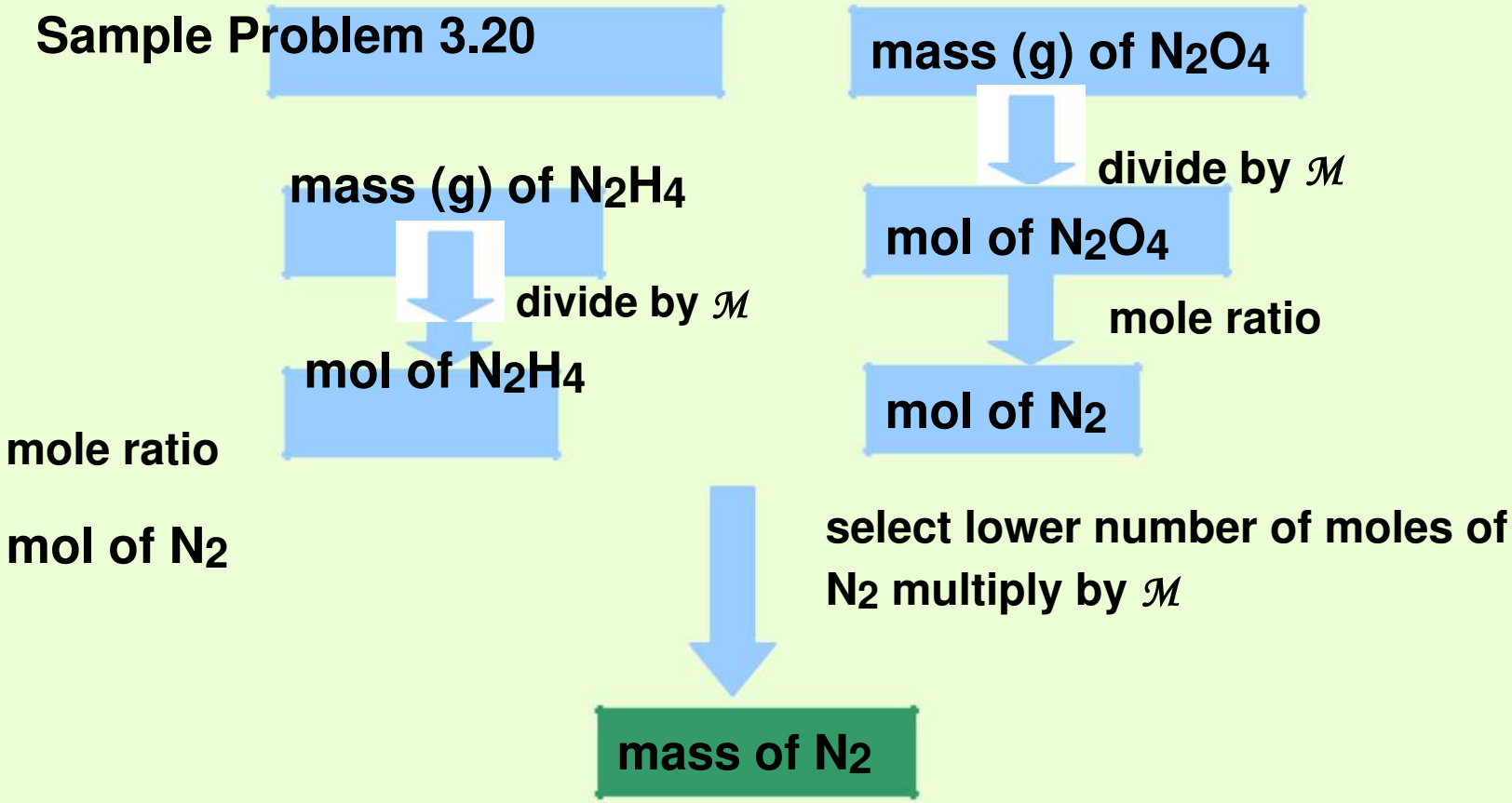
**PROBLEM:** A fuel mixture used in the early days of rocketry consisted of two liquids, hydrazine ($N_2H_4$) and dinitrogen tetraoxide ($N_2O_4$), which ignite on contact to form nitrogen gas and water vapor.

**(a)** How many grams of nitrogen gas form when $1.00 \times 10^2$ g of $N_2H_4$ and $2.00 \times 10^2$ g of $N_2O_4$ are mixed?

**(b)** Write a reaction table for this process.

**PLAN:** Find the limiting reactant by calculating the amount (mol) of $ClF_3$ that can be formed from each given mass of reactant. Use this information to construct a reaction table.

## Sample Problem 3.20



## Sample Problem 3.20

**SOLUTION:** $2N_2H_4\ (l) + N_2O\ (l) \rightarrow 3N_2\ (g) + 4H_2O\ (g)$

For $N_2H_4$: $1.00 \times 10^2\ \text{g } N_2H_4 \times \dfrac{1\ \text{mol } N_2H_4}{32.05\ \text{g } N_2H_4} = 3.12\ \text{mol } N_2H_4$

$$3.12\ \text{mol } N_2H_4 \times \frac{3\ \text{mol } N_2}{2\ \text{mol } N_2H_4} = 4.68\ \text{mol } N_2$$

For N O : $2.00 \times 10^2\ \text{g } N_2O_4 \times \dfrac{1\ \text{mol } N_2O_4}{92.02\ \text{g } N_2O_4} = 2.17\ \text{mol } N_2$

$$2.17\ \text{mol } N_2O_4 \times \frac{3\ \text{mol } N_2}{1\ \text{mol } N_2O_4} = 6.51\ \text{mol } N_2$$

$N_2H_4$ is limiting and only 4.68 mol of $N_2$ can be produced:

$$4.68\ \text{mol } N_2 \times \frac{28.02\ \text{g } N_2}{1\ \text{mol } N_2} = \mathbf{131\ g\ N_2}$$

## Sample Problem 3.20

All the $N_2H_4$ reacts since it is the limiting reactant. For every 2 moles of $N_2H_4$ that react 1 mol of $N_2O_4$ reacts and 3 mol of $N_2$ form:

$$3.12 \text{ mol } N_2H_4 \times \frac{1 \text{ mol } N_2O_4}{2 \text{ mol } N_2H_4} = 1.56 \text{ mol } N_2O_4 \text{ reacts}$$

| **Moles** | **$2N_2H_4$ (l) +** | **$N_2O_4$ (*l*) →** | **$3N_2$ (*g*) +** | **$4H_2O$ (*g*)** |
|---|---|---|---|---|
| Initial | 3.12 | 2.17 | 0 | 0 |
| Change | **-3.12** | **- 1.56** | **+4.68** | +6.24 |
| Final | 0 | 0.61 | 4.68 | 6.24 |

# Reaction Yields

The **theoretical yield** is the amount of product calculated using the molar ratios from the balanced equation.

The **actual yield** is the amount of product actually obtained.

The actual yield is usually less than the theoretical yield.

$$\text{\% yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

**Sample Problem 3.21** **Calculating Percent Yield**

**PROBLEM:** Silicon carbide (SiC) is made by reacting sand(silicon dioxide, SiO2) with powdered carbon at high temperature. Carbon monoxide is also formed. What is the percent yield if 51.4 kg of SiC is recovered from processing 100.0 kg of sand?

**PLAN:**

**write balanced equation**

**find mol reactant**

**find mol product**

**find g product predicted** **percent yield**

**Sample Problem 3.21**

**SOLUTION:** $SiO_2(s) + 3C(s) \rightarrow SiC(s) + 2CO(g)$

$$100.0 \text{ kg } SiO_2 \times \frac{10^3 \text{ g}}{1 \text{ kg}} \times \frac{1 \text{ mol } SiO_2}{60.09 \text{ g } SiO_2} = 1664 \text{ mol } SiO_2$$

$$\text{mol } SiO_2 = \text{mol SiC} = 1664 \text{ mol SiC}$$

$$1664 \text{ mol SiC} \times \frac{40.10 \text{ g SiC}}{1 \text{ mol SiC}} \times \frac{1 \text{ kg}}{10^3 \text{ g}} = 66.73 \text{ kg}$$

$$\frac{51.4 \text{ kg}}{66.73 \text{ kg}} \times 100 = \mathbf{77.0\%}$$

# Solution Stoichiometry

- Many reactions occur in solution.
- A solution consists of one or more solutes dissolved in a ***solvent***.
- The ***amount concentration*** of a solution is given by the amount (mol) of solute present in a given volume of solution (L).

What is the amount concentration of an aqueous solution that contains 0.715 mol of glycine ($H_2NCH_2COOH$) in 495 mL?

$$c_B = \frac{amount\ of\ solute\ B\ (mol)}{volume\ of\ solution\ (L)}$$

**SOLUTION:**

**mol of glycine**

$$\frac{0.715\ \text{mol glycine}}{495\ \text{mL soln}} \times \frac{1000\ \text{mL}}{1\ \text{L}}$$

divide by volume in L

**concentration in mol/L**

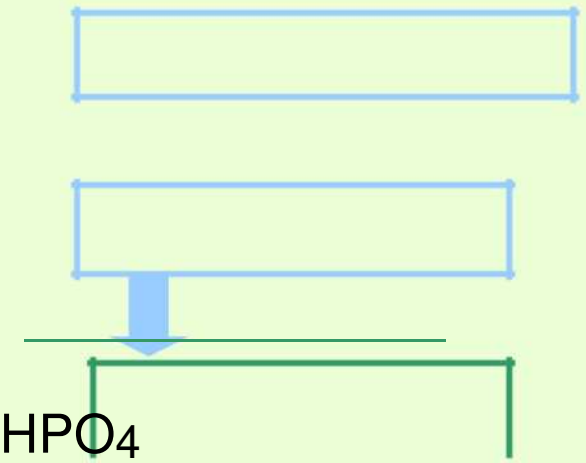
**= 1.44 *mol/L* glycine**

## Sample Problem 3.23

## Calculating Mass of Solute in a Given Volume of Solution

**PROBLEM:** How many grams of solute are in 1.75 L of 0.460 *M* sodium monohydrogen phosphate buffer solution?

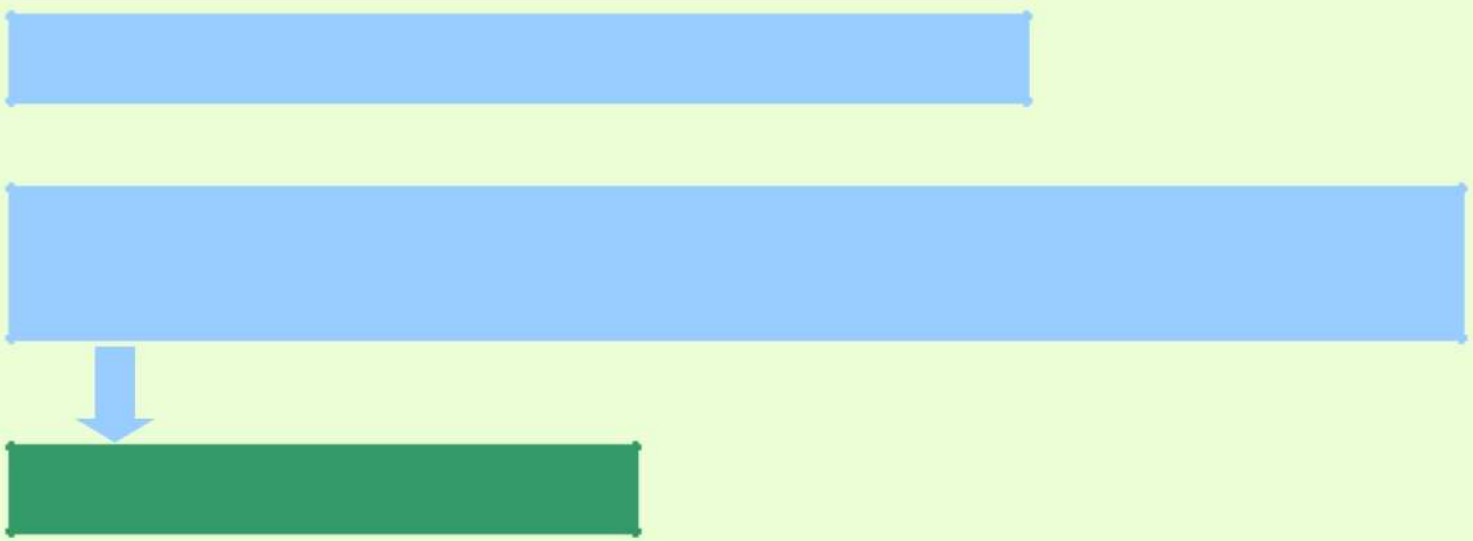
**PLAN:** Calculate the moles of solute using the given molarity and volume. Convert moles to mass using the molar mass of the solute.



**SOLUTION:**

$$1.75\ \text{L} \times \frac{0.460\ \text{moles}}{1\ \text{L}} = 0.805\ \text{mol}\ Na_2HPO_4$$

$$0.805\ \text{mol}\ Na_2HPO_4 \times \frac{141.96\ \text{g}\ Na_2HPO_4}{1\ \text{mol}\ Na_2HPO_4} = \mathbf{114\ g\ Na_2HPO_4}$$



# Chapter 3: Mass Relations in Formulas, Chemical Reactions, and Stoichiometry

## Section 3.1: The Atomic Mass

The atomic mass is the mass of 1 atom. Atoms are very small and their mass is a very small number. A more practical unit is used to describe the mass of an atom. This unit is called the **atomic mass unit** (expressed as **a.m.u**. or just **u**).

The value of 1 atomic mass unit is chosen as 1/12 of the mass of one carbon-12 isotope. Remember that for carbon-12 the mass number A is equal to 12 (that is carbon-12 has 12 nucleons. The mass of a carbon-12 atom is equal to $1.9926 \times 10^{-23}$ g. Hence, we conclude 1 u = $1.6605 \times 10^{-24}$ g. The mass of carbon-12 atom is measured with an instrument called the mass spectrometer.

The atomic masses of elements are generally given in the Periodic Table and are located below the element symbol. For the element carbon, we note that the atomic mass is not 12 but 12.011 u. This is because the element carbon has several isotopes and the number 12.011 u is the average atomic mass of all the isotopes of the element carbon present in a typical sample on earth.

Note: Atomic masses are also called atomic weights.

## Section 3.2: The Atomic Mass in the Periodic Table

Check the Periodic Table and look up the atomic mass of different elements by rolling your mouse over the element's symbol.

## Section 3.3: Avogadro's Number and the Mole

Typical samples of matter contain huge numbers of atoms, often numbers as large as $10^{24}$ or more. The **mole** was established as a unit that is very useful when counting the numbers of atoms, ions or molecules. One mole is equal to the number of carbon atoms in 12 g of carbon-12.

1 atom of carbon-12 has a mass of $1.9927 \times 10^{-23}$ g. Hence, in 12 g of carbon-12, there are:

**(spu) اتحاد سنافر البوليتكنك**

$$\frac{12\ g}{1.9927 \times 10^{-23}\ g/C\ atom} = 6.022 \times 10^{23}\ C\ atoms$$

The number of $6.022 \times 10^{23}$ is called **Avogadro's number**. Avogadro's numberis expressed by the symbol $\mathbf{N_A}$. Hence, one mole of atoms of carbon-12 (i.e. 12g of carbon-12) contains Avogadro's number or $6.022 \times 10^{23}$ atoms of carbon-12.

Note: The term "mole" is analogous to the term "dozen". While a dozen eggs refer to twelve eggs, a mole of particles (atoms, ions or molecules) refers to$6.022 \times 10^{23}$ particles. It follows that while 2 dozen eggs consist of 24 eggs, 2 moles of particles consists of $2 \times (6.022 \times 10^{23})$ particles (i.e. $1.2044 \times 10^{24}$ particles).

## Sections 3.4: Concept of Formula Mass or Molar Mass

The formula mass, or molar mass, is the sum of atomic masses in a chemical formula.

**Examples:**

| Chemical Formula | Formula Mass (a.m.u.) | Molar Mass (g/mol) |
|---|---|---|
| H | 1.0 | 1.0 |
| $H_2$ | 2.0 | 2.0 |
| O | 16.0 | 16.0 |
| $O_2$ | 32.0 | 32.0 |

*Many texts require you to use atomic masses to the nearest hundredth. However, in this DVD you need to always round off to the nearest tenth, unless otherwise specified.*

**Formula mass** is the sum of atomic masses of all atoms in a formula of any molecular or ionic compound. The formula mass is expressed in a.m.u.

**Molar mass** is the sum of atomic masses of all atoms in a mole of pure substance. The molar mass is expressed in g/mol.

It is important that you know how to write chemical formulas from chemical names and vice versa.

**Example: What is the formula mass of tungsten?** Click on the Periodic Table. Roll the mouse and find tungsten, W. You will see the mass in red under W. Hence, the formula mass of tungsten, W, is 183.9 a.m.u.

(spu) اتحاد سنافر البوليتكنك

**Example: Calculate the molar mass of vitamin A, $C_{20}H_{30}O$.**

The vitamin A molecule consists of:

| Element | Number of Atoms | | Atomic Mass |
|---|---|---|---|
| C | 20 | x | 12.0 = 240.0 |
| H | 30 | x | 1.0 = 30.0 |
| O | 1 | x | 16.0 = 16.0 |
| | | | 286.0 |

Hence, the molar mass of vitamin A is 286.0 g/mol.

**Example: Calculate the molar mass of sodium chloride.** Note: Here the chemical name is given but not the chemical formula. Hence, it is important to know the names and formulas of chemicals.

Sodium chloride has the chemical formula NaCl.

| Element | Number of Atoms | | Atomic Mass |
|---|---|---|---|
| Na | 1 | x | 23.0 = 23.0 |
| Cl | 1 | x | 35.5 = 35.5 |
| | | | 58.5 g/mol |

Hence, the molar mass of sodium chloride is 58.5 g/mol.

In Section 2.5, practice Interactive Problems to master these concepts.

## Sections 3.5: Conversion between Number of Moles and Mass

When discussing the amount of a substance, it is common practice to use the word “moles” instead of the more rigorous wording “number of moles”. The symbol “n” is often used to describe the number of moles or “moles” of a substance.

How many moles of a substance are present in a given sample of that substance can be calculated from the mass and the molar mass of that substance, according to the equation.

$$\text{Number of moles of a substance is equal to : } n = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}}$$

Hence, mass = n x molar mass

Thus, if the moles and the chemical formula are given, one can calculate the mass in grams of that chemical.

**Example: Calculate the number of moles of $NH_3$ in 1.0 g of $NH_3$.**

$$n = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}}$$

The mass of $NH_3$ is 1.0 g. However, we need to calculate the molar mass of $NH_3$.

| Element | Number of Atoms | | Atomic Mass |
|---|---|---|---|
| N | 1 | x | 14.0 = 14.0 |
| H | 3 | x | 1.0 = 3.0 |
| | | | 17.0 g/mol |

$$n = \frac{1.0\ \text{g}}{17.0\ \text{g/mol}} = 0.0588235\ \text{mol}$$

In the correct number of significant figures, the answer is 0.059 mol.

Let us continue further with this problem (not covered in the DVD).

**Example: How many moles of H and N atoms are present in 1.0 g of $NH_3$?**

From the previous problem, we know that 1.0 g of $NH_3$ contains 0.059 mol $NH_3$. From the chemical formula of ammonia, we know that for every molecule of $NH_3$ , there is 1 atom of nitrogen and 3 atoms of hydrogen. So, for every mole of $NH_3$, there is 1 mole of nitrogen and 3 moles of hydrogen.

Hence, moles of N = moles $NH_3$ = 0.059 mol N and Moles of H = 3 x moles of $NH_3$
= 3 x 0.059 mol = 0.18 mol H

**Example: Calculate the mass in grams of 12.00 moles of $C_2H_3Cl$.**

$$n = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}} \qquad \text{Hence, mass} = n \times \text{molar mass}$$

Calculate the molar mass of $C_2H_3Cl$.

| Element | Number of Atoms | | Atomic Mass |
|---|---|---|---|
| C | 2 | x | 12.0 = 24.0 |
| H | 3 | x | 1.0 = 3.0 |
| Cl | 1 | x | 35.5 = 35.5 |
| | | | 62.5 g/mol |

(spu) اتحاد سنافر البوليتكنك

mass = 12.00 mol x 62.5 g/mol
mass = 750.g

**Example: Calculate the number of moles in 150.0 g of iron(III) oxide.**

The chemical formula of iron(III) oxide is $Fe_2O_3$. In the problem, the mass $Fe_2O_3$ is given. Hence, the molar mass must be calculated.

| Element | Number of Atoms | Atomic Mass |
|---|---|---|
| Fe | 2 x | 55.8 = 111.6 |
| O | 3 x | 16.0 = 48.0 |
| | | 159.6 g/mol |

$$n = \frac{150.0\ g}{159.6\ g/mol} = 0.93984\ mol$$

In the correct number of significant figures, the answer is 0.9398 mol.

In Section 2.7, visualize one mole for a variety of elements and compounds.

In Section 2.8, practice Interactive Problems.

## Sections 3.6: Problems on Avogadro's Number

In some instances, we want to know how many atoms, ions or molecules are involved in a chemical or physical process. We can calculate this number from the mass of the substance, the molar mass of the substance and Avogadro's number.

Avogadro's number is given by the symbol $N_A = 6.022 \times 10^{23}$ or 6.022E23

$$\text{Remember mole, } n = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}} \text{ and one mole} = 6.022 \times 10^{23}$$

**Example 1: The molecular formula of ethylene glycol is $C_2H_6O_2$. In 13.68 g of ethylene glycol, (a) calculate the number of moles**

| Element | Number of Atoms | Atomic Mass |
|---|---|---|
| C | 2 x | 12.0 = 24.0 |
| H | 6 x | 1.0 = 6.0 |
| O | (spu) اتحاد سنافر البوليتكنك x | 16.0 = 32.0 |
| | | 62.0 |

$$n = \frac{13.68\ g}{62.0\ g/mol}$$

$$n = 0.221\ mol$$

Example 1: The molecular formula of ethylene glycol is $C_2H_6O_2$. In 13.68 g of ethylene glycol, **(b) calculate the number of molecules**

$$\frac{6.022 \times 10^{23}\ molecules}{1\ mol} \times 0.221\ mol = 1.33 \times 10^{23}\ molecules \text{ or } 1.33E23\ molecules$$

Example 1: The molecular formula of ethylene glycol is $C_2H_6O_2$. In 13.68 g of ethylene glycol, **(c) Calculate the number of oxygen atoms**

In each molecule of $C_2H_6O_2$, there are 2 atoms of O. Therefore, in $1.33 \times 10^{23}$ molecules of $C_2H_6O_2$ there are:

$$2 \times 1.33 \times 10^{23} = 2.66 \times 10^{23} \text{ or } 2.66E23 \text{ atoms of oxygen}$$

**Example 2: A sample of the compound, $C_3H_6O$, contains $14.0 \times 10^{14}$ carbon atoms. (a) Calculate the number of $C_3H_6O$ molecules**

Each molecule of $C_3H_6O$ contains 3 carbon atoms. Therefore, the number of $C_3H_6O$ molecules containing $14.0 \times 10^{14}$ C atoms is:

$$\frac{14.0 \times 10^{14}}{3} = 4.67 \times 10^{14}\ \text{molecules of } C_3H_6O$$

Example 2: A sample of the compound, $C_3H_6O$, contains $14.0 \times 10^{14}$ carbon atoms. **(b) Calculate the number of moles of $C_3H_6O$.**

$$\frac{1\ mol}{6.022 \times 10^{23}\ molecules} \times 4.67 \times 10^{14}\ molecules = 7.75 \times 10^{-10} \text{ or } 7.75E-10\ mol$$

Example2: A sample of the compound, $C_3H_6O$, contains $14.0 \times 10^{14}$ carbon atoms. **(c) Calculate the number of grams of $C_3H_6O$.**

| Element | Number of Atoms | Atomic Mass |
|---|---|---|
| C | 3 x | 12.0 = 36.0 |
| H | 6 x | 1.0 = 6.0 |
| O | 1 x | 16.0 = 16.0 |

(spu) البوليتكنك اتحاد سنافر

| | | |
|---|---|---|
| | | 58.0 g/mol |

The molar mass of $C_3H_6O$ is 58.0 g/mol

$$n = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}}$$

Therefore, mass (g) = n x molar mass (g/mol)

Thus, mass = $7.75 \times 10^{-10}$ mol x 58.0 g/mol

mass = $4.50 \times 10^{-8}$ or 4.50E-8 g

In Section 2.10, practice the Interactive Problems.

## Sections 3.7: Percent Composition

The percent composition of a compound is the mass percent of the elements present.

$$\text{Mass percent of element} = \frac{\text{mass of element in sample}}{\text{total mass of sample}} \times 100$$

Knowing the chemical formula of a compound, the mass percent of its constituent elements can be calculated.

Note: the subscripts in a chemical formula allow us to define the atom ratio as well as the mole ratio in which the different elements are combined.

**Example:**

In $H_2O$: the atom ratio is 2 atoms H: 1 atom O

the mole ratio is 2 moles H: 1 mole O

In $Na_2SO_4$: the atom ratio is 2 atoms Na: 1 atom S: 4 atoms O

the mole ratio is 2 moles Na: 1 mole S: 4 moles O

**Example 1: Calculate the mass percent of H and O in $H_2O$.**

| Element | Number of Moles | Molar Mass (g/mol) |
|---|---|---|
| H | 2 x | 1.0 = 2.0 g |
| O | 1 x | 16.0 = 16.0 g |
| | | 18.0 g |

(spu) اتحاد سنافر البوليتكنك

$$\text{Mass \% of H} = \frac{2.0\ \text{g}}{18.0\ \text{g}} \times 100$$
$$= 11\%$$
$$\text{Mass \% O} = \frac{16.0\ \text{g}}{18.0\ \text{g}} \times 100$$
$$= 88.9\ \%$$

**Example 2: Calculate the mass percent of N in $C_3H_3N$**

| Element | Number of Moles | Molar Mass (g/mol) |
|---|---|---|
| C | 3 x | 12.0 = 36.0 g |
| H | 3 x | 1.0 = 3.0 g |
| N | 1 x | 14.0 = 14.0 g |
| | | 53.0 g |

$$\text{Mass \% N} = \frac{14.0\ \text{g}}{53.0\ \text{g}} \times 100$$
$$= 26.4\ \%$$

In Section 2.12, practice the Interactive Problems.

## Sections 3.8: Empirical Formula

When a new compound is formed or discovered, it is important to determine the chemical formula. Most often, this is done by taking a known amount of sample and decomposing, or breaking down this compound into its component elements.

Or

Taking a known amount of sample and reacting it with oxygen to produce $CO_2$ and $H_2O$. The component elements or $CO_2$ and $H_2O$, are then collected and weighed. The results of such analyses give the mass of each element in the compound.

This is used to determine the mass percent of each element in the compound. Knowing the mass percent of each element in the compound makes it possible to determine its chemical formula.

اتحاد سنافر البوليتكنك (spu)

Empirical formula is the simplest chemical formula. The simplest formula gives only the ratios of atoms in a compound.

**Example: A 50.00 g sample contains 13.28 g of potassium, 17.68 g of chromium, and 19.04 g of oxygen. Find the simplest formula.**

*Analyze the problem. The sample mass is given, the masses of elements are also given. Make sure the sum of masses of all elements in the sample is equal to the mass of the sample.*

**Step1:** Calculate the number of moles of K, Cr and O in the given masses.

$$\text{Remember that moles} = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}}$$

$$13.28\ \text{g (K)} \Rightarrow \frac{13.28\ \text{g}}{39.1\ \text{g/mol}} = 0.340\ \text{mol (K)}$$

$$17.68\ \text{g (Cr)} \Rightarrow \frac{17.68\ \text{g}}{52.0\ \text{g/mol}} = 0.340\ \text{mol (Cr)}$$

$$19.04\ \text{g (O)} \Rightarrow \frac{19.04\ \text{g}}{16.0\ \text{g/mol}} = 1.19\ \text{mol (O)}$$

**Step 2:** Divide each of the numbers of moles by the smallest number of moles to obtain the relative amounts in moles of each element in the substance.

$$\text{K}: \frac{0.340\ \text{mol}}{0.340\ \text{mol}} = 1.00$$

$$\text{Cr}: \frac{0.340\ \text{mol}}{0.340\ \text{mol}} = 1.00$$

$$\text{O}: \frac{1.19\ \text{mol}}{0.340\ \text{mol}} = 3.50$$

**Step 3:** Write the formula using these relative numbers of moles of each element. Remember that the subscripts in a formula give the relative numbers of atoms or moles of atoms in that substance.

The results in Step 2 suggest that the simplest formula is: $K_1Cr_1O_{3.5}$

**Step 4:** Write the final formula, ensuring all subscripts are whole numbers.

We multiply each subscript by 2 to get the empirical formula: $\mathbf{K_2Cr_2O_7}$ . This is potassium dichromate. This formula makes sense because the dichromate ion has a -2 charge. The potassium ion has a +1 charge. Hence, this substance has a neutral formula, as it should.

(spu) اتحاد سنافر البوليتكنك

In Section 2.14, practice the Interactive Problem.

## Sections 3.9: More Problems on Empirical Formula

To find the composition of a substance, it is often useful to react that substance with oxygen gas. This is an example of **chemical analysis**. Combustion reactions are useful for the chemical analysis of substances containing carbon and hydrogen as they produce $CO_2$ and $H_2O$. Measuring the amount of $CO_2$ and $H_2O$ produced by a given amount of substance allows the determination of how much carbon and hydrogen are present in that amount of substance.

**Example: When 5.000 g of ibuprofen is burnt with oxygen gas ($O_{2(g)}$), 13.86 g of $CO_{2(g)}$ and 3.926 g of $H_2O_{(l)}$ are formed. Use the following information to determine the empirical formula of ibuprofen. Ibuprofen is known to contain only carbon, oxygen and hydrogen elements.**

**Step 1:** How much carbon is there in 5.000 g of ibuprofen?

All the carbon in ibuprofen ends up in the 13.86 g of $CO_{2(g)}$. So, the question is how many moles of carbon are present in 13.86 g of $CO_{2(g)}$.

$$13.86\ g\,(CO_2) \Rightarrow \frac{13.86\ g\,(CO_2)}{44.01\ g/mol\ \ (CO_2)} = 0.3149\ \ mol\,(CO_2) \Rightarrow 0.3149\ mol\,(C)$$

⇨ mass (C) in 5 g ibuprofen = 0.3149 mol (C) x 12.01 g/mol (C) = 3.782 g (C)

**Step 2:** How much hydrogen is there in 5.000 g of ibuprofen?

All the hydrogen in ibuprofen ends up in the 3.926 g of $H_2O_{(l)}$. So, the question is how many moles of hydrogen are present in 3.926 g of $H_2O_{(l)}$.

$$3.926\ g\,(H_2O) \Rightarrow \frac{3.926\ g\,(H_2O)}{18.00\ \ g/mol\ \ (H_2O)} = 0.2181\ \ mol\,(H_2O) \Rightarrow 0.4362\ mol\,(H)$$

⇨ mass (H) in 5 g ibuprofen = 0.4362 mol (H) x 1.00 g/mol (H) = 0.4362 g (H)

**Step 3:** How much oxygen is there in 5.000 g of ibuprofen?

Since ibuprofen only contains oxygen, carbon and hydrogen and 5.000 g of ibuprofen contain 3.782 g (C) and 0.4362 g (H), then, the mass of oxygen is:

Mass (O) = 5.000 g – 3.782 g – 0.4362 g = 0.7818 g (O).

اتحاد سنافر البوليتكنك (spu)

**Step 4:** Now, we can use the strategy shown in Section 2.13 to determine the formula of ibuprofen.

$$\text{C}: 3.782\ \text{g} \Rightarrow \frac{3.782\ \text{g (C)}}{12.0\ \text{g/mol (C)}} = 0.3149\ \text{mol (C)}$$

$$\text{H}: 0.4362\ \text{g} \Rightarrow \frac{0.4362\ \text{g (H)}}{1.0\ \text{g/mol (H)}} = 0.4362\ \text{mol (H)}$$

$$\text{O}: 0.7818\ \text{g} \Rightarrow \frac{0.7818\ \text{g (O)}}{16.0\ \text{g/mol (O)}} = 0.04886\ \text{mol (O)}$$

$$\text{For Carbon} \Rightarrow \frac{0.3149\ \text{mol (C)}}{0.04886\ \text{mol (O)}} = 6.4\ \text{(C) per (O) approximately 6.5}$$

$$\text{For Hydrogen} \Rightarrow \frac{0.4362\ \text{mol (H)}}{0.04886\ \text{mol (O)}} = 8.9\ \text{(H) per (O) approximately 9}$$

Hence, the formula for ibuprofen is: $C_{6.5}H_9O_1$ or more appropriately, **$C_{13}H_{18}O_2$.**

In Section 2.16, practice the Interactive Problem.

## Sections 3.10: Molecular Formula

The empirical formula of a substance is always written using the smallest possible whole number subscripts to give the relative number of atoms of each element in the substance.

Hence, the empirical formula for sodium chloride is written as NaCl and not $Na_2Cl_2$. NaCl is an ionic compound, not a molecule. Hence, the entity NaCl is called a **formula unit**.

Remember that for ionic compounds, the chemical formula and the empirical formula are always one and the same formula. For molecular (covalent) compounds, however, molecular and empirical formula may be different.

A molecular formula is a whole number multiple of the simplest chemical formula.

Or

A molecular formula is a whole multiple of the empirical formula.

To find the multiple, the molar mass is needed. The empirical formula mass can be calculated from the empirical formula.

اتحاد سنافر البولينكنك (spu)

$$\text{Multiple} = \frac{\text{molar mass}}{\text{empirical formula mass}}$$

**Example: The mass composition of lindane is 24.78% C, 2.08% H and 73.14% Cl. The molar mass of lindane is 290.85 g/mol. Determine the molecular formula.**

| Element | Mass (g) | Molar Mass (g/mol) | Moles | Mole Ratio |
|---|---|---|---|---|
| C | 24.78 | 12.0 | 2.06 | 1 |
| H | 2.08 | 1.0 | 2.08 | 1 |
| Cl | 73.14 | 35.5 | 2.06 | 1 |

Hence, the simplest formula or the empirical formula is CHCl.

The empirical formula mass is 12 + 1 + 35.5 = 48.5 g/mol

$$\text{Multiple} = \frac{290.85\ \text{g/mol}}{48.5\ \text{g/mol}} = 6$$

Hence, the molecular formula is: $C_6H_6Cl_6$

In Section 2.18, practice the Interactive Problem.

## Section 3.11: Balancing Chemical Reactions

A **chemical reaction** is an actual transformation of substances called reactants into substances called products.

To represent a chemical reaction we use a **chemical equation**, a sort of recipe which shows in a symbolic form 1) who the participating substances are (reactants and products), 2) the state or phase these substances are in (solid, liquid, gas, aqueous solution) and 3) the amount in which they must be present (number of atoms, molecules (for covalent compounds) or formula units (for ionic compounds)).

**Example:** Consider the reaction of aluminum metal ($Al_{(s)}$) with solid iron oxide ($Fe_2O_{3\ (s)}$) forming solid aluminum oxide ($Al_2O_{3\ (s)}$) and solid iron ($Fe_{(s)}$). This reaction is represented by the following chemical equation:

$$2\ Al_{(s)} + Fe_2O_{3\ (s)} \rightarrow Al_2O_{3\ (s)} + 2\ Fe_{(s)}$$

اتحادا سنافر البوليتكنك (spu)

REACTANTS PRODUCTS

The arrow (→) shows the direction in which the chemical transformation takes place. The **reactants** (shown on the left-hand side of the arrow) are the substances with which the reaction is started. The **products** (shown on the right-hand side of the arrow) are the substances resulting from the reaction.
In the above reaction all substances are in the solid state, as indicated by the subscript "s" in parentheses.

The **state** or **phase** of a substance is always indicated by a subscript in parentheses after the chemical formula of that substance. The following notations are used for the various phases encountered in chemical reactions:

| | |
|---|---|
| (s) | for solids |
| (l) | for liquids |
| (g) | for gases |
| (aq) | for aqueous solutions |

Finally, one of the most important pieces of information conveyed by a chemical equation is the number of atoms, ions, formula units or molecules associated with each substance. The number in front of each substance is called the **stoichiometric coefficients** or more simply the coefficient. The bulk of this information is often referred to as the **stoichiometry** of the chemical reaction.

For the above reaction, the stoichiometric coefficients are 2, 1, 1 and 2, respectively. Note that when a stoichiometric coefficient is 1, it is not shown (as is the case for $Al_2O_3$ and $Fe_2O_3$).

Stoichiometric coefficients play a very important role in chemical equations. Their presence insures that the number of atoms of each type is the same on the reactants and products sides. For instance, in the above reaction there are two aluminum atoms, two iron atoms and three oxygen atoms on each side of the chemical equation. This observation reflects Dalton's hypothesis that, in a chemical reaction, atoms are neither destroyed nor created.

To keep a chemical equation looking as simple as possible, we will generally ensure that stoichiometric coefficients are written using the smallest possible whole numbers (integers). **Balancing** a chemical equation consists in determining each of the stoichiometric coefficients.

## Section 3.12: Balancing the Chemical Reaction:

$CO_{2(g)} + H_2O_{(l)} \rightarrow C_6H_{12}O_{6(s)} + O_{2(g)}$

To balance a chemical reaction, we always start balancing the elements that are present in the least number of compounds. In the above equation, we can start with either carbon or hydrogen.

**Balancing the element carbon.**

There is one carbon atom on the left-hand side (in $CO_{2(g)}$) and six carbon atoms on the right-hand side (in $C_6\ H_{12}O_{6(s)}$). Hence, we place a coefficient of 6 in front of $CO_{2(g)}$ and 1 in front of $C_6H_{12}O_{6(s)}$. Doing so leads to the balancing of carbon.

When one assigns stoichiometric coefficients to some substances (here, $CO_{2(g)}$ and $C_6H_{12}O_{6(s)}$), one cannot change these coefficients when balancing another element.

**Balancing the element hydrogen.**

$$6\ CO_{2(g)} + H_2O_{(l)} \rightarrow C_6H_{12}O_{6(s)} + O_{2(g)}$$

There are two hydrogen atoms on the left-hand side (in $H_2O_{(l)}$) and 12 hydrogen atoms on the right-hand side (in $C_6H_{12}O_{6(s)}$). Hence, we place a coefficient of 6 in front of $H_2O_{(l)}$. Doing so leads to the balancing of hydrogen.

Assigning a stoichiometric coefficient to $H_2\ O_{(l)}$, implies that this coefficient can no longer be changed when balancing the last element, oxygen.

**Balancing the element oxygen.**

$$6\ CO_{2(g)} + 6\ H_2O_{(l)} \rightarrow C_6H_{12}O_{6(s)} + O_{2(g)}$$

There are 12 + 6 = 18 oxygen atoms on the left-hand side. On the right-hand side here are 6 oxygen atoms in $C_6H_{12}O_{6(s)}$ and 2 oxygen atoms in $O_{2(g)}$. At this stage, we can no longer change the stoichiometric coefficients of $CO_{2(g)}$, $H_2O_{(l)}$, or $C_6H_{12}O_{6(s)}$ since these coefficients have already been assigned when balancing C and H. Hence, to obtain 18 oxygen atoms on the right-hand side, we must assign a stoichiometric coefficient of 6 to $O_{2(g)}$.

The balanced chemical equation is:

اتحاد سنافر البوليتكنك (spu)

$$\mathbf{6}\ CO_{2(g)} + \mathbf{6}\ H_2O_{(l)} \rightarrow C_6H_{12}O_{6(s)} + \mathbf{6}\ O_{2(g)}$$

The sum of the stoichiometric coefficients for reactants and products is **19**.

## Section 3.13: Balancing the Equation of a Combustion Reaction

A combustion reaction is a reaction in which a substance (element or compound) is burnt with oxygen gas ($O_2$). The combustion reactions of organic molecules (molecules based on carbon, hydrogen, oxygen, etc…) lead to the formation of carbon dioxide ($CO_2$) gas and water ($H_2O$).

Consider the combustion reaction of hexane, $C_6H_{14(l)}$. For the combustion of hexane, oxygen gas must be a reactant and $CO_2$ and $H_2O$ must be products.

$$C_6H_{14(l)} + O_{2(g)} \rightarrow CO_{2(g)} + H_2O_{(l)}$$

First, we **balance carbon:**

There are 6 carbon atoms in $C_6H_{14}$ and 1 carbon atom in $CO_2$. Hence, we will use stoichiometric coefficients of 1 (not shown) for $C_6H_{14}$ and 6 for $CO_2$.

The partially balanced equation is: $C_6H_{14(l)} + O_{2(g)} \rightarrow 6\ CO_{2(g)} + H_2O_{(l)}$

Next, we **balance hydrogen:**

There are 14 hydrogen atoms in one molecule of $C_6H_{14}$ and 2 hydrogen atoms in one molecule of $H_2O$. Hence, we will use a coefficient of 7 for $H_2O$.

The partially balanced equation becomes: $C_6H_{14(l)} + O_{2(g)} \rightarrow 6\ CO_{2(g)} + 7\ H_2O_{(l)}$

Finally, we **balance oxygen:**

On the left-hand side there are 2 oxygen atoms. On the right-hand side there are $6 \times 2 + 7 \times 1 = 19$ oxygen atoms. However, oxygen is present as $O_2$ on the reactant side. Hence, we should use 19/2 $O_2$ molecules on the right-hand side.

The balanced equation is: $C_6H_{14(l)} + 19/2\ O_{2(g)} \rightarrow 6\ CO_{2(g)} + 7\ H_2O_{(l)}$

If we decide to use only whole numbers as stoichiometric coefficients in a balanced equation, then, we must multiply all coefficients by 2. Hence, the final form for the balanced combustion reaction of hexane is:

$$\mathbf{2}\ C_6H_{14(l)} + \mathbf{19}\ O_{2(g)} \rightarrow \mathbf{12}\ CO_{2(g)} + \mathbf{14}\ H_2O_{(l)}$$

**(spu) اتحاد سنافر البوليتكنك**

The sum of the stoichiometric coefficients for reactants and products is 47.

## Section 3.14: Balancing the Chemical Reaction:

$$Fe_2(SO_4)_{3(aq)} + NH_{3(g)} + H_2O_{(l)} \rightarrow Fe(OH)_{3(s)} + (NH_4)_2SO_{4(s)}$$

First, we **balance iron:**

There are 2 iron atoms in $Fe_2(SO_4)_3$ and 1 iron atom in $Fe(OH)_3$. Hence, we will use stoichiometric coefficients of 1 (not shown) for $Fe_2(SO_4)_3$ and 2 for $Fe(OH)_3$.

The partially balanced equation is:

$$Fe_2(SO_4)_{3(aq)} + NH_{3(g)} + H_2O_{(l)} \rightarrow \mathbf{2}\ Fe(OH)_{3(s)} + (NH_4)_2SO_{4(s)}$$

Next, we **balance sulfur:**

There are 3 sulfur atoms in one formula unit of $Fe_2(SO_4)_3$ and 1 sulfur atom in one formula unit of $(NH_4)_2SO_4$. Hence, we will use a stoichiometric coefficient of 3 for $(NH_4)_2SO_4$.

The partially balanced equation becomes:

$$Fe_2(SO_4)_{3(aq)} + NH_{3(g)} + H_2O_{(l)} \rightarrow \mathbf{2}\ Fe(OH)_{3(s)} + \mathbf{3}\ (NH_4)_2SO_{4(s)}$$

Next, we **balance nitrogen:**

On the left -hand side there is 1 nitrogen atom in $NH_3$. On the right-hand side there are $3 \times 2 = 6$ nitrogen atoms in 3 formula units of $(NH_4)_2SO_4$. Hence, we will use a stoichiometric coefficient of 6 for $NH_3$.

The partially balanced equation becomes:

$$Fe_2(SO_4)_{3(aq)} + \mathbf{6}\ NH_{3(g)} + H_2O_{(l)} \rightarrow \mathbf{2}\ Fe(OH)_{3(s)} + \mathbf{3}\ (NH_4)_2SO_{4(s)}$$

Next, we **balance oxygen:**

On the left-hand side there are $3 \times 4 = 12$ oxygen atoms in one formula unit of $Fe_2\ (SO_4\ )_3$ and 1 oxygen atom in one molecule of $H_2O_{(l)}$. On the right-hand side there are 18 oxygen atoms [$2 \times 3 = 6$ oxygen atoms in 2 formula units of $Fe(OH)_3$ and $3 \times 4 = 12$ oxygen atoms in 3 formula units of $(NH_4\ )_2SO_4$ ]. The only coefficient that can be modified is the coefficient for $H_2O$ since all other coefficients have already been assigned. Hence, we will use a stoichiometric coefficient of 6 for $H_2O$.

The partially balanced equation becomes:

(spu) اتحاد سنافر البوليتكنك

$$Fe_2(SO_4)_{3(aq)} + \mathbf{6}\ NH_{3(g)} + \mathbf{6}\ H_2O_{(l)} \rightarrow \mathbf{2}\ Fe(OH)_{3(s)} + \mathbf{3}\ (NH_4)_2SO_{4(s)}$$

Finally, we **balance hydrogen:**

Note that we expect the chemical equation to be balanced for hydrogen as all stoichiometric coefficients have already been assigned.
On the left-hand side there are 30 hydrogen atoms [6 x 3 = 18 hydrogen atoms in three formula units of $NH_3$ and 6 x 2 = 12 atoms in two molecules of $H_2O$]. On the right-hand side there are also 30 hydrogen atoms [2x3 = 6 hydrogen atoms in 2 formula units of $Fe(OH)_3$ and 3x4x2 = 24 hydrogen atoms in 3 formula units of $(NH_4)_2SO_4$].

Hence, the fully balanced chemical equation is:

$$Fe_2(SO_4)_{3(aq)} + \mathbf{6}\ NH_{3(g)} + \mathbf{6}\ H_2O_{(l)} \rightarrow \mathbf{2}\ Fe(OH)_{3(s)} + \mathbf{3}\ (NH_4)_2SO_{4(s)}$$

The sum of the stoichiometric coefficients for products and reactants is **18**.

## Sections 3.15: Balancing Chemical Reactions

These are interactive sessions where you are guided to practice what you have learned in the first 6 sections of part 2.19.

## Section 3.16: Practice Problems on Balancing Reactions

Practice, practice, practice.
If you have problems, go back and study Sections 2.19.1 through 2.19.8!

## Sections 3.17: Balancing Chemical Equations in Word Problems

In these sections, you are taught to analyze statements in a word problem and come up with a chemical reaction. Demonstration of the reaction is shown in the video. Following this, a step by step explanation of balancing the chemical reaction is offered.

Chemical reactions are represented by chemical equations. Chemical equations consist of reactants and products.

**Consider the reaction: Solid sodium metal reacts with water giving a solution of sodium hydroxide and releasing hydrogen gas.**

Write a chemical reaction using complete formulas with phases.

Sodium is Na. It exists as a solid. Hence put "(s)" next to $Na_{(s)}$

The formula for water is $H_2O$. Add $H_2O$

$$Na_{(s)} + H_2O$$

$H_2O$ is a liquid. Hence, put "(l)" next to it. $Na_{(s)} + H_2O_{(l)}$

Giving sodium hydroxide means, produces sodium hydroxide.

Draw the line with arrow. $Na_{(s)} + H_2O_{(l)} \rightarrow$

Chemical formula for sodium hydroxide is NaOH.

Solution of NaOH, put "(aq)" next to it.

$$Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)}$$

Remember hydrogen exists as $H_2$.

$$Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)} + H_2$$

Since $H_2$ exists as gas, put "(g)" next to it.

$$Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)} + H_{2(g)}$$

Reactants are on the left-hand side (LHS) of the equation. Products are on the right-hand side (RHS) of the equation.

The coefficients of a balanced equation are positive integers. Note: Integers will be used here only for the sake of convenience. In reality one can use rational numbers (fractions) to balance chemical reactions. To balance a chemical reaction only the coefficients can be changed. The balanced chemical reaction contains the smallest possible coefficients.

$$Na_{(s)} + H_2O_{(l)} \rightarrow NaOH_{(aq)} + H_{2(g)}$$

On the left-hand side of the equation there is one Na. On the right-hand side of the equation there is one Na. Hence, Na is balanced.

On the left-hand side of the equation there are two H. On the right-hand side of the equation there are three H. In this case there should be an even number of H on both sides. The smallest even number that could work would be four. Hence, a coefficient of 2 is placed for $H_2O$ and a coefficient of 2 is placed for NaOH.

$$Na_{(s)} + 2\ H_2O_{(l)} \rightarrow 2\ NaOH_{(aq)} + H_{2(g)}$$

On the left-hand side of the equation there are two O. On the right-hand side of the equation there are two O. Hence, O is balanced.

Now, on the left-hand side of the equation there is one Na. On the right-hand side of the equation there are two Na. Hence, a coefficient of 2 should be placed for Na on the left-hand side.

$$2\ Na_{(s)} + 2\ H_2O_{(l)} \rightarrow 2\ NaOH_{(aq)} + H_{2(g)}$$

Note: When no coefficient appears, it is understood that the coefficient is 1. Hence, the sum of coefficients of reactants and products for the balanced chemical reaction is 7.

## Sections 3.18: Stoichiometry

We are interested in **reaction stoichiometry** whenever we ask questions such as:

1) What is the amount of each reactant required to produce a known amount of product, or,

2) What is the amount of product formed from a known amount of reactants?

Practically speaking, “amounts” of reactants or products are the masses of these compounds measured in the laboratory in grams.

To relate masses of products to masses of reactants requires relating:

1) Masses to moles using the molar masses, and,

2) Moles or reactants to moles of products, using the stoichiometry of the balanced chemical reaction (that is, using the values of the stoichiometric coefficients).

**Example: In the Haber process, nitrogen reacts with hydrogen to produce ammonia gas.**

$$N_{2(g)} + 3\ H_{2(g)} \longrightarrow 2\ NH_{3(g)}$$

The coefficients (1, 3 and 2) represent the number of moles.

Hence, this equation can be represented as: one mole of $N_2$ reacts with three moles of $H_2$ to produce two moles of $NH_3$.

Whenever we carry out stoichiometric calculations (relating masses of products to masses of reactants), we will always follow the 4 steps given below.

$$\text{Mole Ratio} = \frac{\text{Moles Desired}}{\text{Moles Given}}$$

1) Write the balanced chemical equation.

2) Convert masses given for reactants and products to moles, using the molar masses.

3) Write down the **mole ratio** using the stoichiometric coefficients of the balanced chemical equation.

   **Moles Desired**: Stoichiometric coefficient of the substance for which we want to calculate the amount reacted or produced.

   **Moles Given:** Stoichiometric coefficient of the substance for which we know the amount reacted or produced.

4) Multiply the mole ratio by the number of moles given in the problem or calculated in step 2. Then, convert the calculated number of moles to the mass if necessary.

**Example 1: Consider the following reaction: $PH_{3(g)} + O_{2(g)} \rightarrow P_4O_{10(s)} + H_2O_{(l)}$. How many moles of $PH_{3(g)}$ are required for the production of 3.48 mol ($P_4O_{10}$)?**

1) Balance the chemical equation:

$$4\ PH_{3(g)} + 8\ O_{2(g)} \rightarrow P_4O_{10(s)} + 6\ H_2O_{(l)}$$

2) Convert masses to moles (already done).

3) Write the Mole Ratio:

"How many moles of $PH_{3(g)}$" implies what is desired is $PH_{3(g)}$.

"Production of 3.48 mol ($P_4O_{10}$)" implies what is given is $P_4O_{10(s)}$.

$$\text{Mole Ratio} = \frac{4\ \text{mol}\ (PH_3)}{1\ \text{mol}\ (P_4O_{10})}$$

4) Multiply the mole ratio by the number of moles given in the problem:

$$\frac{4\ \text{mol}\ (PH_3)}{1\ \text{mol}\ (P_4O_{10})} \times 3.48\ \text{mol}\ (P_4O_{10}) = 13.92\ \text{mol}\ PH_3$$

**Example 2: Considering the same chemical reaction, how many moles of $O_{2\ (g)}$ react with 16.3 moles of $PH_3$?**

To solve this problem, let us use the conversion factor (mole ratio).

$$\text{mol}\ O_2 = \frac{8\ \text{mol}\ (O_2)}{4\ \text{mol}\ (PH_3)} \times 16.3\ \text{mol}\ (PH_3) = 32.6\ \text{mol}\ O_2$$

**Example 3: Iron reacts with chlorine gas to form iron(III) chloride. Calculate the mass of iron(III) chloride produced from 12.24 moles of iron.**

$$2\ Fe_{(s)} + 3\ Cl_{2\ (g)} \longrightarrow 2\ FeCl_{3\ (s)}$$

The reactant Fe is given in moles:

$$\text{In the chemical equation there are}: \frac{2\ \text{mol}\ FeCl_3}{2\ \text{mol}\ Fe} \times 12.24\ \text{mol}\ Fe$$
$$= 12.24\ \text{mol}\ FeCl_3$$

However, the problem asks for the mass of $FeCl_3$.

$$\text{mass} = 12.24 \text{ mol} \times 162.4 \text{ g/mol} \qquad \text{mass} = \text{moles} \times \text{molar mass}$$

$$\text{mass} = 1988 \text{ g}$$

**Example 4: Silicon dioxide heated with excess of carbon (coke) produces pure silicon and carbon monoxide. How many grams of carbon monoxide are formed when 32.55 g of silicon are produced?**

$$SiO_2 + 2\ C \longrightarrow Si + 2\ CO \qquad \frac{32.55\ g\ Si}{28.1\ g/mol} = 1.158\ mol\ Si$$

$$\text{In the chemical equation there are: } \frac{2\ mol\ CO}{1\ mol\ Si} \times 1.158\ mol\ Si$$
$$= 2.316\ mol\ CO$$

However, the problem asks for the mass of CO.

$$\text{mass} = \text{moles} \times \text{molar mass}$$
$$= 2.316 \text{ mol} \times 28.0 \text{ g/mol}$$
$$= 64.90 \text{ g CO}$$

In Section 2.21, practice the Interactive Problems.

## Sections 3.20: Theoretical Yield, Limiting Reactant and Percent Yield

Consider the chemical reaction:

$$2\ Sb_{(s)} + 3\ Cl_{2(g)} \rightarrow 2\ SbCl_{3(s)}$$

The reaction equation provides a recipe for the preparation of antimony chloride. It indicates that one needs 3 moles of chlorine gas for every two moles of antimony solid to form 2 moles of antimony chloride.

Whenever reactants are present in the relative amount of 3 moles of chlorine for 2 moles of antimony, we say that the reactants are present in the **stoichiometric amount**.

For example, if we are given 0.4 mol Sb with 0.6 mol $Cl_2$, we have reactants present in the stoichiometric amount (ratio of 0.4 to 0.6 = ratio of 2 to 3).

Whenever we have reactants present in the stoichiometric amount, we can use any of the reactants to calculate how many moles of product are formed from a given amount of reactants.

The maximum mass of products that can be formed from a given amount of reactants, assuming the reaction is complete, is called the **Theoretical Yield**.

**For example**: If we react 2 moles of Sb with 3 moles of $Cl_2$ , we obtain 2 moles of $SbCl_3$. The mass of 2 moles of $SbCl_3$ is equal to $2 \times (121.8 + 3 \times 35.5) = 456.6$ g. We say that when the reaction is carried out with 2 moles of Sb and 3 moles of $Cl_2$, the theoretical yield is equal to 456.6 g.

When the reaction is carried out with 4 moles Sb and 6 moles $Cl_2$, the theoretical yield is equal to 913.2 g. When the reaction is carried out with 0.2 moles Sb and 0.3 moles $Cl_2$, the theoretical yield is 45.7 g

When the reactants are present in the stoichiometric amount and the reaction is complete, there is no reactant left at the end of the reaction.

What happens if reactants are not present in the stoichiometric amount? When reactants are not present in the stoichiometric amount, then, some of the reactants are completely consumed in the reaction (**limiting reactant**) and some reactants remain at the end of the reaction (**excess reactant**).

**For example:** Consider the reaction between antimony and chlorine gas, again.

$$2\ Sb_{(s)} + 3\ Cl_{2(g)} \rightarrow 2\ SbCl_{3(s)}$$

When the reaction is started with 4 moles Sb and 6 moles $Cl_2$, we said above that the reactants are present in the stoichiometric amount.

When the reaction is started with 4 moles Sb and 7 moles $Cl_2$, then, at the end of the reaction, there will be one mole of $Cl_2$ left, since to completely react 4 moles Sb, only requires 6 moles $Cl_2$ . We say that $Cl_2$ is in excess and Sb is the limiting reactant (it is the one limiting the amount of product that can be formed).

When the reaction is started with 5 moles Sb and 6 moles $Cl_2$, then, at the end of the reaction, there will be one mole of Sb left, since to completely react 6 moles $Cl_2$, only requires 4 moles Sb. We say that Sb is in excess and $Cl_2$ is the limiting reactant (it is the one limiting the amount of product that can be formed).

Remember that the theoretical yield is always determined by the amount of limiting reactant.

**Example: Assume that the reaction between antimony and chlorine is carried out with 1.5 mol Sb and 1.5 mol $Cl_2$. Determine the limiting reactant, the excess reactant and the theoretical yield.**

To determine which reactant is the limiting reactant, we calculate the amount of product formed from each of the given amounts of reactants. The limiting reactant is the reactant that produces the least amount of products.

1) Write the balanced chemical reaction.

$$2\ Sb_{(s)} + 3\ Cl_{2(g)} \rightarrow 2\ SbCl_{3(s)}$$

2) Determine the amounts of product formed from each reactant:

How much $SbCl_3$ is formed with 1.5 mol Sb?

Using the mole ratio derived from the balanced chemical reaction, we write:

$$\text{moles}\,(SbCl_3) = \frac{2\ \text{mol}\ SbCl_3}{2\ \text{mol}\ Sb} \times 1.5\ \text{mol}\ Sb = 1.5\ \text{mol}\ SbCl_3$$

How much $SbCl_3$ is formed with 1.5 mol $Cl_2$?

Using the mole ratio derived from the balanced chemical reaction, we write:

$$\text{moles}\,(SbCl_3) = \frac{2\ \text{mol}\ SbCl_3}{3\ \text{mol}\ Cl_2} \times 1.5\ \text{mol}\ Cl_2 = 1.0\ \text{mol}\ SbCl_3$$

3) The limiting reactant is the reactant producing the smallest amount of product.

The limiting reactant is $Cl_2$. Hence, the excess reactant is Sb.

The theoretical yield is the amount of products formed from the limiting reactant. 1.5 mol $Cl_2$ produce 1.0 mol $SbCl_3$. The mass of 1.0 mol $SbCl_3$ is equal 228.3 g.

Hence, the theoretical yield is 228.3 g.

We found that the limiting reactant is chlorine gas and the excess reactant is antimony. So we know that all chlorine (1.5 mol) is consumed in the reaction. Now, let us calculate how much antimony is left after the reaction.

(spu) اتحاد سناف البوليتكنك

First, we calculate how much antimony is used to fully react with 1.5 mol $Cl_2$. We do this using the mole ratio.

$$\text{moles (Sb)} = \frac{2 \text{ mol Sb}}{3 \text{ mol } Cl_2} \times 1.5 \text{ mol } Cl_2 = 1.0 \text{ mol Sb}$$

Hence, 1.0 mol Sb is used out of an initial amount of 1.5 mol Sb. We conclude that 0.5 mol Sb remains after the reaction.

**Experimental Yield, Theoretical Yield and Percent Yield**

For a variety of reasons, the vast majority of chemical reactions carried out in laboratories or in chemical plants do not produce the maximum amount of product possible (theoretical yield).

Remember the **Theoretical Yield** is calculated from the amounts of reactants present and from the balanced chemical reaction.

The amount of products actually obtained from a given amount of reactants is called the **Experimental Yield**. The experimental yield is thus always smaller than the theoretical yield.

The **Percent Yield** is defined as the ratio of experimental yield to theoretical yield multiplied by 100.

$$\% \text{ Yield} = 100 \times \frac{\text{Experimental Yield}}{\text{Theoretical Yield}}$$

**Example: 68.5 g of $CO_{(g)}$ is reacted with 8.60 g of $H_{2\,(g)}$ to form 35.7 g of $CH_3OH$ (methanol). Determine if any reactant is in excess. Calculate the theoretical yield and the percent yield.**

1) Write the balanced chemical reaction.

$$CO_{(g)} + 2\ H_{2\,(g)} \rightarrow CH_3OH_{(l)}$$

2) Calculate the amount of methanol formed from 68.5 g CO.

Use the method of conversion factors.

$$g(CH_3OH) = 68.5\ g\ CO \times \frac{1\ mol\ (CO)}{28.0\ g\ (CO)} \times \frac{1\ mol\ (CH_3OH)}{1\ mol\ (CO)} \times \frac{32.0\ g\ (CH_3OH)}{1\ mol\ (CH_3OH)}$$

$$g\ (CH_3OH) = 78.2\ g\ (CH_3OH)$$

3) factors. Calculate the amount of methanol formed from

8.60 g $H_2$. Use the method of conversion

$$g(CH_3OH) = 8.60\ g\ H_2 \times \frac{1\ mol\ (H_2)}{2.0\ g(H_2)} \times \frac{1\ mol\ (CH_3OH)}{2\ mol\ (H_2)} \times \frac{32.0\ g\ (CH_3OH)}{1\ mol\ (CH_3OH)}$$

$$g\ (CH_3OH) = 68.8\ g\ (CH_3OH)$$

4) The mass of methanol formed with 8.60 g hydrogen is less than that formed with 68.5 g carbon monoxide. Hence, hydrogen is the limiting reactant and carbon monoxide is the excess reactant. The theoretical yield is the amount of methanol formed with 8.6 mol $H_2$ (limiting reactant). The theoretical yield is equal to 68.8 g.

5) The percent yield is then expressed as:

$$\%\ Yield = 100 \times \frac{35.7\ g}{68.8\ g} = 51.9\ \%$$

# CH 4 REACTION IN AQUEOUS

## SOLUTION:

- Solute
- Solvent

  - Aqueous solution → solven is water
  - $HCl_{aq} + NaoH_{aq} \rightarrow NaCl_{aq} + H2O_L$

Each solution have a certain concentration

Concentration of solutions (ratio of solute to total solution )

تراكيز المحاليل / نسبة المذاب لكل المحلول

1. Molarity → M= moles of solute /liter of solution = mol/l

2. w/w% percentage by mass

0.1M محلول مخفف diluted solution 1.0M 2.0M محلول مركز most solution

To prepare a solution :

3g of NaCl dissolved in 200ml water , what is conc . of this solution?

Molarity = moles of NaCl/ liter of solution = 0.051mol/0.2l = 0.256 M

- Moles of NaCl Solute = 3g /85.5g/mol = 0.051mol

Liter of solution = 0.20l

**اتحاد سنافر البوليتكنك (spu)**

**How many grams of NaOH are required to prepare a150ml solution of 0.25M ?**

0.25M = moles of NaOH / 0.15L

Moles of NaOH = 0.37 mol* 40g/mol

= 1.48g

**How many grams of NaOH should be used to prepare a150ml solution of 0.03M using a 250 ml flask ?**

M= moles of NaOH /liters of solution → 0.03 = moles /0.25

moles of NaOH = 0.0075mol = $7.5*10^{-3}$

mss of NaOH = $7.5*10^{-3}$*40G/MOL = 0.3g NaOH

prepare a solution of 3M (molars ) of NaOH

3M = moles /1liter → moles of NaOH = 3g /mol

Mass of NaOH = 3*40 = 120g

**نحن نفرض الحجم الذي نريد**

## HOW TO PREPARE SOLUTION : DILUTION

1. To weight **certain** mass of solution in flask of know volume
2. To prepare a diluted solution from a concentrated solution conc- of diluted solution

$$M1V1 = M2V2$$

What is the volume of water that should be added to 50ml of 6M $CaCl_2$ to prepare a solution of 0.5M

**(spu) اتحاد سنافر البوليتكنك**

$$M_1V_1 = M_2V_2$$

6M* 50 ml = 0.9 *V2

V2 = 333ml

V of water = 333- 50 = 283

The volume of new solution is 333ml

So we should add 283ml water ($H_2O$) TO 50ml to produce a solution of 0.9M.

Q :A solution of $NaCl_2$ (M= 0.3m) (v=60ML) of 160ml of water is added to this solution what is the molarity of new solution ?

$$M_1V_1 = M_2V_2$$

$V_2$ = 60+160 = 220ml

0.3 M* 60 = $M_2$ *220

$M_2$ = 0.081M

Q:

What is the volume of 16M $H_2SO_4$ must be used to prepare 1.5l of a 0.10M $H_2SO_4$?

16M*Lt = 0.10M *1.5Lt

X = 0.15/16 = 0.0094Lt = 9.4 ml

Q:

How much of a SM stock solution of NaCl will you need to make 250ml of a1.5M solution ?

Smol/Lt. * (X Lt)= 1.5mol/Lt. * (1.25Lt. )

X Lt = ( 1.5mol /Lt.)* 0.25Lt. * Lt/5mol

X Lt = 0.075 Lt or 75.0ml

**(spu) اتحاد سنافر البوليتكنك**

We Need 0.075Lt. of 5M NaCl

The rest of 0.25Lt. is made up by addition of water

0.25Lt. - 0.075Lt. = 0.175Lt.

## W/W% PERCENTAGE BY MASS

**W/W % = (mass of solute /mass of solution )*%**

**= (mass of solute / mass of solute+ mass of solvent )*%**

**Ex:**

**What is the w/w% of solution that contains 40g of $CaSO_4$ in 60g water ?**

w/w% = 40/40+60*100% = 40/100*100 = 40%

40% = 40g/100g = mass of solute /massof solution

M = moles of solute/L of solution

M=40/136 = 40/136*50

0.005M

**Q:**

**An eques solution contains 700g water and 45g NaOH what is the percentage mass of Sodium hydroxide?**

Mass% NaOH = M NaOH /($M_{NaOH}$ + $M_{H2O}$)%

= 45/45+700*100%

= 6.04%

الكتلة = الكثافة *الحجم

$1.2/cm^3$

60g=1.2ml *v

V = 50ml

Moles = mass/molar mass

40/40+32+64

Moles = 40/136=0.26mol

اتحاد سنافر البوليتكنك (spu)

**Q:**

**What is the mass percentage of hydrogen in water ?**

At.wt of H = 1.008g/mol

At.wt of O = 16.00 g/mol

1 mol . $H_2O$ → 2mol . H

1 mol . $H_2O$ → 2* 1.008g . H

Mass% = (mass of H in 1mol .$H_2O$)/(mass of $H_2O$) *100%

Mass%H = (2*1.008)/22*1.008+16.00)*100%

11.19%

**Q:**

**Calculate the grams of NaCl (5.25% by mass) in 245g of commercial bleach solution ?**

**Mass% = m . solute /m. solution *100%**

**m. so;ute = mass%* m. solution/100**

**5.25*245/100 = 12.9g**

**اتحاد سنافر البوليتكنك (spu)**

## CHAPTER 4. AQUEOUS REACTION CHEMISTRY

**solution** - homogeneous mixture of 2 or more substances; uniform distribution of particles and same properties throughout.

A solution is composed of a solute dissolved in a solvent:
**solute** - substance present in lesser amount
**solvent** - substance present in greatest amount

**aqueous solution** – water is solvent; write (aq) after formula

### *PROPERTIES AND CLASSIFICATION OF AQUEOUS SOLUTIONS*

1) **Strong electrolyte** - a solution that is a good conductor of electricity.
- Substance that totally dissociates or ionizes in water. (Many ions)
- Soluble ionic compounds, strong acids, strong bases. E.g. NaCl, HCl, NaOH

2) **Nonelectrolyte** - solution that does not conduct electricity.
- Neutral molecules are present in solution. (No ions)
- Molecular compounds such as $H_2O$, $C_6H_{12}O_6$ (sugar)

3) **Weak Electrolyte** - solution that conducts electricity poorly.
- Substance that partially ionizes in water. (few ions)
- Weak acids or weak bases such as $HC_2H_3O_2$, $NH_3$

- Understand how ionic compounds & molecular compounds dissolve in water

### *ACID, BASE & SALT DEFINITIONS*

**Acid**: substance that produces $H^+$ ions in aqueous solutions.

- Strong acids completely ionize in water:

E.g. $HNO_3 \rightarrow H^+ + NO_3^-$

only $H^+$ ions and $NO_3^-$ ions are present in water

**KNOW 7 Strong acids**: HCl, HBr, HI, $HNO_3$, $H_2SO_4$, $HClO_4$, $HClO_3$

- All other acids are weak and only partially ionize in water (reversible rxn):

E.g. $HC_2H_3O_2 \rightleftharpoons H^+ + C_2H_3O_2^-$

All 3 species are present - most of the acid molecules have not ionized

**Base**: substance that accepts $H^+$ ions in aqueous solns.; forms $OH^-$ ions

$NaOH_{(aq)} \rightarrow Na^+ + OH^-$

**Common Strong bases**: Group 1A hydroxides (e.g. LiOH, NaOH, KOH) & heavy Group 2A hydroxides ($Ca(OH)_2$, $Ba(OH)_2$, $Sr(OH)_2$)

Weak bases: $NH_3$, several others exist but don't need to recognize yet.

**NEUTRALIZATION RXN** (Type of DR rxn): ACID + BASE $\rightarrow$ SALT + $H_2O$

(spu) اتحاد سنافر البوليتكنك

**Salt** - ionic compound formed from base cation & acid anion during acid-base rxn.

E.g. $HCl + NaOH \rightarrow NaCl + H_2O$

## *DOUBLE REPLACEMENT REACTIONS: AX + BY→ AY + BX*

- Ions trade partners

**DR reactions occur when:**

1) Insoluble product (solid ppt) forms. Refer to solubility rules - will be provided.

2) Weak or nonelectrolyte forms. One of products is a weak acid, weak base or molecular compound.

3) Gas forms.

✱Note: DR rxn does not occur if 2 soluble salts form.

**Precipitation reactions**

**solubility** - maximum amount of solid that will dissolve in a given amount of liquid at a given temperature. Solubility is a physical constant (like BP, density)

E.g. NaCl solubility in water at 20 °C: $\frac{36.0\,g\,NaCl}{100\,g\,H_2O}$

**Solubility Rules** – table 4.1, p. 111

- Know compounds containing 1A or $NH_4^+$ are always soluble; other ions, refer to table
- If insoluble in water, symbol is (s); if soluble in water, symbol is (aq)

E.g. $Na_2CO_{3(aq)} + CaCl_{2(aq)} \rightarrow 2NaCl_{(aq)} + CaCO_{3(s)}$

## *NET IONIC REACTIONS*

**Molecular Equation**- write full formulas of all substances. See example above.

**Total Ionic Equation**: Split up formulas of soluble salts, strong acids and strong bases into ions.

- Don't split apart solids, liquids, gases, weak or non-electrolytes.

**Net Ionic Equation**: An ionic equation in which the spectator ions have been removed - net result after canceling common ions in total ionic equation.

**Spectator ions**: Ions present on both sides of the equation.

- Must include (s), (g) or (l) symbols, but can omit (aq) symbol in writing rxns

Total ionic: $2Na^+ + CO_3^{2-} + Ca^{2+} + 2Cl^- \rightarrow 2Na^+ + 2Cl^- + CaCO_{3(s)}$

Net ionic: $CO_3^{2-} + Ca^{2+} \rightarrow CaCO_{3(s)}$

Exercise. Write molecular, total & net ionic rxns for the following:

1. molecular: $Pb(NO_3)_{2(aq)} + 2HCl_{(aq)} \rightarrow PbCl_{2(s)} + 2HNO_{3(aq)}$

(spu) اتحاد سنافر البوليتكنك

Total: $Pb^{2+} + 2NO_3^- + 2H^+ + 2Cl^- \rightarrow PbCl_{2(s)} + + 2H^+ + 2NO_3^-$

Net: $Pb^{2+} + 2Cl^- \rightarrow PbCl_{2(s)}$

2. Molecular: $2FeBr_{3(aq)} + 3Ca(OH)_{2(aq)} \rightarrow 2Fe(OH)_{3(s)} + 3CaBr_{2(aq)}$

Total: $2Fe^{3+} + 6Br^- + 3Ca^{2+} + 6OH^- \rightarrow 2Fe(OH)_{3(s)} + 3Ca^{2+} + 6Br^-$

Net: $2Fe^{3+} + 6OH^- \rightarrow 2Fe(OH)_{3(s)}$
$Fe^{3+} + 3OH^- \rightarrow Fe(OH)_{3(s)}$

**Acid-base neutralization reaction – nonelectrolyte ($H_2O$) is formed**

Molecular: $H_2SO_{4(aq)} + 2NaOH_{(aq)} \rightarrow 2H_2O_{(l)} + Na_2SO_{4(aq)}$

Total: $2H^+ + SO_4^{2-} + 2Na^+ + 2OH^- \rightarrow 2H_2O_{(l)} + 2Na^+ + SO_4^{2-}$

Net: $2H^+ + 2OH^- \rightarrow 2H_2O_{(l)}$
$H^+ + OH^- \rightarrow H_2O_{(l)}$

- note that formation of water is net result when strong acid & strong base react

**Gas formation during acid-base reactions**

This occurs if
(A) a gas such as $H_2S$ forms

$Na_2S_{(aq)} + 2HNO_{3(aq)} \rightarrow 2NaNO_{3(aq)} + H_2S_{(g)}$

Total: $2Na^+ + S^{2-} + 2H^+ + 2NO_3^- \rightarrow 2Na^+ + 2NO_3^- + H_2S_{(g)}$

Net: $S^{2-} + 2H^+ \rightarrow H_2S_{(g)}$

(B) an unstable product forms & breaks down. E.g. $H_2CO_{3(aq)} \rightarrow CO_{2(g)} + H_2O_{(l)}$

$Na_2CO_{3(aq)} + 2HCl_{(aq)} \rightarrow 2NaCl_{(aq)} + H_2O_{(l)} + CO_{2(g)}$

Total: $2Na^+ + CO_3^{2-} + 2H^+ + 2Cl^- \rightarrow 2Na^+ + 2Cl^- + H_2O_{(l)} + CO_{2(g)}$

Net: $CO_3^{2-} + 2H^+ \rightarrow H_2O_{(l)} + CO_{2(g)}$

**Oxidation numbers- bookkeeping system for keeping track of e-**

- these are values we assign; don't necessarily correspond to true number of e- gained or lost by atom
- write +/- sign before number instead of after

**Rules**

1) Uncombined elements: ox number = 0. E.g. Na or $O_2$
2) monatomic ion: ox number = charge for ion; $S^{2-}$: ox # = -2; $Mg^{2+}$: ox # = +2
3) H: ox # = +1 (except for MH compounds, H = -1)
4) O: ox # = -2 (except $O_2^{2-}$, O = -1,
5) F: ox # = -1; other halogens usually –1 except when combined with O
6) polyatomic: sum of ox #'s = charge on ion
neutral compound: sum of ox #'s = 0

E.g. assign ox #'s to atoms in the following:

(spu) اتحاد سنافر البوليتك

1. $SF_2$ F: -1; S: +2
2. $CO_3^{2-}$ O: -2 $C + 3(-2) = -2 \Rightarrow C + -6 = -2 \Rightarrow C = +4$
3. $ClO_3^-$ O: -2 $Cl + 3(-2) = -1 \Rightarrow Cl + -6 = -1 \Rightarrow Cl = +5$

4. $HNO_3$ H: +1; O: -2; +1 + N + 3(-2) = 0 ⇒ N −5 = 0 ⇒ N = +5

***SINGLE REPLACEMENT (OXIDATION-REDUCTION) REACTIONS: A + BC → B + AC***

**oxidation**: loss of electrons, substance becomes more + (LEO)

$Mg \rightarrow Mg^{2+} + 2e^-$

**reduction**: gain of electrons, substance becomes more - (GER)

$Ag^+ + e^- \rightarrow Ag$

**oxidizing agent**: causes other substance to be oxidized; ox. agent is reduced

**reducing agent**: causes reduction & becomes oxidized

$Ca_{(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + H_{2(g)}$

Net: $Ca_{(s)} + 2H^+_{(aq)} \rightarrow Ca^{2+}{}_{aq)} + H_{2(g)}$

Which is oxidizing agent? reduced – gains e-: $H^+$

Which is reducing agent? oxidized – loses e-: Ca

Many reactions involve oxidation & reduction not just SR:
E.g. For the following reaction $2PbS + 3O_2 \rightarrow 2PbO + 2SO_2$
+2 −2 0 +2 −2 +4 −2

a. Identify the species oxidized $S^{-2}$

b. Identify the species reduced O

c. Oxidized element goes from oxidation number -2 to +4.

d. Reduced element goes from oxidation number 0 to -2.

e. Oxidizing agent $O_2$

f. Reducing agent PbS

**Predict products for SR rxns**

Not all single replacement reactions occur - depends on activity of metal

- Use activity series to decide if reaction occurs (Table on p. 124)
- Elements at top are more reactive (more likely to become oxidized and form ions). An element in the series can replace elements below it but cannot replace elements above it.

Complete the following reactions. Write NR if reaction does not occur.

1. $3Zn_{(s)} + 2H_3PO_4 \rightarrow Zn_3(PO_4)_2 + 3H_{2(g)}$
2. $Ag_{(s)} + HCl_{(aq)} \rightarrow$ NR
3. $Mg_{(s)} + 2AgNO_{3(aq)} \rightarrow$ (Spu) اتحاد وعنصر النيوليتكنك $Mg(NO_3)_{2(aq)} + 2Ag_{(s)}$
4. $2Al_{(s)} + 3H_2SO_{4(aq)} \rightarrow Al_2(SO_4)_{3(aq)} + 3H_{2(g)}$
5. $H_{2(g)} + CuBr_{2(aq)} \rightarrow Cu_{(s)} + 2HBr_{(aq)}$

## *SOLUTION CONCENTRATION*

- Because chemical reactions between solids are usually slow, we carry out these reactions by dissolving the solids in water. (Dissolved ions are more mobile & come into contact more frequently)
- The most common unit for expressing the concentration of the solution is called Molarity.

$$\textbf{Molarity = M = } \frac{\textbf{moles solute}}{\textbf{Liter of solution}} \qquad \text{Units: M} = \frac{moles}{Liter}$$

1) 30.6 g NaOH are dissolved in enough water to make 350 mL of solution. Calculate the molarity.

30.6 g NaOH $\frac{1 mol\ NaOH}{40.0\, g\ NaOH}$ = 0.765 mol NaOH; 350 mL $\frac{1 L}{1000\, mL}$ = 0.35 L

$$M = \frac{mol}{L} = \frac{0.765\, mol\ NaOH}{0.35\, L} = 2.2\ M$$

2) How many mL of 0.256 M KCl will contain 10.0 mol of KCl?

$$10.0\ mol\ KCl \left(\frac{1 L\ KCl}{0.256\, mol\ KCl}\right)\left(\frac{1000\, mL}{1 L}\right) = 39100\ mL$$

### Molarity of ions

Soluble ionic compounds completely dissociate in solution

E.g. What is the molarity of $K^+$ ions in a 0.20 M $K_2CO_3$ solution? What is the molar concentration of the carbonate ions?

$$\frac{0.20\, mol\ K_2CO_3}{1 L}\left(\frac{2\, mol\ K^+}{1 mol\ K_2CO_3}\right) = \frac{0.40\, mol\ K^+}{1 L} = 0.40\ M\ K^+$$

$$\frac{0.20\, mol\ K_2CO_3}{1 L}\left(\frac{1 mol\ CO_3^{2-}}{1 mol\ K_2CO_3}\right) = \frac{0.20\, mol\ CO_3^{2-}}{1 L} = 0.20\ M\ CO_3^{2-}$$

## *SOLUTION STOICHIOMETRY*

Grams or L or M of soln → moles given → moles wanted → Grams or L or M of soln

To go from g ↔ moles use MM = $\frac{g}{mol}$

To go from L (or M) of soln ↔ moles use M = $\frac{mole}{L}$

1. How many L of 0.300 M $CaCl_2$ is needed to react with 4.90 g $H_3PO_4$?

$$3CaCl_{2(aq)} + 2H_3PO_{4(aq)} \rightarrow Ca_3(PO_4)_{2(s)} + 6HCl_{(aq)}$$

g $H_3PO_4$ → mol $H_3PO_4$ → mol $CaCl_2$ → L $CaCl_2$

$$4.90\ g\ H_3PO_4 \left(\frac{mol\ H_3PO_4}{98.0\, g\ H_3PO_4}\right)\left(\frac{3 mol\ CaCl_2}{2 mol\ H_3PO_4}\right)\left(\frac{1 L}{0.300\, mol\ CaCl_2}\right) = 0.250\ L\ CaCl_2$$

2. How many grams of $H_2S$ can be produced from 125 mL of 0.552 M HCl solution?

$$2HCl_{(aq)} + K_2S_{(aq)} \rightarrow H_2S_{(g)} + 2KCl_{(aq)}$$

mL HCl → L HCl → moles HCl → moles $H_2S$ → g $H_2S$

(spu) منافع الجزيئات

$$125\ mL\left(\frac{1\,L}{1000\,mL}\right)\left(\frac{0.552\,mol\,HCl}{1\,L}\right)\left(\frac{1\,mol\,H_2S}{2\,mol\,HCl}\right)\left(\frac{34.1\,g\,H_2S}{mol\,H_2S}\right) = 1.18\ g\ H_2S$$

## *WORKSHEET FOR MOLARITY & SOLUTION STOICHIOMETRY*

1. What is the molarity of $Na^+$ and $PO_4^{3-}$ ions in a 0.25 M solution of $Na_3PO_4$?

$$\left(\frac{0.25\,mol\,Na_3PO_4}{1\,L}\right)\left(\frac{3\,mol\,Na^+}{1\,mol\,Na_3PO_4}\right)=0.75\,M\ Na^+$$

$$\left(\frac{0.25\,mol\,Na_3PO_4}{1\,L}\right)\left(\frac{1\,mol\,PO_4^{3-}}{1\,mol\,Na_3PO_4}\right)=0.25\,M\ PO_4^{3-}$$

2. Find the mass of NaCl present in 150 ml of 1.50 M NaCl.

$$150\ mL\left(\frac{1\,L}{1000\,mL}\right)\left(\frac{1.50\,mol}{1\,L}\right)\left(\frac{58.5\,g\,NaCl}{1\,mol\,NaCl}\right) = 13.2\ g\ NaCl$$

3. How many mL of 0.35 M AgCl solution will contain 44.5 g of AgCl?

$$44.5\ g\ AgCl\left(\frac{1\,mol\,AgCl}{143.4\,g\,AgCl}\right)\left(\frac{1\,L\,AgCl}{0.35\,mol\,AgCl}\right)\left(\frac{1000\,mL}{1\,L}\right)= 8.90\ x10^2\ mL\ AgCl$$

4. What is the molarity of $H_2SO_4$ solution if 50.00 ml of $H_2SO_4$ is required to neutralize 42.00 mL of 0.150 M NaOH?

$$H_2SO_4 + 2NaOH \rightarrow 2H_2O + Na_2SO_4$$

mL NaOH → L NaOH → moles NaOH → moles $H_2SO_4$ → M $H_2SO_4$

$$42.00\ mL\left(\frac{1\,L}{1000\,mL}\right)\left(\frac{0.150\,mol\,NaOH}{1\,L}\right)\left(\frac{1\,mol\,H_2SO_4}{2\,mol\,NaOH}\right) = 0.00315\ moles\ H_2SO_4$$

$$M = \frac{mole}{L} = \left(\frac{0.00315\,mol\,H_2SO_4}{50.00\,mL}\right)\left(\frac{1000\,mL}{1\,L}\right) = 0.0630\ M$$

5. How many L of 0.164 M $Ca(OH)_2$ is needed to neutralize 25.00 mL of 0.458 M HCl solution?

$$Ca(OH)_2 + 2HCl \rightarrow CaCl_2 + 2H_2O$$

$$25.00\ mL\left(\frac{1\,L}{1000\,mL}\right)\left(\frac{0.458\,moles\,HCl}{L}\right)\left(\frac{1\,mol\,Ca(OH)_2}{2\,mol\,HCl}\right)\left(\frac{1\,L}{0.164\,mol\,Ca(OH)_2}\right) = 0.0349\ L$$

**Titration**- procedure used to determine the concentration of a solution by measuring the volume of one reagent required to react with a known mass or volume of another reagent.

**Standard solution** – solution with a known concentration

**Indicator** – substance that changes color at end point of titration

**Equivalence point** – pt at which the reagents are present in stochiometrically equivalent amounts

***DILUTION***

**Dilution -** process of adding more solvent to solution to prepare a more dilute solution.
**concentrated solution** - a solution containing a relatively large amount of solute
**dilute solution** - a solution containing a relatively small amount of solute.

In dilution problems moles before dilution equal moles after dilution:

$$moles_1 = moles_2$$

Dilution formula: $M_1 \times V_1 = M_2 \times V_2$

1) a) How many mL of 16 M $HNO_3$ will you use to prepare 750. ml of 0.69 M $HNO_3$?
b) How many mL of water were added to make this solution?

$M_1$ = 0.69 M $HNO_3$; $V_1$ = 750 mL; $M_2$ = 16 M $HNO_3$; $V_2$ = ?

a) $V_2 = \frac{M_1 V_1}{M_2} = \frac{(0.69\,M)(750\,mL)}{16\,M}$ = 32 mL

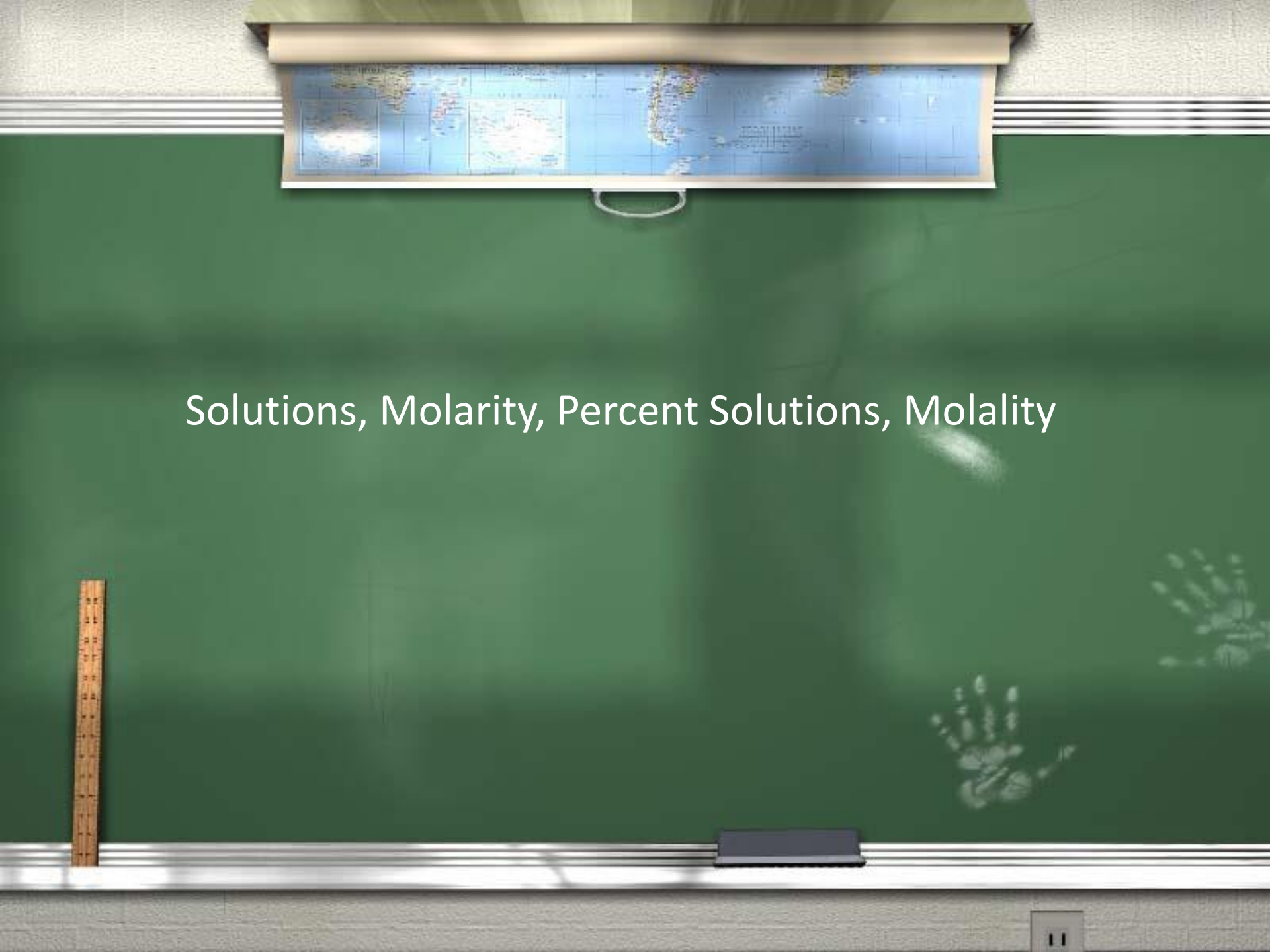
b) 750 mL - 32 mL = 728 mL

2) What is the molarity of a solution prepared by diluting 125 ml of 0.400 M KCl to 1000 mL of KCl solution?

$M_1$ = 0.400 M KCl; $V_1$ = 125 mL; $M_2$ = ?; $V_2$ = 125 mL + 875 mL = 1000 mL

$$M_2 = \frac{M_1 V_1}{V_2} = \frac{(0.400\,M)(125\,mL)}{1000\,mL} = 0.0500\ M$$

**DON'T USE DILUTION FORMULA FOR SOLUTION STOICHIOMETRY PROBLEMS!!!**


Solutions, Molarity, Percent Solutions, Molality

<u>Solution -</u> A homogeneous mixture of 2 or more substances.

Examples: Salt water, Orange juice, "Kool-aid"

All solutions are a mixture of the solute(s) and the solvent.

They are always listed with the lesser constituent (the solute) first.

Solute - The substance that is dissolved. It is the lesser constituent by volume. (Ex: The Salt in the Saltwater.)

Solvent - The substance that does the dissolving. It is the greater constituent by volume. (Ex: The Water in the Saltwater.)

There are 6-types of solutions based on the phases of their components.

| | | |
|---|---|---|
| 1. | Gas - Gas | $O_2$ in air |
| 2. | Liquid - Gas | Water vapor in the air |
| 3. | Gas - Liquid | $CO_2$ in $H_2O$ |
| 4. | Liquid - Liquid | “2-stroke” fuel (oil in gas) |
| 5. | Solid - Liquid | Salt in water |
| 6. | Solid - Solid | Alloys (Cu in Ag, “Sterling Ag”) |

Miscible - Describes two liquids that are mutually soluble in each other.

Examples:

1. Water and alcohol are completely miscible with each other.
2. Water and ether are slightly miscible with each other. About 1.5% water : 98.5% Diethyl ether (m/m)
3. Water and oil are not miscible with each other.

The General Rule with regards to Miscibility:

"Likes dissolve likes."

1. Polar solvents dissolve polar compounds.

2. Non-polar solvents dissolve non-polar compounds.

The “Solution process” involves two actions:

1. Dissolving
2. Crystallizing

<u>Equilibrium (as its relates to solvation)-</u>

The physical state where the two opposing processes of dissolving and crystallizing occur at equal rates.

Le Châtelier's principle -

When stress is applied to a system at equilibrium, the system shifts to relieve the stress.

Example:

As you add salt to water, it dissolves. However, as the concentration of salt approaches saturation, the rate of solvation slows.

## Saturated solution -

A solution in which the dissolved and un-dissolved substances are at equilibrium. (The solvent can no longer dissolve any additional solute. It will simply fall to the bottom of the solution.)

The “solubility” of a substance determines when a solution will become saturated.

## Solubility -

The maximum amount of a solute that can dissolve in a given amount of solvent under specified conditions.

Examples at $20^{\circ C}$ :

1. 222 g of Silver nitrate / 100 g of water
2. 3.89 g of Barium hydroxide / 100 g of water
3. 0.169 g of $CO_2$ / 100 g of water (at SP)
4. 0.0043 g of $O_2$ / 100 g of water (at SP)

**Factors that affect the rate of solution:**

1. Particle size
2. Agitation
3. Temperature (aka: the Heat of Solution)
4. Polarities of solute and solvent
5. Concentration of solution (solvation slows as it nears saturation)

## Heat of Solution -

The difference between the heat content of a solution and it's components.

Heats of solution are either positive or negative.

## Positive Heat of Solution -

Solute + solvent + heat $\Rightarrow$ solution (endothermic)

“Heating helps solvation”

Heat has a positive effect on solvation.

## Negative Heat of Solution -

Solute + solvent $\Rightarrow$ solution + heat (exothermic)

"Heating hinders solvation"

Heat has a negative effect on solvation.

**Concentration units for solutions:**

1. Molarity - The number of moles of solute per liter of solution
2. Molality - The number of moles of solute per kilogram of solvent.
3. Normality - The number of equivalents of a substance dissolved in a liter of solution.
4. Percent solutions - (m/m), (v/v), (m/v)

Molarity examples:

The following examples all utilize the “DIMO” concept as learned in Academic Chemistry.

# 4 Steps to use the DIMO chart.

1 ) Determine the chart mass of the substance you're working with.

2) Deal with the concentration of that substance.

3) Deal with the volume of the solution you're working with.

4) Use "DIMO" to solve.

Example 1: How many grams of NaOH are in 3 liters of a 1M solution of the NaOH?

STEP 1: Determine the chart mass

Example 1 (con't.):

STEP 1: 40 g/mol

STEP 2: Deal with the concentration

Example 1 (con't.):

STEP 1: 40 g/mol

STEP 2: 1 mol/L

STEP 3: Deal with the volume

Example 1 (con't.):

STEP 1: 40 g/mol

STEP 2: 1 mol/L

STEP 3: 3 mol of NaOH in sample

STEP 4: Use "DIMO" to solve

Solution to Example 1:

How many grams of NaOH are in 3 liters of a 1M solution of the NaOH?

There are 120 g of NaOH in 3 liters of a 1M solution of NaOH.

Example 2: 0.75 liters of a 0.5<u>M</u> solution contains how many grams of $CaCl_2$?

STEP 1: Determine the chart mass

Example 2 (con't.):

STEP 1: 111 g/mol

STEP 2: Deal with the concentration

Example 2 (con't.):

STEP 1: 111 g/mol

STEP 2: 0.5 mol/L

STEP 3: Deal with the volume

Example 2 (con't.):

STEP 1: 111 g/mol

STEP 2: 0.5 mol/L

STEP 3: 0.375 mol of $CaCl_2$ in sample

STEP 4: Use "DIMO" to solve

Solution to Example 2:

How many grams of $CaCl_2$ are in 0.75 liter of a 0.5M solution of the $CaCl_2$?

There are 41.625 g of $CaCl_2$ in 0.75 liter of a 0.5M solution of $CaCl_2$.

Notice in the previous examples we calculated the number of grams of a substance.

In the next example, we will determine how many moles of a substance are in a given volume of a solution.

Example 3: What is the Molarity of a solution created by dissolving 150 grams of NaI into 250 mL of distilled water?

STEP 1: Determine the chart mass

Example 3 (con't.):

STEP 1: 150 g/mol

STEP 2: Deal with the concentration

Example 3 (con't.):

STEP 1: 150 g/mol

STEP 2: 150 g / 0.25 L

STEP 3: Deal with the volume

Example 3 (con't.):

: 150 g/mol
L

Solution to Example 3:

What is the Molarity of a solution created by dissolving 150 grams of NaI into 250 mL of distilled water?

600 g of NaI is 4 moles

There are 4 moles of NaI in 1 liter of the solution of NaI.

Therefore the molarity is 4M

Example #4:

A 10.0 mL sample of Potassium iodide solution was analyzed by adding an excess of Silver nitrate solution to produce Silver iodide crystals (which were filtered out of the solution), and Potassium nitrate.

If 2.290 grams of Silver iodide were obtained, what is the Molarity of the original Potassium iodide solution ?

Clues:

1. Write a balanced equation.

2. Determine the theoretical and actual mole ratios of the involved substances.

3. Use the “4-Step method” to solve the Molarity problems.

Molality - The number of moles of solute per kilogram of solvent.

Molality example #1:

5.67g of glucose are dissolved in 25.2g of water.

What is the Molality ?

Step #1: Determine the number of moles of solute.

Molecular weight of glucose = 180.1572 g/mol

Use “DIMO” to determine # of moles.

$$\frac{5.67\text{g}}{180.1572\ \text{g/mol}} = 0.0315\ \text{mol of glucose}$$

Step #2: Determine the mass of the solvent.

Given 25.2g = 0.0252 Kg

Step #3: Set up proportions to solve.

$$\frac{0.0315 \text{ mol glucose}}{0.0252 \text{ Kg solvent}} = \frac{X \text{ mol glucose}}{1 \text{ Kg solvent}}$$

X = 1.25m glucose

Molality example #2:

Toluene, $C_6H_5CH_3$, is a liquid compound similar to Benzene, $C_6H_6$. It is the starting material for other substances, including Trinitrotoluene.

Find the molality of Toluene in a solution that contains 35.6g of Toluene in 125g of Benzene.

Molality example #3:

Fructose, $C_6H_{12}O_6$, is a sugar found in honey and fruits. The sweetest sugar, it is nearly twice as sweet as sucrose.

How much water should be added to 1.75g of fructose to give a 0.125m solution of Fructose ?

Step #1: Determine the number of moles of solute.

Molecular weight of Fructose = 180.1572 g/mol

Use “DIMO” to determine # of moles.

$$\frac{1.75\text{g}}{180.1572\ \text{g/mol}} = 0.00971\ \text{mol of Fructose}$$

Step #2: Determine the mass of the solvent.

Given “X”g = “X” Kg

Step #3: Set up proportions to solve.

$$\frac{0.125 \text{ mol Fructose}}{1 \text{ Kg solvent}} = \frac{0.00971 \text{ mol Fructose}}{\text{“X” Kg solvent}}$$

X = 0.07768 Kg of solvent

Which equals 77.68 g of solvent.

This problem requires an additional step to solve.

Step #4: Use the density of water to convert grams to milliliters.

Percent solutions (con't.)-

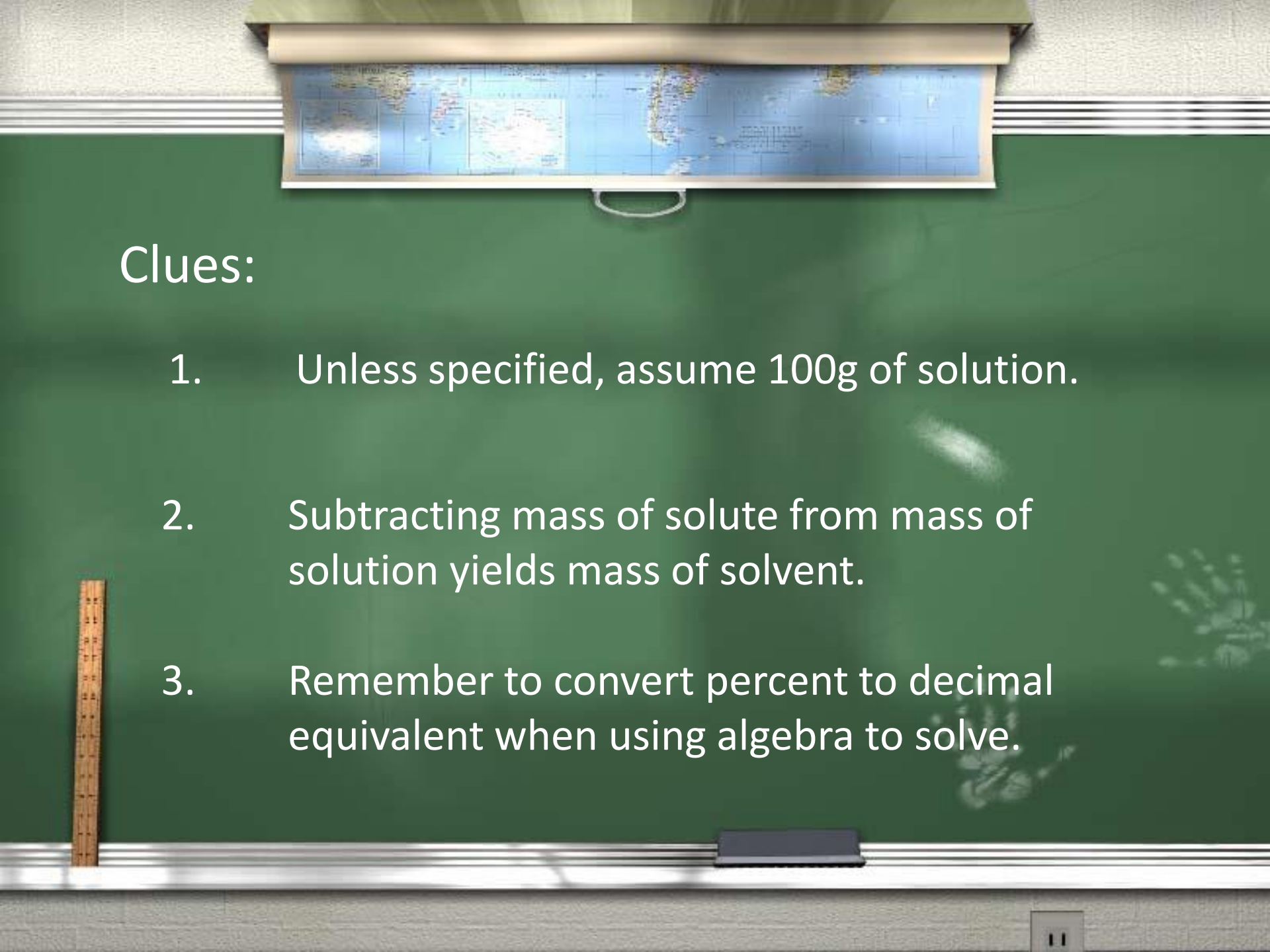
A. Mass percent of solute

$$\text{Mass } \% \text{ Solute} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

Mass percent of solute example #1:

How would you prepare a 3.5% Sodium chloride by mass solution ?

Notice: You are not given a mass or volume of the solution.



# Clues:

1. Unless specified, assume 100g of solution.

2. Subtracting mass of solute from mass of solution yields mass of solvent.

3. Remember to convert percent to decimal equivalent when using algebra to solve.

$$\text{Mass \% Solute} = \frac{\text{Mass of solute}}{\text{Mass of solution}} \times 100$$

$$3.5\% = 0.035 = \frac{\text{grams of NaCl}}{100\text{g of solution}}$$

To prepare dissolve:

3.5g of NaCl into what volume of water ?

3.5g of NaCl into what volume of water ?

$$\frac{\text{3.5g NaCl}}{\text{(100g of solution - 3.5g NaCl)} = \text{mass of solvent}}$$

Mass of solvent = 96.5g of $H_20$

Use the density of water to calculate the volume required.

$\rho_{H20}$ = 1g/mL

Therefore 96.5g of $H_2O$ = 96.5 mL of water

How would you prepare a 3.5% Sodium chloride by mass solution ?

Dissolve 3.5g of NaCl into 96.5mL of water.

Mass percent of solute example #2:

How would you prepare 425g of an aqueous solution that is 2.40% by mass Sodium acetate ($NaC_2H_3O_2$)?

Answer:

Dissolve 10.2g of Sodium acetate into 414.8g of water.

Mass percent of solute example #3:

An experiment requires 35.0g of HCl (aq) that is 20.2% by mass.

1. How many grams of HCl (g) is this ?

2. How many grams of water is this ?

3. What is the molality of this solution ?

1. How many grams of HCl is this ?

20.2% of a total of 35g = 7.07g HCl

2. How many grams of water is this ?

35 total grams of solution

- 7.07g of HCl

27.93g of $H_2O$

3. What is the molality of this solution ?

Formula weight HCl = 36.4609 g/mol

7.07g HCl = 0.194 mol HCl

$$m = \frac{\text{Mol HCl}}{\text{Kg of } H_2O} = \frac{0.194 \text{ mol HCl}}{0.02793 \text{ Kg } H_2O} = 6.95m$$

Mass percent of solute example #4:

What mass of solution containing 6.50% by mass of Sodium sulfate, contains 1.50g of Sodium sulfate ?

Answer:

23.1g of 6.50% Sodium sulfate solution contains 1.50g of Sodium sulfate.

Mass percent of solute example #5:

How would you prepare 455g of an aqueous solution that is 4.45% by mass of Barium hydroxide ?

What is the mass of the water ?

Calculate the molality of this solution.

Percent solutions (con't.)-

B. Percent by volume (v/v) (usually a liquid - liquid)

$$\text{Volume } \% \text{ Solute} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

Percent by volume (v/v) example #1:

What is the percent by volume of Ethanol ($C_2H_6O$) in the final solution when 20.0 mL EtOH are diluted to a volume of 250 mL ?

$$\text{Volume } ^{\%} \text{ Solute} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

$$\text{“X”} = \frac{20.0 \text{ mL EtOH}}{250 \text{ mL solution}} \times 100$$

$$X = 8.00^{\%} \text{ EtOH (v/v)}$$

Percent by volume (v/v) example #2:

Isopropyl alcohol can be purchased as a 90% Isopropyl (v/v) solution. If you dilute 25.0 mL of this solution to 100 mL with water, what will the final percent (v/v) be ?

$$\text{Volume \% Solute} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

$$90\% = \frac{\text{“X” mL Isopropyl}}{\text{25.0 mL solution}} \times 100$$

X = 22.5 mL Isopropyl

Continued on next slide

$$\frac{\text{22.5 mL Isopropyl}}{\text{100 mL of solution}} = \text{22.5\% Isopropyl (v/v)}$$

Percent by volume (v/v) example #3:

A 750 mL bottle of Rum is 40% Ethanol (v/v). How many mL of $C_2H_6O$ can be distilled from this solution (assume 100% efficiency of the "still") ? What is the volume of water that will remain ?

$$\text{Volume } ^{\%} \text{ Solute} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

$$40^{\%} = \frac{\text{“X” mL Ethanol}}{\text{750 mL solution}} \times 100$$

X = 300 mL Ethanol

Therefore, 450 mL of water remain.

Percent by volume (v/v) example #4:

The density of an Ethanol is 0.789 g/mL. What is the Molarity of the solution in the previous example ?

Clues:

1. Using the density of EtOH, determine how many grams of EtOH are in the bottle.

2. Convert grams of EtOH to moles; and, convert mL of water to Liters.

3. Set up a proportion to determine Molarity.

$$\frac{0.789\ \text{g}}{1\ \text{mL}} = \frac{\text{“X”}\ \text{g}}{300\ \text{mL}}$$

“X” = 236.7g EtOH

236.7g EtOH

---

How much $H_2O$ ?

750mL $sol_n$.

Convert grams to Moles.

Convert mL of water to Liters of water.

$$\frac{236.7\text{g EtOH}}{750\text{ mL sol'n.}} = \frac{5.14\text{ Mol EtOH}}{0.750\text{ L of sol'n}}$$

What is the molarity of EtOH in this solution ?

6.85 M

Percent solutions (con't.)-

C. Percent mass/volume (m/v) (usually a solid-liquid)

$$\%\ \text{Mass / Volume} = \frac{\text{Mass of solute (g)}}{\text{Solution volume (mL)}} \times 100$$

Percent mass/volume (m/v) or (w/v) example #1:

A solution contains 4.3g of Iron (II) sulfate in 0.591 liters of solution. What is the percent (m/v) of the solution ?

$$\%\ \text{Mass / Volume} = \frac{\text{Mass of solute (g)}}{\text{Solution volume (mL)}} \times 100$$

$$\%\ \text{Mass / Volume} = \frac{4.30\ \text{g}}{591\ \text{mL}} \times 100$$

$$\%\ (\text{m/v}) = 0.728\%\ (\text{m/v})\ \text{Iron (II) sulfate}$$

What is the molarity of the above solution ?

$$\text{Molarity} = \underline{M} = \frac{\#\text{ of moles of solute}}{\text{Liter of solution}}$$

Iron (II) sulfate = $FeSO_4$ = 151.9046 g/mol

Therefore: 4.3 g $FeSO_4$ = 0.0283 mol $FeSO_4$

$$\frac{0.0283 \text{ mol } FeSO_4}{0.591 \text{ Liters of sol'n}}$$

Molarity = 0.0479 M

What is the molality of the previous solution ? (Assume addition of the 4.3g of solute does not effect the volume of solvent. Density of sol'n = 1 g/mL)

$$\text{Molality} = m = \frac{\text{Moles of solute}}{\text{Kg of solvent}}$$

$$\frac{\text{Moles of solute}}{\text{Kg of solvent}} = \frac{0.0283 \text{ mol}}{0.5867 \text{ Kg}} = 0.0482 \text{ m}$$

Percent mass/volume (m/v) or (w/v) example #2:

How many liters of 2.00% glucose (w/v) solution can you prepare with 40.0 grams of Glucose ?

$$2.00\% = 0.0200 = \frac{40.0\text{g}}{\text{“X” mL}}$$

$$\text{“X”} = 2000\ \text{mL} = 2.00\ \text{L}$$

For example #3, see homework problem involving “toothpaste”.

Percent mass/volume (m/v) or (w/v) example #4:

How many grams of Chloride are in 5.00 liters of seawater if the molal concentration of Chloride is 0.568m ? *(The density of this seawater (the solvent) = 1.024 g/mL.)*

What is the molarity of this solution ?

What is the percent (m/v) of this solution ?

A solution of HCl has a density of 1.16 g/mL, and is labeled as 31.45% by mass (m/m).

What is the molality of the solution?

Clues:

1. Find the moles of solute (HCl)
2. Find the mass of the sovent (water)
3. Substitute values into the formula to solve.

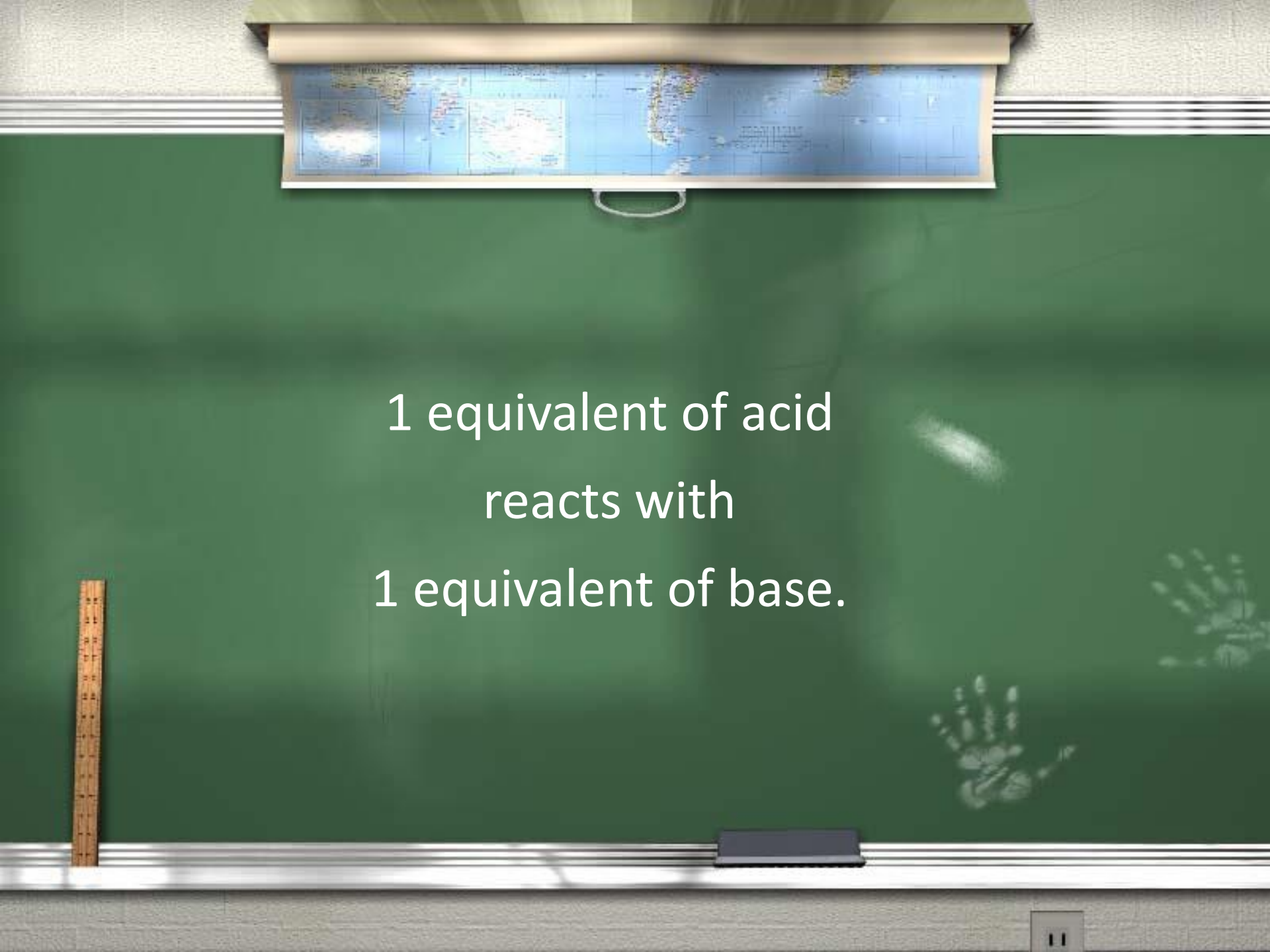
## Equivalent -

The quantity of acid in an acid base reaction that yields 1 mol of $H^+$ ions.

**or**

The quantity of base that reacts with 1 mol $H^+$ ions.

This definition depends on the chemical reaction you are dealing with.


1 equivalent of acid
reacts with
1 equivalent of base.

Equivalent example #1:




1 mol of HCl supplies 1 mol of $H^+$ ions.

Therefore, 1 mol of HCl = 1 equivalent of HCl

AND

1 mol of NaOH reacts with 1 mol of $H^+$ ions.

Therefore, 1 mol of NaOH = 1 equivalent of NaOH.

Equivalent example #2:




1 mol of $H_2SO_4$ supplies 2 mol of $H^+$ ions.

Therefore, 1 mol of $H_2SO_4$ = 2 equivalent of $H_2SO_4$

AND

1 mol of NaOH reacts with 1 mol of $H^+$ ions.

Therefore, 1 mol of NaOH = 1 equivalent of NaOH.

BUT

In this reaction:

1 mol of $H_2SO_4$ = 2 equivalent of $H_2SO_4$

And this requires 2 mol of NaOH,

(remember: 1 mol of NaOH = 1 equivalent of NaOH)

Therefore 2 mol NaOH = 2 equivalents.

2 equiv. $H_2SO_4$ reacts with 2 equiv. NaOH; which in lowest terms is:

1 equiv. $H_2SO_4$ reacts with 1 equiv. NaOH.

Equivalent example #3:




1 mol of $H_2SO_4$ supplies 1 mol of $H^+$ ions.

Therefore, 1 mol of $H_2SO_4$ = 1 equivalent of $H_2SO_4$

AND

1 mol of NaOH reacts with 1 mol of $H^+$ ions.

Therefore, 1 mol of NaOH = 1 equivalent of NaOH.

AND

In this reaction:

1 mol of $H_2SO_4$ = 1 equivalent of $H_2SO_4$

And this requires 1 mol of NaOH,

(remember: 1 mol of NaOH = 1 equivalent of NaOH)

1 equiv. $H_2SO_4$ reacts with 1 equiv. NaOH; which in lowest terms is:

1 equiv. $H_2SO_4$ reacts with 1 equiv. NaOH.

Equivalent mass = # grams acid/eq. acid

The mass of one equivalent.

You can calculate this from the mol:equivalent relationship.

1 mol $H_2SO_4$ = 2 equivalents $H_2SO_4$

And

1 mol $H_2SO_4$ = 98.1 g/mol

$$\frac{98.1\ \text{g } H_2SO_4}{1\ \text{mol } H_2SO_4} \times \frac{1\ \text{mol } H_2SO_4}{2\ \text{equiv. } H_2SO_4} = \frac{98.1\ \text{g } H_2SO_4}{2\ \text{equiv. } H_2SO_4}$$

Therefore: 49.05 g/equiv

## Normality -

The number of equivalents of a solute dissolved in a liter of solution.

$$N = \frac{\text{\# of equivalents}}{\text{Liter of solution}}$$

AND

N = Molarity x number of ionizable $H^+$ or $OH^-$

Normality example #1:

What is the N of:

A. 2M HCl = 2M x 1 = 2 N HCl

B. 0.1M $CH_3COOH$ = 0.1M x 1 = 0.1 N $CH_3COOH$

C. 0.3M $H_3PO_4$ = 0.3M x 3 = 0.9 N $H_3PO_4$

D. 0.1M NaOH = 0.1M x 1 = 0.1 N NaOH

E. 1.0M $Ca(OH)_2$ = 1M x 2 = 2.0 N $Ca(OH)_2$

Normality example #2:

What is N of 20.0g NaOH dissolved in 1.0 L of solution ?

$$20.0\text{g NaOH} \times \frac{1 \text{ equiv. NaOH}}{40.0\text{g}} = 0.5 \text{ equiv.}$$

$$N = \frac{\#\text{ of equivalents}}{\text{Liter of solution}} = \frac{0.5 \text{ equiv.}}{1 \text{ L}} = 0.5 \text{ N}$$

Normality example #3:

What is N of 4.90g $H_2SO_4$ dissolved in 500 mL of solution ?

$$4.90\text{g } H_2SO_4 \times \frac{1 \text{ equiv. } H_2SO_4}{49.0\text{g}} = 0.10 \text{ equiv.}$$

$$N = \frac{\text{\# of equivalents}}{\text{Liter of solution}} = \frac{0.10 \text{ equiv.}}{0.5 \text{ L}} = 0.2 \text{ N}$$

Normality example #4:

What is N of 15.0g HCl dissolved in 0.400 L of solution ?

$$15.0\text{g HCl} \times \frac{1\text{ equiv. HCl}}{36.5\text{g}} = 0.411\text{ equiv.}$$

$$N = \frac{\#\text{ of equivalents}}{\text{Liter of solution}} = \frac{0.411\text{ equiv.}}{0.400\text{ L}} = 1.03\text{ N}$$

Normality example #5:

What is N of 80.0g NaOH dissolved in 1.5 L of solution ?

$$80.0\text{g NaOH} \times \frac{1\text{ equiv. NaOH}}{40.0\text{g}} = 2.0\text{ equiv.}$$

$$N = \frac{\#\text{ of equivalents}}{\text{Liter of solution}} = \frac{2.0\text{ equiv.}}{1.5\text{ L}} = 1.33\text{ N}$$

Normality example #6:

What is N of 196g $H_3PO_4$ dissolved in 2.0 L of solution ?

$$196\text{g } H_3PO_4 \times \frac{1 \text{ equiv. } H_3PO_4}{32.7\text{g}} = 5.99 \text{ equiv.}$$

$$N = \frac{\text{\# of equivalents}}{\text{Liter of solution}} = \frac{5.99 \text{ equiv.}}{2.0 \text{ L}} = 3 \text{ N}$$

Normality example #7:

How many equivalents are in 2.5 L of 0.60 N $H_2SO_4$ ?

$$N = \frac{\text{\# of equivalents}}{\text{Liter of solution}} = 0.60\ N = \frac{\text{“X” equiv.}}{2.5\ L}$$

“X” = 1.5 equivalents

Normality example #8:

How many equivalents are in 0.55 L of 1.8 N NaOH ?

$$N = \frac{\text{\# of equivalents}}{\text{Liter of solution}} = 1.80\ N = \frac{\text{“X” equiv.}}{0.55\ L}$$

“X” = 0.99 equivalents

Normality example #9:

How many equivalents are in 1.60 L of 0.5 N $H_3PO_4$ ?

$$N = \frac{\text{\# of equivalents}}{\text{Liter of solution}} = 0.50\ N = \frac{\text{“X” equiv.}}{1.60\ L}$$

“X” = 0.80 equivalents

Normality example #10:

How many equivalents are in 250 mL of 0.28 N $H_2SO_4$ ?

$$N = \frac{\text{\# of equivalents}}{\text{Liter of solution}} = 0.28\ N = \frac{\text{“X” equiv.}}{0.250\ L}$$

“X” = 0.070 equivalents

Dilutions with Normality:

What if you wished to dilute a more concentrated Normal solution to a specific concentration. How would you do it ?

$$N_i V_i = N_f V_f$$

Normal Dilutions example #1:

A lab requires 500 mL of 0.20 N Sulfuric acid. You have a significant volume of 4.0 N $H_2SO_4$.

How do you prepare the desired solution ?

$$N_i V_i = N_f V_f$$

0.20 N x 0.500 L = 4.0 N x "X"

"X" = 0.025 L

Dilute 25 mL of 4.0 N Sulfuric acid to 500 mL.

Normal Dilutions example #2:

A lab requires 870 mL of 2.0 N Potassium hydroxide. You have a significant volume of 3.0 N KOH.

How do you prepare the desired solution ?

$$N_i V_i = N_f V_f$$

2.0 N x 0.870 L = 3.0 N x "X"

"X" = 0.58 L

Dilute 580 mL of 3.0 N Potassium hydroxide to 870 mL.

Titrations by Normality:

You can determine the Normality of an unknown solution by adding a specific volume of a solution of known Normality until the mixture is neutral.

$$N_{acid} \times Volume_{acid} = N_{base} \times Volume_{base}$$

Both “volumes” must be the same unit.

Titrations by Normality example #1:

If 17 mL of 2.0 N HCl neutralizes 50 mL of $Ca(OH)_2$, what is the Normality of the base ?

$$N_{acid} \times Volume_{acid} = N_{base} \times Volume_{base}$$

$$2.0\ N \times 17\ mL = \text{“X”}\ N\ Ca(OH)_2 \times 50\ mL$$

$$\text{“X”} = 0.68\ N\ Ca(OH)_2$$

Titrations by Normality example #2:

A 25.0 mL sample of $Ba(OH)_2$ solution was neutralized by 45.3 mL of 0.150 N HCl. What is the Normality of the $Ba(OH)_2$ ?

$$N_{acid} \times Volume_{acid} = N_{base} \times Volume_{base}$$

$$0.150\ N \times 45.3\ mL = \text{“X”}\ N\ Ba(OH)_2 \times 25\ mL$$

$$\text{“X”} = 0.272\ N\ Ba(OH)_2$$

What is the molarity of the above determined $Ba(OH)_2$ solution ?

$$N = \underline{M} \times \text{ionizable } H^+ \text{ or } OH^-$$

$$0.272\ N = \text{“X”} \times 2$$

$$\text{“X”} = 0.136\ \underline{M}\ Ba(OH)_2$$

Titrations by Normality example #3:

An antacid tablet contains Magnesium hydroxide and an inert binding agent. The mass of the Magnesium hydroxide in the tablet was determined by titration with HCl. If one tablet required 39.1 mL of 0.2056 N HCl for neutralization; how many equivalents of the base are in the tablet ? How many grams is this ?

Titrations by Normality example #4:

An antacid tablet contains Aluminum hydroxide and an inert binding agent. The mass of the Aluminum hydroxide in the tablet was determined by titration with HCl. If one tablet required 39.1 mL of 0.2056 N HCl for neutralization; how many equivalents of the base are in the tablet ? How many grams is this ?

# Chapter 6
# Principles of Reactivity: Energy and Chemical Reactions

## Thermochemistry

# Goals of Chapter

- Assess heat transfer associated with changes in temperature and changes of state.
- Apply the First Law of Thermodynamics.
- Define and understand the state functions enthalpy (H) and internal energy (E).
- Calculate the energy changes in chemical reactions and learn how these changes are measured.

# Thermochemistry

study of the relationships between energy changes and chemical processes



# Energy

The capacity to do work or to transfer heat

- **Kinetic Energy**

energy of motion; KE = ½ $mv^2$

- **Potential Energy**

**stored energy**: fuel of motor-cars, trains, jets. It is converted into heat and then to work.

**due to relative position**: water at the top of a water wheel. It is converted to mechanical E

**electrostatic**: lightning converts it to light and heat

# Joule

- is the SI unit for energy
- the energy of a 2 kg mass moving at 1 m/s

$KE = \frac{1}{2} mv^2 = \frac{1}{2}(2\ kg)(1\ m/s)^2 = 1\ kg \times m^2/s^2 = 1\ J$

- 1 cal is the amount of energy required to raise the temperature of 1 g water 1°C
- 1 cal = 4.184 J  1 cal = 1 calorie

- 1 Cal = 1000 cal = 1 kcal

1 Cal = dietary Calorie (nutritional calorie)

- 1 kilowatt-hour (kWh) = $3.60 \times 10^6$ J

## How many dietary (nutritional) calories are equivalent to $1.75 \times 10^3$ kJ?

1 cal = 4.184 J  1 Cal = 1000 cal = 1 kcal = 4.184 kJ

1 Cal = dietary Calorie (nutritional calorie)

$$1.75 \times 10^3 \text{ kJ} \times \frac{1000 \text{ J}}{1 \text{ kJ}} \times \frac{1 \text{ cal}}{4.184 \text{ J}} \times \frac{1 \text{ Cal}}{1000 \text{ cal}} = 418 \text{ Cal}$$

$$1.75 \times 10^3 \text{ kJ} \times \frac{1 \text{ Cal}}{4.184 \text{ kJ}} = 418 \text{ Cal}$$

$1.8 \times 10^4$ kJ = _____ kWh ?

# System

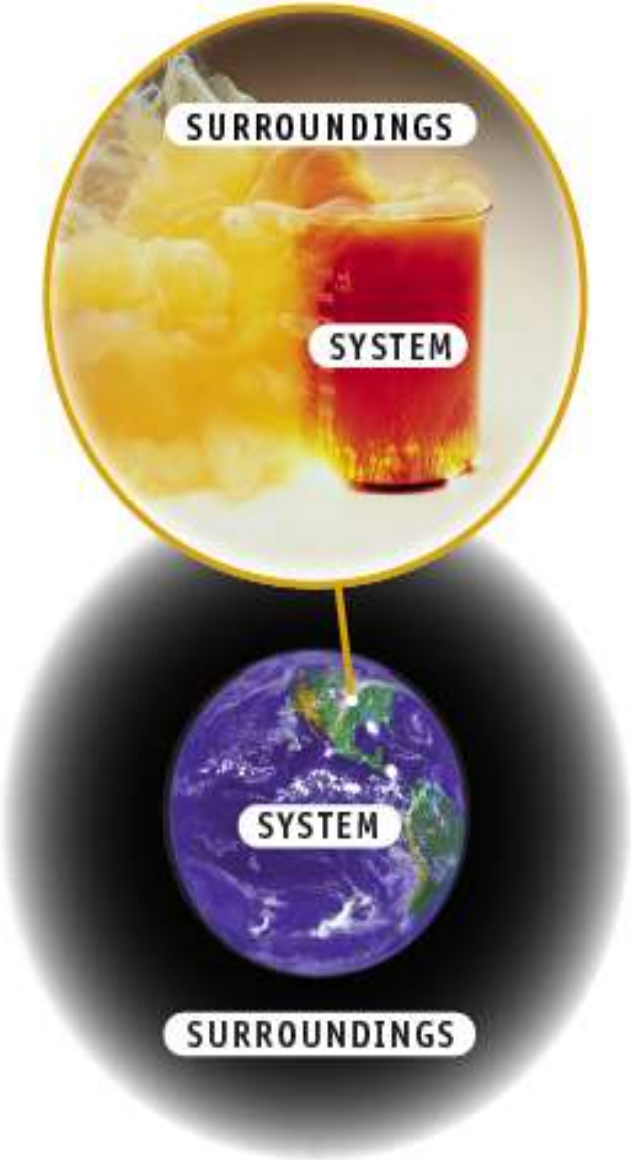
- the part of the universe under study

- the substances involved in the chemical and physical changes under investigation
- in chemistry lab, the system may be the chemicals inside a beaker

# Surroundings

- the rest of the universe
- in chemistry lab, the surroundings are outside the beaker where the chemicals are
- The system plus the surroundings is the universe.

# System and Surroundings




- SYSTEM
  - The object under study
- SURROUNDINGS
  - Everything outside the system

# Thermodynamic State

- The set of conditions that specify all of the properties of the system is called the **thermodynamic state of a system**.
- For example the thermodynamic state could include:
  - The number of moles and identity of each substance.
  - The physical states of each substance.
  - The temperature of the system.
  - The pressure of the system.
  - The volume of the system.
  - The height of a body relative to the ground.

# First Law of Thermodynamics

law of conservation of energy

during any process, energy is neither created nor destroyed, it is merely converted from one form to another*

the mass of a substance is a form of energy

$E = mc^2$ (Albert Einstein)

e.g. in nuclear reactions mass is not conserved,

part of it is transformed into heat (E)

* "The combined amount of energy in the universe is constant."

# Internal Energy (E)

the total energy of a system: Σ of kinetic and potential E of all atoms, molecules, or ions in the system

- E cannot be measured exactly
- E is a state function; change in E does not depend on how change of state happens
- ΔE: change in E. ΔE can be measured
- $\Delta E = E_{final} - E_{initial}$ (of final and initial states)
- ΔE > 0 (+) indicates system gains energy during process (E increases, ↑)
- ΔE < 0 (−) indicates system loses energy during process (E decreases, ↓)

# $\Delta E = q + w$

- **first law of thermodynamics**
- q = heat
- w = work done on the system
- $w > 0$ (+) $\Rightarrow$ work done on system by surroundings (eg. compressing gas); E of system increases
- $w < 0$ (–) $\Rightarrow$ work done by system on surroundings (expanding gas); E decreases
- $q > 0$ (+) $\Rightarrow$ heat flows into system; E $\uparrow$ endo
- $q < 0$ (–) $\Rightarrow$ heat flows out of system; E $\downarrow$ exo
- q and w are not state functions

# Exothermic reactions give off energy in the form of heat (they give off heat). Endothermic reactions absorb heat.

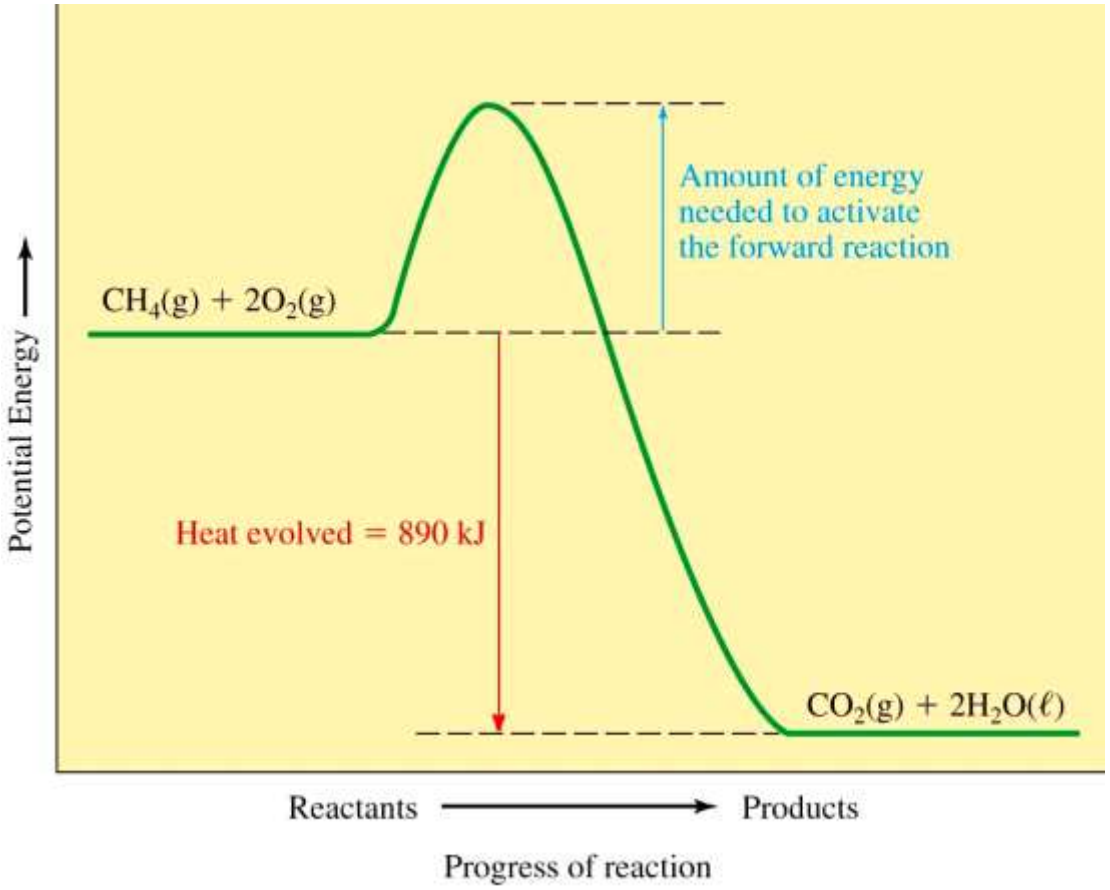
$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l) + 890\ kJ$

exothermic

In this case, heat is given off.

It is released by the system.

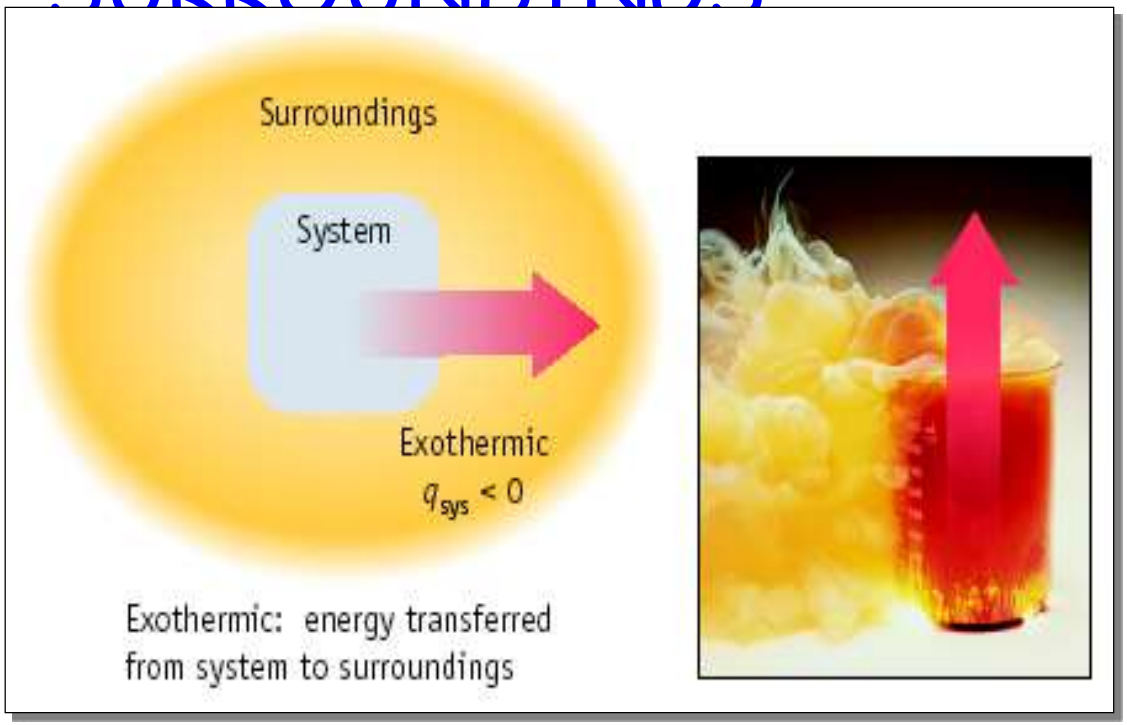
It is a product of the reaction.

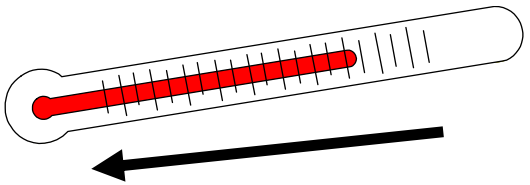



© 2004 Thomson/Brooks Cole

# Directionality of Heat Transfer

- Heat always transfers from hotter object to cooler one.
- EXOthermic: heat transfers from SYSTEM to SURROUNDINGS

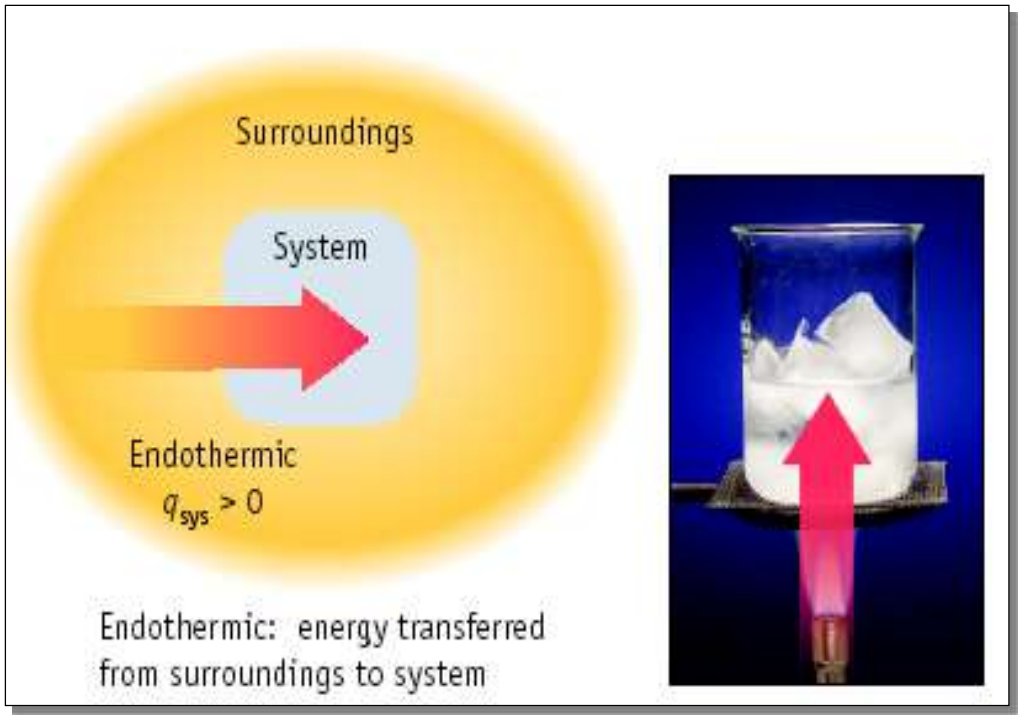


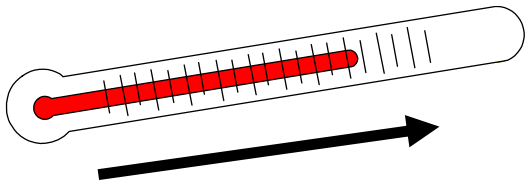



**T(system) goes down**
**T(surr) goes up**

# Directionality of Heat Transfer

- Heat always transfers from hotter object to cooler one.
- ENDOthermic: heat transfers from SURROUNDINGS to the SYSTEM.



**T(system) goes up**
**T (surr) goes down**

# Calculate $\Delta E$ of a system that absorbs 35 J of heat and does 44 J of work on the surroundings.

- q = +35 J (absorbed)
- w = –44 J (the system did it)
- $\Delta E = q + w$
- $\Delta E = 35\ J + (-44\ J)$
- $\Delta E = -9\ J$ (internal E decreases)
- Note that $E_{final}$ and $E_{initial}$ are not calculate, just $\Delta E$

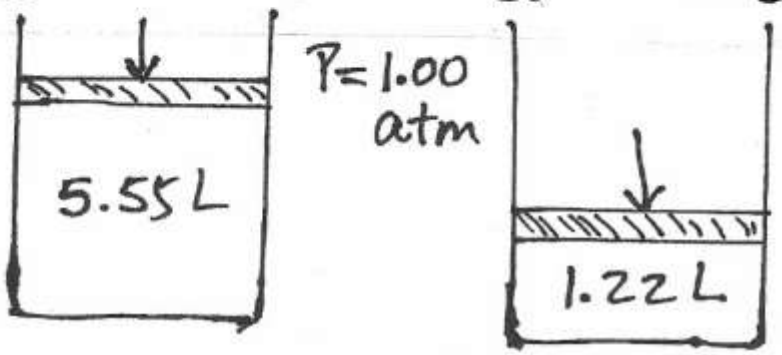
# Work symbol: *w*

- $w$ = force $\times$ distance
- Expansion/compression work at constant P,

  $w = -P\Delta V$ $\Delta V = V_{final} - V_{initial}$
- Then, $\Delta E = q + w$ converts to $\Delta E = q - P\Delta V$
- under conditions of **constant volume**, $P\Delta V = 0$, **w=0**, because $\Delta V = 0$ (no work done on or by the system)
- $\therefore, \Delta E = q - 0$
- $\Delta E = q$
- $\Delta E = q_V$ This provides a way of measuring $\Delta E$; that is in a reactor **at constant V**.

# P-V work and ΔE

**54.** A gas is compressed from an initial volume of 5.55 L to a final volume of 1.22 L by an external pressure of 1.00 atm. During the compression the gas releases 124 J of heat. What is the change in internal energy of the gas?




$q = -124$ J as heat is released.

For a compression w must be positive (it is done on the system → the gas).

$$w = -P \times \Delta V = -1.00\ \text{atm} \times (1.22 - 5.55)\ \text{L} \times \frac{101.3\ \text{J}}{\text{atm} \cdot \text{L}}$$

(the $\frac{101.3\ \text{J}}{\text{atm} \cdot \text{L}}$ conversion factor is given).

$w = +439$ J

$\Delta E = q + w = -124\ \text{J} + 439\ \text{J} = 315\ \text{J}$

# Heat Capacity (C)

the amount of heat energy required to raise the temperature of an object 1 K (or 1°C),

units = J/K, cal/°C, ...

$q = C \times \Delta T$ $\Delta T = T_{final} - T_{initial}$

The amount of heat can be calculated from $\Delta T$

# Specific Heat (c)

the amount of heat energy required to raise the temperature of 1 g of something 1 K (or 1°C)

units = J/gK, J/g°C, cal/g°C, ...

$q = c \times m \times \Delta T$ m: mass (grams)

# Molar Heat Capacity

**The molar heat capacity** **is** the amount of heat energy required to raise the temperature of 1 mol of a substance 1 K (1°C).

units = J/K mol, cal/°C mol

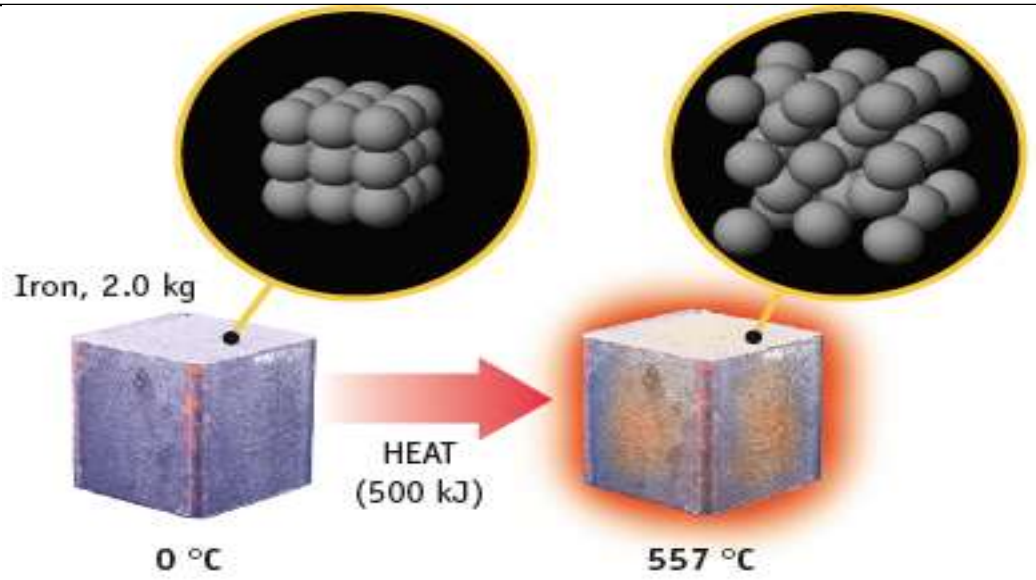
**the specific heat of water is**

**1 cal/g°C = 4.184 J/g°C (KNOW THIS!!!)**

**What is the molar heat capacity?**

$$C = 1\ \frac{\text{cal}}{\text{g °C}} \times \frac{18.0\ \text{g}}{1\ \text{mol}} = 18.0\ \frac{\text{cal}}{\text{mol °C}}$$ **or K**

# Heat/Energy Transfer
## No Change in State (s, ℓ, g)




$$q \text{ transferred} = (\text{sp. ht.}) \times (\text{mass}) \times (\Delta T)$$

$$q = c \times m \times \Delta T$$

$$q = 0.449\ (J/g\ °C) \times 2.0 \times 10^3 g \times (557 - 0)°C$$

$$q = 5.0 \times 10^5\ J = 5.0 \times 10^2\ kJ \quad \text{(2 sig. fig.)}$$

# From the 1st Law of Thermodynamics

**When two bodies, liquids, solutions, solid-liquid, etc.,(*) initially at different temperatures, are put in contact or mixed, the amount of heat absorbed and given off by the two samples have the same absolute value, but one is >0 and the other is <0.**

**That happens until they reach the thermal equilibrium, i.e., same temperature.**

$$q_1 + q_2 = 0 \qquad q_1 = -q_2$$

**If more than two components(*)**

$$q_1 + q_2 + q_3 + \ldots = 0$$

# Example: Calculate the amount of heat energy given off when 45.3 g water cools from 77.9 °C to 14.3 °C

- $\Delta T = T_f - T_i = 14.3\ °C - 77.9\ °C = -63.6\ °C$
- $q = c \times m \times \Delta T$ (know this formula)

$$q = \frac{1\ \text{cal}}{\cancel{\text{g}}\ \cancel{°\text{C}}} \times 45.3\ \cancel{\text{g}} \times (-63.6\ \cancel{°\text{C}}) = -2.88 \times 10^3\ \text{cal}$$

$\Delta T$ is negative because T lowers,

hence q is negative (it is given off)

**After absorbing 1.850 kJ of heat, the temperature of a 0.500-kg block of copper is 37 °C. What was its initial temperature?**

Cu specific heat $c = 0.385 \dfrac{J}{g\ K}$ (it is given)

$$q = m \times c \times \Delta T = m \times c \times (T_{final} - T_{initial})$$

$$1.850\ \cancel{kJ} \times \frac{1000\ \cancel{J}}{1\ \cancel{kJ}} = 500.\ \cancel{g} \times \frac{0.385\ \cancel{J}}{\cancel{g}\ °C} \times (37°C - T_i)$$

$$37°C - T_i = \frac{1.850 \times 1000}{500. \times 0.385} = 9.6\ °C \qquad T_i = 27.4°C$$

A 182-g sample of Au at some temperature is added to 22.1 g of water. The initial water T is 25.0 °C, and the final T of the whole is 27.5 °C. If the specific heat of gold is 0.128 J/g.K, what is the initial T of the gold?

**T of $H_2O$ increased, hence it absorbed heat. Then, Au gave off heat, i.e., its temperature decreased.**

$q_{water}(\text{absorbed}) + q_{Au}(\text{given off}) = 0$ $\quad q = m \times c \times \Delta T$

$>0$ $\quad <0$ $\quad \Delta T = T_f - T_i$

$$22.1\ g \times 4.184\ J/g.°C \times (27.5 - 25.0)°C + 182\ g \times 0.128\ J/g.°C \times (27.5°C - T_i) = 0$$

$$231 + 23.3 \times (27.5\ °C - T_i) = 231\ °C + 641°C - 23.3T_i = 0$$

$$T_i(Au) = \frac{231 + 641}{23.3} = 37.4\ °C$$

One beaker contains 156 g of water at 22 °C and a second contains 85.2 g of water at 95 °C. If the water in the two beakers is mixed, what is the final temperature?

Water in beaker 1 ($w_1$) will absorb heat, its T will ↑.

Water in beaker 2 will give off heat, its T will ↓.

$q_1(\text{absorbed}) + q_2(\text{given off}) = 0$ $\quad q = m \times c \times \Delta T$

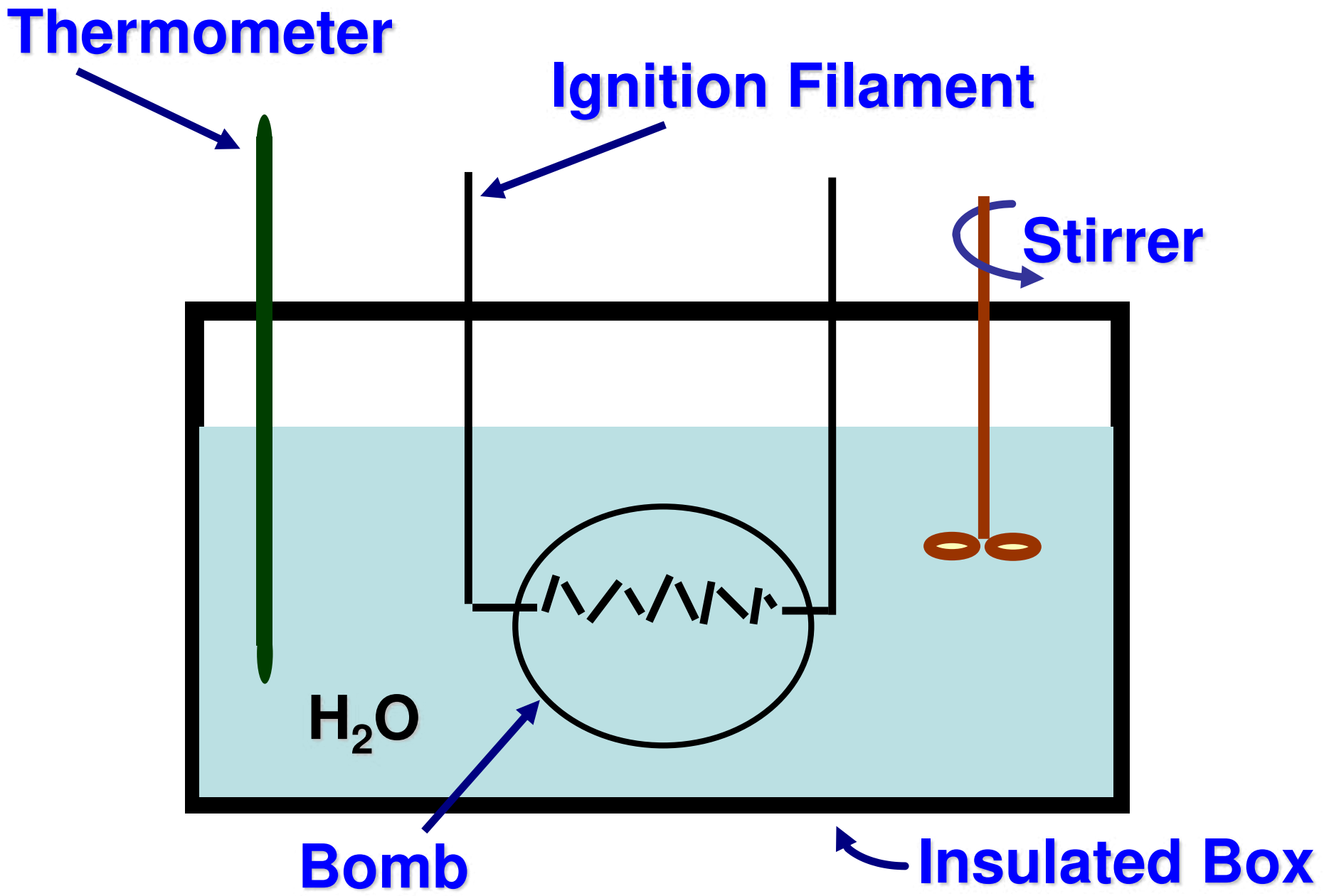
$>0$ $\quad <0$ $\quad \Delta T = T_f - T_i$

$$156\ \text{g} \times 4.184\ \text{J/g.°C} \times (T_f - 22°\text{C}) +$$
$$+\ 85.2\ \text{g} \times 4.184\ \text{J/g.°C} \times (T_f - 95\ °\text{C}) = 0$$

$$156\ T_f - 3432 + 85.2\ T_f - 8094 = 0$$

$$T_f = \frac{8094 + 3432}{156 + 85.2} = 47.8 \approx 48\ °\text{C}$$

# Bomb Calorimeter: constant V



**Example: A 1.50g sample of methane was burned in excess oxygen in a bomb calorimeter with a heat capacity of 11.3kJ/°C. The temperature of the calorimeter increased from 25.0 to 32.3°C. Calculate the ΔE in kJ per gram of methane for this reaction.**

$\Delta T = (32.3 - 25.0)\ ^\circ C = 7.3\ ^\circ C$

$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2\ H_2O(l)$

In a bomb calorimeter V is constant $\rightarrow \Delta E = q$

$q_{calorim} = C \times \Delta T = (11.3\ kJ/^\circ C) \times 7.3^\circ C = 83\ kJ$

$q + q_{calorim} = 0$ Then, $\Delta E = q = -q_{calorim}$, $\Delta E = -83\ kJ$

ΔE is negative because the rxn gives off heat that the calorimeter absorbs

$$\Delta E = \frac{-83\ kJ}{1.50\ g} = -55.3\ \frac{kJ}{g} = -55\ \frac{kJ}{g\ CH_4} \quad \text{(two SF)}$$

**Example: A bomb calorimeter was heated with a heater that supplied a total of 8520 J of heat. The temperature of the calorimeter increased by 2.00°C. A 0.455g sample of sucrose, $C_{12}H_{22}O_{11}$, was then burned in excess oxygen in that calorimeter causing the temperature to increase from 24.49°C to 26.25°C. Calculate the ΔE for this reaction in kJ/mol sucrose and the dietary calories per gram of sucrose.**

$$2C_{12}H_{22}O_{11} + 35\ O_2 \rightarrow 24CO_2 + 22H_2O$$

We will need MW of sucrose = **342.3 g/mol**

## Calorimeter heat capacity (C)

$$C = \frac{\text{heat supplied}}{\Delta T} = \frac{q}{\Delta T} = \frac{8520\ \text{J}}{2.00\ ^\circ\text{C}} = 4260\ \frac{\text{J}}{^\circ\text{C}}$$

**For the reaction,** $\Delta T = (26.25 - 24.49)^\circ\text{C}$

$\Delta V = 0$, then, $\Delta E = q = -q_{calor}$ (**reaction gives off heat**)

$$q_{calorim} = C \times \Delta T = (4260\ \text{J}/^\circ\text{C}) \times 1.76^\circ\text{C} = 7.50 \times 10^3\ \text{J}$$

$$\Delta E = \frac{-7.50 \times 10^3\ \text{J}}{0.455\ \text{g sucrose}} \times \frac{342.3\ \text{g sucrose}}{1\ \text{mol sucrose}} \times \frac{1\ \text{kJ}}{10^3\ \text{J}}$$

$$\Delta E = -5.64 \times 10^3\ \frac{\text{kJ}}{\text{mol sucrose}}$$

# Nutritional (dietary) Calories (Cal)

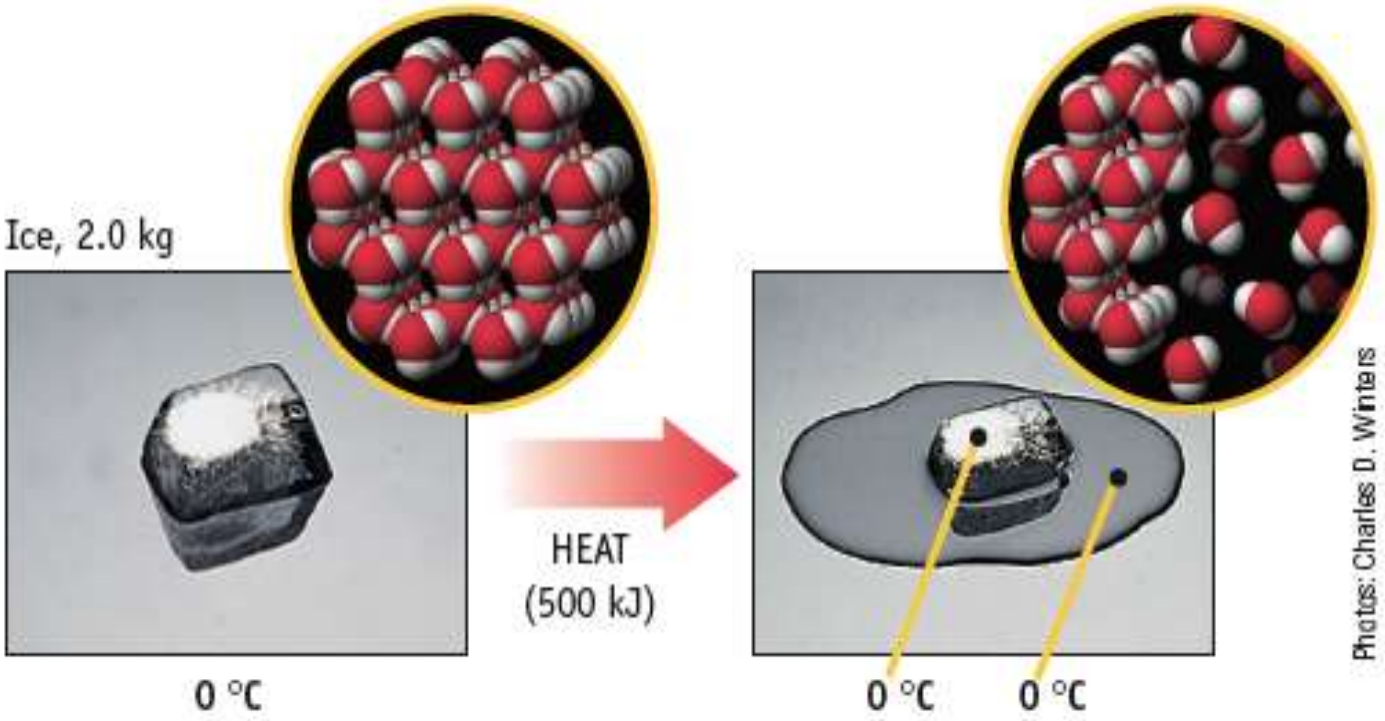
**Strategy:** divide J by grams of sucrose.

Convert J to cal and cal to Cal.

1 Cal = 1000 cal = 1 kcal (see slide # 5)

$$\frac{-7.50 \times 10^{3}\ \text{J}}{0.455\ \text{g sucro}} \times \frac{1\ \text{cal}}{4.184\ \text{J}} \times \frac{1\ \text{Cal}}{1000\ \text{cal}} = \frac{-3.94\ \text{Cal}}{\text{g sucrose}}$$

# Heat Transfer with Change of State (phase, s, ℓ, g)




Changes of state involve energy (at const T)

Ice + 333 J/g (heat of fusion) → Liquid water

$q$ = (heat of fusion)×(mass)

# Heat Transfer and Changes of State

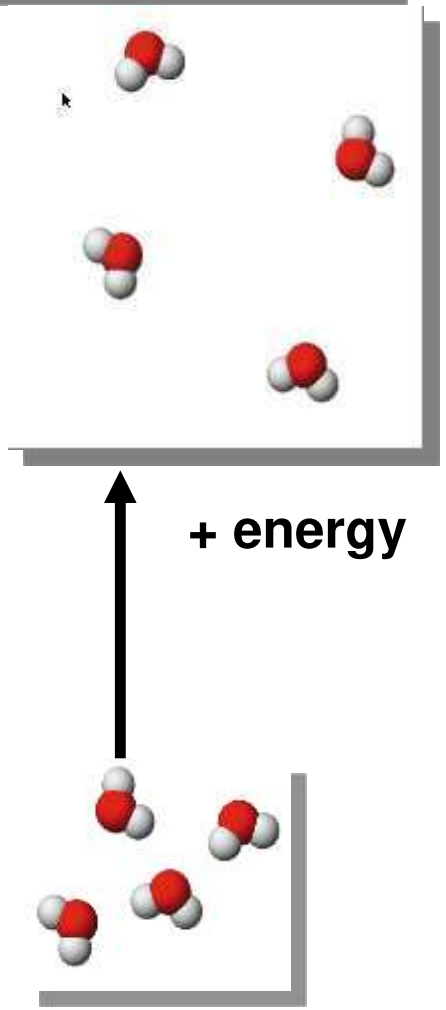
**Liquid → Vapor**

Requires energy (heat).

This is the reason

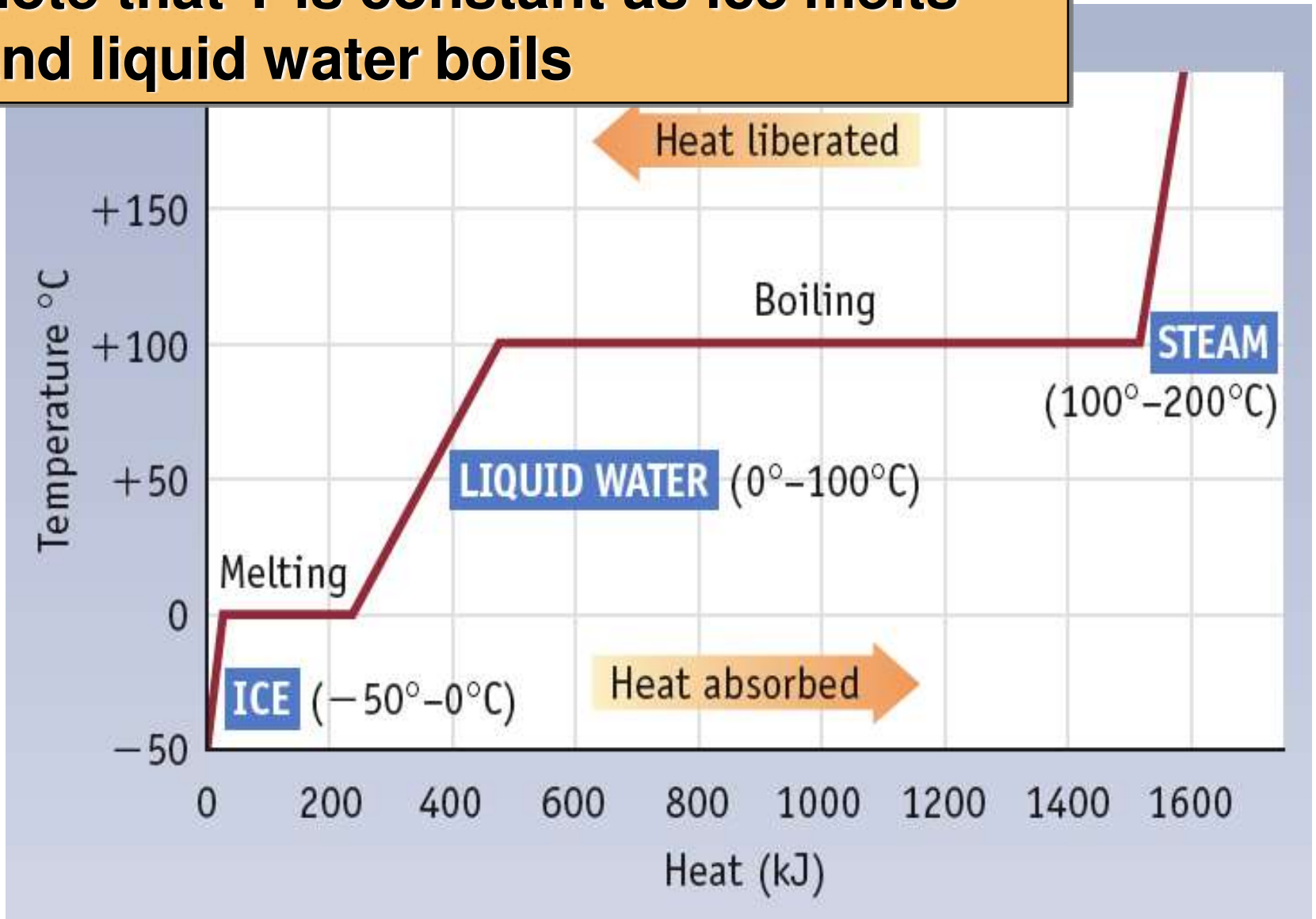
a) you cool down after swimming

b) you use water to put out a fire



# Heating/Cooling Curve for Water

**Note that T is constant as ice melts and liquid water boils**



# Mealting/boiling/Heating/Cooling for Water

**Note that T is constant as ice melts and liquid water boils**

| steps: | I | II | III | IV | V | |
|---|---|---|---|---|---|---|
| Ice, $H_2O(s)$ | $\rightarrow$ $H_2O(s)$ | $\rightarrow$ $H_2O(\ell)$ | $\rightarrow$ $H_2O(\ell)$ | $\rightarrow$ $H_2O(g)$ | $\rightarrow$ | $H_2O(g)$ |
| −50 °C | | 0 °C | 0 °C | 100 °C | 100 °C | 170 °C |

$$q_{total} = q_I + q_{II} + q_{III} + q_{IV} + q_V$$

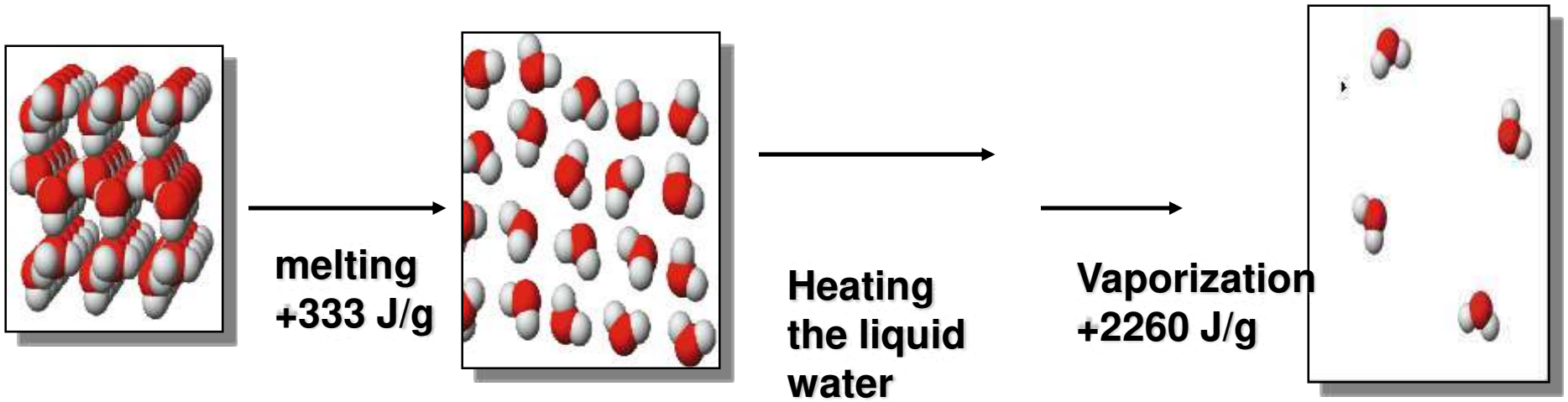
$$q_{total} = m \times c(sol) \times (0-(-50)) + m \times q_{fus} + m \times c(liq) \times (100-0) +$$

$$+ m \times q_{vap} + m \times c(gas) \times (170 - 100)$$

# Heat & Changes of State

What quantity of heat is required to melt 500. g of ice and heat the water to steam at 100 $^{o}$C?

**Heat of fusion of ice = 333 J/g**
**Specific heat of water = 4.2 J/g•K**
**Heat of vaporization = 2260 J/g**



# Heat & Changes of State

What quantity of heat is required to melt 500. g of ice and heat the water to steam at 100 $^{o}$C?

1. To melt ice

$q_1 = (500.\ g)(333\ J/g) = 1.67 \times 10^5\ J$

2. To raise water from 0 $^{o}$C to 100 $^{o}$C

$q_2 = (500.\ g)(4.2\ J/g \cdot K)(100 - 0)K = 2.1 \times 10^5\ J$

3. To evaporate water at 100 $^{o}$C

$q_3 = (500.\ g)(2260\ J/g) = 1.13 \times 10^6\ J$

4. Total heat energy = $q_1 + q_2 + q_3$ =

$1.51 \times 10^6\ J = 1510\ kJ$

**The freezing point of Hg is −38.8°C. What quantity of heat (J) is released to the surroundings if 1.00 mL of Hg is cooled from 23.0 °C to −38.8°C and then frozen to a solid? d = 13.6 g/mL**
**c = 0.140 J/g K  $q_{fus}$ = 11.4 J/g  AW = 200.6 g/mol**

$q = \underset{\text{cooling}}{q_1} + \underset{\text{freezing}}{q_2}$  $1\text{mL} \times \frac{13.6\text{ g}}{1\text{ mL}} = 13.6\text{ g}$

$q = m \times c \times \Delta T + m \times(-q_{fus})$  negative for freezing

$$q = 13.6\text{ g} \times \frac{0.140\text{ J}}{\text{g °C}} \times(-38.8 - 23)\text{°C} + 13.6\text{ g} \times \frac{(-11.4\text{ J})}{\text{g}}$$

$q = -117\text{ J} - 155 = -272\text{ J}$ (given off)

Exothermic process

# Enthalpy, H

- Chemistry is commonly done in open beakers or flasks on a desk top at atmospheric pressure.
  - Because atmospheric pressure only changes by small amounts, this is almost at constant pressure.

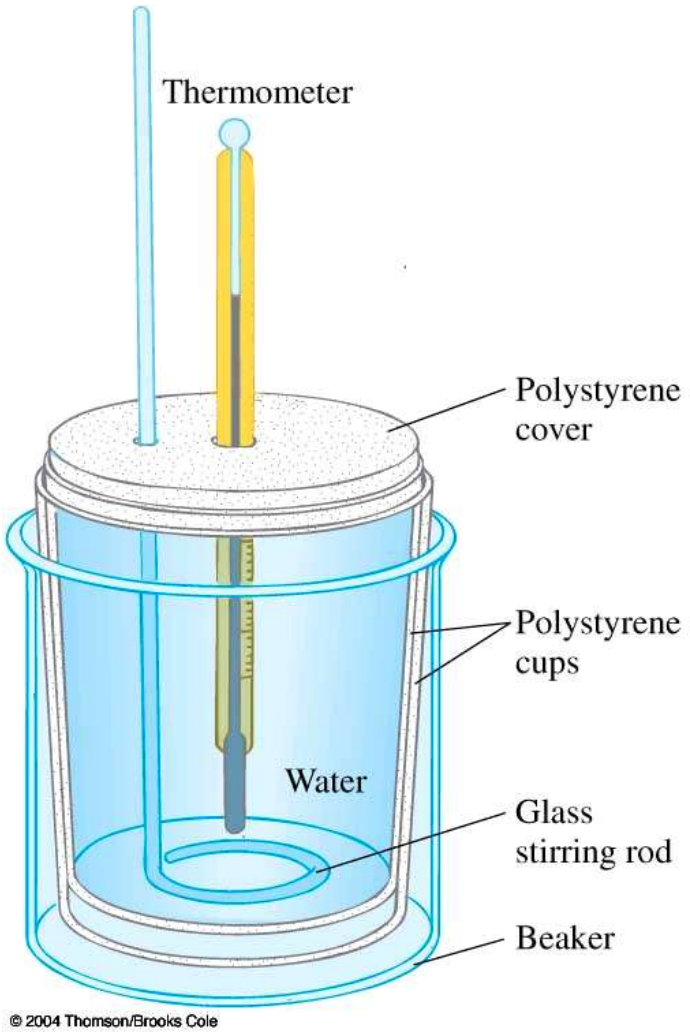
- Because heat at constant pressure is so frequent in chemistry and biology, it is helpful to have a measure of heat transfer under these conditions. That is the **enthalpy change**.

# Enthalpy, H

- heat content
- state function; change in H does not depend on how change of state happens
- $H = E + PV$ (We do not measure
- $\Delta H = H_{final} - H_{initial}$ or calculate $H_{f,i}$ but $\Delta H$)
- $\Delta H = \Delta E + \Delta(PV)$
- **at constant pressure, $\Delta H = q_P$**
- $\Delta H > 0$ (+) $\Rightarrow$ heat flows into system; H $\uparrow$; endothermic
- $\Delta H < 0$ (–) $\Rightarrow$ heat flows out of system; H $\downarrow$; exothermic

# Calorimetry

A coffee-cup calorimeter is used to measure the amount of heat produced (or absorbed) in a reaction at constant P. That is $q_p$ or $\Delta H$. The cup is under constant atmospheric pressure (P) because it is not completely sealed. It is 'isolated': no heat transfer between system and surroundings.



# Product- or Reactant-Favored Reactions

- nature favors processes that decrease energy of the system
- ∴, nature favors exothermic processes

$4\ Fe(s) + 3\ O_2(g) \rightarrow Fe_2O_3(s) \quad \Delta H = -1651\ kJ$

Exothermic. The formation of product is favored.

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g) \quad \Delta H = 179.0\ kJ$

Heat is required for the reaction to occur. The reaction is endothermic. Reagent is favored.

**Is it always like that? Is ΔH the only factor that matters? No, entropy change counts too…**

# Enthalpy of Reaction, Heat of Reaction

- enthalpy of reaction, heat of reaction
- $\Delta H_{reaction}$ when a reaction takes place
- $\Delta H_{fusion}$ when a solid is melted

  $\Delta H_{vaporization}$ when a liquid is boiled up

  $\Delta H_{condensation}$ a gas is condensed to liquid
- $\Delta H_{crystallization}$ when a compound is crystallized from a solution

# Enthalpy of Reaction, Heat of Reaction

100.0 mL of 0.300 M NaOH solution is mixed with 100.0 mL of 0.300 M $HNO_3$ solution in a coffee cup calorimeter. If both solutions were initially at 35.00°C and the temperature of the resulting solution was recorded as 37.00°C, determine the ΔH°rxn (in units of kJ/mol NaOH) for the neutralization reaction between aqueous NaOH and $HNO_3$. Assume 1) that no heat is lost to the calorimeter or the surroundings, and 2) that the density and the heat capacity of the resulting solution are the same as water, 1 g/mL, 4.184 J/g K

$NaOH + HNO_3 \rightarrow NaNO_3(aq) + H_2O$

If $T_f > T_i$, the reaction gave heat off and the calorimeter absorbed it. c(solution) = 4.184 J/g K

# Enthalpy of Reaction, Heat of Reaction

$NaOH + HNO_3 \rightarrow NaNO_3(aq) + H_2O$

Vsolution = 100.0 + 100.0 = 200.0 mL

g solution = V x d = 200.0 mL x 1g/mL = 200.0 g

$\Delta T$ = 37.00°C – 35.00 °C = 2.00 °C = 2.00 K

$q_{sol} = mc\Delta T$ = 200.0g x(4.184J/g K) x 2.00 K = **167**3.6 J

P is constant, $\Delta H = q = -q_{sol}$ = -**1.67**36 kJ (divided $10^3$)

moles of NaOH = 0.1000L x 0.300 mol/L = 0.0300 mol

$$\Delta H = \frac{\text{-1.6736 kJ}}{0.0300 \text{ mol}} = \text{-55.8 kJ/mol NaOH}$$ **3SF**

61. The propane fuel ($C_3H_8$) used in gas barbeques burns according to the thermochemical equation:

$$C_3H_8(g) + 5\,O_2(g) \longrightarrow 3\,CO_2(g) + 4\,H_2O(g)$$
$$\Delta H^\circ_{rxn} = -2217\ \text{kJ}$$

If a pork roast must absorb $1.6 \times 10^3$ kJ to fully cook, and if only 10% of the heat produced by the barbeque is actually absorbed by the roast, what mass of $CO_2$ is emitted into the atmosphere during the grilling of the pork roast?

Molar mass of $CO_2$ = 44.01 g/mol

The 2217 kJ released heat corresponds to the reaction of 1 mol of $C_3H_8$ with 5 moles of $O_2$ to produce 3 moles of $CO_2$ and 4 moles of $H_2O$.

If the efficiency of the barbeque was 100%, we would need $1.6\text{x}10^3$ kJ of heat from it, but being it only 10%, we need (more)

$1.6\text{x}10^3\ \text{kJ} \times \frac{100}{10} = 1.6\text{x}10^4$ kJ produced by the reaction (barbeque). That corresponds to

$$1.6\text{x}10^4\ \text{kJ} \times \frac{3\ \text{moles of } CO_2}{2217\ \text{kJ}} \times \frac{44.01\ \text{g } CO_2}{1\ \text{mol of } CO_2} = 950\ \text{g of } CO_2 \text{ (2 SF)}$$

# Hess's Law

- if a reaction is the sum of two or more other reactions, the $\Delta H$ for the overall reaction is equal to the sum of the $\Delta H$ values of those reactions.
- also applies to $\Delta E$. E and H are state functions

The $\Delta H$ of some reactions can not be measured in a calorimeter, because some other reactions take place at the same time in the reactor.

$C(s) + 2H_2(g) \rightarrow CH_4(g)$; $C_2H_2$, $C_2H_4$, $C_2H_6$, $C_3H_6$, etc., are also produced. In a case like this, the $\Delta H$ of other related reactions can be employed to calculate $\Delta H$

$C_{(s)} + 2H_{2(g)} \rightarrow CH_{4(g)}$ What is the $\Delta H_{rxn}$ = ?

If we know the enthalpy changes for the folowing

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H_1 = -393.5$ kJ

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(l)}$ $\Delta H_2 = -571.6$ kJ

$CO_{2(g)} + 2H_2O_{(l)} \rightarrow CH_{4(g)} + 2O_{2(g)}$ $\Delta H3 = 890.4$ kJ

---

$C(s) + 2H_2(g) \rightarrow CH_4(g)$ $\Delta H_{rxn} = -74.7$ kJ

$\Delta H_r = \Delta H_1 + \Delta H_2 + \Delta H_3$

$\Delta H_r = -393.5$ kJ + (–571.6 kJ) + 890.4 kJ = – 74.7 kJ

Calculate ΔH of the fourth equation out of ΔHs of the first three:

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H = -393.5$ kJ Eq. (1)

$H_{2(g)} + {}^1/_2O_{2(g)} \rightarrow H_2O_{(l)}$ $\Delta H = -285.8$ kJ Eq. (2)

$2C_2H_{2(g)} + 5O_{2(g)} \rightarrow 4CO_{2(g)} + 2H_2O$ $\Delta H = -2598.8$ kJ Eq. (3)

**$2C_{(s)} + H_{2(g)} \rightarrow C_2H_{2(g)}$ $\Delta H_{rxn} =?$ Eq. (4)**

We need to: multiply Eq. (1) by 2; leave Eq. (2) as it is; multiply Eq. (3) by −½, that is to reverse it and times ½.

$2 \times \{ C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H = -393.5$ kJ }

$H_{2(g)} + {}^1/_2O_{2(g)} \rightarrow H_2O_{(l)}$ $\Delta H = -285.8$ kJ

$-½ \times \{ 2C_2H_{2(g)} + 5O_{2(g)} \rightarrow 4CO_{2(g)} + 2H_2O$ $\Delta H = -2598.8$ kJ}

**$2C_{(s)} + 2O_{2(g)} \rightarrow 2CO_{2(g)}$ $\Delta H = 2(-393.5$ kJ)**

**$H_{2(g)} + {}^1/_2O_{2(g)} \rightarrow H_2O_{(l)}$ $\Delta H = -285.8$ kJ**

**$2CO_{2(g)} + H_2O \rightarrow C_2H_{2(g)} + {}^5/_2O_{2(g)}$ $\Delta H = {}^1/_2(+2598.8$ kJ)**

---

**$2C(s) + H_2(g) \rightarrow C_2H_2(g)$ $\Delta H_{rxn} = 226.6$ kJ**

**$\Delta H_{rxn} = 2(-393.5$ kJ) $-285.8$ kJ $+ {}^1/_2(+2598.8$ kJ)**

$2C_{(s)} + H_{2(g)} \rightarrow C_2H_{2(g)}$ $\Delta H_{rxn} = ?$

$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$ $\Delta H = -393.5$ kJ

$H_{2(g)} + {}^1/_2O_{2(g)} \rightarrow H_2O_{(l)}$ $\Delta H = -285.8$ kJ

$2C_2H_{2(g)} + 5O_{2(g)} \rightarrow 4CO_{2(g)} + 2H_2O$

$\Delta H = -2598.8$ kJ

**$2C_{(s)} + 2O_{2(g)} \rightarrow 2CO_{2(g)}$ $\Delta H = 2(-393.5$ kJ)**

**$H_{2(g)} + {}^1/_2O_{2(g)} \rightarrow H_2O_{(l)}$ $\Delta H = -285.8$ kJ**

**$2CO_{2(g)} + H_2O \rightarrow C_2H_{2(g)} + {}^5/_2O_{2(g)}$**

**$\Delta H = {}^1/_2(+2598.8$ kJ)**

---

**$2C(s) + H_2(g) \rightarrow C_2H_2(g)$ $\Delta H_{rxn} = 226.6$ kJ**

**$\Delta H_{rxn} = 2(-393.5$ kJ) $-285.8$ kJ $+ {}^1/_2(+2598.8$ kJ)**

$P_4(s) + 6\ Cl_2(g) \rightarrow 4\ PCl_3(l)$ $\quad \Delta H_{rxn} = ?$

$P_4(s) + 10\ Cl_2(g) \rightarrow 4\ PCl_5(s) \quad \Delta H = -1774.0 kJ$

$PCl_3(l) + Cl_2(g) \rightarrow PCl_5(s) \quad \Delta H = -123.8\ kJ$

**$P_4(s) + 10\ Cl_2(g) \rightarrow 4\ PCl_5(s) \quad \Delta H = -1774.0 kJ$**

**$-4x\{PCl_3(\ell) + Cl_2(g) \rightarrow PCl_5(s) \quad \Delta H = -123.8\ kJ\}$**

**$P_4(s) + 10\ Cl_2(g) \rightarrow 4\ PCl_5(s) \quad \Delta H = -1774.0 kJ$**

**$4PCl_5(s) \rightarrow 4PCl_3(l) + 4Cl_2(g) \quad \Delta H = 4(+123.8\ kJ)$**

---

**$P_4(s) + 6\ Cl_2(g) \rightarrow 4\ PCl_3(l) \quad \Delta H_{rxn} =$** $-1278.8\ kJ$

$\Delta H_{rxn} = -1774.0\ kJ + 495.2\ kJ = -1278.8\ kJ$

# Formation Reaction

Reaction in which **1 mol of a substance** is formed from its elements in their most stable, natural states.

eg. formation reaction for $C_2H_6SO_{(l)}$

$$C_{(s)} + H_{2(g)} + S_{8(s)} + O_{2(g)} \rightarrow C_2H_6SO_{(l)}$$

$$2C_{(s)} + 3H_{2(g)} + {}^{1}/_{8}S_{8(s)} + {}^{1}/_{2}O_{2(g)} \rightarrow C_2H_6SO_{(l)}$$

# Heat (Enthalpy) of Formation

- $\Delta H_f$
- enthalpy change associated with a formation reaction

# Standard Conditions

$T = 25°C = 298\ K$

$P = 1\ atm$

# Standard Heat of Formation or Standard Molar Enthalpy of Formation

**$\Delta H_f°$** is the enthalpy change for the formation of 1 mol of a compound directly from its elements in their standard states

$\Delta H_f°$ may be >0 or <0 for a compound (tables)

**$\Delta H_f° = 0$ for any element in its most stable form,**

e.g. Na(s), Hg(l), $Cl_2$(g), $Br_2$(l), $H_2$(g), $P_4$(s), C(s)

# Enthalpy Change for a Reaction

$$a\,A + b\,B \rightarrow c\,C + d\,D$$

## Hess's Law

$$\Delta H°_{rxn} = \Sigma(\Delta H_f° \text{ products}) - \Sigma(\Delta H_f° \text{ reagents})$$

"means take the sum"

$$\Delta H°_{rxn} = c\times\Delta H_f°(C) + d\times\Delta H_f°(D) - a\times\Delta H_f°(A) - b\times\Delta H_f°(B)$$

**$\Delta H_f°$(element) = 0**

# Example: Calculate the $\Delta H°$ in kJ/mol $B_5H_9$ for the following reaction.

$$2B_5H_9 + 12O_2 \rightarrow 5B_2O_3 + 9H_2O$$

| Compound | $\Delta H_f°$ (kJ/mol) |
|---|---|
| $B_5H_9$ | 73.2 |
| $B_2O_3$ | –1272.8 |
| $H_2O$ | –241.83 |

$$\Delta H°_{Rxn} = [(5(\Delta H_f° B_2O_3) + 9(\Delta H_f° H_2O)] - [(2 (\Delta H_f° B_5H_9) + 12(\Delta H_f° O_2)]$$

$\Delta H°_{Rxn}$= [5mol(–1272.8 kJ/mol) + 9mol(–241.83 kJ/mol)] –[2 mol(73.2 kJ/mol) +12 mol(0 kJ/mol)] = –8686.9 kJ

–8686.9 kJ/2 mol $B_5H_9$ = –4343.5 kJ/mol $B_5H_9$

# Calculate the $\Delta H°$ in kJ/mol Mg for the following reaction.

## $3Mg + SO_2 \rightarrow MgS + 2MgO$

| Compound | $\Delta H_f°$ (kJ/mol) |
|---|---|
| MgO | –601.7 |
| MgS | –598.0 |
| $SO_2$ | –296.8 |

$$\Delta H°_{Rxn} = [1\ mol(\Delta H_f°\ MgS) + 2(\Delta H_f°\ MgO)] - [3(\Delta H_f°\ Mg) + 1\ mol(\Delta H_f°\ SO_2)]$$

$$\Delta H°_{Rxn} = [1mol(–598.0\ kJ/mol) + 2\ mol(–601.7\ kJ/mol)] –[3\ mol(0\ kJ/mol) + 1\ mol(–296.8\ kJ/mol)] = –1504.6\ kJ$$

$$–1504.6\ kJ/3\ mol\ Mg = –501.5\ kJ/\textbf{mol Mg}$$

The enthalpy change for the combustion of styrene, $C_8H_8$, is measured by calorimetry:

$$C_8H_8(l) + 10\ O_2(g) \rightarrow 8\ CO_2(g) + 4\ H_2O(l)$$

$$\Delta H°_{rxn} = -4395.0\ kJ$$

Use this, along with the $\Delta H_f°$ of $CO_2$ and $H_2O$, to calculate the $\Delta H_f°$ $C_8H_8$, in kJ/mol

$\Delta H_f°$ $CO_2(g) = -393.51$ kJ/mol

$H_2O(l) = -285.83$ kJ/mol

$$\Delta H°_{Rxn} = [8(\Delta H_f° CO_2) + 4(\Delta H_f° H_2O(l)] - (\Delta H_f° C_8H_8)$$

$$\Delta H_f° C_8H_8 = [8(\Delta H_f° CO_2) + 4(\Delta H_f° H_2O(l)] - \Delta H°_{rxn}$$

$$\Delta H_f° C_8H_8 = 8(-393.51) + 4(-285.83) - (-4395.0)$$

$$\Delta H_f° C_8H_8 = 103.6\ kJ/mol$$

Nitroglycerin is a powerful explosive that forms four different gases when detonated

**$2C_3H_5(NO_3)_3(\ell) \rightarrow 3N_2(g) + \frac{1}{2}O_2(g) + 6CO_2(g) + 5H_2O(g)$**

Calculate the enthalpy change when 10.0 g of nitroglycerin is detonated. $\Delta H_f°$(kJ/mol) are:

$CO_2(g) = -393.5$ $H_2O(g) = -241.8$

NTG, $C_3H_5(NO_3)_3(\ell) = -364$

$$\Delta H°_{rxn} = [6(\Delta H_f° CO_2) + 5(\Delta H_f° H_2O(g)] - 2(\Delta H_f° NTG)$$

$$\Delta H°_{rxn} = 6(-393.5) + 5(-241.8) - 2(-364) =$$

$$= -2842 \text{ kJ for 2 mol of NTG}$$

For 10.0 g of NTG we need its MW = 227.1 g/mol

–2842 kJ for 2 mol of NTG

for 10.0 g of NTG we need its MW = 227.1 g/mol

**Calculate moles of NTG and use them for $\Delta H°_{rxn}$**

$$\textbf{10.0 g NTG} \times \frac{\textbf{1 mol NTG}}{\textbf{227.1 g NTG}} = \textbf{0.0440 mol of NTG}$$

$$\textbf{0.0440 mol of NTG} \times \frac{\textbf{– 2842 kJ}}{\textbf{2 mol NTG}} = \textbf{–62.6 kJ}$$

How much heat energy is liberated when 11.0 grams of manganese is converted to $Mn_2O_3$ at standard state conditions?

$$4Mn(s) + 3O_2(g) \rightarrow 2Mn_2O_3(s) \quad \Delta H = -1924.6\ kJ$$

$$11.0\ g\ Mn \times \frac{1\ mol\ Mn}{54.94\ g\ Mn} = 0.200\ mol\ Mn$$

The given $\Delta H$ corresponds to the reaction of 4 moles Mn

$$0.200\ mol\ Mn \times \frac{-1924.6\ kJ}{4\ moles\ Mn} = -96.2\ kJ$$

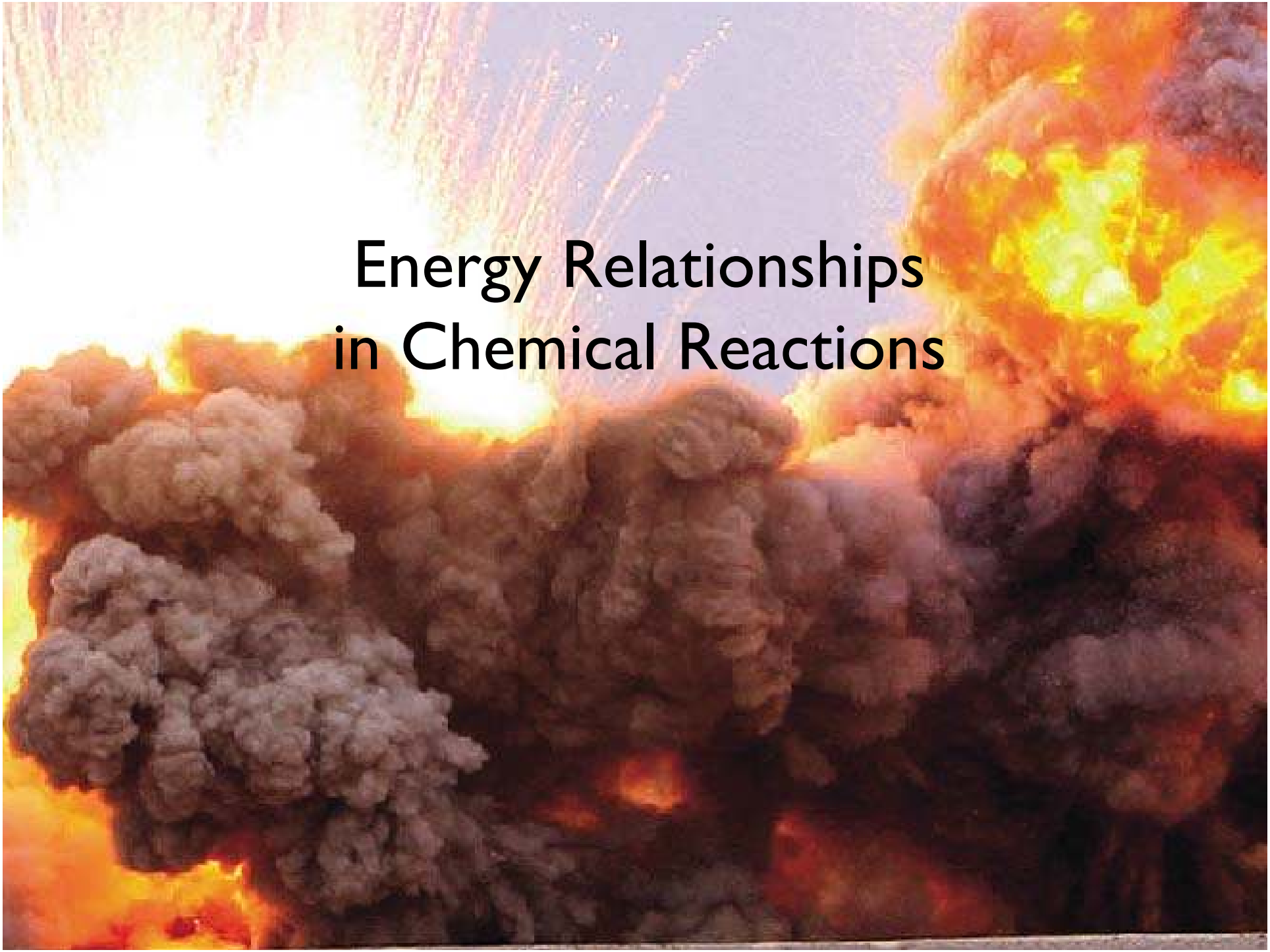
Calculate the amount of heat released in the complete combustion of 8.17 grams of Al to form $Al_2O_3(s)$ at 25°C and 1 atm. $\Delta H_f°$ for $Al_2O_3(s)$ = −1676 kJ/mol

$$4Al(s) + 3O_2(g) \rightarrow 2Al_2O_3(s)$$

$$8.17\ g\ Al \times \frac{1\ mol\ Al}{26.98\ g\ Al} = 0.303\ mol\ Al$$

2 moles mol $Al_2O_3$ are produced out of 4 moles of Al

$$0.303\ mol\ Al \times \frac{2\ mol\ Al_2O_3}{4\ mol\ Al} \times \frac{-1676\ kJ}{1\ mol\ Al_2O_3} = -254\ kJ$$


Energy Relationships
in Chemical Reactions

# Chapter 6 – Energy Relationships in Chemical Reactions

- The nature of energy and types of energy (6.1)
- Energy changes in chemical reactions (6.2)
- Introduction to thermodynamics (6.3)
- Calorimetry (6.5)
- Enthalpy of chemical reactions (6.4)
- Standard enthalpy of formation and reaction (6.6)

# Chapter 6 – Energy Relationships in Chemical Reactions

- System and Surroundings (**6.2**)
- Internal Energy $\Delta U$ (**6.2**)
  - Work
  - Heat
  - First Law of Thermodynamics
  - State Functions
    - $\Delta U$ as a state function
- Enthalpy $\Delta H$ (**6.3** to **6.6**)
  - Exothermic and endothermic processes (6.4, 6.6)
    - Enthalpy and spontaneous change
  - Heat capacity (6.5)
  - Specific heat capacity (6.5)
    - Calorimetry
  - Enthalpy as a state function (6.4, 6.6)
    - Thermochemical cycles
    - $\Delta H_f \rightarrow \Delta H_{rxn}$



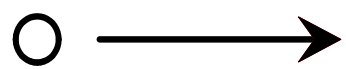
# 6.1 The nature of energy and types of energy

- *Radiant energy* comes from the sun and is earth's primary energy source
- *Thermal energy* is the energy associated with the random motion of atoms and molecules
- *Chemical energy* is the energy stored within the bonds of chemical substances
- *Nuclear energy* is the energy stored within the collection of neutrons and protons in the atom
- *Potential energy* is the energy available by virtue of an object's position

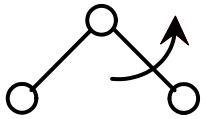
# 6.1 The nature of energy and types of energy

What are the forms of energy *(thinking in terms of particles)?*

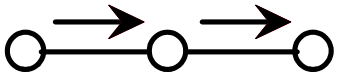
Translational ($E_{K, trans}$)



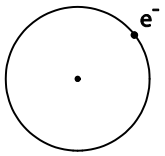
Rotational ($E_{K, rot}$)



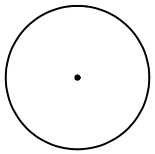
Vibrational ($E_{K, vib}$)



Electronic ($E_{elec}$)



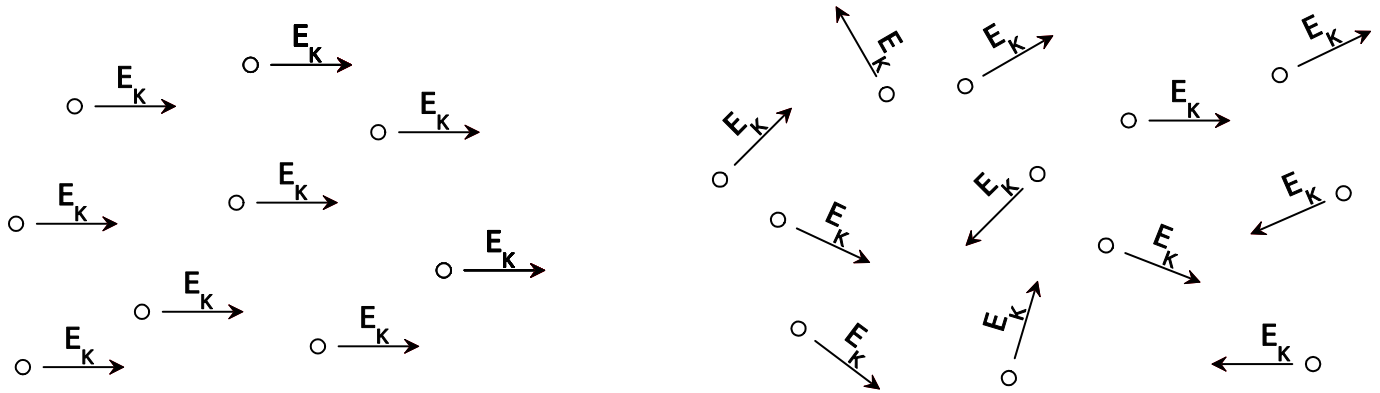

Nuclear ($E_{nucl}$)





# 6.2 Energy changes in chemical reactions

- Key Definition:
  - Energy
    - The capacity to do work
    - The capacity to direct energy change resulting from a process (or to move matter)







Work

Heat

# 6.2 Energy changes in chemical reactions

- Key Definition:
  - Energy

Work + Heat

- Work is moving matter in a concerted fashion (work done by the system by moving matter)

- Heat is moving matter in a non-concerted fashion (work not done by the system, but matter still moved)

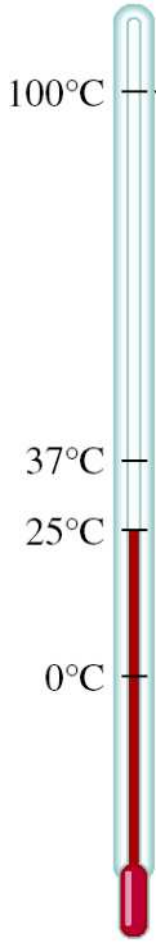


# 6.2 Energy changes in chemical reactions

***Heat*** is the transfer of **thermal energy** between two bodies that are at different temperatures.

***Temperature*** is a measure of the **thermal energy**.

Temperature ≠ Thermal Energy



# 6.2 Energy changes in chemical reactions

- Key Definitions:
  - System
    - The specific part of the universe that is of interest to us (to study and understand).
  - Surroundings
    - The rest of the universe outside of the system
  - Universe
    - The system and the surroundings (everything)

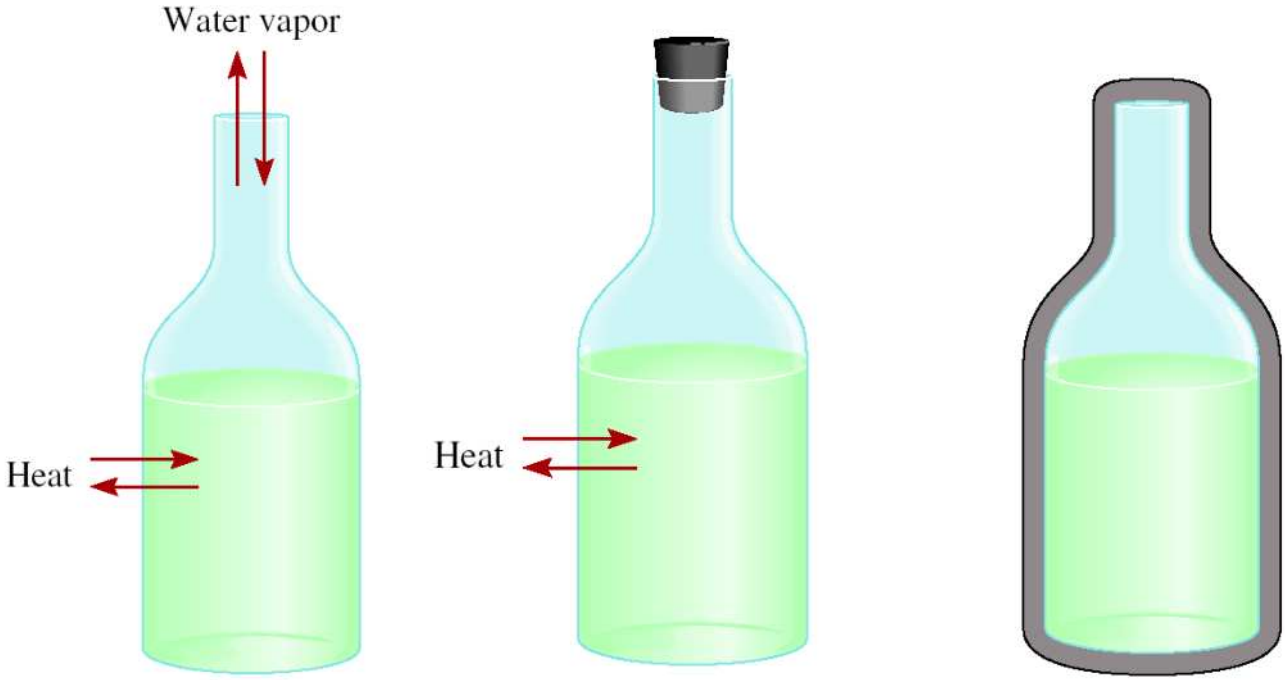
# 6.2 Energy changes in chemical reactions

- Key Definitions:
  - System
    - The specific part of the universe that is of interest to us (to study and understand).
    - Open
      - Can mass and energy (usually in the form of heat) with its surroundings
    - Closed
      - Allows the transfer of energy (usually heat) but not mass
    - Isolated
      - Does not allow the transfer of either mass of energy

# 6.2 Energy changes in chemical reactions

**different systems**

**surrounding**




| | open | closed | isolated |
|---|---|---|---|
| **Exchange:** | mass & energy | energy | nothing |

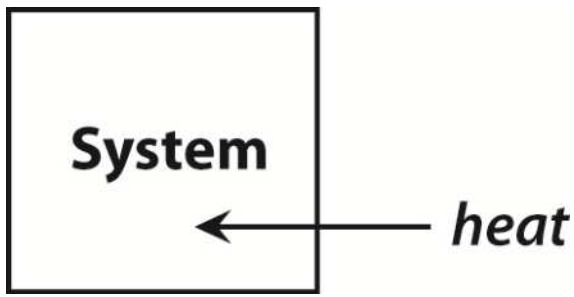
Figure 6.1, p. 180

# 6.2 Energy changes in chemical reactions

- System gains heat (transfer thermal energy from the surroundings to the system)
  - Heat is positive (wrt system)
  - ***Endothermic***

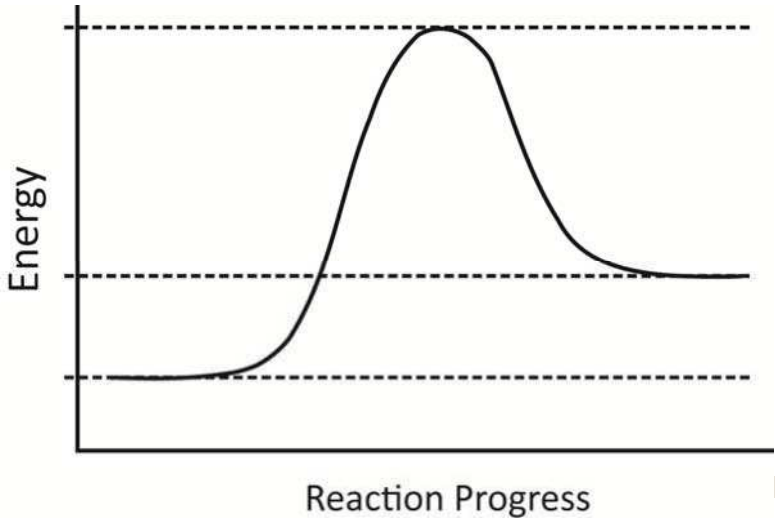
energy + $2HgO(s) \longrightarrow Hg(l) + O_2(g)$

energy + $H_2O(s) \longrightarrow H_2O(l)$







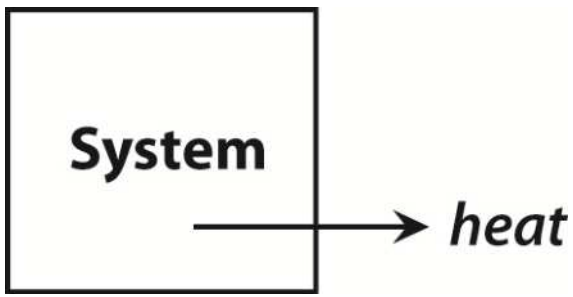


# 6.2 Energy changes in chemical reactions

- System loses heat (transfer thermal energy from the system to the surroundings)
  - Heat is negative (wrt system)
  - ***Exothermic***

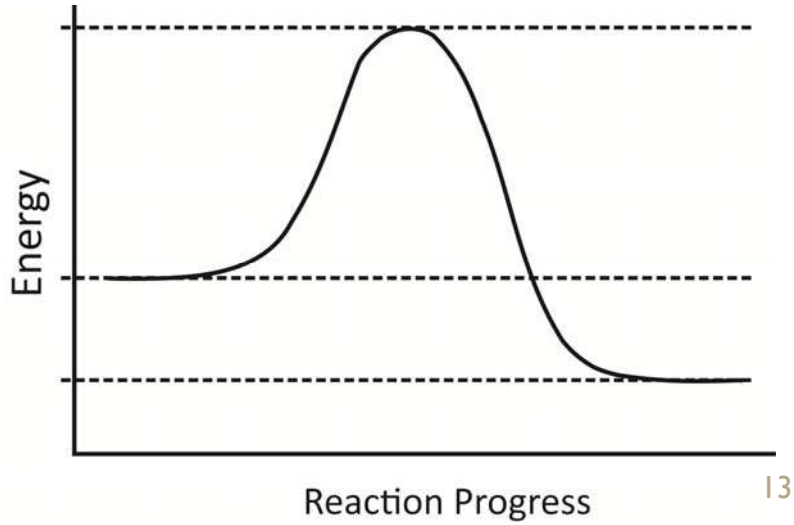
$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l) + \text{energy}$

$H_2O(g) \longrightarrow H_2O(l) + \text{energy}$









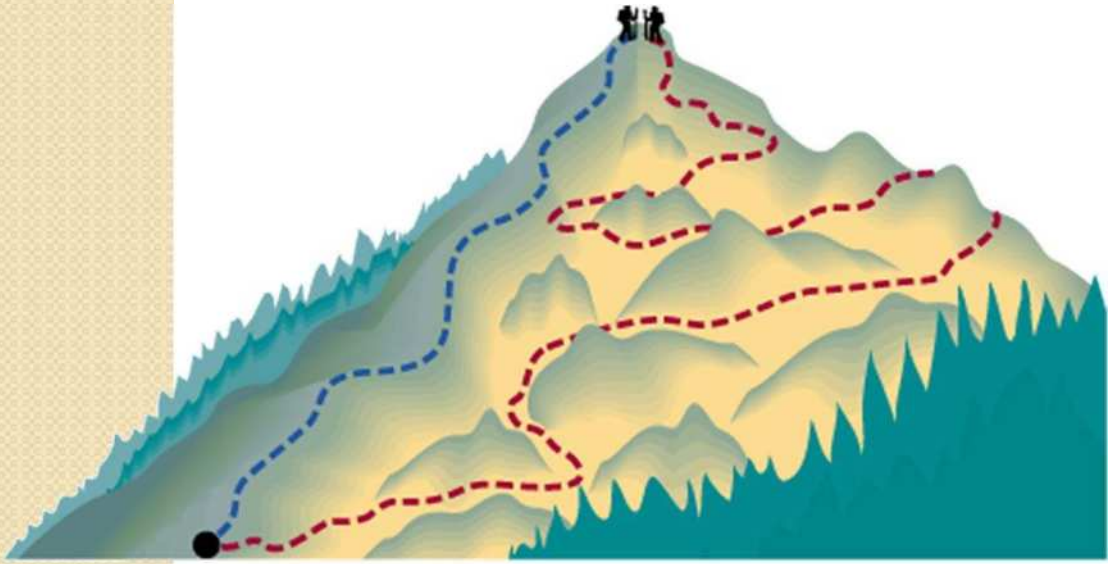
# 6.3 Introduction to thermodynamics

- What is a thermodynamics?
  - scientific study of the interconversion of heat and other kinds of energy
- What is a state function?
  - properties that are determined by the state of the system, regardless of how that condition was achieved.

# 6.3 Introduction to thermodynamics

***State functions*** are properties that are determined by the state of the system, regardless of how that condition was achieved.

energy, pressure, volume, temperature



$$\Delta U = U_{final} - U_{initial}$$

$$\Delta P = P_{final} - P_{initial}$$

$$\Delta V = V_{final} - V_{initial}$$

$$\Delta T = T_{final} - T_{initial}$$

Potential energy of hiker 1 and hiker 2 is the same even though they took different paths.

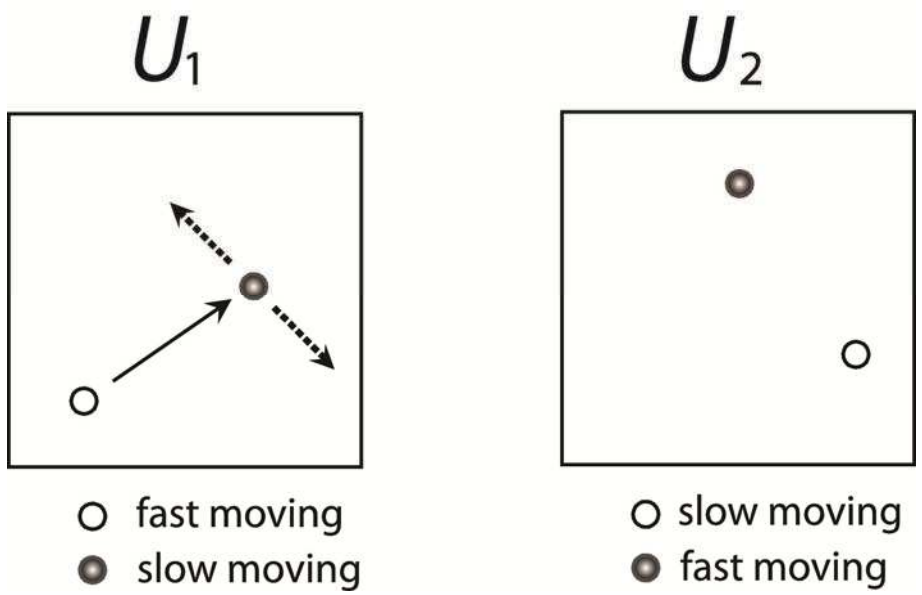
Figure 6.3, p. 182

# 6.3 Introduction to thermodynamics

- What is the first law of thermodynamics?
  - In terms of the energy of the universe?
  - In terms of heat and work?
  - In terms of the energy of the system vs. the surroundings?

# 6.3 Introduction to thermodynamics

## Internal Energy of a System




- $U_1$ vs. $U_2$ (not a particle basis but on a system basis)?
- For an isolated system; $U_1 = U_2$ or $\Delta U = U_2 - U_1 = 0$
- Cannot measure the internal energy of a system, but can measure the change in internal energy
- For chemical reactions: $\Delta U = U_{products} - U_{reactants}$



# 6.3 Introduction to thermodynamics

- What is the first law of thermodynamics?
  - In terms of the energy of the universe?
  - In terms of heat and work?
  - In terms of the energy of the system vs. the surroundings?

**TABLE 6.1 Sign Conventions for Work and Heat**

| Process | Sign |
|---|---|
| Work done by the system on the surroundings | − |
| Work done on the system by the surroundings | + |
| Heat absorbed by the system from the surroundings (endothermic process) | + |
| Heat absorbed by the surroundings from the system (exothermic process) | − |

- What is the law of conservation of energy?

Table 6.1, p. 183

# 6.5 Calorimetry

- Is heat a state function?
- How do we quantify heat for any system?
  - What is heat capacity (C)?
  - The amount of heat (*q*) required to raise the temperature of **a given quantity** of substance (*m*) by **one degree Celsius** for that substance.
- What are the units for heat capacity?
- What is the relationship between heat capacity and change in temperature?

# 6.5 Calorimetry

- How do we quantify heat for any pure substance?
  - What is specific heat (*s*)?
  - The amount of heat (q) required to raise the temperature of **one gram** of the substance **by one degree Celsius**
- What are the units for specific heat?
- What is the relationship between
  - Specific heat and change in temperature?
  - Specific heat and heat capacity?

**Table 6.2**

**The Specific Heats of Some Common Substances**

| Substance | Specific Heat (J/g · °C) |
|---|---|
| Al | 0.900 |
| Au | 0.129 |
| C (graphite) | 0.720 |
| C (diamond) | 0.502 |
| Cu | 0.385 |
| Fe | 0.444 |
| Hg | 0.139 |
| $H_2O$ | 4.184 |
| $C_2H_5OH$ (ethanol) | 2.46 |

Table 6.2, p. 193

# 6.5 Calorimetry

Practice:

What was the initial temperature of water when

1 mol of iron at 550.0°C was placed in a 1.5 L water bath and both substances reached a final temperature of 28.25°C?

The specific heat of iron is 0.444 $J \cdot g^{-1} \cdot {}^{o}C^{-1}$

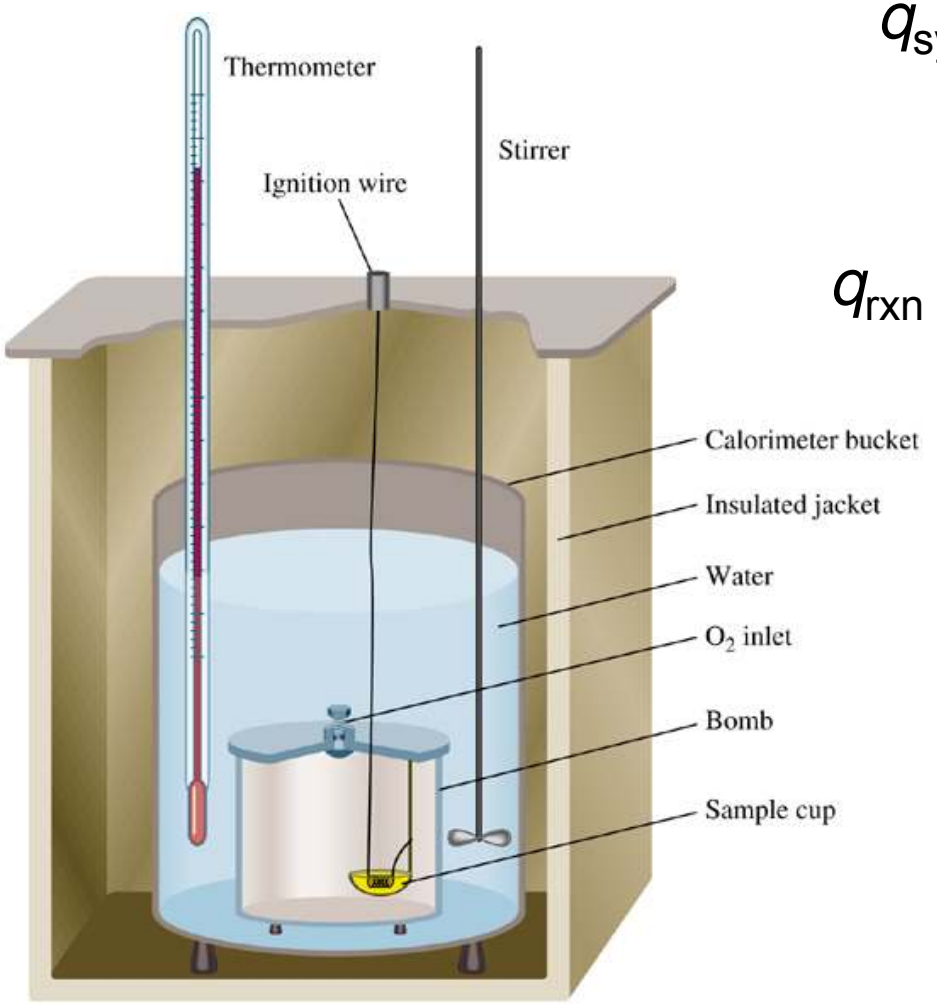
The specific heat of water is 4.18 $J \cdot g^{-1} \cdot {}^{o}C^{-1}$



# 6.5 Calorimetry

- How do we measure heat (or calculate a heat capacity)?
  - What is calorimetry?
    - Different types:
    - Constant Volume
    - Constant Pressure

# 6.5 Calorimetry




$$q_{sys} = q_{water} + q_{bomb} + q_{rxn}$$

$$q_{sys} = 0$$

$$q_{rxn} = -(q_{water} + q_{bomb}) = -q_{cal}$$

$$q_{cal} = C_{cal} \times \Delta t$$

Reaction at Constant $V$

No heat enters or leaves!

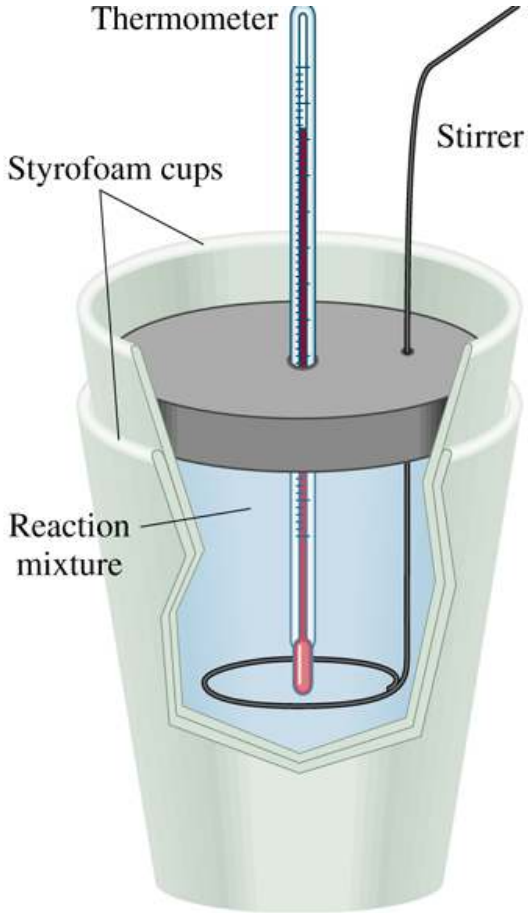
Figure 6.7, p. 194

# 6.5 Calorimetry

Practice:

What is heat of combustion (in $kJ \cdot mol^{-1}$) for graphite when 1.850 g is combusted in a constant volume bomb calorimeter with the water at an initial temperature of 22.02 °C and the final temperature of the water is 27.95 °C.

# 6.5 Calorimetry




$$q_{sys} = q_{water} + q_{rxn}$$

$$q_{sys} = 0$$

$$q_{rxn} = - q_{water}$$

$$q_{water} = m_{water} \times s_{water} \times \Delta t_{water}$$

Reaction at Constant *P*

No heat enters or leaves!

Figure 6.8, p. 196

# 6.5 Calorimetry

**TABLE 6.3 Heats of Some Typical Reactions Measured at Constant Pressure**

| Type of Reaction | Example | $\Delta H$ (kJ/mol) |
|---|---|---|
| Heat of neutralization | $HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$ | −56.2 |
| Heat of ionization | $H_2O(l) \longrightarrow H^+(aq) + OH^-(aq)$ | 56.2 |
| Heat of fusion | $H_2O(s) \longrightarrow H_2O(l)$ | 6.01 |
| Heat of vaporization | $H_2O(l) \longrightarrow H_2O(g)$ | 44.0* |
| Heat of reaction | $MgCl_2(s) + 2Na(l) \longrightarrow 2NaCl(s) + Mg(s)$ | −180.2 |

*Measured at 25°C. At 100°C, the value is 40.79 kJ.




Table 6.3, p. 196

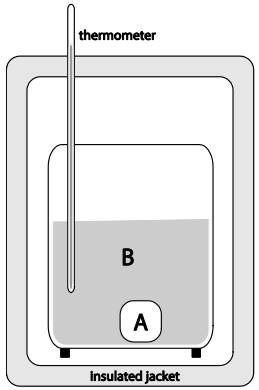
# 6.5 Calorimetry

Practice:

When 10.0 g of ammonium nitrate is dissolved in 100.0 mL of water and the temperature decreases from 25.0°C to 17.1°C. What is the molar heat of solution of ammonium nitrate?

# Chapter 6 Calorimetry – Practice

A reaction occurs in a vessel (part A in the figure) which is submerged in water (part B in the figure) in a constant volume bomb calorimeter. If the observed temperature (on the thermometer) increases,




**A.** the reaction is endothermic, since $q_{cal} = q_{rxn}$.
**B.** the reaction is endothermic, since $q_{cal} = -q_{rxn}$.
**C.** the reaction is exothermic, since $q_{cal} = q_{rxn}$.
**D.** the reaction is exothermic, since $q_{cal} = -q_{rxn}$.

What is the standard molar enthalpy of solution for sodium chloride if 8.00 g of sodium chloride decreases the temperature of 74.90 g of water by 1.81 °C? Assume the heat transfer occurs only between the water and the reaction. The specific heat of water is 4.18 J $\cdot g^{-1} \cdot °C^{-1}$.

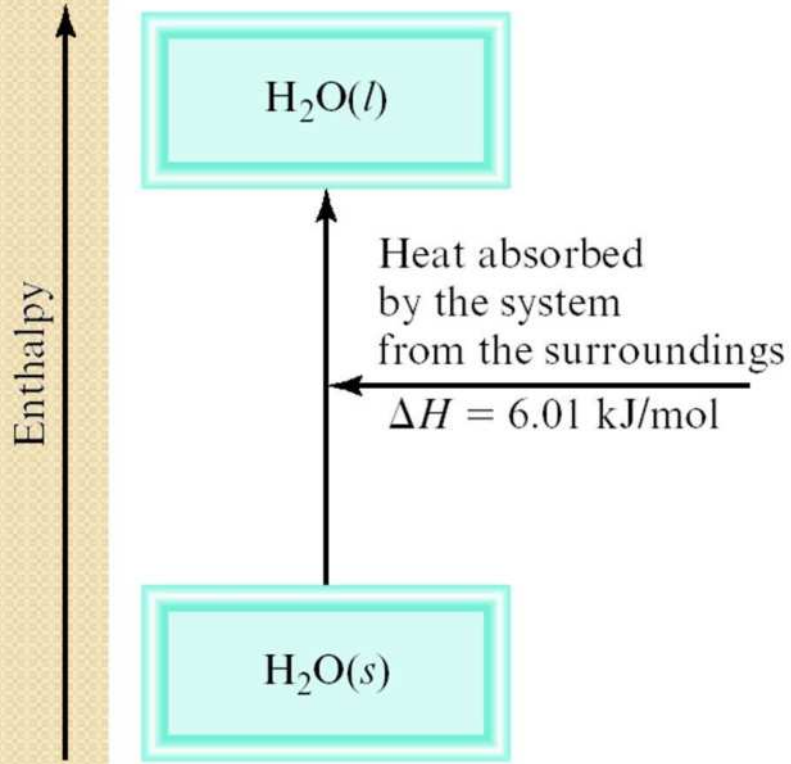
**A.** 4140 kJ $\cdot mol^{-1}$
**B.** 567 kJ $\cdot mol^{-1}$
**C.** 4.14 kJ $\cdot mol^{-1}$
**D.** 0.567 kJ $\cdot mol^{-1}$

# 6.4 and 6.6 Enthalpy

- What is enthalpy?
  - Heat content of a reaction or a quantity to describe the heat flow into or out of a system in a process that occurs at constant pressure
- Is enthalpy a state function?
  - What does this mean in terms of a reaction?
- Is enthalpy extensive?
  - What does this mean in terms of a reaction?
- How can we visualize this?
- How can we calculate this?
  - How are the methods the same/different?



# Thermochemical Equations

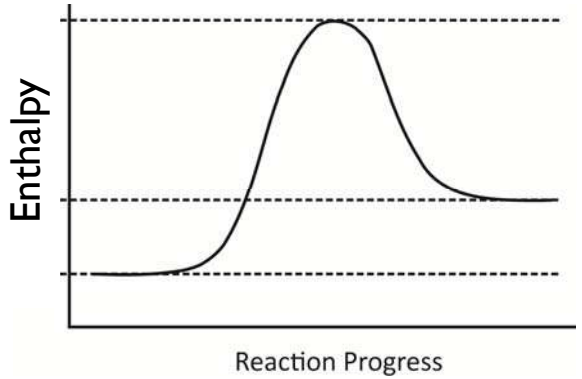



Is $\Delta H$ negative or positive?

System absorbs heat

Endothermic

$\Delta H < 0$ $H_{reactants} < H_{products}$

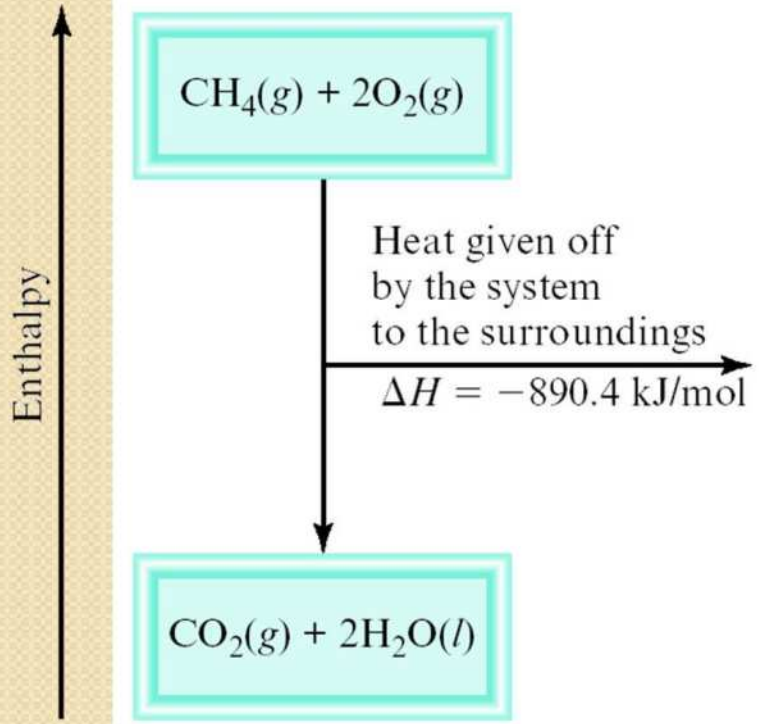



6.01 kJ are absorbed for every 1 mole of ice that melts at $0^0$C and 1 atm.

$$H_2O(s) \longrightarrow H_2O(l) \qquad \Delta H = 6.01 \text{ kJ/mol}$$



# Thermochemical Equations

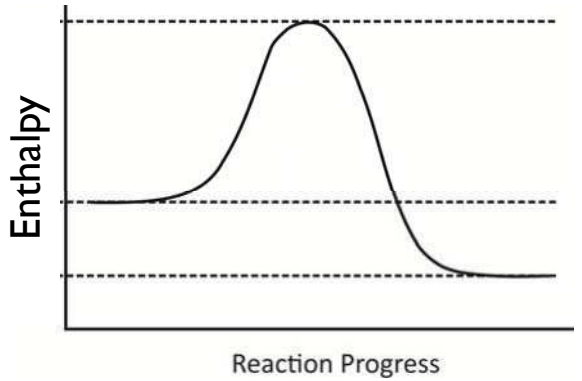


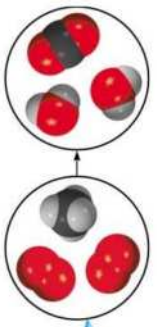

Is $\Delta H$ negative or positive?

System gives off heat

Exothermic

$\Delta H < 0$ $H_{reactants} > H_{products}$





890.4 kJ are released for every 1 mole of methane that is combusted at $25^0$C and 1 atm.

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$ $\Delta H = -890.4$ kJ/mol



# 6.4 and 6.6 Enthalpy

- How are enthalpy and heat related?
- How are enthalpy and energy related?

# 6.4 and 6.6 Enthalpy

- What are the rules for enthalpy?
  1. Writing a thermochemical equation
  2. Direction of the equation (reactants vs. products)
  3. Extensive property – increasing or decreasing the amounts of products or reactants
  4. State property – one reaction or multiple reactions to an overall reaction

# 6.4 and 6.6 Enthalpy

- Key Definitions:
  - Standard conditions
  - Enthalpy of formation
  - Enthalpy of combustion
  - Enthalpy of solution
  - Enthalpy of vaporization (heat of vaporization)
  - Enthalpy of fusion (heat of fusion)

  - What is an energy diagram?



# 6.4 and 6.6 Enthalpy

**Table 6.3** Heats of Some Typical Reactions Measured at Constant Pressure

| Type of Reaction | Example | $\Delta H$ (kJ) |
|---|---|---|
| Heat of neutralization | $HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$ | −56.2 |
| Heat of ionization | $H_2O(l) \longrightarrow H^+(aq) + OH^-(aq)$ | 56.2 |
| Heat of fusion | $H_2O(s) \longrightarrow H_2O(l)$ | 6.01 |
| Heat of vaporization | $H_2O(l) \longrightarrow H_2O(g)$ | 44.0* |
| Heat of reaction | $MgCl_2(s) + 2Na(l) \longrightarrow 2NaCl(s) + Mg(s)$ | −180.2 |

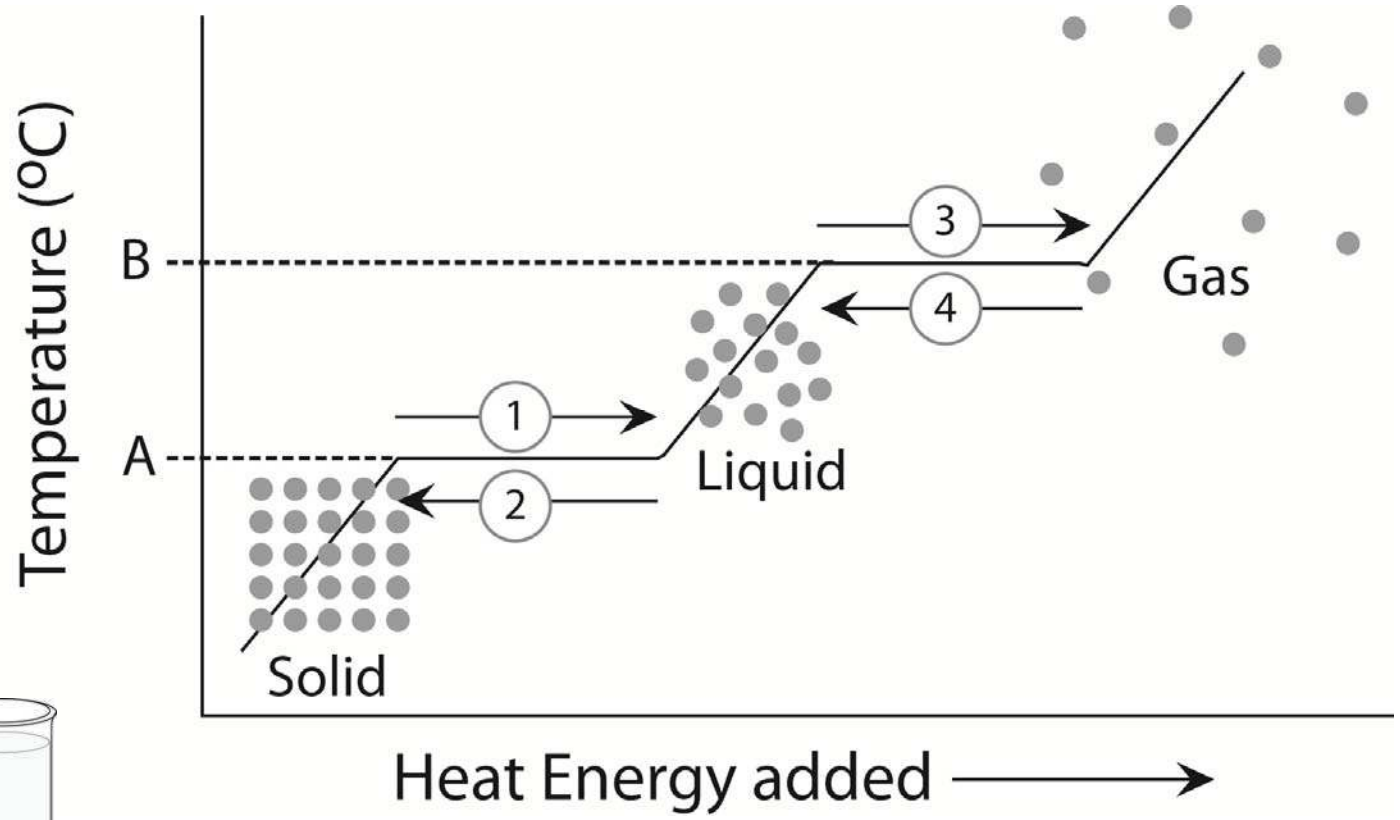
*Measured at 25°C. At 100°C, the value is 40.79 kJ.




Table 6.3, p. 196

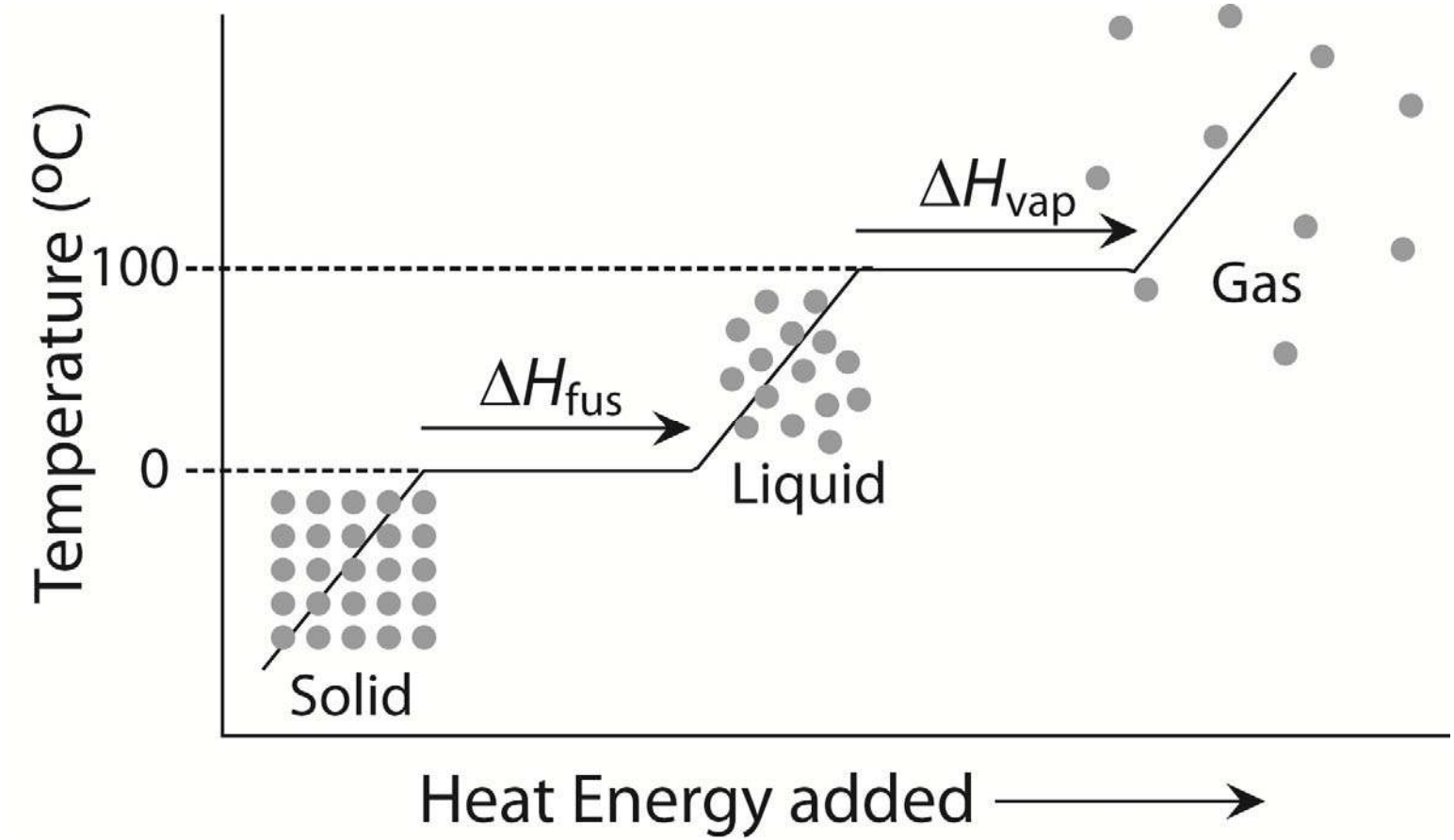
# 6.4 and 6.6 Enthalpy

- What occurs at 1, 2, 3, and 4?
- How do we define this in terms of $\Delta H$






# 6.4 and 6.6 Enthalpy



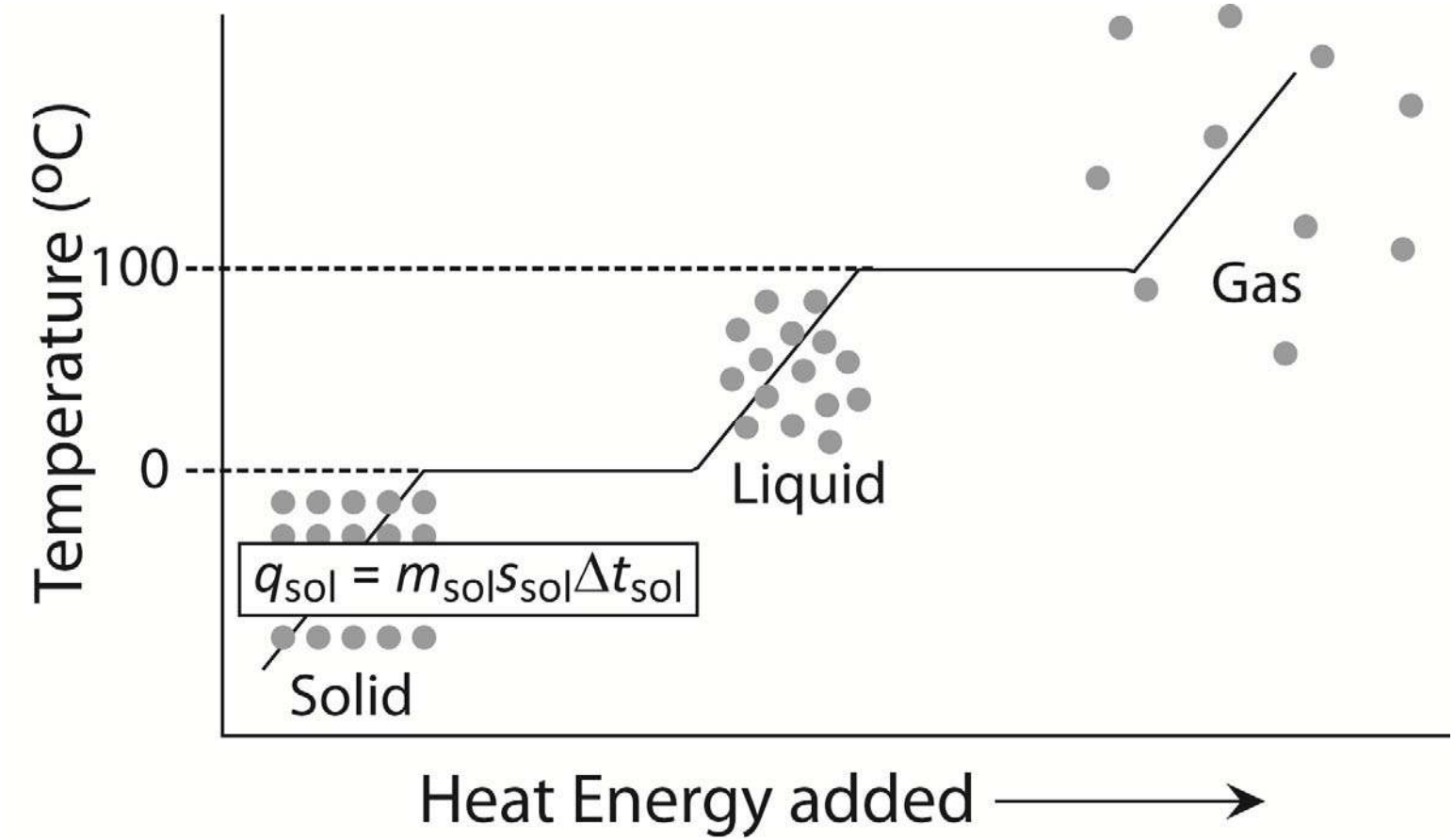



# 6.4 and 6.6 Enthalpy

- How do we incorporate phase changes (melting, boiling) with warming of substances?
- What does this mean on a particle level?
- What is a heating or cooling curve?
- How do we calculate the amount of heat required to warm ice to steam or cool steam to ice?

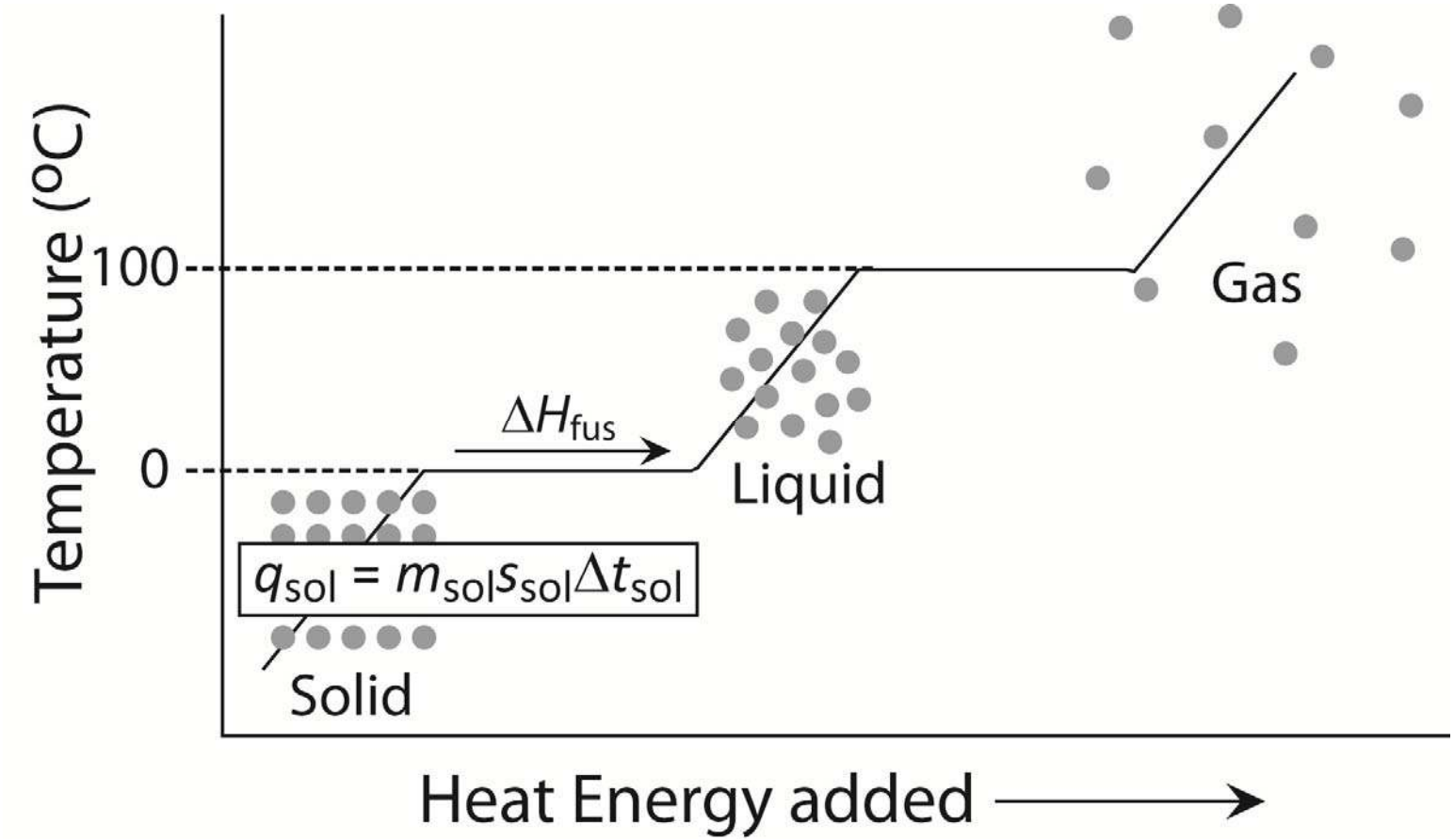


# 6.4 and 6.6 Enthalpy



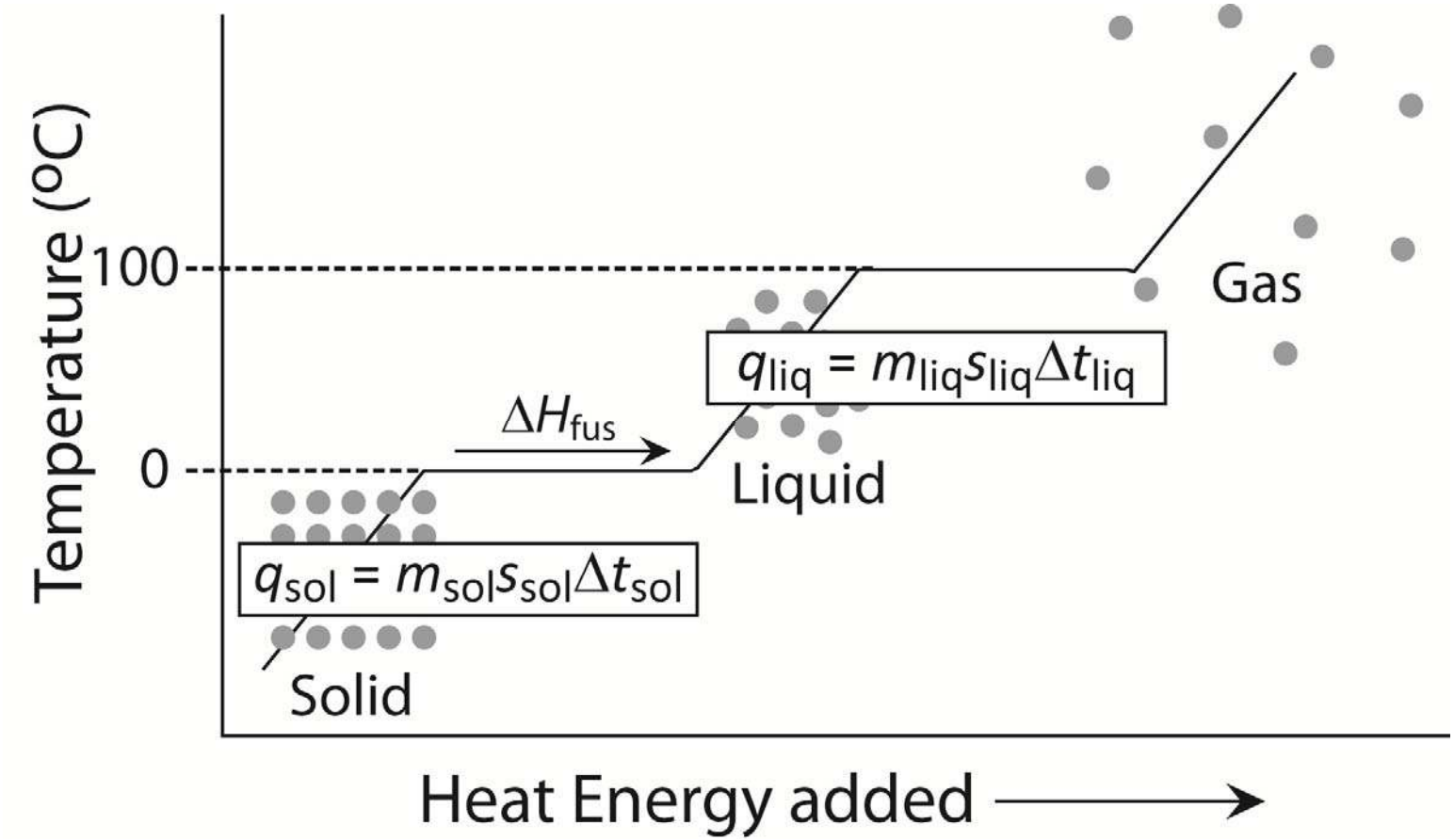



# 6.4 and 6.6 Enthalpy



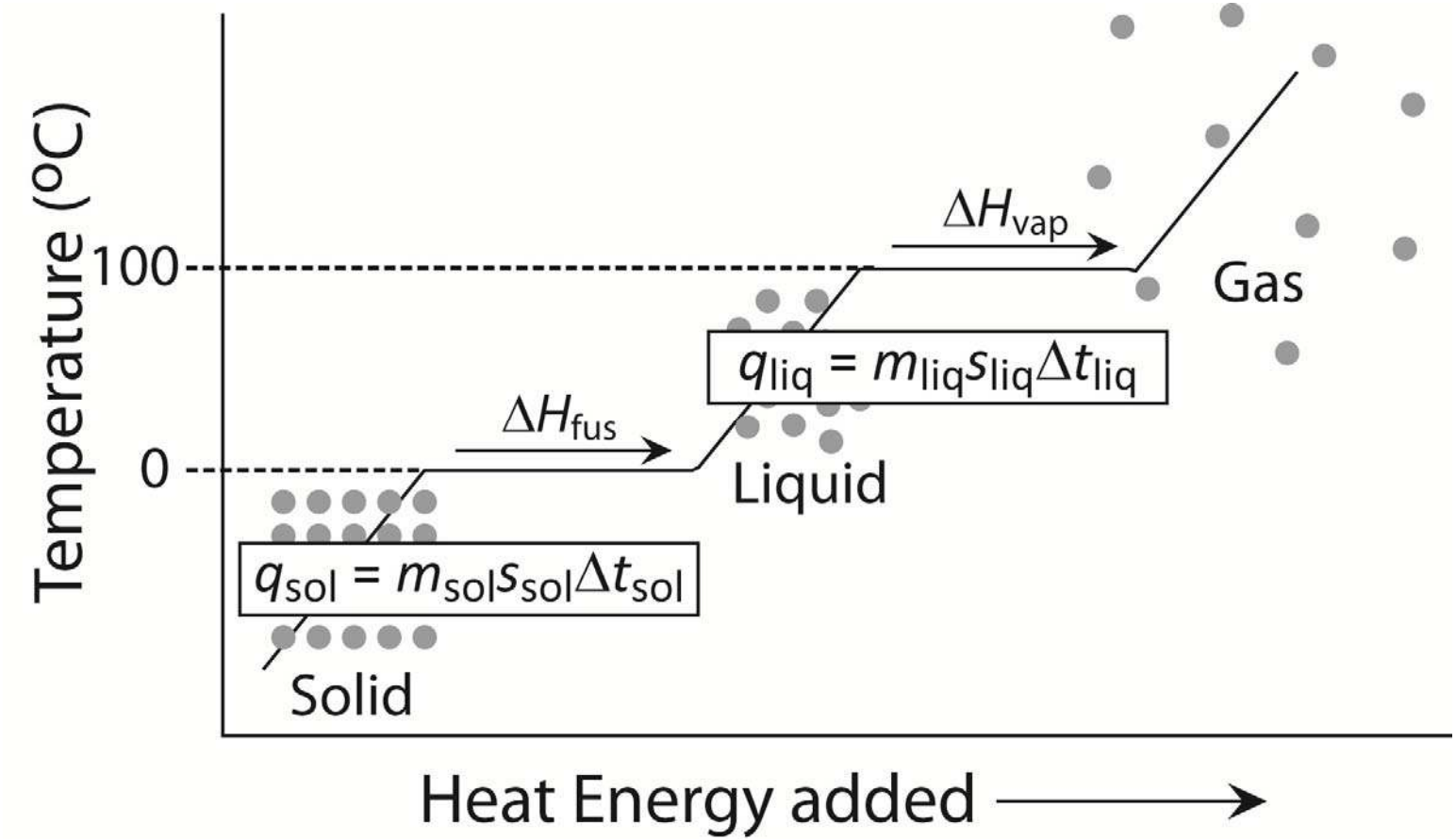



# 6.4 and 6.6 Enthalpy



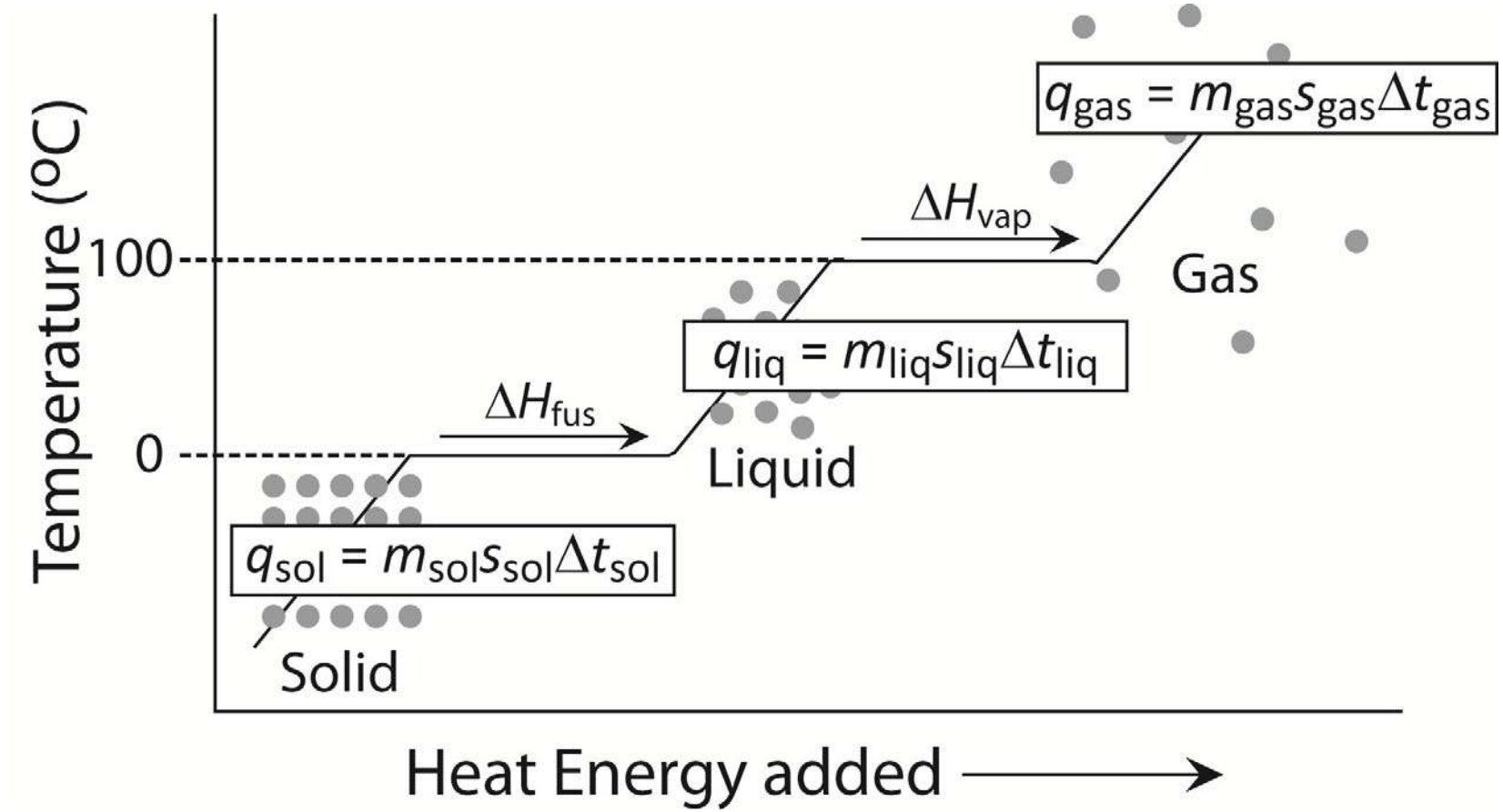



# 6.4 and 6.6 Enthalpy



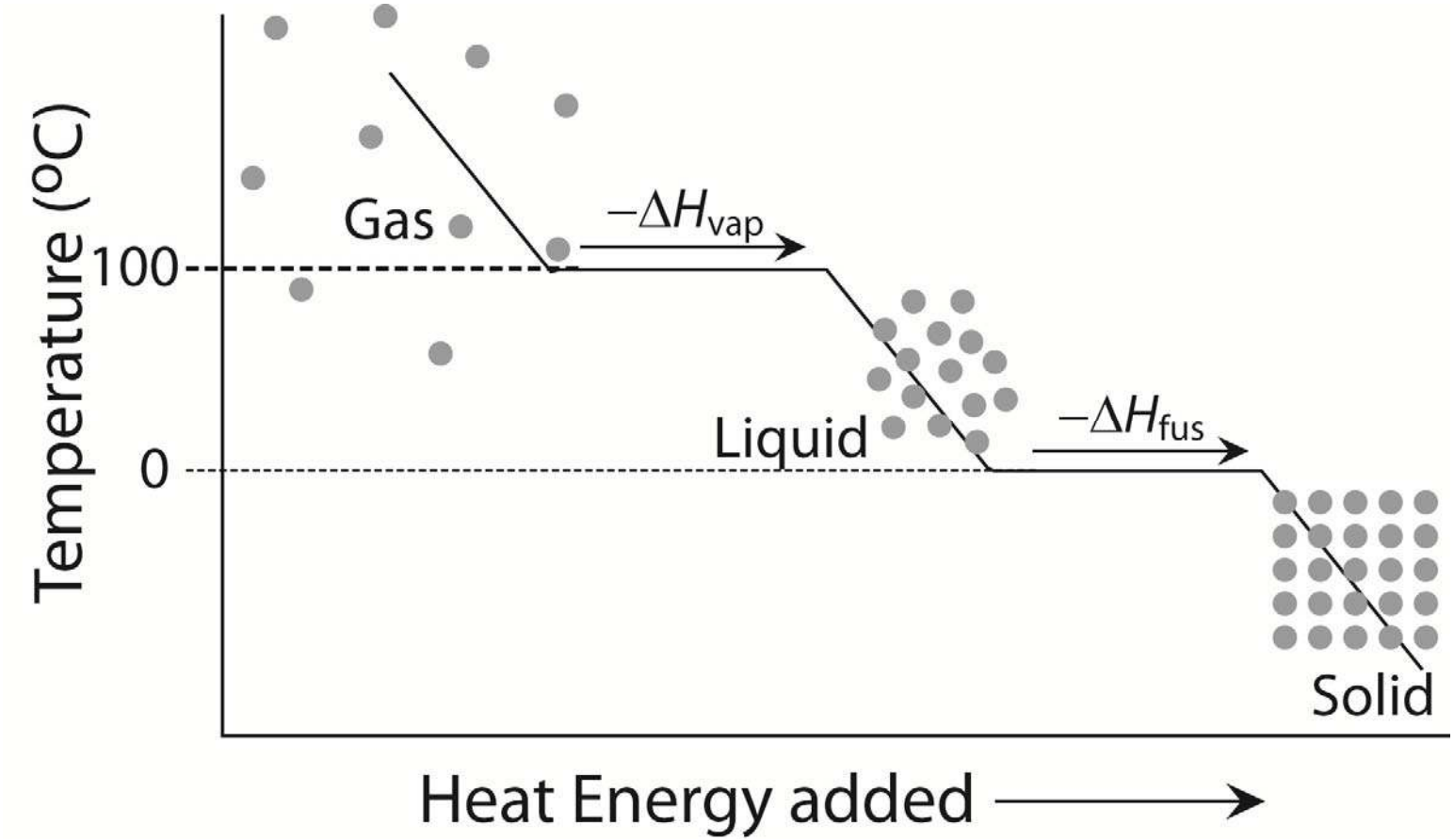



# 6.4 and 6.6 Enthalpy



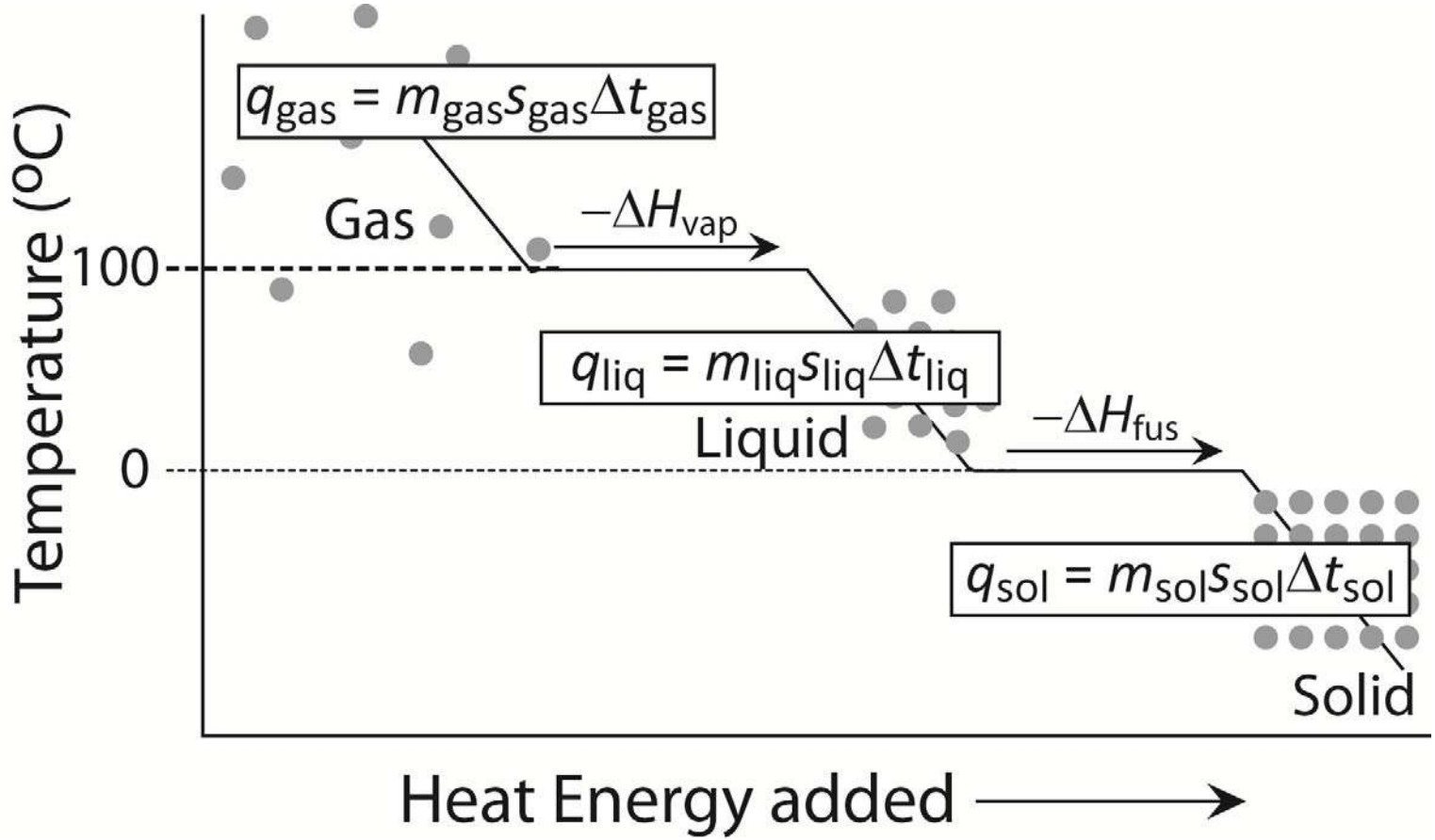



# 6.4 and 6.6 Enthalpy






# 6.4 and 6.6 Enthalpy






# 6.4 and 6.6 Enthalpy

Practice:

What was the initial temperature of water when

100 g of ice at -20°C was placed in a 1.5 L water bath and both substances reached a final temperature of 28.25°C?

The specific heat of ice is 2.06 $J \cdot g^{-1} \cdot {}^{\circ}C^{-1}$

The specific heat of liquid water is 4.18 $J \cdot g^{-1} \cdot {}^{\circ}C^{-1}$

The enthalpy of fusion for water is 6.0 $kJ \cdot mol^{-1}$



# 6.4 and 6.6 Enthalpy

- How do we measure change in enthalpy for a reaction?
  - Direct method
  - Indirect methods
    - Using a series of reactions (Hess's Law)
    - Using tables of data (enthalpy of formation)



**Table 6.4** Standard Enthalpies of Formation of Some Inorganic Substances at 25°C

| Substance | $\Delta H_f^\circ$ (kJ/mol) | Substance | $\Delta H_f^\circ$ (kJ/mol) |
|---|---|---|---|
| $Ag(s)$ | 0 | $H_2O_2(l)$ | −187.6 |
| $AgCl(s)$ | −127.04 | $Hg(l)$ | 0 |
| $Al(s)$ | 0 | $I_2(s)$ | 0 |
| $Al_2O_3(s)$ | −1669.8 | $HI(g)$ | 25.94 |
| $Br_2(l)$ | 0 | $Mg(s)$ | 0 |
| $HBr(g)$ | −36.2 | $MgO(s)$ | −601.8 |
| C(graphite) | 0 | $MgCO_3(s)$ | −1112.9 |
| C(diamond) | 1.90 | $N_2(g)$ | 0 |
| $CO(g)$ | −110.5 | $NH_3(g)$ | −46.3 |
| $CO_2(g)$ | −393.5 | $NO(g)$ | 90.4 |
| $Ca(s)$ | 0 | $NO_2(g)$ | 33.85 |
| $CaO(s)$ | −635.6 | $N_2O_4(g)$ | 9.66 |
| $CaCO_3(s)$ | −1206.9 | $N_2O(g)$ | 81.56 |
| $Cl_2(g)$ | 0 | $O(g)$ | 249.4 |
| $HCl(g)$ | −92.3 | $O_2(g)$ | 0 |
| $Cu(s)$ | 0 | $O_3(g)$ | 142.2 |
| $CuO(s)$ | −155.2 | S(rhombic) | 0 |
| $F_2(g)$ | 0 | S(monoclinic) | 0.30 |
| $HF(g)$ | −268.61 | $SO_2(g)$ | −296.1 |
| $H(g)$ | 218.2 | $SO_3(g)$ | −395.2 |
| $H_2(g)$ | 0 | $H_2S(g)$ | −20.15 |
| $H_2O(g)$ | −241.8 | $ZnO(s)$ | −347.98 |
| $H_2O(l)$ | −285.8 | ZnS(s) | −202.9 |

†In thermodynamics, the standard pressure is defined as 1 bar, where 1 bar = $10^5$ Pa = 0.987 atm. Because 1 bar differs from 1 atm by only 1.3 percent, we will continue to use 1 atm as the standard pressure. Note that the normal melting point and boiling point of a substance are defined in terms of 1 atm.

Table 6.4, p. 199

# 6.4 and 6.6 Enthalpy

Practice:

What is the enthalpy of combustion for methane?

What volume of methane at 1 atm and 25°C is needed to heat 500 mL of water from 5°C to 95°C?

If the current price of natural gas is $0.10 per cubic foot, (and we assume natural gas is all methane) how much does it cost to heat the water?



# Chapter 6 Enthalpy – Practice

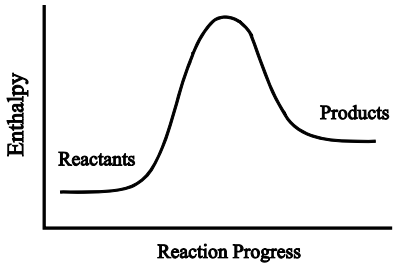
For the reaction energy diagram shown to the right, the reaction is

**A.** exothermic with $\Delta H_{reactants} > \Delta H_{products}$.

**B.** exothermic with $\Delta H_{reactants} < \Delta H_{products}$.

**C.** endothermic with $\Delta H_{reactants} > \Delta H_{products}$.

**D.** endothermic with $\Delta H_{reactants} < \Delta H_{products}$.




Which reaction corresponds to the reaction for the enthalpy of formation for methanol, $CH_3OH(l)$?

**A.** $2CH_3OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 4H_2O(l)$

**B.** $CH_3OH(l) + {}^3/_2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$

**C.** $2C(s, graphite) + O_2(g) + 4H_2(g) \rightarrow 2CH_3OH(l)$

**D.** $C(s, graphite) + ½\ O_2(g) + 2H_2(g) \rightarrow CH_3OH(l)$

# Chapter 6 Enthalpy – Practice

Hess's Law states when reactants are converted to products, the change in enthalpy is the same whether the reaction takes place in one step or in a series of steps. This is valid because change in enthalpy is

**A.** extensive.

**B.** a state function.

**C.** a type of energy.

**D.** equal to energy when no work is done.

**Energy Relations in Chemistry: Thermochemistry**

Enthalpies of Reaction

---

## Heat and Enthalpy Changes

When a chemical reaction occurs in an *open* container most of the energy gained or lost is in the form of ***heat***. Almost no ***work*** is done (i.e. nothing is being moved).

Heat flows between the system and surroundings until the two are at the same temperature.

- When a chemical reaction occurs in which the system absorbs heat, the process is ***endothermic*** (it feels cold)
- When a chemical reaction occurs in which the system produces heat it is ***exothermic*** (it feels hot)

### Enthalpy

Under conditions of *constant pressure* (e.g. most biological processes under constant atmospheric pressure) the heat absorbed or released is termed ***enthalpy*** (or "heat content").

We do not measure enthalpy *directly*, rather we are concerned about ***the heat added or lost*** by the system, which is the *change* in enthalpy (or ΔH).

In formal terms: *The change in enthalpy, ΔH, equals the heat, $q_p$, added to or lost by the system when the process occurs under constant pressure*:

$$\Delta H = q_p$$

ΔH represents the difference between the enthalpy of the system at the beginning of the reaction compared to what it is at the end of the reaction:

$$\Delta H = H_{final} - H_{initial}$$

We are considering the enthalpic state of the ***system***. Thus:

- if the ***system*** has higher enthalpy at the end of the reaction, then it ***absorbed*** heat from the surroundings (***endothermic*** reaction)

- if the system has a lower enthalpy at the end of the reaction, then it ***gave off heat*** during the reaction (***exothermic*** reaction)

Therefore:

- For ***endothermic*** reactions $H_{final} > H_{initial}$ ***and ΔH is positive (+ΔH)***
- For exothermic reactions $H_{final} < H_{initial}$ ***and ΔH is negative (-ΔH)***

---

## Enthalpies of Reaction

Because the enthalpy change for a reaction is described by the final and initial enthalpies:

$$\Delta H = H_{final} - H_{initial}$$

we can also describe ΔH for a reaction by comparing the enthalpies of the products and the reactants:

$$\Delta H = H(products) - H(reactants)$$

The enthalpy change that accompanies a reaction is called ***the enthalpy of reaction*** (***$\Delta H_{rxn}$***).

It is sometimes convenient to provide the value for $\Delta H_{rxn}$ along with the balanced chemical equation for a reaction (also known as a ***thermochemical equation***):

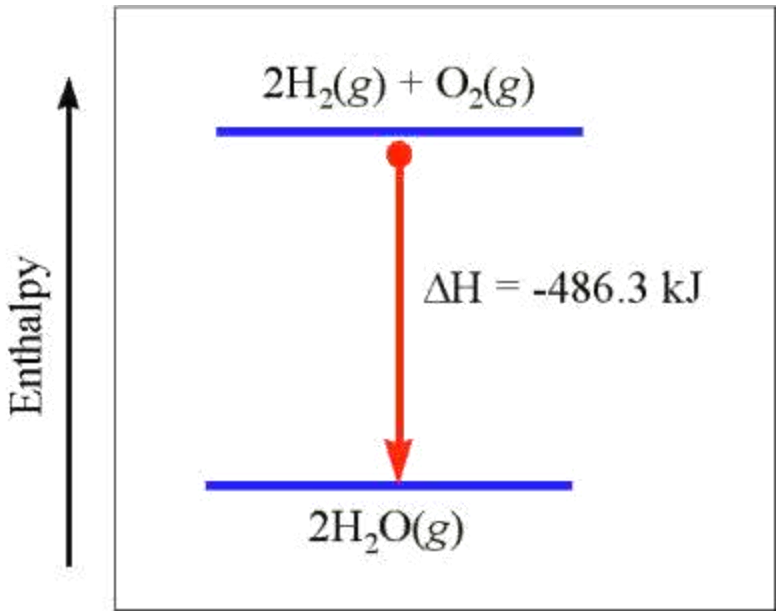
$$2H_2(g) + O_2(g) \rightarrow 2H_2O(g) \quad \Delta H = -483.6 \text{ kJ}$$

Note the following:

- ΔH is negative, indicating that this reaction results in the release of heat (exothermic)
- The reaction gives of 483.6 ***kilo Joules*** of energy when ***2 moles*** of $H_2$ combine with ***1 mole*** of $O_2$ to produce ***2 moles*** of $H_2O$.

The relative enthalpies of the reactants and products can also be shown on an ***energy diagram***:

**(spu) اتحاد سنافر البوليتكنك**




Properties of enthalpy:

1. Enthalpy is an ***extensive*** property. The magnitude of ΔH is dependent upon the amounts of reactants consumed. Doubling the reactants, doubles the amount of enthalpy.
2. Reversing a chemical reaction results in the same magnitude of enthalpy but of the ***opposite sign***. For example, splitting two moles of water to produce 2 moles of $H_2$ and 1 mole of $O_2$ gas requires the ***input*** of +483.6 kJ of energy.
3. The enthalpy change for a reaction depends upon the state of the reactants and products. The states (i.e. *g, l, s or aq*) must be specified.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) \quad \Delta H = -802 \text{ kJ}$$

Given the above thermochemical equation for the combustion of methane, how much heat energy is released when 4.5 grams of methane is burned (in a constant pressure system)?

$$(4.5gCH_4)*\left(\frac{1mole}{16.003g}\right) = 0.281moles$$

$$(0.281 moles CH_4) * \left( \frac{-802 kJ}{1 mole CH_4} \right) = -225.5 kJ$$

The negative sign (exothermic) indicates that 225.5 kJ of energy are given off by the system into the surroundings.

**Energy Relations in Chemistry: Thermochemistry**

Calorimetry

---

## Calorimetry

Experimentally, we can determine the heat flow ($\Delta H_{rxn}$) associated with a chemical reaction *by measuring the temperature change it produces*.

- The measurement of heat flow is called ***calorimetry***
- An apparatus that measures heat flow is called a ***calorimeter***

**Heat capacity and specific heat**

The temperature change experienced by an object when it absorbs a certain amount of energy is determined by its ***heat capacity***.

- The ***heat capacity*** of an object is defined as ***the amount of heat energy required to raise its temperature by 1 K*** (***or °C***)
- The greater the heat capacity of an object, the more heat energy is required to raise the temperature of the object

For pure substances the heat capacity is usually given for a specified amount of the substance

- The heat capacity of 1 mol of a substance is called its ***molar heat capacity***
- The heat capacity of 1 gram of a substance is called its ***specific heat***

The specific heat of a substance can be determined experimentally by measuring the temperature change (***ΔT***) that a known mass (***m***) of the substance undergoes when it gains or loses a specific quantity of heat (***q***):

$$\text{Specific heat} = \frac{q}{m * \Delta T}$$

209 J of energy are required to increase the temperature of 50.0 g of water by 1.00 K. What is the specific heat of water?

$$\text{Specific heat} = \frac{209\ \text{J}}{(50\ \text{g})*(1.00\ ^\circ\text{K})}$$

$$\text{Specific heat} = 4.18\ \text{J g}^{-1}{}^\circ\text{K}^{-1}$$

Specific heat values of some substances (J $g^{-1}K^{-1}$)

| | | | |
|---|---|---|---|
| Al (s) | 0.90 | $CaCO_3$ (s) | 0.85 |
| C (s) | 0.71 | $CCl_4$ (l) | 0.86 |
| Fe (s) | 0.45 | **$H_2O$ (l)** | **4.18** |
| Hg (l) | 0.14 | | |

We can calculate the quantity of heat that a substance has gained or lost by using its specific heat together with its measured mass and temperature change:

$$\text{Specific heat} = \frac{\text{quantity of heat transferred}}{(\text{grams of substance})*(\text{temperature change})}$$

rearrange:

$$\text{quantity of heat transferred} = (\text{Specific heat})*(\text{grams})*(\Delta T)$$

How much heat is required to raise the temperature of 250g of water from 22 °C to 98 °C? (specific heat of water is 4.18 J $g^{-1}$ $K^{-1}$).

$$q = (4.18 \text{ J } g^{-1} \text{ } K^{-1})*(250g)*(371\text{-}295 \text{ K})$$

$$q = \mathbf{79{,}420\ J}\ (79.420 \text{ kJ, or } 7.942 \times 10^4 \text{ J})$$

What is the molar heat capacity of water?

$$\text{Molar heat capacity} = 4.18 \text{ J } g^{-1} \text{ } K^{-1} * (18 \text{ grams}/1.0 \text{ mole})$$

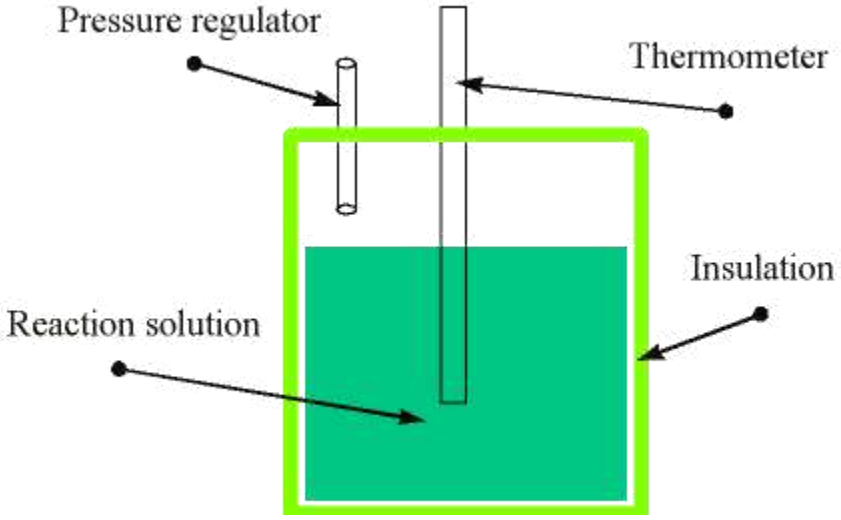
$$= 75.2 \text{ J } \text{mole}^{-1} \text{ } K^{-1}$$

---

**Constant-Pressure Calorimetry**

Recall that ΔH is defined as the quantity of heat transferred under constant pressure ($\Delta H = q_p$).

A calorimeter for such measurements would have the following general construction:




Note that the pressure regulator could be just a vent to allow the pressure to be maintained at atmospheric pressure

- *For reactions which involve dilute aqueous solutions, the specific heat of the solution will be approximately that of water (4.18 J $g^{-1}$ $K^{-1}$)*
- ***The heat absorbed by an aqueous solvent is equal to the heat given off by the reaction of the solutes:***

$$q_{aq\ solvent} = -q_{rxn}$$

Remember: If a reaction *gives off* heat, it is *exothermic* and ΔH is *negative*. The enthalpy of the products is less than the enthalpy of the reactants

$$\Delta H = H_{products} - H_{reactants}$$

In our calorimeter with an aqueous solution, if the reaction of the solutes is exothermic, the solution will absorb this heat and *increase* in temperature.

Thus, for an exothermic reaction:

- The *solutes* have a *lower final enthalpy* after the reaction (ΔH negative)
- The *solution* has a *higher final enthalpy* after the reaction (ΔH positive)

So, to determine the actual $\Delta H_{rxn}$ we would *invert the sign* of the $\Delta H_{soln}$ (the actual value we will measure)

---

50 ml of 1.0 M HCl and 50 ml of NaOH are combined in a constant pressure calorimeter. The temperature of the ***solution*** is observed to rise from 21.0 °C to 27.5 °C. Calculate the enthalpy change for the ***reaction*** (assume density is 1.0 gram/ml, and that the specific heat of the solution is that of water).

$\Delta T_{solution}$ = 27.5 - 21.0 °C = 6.5 °C (K)

specific heat = 4.18 J $g^{-1}$ $K^{-1}$

mass = (100 ml)*(1 gram/ml) = 100 grams

$\Delta H_{solution} = q_p$ = (4.18 J $g^{-1}$ $K^{-1}$)*(100 g)*(6.5 K) = 2,717 J

We know that $\Delta H_{rxn} = -\Delta H_{solution}$ therefore,

$$\Delta H_{rxn} = -\ 2{,}717\ J$$

Note: exothermic reactions have *negative* values for $\Delta H_{rxn}$ thus, if the reaction gave off heat (i.e. raised the temperature of the solution) you know that the sign for $\Delta H_{rxn}$ ***must be negative***.

**اتحاد سنافر البوليتكنك (spu)**

What is the enthalpy change on a molar basis?

$$HCl + NaOH \rightarrow NaCl + H_2O$$

This is the balanced equation, and we combined:

(0.05 liters HCl)*(1.0 mol/liter) = 0.05 moles HCl

and

(0.050 liters NaOH)*(1.0 mol/liter) = 0.05 moles NaOH

The stoichiometry for HCl and NaOH in this reaction is 1:1, so they are combining in stoichiometrically equivalent amounts and will produce 0.05 moles of NaCl (and $H_2O$).

So, the enthalpy change for the production of 0.05 moles of NaCl in the above reaction would be:

-2717 J for each 0.05 moles NaCl, or

-2717 J/.05 moles = 54,340 J/mole = 54.34 kJ/mole

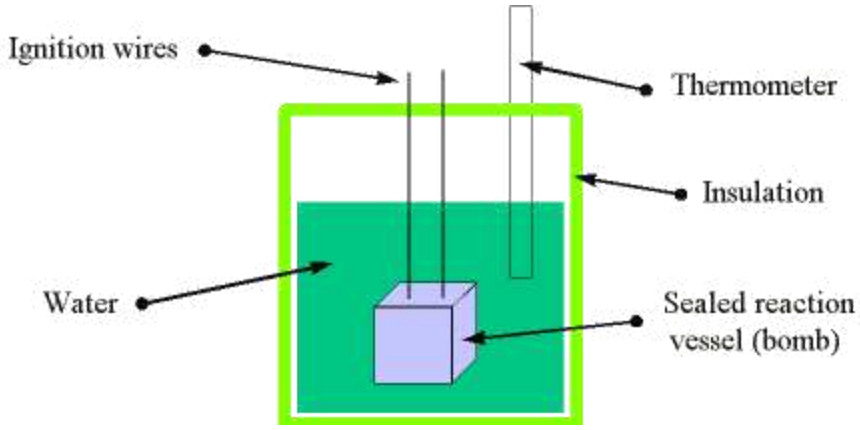
---

**Bomb Calorimetry (constant-volume calorimetry)**

Since combustion reactions involve dramatic increases in pressure, they are typically studied under conditions of ***constant volume*** in a device known as a **bomb calorimeter**.

The bomb calorimeter is essentially a ***sealed*** insulated instrument with no pressure regulation.

The sealed reaction chamber is surrounded by water, and the energy released, or absorbed, by the sample is measured indirectly by monitoring the temperature change of the water.




Analyses under constant volume allow determination of ΔE (energy change at constant volume), not ΔH (energy change at constant pressure).

**Energy Relations in Chemistry: Thermochemistry**

Hess's Law

---

**Hess's Law**

It is often possible to **calculate** ΔH for a reaction from listed ΔH values of **other reactions** (i.e. you can avoid having to do an experiment)

Enthalpy is a ***state function***

- It depends only upon the ***initial and final state*** of the reactants/products and ***not on the specific pathway*** taken to get from the reactants to the products
- Whether one can arrive at the products via either a single step or multi-step mechanism is unimportant as far as the enthalpy of reaction is concerned - they should be equal

Consider the combustion reaction of methane to form $CO_2$ and ***liquid*** $H_2O$

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$$

This reaction can be thought of as occurring in two steps:

- In the first step methane is combusted to produce water vapor:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$$

- In the second step water vapor condenses from the gas phase to the liquid phase:

$$2H_2O(g) \rightarrow 2H_2O(l)$$

Each of these reactions is associated with a specific enthalpy change:

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) \quad \Delta H = -802 \text{ kJ}$$

$$2H_2O(g) \rightarrow 2H_2O(l) \quad \Delta H = -88 \text{ Kj}$$

*(Note: under conditions of standard temperature and pressure the liquid state of water is the normal state. Thus, the gas would be expected to condense. This is an exothermic process under these conditions. In a related process, it should get warmer when it rains)*

Combining these equations yields the following:

$$CH_4(g)+2O_2(g)+\cancel{2H_2O(g)} \rightarrow CO_2(g)+\cancel{2H_2O(g)}+2H_2O(l)$$

$$\Delta H = (-802) \text{ kJ} + (-88) \text{ kJ} = -890 \text{ kJ}$$

**Hess's Law**

***if a reaction is carried out in a series of steps, ΔH for the reaction will be equal to the sum of the enthalpy changes for the individual steps***

*the overall enthalpy change for the process is **independent** of the **number of steps** or the particular nature of the **path** by which the reaction is carried out.*

*Thus we can use information tabulated for a relatively small number of reactions to calculate ΔH for a large number of different reactions*

---

Determining ΔH for the reaction

$$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g)$$

is difficult because some $CO_2$ is also typically produced.

However, complete oxidation of either C or CO to yield $CO_2$ is experimentally pretty easy to do:

$$C(s) + O_2(g) \rightarrow CO_2(g) \quad \Delta H = -393.5\ kJ$$

$$CO(g) + \frac{1}{2}O_2(g) \rightarrow CO_2(g) \quad \Delta H = -283.0\ kJ$$

We can invert reaction number 2 (making it ***endothermic***) and have CO(g) as a ***product.*** (This describes the decomposition of $CO_2$ to produce CO and $O_2$)

$$CO_2(g) \rightarrow CO(g) + \frac{1}{2}O_2(g) \quad \Delta H = +283.0\ kJ$$

Thus, we now have two equations with known enthalpies of reaction: the first describes the combustion of carbon and oxygen to produce $CO_2$ and the second describes how $CO_2$ can be decomposed to produce carbon monoxide (and oxygen). We can combine these together to describe the production of carbon monoxide from the combustion of carbon and oxygen:

$$C(s) + O_2(g) \xrightarrow{\Delta H = -393.5\ kJ} CO_2(g) \xrightarrow{\Delta H = +283.0\ kJ} CO(g) + \frac{1}{2}O_2(g)$$

The overall reaction, going from left-hand side reactant(s) to the right-hand side product(s) would be:

$$C(s) + O_2(g) \longrightarrow CO(g) + \frac{1}{2}O_2(g)$$

We can algebraically subtract the one-half $O_2$ from both sides to yield the following equation with the associated overall enthalpy:

$$C(s) + \frac{1}{2}O_2(g) \longrightarrow CO(g) \quad \Delta H = -110.5\ kJ$$

(spu) اعداد صفاء البوليني

Another way to look at the method of combining reactions would be as follows:

$$C(s) + O_2(g) \rightarrow CO_2(g) \quad \Delta H = -393.5\ kJ$$

plus

$$CO_2(g) \rightarrow CO(g) + \frac{1}{2}O_2(g) \quad \Delta H = +283.0\ kJ$$

gives:

$$C(s) + O_2(g) + CO_2(g) \rightarrow CO_2(g) + CO(g) + \frac{1}{2}O_2(g)$$

$$\Delta H = (-393.5\ kJ) + (283.0\ kJ) = -110.5\ kJ$$

canceling out identical compounds from the left and right hand sides of this reaction gives

$$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g) \quad \text{and } \Delta H = -110.5\ kJ$$

---

Carbon occurs in two forms: graphite and diamond. The enthalpy of combustion of graphite is -393.5 kJ, and that of diamond is -395.4 kJ

$$C(graphite) + O_2(g) \rightarrow CO_2(g) \quad \Delta H = -393.5\ kJ$$

$$C(diamond) + O_2(g) \rightarrow CO_2(g) \quad \Delta H = -395.4\ kJ$$

Calculate ΔH for the conversion of graphite to diamond

What we want is ΔH for the reaction:

$$C(graphite) \rightarrow C(diamond)$$

اتحاد سنافر البوليتكنك (spu)

$$C(graphite) + O_2(g) \rightarrow CO_2(g) \quad \Delta H = -393.5\ kJ$$

~~$CO_2(g)$~~ -> C(diamond) + ~~$O_2(g)$~~ $\Delta H = +395.4$ kJ

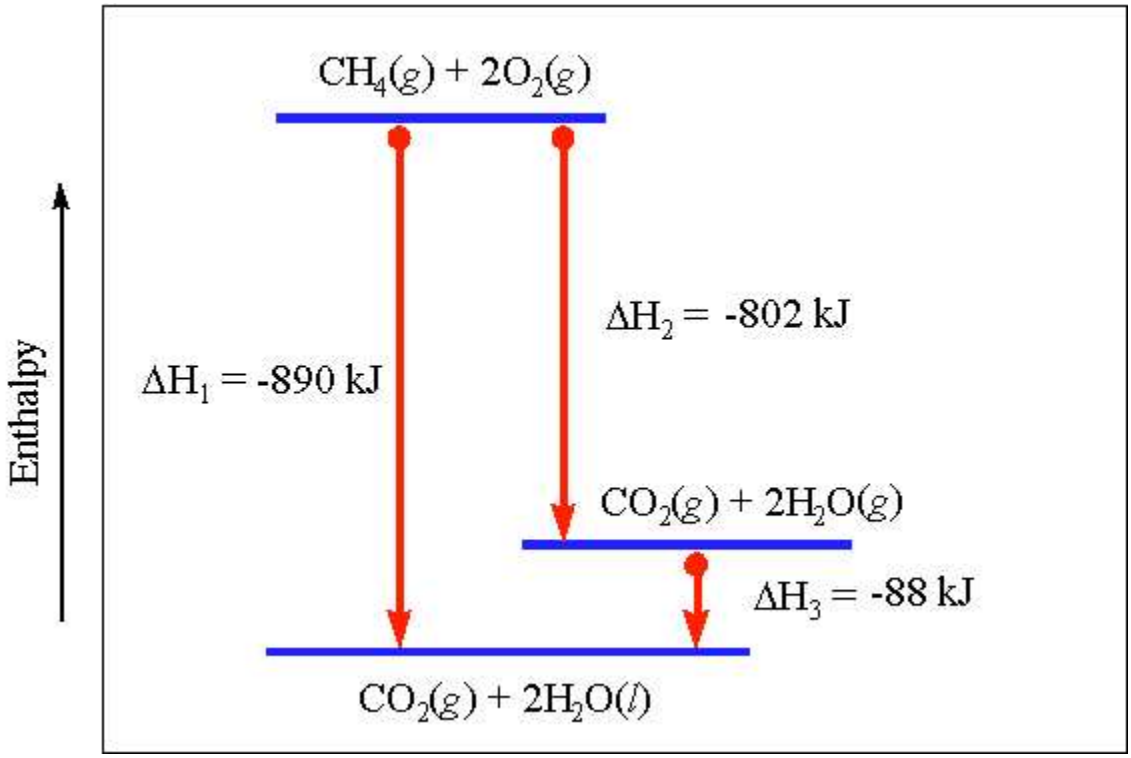
C(graphite) -> C(diamond) $\Delta H = +1.9$ kJ

---

We can never expect to obtain more or less energy from a chemical reaction by changing the method of carrying out the reaction ("conservation of energy").

Another way of saying this is that the particular pathway chosen to arrive at the same reactants yields the same $\Delta H$ for the overall reaction.

Consider the previous example of the combustion of methane to produce gaseous $H_2O$ and then the condensation of the gaseous $H_2O$ to the liquid state. How is this represented in an **Enthalpy Diagram**?




The key features are the following:

1. Each line represents a set of reactants or products for a **balanced** chemical reaction. When going from one line to another, the atoms must balance. For example, if we were to ask what is the enthalpy associated with the condensation of water we would have (from the above data):

(spu) اتحاد سنافر البولتيكنك

$$CO_2(g) + 2H_2O(g) \rightarrow CO_2(g) + 2H_2O(l) \quad \Delta H = -88 \text{ kJ}$$

The $CO_2(g)$ on both sides will cancel to yield:

$$2H_2O(g) \rightarrow 2H_2O(l) \quad \Delta H = -88 \text{ kJ}$$

2. The relative distance of each line must reflect the relative enthalpy difference (ΔH) between the reactants/products. If the enthalpy change in going from reactants to products is negative, then the line for the products must be below the reactants. Furthermore, the length of the distance must be proportional. For example, the distance reflecting the enthalpy associated with the condensation of water (ΔH = -88 kJ)is only about 10% as long as the distance between the reactants and products for the combustion of methane to $CO_2$ and liquid water (ΔH = -890 kJ)

**Energy Relations in Chemistry: Thermochemistry**

Enthalpies of Formation

---

## Enthalpies of Formation

Using Hess's Law we can calculate reaction enthalpies for a variety of reactions using tables of known enthalpies

Many experimentally determined enthalpies are listed by ***the type of process***

- ΔH for converting various *liquids to the gas phase* are listed in tables of ***enthalpies of vaporization***
- ΔH for *melting solids to liquids* are listed in tables of ***enthalpies of fusion***
- ΔH for *for combusting a substance in oxygen* are listed in tables of ***enthalpies of combustion***

The enthalpy change associated with the formation of a compound from its constituent elements is called the ***enthalpy of formation ($\Delta H_f$)***

Conditions which influence enthalpy changes include:

**(spu) اتحاد سنافر البوليتكنك**

- temperature

- pressure
- state of reactants and products (*s, g, l, aq*)

The ***standard state*** of a substance is the form most stable at 298 °K (25 °C, or standard "room temperature") and 1 atmosphere (1 atm) of pressure

When a reaction occurs with all reactants and products in their standard states, the enthalpy change is ***the standard enthalpy of reaction (ΔH°)***

Thus, *the* ***standard enthalpy of formation*** ***($\Delta H^\circ_f$)****of a compound is the change in enthalpy that accompanies the* ***formation of 1 mole*** *of that substance from its elements, with all substances in their* ***standard states***

The standard enthalpy of formation for ethanol ($C_2H_5OH$) is the enthalpy change for the following reaction

$$2C(graphite) + 3H_2(g) + \frac{1}{2}O_2(g) \rightarrow C_2H_5OH(l) \quad \Delta H^\circ_f = -277.7\ \text{kJ}$$

Notes:

- Elemental source of oxygen is $O_2$ and not O because $O_2$ is the stable form of oxygen at 25 °C and 1 atm, likewise with $H_2$
- Elemental source of carbon is specified as *graphite* (and not, for example, diamond) because graphite is the lowest energy form of carbon at room temp and 1 atm
- Why is the $O_2$ stoichiometry left at "1/2"? The stoichiometry of formation reactions *always indicates the formation of 1 mol of product.* Thus, $\Delta H^\circ_f$ values are reported as kJ / mole of the substance produced

If C(*graphite*) is the lowest energy form of carbon under standard conditions, then what is the $\Delta H^\circ_f$ for C(*graphite*)?

- By definition, *the standard enthalpy of formation of the most stable form of any* ***element*** *is zero* because there is no formation reaction needed when the element is *already* in its standard state
- $\Delta H^\circ_f$ for C(*graphite*), $H_2(g)$ and $O_2(g)$ = 0

**Using enthalpies of formation ($\Delta H^\circ_f$) to calculate enthalpies of reaction under standard conditions ($\Delta H^\circ_{rxn}$)** (spu) اتحاد سنافر البوليتكنك

*We can determine the standard enthalpy change for any reaction ($\Delta H°_{rxn}$) by using standard enthalpies of formation ($\Delta H°_f$) and Hess's Law*

Consider the following combustion reaction of propane:

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$$

**The reactants**:

- The standard heat of formation ($\Delta H°_f$) of propane gas from its elemental constituents in the standard state is 103.85 kJ/mole

$$3C(graphite) + 4H_2(g) \rightarrow C_3H_8(g)\ \Delta H°_f = -103.85\ kJ$$

- The standard heat of formation ($\Delta H°_f$) for $O_2(g)$ is zero

$$O_2(g) \rightarrow O_2(g) \square \Delta H°_f = 0\ kJ$$

and so...

$$5O_2(g) \rightarrow 5O_2(g) \square \Delta H°_f = 0\ kJ$$

Overall, therefore, the standard heat of formation ($\Delta H°_f$) for the reactants is:

$$3C(graphite)+4H_2(g)+5O_2(g) \rightarrow C_3H_8(g)+5O_2(g)$$

$$\Delta H°_f = -103.85\ kJ$$

**The products:**

- The standard heat of formation ($\Delta H°_f$) of $CO_2(g)$ from its elemental constituents in the standard state is 393.5 kJ/mole

$$C(graphite) + O_2(g) \rightarrow CO_2(g)\ \Delta H°_f = 393.5\ kJ$$

so, for **3** moles of $CO_2$ molecules the standard heat of formation would be:

$$3C(graphite) + 3O_2(g) \rightarrow 3CO_2(g)\ \Delta H°_f = 1180.5\ kJ$$

- The standard heat of formation ($\Delta H°_f$) of $H_2O(l)$ from its elemental constituents in the standard state is 285.8 kJ/mole

(spu) اتحاد سنافر البولتيكنك

$$H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l) \quad \Delta H°_f = -285.8\ kJ$$

and so the $\Delta H°_f$ for 4 waters would be:

$$4H_2(g) + 2O_2(g) \rightarrow 4H_2O(l) \quad \Delta H°_f = -1143.2\ kJ$$

combining the $\Delta H°_f$ for both products yields:

$$3C(graphite)+4H_2(g)+5O_2(g) \rightarrow 3CO_2(g)+4H_2O(l)$$

with $\Delta H°_f = (-1180.5) + (-1143.2) = -2323.7$ kJ

Let's summarize what we have determined so far:

Overall the standard heat of formation ($\Delta H°_f$) for **the reactants** is:

$$3C(graphite)+4H_2(g)+5O_2(g) \rightarrow C_3H_8(g)+5O_2(g)$$

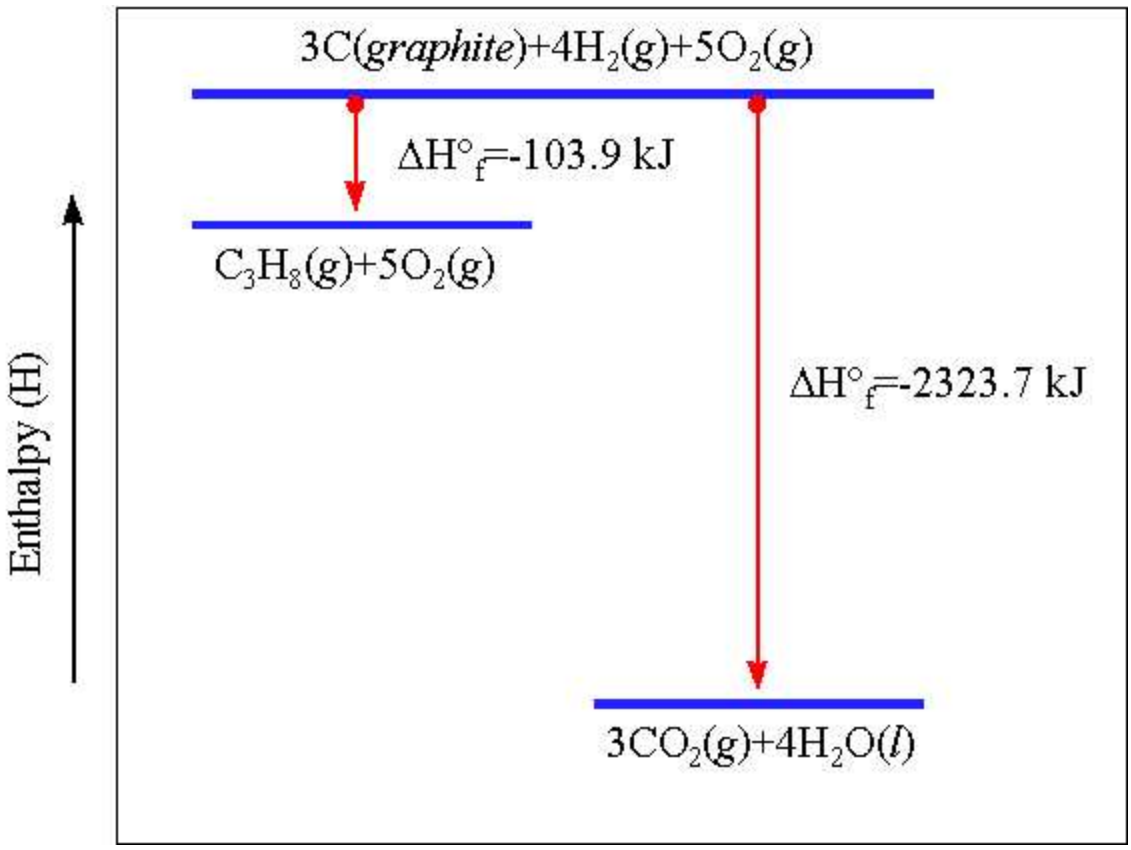
$$\Delta H°_f = -103.9\ kJ$$

Overall the standard heat of formation ($\Delta H°_f$) for **the products** is:

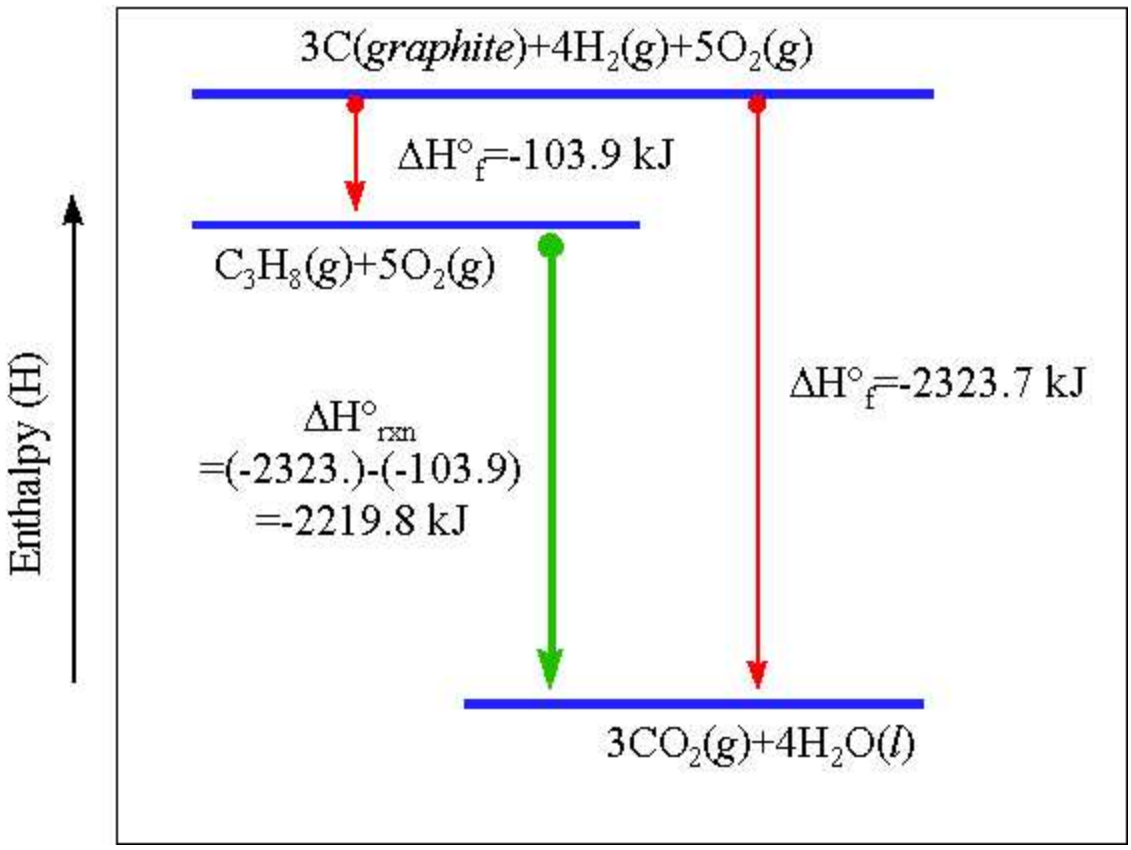
$$3C(graphite)+4H_2(g)+5O_2(g) \rightarrow 3CO_2(g)+4H_2O(l)$$

$$\Delta H°_f = -2323.7\ kJ$$

Lets plot these on a relative scale of enthalpy:




This information can be used to determine the relative enthalpy difference under standard conditions (ΔH°) between the reactants and products:




This enthalpy difference (-2219.8 kJ) is the enthalpy of the reaction for the combustion of propane under standard conditions ($\Delta H°_{rxn}$)

Calculate the enthalpy change ($\Delta H°_{rxn}$) for the combustion of 1 mol of ethanol

$$C_2H_5OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$$

**heat of formation for reactants**

$$2C(graphite)+3H_2(g)+(1/2)O_2(g) \rightarrow C_2H_5OH(l)$$

$$\Delta H°_f = -277.7 \text{ kJ}$$

plus

$$3O_2(g) \rightarrow 3O_2(g) \quad \Delta H°_f = 0 \text{ kJ}$$

gives:

اتحاد سنافر البوليتكنك (spu)

$$2C(graphite)+3H_2(g)+(7/2)O_2(g) \rightarrow C_2H_5OH(l)+ 3O_2(g)$$

$$\Delta H^\circ_f = -277.7 \text{ kJ}$$

**heat of formation for products**

$$C(graphite)+O_2(g) \rightarrow CO_2(g) \quad \Delta H^\circ_f = -393.5 \text{ kJ}$$

therefore

$$2C(graphite)+2O_2(g) \rightarrow 2CO_2(g) \quad \Delta H^\circ_f = -787 \text{ kJ}$$

$$H_2(g) + (1/2)O_2(g) \rightarrow H_2O(l) \quad \Delta H^\circ_f = -285.8 \text{ kJ}$$

therefore

$$3H_2(g) + (3/2)O_2(g) \rightarrow 3H_2O(l) \quad \Delta H^\circ_f = -857.4 \text{ kJ}$$

combining gives:

$$2C(graphite)+3H_2(g)+(7/2)O_2(g) \rightarrow 2CO_2(g)+ 3H_2O(l)$$

$$\Delta H^\circ_f = (-857.4)+(-787) = -1644.4\text{kJ}$$

**$\Delta H^\circ_{rxn}$**

$$\Delta H^\circ_{rxn} = \Delta H^\circ_f \text{ (products)} - \Delta H^\circ_f \text{ (reactants)}$$

$$(-1644.4) - (-277.7) = \mathbf{-1366.7\ kJ}$$

# THERMOCHEMISTRY

- Thermochemistry is the study of the quantity of heat energy released or absorbed in a chemical reaction.
  Example: the burning of fuel: is a heat-evolving reaction

- Heat : is a form of energy

- Energy: -the potential to do work (to move matter)
  -exists in many different forms:
  - Electrical energy
  - Kinetic Energy (energy of motion)
  - Light energy
  - Heat energy
  - Chemical energy (energy of substances)

- Different forms of energy can be interconverted
  Examples: Light bulb: electrical energy is converted to light energy and heat energy

  Car engine: chemical energy (from gasoline) → heat energy → kinetic energy

## KINETIC ENERGY (Ek)

- Definition: is the energy associated with a moving object

- Formula: $Ek = \frac{1}{2} mv^2$ , where m = mass of the moving object (in kg)
  v = speed of the moving object (in m/s)

**NOTE**: Two objects moving at different speeds may have the same kinetic energy
Example: slow moving truck and fast moving sports' car

**SI unit of energy** = from formula of energy = **$kg \cdot m^2/s^2$** = **joule (J)**
The **joule (J)** is a very a very small unit

Non-SI unit of heat energy : **The calorie (cal)**
**amount of energy required to raise the temperature of one gram of water by one degree Celsius**

**1 cal = 4.184 J**

**اتحاد سنافر البوليتكنك (spu)**

## POTENTIAL ENERGY (Ep)

- Definition: - the energy of an object has because of its position in a field of force
  - is stored energy

- Example: Any object above ground level possesses Potential Energy since it is attracted by the gravitational force. As such, it has the potential to fall, and therefore to move. (Potential Energy changes into Kinetic Energy)

- Formula: **Ep = mgh,** where m = mass of the object
  g = constant acceleration of gravity
  h = height (level) of the object aboveground level

**NOTE**: differences of level are usually more important

As an object falls toward ground level:

- its Ep decreases (h decreases)
- its Ek increases ( object speeds up; v increases)

## INTERNAL ENERGY (U)

- Definition: the sum of the kinetic and potential energies of the particles making up a substance.

In general: Total Energy of a Substance = **U + Ek + Ep**

**NOTE**: A substance at rest, in a flask in the laboratory

- possesses no kinetic energy
- potential energy is ignored
- **possesses Internal Energy**

**Total Energy of the Substance = Internal Energy of the Substance = U**

Conversions of different forms of energy are governed by :

**The Law of Conservation of Energy:**

- **Energy may be converted from one form to another, but the total quantity of energy remains constant.**

**(spu) اتحاد سنافر البوليتكنك**

# HEAT OF REACTION

- Thermodynamics: is the science of the relationships between heat energy and other forms of energy

  Ek , Ep , U :are thermodynamic properties of matter

- Thermodynamic System: the substance or mixture of substance undergoing a physical or a chemical change

- The Surroundings: everything around the thermodynamic system

**Definition of Heat:**

- **The energy that flows into, or out of a system because of a difference in temperature between the thermodynamic system and its surroundings.**

This assumes that the system and its surroundings are in thermal contact, (that is, they are not thermally insulated).

Example: hot coffee in a thermos is thermally insulated from its surroundings

**Direction of Heat Flow :**

- **Heat always flows from a region of higher temperature to a region of lower temperature until the temperatures of the two regions become equal (thermal equilibrium).**

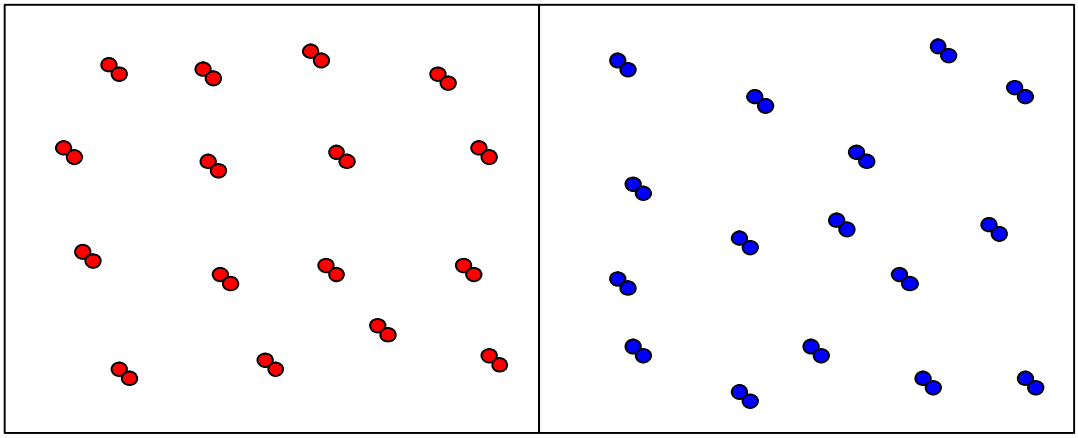
1. If heat energy flows into a system from its surroundings, its internal energy increases:

   Example: A glass of ice water slowly reaches room temperature.
   The internal energy (U) of the water increases, as it warms up.

2. If heat energy flows out from a system into the surroundings, its internal energy decreases

   Example: A cup of hot tea slowly reaches room temperature
   The internal energy (U) of the tea decreases, as it cools down.

## EXPLANATION OF HEAT FLOW



**HOTTER GAS**
**FASTER MOVING MOLECULES**

**COLDER GAS**
**SLOWER MOVING MOLECULES**

**Average Molecular Speed of Hotter Gas** > **Average Molecular Speed of Colder Gas**

**Faster moving molecules slow down**

**Slower moving molecules speed up**

- The Average Molecular Speed in the two containers becomes equal.
- The temperatures of the two gases become equal.
- **A given quantity of heat will raise the temperature of a gas more if the sample is small.**

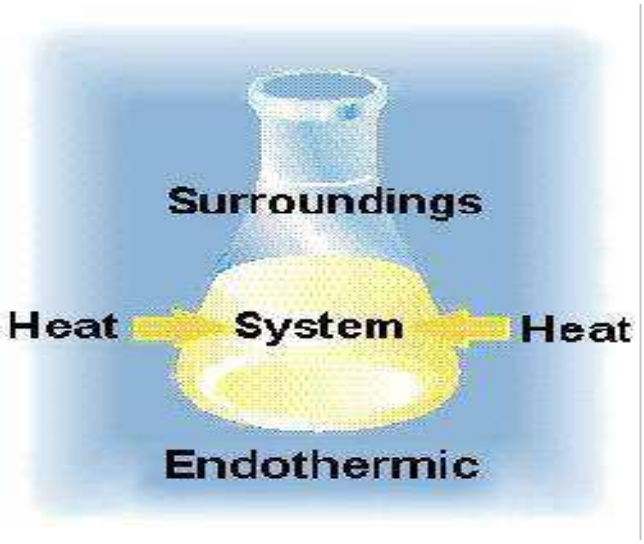
  Reason: The Absolute Temperature is directly proportional to the Average Kinetic Energy

  - When you add heat to a gas, you increase its total Ek
  - The increase in $E_k$ will be distributed over all the molecules in the sample
  - The increase in average Ek per individual molecule (the increase in temperature) will be greater if there are fewer molecules.

**(spu) اتحاد سنافر البوليتكنك**

## SYMBOL FOR HEAT

- The symbol for heat is q.
- **In a chemical equation, q is always considered a reactant**




**heat is absorbed by the system from surroundings**

**Reactants + q ⟶ Products**




**heat is evolved by the system to the surroundings**

**Reactants ⟶ Products + q**

**By convention, in a chemical equation q (the Heat term) is always considered a reactant.**

**Reactants + q ⟶ Products**
**q is positive**

- Temperature of the surroundings decreases.
- **Reaction vessel cools**

**Reactants - q ⟶ Products**
**q is negative**

- Temperature of the surroundings increases.
- **Reaction vessel warms**

**Example:**
Ammonia burns in the presence of a platinum catalyst to produce nitrogen monoxide.

$$4\,NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(l)$$

In an experiment, 4 mol of $NH_3$ is burned and evolves 1170 kJ of heat. Is the reaction endothermic or exothermic? What is the value of q?

# ENTHALPY AND ENTHALPY CHANGE

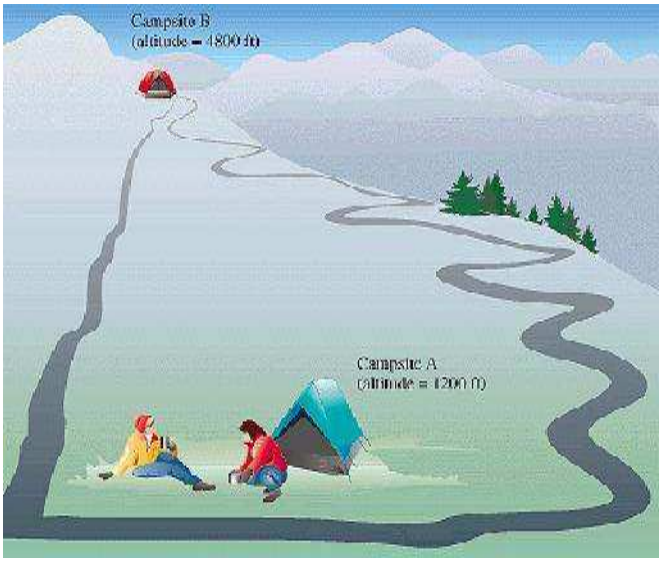
- **Enthalpy is an extensive property of a substance that can be used to obtain the heat absorbed or evolved in a chemical reaction.**

- (extensive property = property that depends on the amount of substance)

**Symbol for Enthalpy: H**

**Enthalpy (H) is a State Function:**

- **A State Function is a property of a system that depends only on its present state and independent of any previous history of the system**




Ex: Hiking in the mountains
You may take several alternative routes to reach certain peak

**altitude** reached
**is** analogous to **a state function**
**(difference in altitude = 3600 ft)**

**distance** traveled
**is not** analogous to **a state function**

# ENTHALPY OF REACTION

Consider a chemical reaction: **REACTANTS** ⟶ PRODUCTS

At first: **Reactants only** No Products

The Enthalpy of the system = The Enthalpy of the Reactants = **H (reactants)**

At the end: REACTANTS ⟶ **PRODUCTS**

No Reactants **Products only**

The Enthalpy of the system = The Enthalpy of the Products = **H (products)**

**The change in enthalpy = ΔH = Enthalpy of Reaction = H(products) - H(reactants)**

The value of ΔH: - is independent of the details of the reaction
- depends only on the initial state (reactants) and the final state (products)

Recall that H is a state function!

- THE ENTHALPY OF REACTION EQUALS THE HEAT OF REACTION AT CONSTANT PRESSURE:

$$\Delta H = q_p$$

where: ΔH = Enthalpy of reaction
$q_p$ = Heat of reaction at constant pressure

# THERMOCHEMICAL EQUATIONS

- In a thermochemical equation, the enthalpy of a reaction is indicated after the equation and is associated with the molar coefficients of the equation.

**Example:**

$2H_2(g) + O_2(g) \longrightarrow 2H_2O\ (g)$ **$\Delta H = -483.7$ kJ**

2 mol of $H_2$ gas reacts with 1 mole $O_2$ to produce 2 mol of **water vapor** and **483.7 kJ evolves**

**NOTE: Indicating phase labels in thermochemical equations is very important**

Consider:

$2H_2(g) + O_2(g) \longrightarrow 2H_2O\ (l)$ **$\Delta H = -571.7$ kJ**

This equation says: 2 mol of hydrogen gas reacts with one mole of oxygen gas to produce 2 mole of **liquid** water, and **571.7 kJ** of heat evolves.
(additional heat is released when water vapor condenses to liquid)

## MANIPULATING THERMOCHEMICAL EQUATIONS

1. Changing the coefficients of the balanced chemical equation, changes the **value** of **ΔH** accordingly

**Example:**
Liquid carbon disulfide burns in air, producing carbon dioxide gas and sulfur dioxide gas.

$$1\ CS_2(l) + 3\ O_2(g) \longrightarrow 1\ CO_2(g) + 2SO_2(g) \qquad \Delta H = -1075\ kJ$$

What is ΔH for the following equation ?

$$\tfrac{1}{2}\ CS_2(l) + 3/2\ O_2(g) \longrightarrow \tfrac{1}{2}\ CO_2(g) + 1\ SO_2(g) \qquad \Delta H = ?$$

**Note: All the coefficients are halved**

Therefore: $$\Delta H = -\frac{1075\ kJ}{2} = -537.5\ kJ$$

**RULE 1**

- **When a thermochemical equation is multiplied (or divided) by any factor, the value of ΔH for the new equation is obtained by multiplying(or dividing) the value of ΔH in the original equation by that same factor.**

2. **Reversing the thermochemical equation, changes the sign of ΔH accordingly**

**Example:**

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(l) \qquad \Delta H = -571.7\ kJ\ (exothermic)$$

$$2H_2O(l) \longrightarrow 2H_2(g) + O_2(g) \qquad \Delta H = +571.7\ kJ\ (endothermic)$$

**RULE 2**

- **When a thermochemical equation is reversed, the value of ΔH is reversed in sign**

**Examples:**

1. With a platinum catalyst, ammonia gas will burn in oxygen gas to give gaseous nitrogen monoxide and water vapor:




What is the enthalpy change for the following reaction?




**Step 1: First reverse the given equation:**

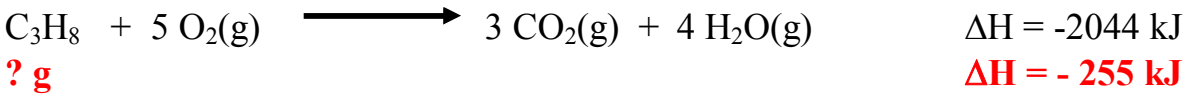



**Step 2: Divide the coefficients and ΔH by 4:**




**APPLYING STOICHIOMETRY TO HEATS OF REACTION**

2. Propane, $C_3H_8$, is a common fuel gas. Use the following to calculate the **grams of propane** you would need to provide **255 kJ of heat.**




$$\text{? g } C_3H_8 = 255 \text{ kJ} \times \frac{1 \text{ mol } C_3H_8}{2044 \text{ kJ}} \times \frac{44.06 \text{ g}}{1 \text{ mol } C_3H_8} = 5.50 \text{ g}$$

**اتحاد سنافر البوليتكنك (spu)**

# HEAT CAPACITY AND SPECIFIC HEAT

## HEAT CAPACITY , C

- The quantity of heat needed to raise the temperature of a sample of a substance one degree Celsius (or one degree Kelvin)

Recall : $K = {}^0C + 273$

NOTE: $\Delta K = \Delta {}^0C$

(Change in temperature in Kelvin) = Change in temperature in $^0C$)

**$q = C\,\Delta t$** where **q** = quantity of heat required to produce the temperature change

**$\Delta t = t_f - t_i$** = change in temperature from an initial temp., $t_i$, to a final temp., $t_f$

**C** = Heat capacity

- **HEAT CAPACITY(C) is directly proportional to the mass of substance**

- **MOLAR HEAT CAPACITY** = Heat capacity for one mole of substance

- **SPECIFIC HEAT CAPACITY or SPECIFIC HEAT, s**
  Quantity of heat required to raise the temperature of one gram of substance by one degree Celsius (or one Kelvin) at constant pressure

**$q = m \times s \times \Delta t$** where m = mass
s = specific heat
$\Delta t$ = temperature difference

**Specific Heat** and **Molar Heat Capacity**:

- are different for different substances
- are **physical constants, characteristic for a specific substance**
- are slightly temperature dependent (usually given at $25^0$ C)
- are available in Tables in textbooks

Specific Heat of water = $1.00 \frac{cal}{g \times {}^0C} = 4.18 \frac{J}{g \times {}^0C}$

**(spu) تعادل سنافر البوليتكنك**

**Examples:**

1. How much heat (in joules) must be used to raise the temperature of 185 g of water from 15 $^0$C to 96 $^0$C ?

$m = 185$ g
$t_i = 15$ °C
$t_f = 96$ °C
$\Delta t = 96$ °C - 15 °C = 81 °C

$$s = 4.18 \ \frac{J}{g \times °C}$$

**q = ?**

**q = m x s x Δt**

$$q = (185 \text{ g})\left(4.18 \ \frac{J}{g \times °C}\right)(81°C)$$

$q = 6.3 \times 10^4$ J = **63 kJ**

2. A sample of aluminum with a mass of 40.0 g requires 726 J of heat to raise its temperature from 30.0°C to 50.0 °C. What is the specific heat of aluminum?

3. What is the heat capacity of 25.0 g of ethyl alcohol if its specific heat is 2.42 J/g°C?

4. Molar heat capacity of iron is 25.1 J/mol°C. What is the specific heat of iron?

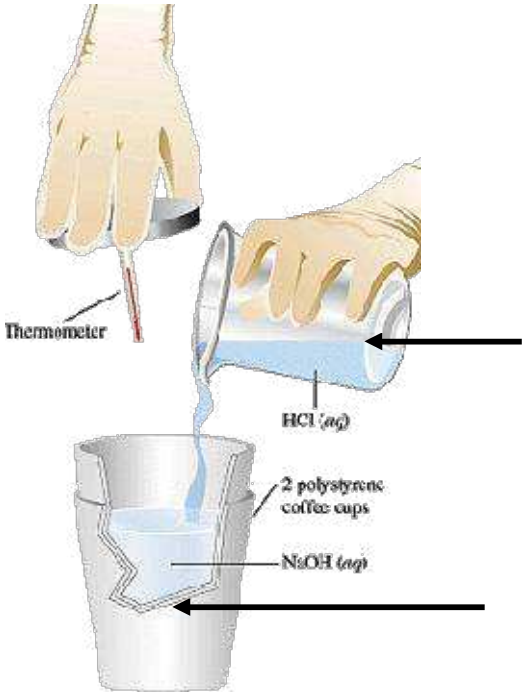
# CALORIMETRY

- The experimental method for measurement of heat is called **calorimetry**.
- A calorimeter is a device used to measure the heat absorbed or evolved during a physical or chemical change.
- A calorimeter is an insulated container (styrofoam coffee cup or a Snack-Jar equipped with a thermometer.
- It is used to measure temperature changes which are then related to changes in heat energy.

**Sample Problem 1:**
50.0 mL each of 1.0 M HCl and 1.0 M NaOH, at room temperature (20.0 $^{O}$ C) are mixed. The temperature of the resulting NaCl solution increases to 27.5 $^{O}$ C.

The density of the resulting NaCl solution is 1.02 g/mL.
The specific heat of the resulting NaCl solution is 4.06 J/g $^{0}$ C

Calculate the **Heat of Neutralization of HCl(aq) and NaOH(aq) in KJ/mol NaCl produced**




**50.0 mL HCl, 1.0 M (20.0$^{0}$C)**

**50.0 mL NaOH, 1.0 M (20.0$^{0}$C)**

**$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$ + Heat of Neutralization (ΔH)**

Heat given off by the Neutralization Rxn = Heat gained by the NaCl solution

**$HCl(aq) + NaOH(aq) \longrightarrow NaCl(aq) + H_2O(l)$** **ΔH = ???**

**ΔH = exothermic (temperature of solution increased)**

mass of solution: V x d = (100.0 mL) x (1.02 g/mL) = 102 g
m = 102 g
Specific heat = 4.06 J/g °C
Δt = 7.5 °C

q = **– m x s x Δt**

ΔH = – (102 g) (4.06 J/g °C) (7.5 °C) = –3100 J = **–3.1 kJ (exothermic reaction)**

$$\frac{\text{–3.1 kJ}}{\text{mol of NaCl produced}} = ?$$

**mol NaCl produced = ?**

| **$HCl(aq)$** | **+** | **$NaOH(aq)$** | **⟶** | **$NaCl(aq)$** | **+** | **$H_2O(l)$** |
|---|---|---|---|---|---|---|
| **50.0 mL** | | **50.0 mL** | | **? mol** | | |
| **1.0 M** | | **1.0 M** | | | | |

**NOTE**: Reactants are in stoichiometric ratio (No Limiting Reactant, No Reactant in excess)

Therefore: The resulting moles of NaCl can be calculated from the volumes of either NaOH solution or HCl solution.

$$? \text{ mol NaCl} = 50.0\text{mL HCl} \times \frac{1.0 \text{ mol HCl}}{1000 \text{ mL HCl}} \times \frac{1 \text{ mol NaCl}}{1 \text{ mol HCl}} = \textbf{0.0050 mol NaCl}$$

$$\Delta H = - \frac{\text{–3.1 KJ}}{\text{0.050 mol NaCl produced}} = \textbf{- 62 KJ/mol}$$

**Sample Problem 2:**

2.08 g of NaOH are dissolved in 50.0 mL of water whose temperature is 20.0 °C. The temperature of the NaOH solution after mixing increases to 28.8 °C.

The specific heat of the resulting NaOH solution is about 3.93 J/g °C

**Calculate the Heat of Solution of NaOH in KJ/mol**

| | | | | | |
|---|---|---|---|---|---|
| **NaOH(s)** | + | **$H_2O$(l)** | ⟶ | **NaOH(aq)** | **ΔH = ???** |
| 2.08 g | | 50.0 g | | 52.1 g | |

**Heat of solution of NaOH = ΔH = exothermic (temperature of solution increased)**

Mass of solution = 52.1 g
Initial Temperature = 20.0 °C.
Final Temperature = 28.8 °C
Δt = 8.8 °C
Specific heat = 3.93 J/g °C

**q = m x s x Δt**

q = – (52.1 g) (3.93 J/g °C) (8.8 °C) = –1800 J = **–1.8 KJ**

$$\text{? mol NaOH} = 2.08\ \text{g} \times \frac{1\ \text{mol}}{40.0\ \text{g}} = \textbf{0.052 mol NaOH}$$

$$\textbf{Heat of solution} = \Delta H = \text{?}\ \frac{\textbf{kJ}}{\textbf{mol NaOH}} \qquad \Delta H = \frac{\textbf{–1.8 kJ}}{\textbf{0.052 mol NaOH}} = \textbf{–35 KJ/mol}$$

**اتحاد سنافر البوليتكنك (spu)**

**Sample Problem 3:**

0.562 g of graphite is placed in a calorimeter with excess oxygen at 25.00 °C and 1 atm pressure. The graphite is ignited and burns completely to form $CO_2$ as shown below:

$$C\ (graphite) + O_2 \rightarrow CO_2\ (g)$$

After the completion of the reaction, the calorimeter temperature rises to 25.90 °C. The heat capacity of the calorimeter is found to be 20.7 kJ/°C. Based on this information calculate the heat of the reaction.

**Examples:**

1. Two solutions, 100.0 mL of 1.00 M $AgNO_3$ and 100.0 mL of 1.00 M NaCl, both initially at 22.4 °C, are added to a calorimeter and allowed to react. The temperature rises to 30.2 °C. Calculate the heat of reaction in kJ/mol AgCl produced. (Assume density and specific heat of solution to be the same as water)

2. A 0.205-g pellet of potassium hydroxide, KOH, is dissolved in 56.0 g of water in a calorimeter. The water temperature rises from 23.5 to 24.4 °C. Calculate $\Delta H_{soln}$ in kJ/mol KOH. (Assume the specific heat of the solution to be the same as water, 4.184 J/g°C)

3. When 23.6 g of $CaCl_2$ was dissolved in water in a calorimeter, the temperature rose from 25.0°C to 38.7°C. If the heat capacity of the calorimeter and the solution is 1258 J/°C, what is the $\Delta H_{soln}$ for $CaCl_2$ in kJ/mol?

# HESS'S LAW

**Hess's Law of Summation**

- **For a chemical equation that can be written as the sum of two or more steps, the enthalpy change for the overall equation equals the sum of the enthalpy changes for the individual steps.**

**Example:**
Ammonia will burn in the presence of a platinum catalyst to produce nitric oxide, NO

$$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g) \qquad \Delta H = ?$$

**What is the heat of reaction at constant pressure?**

Use the following thermochemical equations:

$$N_2(g) + O_2(g) \longrightarrow 2NO(g) \qquad \Delta H_1 = +\ 180.6\ kJ \quad (Eq\ 1)$$

$$N_2(g) + 3H_2(g) \longrightarrow 2NH_3(g) \qquad \Delta H_2 = -\ 91.8\ kJ \quad (Eq\ 2)$$

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g) \qquad \Delta H_3 = -\ 483.7\ kJ \quad (Eq\ 3)$$

$$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g) \qquad \Delta H = ?$$

---

- **Step 1:** To get 4 $NH_3(g)$ on the reactant side: **Multiply Eq 2 by 2,** and reverse its its direction:

$$N_2(g) + O_2(g) \longrightarrow 2NO(g) \qquad \Delta H_1 = +180.6\ kJ \qquad (Eq\ 1)$$

$$4NH_3(g) \longrightarrow 2N_2(g) + 6H_2(g) \qquad \Delta H_2 = (-91.8\ kJ)(-2) = +184\ kJ \quad (Eq\ 2)$$

$$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g) \qquad \Delta H_3 = -\ 483.7\ kJ \qquad (Eq\ 3)$$

---

**$4NH_3(g) + 5O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$ $\Delta H = ?$**

---

- **Step 2**: To get 4 NO (g) on the product side: Multiply **Eq 1 by 2:**

**$2N_2(g) + 2O_2(g) \longrightarrow 4NO(g)$ $\Delta H_1 = (+180.6\ kJ)(2) = +361.2\ kJ$ (Eq 1)**

$4NH_3(g) \longrightarrow 2N_2(g) + 6H_2(g)$ $\Delta H_2 = +184\ kJ$ (Eq 2)

$2H_2(g) + O_2(g) \longrightarrow 2H_2O(g)$ $\Delta H_3 = -483.7\ kJ$ (Eq 3)

---

- **Step 3:** To get 6 $H_2O$ (g) on the product side: **Multiply Eq 3 by 3:**

$2N_2(g) + 2O_2(g) \longrightarrow 4NO(g)$ $\Delta H_1 = +361.2\ kJ$ (Eq 1)

$4NH_3(g) \longrightarrow 2N_2(g) + 6H_2(g)$ $\Delta H_2 = +184\ kJ$ (Eq 2)

**$6H_2(g) + 3O_2(g) \longrightarrow 6H_2O(g)$ $\Delta H_3 = -483.7\ kJ\ (3) = -1450\ kJ$ (Eq 3)**

---

- **Step 4**: Add the 3 steps (equations) and cancel out $2N_2(g)$ and $6H_2(g)$

~~$2N_2(g)$~~ $+ 2O_2(g) \longrightarrow 4NO(g)$ $\Delta H_1 = +361.2\ kJ$ (Eq 1)

$4NH_3(g) \longrightarrow$ ~~$2N_2(g)$~~ $+$ ~~$6H_2(g)$~~ $\Delta H_2 = +184\ kJ$ (Eq 2)

~~$6H_2(g)$~~ $+ 3O_2(g) \longrightarrow 6H_2O(g)$ $\Delta H_3 = -1450\ kJ$ (Eq 3)

**$2O_2(g) + 4NH_3(g) + 3O_2(g) \longrightarrow 4NO(g) + 6H_2O(g)$** $\Delta H_T = \Delta H_1 + \Delta H_2 + \Delta H_3$

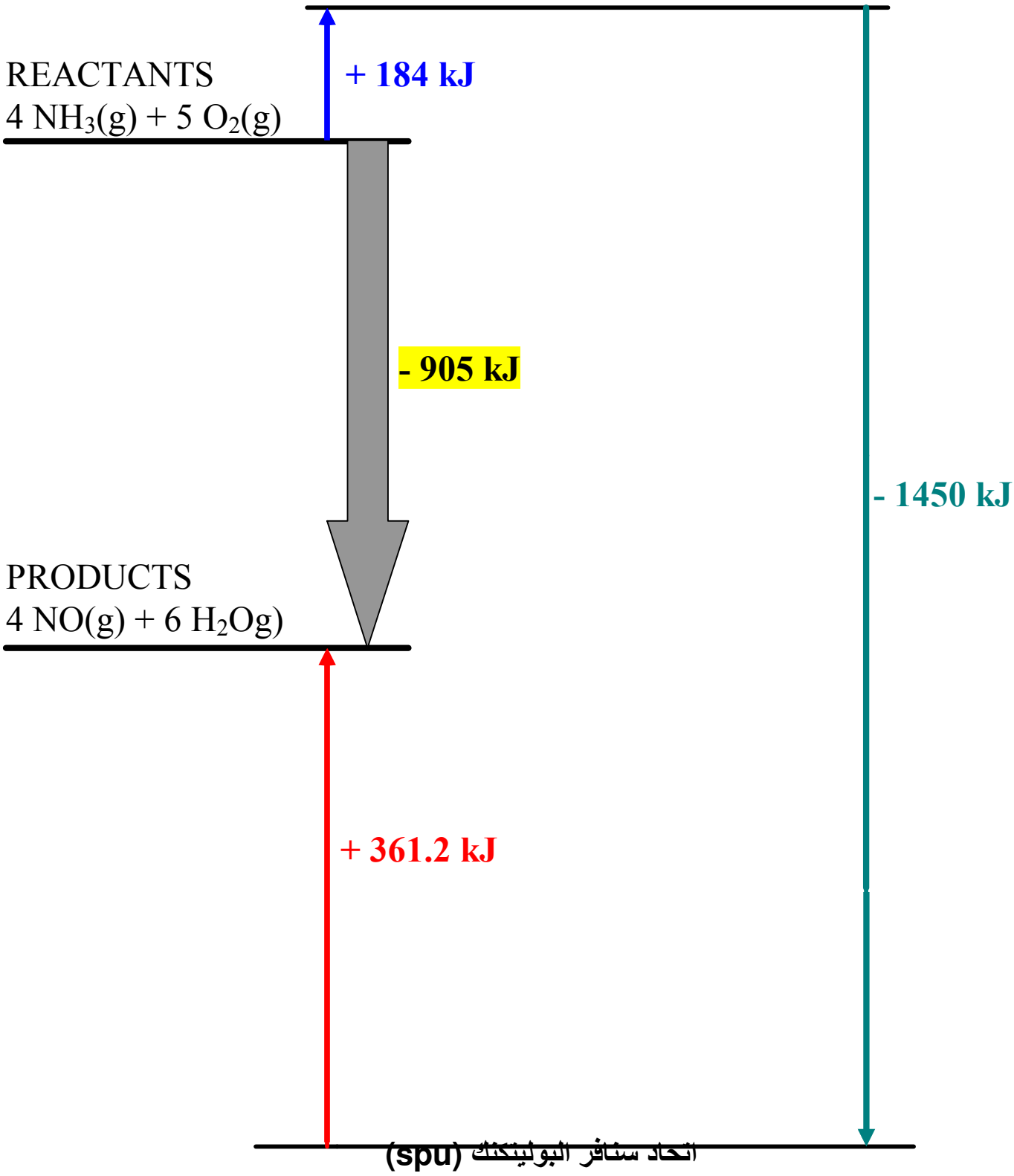
$\Delta H_T = +361.2\ kJ + 184\ kJ - 1450\ kJ = -905\ kJ$

**SUM:**

**$5O_2(g) + 4NH_3(g) \longrightarrow 4NO(g) + 6H_2O(g)$** $\Delta H_T = -905\ kJ$

**(spu) اتحاد سنافر البوليتكنك**

- An Enthalpy diagram illustrating how Hess's law applies to this example is shown below.



**Examples:**

1. Calculate the enthalpy of reaction for the reaction shown below:

$$C_2H_4\,(g) + H_2\,(g) \rightarrow C_2H_6\,(g) \qquad \Delta H = ???$$

from the following:

$$C_2H_4\,(g) + 3\,O_2\,(g) \rightarrow 2\,CO_2\,(g) + 2\,H_2O\,(l) \qquad \Delta H = -1401\ kJ$$

$$2\,C_2H_6\,(g) + 7\,O_2\,(g) \rightarrow 4\,CO_2\,(g) + 6\,H_2O\,(l) \qquad \Delta H = -3100\ kJ$$

$$2\,H_2\,(g) + O_2\,(g) \rightarrow 2\,H_2O\,(l) \qquad \Delta H = -572$$

2. Calculate the enthalpy of reaction for the reaction shown below:

$$CH_4\,(g) + NH_3\,(g) \rightarrow HCN\,(g) + 3\,H_2\,(g)$$

from the following equations:

$$N_2\,(g) + 3\,H_2\,(g) \rightarrow 2\,NH_3\,(g) \qquad \Delta H = -91.8\ kJ$$

$$C(\text{graphite}) + 2\,H_2\,(g) \rightarrow CH_4\,(g) \qquad \Delta H = -74.9\ kJ$$

$$H_2\,(g) + 2\,C(\text{graphite}) + N_2\,(g) \rightarrow 2\,HCN\,(g) \qquad \Delta H = 270.3\ kJ$$

# STANDARD ENTHALPIES OF FORMATION

**Standard State** = **Standard Thermodynamic Conditions**

Indicated by a superscript: ($^o$) (1 atm pressure and 25$^o$ C)

**Standard Enthalpy of Reaction, $\Delta H^o$** (Read delta H zero)

- The Enthalpy change for a reaction in which reactants in their standard states yield products in their standard state.
- Tables usually give Enthalpy changes for **Formation Reactions** only.

**Formation Reactions:**

Reactions in which compounds are formed from their elements

$$A + B \longrightarrow AB$$

**Example:**

$$C(\text{graphite}) + 2Cl_2(g) \longrightarrow CCl_4(l) \qquad \Delta H^o_f = -139 \text{ kJ/mol}$$

- Standard Enthalpy of formation for liquid $CCl_4$ from graphite = 139 kJ/mol

**Note:**

- It is important to indicate the form in which the element participates in a formation reaction, since the value of $\Delta H^o_f$ is determined by this.
- Carbon may exist in other forms as well, referred to as allotropes (ex: diamond).
- However, graphite is the most stable form of carbon (Graphite is referred to as the reference form of carbon)

**Reference form of an element :**

- The most stable form of the element under standard thermodynamic conditions (1 atm pressure and 25° C)

$\Delta H^o_f$ of an element in its reference form = 0

For example:

$\Delta H^o_f$ (graphite) = 0
$\Delta H^o_f$ (diamond) = 1.9 kJ/mol

Meaning:

$$C(\text{graphite}) \longrightarrow C(\text{diamond}) \qquad \Delta H^o_f = +1.9 \text{ kJ/mol (endothermic)}$$

**$\Delta H^o_f$ = STANDARD ENTHALPY OF FORMATION**
**= STANDARD HEAT OF FORMATION**
**= is the enthalpy change for the formation of one mole of the substance in its standard state from its elements in their reference form and in their standard states.**
**(For Values, see Table 6.2, and in Appendix C)**

**Applications:**

- Using Hess's Law, the given standard enthalpies of formation can be used to find the standard enthalpy change for a reaction.

**Examples:**

1. The cooling effect of alcohol on the skin is due to its evaporation. Calculate **the heat of vaporization of ethanol** (ethyl alcohol), $C_2H_5OH$.

$$C_2H_5OH\ (l) \longrightarrow C_2H_5OH\ (g) \qquad \Delta H^o \text{ vaporization} = ?$$

From Table 6.2 or Appendix C:

$\Delta H^o_f[C_2H_5OH(l)]$ = –277.6 kJ/mol

$\Delta H^o_f[C_2H_5OH(g)]$ = –235.4 kJ/mol

| | $C_2H_5OH(l)$ | ⟶ | $C_2H_5OH(g)$ | $\Delta H^o$ vaporization = ? |
|---|---|---|---|---|
| $\Delta H^o_f$ | –277.6 kJ/mol | | –235.4 kJ/mol | |

$\Delta H^o$ vaporization = $\Delta H^o_f[C_2H_5OH(g)] - \Delta H^o_f[C_2H_5OH(l)]$

$\Delta H^o$ vaporization = (-235.4 kJ/mol) - (-277 kJ/mol) = + 42. 2 kJ/mol

اتحاد سنافر البوليتكنك (spu)

**Note**: Positive sign indicates that reaction is endothermic

- **IN GENERAL, $\Delta H^o$ for a reaction can be calculated as:**

**$\Delta H^o$ = Sum of $\Delta H^o_f$ of products – Sum of $\Delta H^o_f$ of reactants**

**$\Delta H^o = \Sigma\ n\ \Delta H^o_f$ (products) – $\Sigma\ m\ \Delta H^o_f$ (reactants)**

**where: $\Sigma$ = mathematical symbol for "the sum of"**
**n, m, = the coefficients of the substances in the chemical equation**

**Examples:**

2. Iron is obtained from iron ore by reduction with carbon monoxide. The overall reaction is shown below. Calculate the standard enthalpy change for this equation. See Appendix C for $\Delta H°_f$ data.

$$Fe_2O_3\,(s) + 3\,CO\,(g) \longrightarrow 2\,Fe\,(s) + 3\,CO_2\,(g)$$

| | | | | |
|---|---|---|---|---|
| **$\Delta H^o_f$ (kJ/mol)** | –825.5 | –110.5 | 0 | –393.5 |

**$\Delta H^o = \Sigma\ n\ \Delta H^o_f$ (products) – $\Sigma\ m\ \Delta H^o_f$ (reactants)**

**= [2 (0) + 3 (–395.5)] – [(–825.5) + 3 (–110.5) = –23.5 kJ**

3. Calculate the standard enthalpy of formation for ethylene ($C_2H_2$) from the standard enthalpies of reaction shown below:

$$2\,C_2H_2\,(g) + 5\,O_2\,(g) \rightarrow 4\,CO_2\,(g) + 2\,H_2O\,(g) \qquad \Delta H° = -2512\ kJ$$

| | | | | |
|---|---|---|---|---|
| $\Delta H_f°$ (kJ/mol) | ?? | 0 | –393.5 | –242 |

**اتحاد سنافر البوليتكنك (spu)**

# Electronic Structure of Atoms (Electron Configurations)

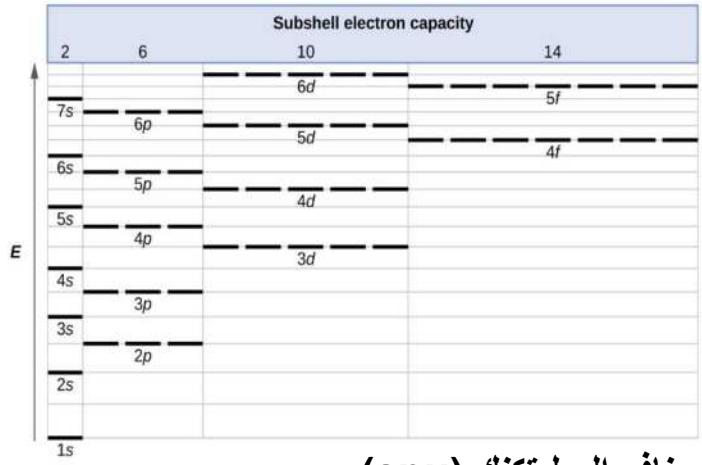
Learning Objectives

- Derive the predicted ground-state electron configurations of atoms
- Identify and explain exceptions to predicted electron configurations for atoms and ions
- Relate electron configurations to element classifications in the periodic table

Having introduced the basics of atomic structure and quantum mechanics, we can use our understanding of quantum numbers to determine how atomic orbitals relate to one another. This allows us to determine which orbitals are occupied by electrons in each atom. The specific arrangement of electrons in orbitals of an atom determines many of the chemical properties of that atom.

## Orbital Energies and Atomic Structure

The energy of atomic orbitals increases as the principal quantum number, nn, increases. In any atom with two or more electrons, the repulsion between the electrons makes energies of subshells with different values of ll differ so that the energy of the orbitals increases within a shell in the order $s < p < d < f$. Figure 6.4.16.4.1 depicts how these two trends in increasing energy relate. The 1*s* orbital at the bottom of the diagram is the orbital with electrons of lowest energy. The energy increases as we move up to the 2*s* and then 2*p*, 3*s*, and 3*p* orbitals, showing that the increasing *n* value has more influence on energy than the increasing *l* value for small atoms. However, this pattern does not hold for larger atoms. The 3*d* orbital is higher in energy than the 4*s* orbital. Such overlaps continue to occur frequently as we move up the chart.




اتحاد سنافر البوليتكنك (spu)

Figure : Generalized energy-level diagram for atomic orbitals in an atom with two or more electrons (not to scale).

Electrons in successive atoms on the periodic table tend to fill low-energy orbitals first. Thus, many students find it confusing that, for example, the 5*p* orbitals fill immediately after the 4*d*, and immediately before the 6*s*. The filling order is based on observed experimental results, and has been confirmed by theoretical calculations. As the principal quantum number, *n*, increases, the size of the orbital increases and the electrons spend more time farther from the nucleus. Thus, the attraction to the nucleus is weaker and the energy associated with the orbital is higher (less stabilized). But this is not the only effect we have to take into account. Within each shell, as the value of *l* increases, the electrons are less penetrating (meaning there is less electron density found close to the nucleus), in the order $s > p > d > f$. Electrons that are closer to the nucleus slightly repel electrons that are farther out, offsetting the more dominant electron–nucleus attractions slightly (recall that all electrons have −1 charges, but nuclei have +*Z* charges). This phenomenon is called shielding and will be discussed in more detail in the next section. Electrons in orbitals that experience more shielding are less stabilized and thus higher in energy. For small orbitals (1*s* through 3*p*), the increase in energy due to *n* is more significant than the increase due to *l*; however, for larger orbitals the two trends are comparable and cannot be simply predicted. We will discuss methods for remembering the observed order.

The arrangement of electrons in the orbitals of an atom is called the electron configuration of the atom. We describe an electron configuration with a symbol that contains three pieces of information ( Figure 6.4.26.4.2):

1. The number of the principal quantum shell, *n*,
2. The letter that designates the orbital type (the subshell, *l*), and
3. A superscript number that designates the number of electrons in that particular subshell.

For example, the notation $2p^4$ (read "two–p–four") indicates four electrons in a *p* subshell ($l = 1$) with a principal quantum number (*n*) of 2. The notation $3d^8$ (read "three–d–eight") indicates eight electrons in the *d* subshell (i.e., $l = 2$) of the principal shell for which $n = 3$.

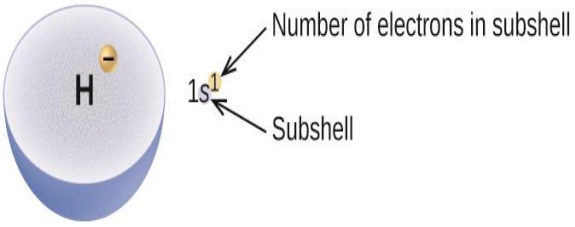



Figure 6.4.26.4.2: The diagram of an electron configuration specifies the subshell (n and l value, with letter symbol) and superscript number of electrons.

## The Aufbau Principle

To determine the electron configuration for any particular atom, we can “build” the structures in the order of atomic numbers. Beginning with hydrogen, and continuing across the periods of the periodic table, we add one proton at a time to the nucleus and one electron to the proper subshell until we have described the electron configurations of all the elements. This procedure is called

(spu) اتحاد سنافر البوليتكنك

the Aufbau principle, from the German word *Aufbau* (“to build up”). Each added electron occupies the subshell of lowest energy available (in the order shown in Figure 6.4.36.4.3), subject to the limitations imposed by the allowed quantum numbers according to the Pauli exclusion principle. Electrons enter higher-energy subshells only after lower-energy subshells have been filled to capacity. Figure 6.4.36.4.3 illustrates the traditional way to remember the filling order for atomic orbitals.

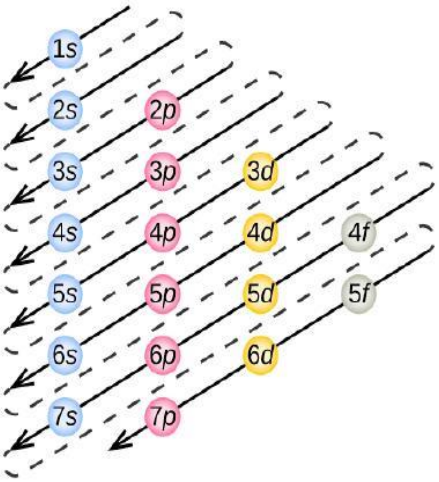



Figure 6.4.36.4.3: The arrow leads through each subshell in the appropriate filling order for electron configurations. This chart is straightforward to construct. Simply make a column for all the s orbitals with each n shell on a separate row. Repeat for p, d, and f. Be sure to only include orbitals allowed by the quantum numbers (no 1p or 2d, and so forth). Finally, draw diagonal lines from top to bottom as shown.

Since the arrangement of the periodic table is based on the electron configurations, Figure 6.4.46.4.4 provides an alternative method for determining the electron configuration. The filling order simply begins at hydrogen and includes each subshell as you proceed in increasing *Z* order. For example, after filling the 3*p* block up to Ar, we see the orbital will be 4s (K, Ca), followed by the 3*d* orbitals.

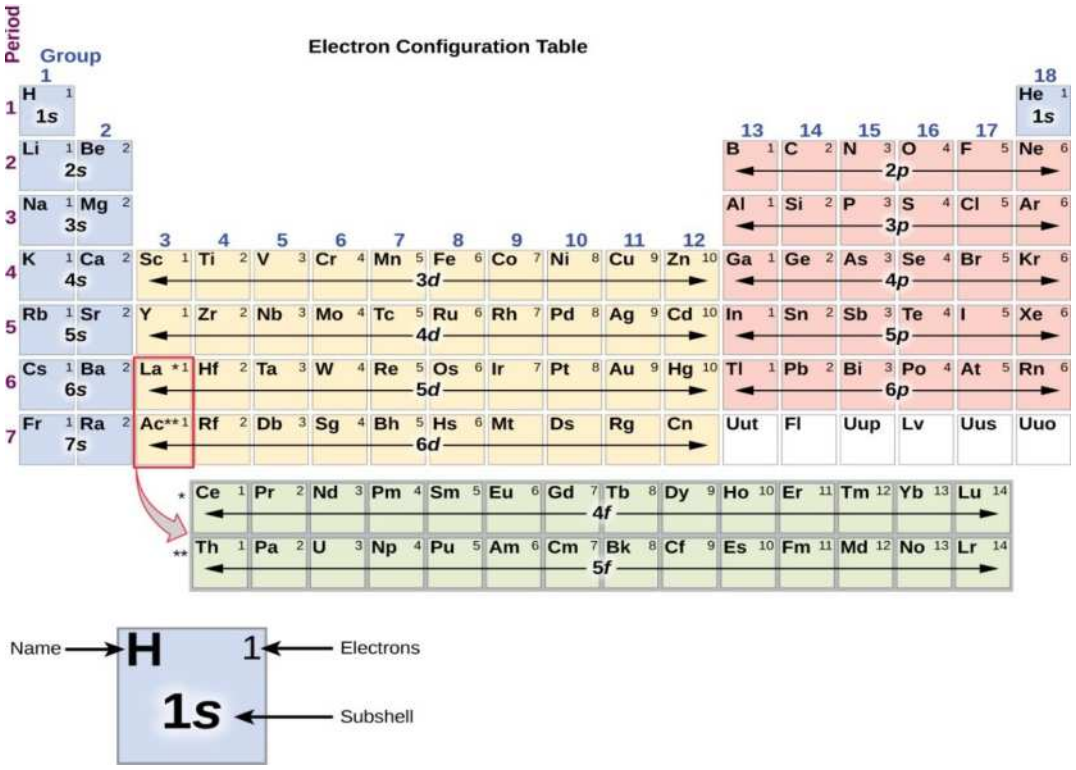
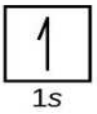



Figure : This periodic table shows the electron configuration for each subshell. By "building up" from hydrogen, this table can be used to determine the electron configuration for any atom on the periodic table.

We will now construct the ground-state electron configuration and orbital diagram for a selection of atoms in the first and second periods of the periodic table. Orbital diagrams are pictorial representations of the electron configuration, showing the individual orbitals and the pairing arrangement of electrons. We start with a single hydrogen atom (atomic number 1), which consists of one proton and one electron. Referring to either Figure 6.4.36.4.3 or 6.4.46.4.4, we would expect to find the electron in the 1*s* orbital. By convention, the $m_s=+12$ms=+12 value is usually filled first. The electron configuration and the orbital diagram are:

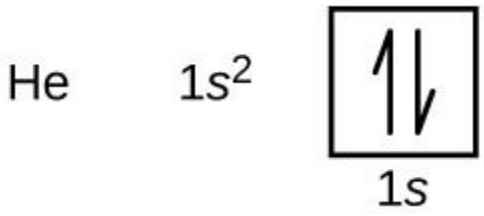
H $1s^1$




Following hydrogen is the noble gas helium, which has an atomic number of 2. The helium atom contains two protons and two electrons. The first electron has the same four quantum numbers as the hydrogen atom electron ($n = 1$, $l = 0$, $m_l = 0$, $m_s=+12$ms=+12). The second electron also goes into the 1*s* orbital and fills that orbital. The second electron has the same $n$, $l$, and $m_l$ quantum numbers, but must have the opposite spin quantum number, $m_s=-12$ms=−12. This is in accord with the Pauli exclusion principle, which states two electrons in the same atom can have the same set of four quantum numbers. For orbital diagrams, this means two arrows go in each box (representing

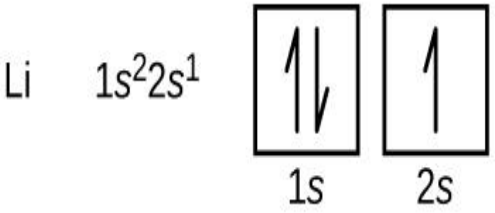
(spu) اتحاد سنافر البوليتكنك

two electrons in each orbital) and the arrows must point in opposite directions (representing paired spins). The electron configuration and orbital diagram of helium are:

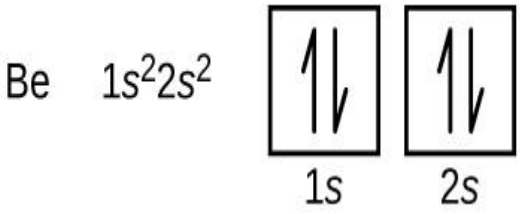



The $n = 1$ shell is completely filled in a helium atom.

The next atom is the alkali metal lithium with an atomic number of 3. The first two electrons in lithium fill the 1*s* orbital and have the same sets of four quantum numbers as the two electrons in helium. The remaining electron must occupy the orbital of next lowest energy, the 2*s* orbital (Figure 6.4.36.4.3 or 6.4.46.4.4). Thus, the electron configuration and orbital diagram of lithium are:

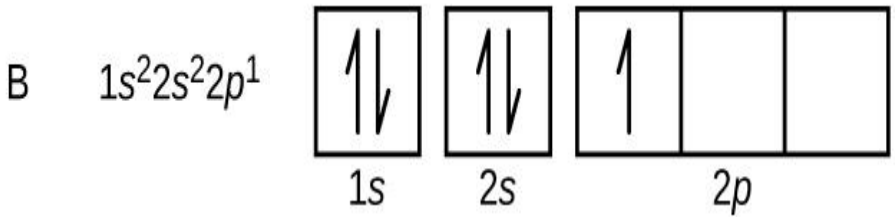



An atom of the alkaline earth metal beryllium, with an atomic number of 4, contains four protons in the nucleus and four electrons surrounding the nucleus. The fourth electron fills the remaining space in the 2*s* orbital.

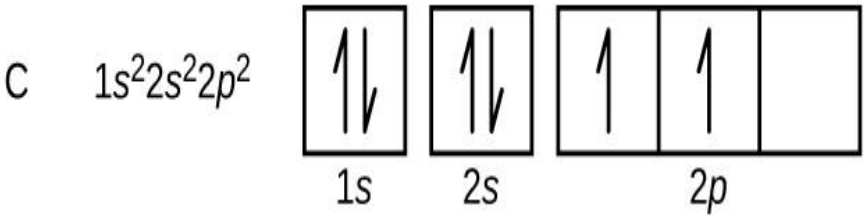



An atom of boron (atomic number 5) contains five electrons. The $n = 1$ shell is filled with two electrons and three electrons will occupy the $n = 2$ shell. Because any *s* subshell can contain only two electrons, the fifth electron must occupy the next energy level, which will be a 2*p* orbital. There are three degenerate 2*p* orbitals ($m_l = -1, 0, +1$) and the electron can occupy any one of these *p* orbitals. When drawing orbital diagrams, we include empty boxes to depict any empty orbitals in the same subshell that we are filling.

**اتحاد سنافر البوليتكنك (spu)**




Carbon (atomic number 6) has six electrons. Four of them fill the 1*s* and 2*s* orbitals. The remaining two electrons occupy the 2*p* subshell. We now have a choice of filling one of the 2*p* orbitals and pairing the electrons or of leaving the electrons unpaired in two different, but degenerate, *p* orbitals. The orbitals are filled as described by Hund's rule: the lowest-energy configuration for an atom with electrons within a set of degenerate orbitals is that having the maximum number of unpaired electrons. Thus, the two electrons in the carbon 2*p* orbitals have identical $n$, $l$, and $m_s$ quantum numbers and differ in their $m_l$ quantum number (in accord with the Pauli exclusion principle). The electron configuration and orbital diagram for carbon are:




Nitrogen (atomic number 7) fills the 1*s* and 2*s* subshells and has one electron in each of the three 2*p* orbitals, in accordance with Hund's rule. These three electrons have unpaired spins. Oxygen (atomic number 8) has a pair of electrons in any one of the 2*p* orbitals (the electrons have opposite spins) and a single electron in each of the other two. Fluorine (atomic number 9) has only one 2*p* orbital containing an unpaired electron. All of the electrons in the noble gas neon (atomic number 10) are paired, and all of the orbitals in the $n = 1$ and the $n = 2$ shells are filled. The electron configurations and orbital diagrams of these four elements are:

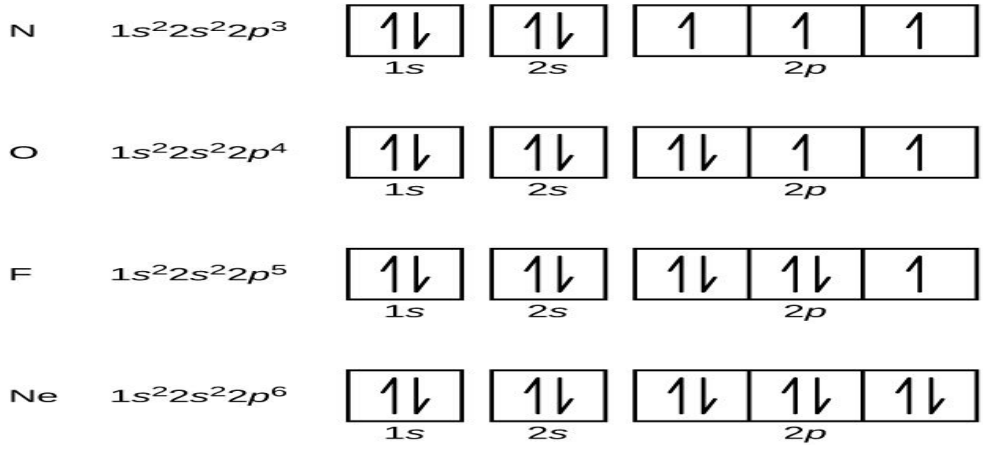



Figure 6.4.56.4.5: Since the core electrons in all of the noble gas electron configurations, we can abbreviate electron configurations by writing the noble gas that matches the core electron configuration, along with the valence electrons in a condensed format. For our sodium example,

(spu) نحاد سنفر البوليتك

the symbol [Ne] represents core electrons, ($1s^2 2s^2 2p^6$) and our abbreviated or condensed configuration is [Ne]$3s^1$.

The alkali metal sodium (atomic number 11) has one more electron than the neon atom. This electron must go into the lowest-energy subshell available, the 3*s* orbital, giving a $1s^2 2s^2 2p^6 3s^1$ configuration. The electrons occupying the outermost shell orbital(s) (highest value of *n*) are called valence electrons, and those occupying the inner shell orbitals are called core electrons ( Figure \PageIndex5\PageIndex5). Since the core electron shells correspond to noble gas electron configurations, we can abbreviate electron configurations by writing the noble gas that matches the core electron configuration, along with the valence electrons in a condensed format. For our sodium example, the symbol [Ne] represents core electrons, ($1s^2 2s^2 2p^6$) and our abbreviated or condensed configuration is [Ne]$3s^1$.




Figure 6.4.56.4.5: A core-abbreviated electron configuration (right) replaces the core electrons with the noble gas symbol whose configuration matches the core electron configuration of the other element.

Similarly, the abbreviated configuration of lithium can be represented as [He]$2s^1$, where [He] represents the configuration of the helium atom, which is identical to that of the filled inner shell of lithium. Writing the configurations in this way emphasizes the similarity of the configurations of lithium and sodium. Both atoms, which are in the alkali metal family, have only one electron in a valence *s* subshell outside a filled set of inner shells.

Li:[He]2s1Na:[Ne]3s1(6.4.1)(6.4.1)Li:[He]2s1Na:[Ne]3s1

The alkaline earth metal magnesium (atomic number 12), with its 12 electrons in a [Ne]$3s^2$ configuration, is analogous to its family member beryllium, [He]$2s^2$. Both atoms have a filled *s* subshell outside their filled inner shells. Aluminum (atomic number 13), with 13 electrons and the electron configuration [Ne]$3s^2 3p^1$, is analogous to its family member boron, [He]$2s^2 2p^1$.

The electron configurations of silicon (14 electrons), phosphorus (15 electrons), sulfur (16 electrons), chlorine (17 electrons), and argon (18 electrons) are analogous in the electron configurations of their outer shells to their corresponding family members carbon, nitrogen, oxygen, fluorine, and neon, respectively, except that the principal quantum number of the outer shell of the heavier elements has increased by one to $n = 3$. Figure 6.4.66.4.6 shows the lowest energy, or ground-state, electron configuration for these elements as well as that for atoms of each of the known elements.

**اتحاد سنافر البوليتكنك (spu)**

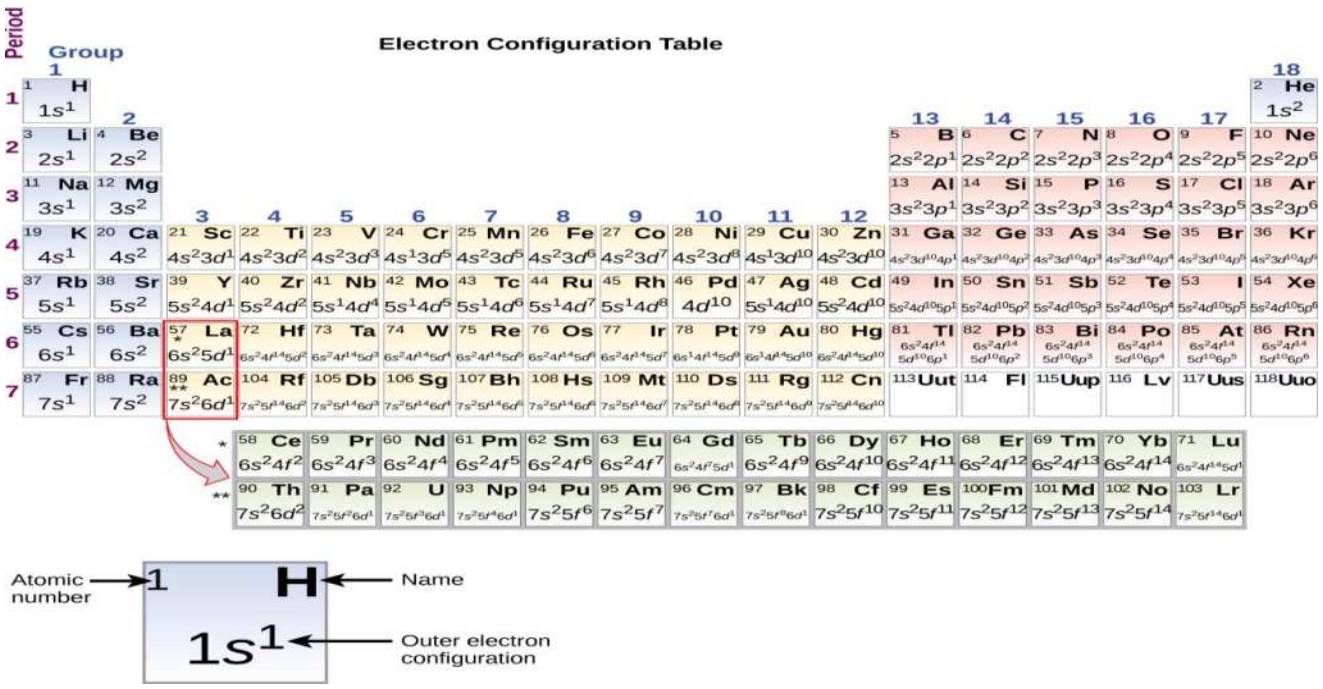



Figure : This version of the periodic table shows the outer-shell electron configuration of each element. Note that down each group, the configuration is often similar.

When we come to the next element in the periodic table, the alkali metal potassium (atomic number 19), we might expect that we would begin to add electrons to the 3*d* subshell. However, all available chemical and physical evidence indicates that potassium is like lithium and sodium, and that the next electron is not added to the 3*d* level but is, instead, added to the 4*s* level . As discussed previously, the 3*d* orbital with no radial nodes is higher in energy because it is less penetrating and more shielded from the nucleus than the 4*s*, which has three radial nodes. Thus, potassium has an electron configuration of [Ar]$4s^1$. Hence, potassium corresponds to Li and Na in its valence shell configuration. The next electron is added to complete the 4*s* subshell and calcium has an electron configuration of [Ar]$4s^2$. This gives calcium an outer-shell electron configuration corresponding to that of beryllium and magnesium.

Beginning with the transition metal scandium (atomic number 21), additional electrons are added successively to the 3*d* subshell. This subshell is filled to its capacity with 10 electrons (remember that for $l = 2$ [*d* orbitals], there are $2l + 1 = 5$ values of $m_l$, meaning that there are five *d* orbitals that have a combined capacity of 10 electrons). The 4*p* subshell fills next. Note that for three series of elements, scandium (Sc) through copper (Cu), yttrium (Y) through silver (Ag), and lutetium (Lu) through gold (Au), a total of 10 *d* electrons are successively added to the $(n - 1)$ shell next to the *n* shell to bring that $(n - 1)$ shell from 8 to 18 electrons. For two series, lanthanum (La) through lutetium (Lu) and actinium (Ac) through lawrencium (Lr), 14 *f* electrons ($l = 3$, $2l + 1 = 7$ $m_l$ values; thus, seven orbitals with a combined capacity of 14 electrons) are successively added to the $(n - 2)$ shell to bring that shell from 18 electrons to a total of 32 electrons.

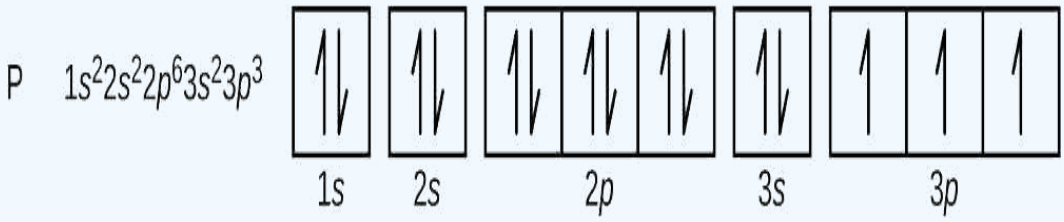
اتحاد شافر البوليتكنك (spu)

## Example : Quantum Numbers and Electron Configurations

What is the electron configuration and orbital diagram for a phosphorus atom? What are the four quantum numbers for the last electron added?

**Solution**

The atomic number of phosphorus is 15. Thus, a phosphorus atom contains 15 electrons. The order of filling of the energy levels is 1*s*, 2*s*, 2*p*, 3*s*, 3*p*, 4*s*, . . . The 15 electrons of the phosphorus atom will fill up to the 3*p* orbital, which will contain three electrons:




The last electron added is a 3*p* electron. Therefore, $n = 3$ and, for a *p*-type orbital, $l = 1$. The $m_l$ value could be –1, 0, or +1. The three *p* orbitals are degenerate, so any of these $m_l$ values is correct. For unpaired electrons, convention assigns the value of $+12$+12 for the spin quantum number; thus, $m_s=+12$ms=+12.

## Exercise

Identify the atoms from the electron configurations given:

a. [Ar]$4s^2 3d^5$
b. [Kr]$5s^2 4d^{10} 5p^6$

The periodic table can be a powerful tool in predicting the electron configuration of an element. However, we do find exceptions to the order of filling of orbitals. For instance, the electron configurations of the transition metals chromium (Cr; atomic number 24) and copper (Cu; atomic number 29), among others, are not those we would expect. In general, such exceptions involve subshells with very similar energy, and small effects can lead to changes in the order of filling.

In the case of Cr and Cu, we find that half-filled and completely filled subshells apparently represent conditions of preferred stability. This stability is such that an electron shifts from the 4*s* into the 3*d* orbital to gain the extra stability of a half-filled 3*d* subshell (in Cr) or a filled 3*d* subshell (in Cu). Other exceptions also occur. For example, niobium (Nb, atomic number 41) is predicted to have the electron configuration [Kr]$5s^2 4d^3$. Experimentally, we observe that its ground-state electron configuration is actually [Kr]$5s^1 4d^4$. We can rationalize this observation by saying that the electron–electron repulsions experienced by pairing the electrons in the 5*s* orbital are larger than the gap in energy between the 5*s* and 4*d* orbitals. There is no simple method to

predict the exceptions for atoms where the magnitude of the repulsions between electrons is greater than the small differences in energy between subshells.

## Electron Configurations and the Periodic Table

As described earlier, the periodic table arranges atoms based on increasing atomic number so that elements with the same chemical properties recur periodically. When their electron configurations we also see a periodic recurrence of similar electron configurations in the outer shells of these elements. Because they are in the outer shells of an atom, valence electrons play the most important role in chemical reactions. The outer electrons have the highest energy of the electrons in an atom and are more easily lost or shared than the core electrons. Valence electrons are also the determining factor in some physical properties of the elements.

Elements in any one group (or column) have the same number of valence electrons; the alkali metals lithium and sodium each have only one valence electron, the alkaline earth metals beryllium and magnesium each have two, and the halogens fluorine and chlorine each have seven valence electrons. The similarity in chemical properties among elements of the same group occurs because they have the same number of valence electrons. It is the loss, gain, or sharing of valence electrons that defines how elements react.

It is important to remember that the periodic table was developed on the basis of the chemical behavior of the elements, well before any idea of their atomic structure was available. Now we can understand why the periodic table has the arrangement it has—the arrangement puts elements whose atoms have the same number of valence electrons in the same group., elements classified by the orbitals being filled: main group, transition, and inner transition elements. These classifications determine which orbitals are counted in the valence shell, or highest energy level orbitals of an atom.

1. **Main group elements** (sometimes called **representative elements**) are those in which the last electron added enters an *s* or a *p* orbital in the outermost shell.This category includes all the nonmetallic elements, as well as many metals and the intermediate semimetallic elements. The valence electrons for main group elements are those with the highest *n* level. For example, gallium (Ga, atomic number 31) has the electron configuration [Ar]$\underline{4s^2}3d^{10}\underline{4p^1}$, which contains three valence electrons (underlined). The completely filled *d* orbitals count as core, not valence, electrons.
2. **Transition elements or transition metals**. These are metallic elements in which the last electron added enters a *d* orbital. The valence electrons (those added after the last noble gas configuration) in these elements include the *ns* and $(n - 1)$ *d* electrons. The official IUPAC definition of transition elements specifies those with partially filled *d* orbitals. Thus, the elements with completely filled orbitals (Zn, Cd, Hg, as well as Cu, Ag, and Au. However, the term is frequently used to refer to the entire *d* block.
3. **Inner transition elements** are metallic elements in which the last electron added occupies an *f* orbital. They are shown in green in Figure 6.4.66.4.6. The valence shells of the inner transition elements consist of the $(n - 2)f$, the $(n - 1)d$, and the *ns* subshells. There are two inner transition series:

(spu) اتحاد صنافي البوليتكنك

1. The lanthanide series: lanthanide (La) through lutetium (Lu)
2. The actinide series: actinide (Ac) through lawrencium (Lr)

Lanthanum and actinium, because of their similarities to the other members of the series, are included and used to name the series, even though they are transition metals with no *f* electrons.

## Electron Configurations of Ions

We have seen that ions are formed when atoms gain or lose electrons. A cation (positively charged ion) forms when one or more electrons are removed from a parent atom. For main group elements, the electrons that were added last are the first electrons removed. For transition metals and inner transition metals, however, electrons in the *s* orbital are easier to remove than the *d* or *f* electrons, and so the highest *ns* electrons are lost, and then the $(n-1)d$ or $(n-2)f$ electrons are removed. An anion (negatively charged ion) forms when one or more electrons are added to a parent atom. The added electrons fill in the order predicted by the Aufbau principle.

### Example : Predicting Electron Configurations of Ions

What is the electron configuration and orbital diagram of:

a. $Na^+$
b. $P^{3-}$
c. $Al^{2+}$
d. $Fe^{2+}$
e. $Sm^{3+}$

**Solution**

First, write out the electron configuration for each parent atom. We have chosen to show the full, unabbreviated configurations to provide more practice for students who want it, but listing the core-abbreviated electron configurations is also acceptable.

Next, determine whether an electron is gained or lost. Remember electrons are negatively charged, so ions with a positive charge have *lost* an electron. For main group elements, the last orbital gains or loses the electron. For transition metals, the last *s* orbital loses an electron before the *d* orbitals.

a. Na: $1s^2 2s^2 2p^6 3s^1$. Sodium cation loses one electron, so $Na^+$: $1s^2 2s^2 2p^6 3s^1$ = $Na^+$: $1s^2 2s^2 2p^6$.
b. P: $1s^2 2s^2 2p^6 3s^2 3p^3$. Phosphorus trianion gains three electrons, so $P^{3-}$: $1s^2 2s^2 2p^6 3s^2 3p^6$.
c. Al: $1s^2 2s^2 2p^6 3s^2 3p^1$. Aluminum dication loses two electrons $Al^{2+}$: $1s^2 2s^2 2p^6 3s^2 3p^1$ = $Al^{2+}$: $1s^2 2s^2 2p^6 3s^1$.
d. Fe: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$. Iron(II) loses two electrons and, since it is a transition metal, they are removed from the 4*s* orbital: $Fe^{2+}$: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ = $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$.
e. Sm: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^6$. Samarium trication loses three electrons. The first two will be lost from the 6*s* orbital, and the final one is removed from the

اتحاد سفارة البوليتكنك (spu)

4*f* orbital. $Sm^{3+}$: $1s^22s^22p^63s^23p^64s^23d^{10}4p^65s^24d^{10}5p^66s^24f^6$ = $1s^22s^22p^63s^23p^64s^23d^{10}4p^65s^24d^{10}5p^64f^5$.

Exercise

a. Which ion with a +2 charge has the electron configuration $1s^22s^22p^63s^23p^63d^{10}4s^24p^64d^5$?
b. Which ion with a +3 charge has this configuration?

## Summary

The relative energy of the subshells determine the order in which atomic orbitals are filled (1*s*, 2*s*, 2*p*, 3*s*, 3*p*, 4*s*, 3*d*, 4*p*, and so on). Electron configurations and orbital diagrams can be determined by applying the Pauli exclusion principle (no two electrons can have the same set of four quantum numbers) and Hund's rule (whenever possible, electrons retain unpaired spins in degenerate orbitals).

Electrons in the outermost orbitals, called valence electrons, are responsible for most of the chemical behavior of elements. In the periodic table, elements with analogous valence electron configurations usually occur within the same group. There are some exceptions to the predicted filling order, particularly when half-filled or completely filled orbitals can be formed. The periodic table can be divided into three categories based on the orbital in which the last electron to be added is placed: main group elements (*s* and *p* orbitals), transition elements (*d* orbitals), and inner transition elements (*f* orbitals).

# Electronic Structure and Periodic Properties



## Electron Orbitals

- The properties of elements result from the **electronic structure** of the atom
- Each electron in an tom is in an **orbital** of fixed energy; that is the orbitals are said to be **quantized**
- Each electron is described by **3 quantum numbers**

**i) Principle quantum number n:**

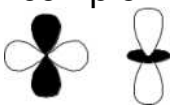
n = 1,2,3, …. This quantum number determines the energy of the orbital

ii) **Angular momentum quantum number, ℓ:**

ℓ = 0, 1, 2, …., n-1 This quantum number determines the shape of various orbitals having the energy of quantum number n



اتحاد سنافر البوليتكنك (spu)

| ℓ | orbital name | shape |
|---|---|---|
| 0 | s | spherical |
| 1 | p | dumbbell  |
| 2 | d | complex  |
| 3 | f | complex |



**iii) Magnetic quantum number mℓ:**

mℓ = -ℓ, …,0, …+ℓ in integer steps.

This quantum number determines the number of orbitals of shape ℓ and energy n = (2ℓ + 1)

∴ For a given n

| orbital name | # of orbitals |
|---|---|
| s | 1 |
| p | 3 |
| d | 5 |
| f | 7 |



(spu) اتحاد سنافر البولبكنك

iv) **Electron spin quantum number, $m_s$:**

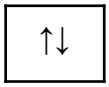
Electrons act as though they have an intrinsic **spin** which gives them a magnetic moment

Thus, unpaired electrons are acted on by magnetic fields and are said to be **paramagnetic**
substances that are unaffected by magnetic fields are said to be **diamagnetic**

The two possible spin values for an electron are a consequence of the electron spin quantum number $m_s$ where $m_s = \pm ½$

These spins are often labeled "up" (↑) and "down" (↓) (although this is actually incorrect; electron spin doesn't depend on coordinates!)

A maximum of 2 electrons can occupy each electron orbital, and then, only when their spins are **paired:**





This is a consequence of the **Pauli Exclusion Principle**:

only 2 electrons maximum can be in any given atomic orbital;

two electrons in the same orbital can not have the same quantum numbers

The maximum number of electrons for any principle quantum number n, is given by $2n^2$.

For example: if $n = 2$ , $2n^2 = 2(2)^2 = 8$: 2 in the one s-orbital + 6 in the three p-orbitals



اتحاد سنافر للبوليتكنك (spu)

## Shape of the orbitals

In quantum mechanics the "shape" of an orbital is defined as 90% probability maps of where spatially one could expect to find the electron in question.

It says nothing about the actually motion (trajectory) of the electron.

We don't indicate 100% probability maps because if the electron exists it must be in the universe somewhere and that's too big to draw!

Luckily, electrons tend to stick "close" to the nucleus!

**i) s-orbital**

If $\ell = 0$, $m_\ell = 0$ which means there can only be 1 s-orbital for any given principle quantum number n.



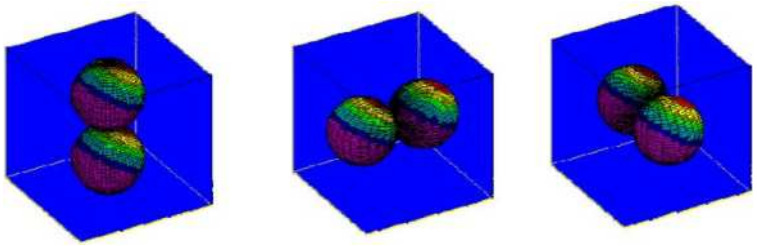
Notation: if, for example the s-orbital for n = 2 has just one electron we denote this by writing: $2s^1$. If it has two electrons (with opposite spins) then $2s^2$.



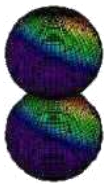
**ii) p-orbitals**

For a given n, if $\ell = 1$, $m_\ell = +1, 0, -1$.

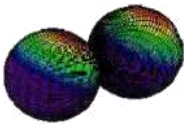
Note the rules indicate that there are no p-orbitals for n = 1 since $\ell_{max} = n-1$.



$\ell = 1$, $m_\ell = 0$ $\ell = 1$, $m_\ell = 1$ $\ell = 1$, $m_\ell = -1$



$\ell = 1, m_\ell = 0$



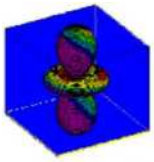
$\ell = 1, m_\ell = +1$ or -1



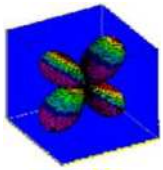
**iv) d-orbitals**

For a given n, if $\ell = 2$, $m_\ell = +2, +1, 0, -1, -2$.

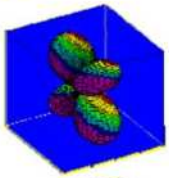
Note the rules indicate that there are no d-orbitals for n = 1 and 2 since $\ell_{max} = n-1$.



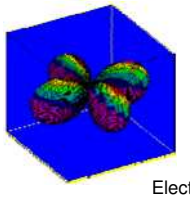
$\ell = 2, m_\ell = 0$



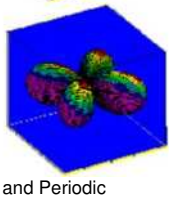
$\ell = 2, m_\ell = 1$



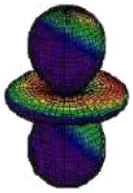
$\ell = 2, m_\ell = -1$



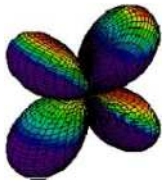
$\ell = 2, m_\ell = 2$



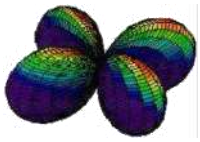
$\ell = 2, m_\ell = -2$



$\ell = 2, m_\ell = 0$



$\ell = 2, m_\ell = 1$ or -1



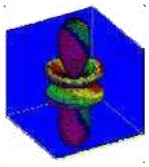
$\ell = 2, m_\ell = 2$ or -2



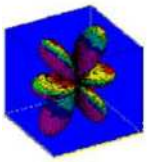
**v) f-orbitals**

For a given n, if $\ell = 3$, $m_\ell = +3, +2, +1, 0, -1, -2, -3$.

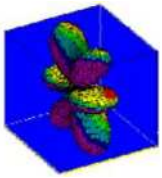
Note the rules indicate that there are no f-orbitals for n = 1, 2, and 3 since $\ell_{max} = n-1$.



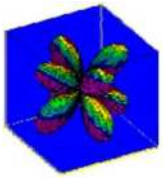
$\ell = 3, m_\ell = 0$



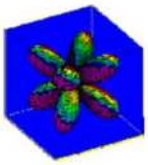
$\ell = 3, m_\ell = 1$



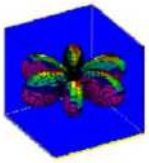
$\ell = 3, m_\ell = -1$



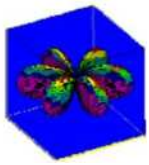
$\ell = 3, m_\ell = 2$



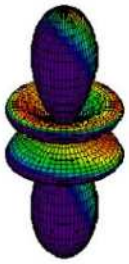
$\ell = 3, m_\ell = -2$



$\ell = 3, m_\ell = 3$



$\ell = 3, m_\ell = -3$



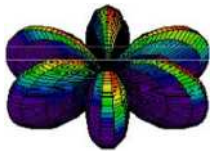
$\ell = 3, m_\ell = 0$



$\ell = 3, m\ell = 1$ or -1



$\ell = 3, m\ell = 2$ or -2



$\ell = 3, m\ell = 3$ or -3



## Electronic Configuration of the Elements

As the number of electrons increases with the number of protons of a **neutral atom**, they occupy orbitals of increasing energy:

The possibilities are:

Energy ↓

| n | ℓ | $m_\ell$ | $m_s$ | name | max # $e^-$ | $2n^2$ |
|---|---|---|---|---|---|---|
| 1 | 0 | 0 | ±½ | 1s | 2 | 2 |
| 2 | 0 | 0 | ±½ | 2s | 2 | |
| | 1 | -1,0,+1 | ±½ | 2p | 6 | 8 |
| 3 | 0 | 0 | ±½ | 3s | 2 | |
| | 1 | -1,0,+1 | ±½ | 3p | 6 | 18 |
| | 2 | -2,-1,0,+1,+2 | ±½ | 3d | 10 | |
| 4 | 0 | 0 | ±½ | 4s | 2 | |
| | 1 | -1,0,+1 | ±½ | 4p | 6 | |
| | 2 | -2,-1,0,+1,+2 | ±½ | 4d | 10 | 32 |
| | 3 | -3,-2,-1,0,+1,+2,+3 | ±½ | 4f | 14 | |
| … | … | … | … | … | … | |

For orbital energies:
n = 4 > n = 3 > n= 2 > n = 1
and for ℓ:
f > d > p > s

In general for orbital energies:

1) Orbital energies increase with increasing (n+ℓ)

2) For (n+ℓ) of equal value, that of lower n is more stable

For example:

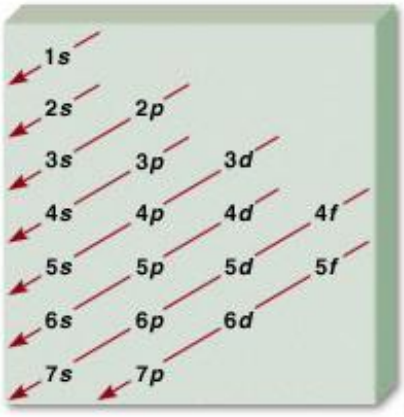
**4s** n = 4ℓ = 0 n+ℓ = 4
**3d** n = 3ℓ = 2 n+ℓ = 5

- Order of energy is 3d > 4s ≡ "(n+ℓ)" rule

A simple guide to the order in which electrons fill orbitals is:



**Order in Which Atomic Subshells are Filled**




**اتحاد متأثر البوليمك (spd)**

Since each orbital can hold a maximum of $2e^-$, it is possible to construct the electronic configuration of elements as follows:

Example: for $^{13}Al = 1s^2 2s^2 2p^6 3s^2 3p^1 \equiv 13\ e^-$

Here $1s^2$ means 2 electrons in the 1s orbital; $2p^6$ means 6 electrons in the 2p orbitals, etc.

This notation is termed **spectroscopic notation**

A shorter form of the notation gives the symbol for the rare gas whose configuration is identical to that of the closed shells of an element

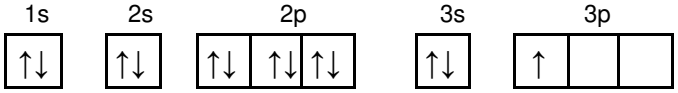
- for $^{13}Al$ above: $1s^2 2s^2 2p^6 \equiv$ configuration for $^{10}Ne$
- notation becomes $^{13}Al = [Ne]3s^2 3p^1$

Note: electrons in the orbital of highest n are called **valence electrons**; the rest are **core electrons**



If you wish to illustrate orbitals occupied by electrons more clearly you can use an **orbital box** diagram:

Example: $^{13}Al \equiv$




Question: Use spectroscopic notation to describe the electron configuration for $^{50}Sn$.
What are the quantum numbers of the highest energy electron?

$^{50}Sn = [1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6] 5s^2 4d^{10} 5p^2$

$= [Kr] 5s^2 4d^{10} 5p^2$

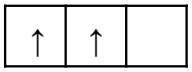
Main question is how to distribute the 2 highest energy $e^-$ in the 3 5p-orbitals.

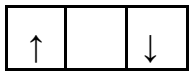
Filling of orbitals to give ground state configurations follows **Hund's Rule.**



اتحاد سنافر للبوليتكنك (spu)

Hund's rule states that electrons occupy orbitals of equal energy in such a way that the number of unpaired electrons is at a maximum, thus minimizing inter-electronic repulsion

For $^{50}Sn$ this means $5p^2$ ≡ 

but not:  or 

- quantum numbers of the highest energy electron are n = 5, ℓ = 1, $m_ℓ$ = -1, 0, or +1, $m_S$ = +½ or -½



Note : 3d/4s and 4d/5s orbitals are very close in energy.
Therefore, sometimes the scheme does not work for transition metals.

Consider the electronic configuration of $^{24}Cr$

Might expect electronic configuration = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^4$
$=[Ar]4s^2 3d^4$
Not the case.
Instead it is energetically favorable to have both the
3d and 4s orbitals half filled to minimize electron repulsions.
≡ Hund's Rule

Similarly complete 3d orbitals are favorable eg $^{29}Cu$.
The energy of 3d falls below that of 4s because of the increased nuclear charge.
Electronic configuration of $^{29}Cu$ = $[Ar]4s^1 3d^{10}$

Cu and Cr belong to the first series of the transition metals.
Second series metals behave similarly.



اتحاد منافر انبوبكنك (spu)

## Atomic Ions

When an atom loses electrons to for a cation, the electrons are always lost from the valence shale of **highest n**

Thus, for transition metals, the electrons remaining over the noble gas core are always found in the 3d orbitals



## The Periodic Table

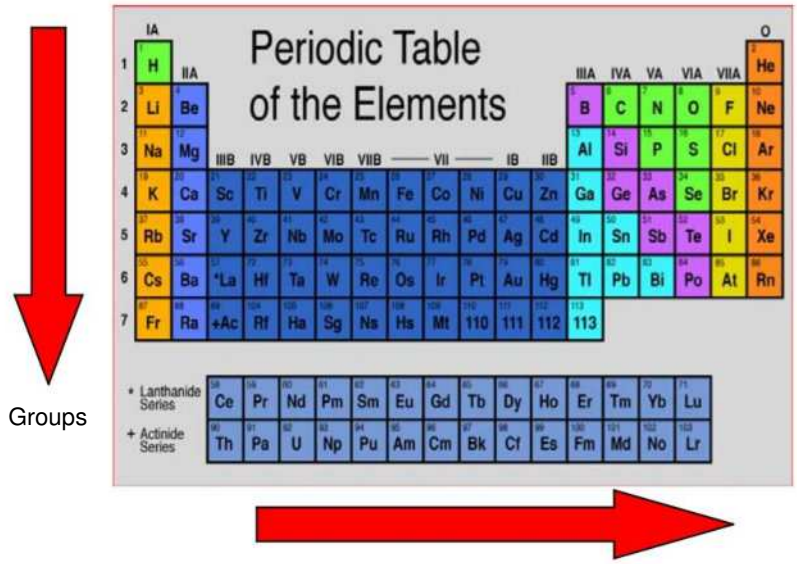
When the elements are grouped together on the basis of their physical and chemical properties, they fall into groups illustrated by the **periodic table**

First devised by D. Mendeleev long before the advent of quantum theory





**اتحاد سنافر البولينيكنك (spu)**






20th century chemistry has shown that these groups have **similar electronic configurations**

This demonstrating that the behavior of the elements is a consequence of the **number and configuration of their electrons only**:

The atomic nucleus is unimportant chemically

The electrons that are the most important in determining the properties of an element are those in the **highest energy**, **incompletely filled orbitals**:

≡ the **valence electrons**

**Main group elements** are those with s or p subshells being filled, with other subshells being full or empty

Those with d orbitals being filled are termed **transition metal** elements. Each d subshell holds 10 electrons

Elements in the f-block are those with f-orbitals being filled. These orbitals hold 14 electrons.
The elements with the 4f orbitals being filled are called the **lanthanides**
Those with the 5-f orbitals being filled are called the **actinides**



اتحاد سنافر البوليتكنك (spu)

Since elements in any one group have the same number of valence electrons, it is no surprise that they have similar chemical properties

| | | |
|---|---|---|
| **Group 1A** Li, Na, K, Rb, Cs, Fr | Alkali metals | valence config. = $ns^1$ |
| **Group 2A** Be, Mg, Ca, Sr, Ba, Ra | Alkaline Earths | valence config. = $ns^2$ |
| **Group 3A** (13) B, Al, Ga, In, Tl | | valence config. = $ns^2np^1$ |
| **Group 4A (14)** C, Si, Ge, Sn, Pb | | valence config. = $ns^2np^2$ |
| **Group 5A (15)** N, P, As, Sb, Bi | | valence config. = $ns^2np^3$ |
| **Group 6A (16)** O, S, Se, Te, Po | Chalcogens | valence config. = $ns^2np^4$ |
| **Group 7A (17)** F, C, Br, I, At | Halogens | valence config. = $ns^2np^5$ |
| **Group 8A (18)** He($1s^2$), Ne, Ar, Kr, Xe, Rn | Noble gases | valence config. = $ns^2np^6$ |



Noble gases are extremely stable and not very reactive due to their full valence shell.

Other elements can lose or gain electron to attain the same electronic configuration as these noble gases.

For example:
K tends to give away one electron and Cl to accept one electron to attain the Ar configuration



اتحاد سنافر للبوليتكنك (spu)

## Periodic Trends in Atomic Properties

**1. Atomic Size**

The atomic radius reflects the distance from the nucleus to the highest occupied electronic orbital.
Therefore we expect the following:

i) Atomic radii increase from top to bottom within a group.
For example: r(Li) < r(Na) < r(K) < r(Rb) < r(Cs)

ii) Atomic radii increase from left to right across a period
For example: r(Li) > r(Be > r(B) > r(C) > r(N) >r(O) > r(F)

This is because the electrons in the same n-orbital (here n = 2) are more strongly attracted to the greater number of protons in the nuclei of the heavier elements

Cations are smaller than the neutral parent atom. For example $r(Li^+) < r(Li)$
Anions are larger than the neutral parent atom. For example $r(Cl^-) > r(Cl)$








اتحاد سنافر للبولايتكنك (spu)

## Ionization Energies

Ionization energy, IE, is the energy required to remove one electron from an atom or ion;
an **endothermic** process

that is, $A \rightarrow A^+ + 1\ e^-$ $\Delta H = +ve$

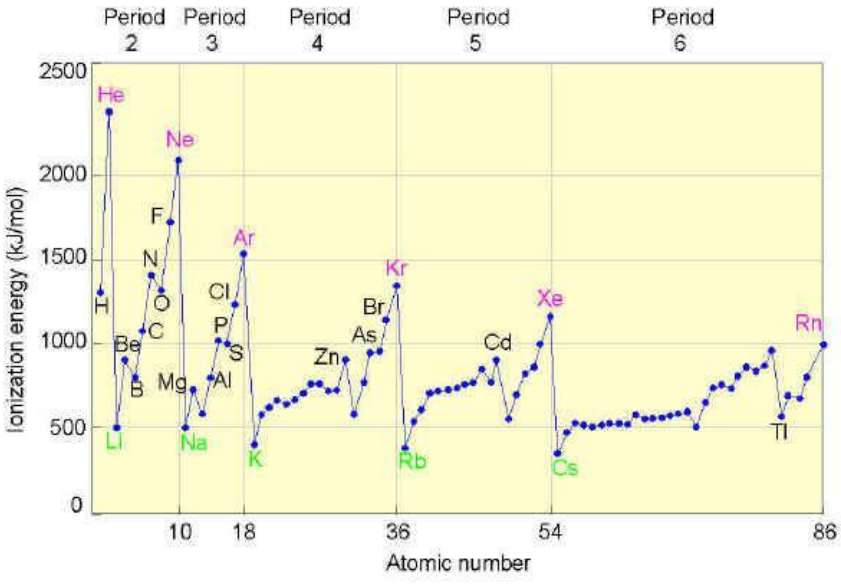
The energy, in kJ $mol^{-1}$, required to remove 1 electron from the neutral atom ($1^{st}$ ionization limit) depends on the orbital in which the electron resides

Two general trends:

1.) IEs **decrease down a group**
(an electron removed from an orbital more distant from the nucleus is less tightly bound)

2.) IEs **increase from left to right** across a period
(electrons being removed from orbitals of equal n are subject to increasing nuclear attraction)








اتحاد سنافر للبولينكنك (spu)

There are some exceptions to generality 2).

They occur whenever the removal of $1e^-$ leads to a filled or half-filled orbital.

The extra stability of filled or half-filled orbitals leads to a lower IE for those elements which are one electron away from such a state

Example: IE of B < Be

B ≡ $1s^2 2s^2 2p^1$  Be ≡ $1s^2 2s^2$

$2p^1$ electron more easily lost than 1 electron from the filled 2s orbital

Example: IE O < N

O ≡ $1s^2 2s^2 2p^4$  N ≡ $1s^2 2s^2 2p^3$

By losing $1e^-$ O attains half-filled 2p-orbitals



## Electron Affinities

Electron affinity, EA, is the energy change that results from the **capture of an electron** by a neutral atom

that is, $A + e^- \rightarrow A^-$  EA (=$\Delta$H which can +ve or -ve)

Again there are two general trends:

1.) EAs decrease down a group

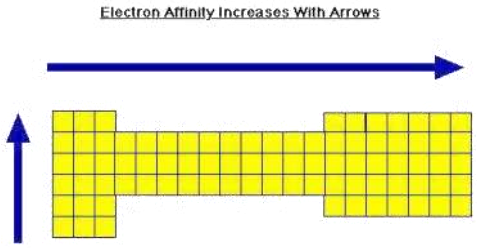
(because $e^-$s are being captured in orbitals more distant from the nucleus)

2.) EAs increase across a period

(because $e^-$s are being captured in orbitals exposed to a greater nuclear charge)



اتحاد سنافر للبوليتكنك (spu)

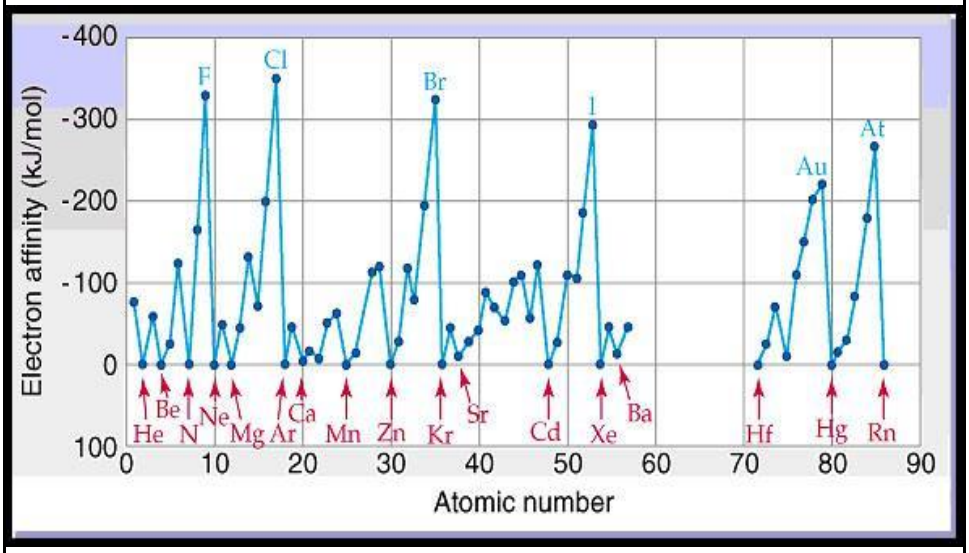



EAs are especially large (large and –ve) when capture of an $e^-$ leads to filled or half-filled orbitals.

Example: $C + e^- \rightarrow C^-$

$C \equiv 1s^2 2s^2 2p^2$ $C^- \equiv 1s^2 2s^2 2p^3$





## Electronegativity

Electronegativity is the tendency of an atom in a covalently bonded molecule to attract the bonding electron pair to itself

It follows the same trend as electron affinity;

that is, it increases going up a group and across a period.

Values are relative and not absolute

The greatest electronegativity values are found with small non-metals.

F is the most electronegative element.

Electronegativity is important in bonding because a bond between two atoms of different electronegativity is polarized, with the $e^-$ pair closer to the more electronegative atom



| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| H 2.1 | | | | | | | | | | | | | | | | | He |
| Li 1.0 | Be 1.5 | | | | | | | | | | | B 2.0 | C 2.5 | N 3.0 | O 3.5 | F 4.0 | Ne |
| Na 0.9 | Mg 1.2 | | | | | | | | | | | Al 1.5 | Si 1.8 | P 2.1 | S 2.5 | Cl 3.0 | Ar |
| K 0.8 | Ca 1.0 | Sc 1.3 | Ti 1.5 | V 1.6 | Cr 1.6 | Mn 1.5 | Fe 1.8 | Co 1.8 | Ni 1.8 | Cu 1.9 | Zn 1.6 | Ga 1.6 | Ge 1.8 | As 2.0 | Se 2.4 | Br 2.8 | Kr 3.0 |
| Rb 0.8 | Sr 1.0 | Y 1.2 | Zr 1.4 | Nb 1.6 | Mo 1.8 | Tc 1.9 | Ru 2.2 | Rh 2.2 | Pd 2.2 | Ag 1.9 | Cd 1.7 | In 1.7 | Sn 1.8 | Sb 1.9 | Te 2.1 | I 2.5 | Xe 2.6 |
| Cs 0.7 | Ba 0.9 | La 1.1 | Hf 1.3 | Ta 1.5 | W 1.7 | Re 1.9 | Os 2.2 | Ir 2.2 | Pt 2.2 | Au 2.4 | Hg 1.9 | Ti 1.8 | Pb 1.8 | Bi 1.9 | Po 2.0 | At 2.2 | Rn 2.4 |
| Fr 0.7 | Ra 0.7 | Ac 1.1 | Unq | Unp | Unh | Uns | Uno | Une | | | | | | | | | |

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ce 1.1 | Pr 1.1 | Nd 1.1 | Pm 1.1 | Sm 1.1 | Eu 1.1 | Gd 1.1 | Tb 1.1 | Dy 1.1 | Ho 1.1 | Er 1.1 | Tm 1.1 | Yb 1.1 | Lu 1.2 |
| Th 1.3 | Pa 1.5 | U 1.7 | Np 1.3 | Pu 1.3 | Am 1.3 | Cm 1.3 | Bk 1.3 | Cf 1.3 | Es 1.3 | Fm 1.3 | Md 1.3 | No 1.3 | Lr |



اتحاد سنافر للبولينكنك (spu)

## Chapter 7. Electronic structure of atoms

Much information about atomic electronic structure was obtained from studies on the interaction of electromagnetic radiation with matter.

- Electromagnetic radiation carries energy through space & has a wavelike nature. E.g. light, x-rays, microwaves
- Each wave has a characteristic wavelength and frequency.

**wavelength, $\lambda$** : distance between wave peaks. Units: m

**frequency, $\nu$** : # of cycles (complete waves) that pass a point in one second. Units are hertz. 1 Hz = 1 $s^{-1}$

In a vacuum, all electromagnetic radiation travels at a speed of 3.00 x $10^8$ m/s. This is the speed of light, c.

$$c = \nu \lambda$$

units: m/s = $s^{-1}$m

- frequency and wavelength are inversely proportional

**EM spectrum**

| Gamma | X-rays | UV | Visible | Infrared | Microwaves | radio waves |
|---|---|---|---|---|---|---|

$\sim 10^{-12}$ m → wavelength increases → $10^4$ m

$\sim 10^{20}$ $s^{-1}$ ← frequency increases ← $10^4$ $s^{-1}$

- Energy of radiation is proportional to $\nu$ and inversely proportional to $\lambda$.
- Visible rays: ROY G BIV represents colors in spectrum
  R has highest $\lambda$ (~750 nm), V has lowest (~400 nm)

### *6.2 QUANTIZED ENERGY*

1900 - Max Planck introduced the theory of "quantum packets of energy". This theory states that energy can only be absorbed or released from atoms in certain amounts of Energy. Thus, E is quantized, not continuous.

$E = h\nu$ h = Planck's constant = 6.63 x $10^{-34}$ J-s

- For a given frequency, only "chunks" of energy equaling $h\nu$, $2h\nu$, $3h\nu$, etc. are observed.
- $h\nu$ = smallest amount of energy (This is a **quantum**).

1905 - Einstein used Plank's theory to explain the photoelectric effect. He assumed light traveled in energy packets called photons.

1 **photon** = smallest increment of radiant energy

- Energy of 1 photon: $E = h\nu$
- Thus light has both wave-like characteristics (EM studies) & particle nature (Planck & Einstein)

***Problem.*** Calculate the smallest increment of energy that can be emitted or absorbed at a wavelength of 645 nm.

$$\lambda = 645 \text{ nm} \frac{10^{-9} m}{1 nm} = 6.45 \times 10^{-7} \text{ m} \qquad \nu = \frac{c}{\lambda} = \frac{3.0x10^8 \, m/s}{6.45x10^{-7} \, m} = 4.65x10^{14} \text{ s}^{-1}$$

**(spu) إتحاد سنافر البوليتكنك**

$E = h\nu = (6.63x10^{-34} \text{ J·s})(4.65x10^{14} s^{-1}) =$ **$3.08x10^{-19}$ J**

***Problem.*** What frequency and wavelength of radiation has photons of energy $8.23 \times 10^{-20}$ J? What type of electromagnetic radiation is this (refer to EM spectrum)?

$E = h\nu$ $\quad \nu = E/h = 8.23\times10^{-20}\ J/(6.63\times10^{-34}\ J\cdot s) = 1.24\times10^{14}\ s^{-1}$

$c = \lambda\nu$ $\quad \lambda = c/\nu = \dfrac{3.0\times10^{8}\ \frac{m}{s}}{1.24\times10^{14}\ s^{-1}}$ = **$2.41\times10^{-6}$ m (infrared portion)**

**Line spectra**

- When white light is passed through a prism, it separates into a **continuous spectrum** of all wavelengths of visible light. (ROYGBIV)
- When the light from a heated element passes through a prism, **a line spectrum** with distinct lines is observed. Each line corresponds to a specific wavelength of visible light.
- Each atom has its own unique line spectrum.

***BOHR MODEL ~1913***

- Bohr tried to explain observed line spectra based on movement of electrons.
- - charged e-'s travel rapidly in orbits around the tiny + charged nucleus

1) Electrons are contained in specific energy levels called orbits. These energy levels are quantized which means only certain energies are allowed. An e- in a permitted orbit has a specific energy.

- Energy levels are designated by the principal quantum number, n. n = 1,2,3… n = 1 is **ground state** level - this is level closest to nucleus (lowest in E).

2) Electrons circle the nucleus at specific radii. ($r \propto n^2$)

3) Electrons can jump from one level to another by absorbing or emitting photons of specific frequencies. (e- must gain energy from heat, radiation, etc. to jump to higher level.)

- Bohr's theory explains 4 observed lines in line spectra for hydrogen. Lines correspond to emitted radiation in visible portion of the EM spectrum when e-jumps from 1 level to another.
- This process is responsible for colors of fireworks & neon signs. Electrons are excited by heat or electricity and electrons jump to higher energy levels. Light is emitted when electrons lose energy and drop back down to lower energy levels. The colors correspond to wavelengths of emitted light waves.

***WAVE BEHAVIOR OF MATTER***

DeBroglie proposed that e-`s move in wave-like manner around the nucleus and he derived an equation relating the momentum of a particle to its wavelength.

**Heisenberg's Uncertainty Principle** – It is impossible to know both the position and the momentum of a particle simultaneously.

**(spu) نادي سفار البوليتكنك**

**Reason's**

1) due to tiny size e- is affected by many forces (attraction to nucleus, $e^-$ - $e^-$repulsions, interactions with equipment employed to examine e-'s)
2) e- is oscillating about nucleus

## Schrodinger's model

In 1920's, Schrodinger realized that previous models were oversimplified. He set-up and solved a series of complex equations that took into account:

1) KE of an e-
2) $\lambda$ of an e-
3) attraction of e- for nucleus
4) repulsion's between e-'s

- The solutions to the wave equations are called $\Psi$ (psi), wave functions.
- $\Psi^2$ gives the probability of finding an e-. This is called the e- density. Regions of high probability have high e- density - these areas represent orbitals.

**Schrodinger used 3 quantum number's to describe electronic structure**:

1) **n = principal quantum number**, where n is energy shell. Values for n = 1,2,3,4… (n = 1 closest shell to nucleus)
2) **ℓ = azimuthal quantum number**, where ℓ is type of subshell. Know general shapes of s, p, d orbitals. ℓ values: s, ℓ = 0; p, ℓ = 1;d, ℓ = 2; f, ℓ= 3
3) **mℓ = magnetic quantum number**, where mℓ is specific orbital. mℓ describes orientation of orbital in space. Values are -ℓ·· 0·· +ℓ

**Important Results from Schrodinger's model:**

1) Each n shell has n subshells

| | | |
|---|---|---|
| n = 1 | ℓ = 0 | s |
| n = 2 | ℓ = 0, 1 | s, p |
| n = 3 | ℓ = 0, 1, 2 | s, p, d |

2) Each subshell contains a specific number of orbitals: Orbitals are regions in space where there is a high probability of finding an e-. Orbitals don't have set boundaries unlike Bohr's orbits which had a fixed radius.

| Subshell | ℓ value | Region on chart | max # of e- | # of orbitals |
|---|---|---|---|---|
| s | 0 | 1A & 2A | 2 | 1 |
| p | 1 | 3A - 8A | 6 | 3 |
| d | 2 | Transition metals | 10 | 5 |
| f | 3 | Lanthanides & Actinides | 14 | 7 |

3) # of orbitals in a shell = $n^2$

e.g. n = 1 # orbitals = 1; n = 3 # orbitals = 9

E level diagram shows how shells and subshells are ranked according to their energy.

- For atoms other than H, more than 1 e- is present and due to shape of orbitals & e-e- repulsions, subshells within a shell have different energies. (s < p < d < f)

(spu) اتحاد سنافر اليم ليتكنك

- Also e- in a many electron atom experience an effective nuclear charge, $Z_{eff}$. This is the net charge felt by an e-. $Z_{eff}$ is less than the full nuclear charge, due to screening or shielding by inner shell electrons.

***ELECTRON CONFIGURATIONS AND ORBITAL DIAGRAMS***

**4th quantum #**: **$m_s$ = spin quantum number**, where $m_s$ indicates direction of electron spin. Values are + 1/2, -1/2

1) **Pauli Exclusion principle**: No two electrons may have same 4 quantum numbers. Only 2 e- can occupy an orbital.
2) **Aufbau principle**: Orbitals are filled in order of increasing energy.
3) **Hund's rule**: When filling degenerate orbitals (same subshell), 1 e- is placed in each orbital before pairing up.

An electron configuration describes how many e- an atom has in each of its subshells.

| | | | |
|---|---|---|---|
| H (Z = 1) | $1s^1$ | He (Z = 2) | $1s^2$ |
| C (Z = 6) | $1s^2 2s^2 2p^2$ | F (Z = 9) | $1s^2 2s^2 2p^5$ |
| Mg (Z = 12) | $1s^2 2s^2 2p^6 3s^2$ | S (Z = 16) | $1s^2 2s^2 2p^6 3s^2 3p^4$ |

**Noble gas core** - can use noble gas from previous row as short cut.

| | | |
|---|---|---|
| Na (Z = 11) | $1s_2 2s_2 2p_6 3s_1$ | [Ne]$3s_1$ |
| P (Z = 15) | [Ne]$3s_2 3p_3$ | |
| Cl (Z = 17) | [Ne]$3s_2 3p_5$ | |

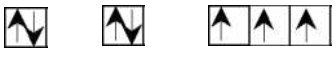
- Energies of some levels overlap. Consequently, 4s level fills before 3d.
- Note that the 3d subshell is located in 4th period of periodic table.
- d subshells: (n-1)d
- f subshells: (n-2)f

| | | |
|---|---|---|
| Fe (Z = 26) | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ | or [Ar] $4s^2 3d^6$ |
| As (Z = 33) | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$ | or [Ar] $4s^2 3d^{10} 4p^3$ |
| Ir (Z = 77) | [Xe] $6s^2 4f^{14} 5d^7$ | |

**Orbital diagrams - box and arrow representation**

**Complete orbital diagram** - based on complete electron configuration

**N**



1s 2s 2p

**Outer orbital diagram** - based on noble gas core method:

**N**



2s 2p

Fe



4s 3d

# CHAPTER 7: MODELS OF THE ATOM

## 7.1 DALTON MODEL OF THE ATOM

1. An element is composed of tiny, indivisible, indestructible particles called atoms.
2. All atoms of an element are identical and have the same properties.
3. Atoms of different elements combine to form compounds.
4. Compounds contain atoms in small whole number ratios.
   e.g. Each $H_2O$ molecule consists of one O and 2 H atoms
5. Atoms can combine to form different compounds.
   e.g. carbon and oxygen can form $CO_2$ or CO

## 7.2 THOMSON MODEL OF THE ATOM

J.J. Thomson was given credit for discovery of electron (1897)

⇒ Cathode rays are deflected by electric fields & magnetic fields, so cathode rays are composed of tiny, negatively charged subatomic particles called electrons **($e^-$)**

E. Goldstein (1886) discovered *positively* charged subatomic particles called protons **(p+)**

**Plum-pudding Model of the Atom - very short lived model!**

J.J. Thomson proposed a model where an atom is a positively, charged sphere with electrons scattered about like raisins in plum pudding.

James Chadwick discovered neutrons (1932): he won Nobel Prize for this in 1935

⇒ **(n) =** *neutral* subatomic particle

## 7.3 RUTHERFORD'S MODEL OF THE ATOM Rutherford's

- Alpha ($\alpha$) particles shot at a thin gold foil surrounded by a detector
- Most of $\alpha$ particles went straight through, **but** a few actually bounced backwards
  - It was almost like a bullet ricocheting off a piece of *tissue paper*!

**Rutherford's interpretation of the experimental results**

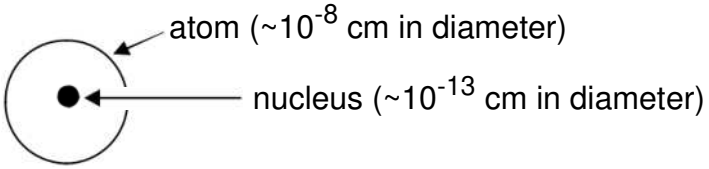
1. Most alpha ($\alpha$) particles passed through foil
   ⇒ The atom is mostly empty space with electrons moving around the space.
2. Some $\alpha$ particles were deflected or bounced back
   ⇒ Atom contains a *small dense region* and when $\alpha$ particles strike this region they recoil.
   ⇒ dense region = **atomic nucleus** (contains atom's protons and neutrons)

---

**Rutherford's Planetary Model of the Atom**

⇒ *Negatively charged* **$e^-$'s** move around the *positively charged* nucleus

Rutherford also estimated the size of the atom and its nucleus:




If nucleus = size of a small marble, then atom ≅ Cardinal's new stadium!

| Particle | Symbol | Location | Charge | Relative Mass (amu) |
|---|---|---|---|---|
| **electron** | $e^-$ | outside nucleus | –1 | 1/1836 ≈ 0 |
| **proton** | $p^+$ | inside nucleus | +1 | 1 |
| **neutron** | n | inside nucleus | 0 | 1 |

## 7.4 ATOMIC NOTATION

Every atom of an element has the same number of protons.

**Atomic Notation (also called "Nuclear Symbol"):**

– shorthand for keeping track of protons and neutrons in the nucleus

**atomic number**: total number of $p^+$

**mass number**: total number of $p^+$ and n in an atom's nucleus

$$\text{mass number} = \mathbf{A}\ \ E \quad E = \text{element symbol}$$

$$\text{atomic number} = \mathbf{Z}$$

total number of n = A - Z

For neutral atoms: total number of $p^+$ = total number of $e^-$

**Isotopes:** Atoms of an element that have a different number of neutrons. Isotopes of an element have the same atomic number, but a different mass number.

⇒ We often refer to a specific isotope of an element by giving the name of the element followed by the mass number.

e.g. carbon-12 (C-12), carbon-13 (C-13) and carbon-14 (C-14) are isotopes of carbon

Ex. 1: a. Write the atomic notation for sodium-23.

b. How many neutrons are in each neutral sodium-23 atom? _____

Ex. 2:a. Write the atomic notation for chlorine-37.

(spu) اتحاد سنافر الجزيرة

b. How many neutrons are in each neutral chlorine-37 atom? _____

Ex. 3: Indicate the mass number, number of protons, neutrons, and electrons:

| Isotope of carbon | mass # | # of protons | # of neutrons | # of electrons |
|---|---|---|---|---|
| carbon-12 | | | | |
| carbon-13 | | | | |
| carbon-14 | | | | |

*Note: Z is sometimes omitted from the atomic notation:*

e.g. $^{13}_{6}C$ or $^{13}C$ can be used to represent carbon-

## 7.5 ATOMIC MASS

Masses of atoms are so small that we define the atomic mass unit (amu) to scale up the numbers.

***Carbon-12*** was chosen as the **reference** and given a mass value of exactly **12 amu**

⇒ Mass of all other atoms scaled relative to mass of C-12

**Atomic Mass of an Element** is ***weighted average*** of all naturally occurring isotopes for that element.

For example, there are two naturally occurring isotopes of carbon: C-12 and C-13
More carbon exists as carbon-12 (98.89%) compared with carbon-13 (1.11%), so the atomic mass reported for carbon (12.01 amu) is closer to carbon-12.

Example: Use the atomic weight reported on the Periodic Table to determine the most abundant naturally occurring isotope for each of the following:

a. lithium-6 or lithium-7 (Circle one)

b. chlorine-35 or chlorine-37 (Circle one)

Some elements are ***radioactive*** and ***unstable.***

⇒ distinguished on the Periodic Table with parentheses around the ***mass number (not the reported atomic mass)*** of the most common isotope for the radioactive element. e.g. Element #96, curium (Cm), has a mass number of 247.

## 7.6 THE WAVE NATURE OF LIGHT

Light has two components: **wavelength** and **frequency**.

**Wavelength** (**λ**) is the distance between peaks on adjacent waves.

**Frequency** (**ν**) is the number of wave cycles completed in one second.

اتحاد سنافر البوليتكنك (spu)

⇒ As wavelength ↑, the frequency ↓, and the energy ↓

⇒ As wavelength ↓, the frequency ↑, and the energy ↑

**Radiant Energy Spectrum**

⇒ Continuous spectrum from gamma rays to radio waves

**Light** we observe with the naked eye falls within the ***visible spectrum***.

⇒ When white light is passed through a prism, it separates into a continuous spectrum of all wavelengths of visible light. (ROYGBIV)

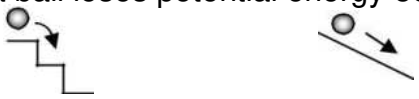
⇒ Range of visible spectrum is from violet (400 nm) to red (700nm).

## 7.7 THE QUANTUM CONCEPT

In 1900, Max Planck proposed the controversial idea that **energy was emitted in small bundles called quanta.** (Energy is not continuously emitted).

⇒ an individual unit of light energy is called a photon

Analogy: A ball loses potential energy in **quantized amounts** when it bounces down a stairway whereas a ball loses potential energy **continuously** if it rolls down a ramp.

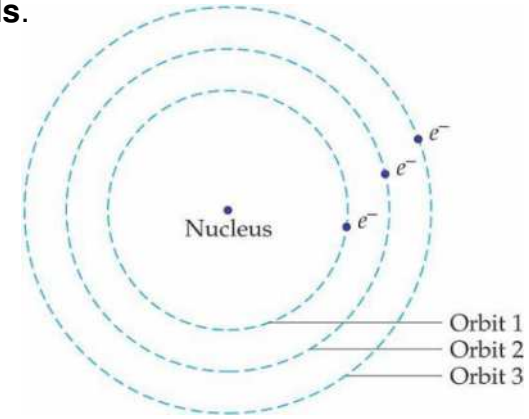


## 7.8 BOHR MODEL OF THE ATOM

In 1913, Neils Bohr proposed that electrons orbit rapidly around the nucleus, occupying circular orbits with distinct **energy levels**.

**Bohr's model of the atom (1913)**

- The electrons orbit around the nucleus like planets orbit around the sun.
- Each electron occupies a specific orbit referred to as an **energy level**.
- Each orbit has a specific radius and a specific energy.
  - ⇒ The orbit closest to the nucleus is lowest in energy; the energy of the orbit increases with distance from the nucleus.



A gas is sealed in a gas discharge tube and energized by electricity. When the light from the heated gas passes through a prism, an **emission line spectrum** with distinct bands of color is observed. The colors correspond to wavelengths of emitted light.

¾ Each atom has its own unique line spectrum – this is virtually an **"atomic fingerprint"** that can be used identify the element.

Bohr used his model to explain the observed line emission spectra for hydrogen gas.

⇒ Electrons gain energy from heat or electricity and jump to a higher energy level. These "excited" electrons ultimately lose energy and drop to lower energy levels, which causes light to be emitted.

⇒ Bohr's theory explains 3 observed lines in line spectra for hydrogen. Each line corresponds to emitted radiation in the visible spectrum when an $e^-$ drops back down to a lower energy level.

*This process is also responsible for the colors we see when fireworks explode & the colors of neon signs.*

## 7.9 ENERGY LEVELS AND SUBLEVELS

Bohr's model could not explain the emission spectra for other elements. It later emerged that each **principal energy level** (numbered 1, 2, 3,…) could be divided into **energy sublevels**.

**principal energy level (n)**: n = 1,2,3,...

**energy sublevels**: s, p, d, and f

Each n level has n sublevels. The sublevels for the first four energy levels are provided below:

| | | |
|---|---|---|
| n = 1 | one sublevel | 1s |
| n = 2 | two sublevels | 2s 2p |
| n = 3 | three sublevels | 3s 3p 3d |
| n = 4 | four sublevels | 4s 4p 4d 4f |

## 7.10 QUANTUM MECHANICAL MODEL OF THE ATOM

In reality, the electron does not move in a fixed orbit. Instead, an electron has a high probability of being found within a given **volume.**

**orbital**: region in space where there is a high probability of finding an electron.

⇒ Each orbital can hold a maximum of 2 electrons.

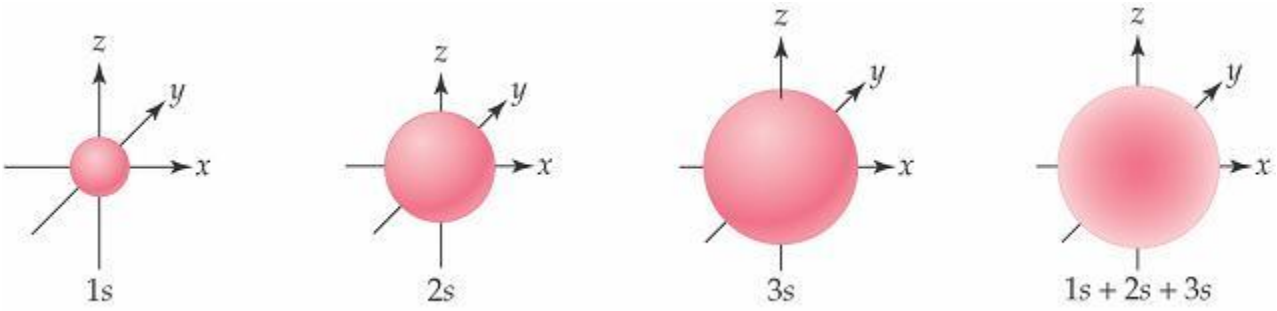
Each sublevel contains a specific number of orbitals.

| Sublevel | Max # of $e^-$ in sublevel | # of orbitals |
|---|---|---|
| s | 2 | 1 |
| p | 6 | 3 |
| d | 10 | 5 |
| f | 14 | 7 |

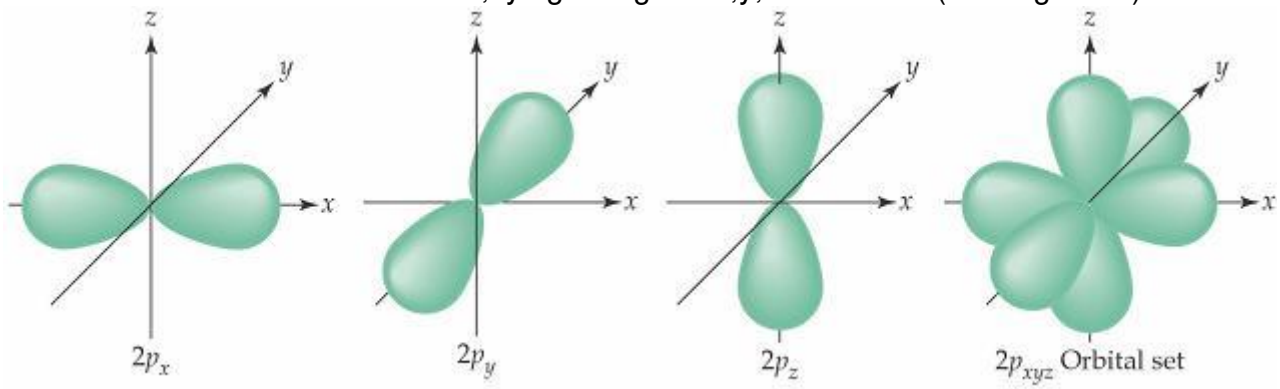
**Sizes and Shapes of Orbitals**

**S orbitals are spherical** (see Fig. 5.17)

⇒ As principal number (n) ↑ , the size and energy of the orbitals ↑




**P orbitals resemble dumbbells**, lying along the x,y, and z axes (see Fig. 5.18)




**Note: You do not need to know about the d orbitals.**

## 5.10 ELECTRON CONFIGURATION

**Electron Configuration:** Shorthand description of the arrangement of electrons by sublevel.

¾ Sublevels are filled in order of increasing energy.

1s < 2s < 2p < 3s < 3p < 4s

⇒ You will only need to know these sublevels to write electron configurations for the 1st 20 elements.

¾ Once a sublevel has the maximum number of electrons it can hold, it is considered "filled." Remaining electrons must then be placed into the next highest energy sublevel, and so on.

***Suggestions for writing electron configurations***

1) Find the atomic number (Z) for the element on the periodic table
   ⇒ this corresponds to the number of electrons in a neutral atom of the element.
2) List all the sublevels for the element in order of increasing energy.
3) Use superscript numbers to indicate number of e⁻'s in each sublevel.
   ⇒ Note: Sum of superscript numbers must = Z

**(spu) اتحاد سنافر البوليتكنك**

Ex. 1 He → atomic #=2 ⇒ _____ e-

electron configuration for He: _____

Ex. 2 C → _____ e-

electron configuration for C: _____

Ex. 3 S → _____ e-

electron configuration for S: _____

Ex. 4 K → _____ e-

electron configuration for K: _____

## 7.11 BLOCKS OF ELEMENTS

The Periodic Table actually corresponds to the order of energy sublevels. The sublevels correspond to blocks of elements on the periodic table.

⇒ Fig. 6.6 on p. 152 shows the energy sublevels for the s, p, d and f block elements.

Elements in the last column of the Periodic Table are called "Noble gases."
All of the electrons in the Noble gases (Group 8A) occupy completely filled orbitals.

⇒ Such electrons are called "**core electrons**" since they are more stable (less reactive) when they belong to completely filled orbitals.

Noble gas electron configurations can be used to abbreviate the "core electrons".
An electron configuration using the Noble gas from the previous row is called "**core notation**".

[He] = $1s^2$
[Ne] = $1s^2\ 2s^2\ 2p^6$
[Ar] = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

**Electron Configurations using Core Notation:**

a. Electron configuration for K using full notation: **$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$** $4s^1$

Electron configuration for K using core notation: **[Ar]** $4s^1$

b. Electron configuration for Cl using full notation: _____

Electron configuration for Cl using core notation: _____

c. Electron configuration for Mg using full notation: _____

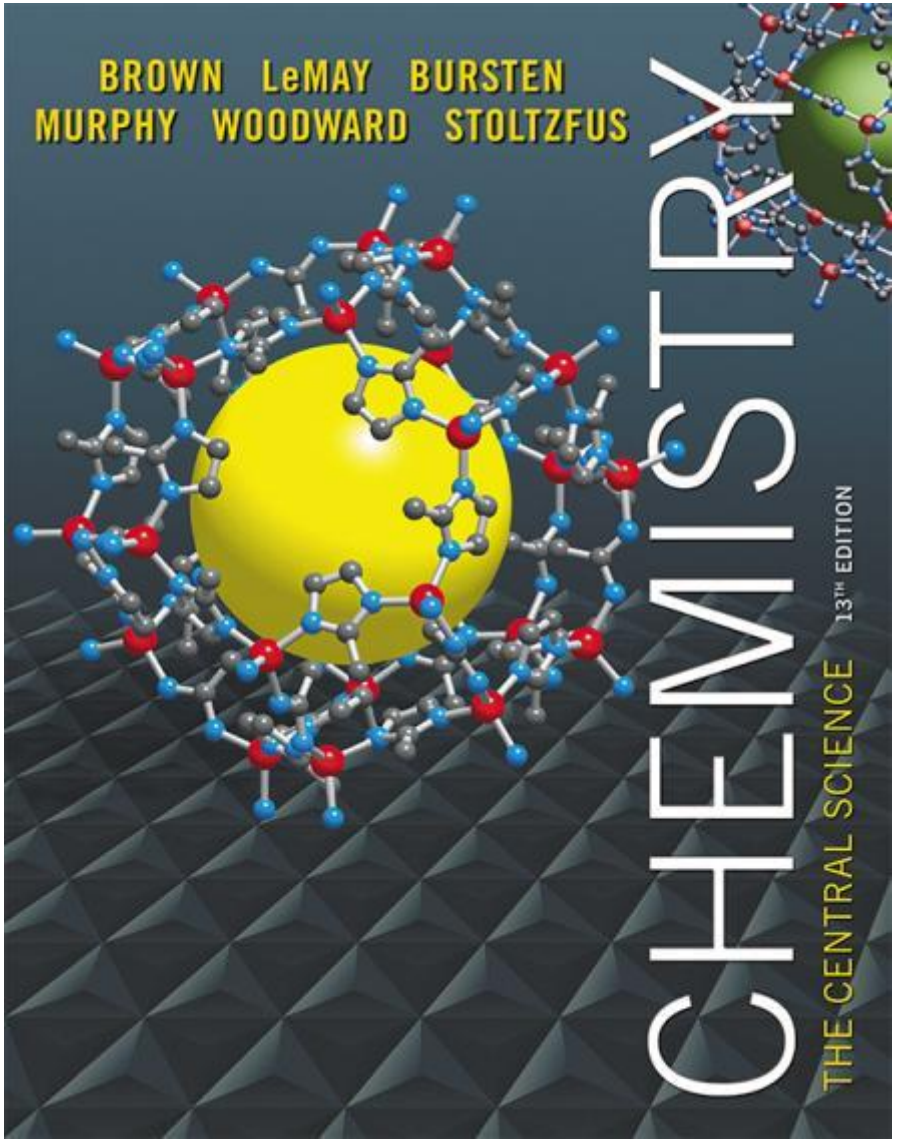
Electron configuration for Mg using core notation: _____

d. Electron configuration for Si using full notation: _____

Electron configuration for Si using core notation: _____

اتحاد سنافر البوليتكنك (spu)

**Note: Be able to write electron configurations for elements #1-20.**



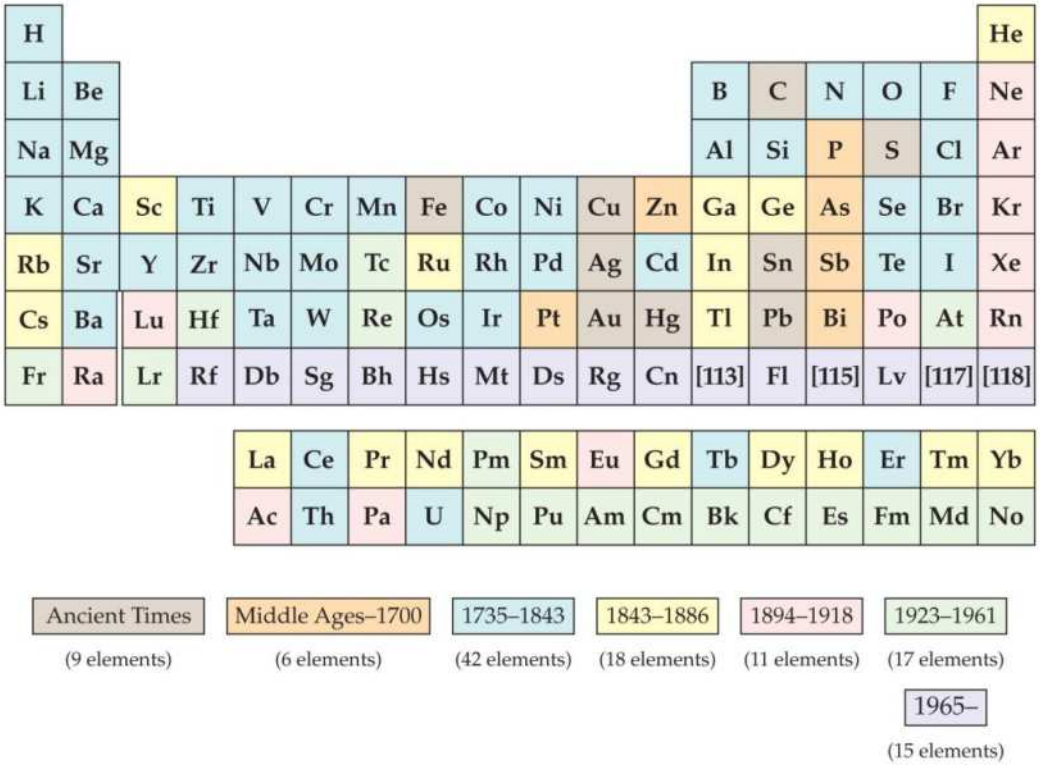

Lecture Presentation

# Chapter 8

# Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Development of the Periodic Table




Dmitri Mendeleev and Lothar Meyer independently came to the same conclusion about how elements should be grouped.

Periodic Properties of the Elements

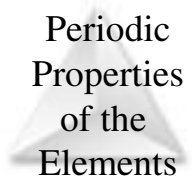
© 2015 Pearson Education, Inc.

# Mendeleev and the Periodic Table

**Table 7.1 Comparison of the Properties of Eka-Silicon Predicted by Mendeleev with the Observed Properties of Germanium**

| Property | Mendeleev's Predictions for Eka-Silicon (made in 1871) | Observed Properties of Germanium (discovered in 1886) |
|---|---|---|
| Atomic weight | 72 | 72.59 |
| Density ($g/cm^3$) | 5.5 | 5.35 |
| Specific heat (J/g-K) | 0.305 | 0.309 |
| Melting point (°C) | High | 947 |
| Color | Dark gray | Grayish white |
| Formula of oxide | $XO_2$ | $GeO_2$ |
| Density of oxide ($g/cm^3$) | 4.7 | 4.70 |
| Formula of chloride | $XCl_4$ | $GeCl_4$ |
| Boiling point of chloride (°C) | A little under 100 | 84 |

Chemists mostly credit Mendeleev because he also used chemical properties to organize the table and predicted some missing elements and their expected properties, including germanium.




© 2015 Pearson Education, Inc.

# Atomic Number

- Mendeleev's table was based on atomic masses. It was the most fundamental property of elements known at the time.
- About 35 years later, the nuclear atom was discovered by Ernest Rutherford.
- Henry Moseley developed the concept of atomic number experimentally. The number of protons was considered the basis for the periodic property of elements.



© 2015 Pearson Education, Inc.

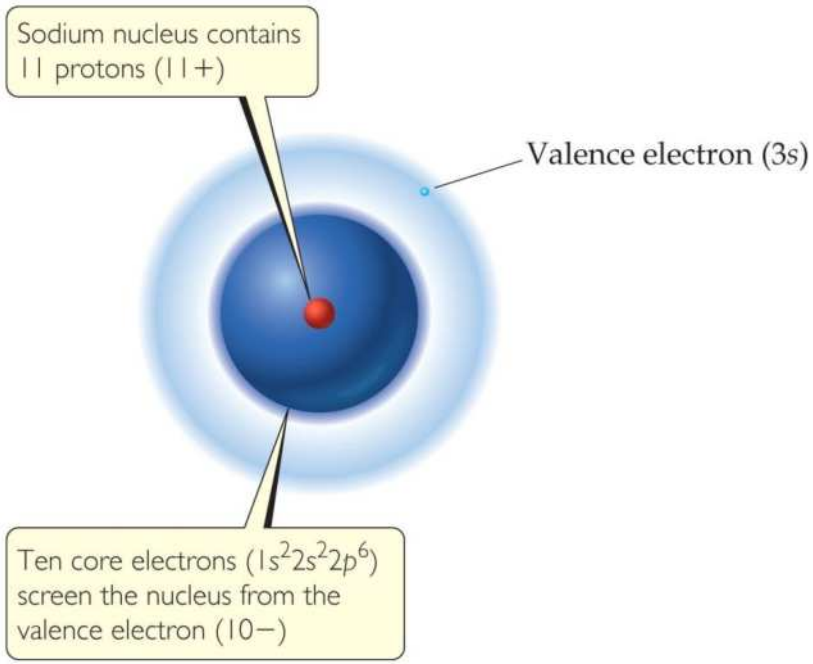
# Periodicity

- Periodicity is the repetitive pattern of a property for elements based on atomic number.
- The following properties are discussed in this chapter:
  - Sizes of atoms and ions
  - Ionization energy
  - Electron affinity
  - Some group chemical property trends
- First, we will discuss a fundamental property that leads to may of the trends, effective nuclear charge.

Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Effective Nuclear Charge




- Many properties depend on attractions between valence electrons and the nucleus.
- Electrons are both attracted to the nucleus and repelled by other electrons.
- The forces an electron experiences depend on both factors.

Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Effective Nuclear Charge

- The effective nuclear charge, $Z_{eff}$, is found this way:

$$Z_{eff} = Z - S$$

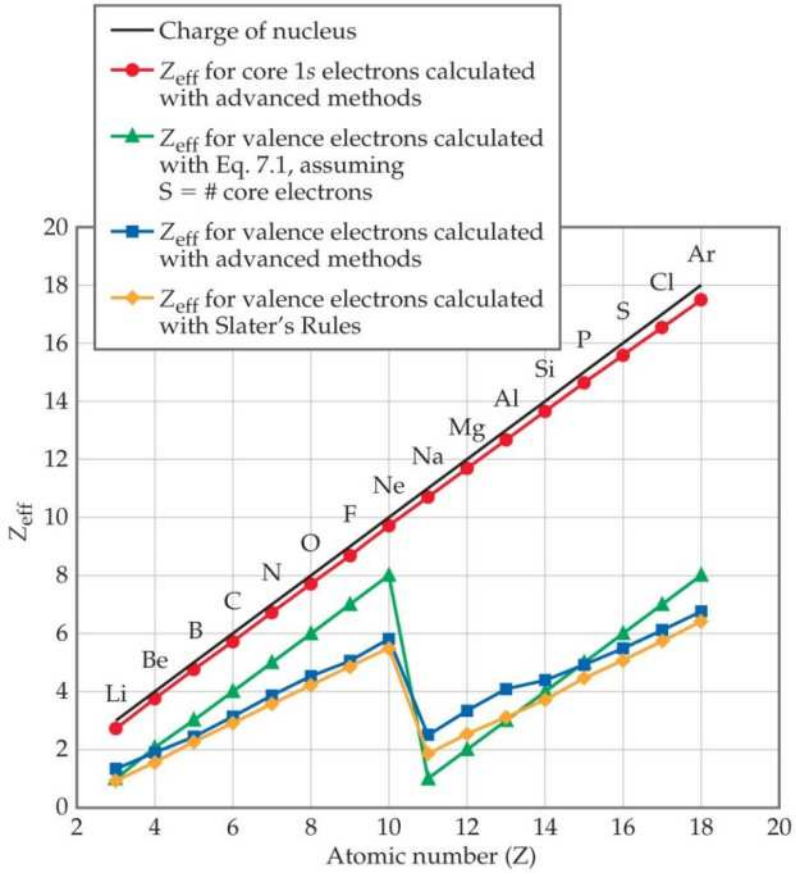
where $Z$ is the atomic number and $S$ is a screening constant, usually close to the number of inner electrons.

- Effective nuclear charge is a periodic property:
  - It increases across a period.
  - It decreases down a group.



© 2015 Pearson Education, Inc.

# Effective Nuclear Charge Increases across a Period

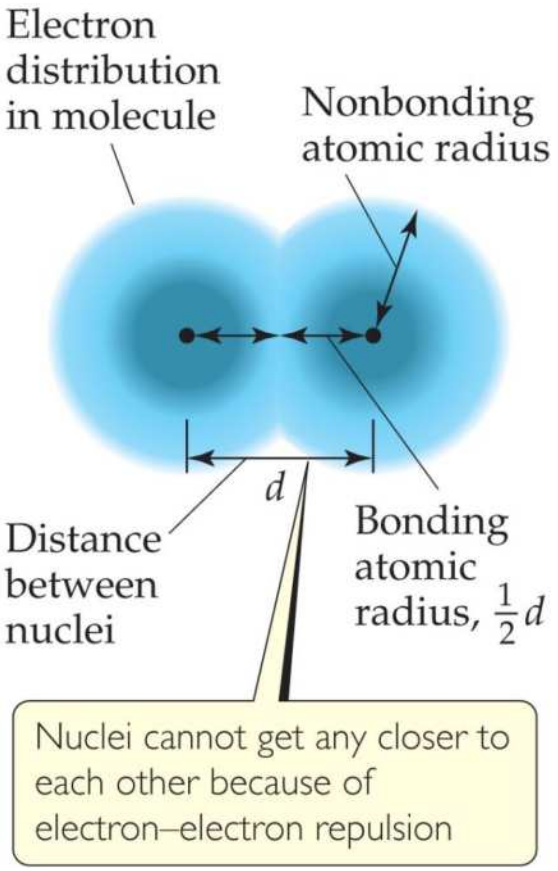





© 2015 Pearson Education, Inc.

# What Is the Size of an Atom?

The **nonbonding atomic radius** or **van der Waals radius** is half of the shortest distance separating two nuclei during a collision of atoms.

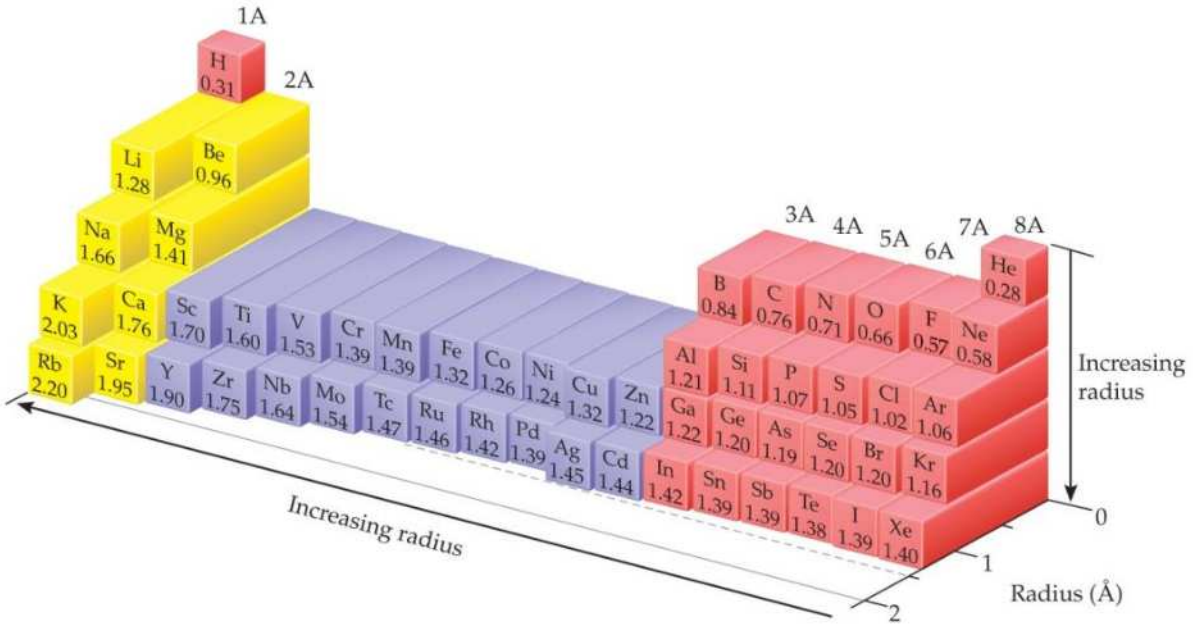





© 2015 Pearson Education, Inc.

# Sizes of Atoms

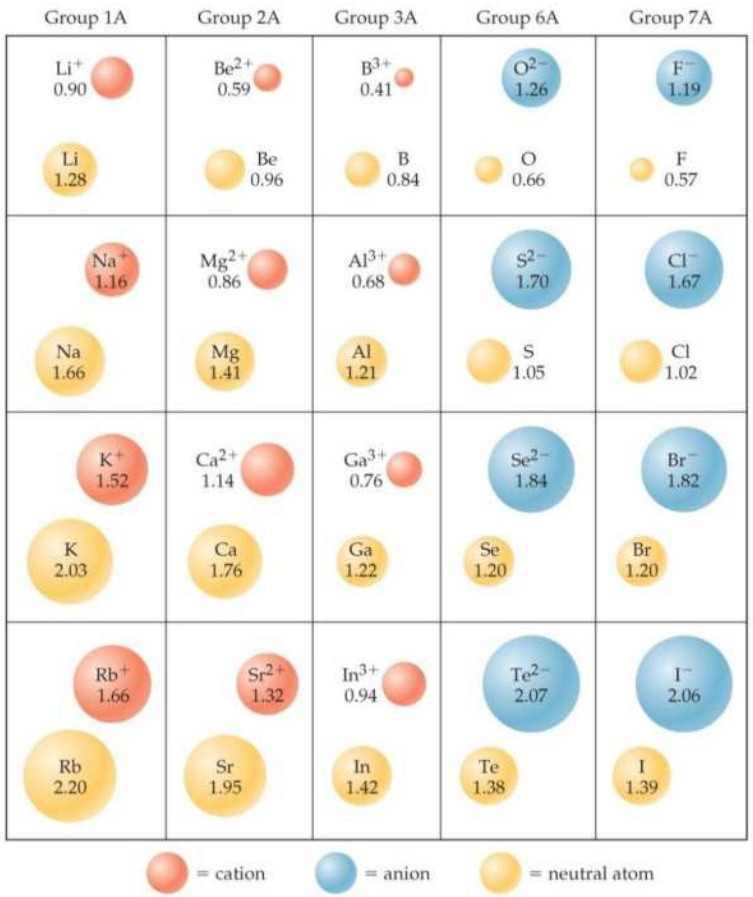
- The bonding atomic radius is half the internuclear distance when atoms are bonded.
- The bonding atomic radius tends to
  - — decrease from left to right across a period ($Z_{eff}$ ↑).
  - — increase from top to bottom of a group ($n$ ↑).




Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Sizes of Ions

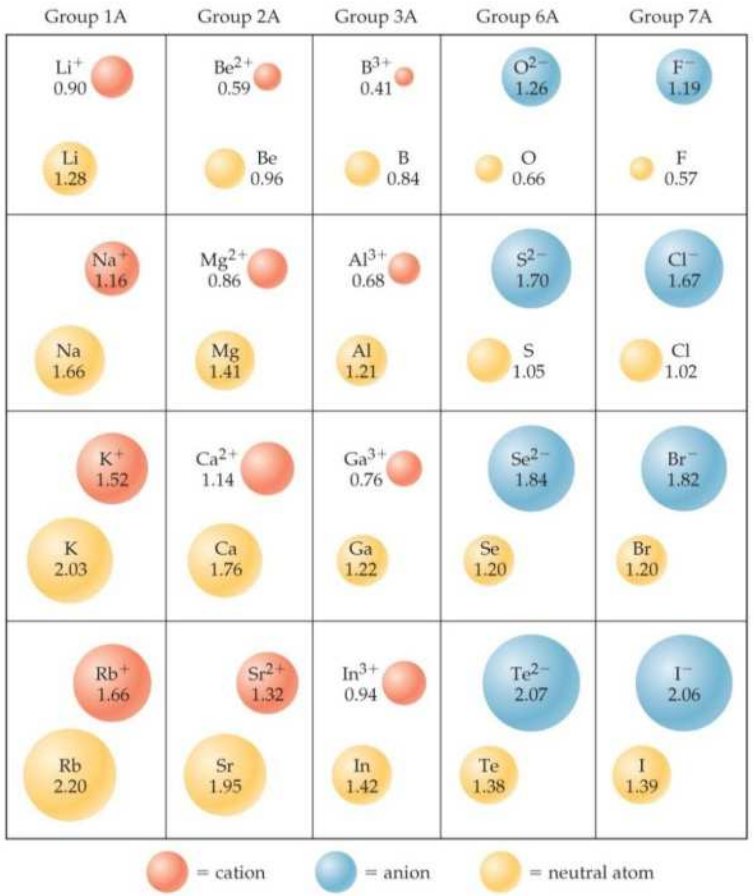



- Determined by interatomic distances in ionic compounds
- Ionic size depends on
  - the nuclear charge.
  - the number of electrons.
  - the orbitals in which electrons reside.

Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Sizes of Ions




- Cations are smaller than their parent atoms:
  - The outermost electron is removed and repulsions between electrons are reduced.
- Anions are larger than their parent atoms:
  - Electrons are added and repulsions between electrons are increased.



© 2015 Pearson Education, Inc.

# Size of Ions—Isoelectronic Series

- In an **isoelectronic series**, ions have the same number of electrons.
- Ionic size decreases with an increasing nuclear charge.

➢ **An Isoelectronic Series (10 electrons)**

- Note increasing nuclear charge with decreasing ionic radius as atomic number increases

| $O^{2-}$ | $F^{-}$ | $Na^{+}$ | $Mg^{2+}$ | $Al^{3+}$ |
|---|---|---|---|---|
| **1.26 Å** | **1.19 Å** | **1.16 Å** | **0.86 Å** | **0.68 Å** |



© 2015 Pearson Education, Inc.

# Ionization Energy (*I*)

- The **ionization energy** is the minimum energy required to remove an electron from the ground state of a gaseous atom or ion.
  - The first ionization energy is that energy required to remove the first electron.
  - The second ionization energy is that energy required to remove the second electron, etc.
- Note: the higher the ionization energy, the more difficult it is to remove an electron!



© 2015 Pearson Education, Inc.

# Ionization Energy

- It requires more energy to remove each successive electron.
- When all valence electrons have been removed, it takes a great deal more energy to remove the next electron.

Table 7.2 Successive Values of Ionization Energies, $I$, for the Elements Sodium through Argon (kJ/mol)

| Element | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ |
|---|---|---|---|---|---|---|---|
| Na | 496 | 4562 | (inner-shell electrons) | | | | |
| Mg | 738 | 1451 | 7733 | | | | |
| Al | 578 | 1817 | 2745 | 11,577 | | | |
| Si | 786 | 1577 | 3232 | 4356 | 16,091 | | |
| P | 1012 | 1907 | 2914 | 4964 | 6274 | 21,267 | |
| S | 1000 | 2252 | 3357 | 4556 | 7004 | 8496 | 27,107 |
| Cl | 1251 | 2298 | 3822 | 5159 | 6542 | 9362 | 11,018 |
| Ar | 1521 | 2666 | 3931 | 5771 | 7238 | 8781 | 11,995 |



© 2015 Pearson Education, Inc.

# Periodic Trends in First Ionization Energy ($I_1$)

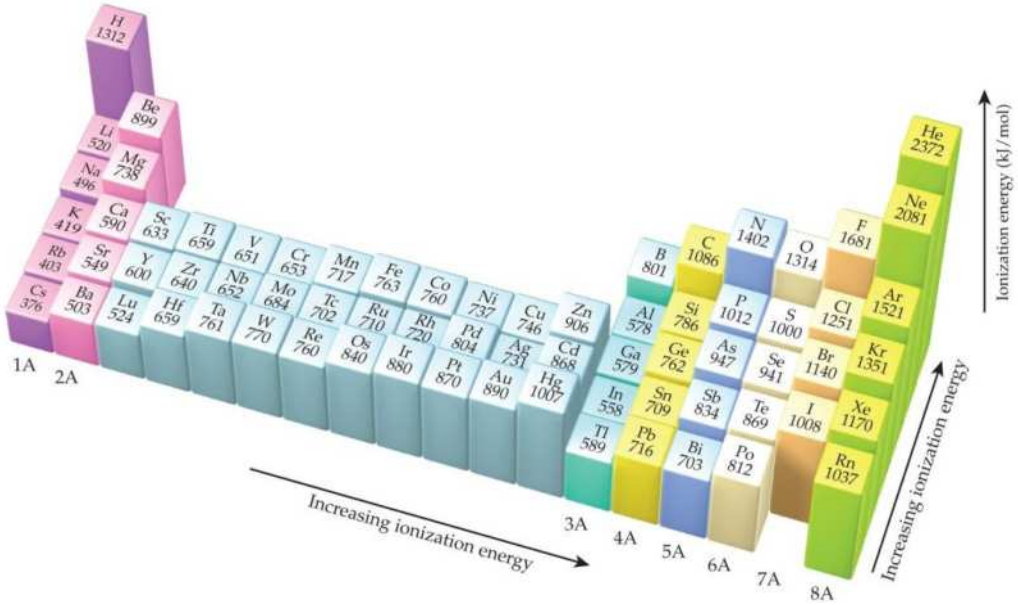
1) $I_1$ generally increases across a period.
2) $I_1$ generally decreases down a group.
3) The *s*- and *p*-block elements show a larger range of values for $I_1$. (The *d*-block generally increases slowly across the period; the *f*-block elements show only small variations.)



© 2015 Pearson Education, Inc.

# Factors that Influence Ionization Energy

- Smaller atoms have higher *I* values.
- *I* values depend on effective nuclear charge and average distance of the electron from the nucleus.




Periodic Properties of the Elements

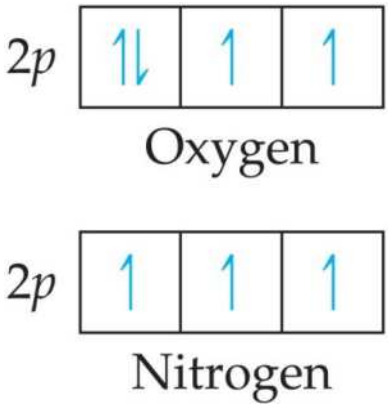
© 2015 Pearson Education, Inc.

# Irregularities in the General Trend

- The trend is **not** followed when the added valence electron in the next element

➢ enters a new sublevel (higher energy sublevel);

➢ is the first electron to pair in one orbital of the sublevel (electron repulsions lower energy).






© 2015 Pearson Education, Inc.

# Electron Configurations of Ions

- Cations: The electrons are lost from the *highest* energy level (*n* value).
  - $Li^+$ is $1s^2$ (losing a $2s$ electron).
  - $Fe^{2+}$ is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$ (losing two $4s$ electrons).
- Anions: The electron configurations are filled to $ns^2 np^6$; e.g., $F^-$ is $1s^2 2s^2 2p^6$ (gaining one electron in $2p$).



© 2015 Pearson Education, Inc.

# Electron Affinity

- **Electron affinity** is the energy change accompanying the addition of an electron to a gaseous atom:

$$Cl + e^- \longrightarrow Cl^-$$

- It is typically *exothermic*, so, for most elements, it is negative!



© 2015 Pearson Education, Inc.

# General Trend in Electron Affinity

- Not much change in a group.
- Across a period, it generally increases. *Three* notable exceptions include the following:

1) Group 2A: *s* sublevel is full!
2) Group 5A: *p* sublevel is half-full!
3) Group 8A: *p* sublevel is full!

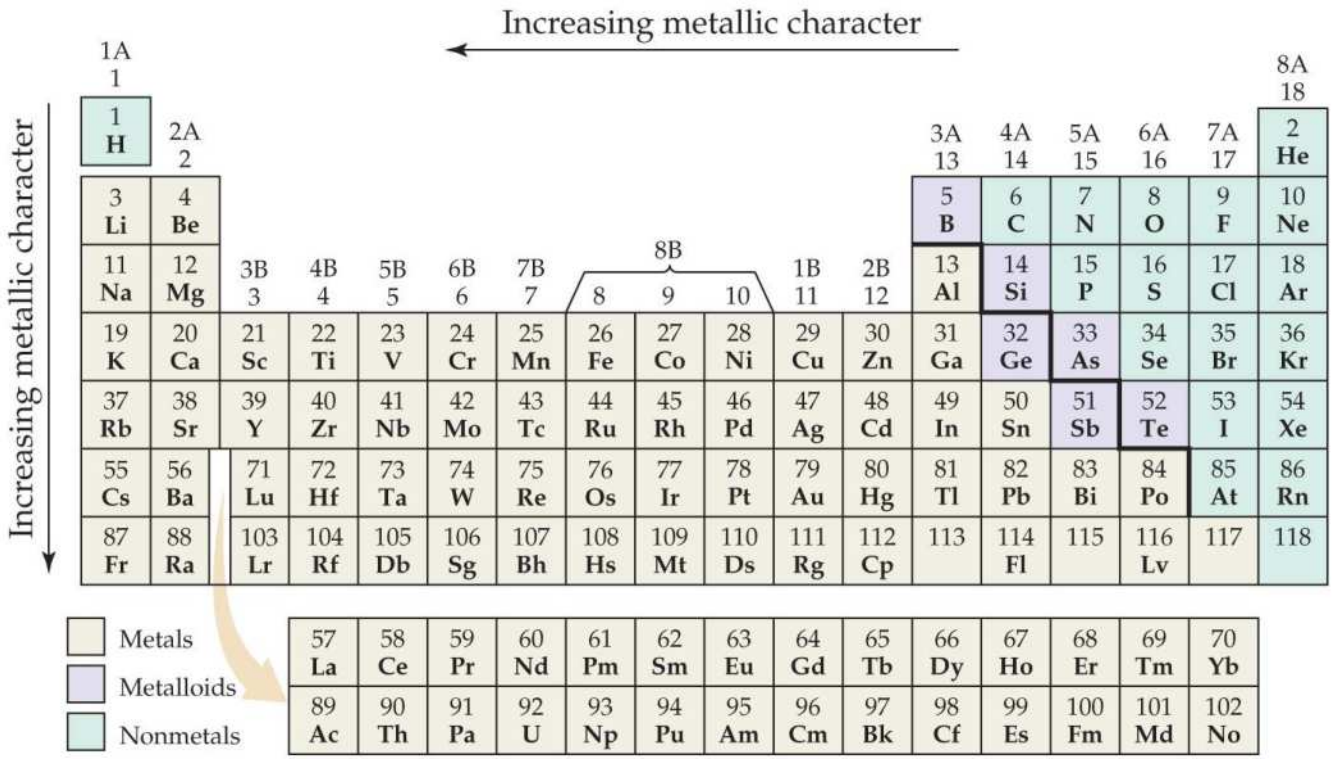
Note: the electron affinity for many of these elements is *positive* ($X^-$ is unstable).

| 1A | | | | | | | 8A |
|---|---|---|---|---|---|---|---|
| H −73 | 2A | 3A | 4A | 5A | 6A | 7A | He > 0 |
| Li −60 | Be > 0 | B −27 | C −122 | N > 0 | O −141 | F −328 | Ne > 0 |
| Na −53 | Mg > 0 | Al −43 | Si −134 | P −72 | S −200 | Cl −349 | Ar > 0 |
| K −48 | Ca −2 | Ga −30 | Ge −119 | As −78 | Se −195 | Br −325 | Kr > 0 |
| Rb −47 | Sr −5 | In −30 | Sn −107 | Sb −103 | Te −190 | I −295 | Xe > 0 |



© 2015 Pearson Education, Inc.

# Metal, Nonmetals, and Metalloids

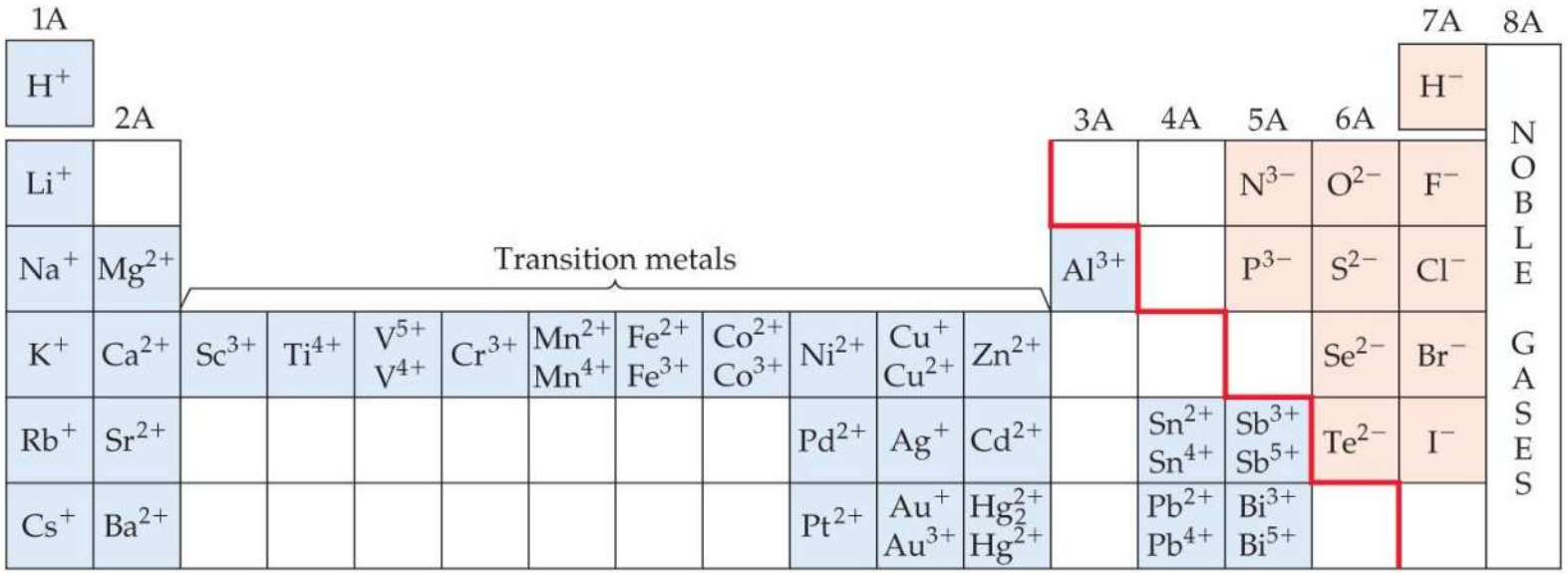



Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Metals Differ from Nonmetals

- Metals tend to form cations.
- Nonmetals tend to form anions.




Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Metals

- Most of the elements in nature are metals.
- Properties of metals:
  - Shiny luster
  - Conduct heat and electricity
  - Malleable and ductile
  - Solids at room temperature (except mercury)
  - Low ionization energies/form cations easily





© 2015 Pearson Education, Inc.

# Metal Chemistry

- Compounds formed between metals and nonmetals tend to be ionic.
- Metal oxides tend to be basic.




Nickle oxide (NiO), nitric acid ($HNO_3$), and water




NiO is insoluble in water but reacts with $HNO_3$ to give a green solution of the salt $Ni(NO_3)_2$

Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Nonmetals

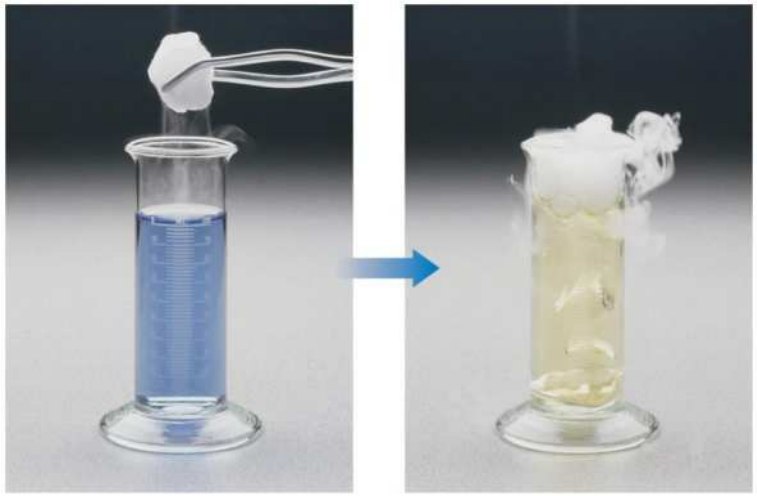
- Nonmetals are found on the right hand side of the periodic table.
- Properties of nonmetals include the following:
- Solid, liquid, or gas (depends on element)
- Solids are dull, brittle, poor conductors
- Large negative electronegativity/form anions readily





© 2015 Pearson Education, Inc.

# Nonmetal Chemistry



- Substances containing only nonmetals are molecular compounds.
- Most nonmetal oxides are acidic.



© 2015 Pearson Education, Inc.

# Recap of a Comparison of the Properties of Metals and Nonmetals

**Table 7.3 Characteristic Properties of Metals and Nonmetals**

| Metals | Nonmetals |
|---|---|
| Have a shiny luster; various colors, although most are silvery | Do not have a luster; various colors |
| Solids are malleable and ductile | Solids are usually brittle; some are hard, and some are soft |
| Good conductors of heat and electricity | Poor conductors of heat and electricity |
| Most metal oxides are ionic solids that are basic | Most nonmetal oxides are molecular substances that form acidic solutions |
| Tend to form cations in aqueous solution | Tend to form anions or oxyanions in aqueous solution |



© 2015 Pearson Education, Inc.

# Metalloids

- Metalloids have some characteristics of metals and some of nonmetals.
- Several metalloids are electrical semiconductors (computer chips).





© 2015 Pearson Education, Inc.

# Group Trends

- Elements in a group have similar properties.
- Trends also exist within groups.
- Groups Compared:

➢ Group 1A: The Alkali Metals

➢ Group 2A: The Alkaline Earth Metals

➢ Group 6A: The Oxygen Group

➢ Group 7A: The Halogens

➢ Group 8A: The Noble Gases



© 2015 Pearson Education, Inc.

# Alkali Metals

- **Alkali metals** are soft, metallic solids.
- They are found only in compounds in nature, not in their elemental forms.
- Typical metallic properties (luster, conductivity) are seen in them.






© 2015 Pearson Education, Inc.

# Alkali Metal Properties

- They have low densities and melting points.
- They also have low ionization energies.

**Table 7.4 Some Properties of the Alkali Metals**

| Element | Electron Configuration | Melting Point (°C) | Density ($g/cm^3$) | Atomic Radius (Å) | $I_1$ (kJ/mol) |
|---|---|---|---|---|---|
| Lithium | $[He]2s^1$ | 181 | 0.53 | 1.28 | 520 |
| Sodium | $[Ne]3s^1$ | 98 | 0.97 | 1.66 | 496 |
| Potassium | $[Ar]4s^1$ | 63 | 0.86 | 2.03 | 419 |
| Rubidium | $[Kr]5s^1$ | 39 | 1.53 | 2.20 | 403 |
| Cesium | $[Xe]6s^1$ | 28 | 1.88 | 2.44 | 376 |

Periodic Properties of the Elements

© 2015 Pearson Education, Inc.

# Alkali Metal Chemistry



Li Na K

Their reactions with water are famously exothermic.



© 2015 Pearson Education, Inc.

# Differences in Alkali Metal Chemistry

- Lithium reacts with oxygen to make an oxide:

$$4\ Li + O_2 \longrightarrow 2\ Li_2O$$

- Sodium reacts with oxygen to form a peroxide:

$$2\ Na + O_2 \longrightarrow Na_2O_2$$

- K, Rb, and Cs also form superoxides:

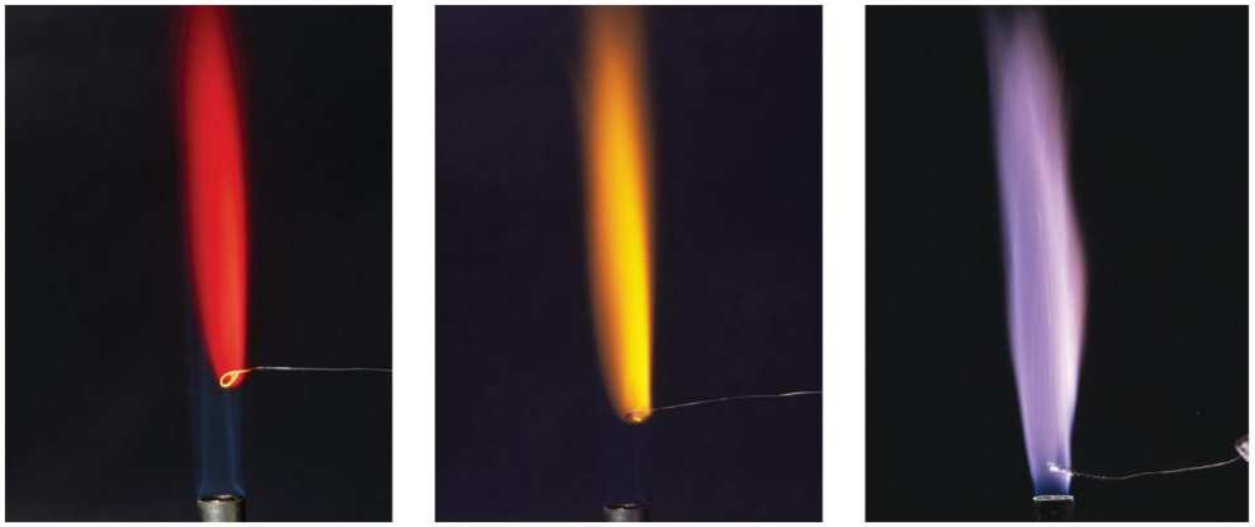
$$M + O_2 \longrightarrow MO_2$$



© 2015 Pearson Education, Inc.

# Flame Tests

- Qualitative tests for alkali metals include their characteristic colors in flames.



Li Na K



© 2015 Pearson Education, Inc.

# Alkaline Earth Metals—Compare to Alkali Metals

Table 7.5 Some Properties of the Alkaline Earth Metals

| Element | Electron Configuration | Melting Point (°C) | Density ($g/cm^3$) | Atomic Radius (Å) | $I_1$ (kJ/mol) |
|---|---|---|---|---|---|
| Beryllium | $[He]2s^2$ | 1287 | 1.85 | 0.96 | 899 |
| Magnesium | $[Ne]3s^2$ | 650 | 1.74 | 1.41 | 738 |
| Calcium | $[Ar]4s^2$ | 842 | 1.55 | 1.76 | 590 |
| Strontium | $[Kr]5s^2$ | 777 | 2.63 | 1.95 | 549 |
| Barium | $[Xe]6s^2$ | 727 | 3.51 | 2.15 | 503 |

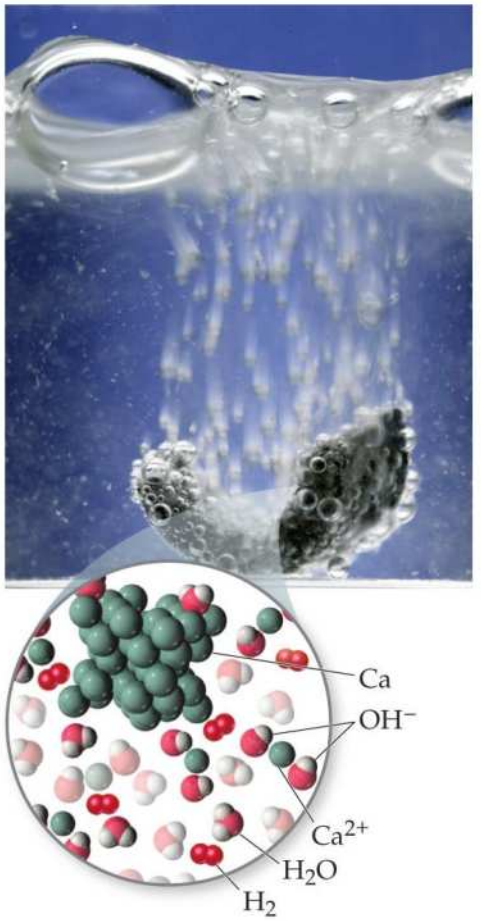
- Alkaline earth metals have higher densities and melting points than alkali metals.
- Their ionization energies are low, but not as low as those of alkali metals.



© 2015 Pearson Education, Inc.

# Alkaline Earth Metals

- Beryllium does not react with water, and magnesium reacts only with steam, but the other alkaline earth metals react readily with water.
- Reactivity tends to increase as you go down the group.






© 2015 Pearson Education, Inc.

# Group 6A—Increasing in Metallic Character down the Group

**Table 7.6 Some Properties of the Group 6A Elements**

| Element | Electron Configuration | Melting Point (°C) | Density | Atomic Radius (Å) | $I_1$ (kJ/mol) |
|---|---|---|---|---|---|
| Oxygen | $[He]2s^2 2p^4$ | −218 | 1.43 g/L | 0.66 | 1314 |
| Sulfur | $[Ne]3s^2 3p^4$ | 115 | 1.96 g/cm$^3$ | 1.05 | 1000 |
| Selenium | $[Ar]3d^{10} 4s^2 4p^4$ | 221 | 4.82 g/cm$^3$ | 1.20 | 941 |
| Tellurium | $[Kr]4d^{10} 5s^2 5p^4$ | 450 | 6.24 g/cm$^3$ | 1.38 | 869 |
| Polonium | $[Xe]4f^{14} 5d^{10} 6s^2 6p^4$ | 254 | 9.20 g/cm$^3$ | 1.40 | 812 |

- Oxygen, sulfur, and selenium are nonmetals.
- Tellurium is a metalloid.
- The radioactive polonium is a metal.



© 2015 Pearson Education, Inc.

# Group 7A—Halogens

**Table 7.7 Some Properties of the Halogens**

| Element | Electron Configuration | Melting Point (°C) | Density | Atomic Radius (Å) | $I_1$ (kJ/mol) |
|---|---|---|---|---|---|
| Fluorine | $[He]2s^2 2p^5$ | −220 | 1.69 g/L | 0.57 | 1681 |
| Chlorine | $[Ne]3s^2 3p^5$ | −102 | 3.12 g/L | 1.02 | 1251 |
| Bromine | $[Ar]4s^2 3d^{10} 4p^5$ | −7.3 | 3.12 g/cm$^3$ | 1.20 | 1140 |
| Iodine | $[Kr]5s^2 4d^{10} 5p^5$ | 114 | 4.94 g/cm$^3$ | 1.39 | 1008 |

- The **halogens** are typical nonmetals.
- They have highly negative electron affinities, so they exist as anions in nature.
- They react directly with metals to form metal halides.



© 2015 Pearson Education, Inc.

# Group 8A—Noble Gases

**Table 7.8 Some Properties of the Noble Gases**

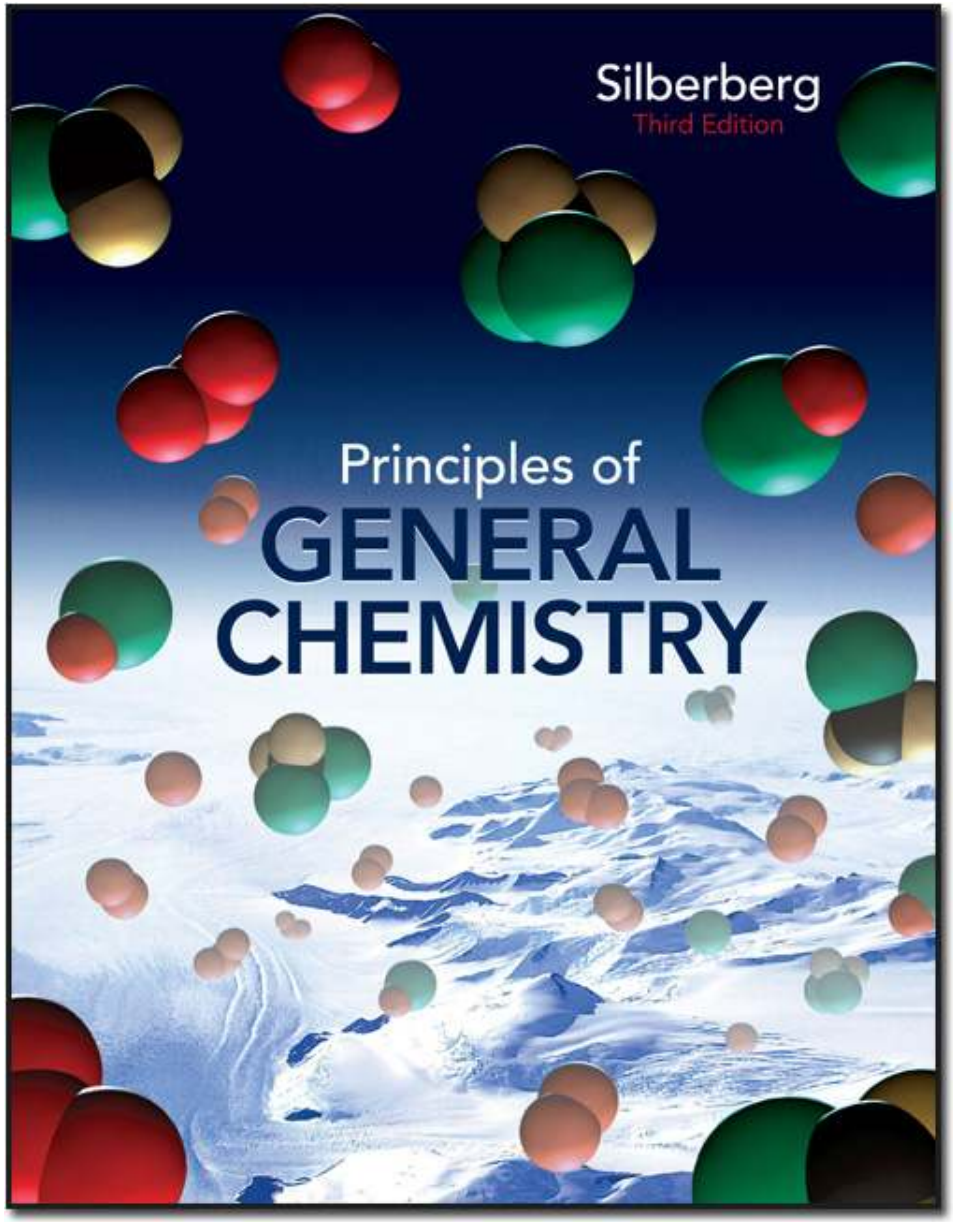
| Element | Electron Configuration | Boiling Point (K) | Density (g/L) | Atomic Radius* (Å) | $I_1$ (kJ/mol) |
|---|---|---|---|---|---|
| Helium | $1s^2$ | 4.2 | 0.18 | 0.28 | 2372 |
| Neon | $[He]2s^22p^6$ | 27.1 | 0.90 | 0.58 | 2081 |
| Argon | $[Ne]3s^23p^6$ | 87.3 | 1.78 | 1.06 | 1521 |
| Krypton | $[Ar]4s^23d^{10}4p^6$ | 120 | 3.75 | 1.16 | 1351 |
| Xenon | $[Kr]5s^24d^{10}5p^6$ | 165 | 5.90 | 1.40 | 1170 |
| Radon | $[Xe]6s^24f^{14}5d^{10}6p^6$ | 211 | 9.73 | 1.50 | 1037 |

*Only the heaviest of the noble-gas elements form chemical compounds. Thus, the atomic radii for the lighter noble gas elements are estimated values.

- The noble gases have very large ionization energies.
- Their electron affinities are positive (can't form stable anions).
- Therefore, they are relatively unreactive.
- They are found as monatomic gases.



© 2015 Pearson Education, Inc.




# Chapter 08

# Lecture Outline

**See separate *Image PowerPoint* slides for all figures and tables pre-inserted into PowerPoint without notes.**



Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

اتحاد سنافر البوليتكنك (spu)

# Chapter 8

## Electron Configuration and Chemical Periodicity

# Electron Configuration and Chemical Periodicity

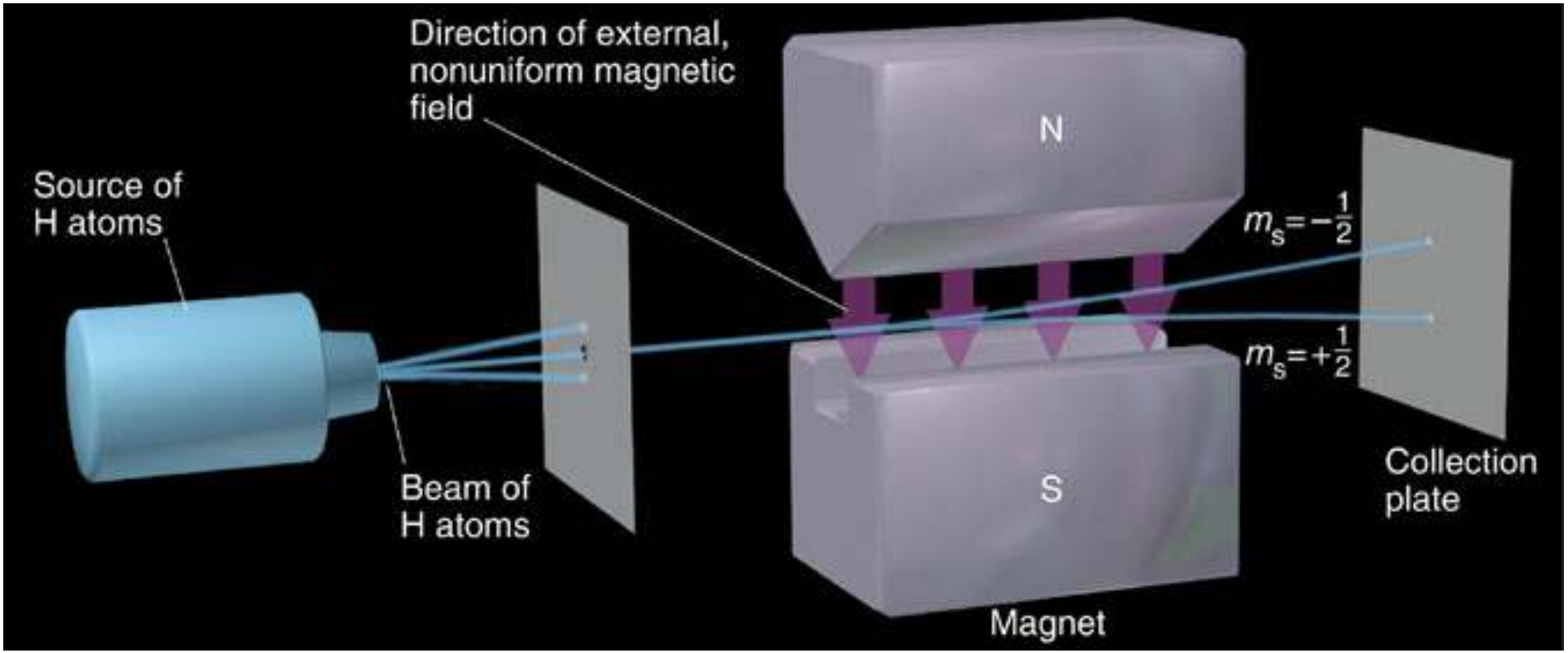
## 8.1 Characteristics of Many-Electron Atoms

## 8.2 The Quantum-Mechanical Model and the Periodic Table

## 8.3 Trends in Three Atomic Properties

## 8.4 Atomic Properties and Chemical Reactivity

**Figure 8.1 The effect of electron spin.**



## Table 8.1 Summary of Quantum Numbers of Electrons in Atoms

| Name | Symbol | Permitted Values | Property |
|---|---|---|---|
| principal | $n$ | positive integers (1, 2, 3, …) | orbital energy (size) |
| angular momentum | $l$ | integers from 0 to $n$-1 | orbital shape (The $l$ values 0, 1, 2, and 3 correspond to *s*, *p*, *d*, and *f* orbitals, respectively.) |
| magnetic | $m_l$ | integers from -$l$ to 0 to +$l$ | orbital orientation |
| spin | $m_s$ | +½ or -½ | direction of $e^-$ spin |




# Quantum Numbers and The Exclusion Principle

Each electron in any atom is described completely by a set of ***four*** quantum numbers.

The first three quantum numbers describe the orbital, while the fourth quantum number describes electron spin.

Pauli's **exclusion principle** states that *no two electrons in the same atom can have the same four quantum numbers*.

An atomic orbital can hold a ***maximum of two electrons*** and they must have ***opposing spins***.





# Factors Affecting Atomic Orbital Energies

- The energies of atomic orbitals are affected by
  - nuclear charge ($Z$) and
  - shielding by other electrons.
- A **higher nuclear charge** increases nucleus-electron interactions and lowers sublevel energy.
- **Shielding** by other electrons reduces the full nuclear charge to an ***effective nuclear charge*** ($\boldsymbol{Z_{eff}}$).
  - $Z_{eff}$ is the nuclear charge an electron actually experiences.
- **Orbital shape** also affects sublevel energy.



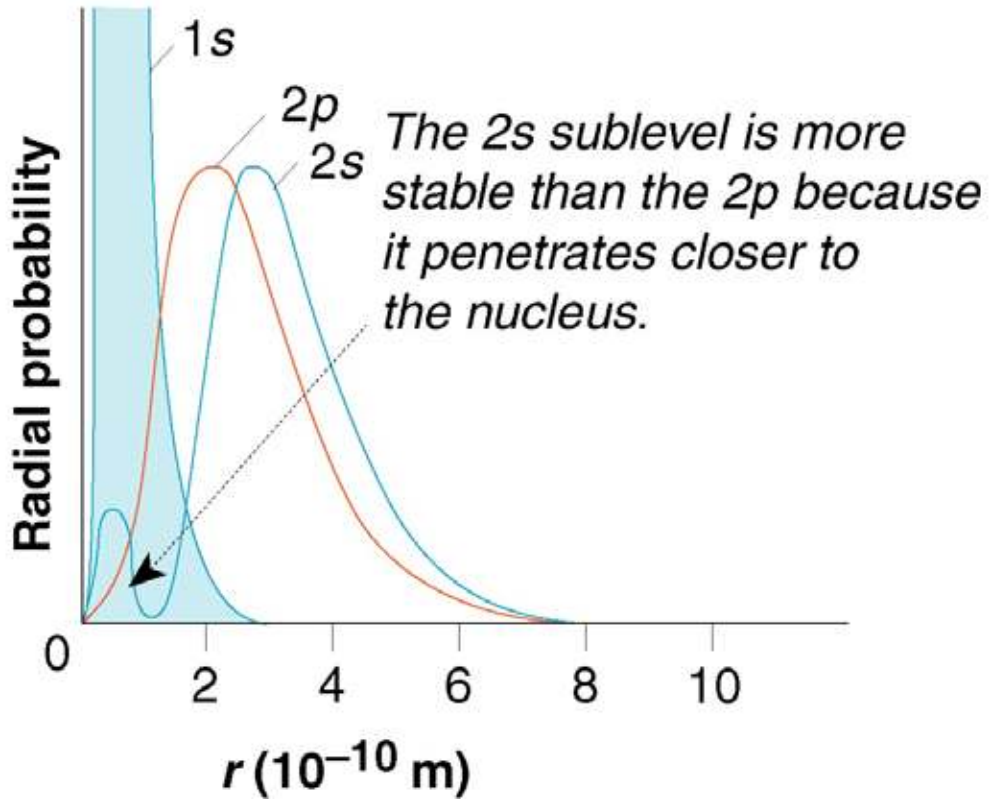


# Shielding and Orbital Energy

- Electrons in the ***same*** energy level shield each other to some extent.
- Electrons in ***inner*** energy levels shield the outer electrons very effectively.
  - The farther from the nucleus an electron is, the lower the $Z_{eff}$ for that particular electron.

## Figure 8.2 Penetration and sublevel energy.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.




Orbital shape causes electrons in some orbitals to "penetrate" close to the nucleus.

Penetration increases nuclear attraction and decreases shielding.

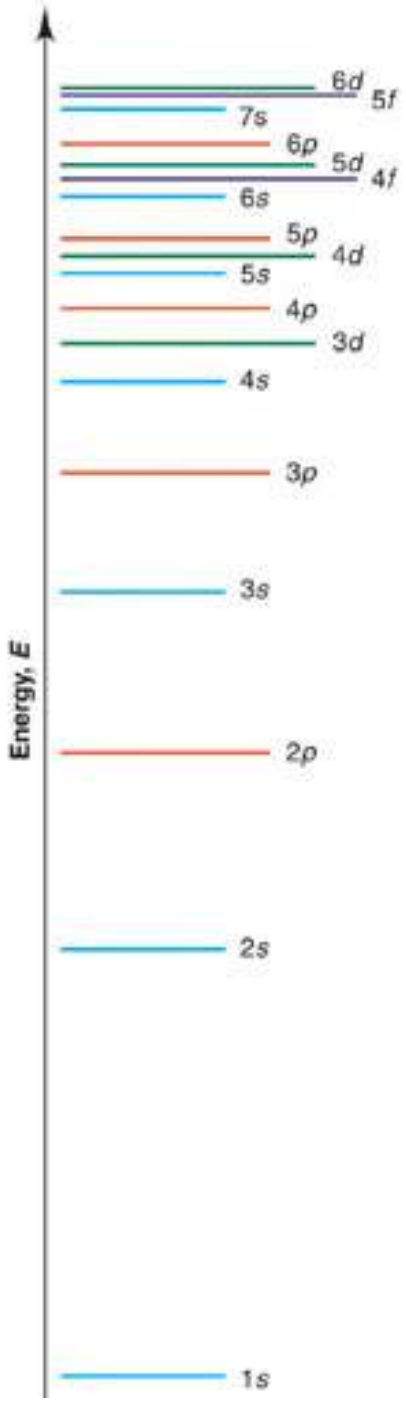
# Splitting of Levels into Sublevels

Each energy level is split into ***sublevels*** of differing energy.

Splitting is caused by penetration and its effect on shielding.

For a given *n* value, a lower *l* value indicates a lower energy sublevel.

**Order of sublevel energies: *s* < *p* < *d* < *f***




## Figure 8.3

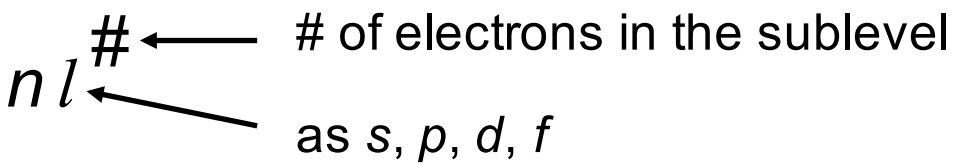
**Order for filling energy sublevels with electrons.**

In general, energies of sublevels increase as $n$ increases ($1 < 2 < 3$, etc.)
and as $l$ increases ($s < p < d < f$).

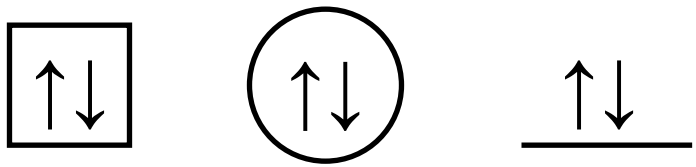
As $n$ increases, some sublevels overlap.

# Electron Configurations and Orbital Diagrams

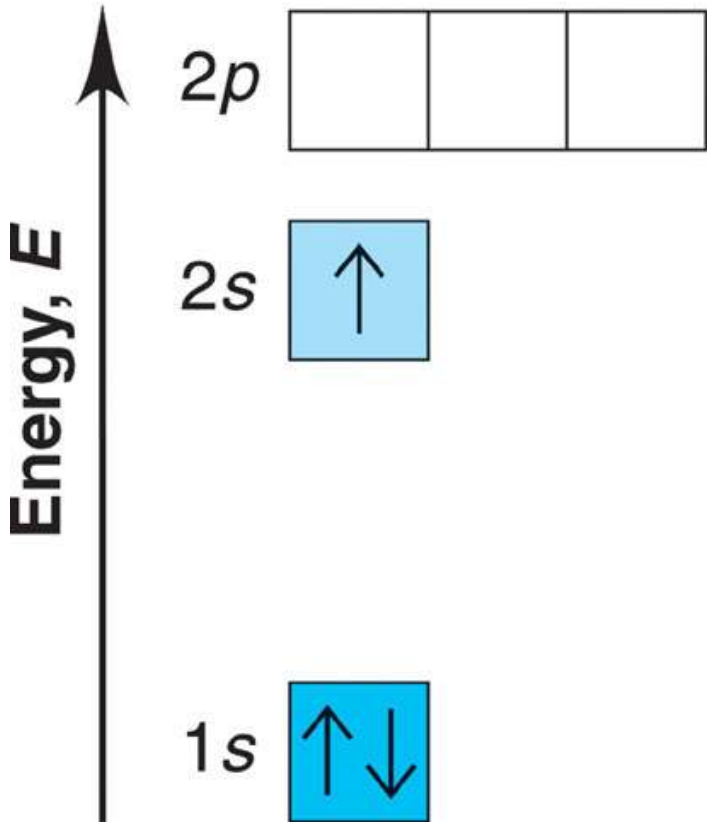
Electron configuration is indicated by a shorthand notation:




Orbital diagrams make use of a box, circle, or line for each orbital in the energy level. An arrow is used to represent an electron *and* its spin.

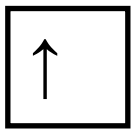


**A vertical orbital diagram for the Li ground state.**



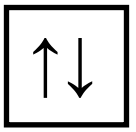
# Building Orbital Diagrams

The **aufbau principle** is applied – electrons are always placed in the lowest energy sublevel available.

H ($Z$ = 1) $1s^1$ 

$1s$

The **exclusion principle** states that each orbital may contain a maximum of 2 electrons, which must have opposite spins.

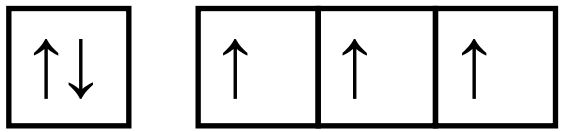
He ($Z$ = 2) $1s^2$ 

$1s$

# Building Orbital Diagrams

**Hund's rule** specifies that when orbitals of equal energy are available, the lowest energy electron configuration has the maximum number of unpaired electrons with parallel spins.

N ($Z$ = 7) $1s^2 2s^2 2p^3$



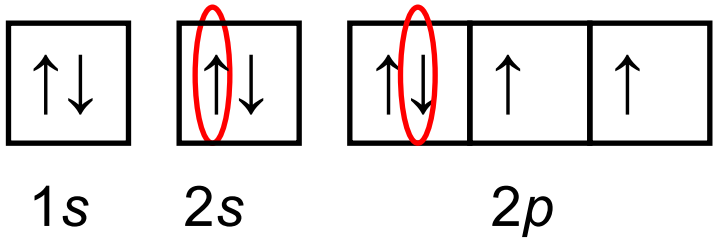
**Sample Problem 8.1** **Determining Quantum Numbers from Orbital Diagrams**

**PROBLEM:** Write a set of quantum numbers for the third electron and a set for the eighth electron of the F atom.

**PLAN:** Identify the electron of interest and note its level ($n$), sublevel, ($l$), orbital ($m_l$) and spin ($m_s$). Count the electrons in the order in which they are placed in the diagram.

**SOLUTION:**

F ($Z$ = 9) $1s^2 2s^2 2p^3$



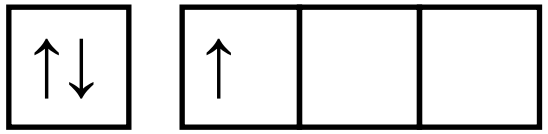

**For the 3rd electron: $n = 2$, $l = 0$, $m_l = 0$, $m_s = +½$**

**For the 8th electron: $n = 2$, $l = 1$, $m_l = -1$, $m_s = -½$**

# Partial Orbital Diagrams and Condensed Configurations

A ***partial orbital diagram*** shows only the highest energy sublevels being filled.

Al ($Z$ = 13) $1s^2 2s^2 2p^6 3s^2 3p^1$




A ***condensed electron configuration*** has the element symbol of the ***previous*** noble gas in square brackets.

Al has the condensed configuration [Ne]$3s^2 3p^1$

## Table 8.2 Partial Orbital Diagrams and Electron Configurations* for the Elements in Period 3.

| Atomic Number | Element | Partial Orbital Diagram (3s and 3p Sublevels Only) | Full Electron Configuration† | Condensed Electron Configuration |
|---|---|---|---|---|
| | | 3s 3p | | |
| 11 | Na | ↑ | $[1s^2 2s^2 2p^6]\ 3s^1$ | [Ne] $3s^1$ |
| 12 | Mg | ↑↓ | $[1s^2 2s^2 2p^6]\ 3s^2$ | [Ne] $3s^2$ |
| 13 | Al | ↑↓ ↑ | $[1s^2 2s^2 2p^6]\ 3s^2 3p^1$ | [Ne] $3s^2 3p^1$ |
| 14 | Si | ↑↓ ↑ ↑ | $[1s^2 2s^2 2p^6]\ 3s^2 3p^2$ | [Ne] $3s^2 3p^2$ |
| 15 | P | ↑↓ ↑ ↑ ↑ | $[1s^2 2s^2 2p^6]\ 3s^2 3p^3$ | [Ne] $3s^2 3p^3$ |
| 16 | S | ↑↓ ↑↓ ↑ ↑ | $[1s^2 2s^2 2p^6]\ 3s^2 3p^4$ | [Ne] $3s^2 3p^4$ |
| 17 | Cl | ↑↓ ↑↓ ↑↓ ↑ | $[1s^2 2s^2 2p^6]\ 3s^2 3p^5$ | [Ne] $3s^2 3p^5$ |
| 18 | Ar | ↑↓ ↑↓ ↑↓ ↑↓ | $[1s^2 2s^2 2p^6]\ 3s^2 3p^6$ | [Ne] $3s^2 3p^6$ |

*Colored type indicates the sublevel to which the last electron is added.

†The full configuration is not usually written with square brackets; they are included here to show how the [Ne] designation arises.

# Electron Configuration and Group

Elements in the same group of the periodic table have the same outer electron configuration.

Elements in the same group of the periodic table exhibit similar chemical behavior.

***Similar outer electron configurations correlate with similar chemical behavior.***

**Figure 8.4** **Condensed electron configurations in the first three periods.**

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

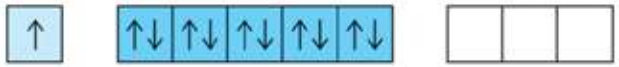
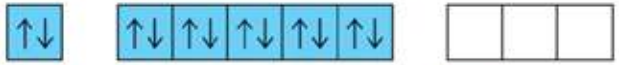
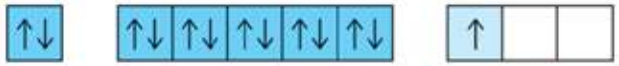
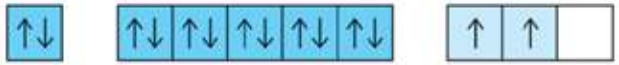
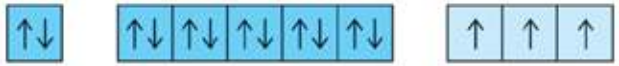
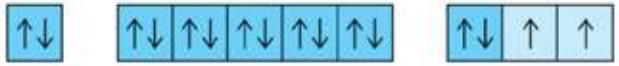
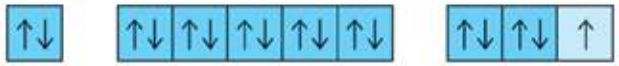
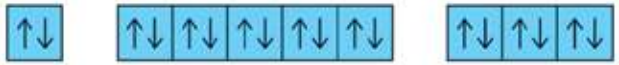
| Period | 1A (1) | 2A (2) | 3A (13) | 4A (14) | 5A (15) | 6A (16) | 7A (17) | 8A (18) |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 **H** $1s^1$ | | | | | | | 2 **He** $1s^2$ |
| 2 | 3 **Li** [He] $2s^1$ | 4 **Be** [He] $2s^2$ | 5 **B** [He] $2s^2 2p^1$ | 6 **C** [He] $2s^2 2p^2$ | 7 **N** [He] $2s^2 2p^3$ | 8 **O** [He] $2s^2 2p^4$ | 9 **F** [He] $2s^2 2p^5$ | 10 **Ne** [He] $2s^2 2p^6$ |
| 3 | 11 **Na** [Ne] $3s^1$ | 12 **Mg** [Ne] $3s^2$ | 13 **Al** [Ne] $3s^2 3p^1$ | 14 **Si** [Ne] $3s^2 3p^2$ | 15 **P** [Ne] $3s^2 3p^3$ | 16 **S** [Ne] $3s^2 3p^4$ | 17 **Cl** [Ne] $3s^2 3p^5$ | 18 **Ar** [Ne] $3s^2 3p^6$ |

## Table 8.3 Partial Orbital Diagrams and Electron Configurations* for the Elements in Period 4.

| Atomic Number | Element | Partial Orbital Diagram (4*s*, 3*d*, and 4*p* Sublevels Only) | Full Electron Configuration | Condensed Electron Configuration |
|---|---|---|---|---|
| | | 4*s* 3*d* 4*p* | | |
| 19 | K | ↑ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$ | [Ar] $4s^1$ |
| 20 | Ca | ↑↓ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ | [Ar] $4s^2$ |
| 21 | Sc | ↑↓ ↑ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^1$ | [Ar] $4s^2 3d^1$ |
| 22 | Ti | ↑↓ ↑ ↑ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$ | [Ar] $4s^2 3d^2$ |
| 23 | V | ↑↓ ↑ ↑ ↑ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$ | [Ar] $4s^2 3d^3$ |
| 24 | Cr | ↑ ↑ ↑ ↑ ↑ ↑ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$ | [Ar] $4s^1 3d^5$ |
| 25 | Mn | ↑↓ ↑ ↑ ↑ ↑ ↑ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$ | [Ar] $4s^2 3d^5$ |
| 26 | Fe | ↑↓ ↑↓ ↑ ↑ ↑ ↑ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$ | [Ar] $4s^2 3d^6$ |
| 27 | Co | ↑↓ ↑↓ ↑↓ ↑ ↑ ↑ | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$ | [Ar] $4s^2 3d^7$ |

*Colored type indicates the sublevel to which the last electron is added.

# Table 8.3 Partial Orbital Diagrams and Electron Configurations* for the Elements in Period 4.

| Atomic Number | Element | Partial Orbital Diagram (4*s*, 3*d*, and 4*p* Sublevels Only) | Full Electron Configuration | Condensed Electron Configuration |
|---|---|---|---|---|
| 28 | Ni |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$ | [Ar] $4s^2 3d^8$ |
| 29 | Cu |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$ | [Ar] $4s^1 3d^{10}$ |
| 30 | Zn |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$ | [Ar] $4s^2 3d^{10}$ |
| 31 | Ga |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^1$ | [Ar] $4s^2 3d^{10} 4p^1$ |
| 32 | Ge |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$ | [Ar] $4s^2 3d^{10} 4p^2$ |
| 33 | As |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^3$ | [Ar] $4s^2 3d^{10} 4p^3$ |
| 34 | Se |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^4$ | [Ar] $4s^2 3d^{10} 4p^4$ |
| 35 | Br |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^5$ | [Ar] $4s^2 3d^{10} 4p^5$ |
| 36 | Kr |  | $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$ | [Ar] $4s^2 3d^{10} 4p^6$ |

*Colored type indicates the sublevel to which the last electron is added.

# Figure 8.5 A periodic table of partial ground-state electron configurations.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

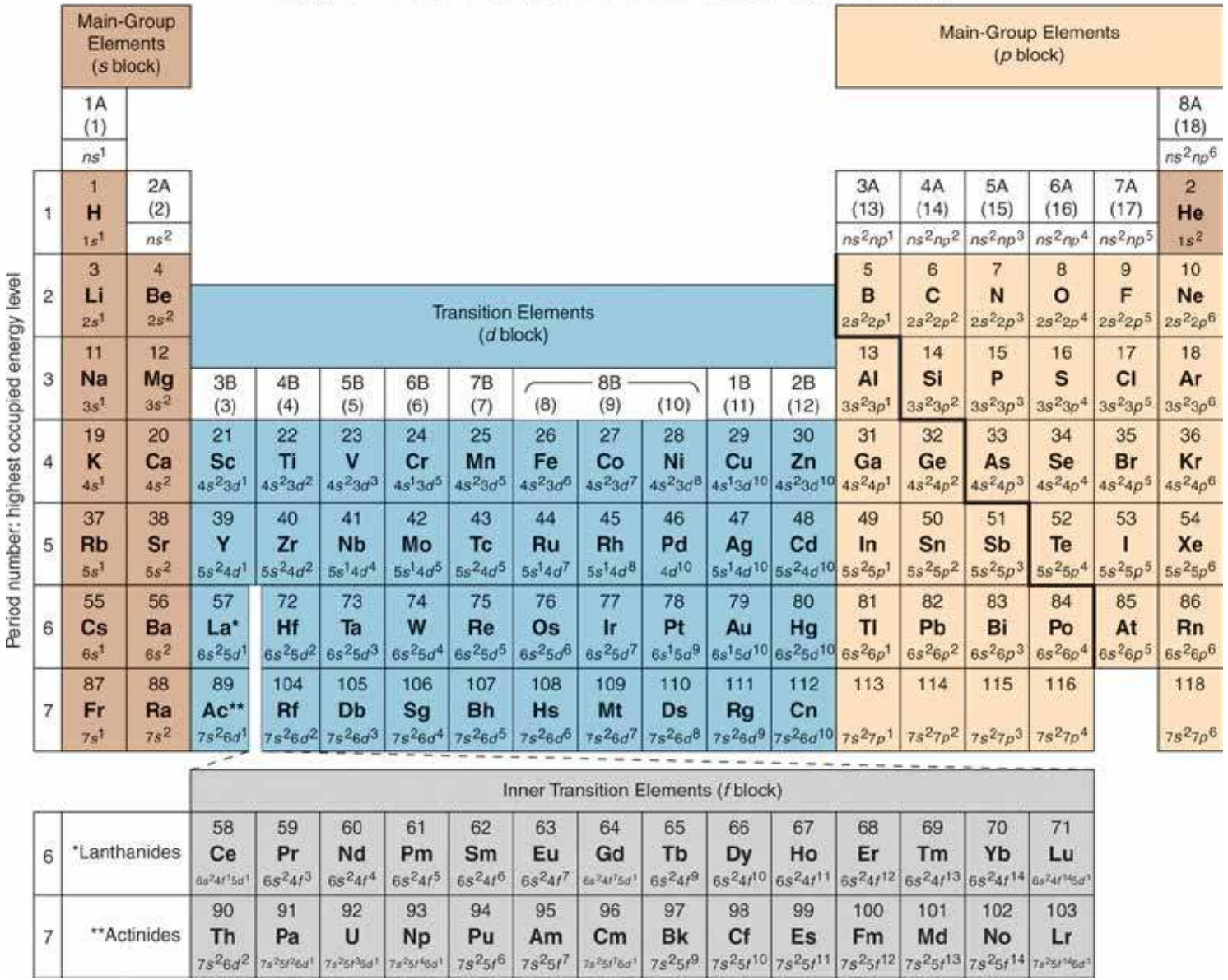
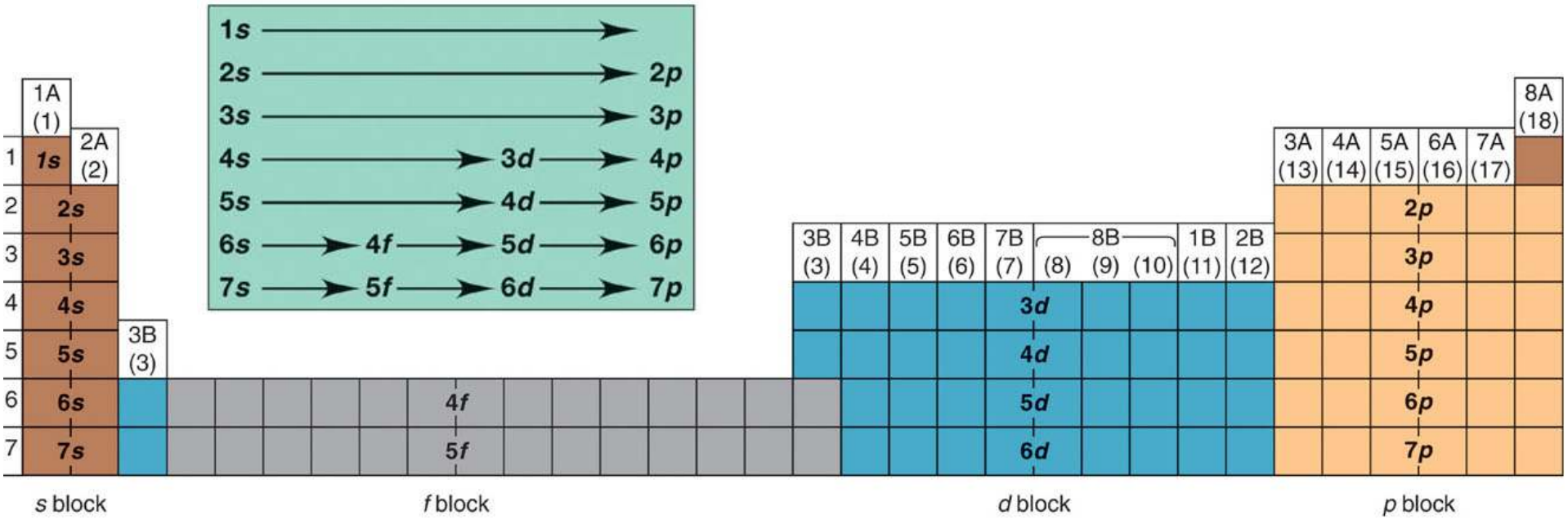


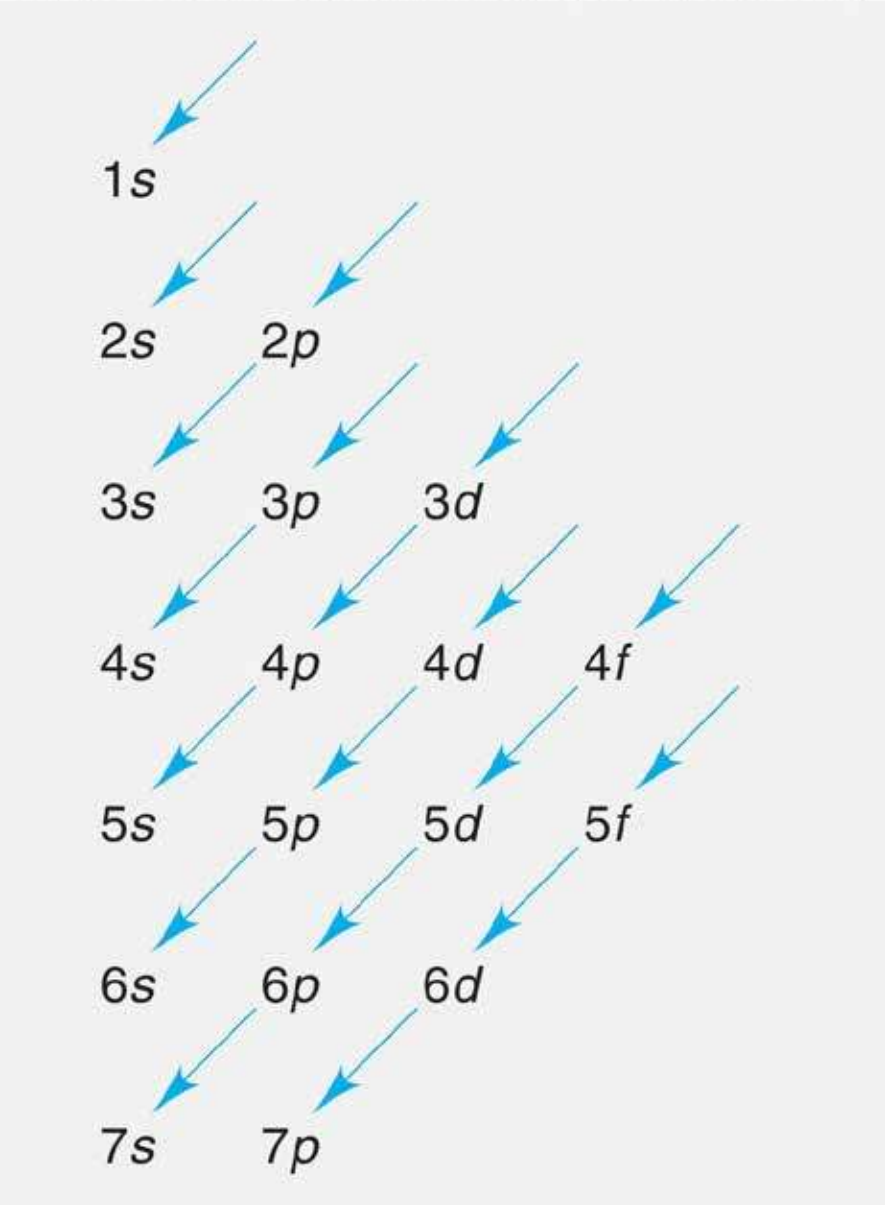
**Figure 8.6 Orbital filling and the periodic table.**




The order in which the orbitals are filled can be obtained directly from the periodic table.

# Aid to memorizing sublevel filling order.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.




The *n* value is constant horizontally.
The *l* value is constant vertically.
*n* + *l* is constant diagonally.

# Categories of Electrons

**Inner (core) electrons** are those an atom has in common with the pervious noble gas and any ***completed*** transition series.

**Outer electrons** are those in the ***highest*** energy level (highest *n* value).

**Valence electrons** are those involved in forming compounds.

For **main group** elements, the valence electrons ***are*** the outer electrons.

For **transition elements**, the valence electrons include the outer electrons and any (*n*-1)*d* electrons.





## Sample Problem 8.2 Determining Electron Configurations

**PROBLEM:** Using the periodic table on the inside cover of the text (not Figure 8.5 or Table 8.3), give the full and condensed electron configurations, partial orbital diagrams showing valence electrons only, and number of inner electrons for the following elements:

**(a)** potassium (K; $Z = 19$)

**(b)** technetium (Tc; $Z = 43$)

**(c)** lead (Pb; $Z = 82$)

**PLAN:** The atomic number gives the number of electrons, and the periodic table shows the order for filling orbitals. The partial orbital diagram includes all electrons added after the previous noble gas except those in filled inner sublevels.





**Sample Problem 8.2**

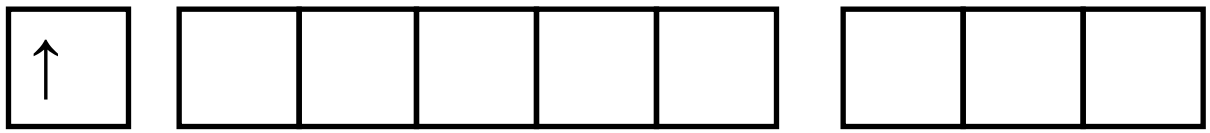
**SOLUTION:**

**(a)** For K ($Z = 19$)

full configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

condensed configuration [Ar] $4s^1$

partial orbital diagram




**There are 18 inner electrons.**

**Sample Problem 8.2**

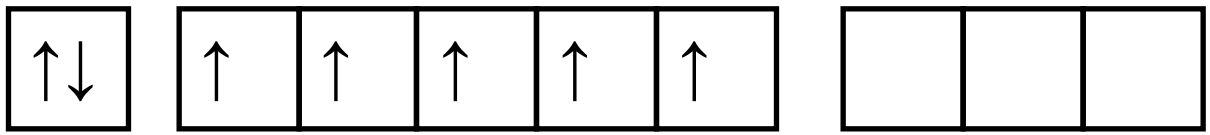
**SOLUTION:**

**(b)** For Tc ($Z$ = 43)

full configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^5$

condensed configuration $[Kr]5s^2 4d^5$

partial orbital diagram




**There are 36 inner electrons.**

**Sample Problem 8.2**

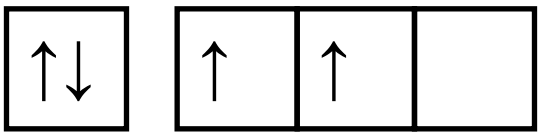
**SOLUTION:**

**(a)** For Pb ($Z$ = 82)

full configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^{10} 6p^2$

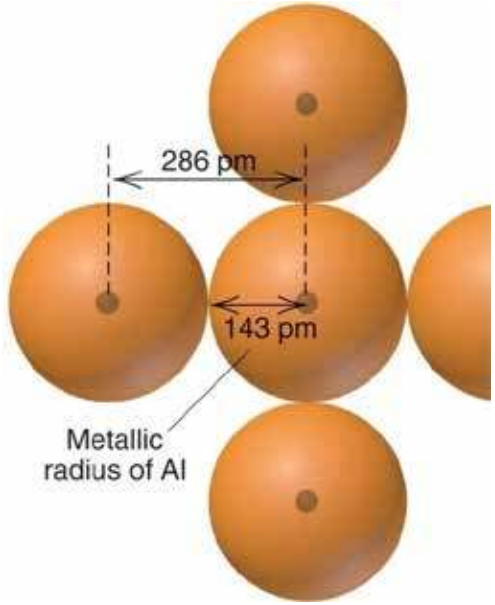
condensed configuration [Xe] $6s^2 4f^{14} 5d^{10} 6p^2$

partial orbital diagram

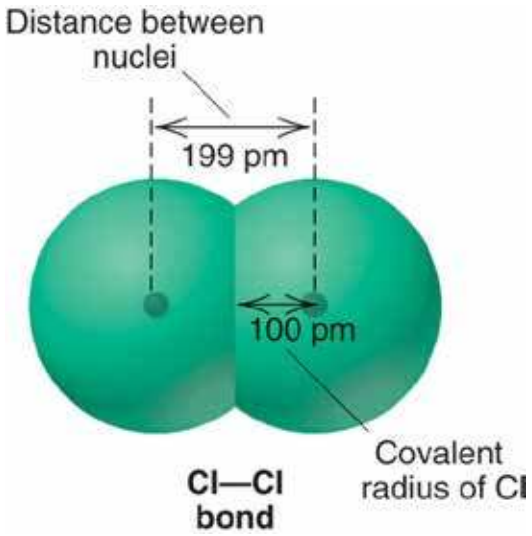



**There are 78 inner electrons.**

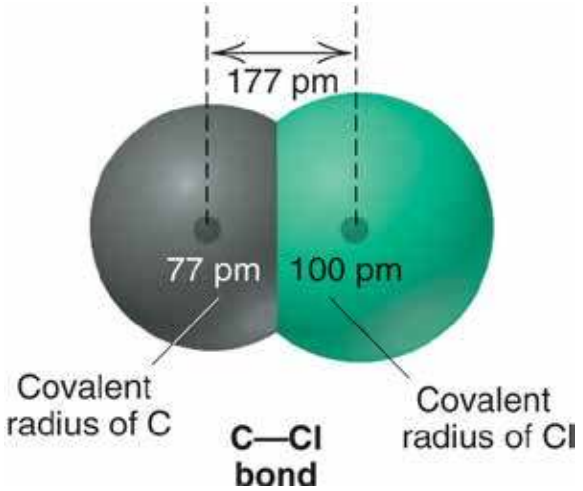
## Figure 8.7 Defining atomic size.




A. The metallic radius of aluminum.




B. The covalent radius of chlorine.




C. Known covalent radii and distances between nuclei can be used to find unknown radii.

# Trends in Atomic Size

Atomic size ***increases*** as the principal quantum number *n* ***increases***.

As *n* increases, the probability that the outer electrons will be farther from the nucleus increases.

Atomic size ***decreases*** as the effective nuclear charge $Z_{eff}$ ***increases***.

As $Z_{eff}$ increases, the outer electrons are pulled closer to the nucleus.

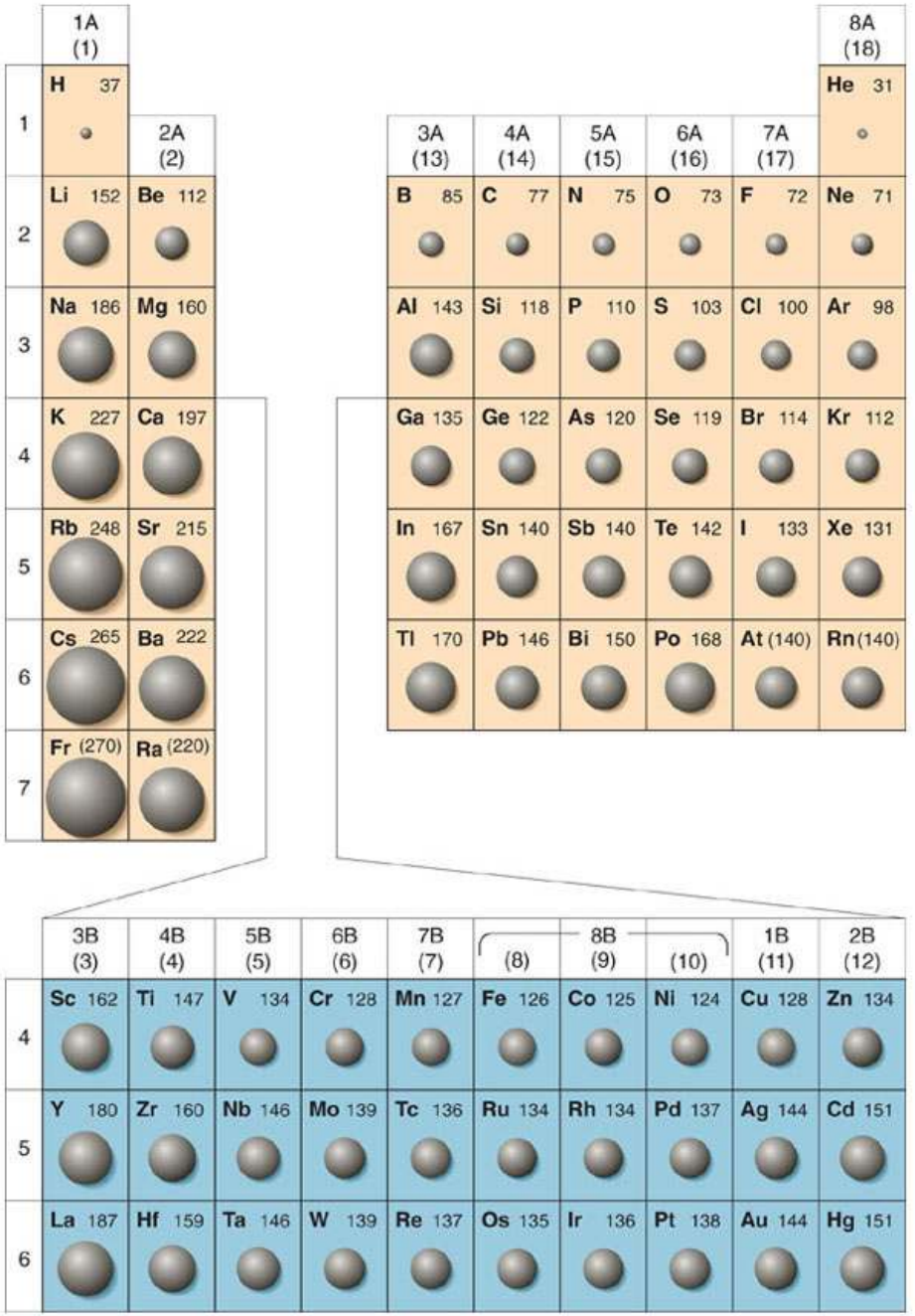
For ***main group elements***:
atomic size ***increases*** down a group in the periodic table and ***decreases*** across a period.



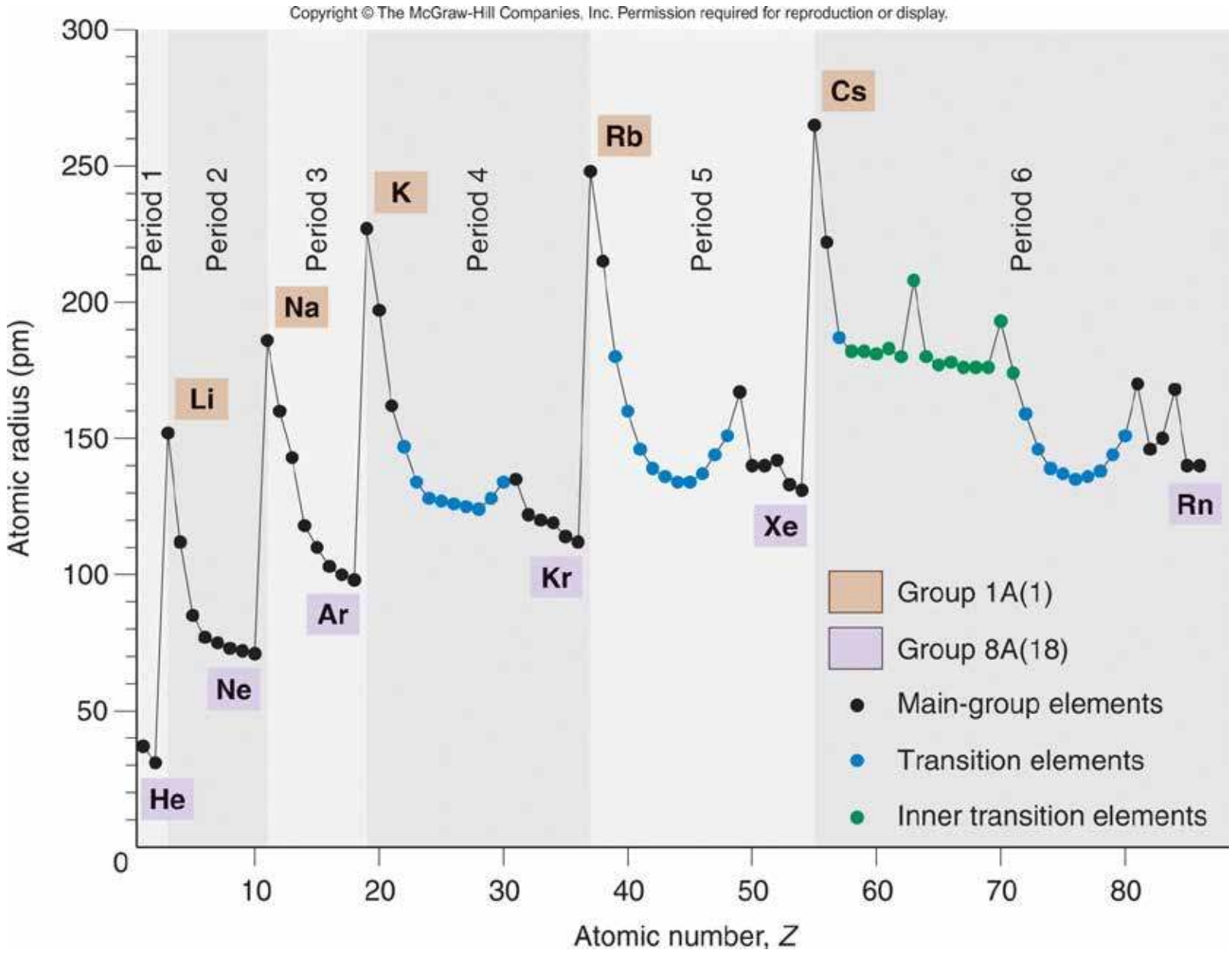

## Figure 8.8

**Atomic radii of the main-group and transition elements.**

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



## Figure 8.9 Periodicity of atomic radius.



**Sample Problem 8.3** **Ranking Elements by Atomic Size**

**PROBLEM:** Using only the periodic table (not Figure 8.8), rank each set of main-group elements in order of *decreasing* atomic size:

**(a)** Ca, Mg, Sr **(b)** K, Ga, Ca

**(c)** Br, Rb, Kr **(d)** Sr, Ca, Rb

**PLAN:** Locate each element on the periodic table. Main-group elements increase in size down a group and decrease in size across the period.





**Sample Problem 8.3**

**SOLUTION:**

**(a) Sr > Ca > Mg**

Ca, Mg, and Sr are in Group 2A. Size increases down the group.

**(b) K > Ca > Ga**

K, Ga, and Ca are all in Period 4. Size decreases across the period.

**(c) Rb > Br > Kr**

Rb is the largest because it has one more energy level than the other elements. Kr is smaller than Br because Kr is further to the right in the same period.

**(d) Rb > Sr > Ca**

Ca is the smallest because it has one fewer energy level. Sr is smaller than Rb because it is further to the right in the same period.





# Trends in Ionization Energy

***Ionization energy (IE)*** is the energy required for the ***complete removal*** of 1 mol of electrons from 1 mol of gaseous atoms or ions.

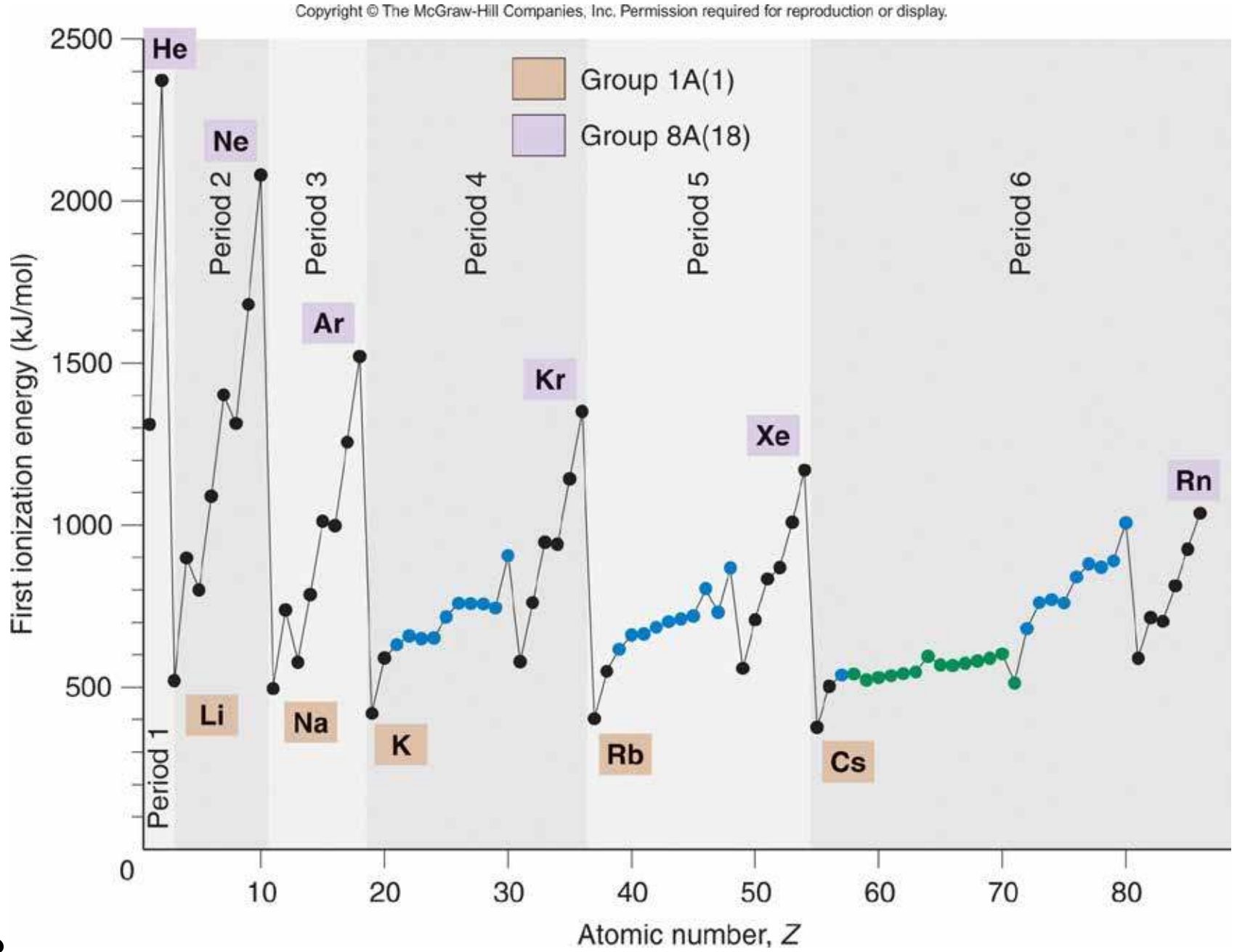
Atoms with a ***low IE*** tend to form ***cations***.
Atoms with a ***high IE*** tend to form ***anions*** (except the noble gases).

Ionization energy tends to ***decrease*** down a group and ***increase*** across a period.



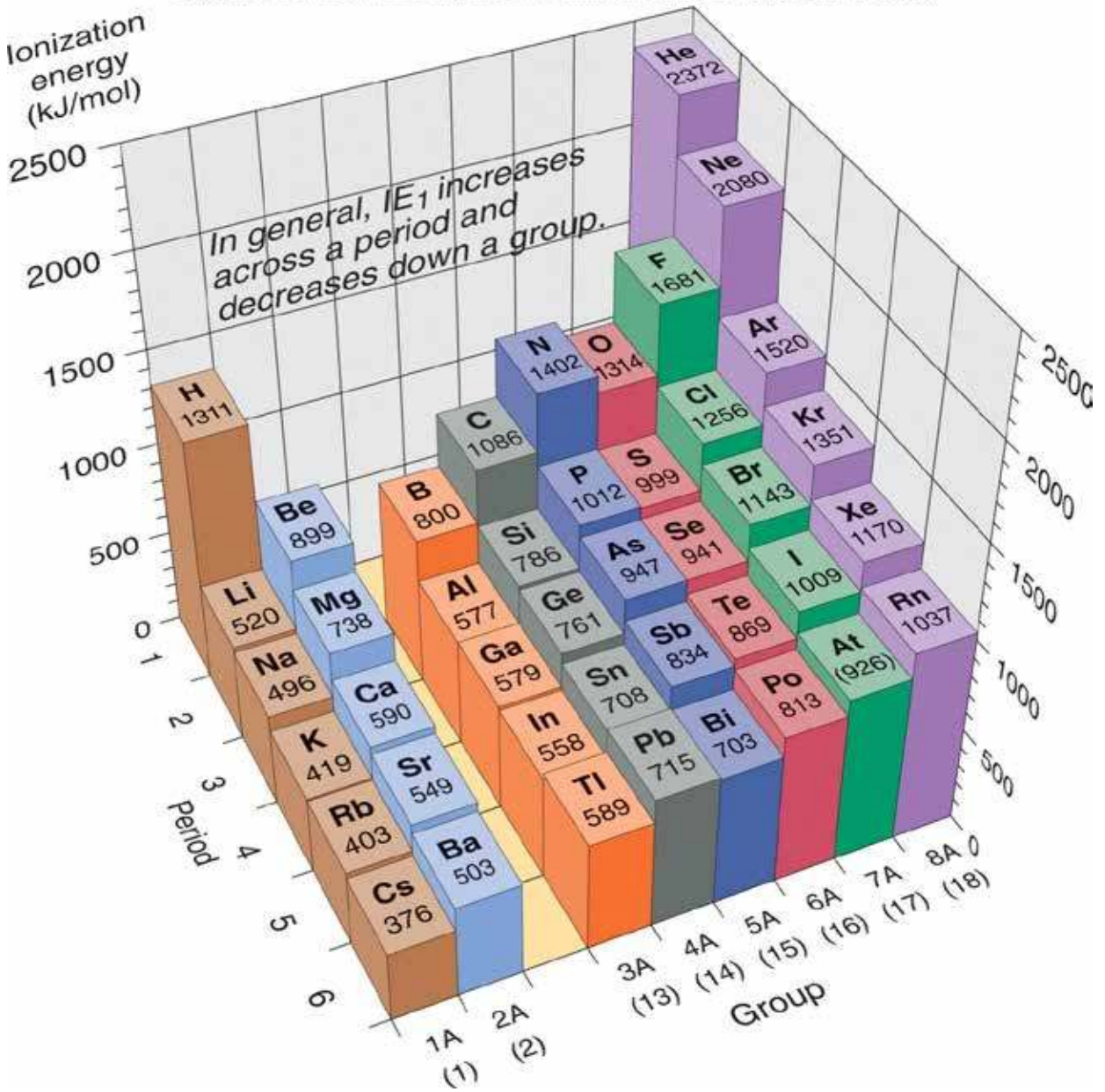


# Figure 8.10 Periodicity of first ionization energy ($IE_1$).



## Figure 8.11 First ionization energies of the main-group elements.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



**Sample Problem 8.4** **Ranking Elements by First Ionization Energy**

**PROBLEM:** Using the periodic table only, rank the elements in each of the following sets in order of *decreasing* $IE_1$:

**(a)** Kr, He, Ar **(b)** Sb, Te, Sn

**(c)** K, Ca, Rb **(d)** I, Xe, Cs

**PLAN:** Find each element on the periodic table. $IE_1$ generally decreases down a group and increases across a period.

**SOLUTION:**

**(a) He > Ar > Kr**

Kr, He, and Ar are in Group 8A. $IE_1$ decreases down the group.





## Sample Problem 8.4

**SOLUTION:**

**(b) Te > Sb > Sn**

Sb, Te, and Sn are in Period 5. $IE_1$ increases across a period.

**(c) Ca > K > Rb**

K has a higher $IE_1$ than Rb because K is higher up in Group 1A. Ca has a higher $IE_1$ than K because Ca is further to the right in Period 4.

**(d) Xe > I > Cs**

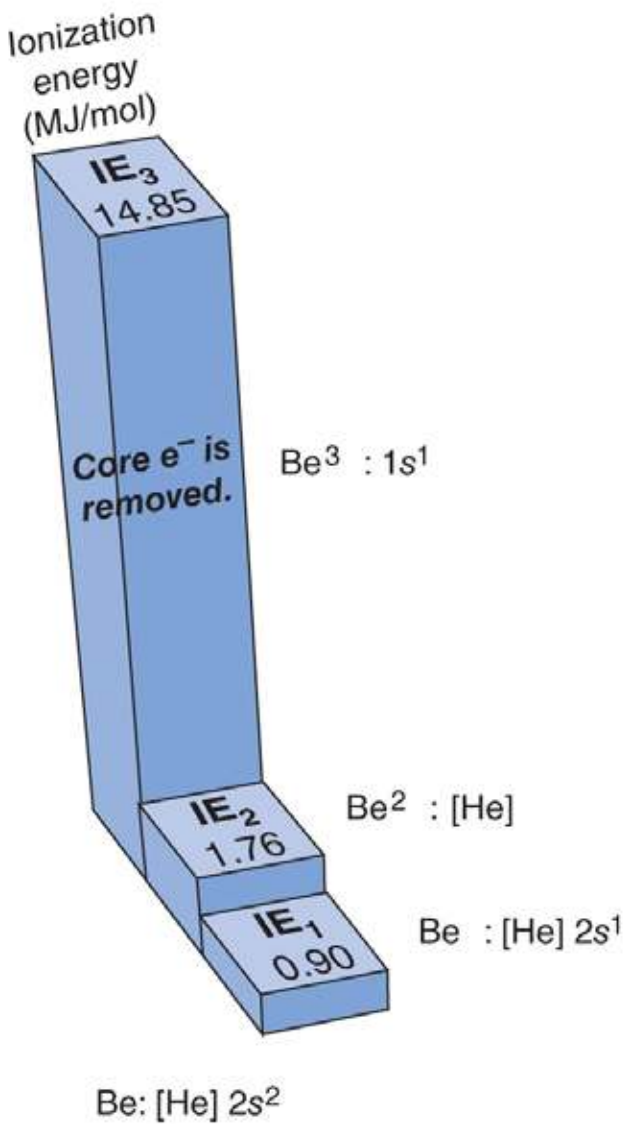
Xe has a higher $IE_1$ than I because Xe is further to the right in the same period. Cs has a lower $IE_1$ than I because it is further to the left in a higher period.





## Figure 8.12 The first three ionization energies of beryllium.

Copyright © McGraw-Hill Education. All rights reserved. No reproduction or distribution without the prior written consent of McGraw-Hill Education.




Beryllium has 2 valence electrons, so $IE_3$ is much larger than $IE_2$.

# Table 8.4 Successive Ionization Energies of the Elements Lithium Through Sodium

| Z | Element | Number of Valence Electrons | Ionization Energy (MJ/mol)* | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | $IE_1$ | $IE_2$ | $IE_3$ | $IE_4$ | $IE_5$ | $IE_6$ | $IE_7$ | $IE_8$ | $IE_9$ | $IE_{10}$ |
| 3 | Li | 1 | 0.52 | 7.30 | 11.81 | | | | | | | |
| 4 | Be | 2 | 0.90 | 1.76 | 14.85 | 21.01 | | | CORE ELECTRONS | | | |
| 5 | B | 3 | 0.80 | 2.43 | 3.66 | 25.02 | 32.82 | | | | | |
| 6 | C | 4 | 1.09 | 2.35 | 4.62 | 6.22 | 37.83 | 47.28 | | | | |
| 7 | N | 5 | 1.40 | 2.86 | 4.58 | 7.48 | 9.44 | 53.27 | 64.36 | | | |
| 8 | O | 6 | 1.31 | 3.39 | 5.30 | 7.47 | 10.98 | 13.33 | 71.33 | 84.08 | | |
| 9 | F | 7 | 1.68 | 3.37 | 6.05 | 8.41 | 11.02 | 15.16 | 17.87 | 92.04 | 106.43 | |
| 10 | Ne | 8 | 2.08 | 3.95 | 6.12 | 9.37 | 12.18 | 15.24 | 20.00 | 23.07 | 115.38 | 131.43 |
| 11 | Na | 1 | 0.50 | 4.56 | 6.91 | 9.54 | 13.35 | 16.61 | 20.11 | 25.49 | 28.93 | 141.37 |

*MJ/mol, or megajoules per mole = $10^3$ kJ/mol.

**Sample Problem 8.5** **Identifying an Element from Its Ionization Energies**

**PROBLEM:** Name the Period 3 element with the following ionization energies (in kJ/mol) and write its electron configuration:

| $IE_1$ | $IE_2$ | $IE_3$ | $IE_4$ | $IE_5$ | $IE_6$ |
|---|---|---|---|---|---|
| 1012 | 1903 | 2910 | 4956 | 6278 | 22,230 |

**PLAN:** Look for a large increase in IE, which occurs after all valence electrons have been removed.

**SOLUTION:**

The largest increase occurs after $IE_5$, that is, after the 5th valence electron has been removed. The Period 3 element with 5 valence electrons is **phosphorus (P; *Z* = 15)**.

The complete electron configuration is **$1s^2 2s^2 2p^6 3s^2 3p^3$**.





# Trends in Electron Affinity

***Electron Affinity (EA)*** is the energy change that occurs when 1 mol of electrons is ***added*** to 1 mol of gaseous atoms or ions.

Atoms with a ***low EA*** tend to form ***cations***.
Atoms with a ***high EA*** tend to form ***anions***.

The trends in electron affinity are not as regular as those for atomic size or IE.

## Figure 8.13 Electron affinities of the main-group elements (in kJ/mol).

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

| 1A (1) | 2A (2) | 3A (13) | 4A (14) | 5A (15) | 6A (16) | 7A (17) | 8A (18) |
|---|---|---|---|---|---|---|---|
| H −72.8 | | | | | | | He (0.0) |
| Li −59.6 | Be ≤0 | B −26.7 | C −122 | N +7 | O −141 | F −328 | Ne (+29) |
| Na −52.9 | Mg ≤0 | Al −42.5 | Si −134 | P −72.0 | S −200 | Cl −349 | Ar (+35) |
| K −48.4 | Ca −2.37 | Ga −28.9 | Ge −119 | As −78.2 | Se −195 | Br −325 | Kr (+39) |
| Rb −46.9 | Sr −5.03 | In −28.9 | Sn −107 | Sb −103 | Te −190 | I −295 | Xe (+41) |
| Cs −45.5 | Ba −13.95 | Tl −19.3 | Pb −35.1 | Bi −91.3 | Po −183 | At −270 | Rn (+41) |

# Behavior Patterns for IE and EA

***Reactive nonmetals*** have high IEs and highly negative EAs.

These elements attract electrons strongly and tend to form negative ions in ionic compounds.

***Reactive metals*** have low IEs and slightly negative EAs.

These elements lose electrons easily and tend to form positive ions in ionic compounds.

***Noble gases*** have very high IEs and slightly positive EAs.

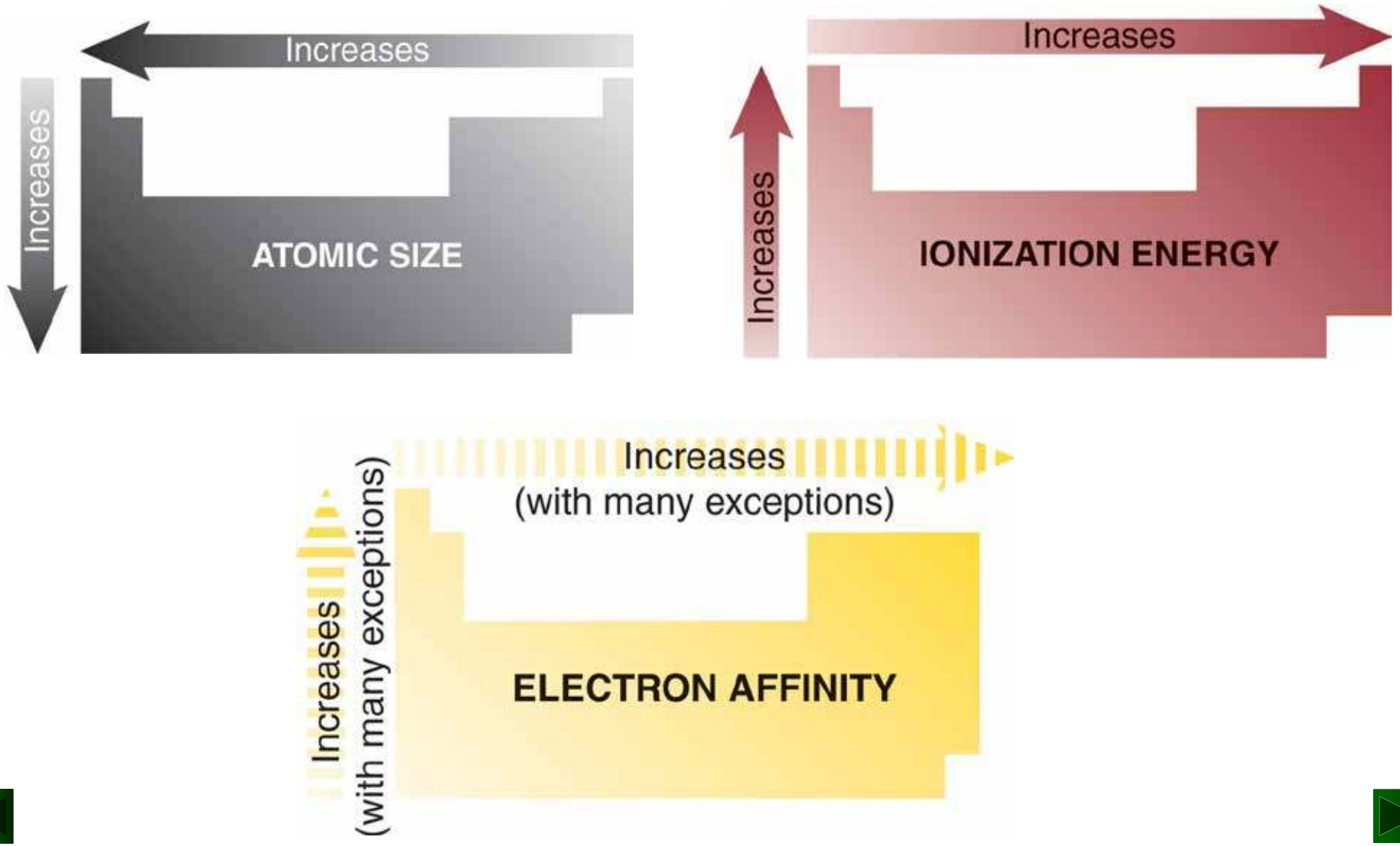
These elements tend to neither lose nor gain electrons.





# Figure 8.14 Trends in three atomic properties.

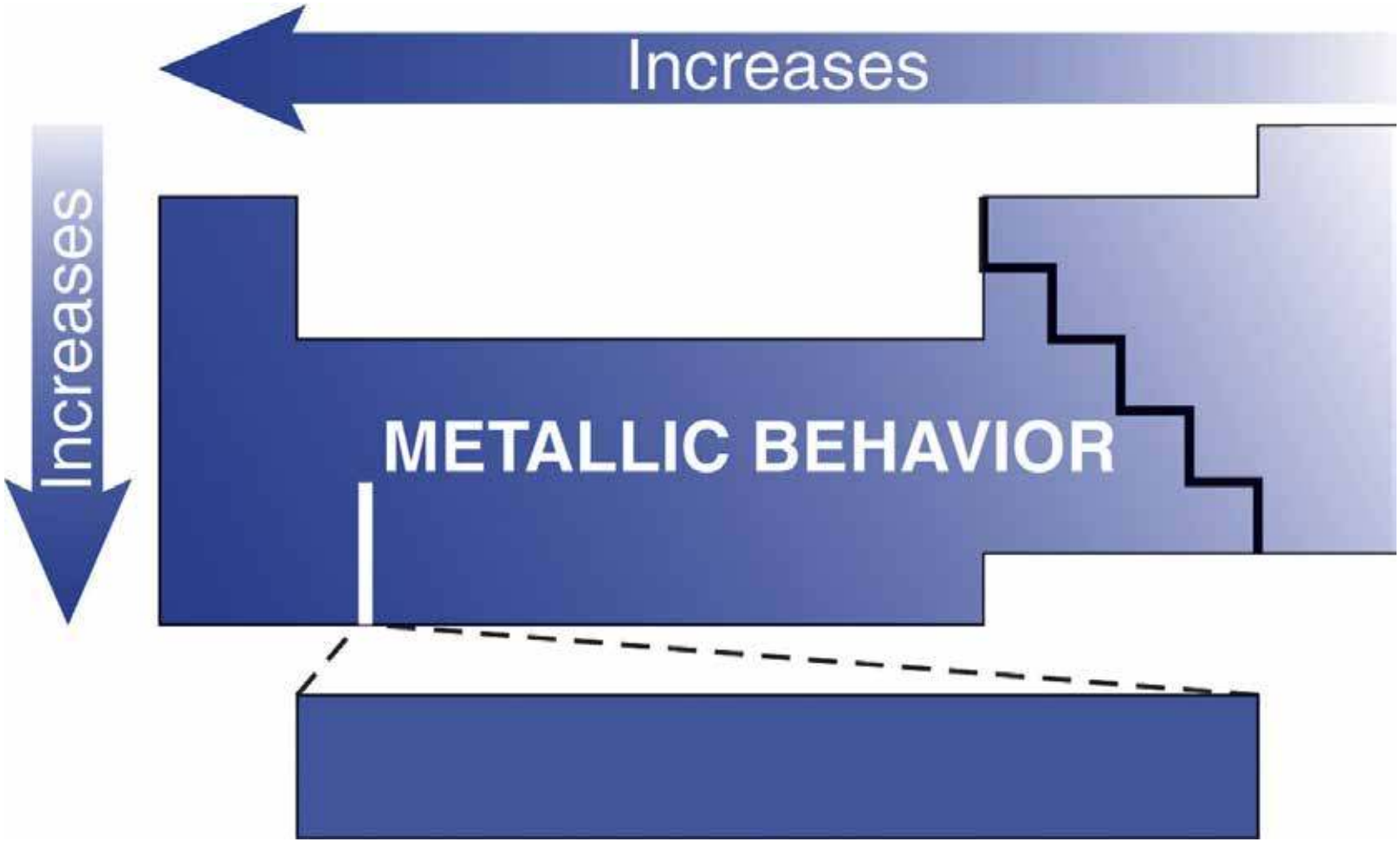
` Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



# Metallic Behavior

- Metals are typically shiny solids with moderate to high melting points.
- Metals are good conductors of heat and electricity, and can easily be shaped.
- Metals tend to lose electrons and form cations, i.e., they are easily ***oxidized***.
- Metals are generally ***strong reducing agents***.
- Most metals form ionic oxides, which are ***basic*** in aqueous solution.

**Figure 8.15 Trends in metallic behavior.**



# Acid-Base Behavior of Oxides

Main-group metals form ***ionic oxides***, which are ***basic*** in aqueous solution.

Main-group nonmetals form ***covalent oxides***, which are ***acidic*** in aqueous solution.

Some metals and metalloids form ***amphoteric oxides***, which can act as acids or bases in water:

$$Al_2O_3\ (s) + 6HCl\ (aq) \rightarrow 2AlCl_3\ (aq) + 3H_2O\ (l)$$

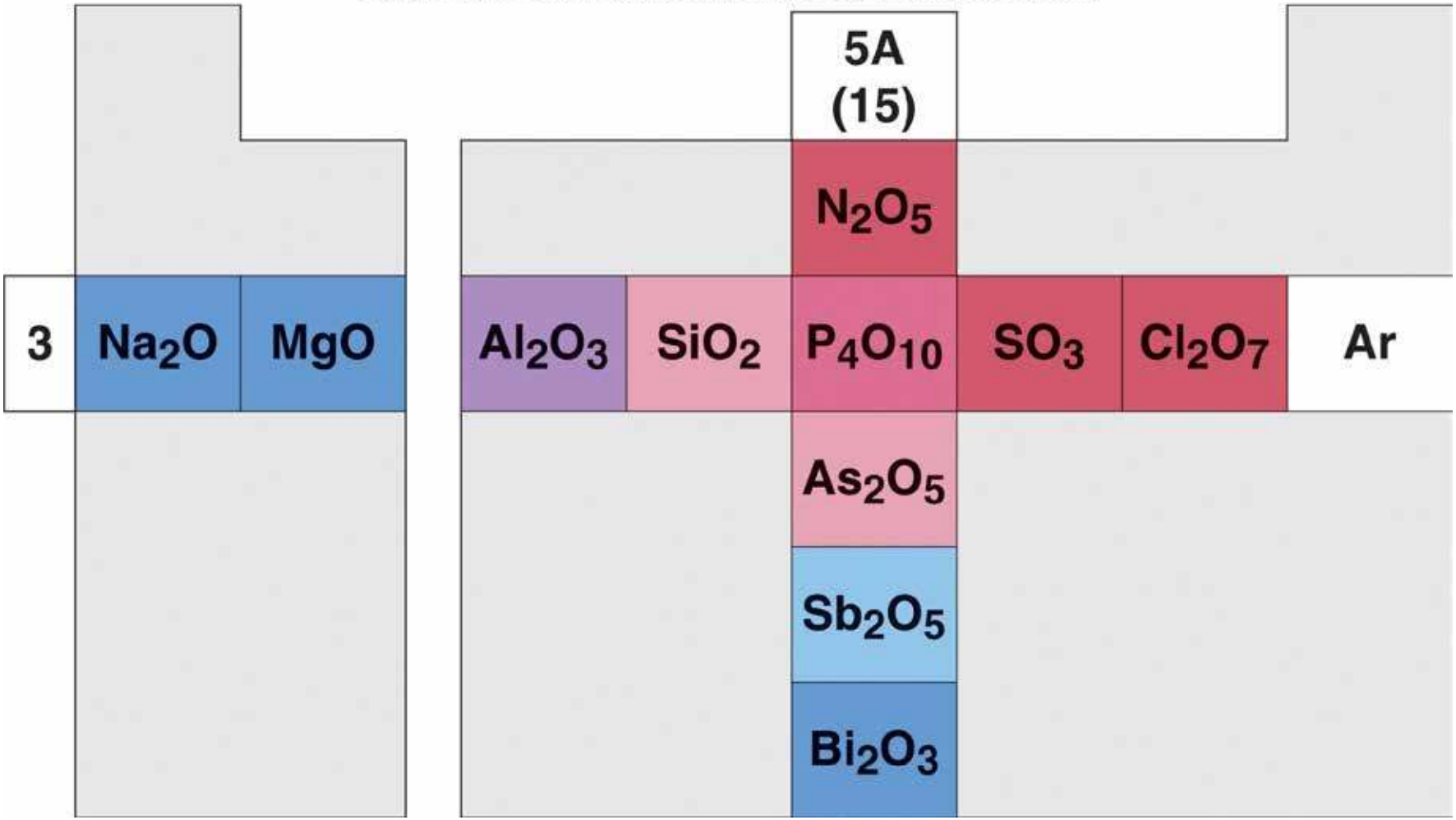
$$Al_2O_3\ (s) + 2NaOH\ (aq) + 3H_2O\ (l) \rightarrow 2NaAl(OH)_4\ (aq)$$





## Figure 8.16 Acid-base behavior of some element oxides.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.




Oxides become more basic down a group and more acidic across a period.

# Electron configurations of Monatomic Ions

Elements at either end of a period gain or lose electrons to attain a filled outer level. The resulting ion will have a ***noble gas electron configuration*** and is said to be ***isoelectronic*** with that noble gas.

$Na(1s^2 2s^2 2p^6 \mathbf{3s^1}) \rightarrow \mathbf{e^-} + Na^+([He]2s^2 2p^6)$

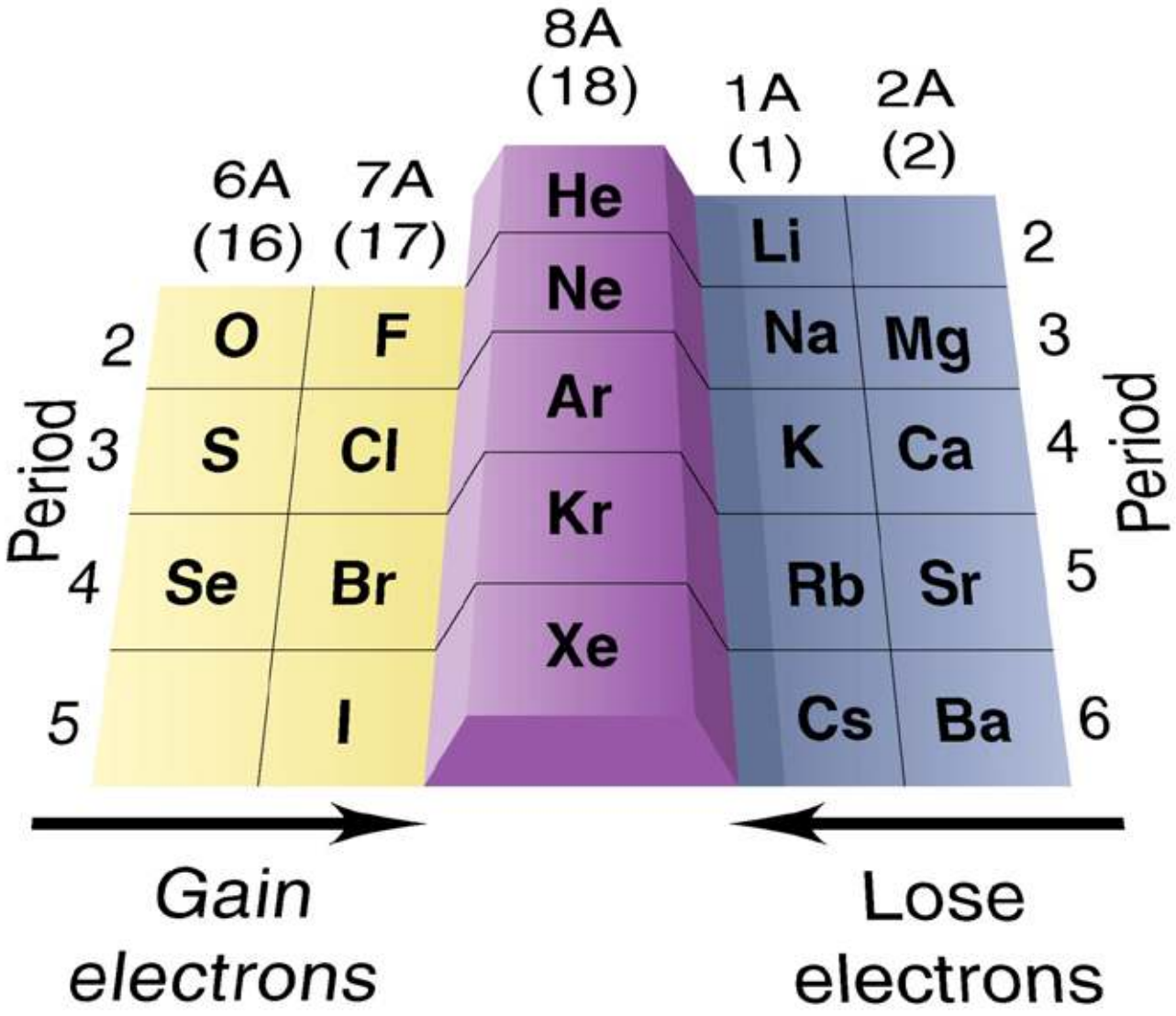
[isoelectronic with Ne]

$Br([Ar]4s^2 3d^{10} \mathbf{4p^5}) + \mathbf{e^-} \rightarrow Br^- ([Ar]4s^2 3d^{10} 4p^6)$

[isoelectronic with Kr]

**Figure 8.17** **Main-group elements whose ions have noble gas electron configurations.**

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



# Electron Configurations of Monatomic Ions

A ***pseudo-noble gas configuration*** is attained when a metal atom empties its highest energy level.
The ion attains the stability of empty *ns* and *np* sublevels and a filled (*n*-1)*d* sublevel.

$$Sn([Kr]\mathbf{5}s^{\mathbf{2}}4d^{10}\mathbf{5}p^{\mathbf{2}}) \rightarrow \mathbf{4e^-} + Sn^{4+}([Kr]4d^{10})$$

A metal may lose only the *np* electrons to attain an ***inert pair configuration***.
The ion attains the stability of a filled *ns* and (*n*-1)*d* sublevels.

$$Sn([Kr]5s^{2}4d^{10}\mathbf{5}p^{\mathbf{2}}) \rightarrow \mathbf{2e^-} + Sn^{2+}([Kr]5s^{2}4d^{10})$$




## Sample Problem 8.6 Writing Electron Configurations of Main-Group Ions

**PROBLEM:** Using condensed electron configurations, write reactions for the formation of the common ions of the following elements:

**(a)** Iodine ($Z = 53$) **(b)** Potassium ($Z = 19$) **(c)** Indium ($Z = 49$)

**PLAN:** Identify the position of each element on the periodic table and recall that:

- Ions of elements in Groups 1A(1), 2A(2), 6A(16), and 7A(17) are usually isoelectronic with the nearest noble gas.
- Metals in Groups 3A(13) to 5A(15) can lose the *ns* and *np* electrons or just the *np* electrons.






اتحاد سنافر البوليتكنك (spu)

**Sample Problem 8.6**

**SOLUTION:**

**(a)** Iodine (Z = 53) is in Group 7A(17) and will gain one electron to be isoelectronic with Xe: $I\ ([Kr]\ 5s^2 4d^{10} 5p^5) + e^- \longrightarrow I^-\ ([Kr]\ 5s^2 4d^{10} 5p^6)$

**(b)** Potassium (Z = 19) is in Group 1A(1) and will lose one electron to be isoelectronic with Ar: $K\ ([Ar]\ 4s^1) \longrightarrow K^+\ ([Ar]) + e^-$

**(c)** Indium (Z = 49) is in Group 3A(13) and can lose either one electron or three electrons: $In\ ([Kr]\ 5s^2 4d^{10} 5p^1) \longrightarrow In^+\ ([Kr]\ 5s^2 4d^{10}) + e^-$

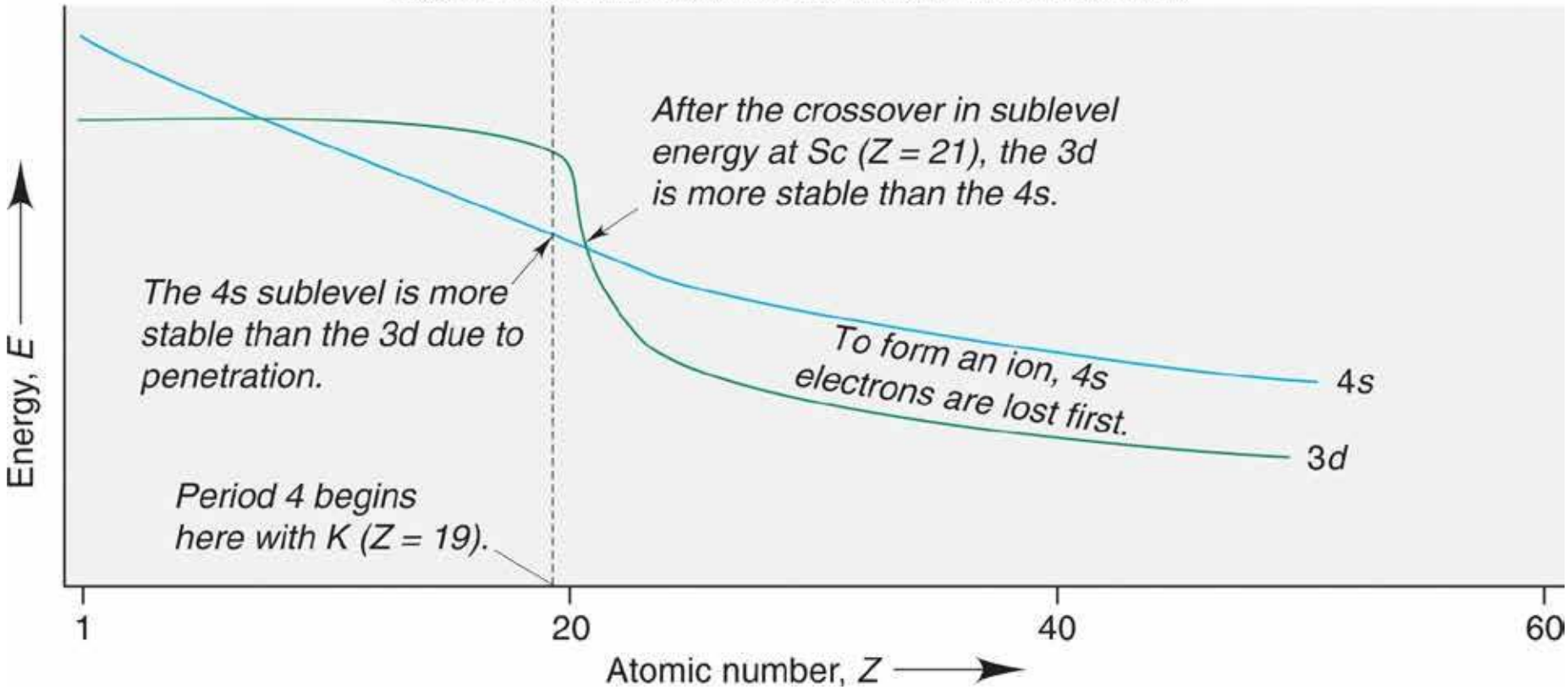
$In\ ([Kr]\ 5s^2 4d^{10} 5p^1) \longrightarrow In^{3+}([Kr]\ 4d^{10}) + 3e^-$





## Figure 8.18 The crossover of sublevel energies in Period 4.

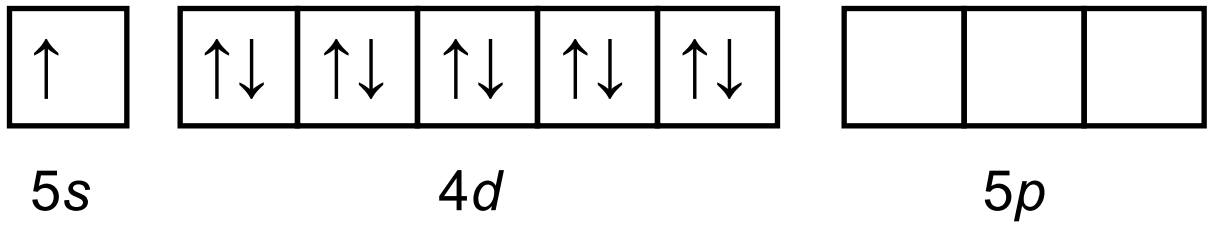
Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



# Magnetic Properties of Transition Metal Ions

A species with one or more unpaired electrons exhibits ***paramagnetism*** – it is attracted by a magnetic field.

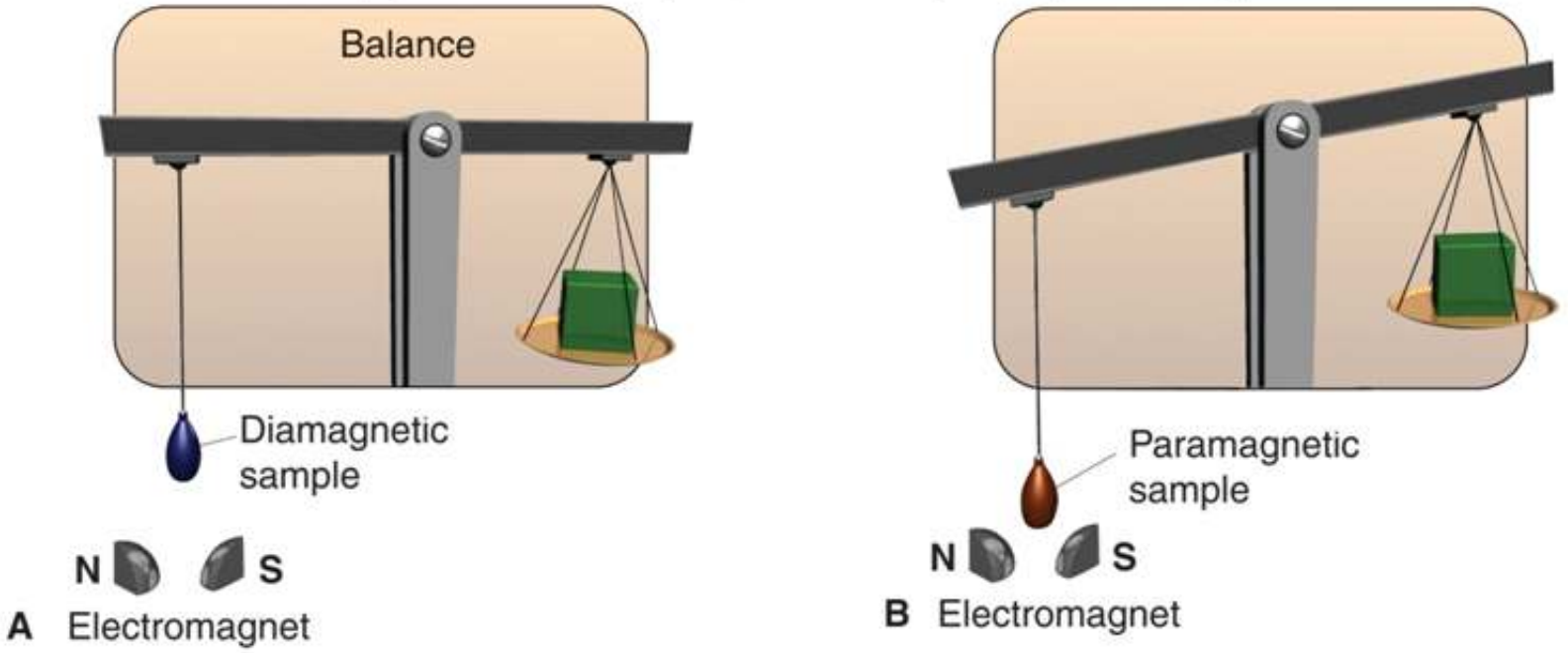
Ag ($Z$ = 47)




A species with all its electrons paired exhibits ***diamagnetism*** – it is not attracted (and is slightly repelled) by a magnetic field.

## Figure 8.19 Measuring the magnetic behavior of a sample.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.

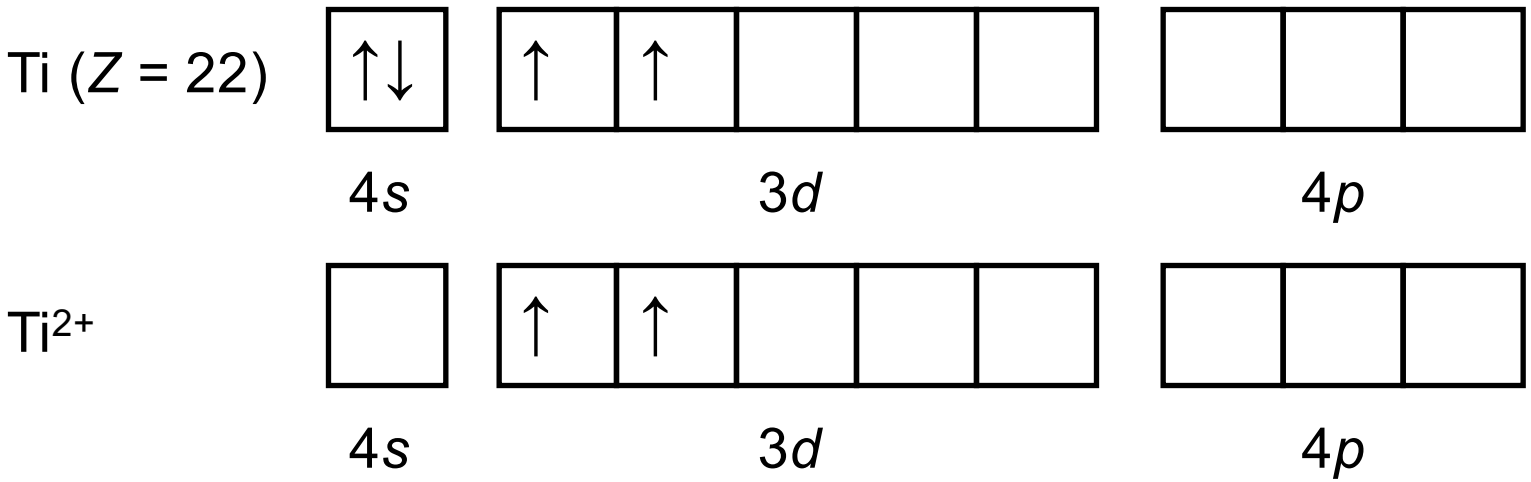



The apparent mass of a diamagnetic substance is unaffected by the magnetic field.

The apparent mass of a paramagnetic substance increases as it is attracted by the magnetic field.

# Magnetic Properties of Transition Metal ions

Magnetic behavior can provide evidence for the electron configuration of a given ion.




$Ti^{2+}$ has 2 unpaired electrons and is paramagnetic, providing evidence that the 4*s* electrons are lost before the 3*d* electrons.

**Sample Problem 8.7** **Writing Electron Configurations and Predicting Magnetic Behavior of Transition Metal Ions**

**PROBLEM:** Use condensed electron configurations to write the reaction for the formation of each transition metal ion, and predict whether the ion is paramagnetic or diamagnetic.

**(a)** $Mn^{2+}(Z = 25)$ **(b)** $Cr^{3+}(Z = 24)$ **(c)** $Hg^{2+}(Z = 80)$

**PLAN:** Write the condensed electron configuration for each atom, recalling the irregularity for Cr. Remove electrons, beginning with the *ns* electrons, and determine if there are any unpaired electrons.





**Sample Problem 8.7**

**SOLUTION:**

**(a)** $Mn^{2+}(Z = 25)$ Mn ([Ar] $4s^2 3d^5$) $\longrightarrow$ $Mn^{2+}$ ([Ar] $3d^5$) + $2e^-$

Since there are 5 *d* electrons they are all unpaired. $Mn^{2+}$ is **paramagnetic**.

**(b)** $Cr^{3+}(Z = 24)$ Cr ([Ar] $4s^1 3d^5$) $\longrightarrow$ $Cr^{3+}$ ([Ar] $3d^3$) + $3e^-$

Since there are 3 *d* electrons they are all unpaired. $Cr^{3+}$ is **paramagnetic.**

**(c)** $Hg^{2+}(Z = 80)$ Hg ([Xe] $6s^2 4f^{14} 5d^{10}$) $\longrightarrow$ $Hg^{2+}$ ([Xe] $4f^{14} 5d^{10}$) + $2e^-$

The 4*f* and the 5*s* sublevels are filled, so there are no unpaired electrons. $Hg^{2+}$ is **diamagnetic**.





# Ionic Size vs. Atomic Size

Cations are ***smaller*** than their parent atoms while anions are ***larger***.

Ionic radius ***increases*** down a group as *n* increases.

Cation size ***decreases*** as charge ***increases***.

An ***isoelectronic series*** is a series of ions that have the same electron configuration. Within the series, ion size ***decreases*** with increasing nuclear charge.

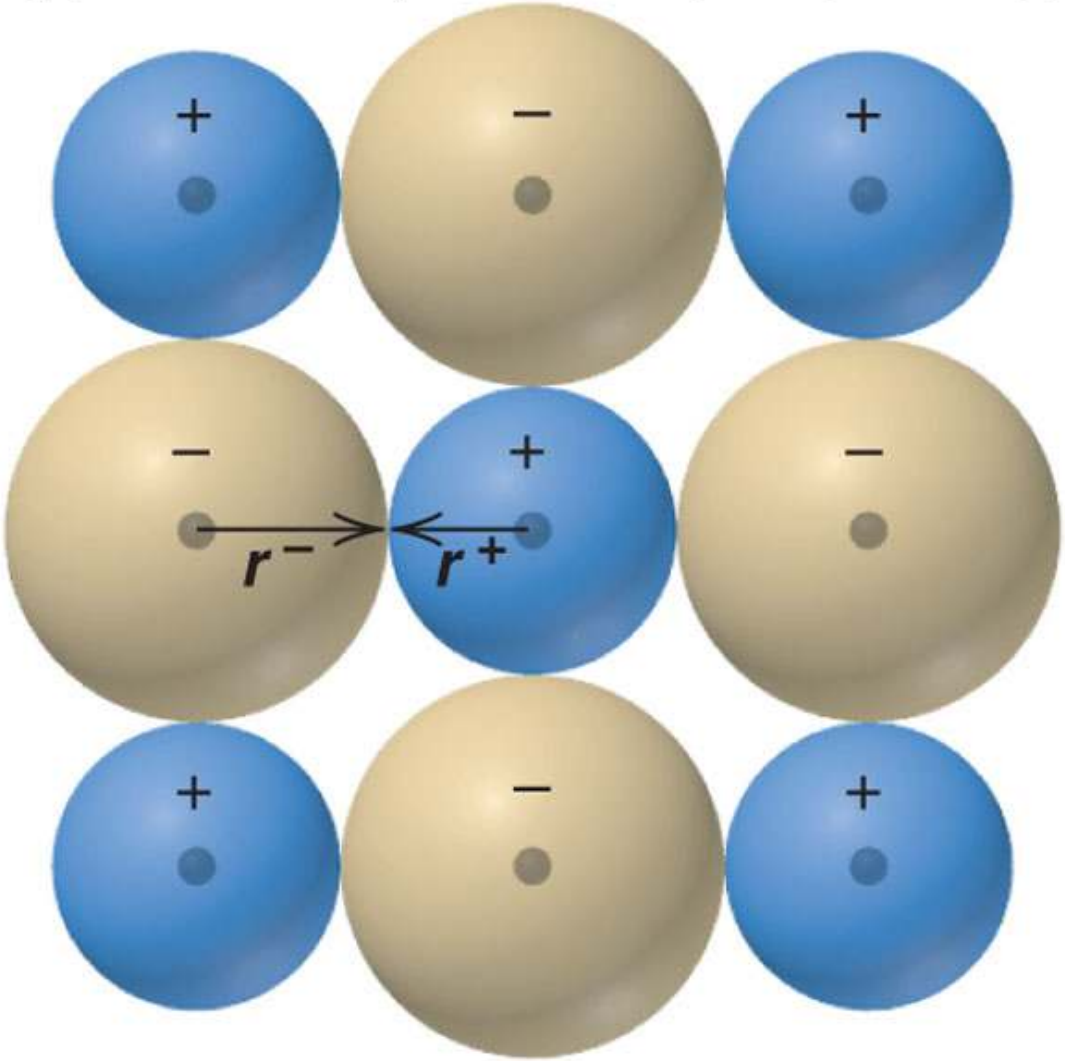
**3- > 2- > 1- > 1+ > 2+ > 3+**





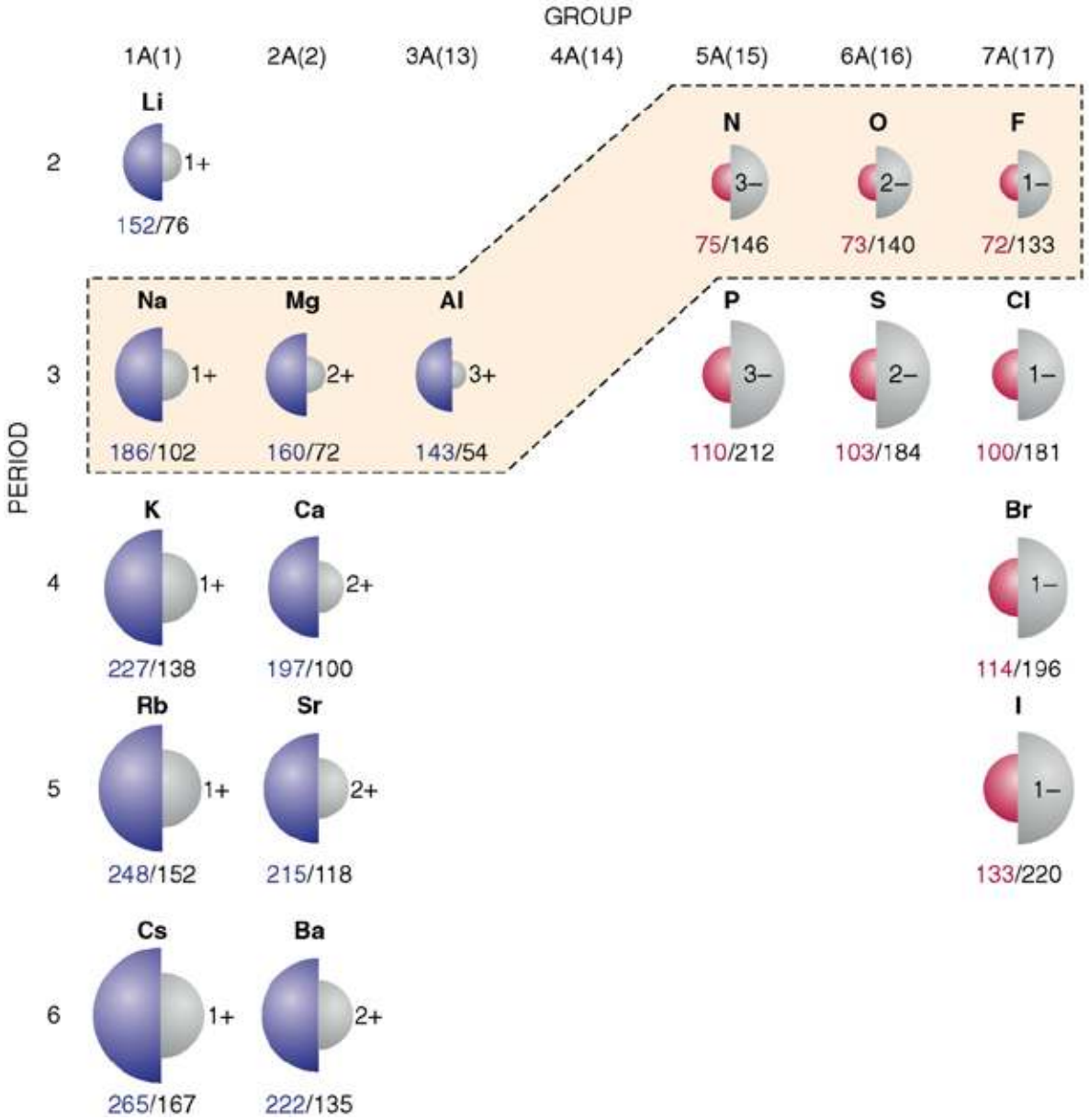
## Figure 8.20 Ionic radius.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



# Figure 8.21 Ionic vs. atomic radii.

Copyright © The McGraw-Hill Companies, Inc. Permission required for reproduction or display.



**Sample Problem 8.8** **Ranking Ions by Size**

**PROBLEM:** Rank each set of ions in order of *decreasing* size, and explain your ranking:

**(a)** $Ca^{2+}$, $Sr^{2+}$, $Mg^{2+}$ **(b)** $K^{+}$, $S^{2-}$, $Cl^{-}$ **(c)** $Au^{+}$, $Au^{3+}$

**PLAN:** Find the position of each element on the periodic table and apply the trends for ionic size.

**SOLUTION:**

**(a) $Sr^{2+}$ > $Ca^{2+}$ > $Mg^{2+}$**

**All these ions are from Group 2A, so size increases down the group.**



## Sample Problem 8.8

**SOLUTION:**

**(b) $S^{2-} > Cl^{-} > K^{+}$**

**These ions are isoelectronic, so size decreases as nuclear charge increases.**

**(c) $Au^{+} > Au^{3+}$**

**Cation size decreases as charge increases.**

# 7 Atomic Structure and Periodicity

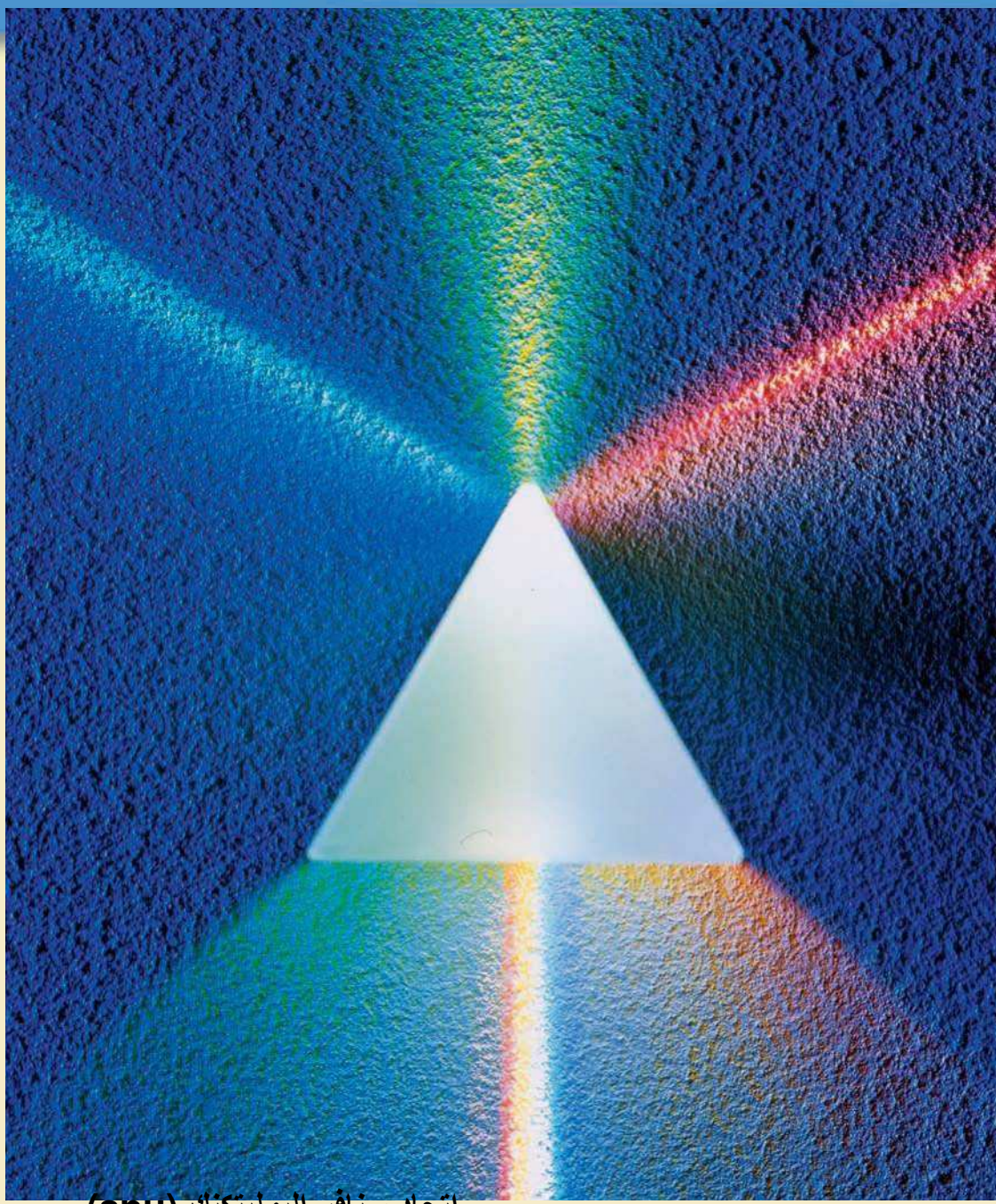
## Contents


**7.1 Electromagnetic Radiation**
**7.2 The Nature of Matter**
- The Photoelectric Effect

**7.3 The Atomic Spectrum of Hydrogen**
**7.4 The Bohr Model**
**7.5 The Quantum Mechanical Model of the Atom**
- The Physical Meaning of a Wave Function

**7.6 Quantum Numbers**
**7.7 Orbital Shapes and Energies**
**7.8 Electron Spin and the Pauli Principle**
**7.9 Polyelectronic Atoms**
**7.10 The History of the Periodic Table**
**7.11 The Aufbau Principle and the Periodic Table**
**7.12 Periodic Trends in Atomic Properties**
- Ionization Energy
- Electron Affinity
- Atomic Radius

**7.13 The Properties of a Group: The Alkali Metals**
- Information Contained in the Periodic Table
- The Alkali Metals

*Light refracted through a prism.*

اتحاد سنافر البوليتكنك (spu)



In the past 200 years, a great deal of experimental evidence has accumulated to support the atomic model. This theory has proved to be both extremely useful and physically reasonable. When atoms were first suggested by the Greek philosophers Democritus and Leucippus about 400 B.C., the concept was based mostly on intuition. In fact, for the following 20 centuries, no convincing experimental evidence was available to support the existence of atoms. The first real scientific data were gathered by Lavoisier and others from quantitative measurements of chemical reactions. The results of these stoichiometric experiments led John Dalton to propose the first systematic atomic theory. Dalton's theory, although crude, has stood the test of time extremely well.

Once we came to "believe in" atoms, it was logical to ask: What is the nature of an atom? Does an atom have parts, and if so, what are they? In Chapter 2 we considered some of the experiments most important for shedding light on the nature of the atom. Now we will see how the atomic theory has evolved to its present state.

One of the most striking things about the chemistry of the elements is the periodic repetition of properties. There are several groups of elements that show great similarities in chemical behavior. As we saw in Chapter 2, these similarities led to the development of the periodic table of the elements. In this chapter we will see that the modern theory of atomic structure accounts for periodicity in terms of the electron arrangements in atoms.

However, before we examine atomic structure, we must consider the revolution that took place in physics in the first 30 years of the twentieth century. During that time, experiments were carried out, the results of which could not be explained by the theories of classical physics developed by Isaac Newton and many others who followed him. A radical new theory called *quantum mechanics* was developed to account for the behavior of light and atoms. This "new physics" provides many surprises for humans who are used to the macroscopic world, but it seems to account flawlessly (within the bounds of necessary approximations) for the behavior of matter.

As the first step in our exploration of this revolution in science we will consider the properties of light, more properly called *electromagnetic radiation*.

## 7.1 Electromagnetic Radiation

One of the ways that energy travels through space is by **electromagnetic radiation.** The light from the sun, the energy used to cook food in a microwave oven, the X rays used by dentists, and the radiant heat from a fireplace are all examples of electromagnetic radiation. Although these forms of radiant energy seem quite different, they all exhibit the same type of wavelike behavior and travel at the speed of light in a vacuum.

Waves have three primary characteristics: wavelength, frequency, and speed. **Wavelength** (symbolized by the lowercase Greek letter lambda, $\lambda$) is the *distance between two consecutive peaks or troughs in a wave,* as shown in Fig. 7.1. The **frequency** (symbolized by the lowercase Greek letter nu, $\nu$) is defined as the *number of waves (cycles) per second that pass a given point in space*. Since all types of electromagnetic radiation travel at the speed of light, short-wavelength radiation must have a high frequency. You can see this in Fig. 7.1, where three waves are shown traveling between two points at constant speed. Note that the wave with the shortest wavelength ($\lambda_3$) has the highest frequency and the wave with the longest wavelength ($\lambda_1$) has the lowest frequency. This implies an inverse relationship between wavelength and frequency, that is, $\lambda \propto 1/\nu$, or

$$\lambda\nu = c$$

Wavelength $\lambda$ and frequency $\nu$ are inversely related.

$c$ = speed of light
= $2.9979 \times 10^8$ m/s

(spu) اتحاد سنافر البولنيتكنك

Visualization: Electromagnetic Wave

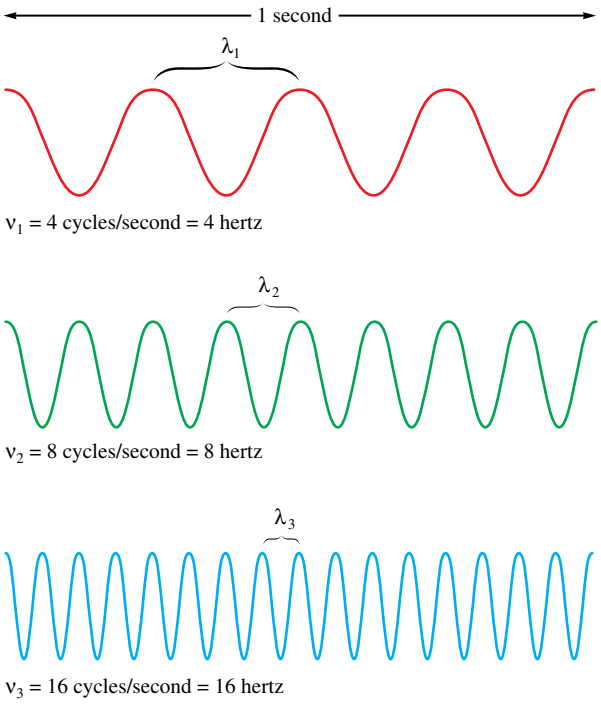



**FIGURE 7.1**
The nature of waves. Note that the radiation with the shortest wavelength has the highest frequency.



Although the waves associated with light are not obvious to the naked eye, ocean waves provide a familiar source of recreation.

where $\lambda$ is the wavelength in meters, $\nu$ is the frequency in cycles per second, and $c$ is the speed of light ($2.9979 \times 10^8$ m/s). In the SI system, cycles is understood, and the unit per second becomes 1/s, or $s^{-1}$, which is called the *hertz* (abbreviated Hz).

Electromagnetic radiation is classified as shown in Fig. 7.2. Radiation provides an important means of energy transfer. For example, the energy from the sun reaches the earth mainly in the form of visible and ultraviolet radiation, whereas the glowing coals of a fireplace transmit heat energy by infrared radiation. In a microwave oven the water molecules in food absorb microwave radiation, which increases their motions. This energy is then transferred to other types of molecules via collisions, causing an increase in the food's temperature. As we proceed in the study of chemistry, we will consider many of the classes of electromagnetic radiation and the ways in which they affect matter.

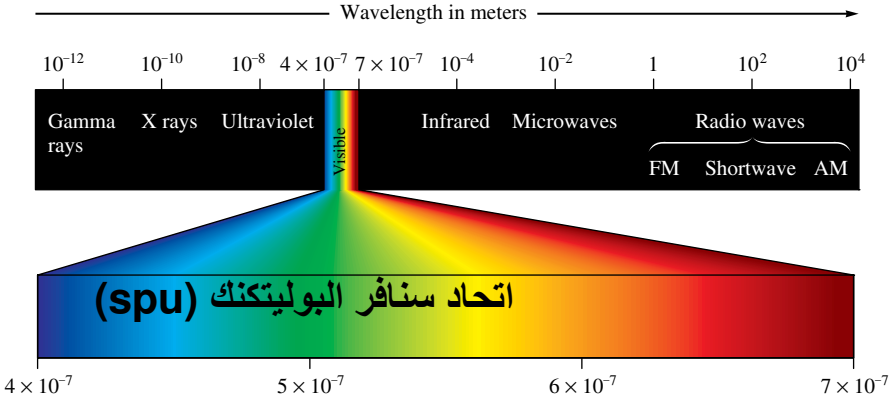



**FIGURE 7.2**
Classification of electromagnetic radiation.
Spectrum adapted by permission from C. W. Keenan, D. C. Kleinfelter, and J. H. Wood, *General College Chemistry,* 6th ed. (New York: Harper & Row, 1980).



## CHEMICAL IMPACT

### Flies That Dye

Mediterranean and Mexican fruit flies are formidable pests that have the potential to seriously damage several important fruit crops. Because of this, there have been several widely publicized sprayings of residential areas in southern California with the pesticide malathion to try to control fruit flies. Now there may be a better way to kill fruit flies—with a blend of two common dyes (red dye no. 28 and yellow dye no. 8) long used to color drugs and cosmetics. One of the most interesting things about this new pesticide is that it is activated by light. After an insect eats the blend of dyes, the molecules absorb light (through the insect's transparent body), which causes them to generate oxidizing agents that attack the proteins and cell membranes in the bug's body. Death occurs within 12 hours.

The sunlight that turns on the dye's toxicity after the fly ingests it also degrades the dye in the environment, making it relatively safe. It appears likely that in the near future the fruit fly will "dye" with little harm to the environment.

*Sample Exercise 7.1*

### Frequency of Electromagnetic Radiation



When a strontium salt is dissolved in methanol (with a little water) and ignited, it gives a brilliant red flame. The red color is produced by emission of light when electrons, excited by the energy of the burning methanol, fall back to their ground states.

The brilliant red colors seen in fireworks are due to the emission of light with wavelengths around 650 nm when strontium salts such as $Sr(NO_3)_2$ and $SrCO_3$ are heated. (This can be easily demonstrated in the lab by dissolving one of these salts in methanol that contains a little water and igniting the mixture in an evaporating dish.) Calculate the frequency of red light of wavelength $6.50 \times 10^2$ nm.

***Solution***

We can convert wavelength to frequency using the equation

$$\lambda\nu = c \quad \text{or} \quad \nu = \frac{c}{\lambda}$$

where $c = 2.9979 \times 10^8$ m/s. In this case $\lambda = 6.50 \times 10^2$ nm. Changing the wavelength to meters, we have

$$6.50 \times 10^2 \cancel{\text{nm}} \times \frac{1 \text{ m}}{10^9 \cancel{\text{nm}}} = 6.50 \times 10^{-7} \text{ m}$$

and

$$\nu = \frac{c}{\lambda} = \frac{2.9979 \times 10^8 \cancel{\text{m}}/\text{s}}{6.50 \times 10^{-7} \cancel{\text{m}}} = 4.61 \times 10^{14} \text{ s}^{-1} = 4.61 \times 10^{14} \text{ Hz}$$

*See Exercises 7.31 and 7.32.*

## 7.2 The Nature of Matter

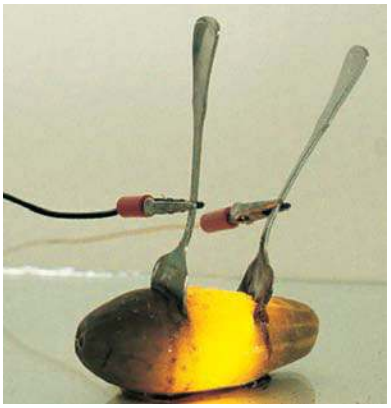
It is probably fair to say that at the end of the nineteenth century, physicists were feeling rather smug. Theories could explain phenomena as diverse as the motions of the planets and the dispersion of visible light by a prism. Rumor has it that students were being discouraged from pursuing physics as a career because it was felt that all the major problems had been solved, or at least described in terms of the current physical theories.



Visualization: Electrified Pickle

At the end of the nineteenth century, the idea prevailed that matter and energy were distinct. Matter was thought to consist of particles, whereas energy in the form of light (electromagnetic radiation) was described as a wave. Particles were things that had mass and whose position in space could be specified. Waves were described as massless and delocalized; that is, their position in space could not be specified. It also was assumed that there was no intermingling of matter and light. Everything known before 1900 seemed to fit neatly into this view.

(spu) اتحاد سنافر ابوليتكاك



When alternating current at 110 volts is applied to a dill pickle, a glowing discharge occurs. The current flowing between the electrodes (forks), which is supported by the $Na^+$ and $Cl^-$ ions present, apparently causes some sodium atoms to form in an excited state. When these atoms relax to the ground state, they emit visible light at 589 nm, producing the yellow glow reminiscent of sodium vapor lamps.

At the beginning of the twentieth century, however, certain experimental results suggested that this picture was incorrect. The first important advance came in 1900 from the German physicist Max Planck (1858–1947). Studying the radiation profiles emitted by solid bodies heated to incandescence, Planck found that the results could not be explained in terms of the physics of his day, which held that matter could absorb or emit any quantity of energy. Planck could account for these observations only by postulating that energy can be gained or lost only in *whole-number multiples* of the quantity $h\nu$, where $h$ is a constant called **Planck's constant,** determined by experiment to have the value $6.626 \times 10^{-34}$ J · s. That is, the change in energy for a system $\Delta E$ can be represented by the equation

$$\Delta E = nh\nu$$

where $n$ is an integer (1, 2, 3, . . .), $h$ is Planck's constant, and $\nu$ is the frequency of the electromagnetic radiation absorbed or emitted.

Energy can be gained or lost only in integer multiples of $h\nu$.

Planck's constant = $6.626 \times 10^{-34}$ J · s.

Planck's result was a real surprise. It had always been assumed that the energy of matter was continuous, which meant that the transfer of any quantity of energy was possible. Now it seemed clear that energy is in fact **quantized** and can occur only in discrete units of size $h\nu$. Each of these small "packets" of energy is called a *quantum.* A system can transfer energy only in whole quanta. Thus energy seems to have particulate properties.

## Sample Exercise 7.2 The Energy of a Photon

The blue color in fireworks is often achieved by heating copper(I) chloride (CuCl) to about 1200°C. Then the compound emits blue light having a wavelength of 450 nm. What is the increment of energy (the quantum) that is emitted at $4.50 \times 10^2$ nm by CuCl?

### Solution

The quantum of energy can be calculated from the equation

$$\Delta E = h\nu$$

The frequency $\nu$ for this case can be calculated as follows:

$$\nu = \frac{c}{\lambda} = \frac{2.9979 \times 10^8 \text{ m/s}}{4.50 \times 10^{-7} \text{ m}} = 6.66 \times 10^{14} \text{ s}^{-1}$$

So

$$\Delta E = h\nu = (6.626 \times 10^{-34} \text{ J} \cdot \text{s})(6.66 \times 10^{14} \text{ s}^{-1}) = 4.41 \times 10^{-19} \text{ J}$$

A sample of CuCl emitting light at 450 nm can lose energy only in increments of $4.41 \times 10^{-19}$ J, the size of the quantum in this case.

*See Exercises 7.33 and 7.34.*

The next important development in the knowledge of atomic structure came when Albert Einstein (see Fig. 7.3) proposed that electromagnetic radiation is itself quantized. Einstein suggested that electromagnetic radiation can be viewed as a stream of "particles" called **photons.** The energy of each photon is given by the expression

$$E_{photon} = h\nu = \frac{hc}{\lambda}$$

اتحاد سنافر البوليتكنك (spu)



Visualization: Photoelectric Effect

where $h$ is Planck's constant, $\nu$ is the frequency of the radiation, and $\lambda$ is the wavelength of the radiation.

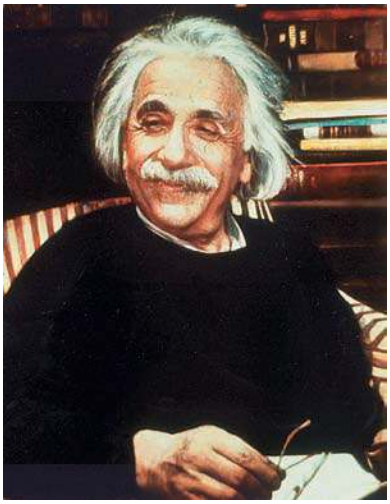


**FIGURE 7.3**
Albert Einstein (1879–1955) was born in Germany. Nothing in his early development suggested genius; even at the age of 9 he did not speak clearly, and his parents feared that he might be handicapped. When asked what profession Einstein should follow, his school principal replied, "It doesn't matter; he'll never make a success of anything." When he was 10, Einstein entered the Luitpold Gymnasium (high school), which was typical of German schools of that time in being harshly disciplinarian. There he developed a deep suspicion of authority and a skepticism that encouraged him to question and doubt—valuable qualities in a scientist. In 1905, while a patent clerk in Switzerland, Einstein published a paper explaining the photoelectric effect via the quantum theory. For this revolutionary thinking he received a Nobel Prize in 1921. Highly regarded by this time, he worked in Germany until 1933, when Hitler's persecution of the Jews forced him to come to the United States. He worked at the Institute for Advanced Studies in Princeton, New Jersey, until his death in 1955.

Einstein was undoubtedly the greatest physicist of our age. Even if someone else had derived the theory of relativity, his other work would have ensured his ranking as the second greatest physicist of his time. Our concepts of space and time were radically changed by ideas he first proposed when he was 26 years old. From then until the end of his life, he attempted unsuccessfully to find a single unifying theory that would explain all physical events.

## The Photoelectric Effect

Einstein arrived at this conclusion through his analysis of the **photoelectric effect** (for which he later was awarded the Nobel Prize). The photoelectric effect refers to the phenomenon in which electrons are emitted from the surface of a metal when light strikes it. The following observations characterize the photoelectric effect.

1. Studies in which the frequency of the light is varied show that no electrons are emitted by a given metal below a specific threshold frequency $\nu_0$.
2. For light with frequency lower than the threshold frequency, no electrons are emitted regardless of the intensity of the light.
3. For light with frequency greater than the threshold frequency, the number of electrons emitted increases with the intensity of the light.
4. For light with frequency greater than the threshold frequency, the kinetic energy, of the emitted electrons increases linearly with the frequency of the light.

These observations can be explained by assuming that electromagnetic radiation is quantized (consists of photons), and that the threshold frequency represents the minimum energy required to remove the electron from the metal's surface.

$$\text{Minimum energy required to remove an electron} = E_0 = h\nu_0$$

Because a photon with energy less than $E_0$ ($\nu < \nu_0$) cannot remove an electron, light with a frequency less than the threshold frequency produces no electrons. On the other hand, for light where $\nu > \nu_0$, the energy in excess of that required to remove the electron is given to the electron as kinetic energy (KE):

$$\text{KE}_{\text{electron}} = \tfrac{1}{2}m\nu^2 = h\nu - h\nu_0$$

Mass of electron; Velocity of electron; Energy of incident photon; Energy required to remove electron from metal's surface

Because in this picture the intensity of light is a measure of the number of photons present in a given part of the beam, a greater intensity means that more photons are available to release electrons (as long as $\nu > \nu_0$ for the radiation).

In a related development, Einstein derived the famous equation

$$E = mc^2$$

in his *special theory of relativity* published in 1905. The main significance of this equation is that *energy has mass*. This is more apparent if we rearrange the equation in the following form:

(spu) اتحاد سنافر البوليتكنك

$$m = \frac{E}{c^2}$$

$E$: Energy; $m$: Mass; $c^2$: Speed of light



## CHEMICAL IMPACT

### Chemistry That Doesn't Leave You in the Dark

In the animal world, the ability to see at night provides predators with a distinct advantage over their prey. The same advantage can be gained by military forces and law enforcement agencies around the world through the use of recent advances in night vision technology.

All types of night vision equipment are electro-optical devices that amplify existing light. A lens collects light and focuses it on an image intensifier. The image intensifier is based on the photoelectric effect—materials that give off electrons when light is shined on them. Night vision intensifiers use semiconductor-based materials to produce large numbers of electrons for a given input of photons. The emitted electrons are then directed onto a screen covered with compounds that phosphoresce (glow when struck by electrons). While television tubes use various phosphors to produce color pictures, night vision devices use phosphors that appear green, because the human eye can distinguish more shades of green than any other color. The viewing screen shows an image that otherwise would be invisible to the naked eye during nighttime viewing.

Current night vision devices use gallium arsenide (GaAs)–based intensifiers that can amplify input light as much as 50,000 times. These devices are so sensitive they can use starlight to produce an image. It is also now possible to use light (infrared) that cannot be sensed with the human eye to create an image.

This technology, while developed originally for military and law enforcement applications, is now becoming available to the consumer. For example, Cadillac included night vision as an option on its cars for the year 2000. As night-imaging technology improves and costs become less prohibitive, a whole new world is opening up for the technophile—after the sun goes down.



A night vision photo of the midair refueling of a U.S. Air Force plane.

Using this form of the equation, we can calculate the mass associated with a given quantity of energy. For example, we can calculate the *apparent* mass of a photon. For electromagnetic radiation of wavelength $\lambda$, the energy of each photon is given by the expression

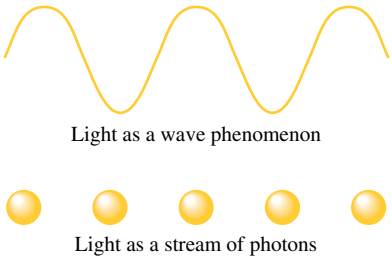
$$E_{\text{photon}} = \frac{hc}{\lambda}$$

Note that the apparent mass of a photon depends on its wavelength. The mass of a photon at rest is thought to be zero, although we never observe it at rest.

Then the apparent mass of a photon of light with wavelength $\lambda$ is given by

$$m = \frac{E}{c^2} = \frac{hc/\lambda}{c^2} = \frac{h}{\lambda c}$$

Does a photon really have mass? The answer *appears to be* yes. In 1922 American physicist Arthur Compton (1892–1962) performed experiments involving collisions of X rays and electrons that showed that photons do exhibit the apparent mass calculated from the preceding equation. However, it is clear that photons do not have mass in the classical sense. A photon has mass only in a relativistic sense—it has no rest mass.




اتحاد سنافر البوليتكنك (spu)

**FIGURE 7.4**
Electromagnetic radiation exhibits wave properties and particulate properties. The energy of each photon of the radiation is related to the wavelength and frequency by the equation $E_{\text{photon}} = h\nu = hc/\lambda$.

We can summarize the important conclusions from the work of Planck and Einstein as follows:

Energy is quantized. It can occur only in discrete units called quanta.

Electromagnetic radiation, which was previously thought to exhibit only wave properties, seems to show certain characteristics of particulate matter as well. This phenomenon is sometimes referred to as the **dual nature of light** and is illustrated in Fig. 7.4.

Thus light, which previously was thought to be purely wavelike, was found to have certain characteristics of particulate matter. But is the opposite also true? That is, does matter that is normally assumed to be particulate exhibit wave properties? This question was raised in 1923 by a young French physicist named Louis de Broglie (1892–1987). To see how de Broglie supplied the answer to this question, recall that the relationship between mass and wavelength for electromagnetic radiation is $m = h/\lambda c$. For a particle with velocity $v$, the corresponding expression is

Do not confuse $\nu$ (frequency) with $v$ (velocity).

$$m = \frac{h}{\lambda v}$$

Rearranging to solve for $\lambda$, we have

$$\lambda = \frac{h}{mv}$$

This equation, called *de Broglie's equation,* allows us to calculate the wavelength for a particle, as shown in Sample Exercise 7.3.

## Sample Exercise 7.3 Calculations of Wavelength

Compare the wavelength for an electron (mass = $9.11 \times 10^{-31}$ kg) traveling at a speed of $1.0 \times 10^7$ m/s with that for a ball (mass = 0.10 kg) traveling at 35 m/s.

### Solution

We use the equation $\lambda = h/mv$, where

$$h = 6.626 \times 10^{-34}\ \text{J} \cdot \text{s} \quad \text{or} \quad 6.626 \times 10^{-34}\ \text{kg} \cdot \text{m}^2/\text{s}$$

since

$$1\ \text{J} = 1\ \text{kg} \cdot \text{m}^2/\text{s}^2$$

For the electron,

$$\lambda_e = \frac{6.626 \times 10^{-34}\ \dfrac{\cancel{\text{kg}} \cdot \cancel{\text{m}} \cdot \text{m}}{\cancel{\text{s}}}}{(9.11 \times 10^{-31}\ \cancel{\text{kg}})(1.0 \times 10^7\ \cancel{\text{m}}/\cancel{\text{s}})} = 7.27 \times 10^{-11}\ \text{m}$$

For the ball,

$$\lambda_b = \frac{6.626 \times 10^{-34}\ \dfrac{\cancel{\text{kg}} \cdot \cancel{\text{m}} \cdot \text{m}}{\cancel{\text{s}}}}{(0.10\ \cancel{\text{kg}})(35\ \cancel{\text{m}}/\cancel{\text{s}})} = 1.9 \times 10^{-34}\ \text{m}$$

*See Exercises 7.41 through 7.44.*

Notice from Sample Exercise 7.3 that the wavelength associated with the ball is incredibly short. On the other hand, the wavelength of the electron, although still quite small, happens to be on the same order as the spacing between the atoms in a typical crystal. This is important because, as we will see presently, it provides a means for testing de Broglie's equation.

(spu) اتحاد سافر البوليتكنك



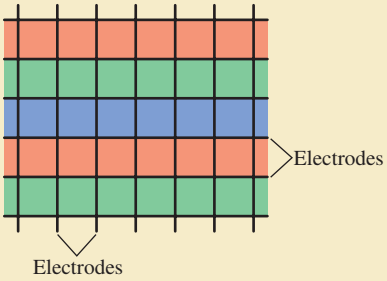
## CHEMICAL IMPACT

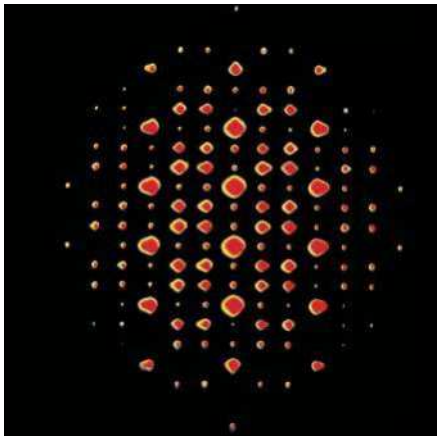
### Thin Is In

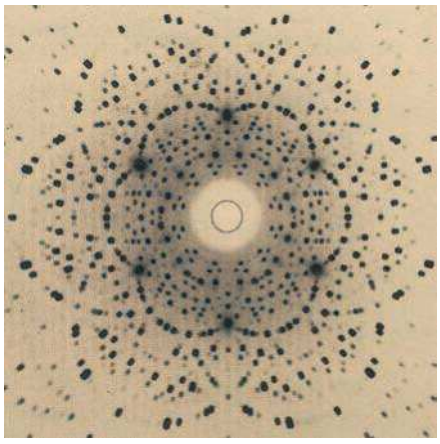
Since the beginning of television about 75 years ago, TV sets have been built around cathode ray tubes (CRTs) in which a "gun" fires electrons at a screen containing phosphors (compounds that emit colored light when excited by some energy source). Although CRT televisions produce excellent pictures, big-screen TVs are very thick and very heavy. Several new technologies are now being used that reduce the bulk of color monitors. One such approach involves a plasma flat-panel display. As the name suggests, the major advantage of these screens is that they are very thin and relatively light.

All color monitors work by manipulating millions of pixels, each of which contains red, blue, and green color-producing phosphors. By combining these three fundamental colors with various weightings, all colors of the rainbow can be generated, thereby producing color images on the monitor. The various types of monitors differ in the energy source used to excite the phosphors. Whereas a CRT monitor uses an electron gun as the energy source, a plasma monitor uses an applied voltage to produce gas-phase ions and electrons, which, when they recombine, emit ultraviolet light. This light, in turn, excites the phosphors.

Plasma monitors have pixel compartments that contain xenon and neon gas. Each pixel consists of three subpixels: one containing a red phosphor, one with a green phosphor, and one with a blue phosphor. Two perpendicular sets of electrodes define a matrix around the subpixels:





**Diffraction** results when light is scattered from a regular array of points or lines. You may have noticed the diffraction of light from the ridges and grooves of a compact disc. The colors result because the various wavelengths of visible light are not all scattered in the same way. The colors are "separated," giving the same effect as light passing through a prism. Just as a regular arrangement of ridges and grooves produces diffraction, so does a regular array of atoms or ions in a crystal, as shown in the photographs below. For example, when X rays are directed onto a crystal of sodium chloride, with its regular array of $Na^+$ and $Cl^-$ ions, the scattered radiation produces a **diffraction pattern** of bright spots and dark areas on a photographic plate, as shown in Fig. 7.5(a). This occurs because the scattered light can interfere constructively (the peaks and troughs of the beams are in phase) to produce a bright spot [Fig. 7.5(b)] or destructively (the peaks and troughs are out of phase) to produce a dark area [Fig. 7.5(c)].

A diffraction pattern can only be explained in terms of waves. Thus this phenomenon provides a test for the postulate that particles such as electrons have wavelengths. As we saw in Sample Exercise 7.3, an electron with a velocity of $10^7$ m/s (easily achieved by acceleration of the electron in an electric field) has a wavelength of about $10^{-10}$ m, which is roughly the distance between the ions in a crystal such as sodium chloride. This is important because diffraction occurs most efficiently when the spacing between the scattering points is about the same as the wavelength of the wave being diffracted. Thus, if electrons really do have an associated wavelength, a crystal should diffract electrons. An experiment to test this idea was carried out in 1927 by C. J. Davisson and

(top) The pattern produced by electron diffraction of a titanium/nickel alloy. (bottom) Pattern produced by X-ray diffraction of a beryl crystal.

One set of the electrodes is above the pixels, and the perpendicular set is below the pixels. When the computer managing the image places a voltage difference across a given subpixel, electrons are removed from the xenon and neon atoms present to form a plasma (cations and electrons). When the cations recombine with the electrons, photons of light are emitted that are absorbed by the phosphor compound, which then emits red, green, or blue light. By controlling the size of the voltage on a given subpixel, a given pixel can produce a variety of colors. When all of the pixels are excited appropriately, a color image is produced.

The plasma display makes it possible to have a large, yet relatively thin screen. Since each pixel is energized individually, this display looks bright and clear from almost any angle. The main disadvantage of this technology is its relatively high cost. However, as advances are being made, the price is falling significantly. CRT monitors may soon be of interest only to antique collectors.



A plasma display from Sony.

L. H. Germer at Bell Laboratories. When they directed a beam of electrons at a nickel crystal, they observed a diffraction pattern similar to that seen from the diffraction of X rays. This result verified de Broglie's relationship, at least for electrons. Larger chunks of matter, such as balls, have such small wavelengths (see Sample Exercise 7.3) that they are impossible to verify experimentally. However, we believe that all matter obeys de Broglie's equation.

Now we have come full circle. Electromagnetic radiation, which at the turn of the twentieth century was thought to be a pure waveform, was found to possess particulate properties. Conversely, electrons, which were thought to be particles, were found to have a wavelength associated with them. The significance of these results is that matter and energy are not distinct.

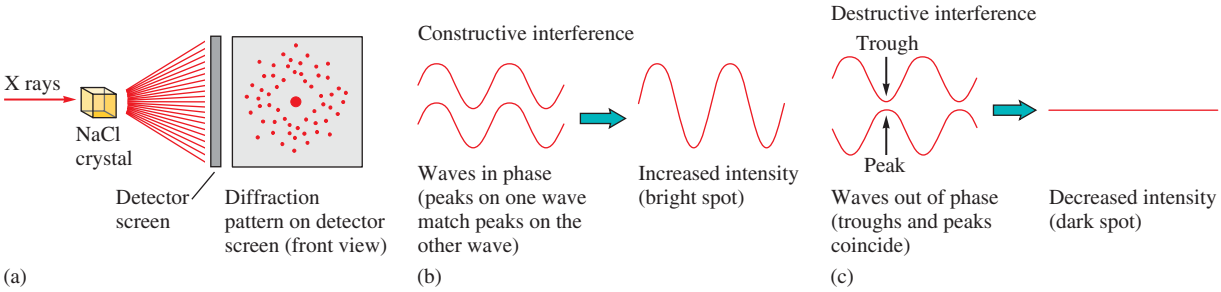



**FIGURE 7.5**
(a) Diffraction occurs when electromagnetic radiation is scattered from a regular array of objects, such as the atoms in a crystal of sodium chloride. The large spot in the center is from the main incident beam of X rays. (b) Bright spots in the diffraction pattern result from *constructive interference* of waves. The waves are in phase; that is, their peaks match. (c) Dark areas result from *destructive interference* of waves. The waves are out of phase; the peaks of one wave coincide with the troughs of another wave.

اتحاد شاقر البوليتكنك (spu)

Energy is really a form of matter, and all matter shows the same types of properties. That is, *all matter exhibits both particulate and wave properties.* Large pieces of matter, such as baseballs, exhibit predominantly particulate properties. The associated wavelength is so small that it is not observed. Very small "bits of matter," such as photons, while showing some particulate properties, exhibit predominantly wave properties. Pieces of matter with intermediate mass, such as electrons, show clearly both the particulate and wave properties of matter.

## 7.3 The Atomic Spectrum of Hydrogen



A beautiful rainbow.

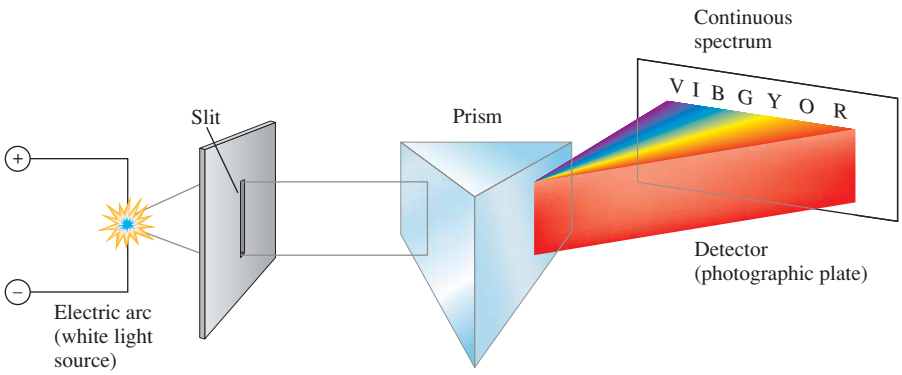
As we saw in Chapter 2, key information about the atom came from several experiments carried out in the early twentieth century, in particular Thomson's discovery of the electron and Rutherford's discovery of the nucleus. Another important experiment was the study of the emission of light by excited hydrogen atoms. When a sample of hydrogen gas receives a high-energy spark, the $H_2$ molecules absorb energy, and some of the H—H bonds are broken. The resulting hydrogen atoms are *excited;* that is, they contain excess energy, which they release by emitting light of various wavelengths to produce what is called the *emission spectrum* of the hydrogen atom.

To understand the significance of the hydrogen emission spectrum, we must first describe the **continuous spectrum** that results when white light is passed through a prism, as shown in Fig. 7.6(a). This spectrum, like the rainbow produced when sunlight is

Visualization: Refraction of White Light

Visualization: The Line Spectrum of Hydrogen

Visualization: Flame Tests



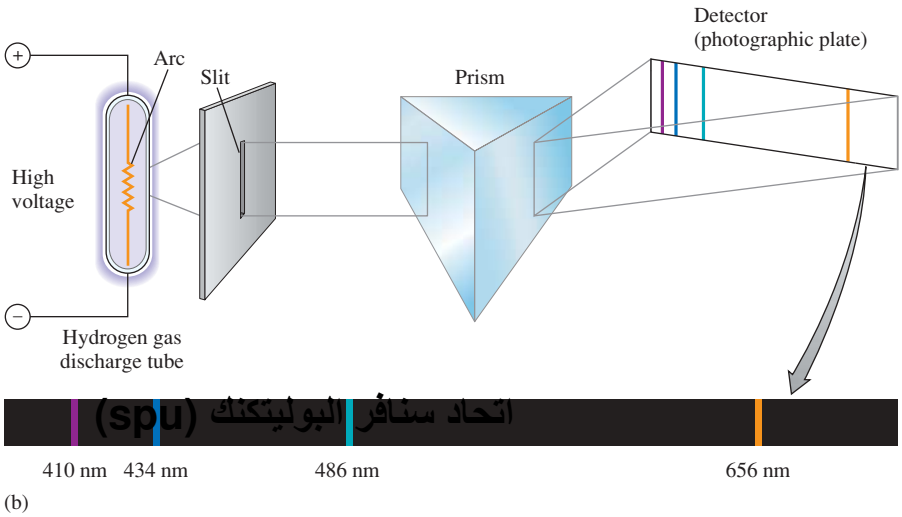




**FIGURE 7.6**
(a) A continuous spectrum containing all wavelengths of visible light (indicated by the initial letters of the colors of the rainbow). (b) The hydrogen line spectrum contains only a few discrete wavelengths.
Spectrum adapted by permission from C. W. Keenan, D. C. Kleinfelter, and J. H. Wood, *General College Chemistry,* 6th ed. (New York: Harper & Row, 1980).

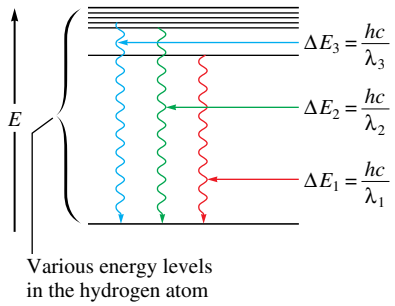



**FIGURE 7.7**
A change between two discrete energy levels emits a photon of light.

dispersed by raindrops, contains *all* the wavelengths of visible light. In contrast, when the hydrogen emission spectrum in the visible region is passed through a prism, as shown in Fig. 7.6(b), we see only a few lines, each of which corresponds to a discrete wavelength. The hydrogen emission spectrum is called a **line spectrum.**

What is the significance of the line spectrum of hydrogen? It indicates that *only certain energies are allowed for the electron in the hydrogen atom.* In other words, the energy of the electron in the hydrogen atom is *quantized.* This observation ties in perfectly with the postulates of Max Planck discussed in Section 7.2. Changes in energy between discrete energy levels in hydrogen will produce only certain wavelengths of emitted light, as shown in Fig. 7.7. For example, a given change in energy from a high to a lower level would give a wavelength of light that can be calculated from Planck's equation:

$$\Delta E = h\nu = \frac{hc}{\lambda}$$

Change in energy; Frequency of light emitted; Wavelength of light emitted

The discrete line spectrum of hydrogen shows that only certain energies are possible; that is, the electron energy levels are quantized. In contrast, if any energy level were allowed, the emission spectrum would be continuous.

## 7.4 The Bohr Model

In 1913, a Danish physicist named Niels Bohr (1885–1962), aware of the experimental results we have just discussed, developed a **quantum model** for the hydrogen atom. Bohr proposed that the *electron in a hydrogen atom moves around the nucleus only in certain allowed circular orbits.* He calculated the radii for these allowed orbits by using the theories of classical physics and by making some new assumptions.

From classical physics Bohr knew that a particle in motion tends to move in a straight line and can be made to travel in a circle only by application of a force toward the center of the circle. Thus Bohr reasoned that the tendency of the revolving electron to fly off the atom must be just balanced by its attraction for the positively charged nucleus. But classical physics also decreed that a charged particle under acceleration should radiate energy. Since an electron revolving around the nucleus constantly changes its direction, it is constantly accelerating. Therefore, the electron should emit light and lose energy—and thus be drawn into the nucleus. This, of course, does not correlate with the existence of stable atoms.

Clearly, an atomic model based solely on the theories of classical physics was untenable. Bohr also knew that the correct model had to account for the experimental spectrum of hydrogen, which showed that only certain electron energies were allowed. The experimental data were absolutely clear on this point. Bohr found that his model would fit the experimental results if he assumed that the angular momentum of the electron (angular momentum equals the product of mass, velocity, and orbital radius) could occur only in certain increments. It was not clear why this should be true, but with this assumption, Bohr's model gave hydrogen atom energy levels consistent with the hydrogen emission spectrum. The model is represented pictorially in Fig. 7.8.

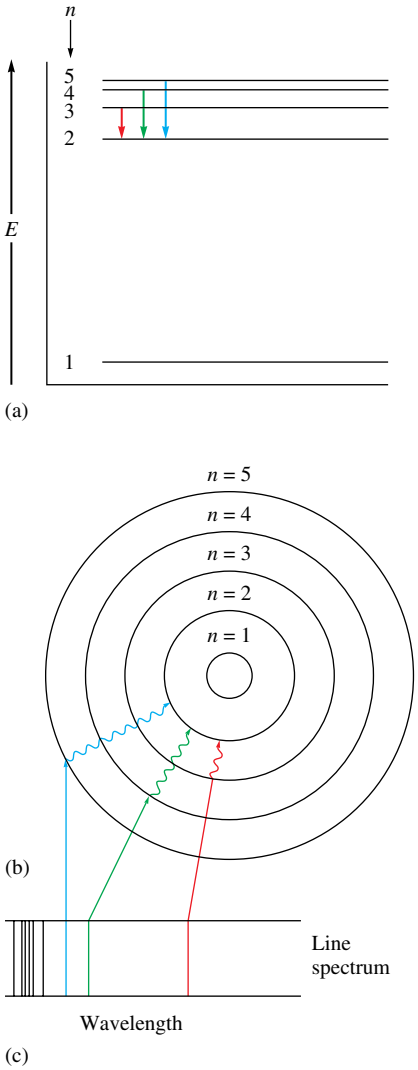
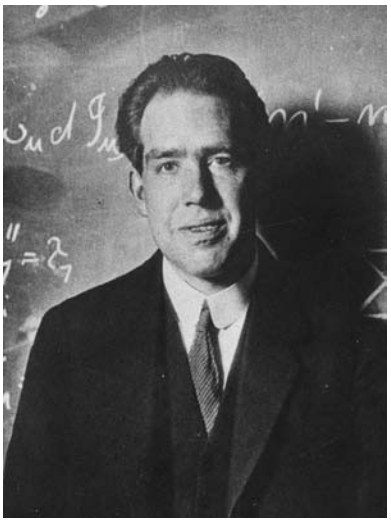



**FIGURE 7.8**
Electronic transitions in the Bohr model for the hydrogen atom. (a) An energy-level diagram for electronic transitions. (b) An orbit-transition diagram, which accounts for the experimental spectrum. (Note that the orbits shown are schematic. They are not drawn to scale.) (c) The resulting line spectrum on a photographic plate. Note that the lines in the visible region of the spectrum correspond to transitions from higher levels to the $n = 2$ level.

ابحاد سناف البوليتكنك (spu)

The J in Equation (7.1) stands for joules.



Niels Hendrik David Bohr (1885–1962) as a boy lived in the shadow of his younger brother Harald, who played on the 1908 Danish Olympic Soccer Team and later became a distinguished mathematician. In school, Bohr received his poorest marks in composition and struggled with writing during his entire life. In fact, he wrote so poorly that he was forced to dictate his Ph.D. thesis to his mother. Nevertheless, Bohr was a brilliant physicist. After receiving his Ph.D. in Denmark, he constructed a quantum model for the hydrogen atom by the time he was 27. Even though his model later proved to be incorrect, Bohr remained a central figure in the drive to understand the atom. He was awarded the Nobel Prize in physics in 1922.

Although we will not show the derivation here, the most important equation to come from Bohr's model is the expression for the *energy levels available to the electron in the hydrogen atom:*

$$E = -2.178 \times 10^{-18}\ \text{J}\left(\frac{Z^2}{n^2}\right) \tag{7.1}$$

in which $n$ is an integer (the larger the value of $n$, the larger is the orbit radius) and $Z$ is the nuclear charge. Using Equation (7.1), Bohr was able to calculate hydrogen atom energy levels that exactly matched the values obtained by experiment.

The negative sign in Equation (7.1) simply means that the energy of the electron bound to the nucleus is lower than it would be if the electron were at an infinite distance ($n = \infty$) from the nucleus, where there is no interaction and the energy is zero:

$$E = -2.178 \times 10^{-18}\ \text{J}\left(\frac{Z^2}{\infty}\right) = 0$$

The energy of the electron in any orbit is negative relative to this reference state.

Equation (7.1) can be used to calculate the change in energy of an electron when the electron changes orbits. For example, suppose an electron in level $n = 6$ of an excited hydrogen atom falls back to level $n = 1$ as the hydrogen atom returns to its lowest possible energy state, its **ground state.** We use Equation (7.1) with $Z = 1$, since the hydrogen nucleus contains a single proton. The energies corresponding to the two states are as follows:

$$\text{For } n = 6: \qquad E_6 = -2.178 \times 10^{-18}\ \text{J}\left(\frac{1^2}{6^2}\right) = -6.050 \times 10^{-20}\ \text{J}$$

$$\text{For } n = 1: \qquad E_1 = -2.178 \times 10^{-18}\ \text{J}\left(\frac{1^2}{1^2}\right) = -2.178 \times 10^{-18}\ \text{J}$$

Note that for $n = 1$ the electron has a more negative energy than it does for $n = 6$, which means that the electron is more tightly bound in the smallest allowed orbit.

The change in energy $\Delta E$ when the electron falls from $n = 6$ to $n = 1$ is

$$\begin{aligned}\Delta E &= \text{energy of final state} - \text{energy of initial state}\\ &= E_1 - E_6 = (-2.178 \times 10^{-18}\ \text{J}) - (-6.050 \times 10^{-20}\ \text{J})\\ &= -2.117 \times 10^{-18}\ \text{J}\end{aligned}$$

The negative sign for the *change* in energy indicates that the atom has *lost* energy and is now in a more stable state. The energy is carried away from the atom by the production (emission) of a photon.

The wavelength of the emitted photon can be calculated from the equation

$$\Delta E = h\left(\frac{c}{\lambda}\right) \quad \text{or} \quad \lambda = \frac{hc}{\Delta E}$$

where $\Delta E$ represents the change in energy of the atom, which equals the energy of the emitted photon. We have

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34}\ \text{J}\cdot\text{s})(2.9979 \times 10^{8}\ \text{m/s})}{2.117 \times 10^{-18}\ \text{J}} = 9.383 \times 10^{-8}\ \text{m}$$

اتحاد سنافر البوليتكنك (spu)

Note that for this calculation the absolute value of $\Delta E$ is used (we have not included the negative sign). In this case we indicate the direction of energy flow by saying that a photon

of wavelength $9.383 \times 10^{-8}$ m has been *emitted* from the hydrogen atom. Simply plugging the negative value of $\Delta E$ into the equation would produce a negative value for $\lambda$, which is physically meaningless.

**Sample Exercise 7.4**

## Energy Quantization in Hydrogen

Calculate the energy required to excite the hydrogen electron from level $n = 1$ to level $n = 2$. Also calculate the wavelength of light that must be absorbed by a hydrogen atom in its ground state to reach this excited state.*

***Solution***

Using Equation (7.1) with $Z = 1$, we have

$$E_1 = -2.178 \times 10^{-18}\,\text{J}\left(\frac{1^2}{1^2}\right) = -2.178 \times 10^{-18}\,\text{J}$$

$$E_2 = -2.178 \times 10^{-18}\,\text{J}\left(\frac{1^2}{2^2}\right) = -5.445 \times 10^{-19}\,\text{J}$$

$$\Delta E = E_2 - E_1 = (-5.445 \times 10^{-19}\,\text{J}) - (-2.178 \times 10^{-18}\,\text{J}) = 1.633 \times 10^{-18}\,\text{J}$$

The positive value for $\Delta E$ indicates that the system has gained energy. The wavelength of light that must be *absorbed* to produce this change is

Note from Fig. 7.2 that the light required to produce the transition from the n = 1 to n = 2 level in hydrogen lies in the ultraviolet region.

$$\lambda = \frac{hc}{\Delta E} = \frac{(6.626 \times 10^{-34}\,\text{J}\cdot\text{s})(2.9979 \times 10^{8}\,\text{m/s})}{1.633 \times 10^{-18}\,\text{J}}$$
$$= 1.216 \times 10^{-7}\,\text{m}$$

***See Exercises 7.45 and 7.46.***

At this time we must emphasize two important points about the Bohr model:

1. The model correctly fits the quantized energy levels of the hydrogen atom and postulates only certain allowed circular orbits for the electron.
2. As the electron becomes more tightly bound, its energy becomes more negative relative to the zero-energy reference state (corresponding to the electron being at infinite distance from the nucleus). As the electron is brought closer to the nucleus, energy is released from the system.

Using Equation (7.1), we can derive a general equation for the electron moving from one level ($n_{\text{initial}}$) to another level ($n_{\text{final}}$):

$$\begin{aligned}\Delta E &= \text{energy of level } n_{\text{final}} - \text{energy of level } n_{\text{initial}}\\ &= E_{\text{final}} - E_{\text{initial}}\\ &= (-2.178 \times 10^{-18}\,\text{J})\left(\frac{1^2}{n_{\text{final}}^{\,2}}\right) - (-2.178 \times 10^{-18}\,\text{J})\left(\frac{1^2}{n_{\text{initial}}^{\,2}}\right)\\ &= -2.178 \times 10^{-18}\,\text{J}\left(\frac{1}{n_{\text{final}}^{\,2}} - \frac{1}{n_{\text{initial}}^{\,2}}\right)\end{aligned} \tag{7.2}$$

Equation (7.2) can be used to calculate the energy change between *any* two energy levels in a hydrogen atom, as shown in Sample Exercise 7.5.

اتحاد سنافر البوليتكنك (spu)

*After this exercise we will no longer show cancellation marks. However, the same process for canceling units applies throughout this text.



## CHEMICAL IMPACT

### Fireworks

The art of using mixtures of chemicals to produce explosives is an ancient one. Black powder—a mixture of potassium nitrate, charcoal, and sulfur—was being used in China well before 1000 A.D. and has been used subsequently through the centuries in military explosives, in construction blasting, and for fireworks. The DuPont Company, now a major chemical manufacturer, started out as a manufacturer of black powder. In fact, the founder, Eleuthère duPont, learned the manufacturing technique from none other than Lavoisier.

Before the nineteenth century, fireworks were confined mainly to rockets and loud bangs. Orange and yellow colors came from the presence of charcoal and iron filings. However, with the great advances in chemistry in the nineteenth century, new compounds found their way into fireworks. Salts of copper, strontium, and barium added brilliant colors. Magnesium and aluminum metals gave a dazzling white light. Fireworks, in fact, have changed very little since then.

How do fireworks produce their brilliant colors and loud bangs? Actually, only a handful of different chemicals are responsible for most of the spectacular effects. To produce the noise and flashes, an oxidizer (an oxidizing agent) and a fuel (a reducing agent) are used. A common mixture involves potassium perchlorate ($KClO_4$) as the oxidizer and aluminum and sulfur as the fuel. The perchlorate oxidizes the fuel in a very exothermic reaction, which produces a brilliant flash, due to the aluminum, and a loud report from the rapidly expanding gases produced. For a color effect, an element with a colored emission spectrum is included. Recall that the electrons in atoms can be raised to higher-energy orbitals when the atoms absorb energy. The excited atoms can then release this excess energy by emitting light of specific wavelengths, often in the visible region. In fireworks, the energy to excite the electrons comes from the reaction between the oxidizer and fuel.

Yellow colors in fireworks are due to the 589-nm emission of sodium ions. Red colors come from strontium salts emitting at 606 nm and from 636 to 688 nm. This red color is familiar from highway safety flares. Barium salts give a green color in fireworks, due to a series of emission lines between 505 and 535 nm. A really good blue color, however, is hard to obtain. Copper salts give a blue color, emitting in the 420- to 460-nm region. But difficulties occur because the oxidizing agent, potassium chlorate ($KClO_3$), reacts with copper salts to form copper chlorate, a highly explosive compound that is dangerous to store. (The use of $KClO_3$ in fireworks has been largely abandoned because of its explosive hazards.) Paris green, a copper salt containing arsenic, was once used extensively but is now considered to be too toxic.

In recent years the colors produced by fireworks have become more intense because of the formation of metal chlorides during the burning process. These gaseous metal chloride molecules produce colors much more brilliant than do the metal atoms by themselves. For example, strontium chloride produces a much brighter red than do strontium atoms.

اتحاد سنافر البوليتكنك (spu)



Fireworks in Washington, D.C.

Thus, chlorine-donating compounds are now included in many fireworks shells.

A typical aerial shell is shown in the diagram. The shell is launched from a mortar (a steel cylinder) using black powder as the propellant. Time-delayed fuses are used to fire the shell in stages. A list of chemicals commonly used in fireworks is given in the table.

Although you might think that the chemistry of fireworks is simple, the achievement of the vivid white flashes and the brilliant colors requires complex combinations of chemicals. For example, because the white flashes produce high flame temperatures, the colors tend to wash out. Thus oxidizers such as $KClO_4$ are commonly used with fuels that produce relatively low flame temperatures. An added difficulty, however, is that perchlorates are very sensitive to accidental ignition and are therefore quite hazardous. Another problem arises from the use of sodium salts. Because sodium produces an extremely bright yellow emission, sodium salts cannot be used when other colors are desired. Carbon-based fuels also give a yellow flame that masks other colors, and this limits the use of organic compounds as fuels. You can see that the manufacture of fireworks that produce the desired effects and are also safe to handle requires careful selection of chemicals. And, of course, there is still the dream of a deep blue flame.

**Chemicals Commonly Used in the Manufacture of Fireworks**

| Oxidizers | Fuels | Special Effects |
|---|---|---|
| Potassium nitrate | Aluminum | Red flame: strontium nitrate, strontium carbonate |
| Potassium chlorate | Magnesium | Green flame: barium nitrate, barium chlorate |
| Potassium perchlorate | Titanium | Blue flame: copper carbonate, copper sulfate, copper oxide |
| Ammonium perchlorate | Charcoal | Yellow flame: sodium oxalate, cryolite ($Na_3AlF_6$) |
| Barium nitrate | Sulfur | White flame: magnesium, aluminum |
| Barium chlorate | Antimony sulfide | Gold sparks: iron filings, charcoal |
| Strontium nitrate | Dextrin | White sparks: aluminum, magnesium, aluminum–magnesium alloy, titanium |
| | Red gum | Whistle effect: potassium benzoate or sodium salicylate |
| | Polyvinyl chloride | White smoke: mixture of potassium nitrate and sulfur |
| | | Colored smoke: mixture of potassium chlorate, sulfur, and organic dye |

اتحاد سنافر البوليتكنك (spu)

**Sample Exercise 7.5**

## Electron Energies

Calculate the energy required to remove the electron from a hydrogen atom in its ground state.

***Solution***

Removing the electron from a hydrogen atom in its ground state corresponds to taking the electron from $n_{initial} = 1$ to $n_{final} = \infty$. Thus

$$\Delta E = -2.178 \times 10^{-18}\,\text{J}\left(\frac{1}{n_{final}^2} - \frac{1}{n_{initial}^2}\right)$$

$$= -2.178 \times 10^{-18}\,\text{J}\left(\frac{1}{\infty} - \frac{1}{1^2}\right)$$

$$= -2.178 \times 10^{-18}\,\text{J}(0 - 1) = 2.178 \times 10^{-18}\,\text{J}$$

The energy required to remove the electron from a hydrogen atom in its ground state is $2.178 \times 10^{-18}$ J.

***See Exercises 7.51 and 7.52.***

Visualization: Flame Tests

Although Bohr's model fits the energy levels for hydrogen, it is a fundamentally incorrect model for the hydrogen atom.

At first Bohr's model appeared to be very promising. The energy levels calculated by Bohr closely agreed with the values obtained from the hydrogen emission spectrum. However, when Bohr's model was applied to atoms other than hydrogen, it did not work at all. Although some attempts were made to adapt the model using elliptical orbits, it was concluded that Bohr's model is fundamentally incorrect. The model is, however, very important historically, because it showed that the observed quantization of energy in atoms could be explained by making rather simple assumptions. Bohr's model paved the way for later theories. It is important to realize, however, that the current theory of atomic structure is in no way derived from the Bohr model. Electrons do *not* move around the nucleus in circular orbits, as we shall see later in this chapter.

# 7.5 The Quantum Mechanical Model of the Atom

By the mid-1920s it had become apparent that the Bohr model could not be made to work. A totally new approach was needed. Three physicists were at the forefront of this effort: Werner Heisenberg (1901–1976), Louis de Broglie (1892–1987), and Erwin Schrödinger (1887–1961). The approach they developed became known as *wave mechanics* or, more commonly, *quantum mechanics.* As we have already seen, de Broglie originated the idea that the electron, previously considered to be a particle, also shows wave properties. Pursuing this line of reasoning, Schrödinger, an Austrian physicist, decided to attack the problem of atomic structure by giving emphasis to the wave properties of the electron. To Schrödinger and de Broglie, the electron bound to the nucleus seemed similar to a **standing wave,** and they began research on a wave mechanical description of the atom.

The most familiar example of standing waves occurs in association with musical instruments such as guitars or violins, where a string attached at both ends vibrates to produce a musical tone. The waves are described as "standing" because they are stationary;

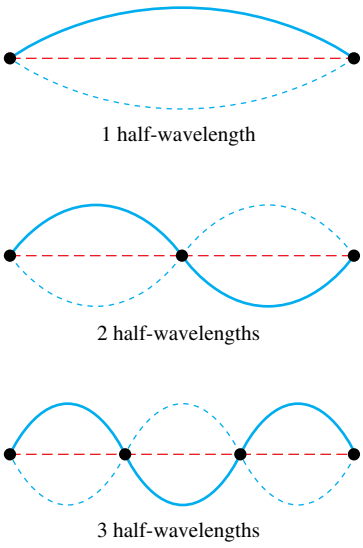



**FIGURE 7.9**
The standing waves caused by the vibration of a guitar string fastened at both ends. Each dot represents a node (a point of zero displacement).




Wave-generating apparatus.

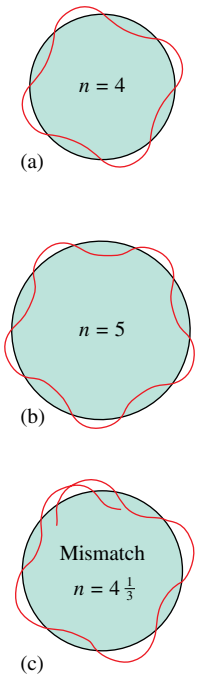



**FIGURE 7.10**
The hydrogen electron visualized as a standing wave around the nucleus. The circumference of a particular circular orbit would have to correspond to a whole number of wavelengths, as shown in (a) and (b), or else destructive interference occurs, as shown in (c). This is consistent with the fact that only certain electron energies are allowed; the atom is quantized. (Although this idea encouraged scientists to use a wave theory, it does not mean that the electron really travels in circular orbits.)

the waves do not travel along the length of the string. The motions of the string can be explained as a combination of simple waves of the type shown in Fig. 7.9. The dots in this figure indicate the nodes, or points of zero lateral (sideways) displacement, for a given wave. Note that there are limitations on the allowed wavelengths of the standing wave. Each end of the string is fixed, so there is always a node at each end. This means that there must be a whole number of *half* wavelengths in any of the allowed motions of the string (see Fig. 7.9). Standing waves can be illustrated using the wave generator shown in the photo below.

A similar situation results when the electron in the hydrogen atom is imagined to be a standing wave. As shown in Fig. 7.10, only certain circular orbits have a circumference into which a whole number of wavelengths of the standing electron wave will "fit." All other orbits would produce destructive interference of the standing electron wave and are not allowed. This seemed like a possible explanation for the observed quantization of the hydrogen atom, so Schrödinger worked out a model for the hydrogen atom in which the electron was assumed to behave as a standing wave.

It is important to recognize that Schrödinger could not be sure that this idea would work. The test had to be whether or not the model would correctly fit the experimental data on hydrogen and other atoms. The physical principles for describing standing waves were well known in 1925 when Schrödinger decided to treat the electron in this way. His mathematical treatment is too complicated to be detailed here. However, the form of Schrödinger's equation is

$$\hat{H}\psi = E\psi$$

where $\psi$, called the **wave function,** is a function of the coordinates ($x$, $y$, and $z$) of the electron's position in three-dimensional space and $\hat{H}$ represents a set of mathematical instructions called an *operator.* In this case, the operator contains mathematical terms that produce the total energy of the atom when they are applied to the wave function. $E$ represents the total energy of the atom (the sum of the potential energy due to the attraction between the proton and electron and the kinetic energy of the moving electron). When this equation is analyzed, many solutions are found. Each solution consists of a wave function $\psi$ that is characterized by a particular value of $E$. A specific wave function is often called an **orbital.**

To illustrate the most important ideas of the **quantum (wave) mechanical model** of the atom, we will first concentrate on the wave function corresponding to the lowest energy for the hydrogen atom. This wave function is called the 1*s* orbital. The first point of interest is to explore the meaning of the word *orbital.* As we will see, this is not a trivial matter. One thing is clear: An orbital is *not* a Bohr orbit. The electron in the hydrogen 1*s* orbital is not moving around the nucleus in a circular orbit. How, then, is the electron moving? The answer is quite surprising: *We do not know.* The wave function gives us no information about the detailed pathway of the electron. This is somewhat disturbing. When we solve problems involving the motions of particles in the macroscopic world, we are able to predict their pathways. For example, when two billiard balls with known velocities collide, we can predict their motions after the collision. However, we cannot predict the electron's motion from the 1*s* orbital function. Does this mean that the theory is wrong? Not necessarily: We have already learned that an electron does not behave much like a billiard ball, so we must examine the situation closely before we discard the theory.

To help us understand the nature of an orbital, we need to consider a principle discovered by Werner Heisenberg, one of the primary developers of quantum mechanics. Heisenberg's mathematical analysis led him to a surprising conclusion: *There is a fundamental limitation to just how precisely we can know both the position and momentum of a particle at a given time.* This is a statement of the **Heisenberg uncertainty principle.** Stated mathematically, the uncertainty principle is

$$\Delta x \cdot \Delta(mv) \geq \frac{h}{4\pi}$$

(spu) اتحاد سنافر البوليتكنك

where $\Delta x$ is the uncertainty in a particle's position, $\Delta(mv)$ is the uncertainty in a particle's momentum, and $h$ is Planck's constant. Thus the minimum uncertainty in the product $\Delta \cdot \Delta(mv)$ is $h/4\pi$. What this equation really says is that the more accurately we know a particle's position, the less accurately we can know its momentum, and vice versa. This limitation is so small for large particles such as baseballs or billiard balls that it is unnoticed. However, for a small particle such as the electron, the limitation becomes quite important. Applied to the electron, the uncertainty principle implies that we cannot know the exact motion of the electron as it moves around the nucleus. It is therefore not appropriate to assume that the electron is moving around the nucleus in a well-defined orbit, as in the Bohr model.

## The Physical Meaning of a Wave Function

Probability is the likelihood, or odds, that something will occur.

Given the limitations indicated by the uncertainty principle, what then is the physical meaning of a wave function for an electron? That is, what is an atomic orbital? Although the wave function itself has no easily visualized meaning, the square of the function does have a definite physical significance. *The square of the function indicates the probability of finding an electron near a particular point in space.* For example, suppose we have two positions in space, one defined by the coordinates $x_1$, $y_1$, and $z_1$ and the other by the coordinates $x_2$, $y_2$, and $z_2$. The relative probability of finding the electron at positions 1 and 2 is given by substituting the values of $x$, $y$, and $z$ for the two positions into the wave function, squaring the function value, and computing the following ratio:

$$\frac{[\psi(x_1, y_1, z_1)]^2}{[\psi(x_2, y_2, z_2)]^2} = \frac{N_1}{N_2}$$

The quotient $N_1/N_2$ is the ratio of the probabilities of finding the electron at positions 1 and 2. For example, if the value of the ratio $N_1/N_2$ is 100, the electron is 100 times more likely to be found at position 1 than at position 2. The model gives no information concerning when the electron will be at either position or how it moves between the positions. This vagueness is consistent with the concept of the Heisenberg uncertainty principle.

The square of the wave function is most conveniently represented as a **probability distribution,** in which the intensity of color is used to indicate the probability value near a given point in space. The probability distribution for the hydrogen 1*s* wave function (orbital) is shown in Fig. 7.11(a). The best way to think about this diagram is as a three-dimensional time exposure with the electron as a tiny moving light. The more times the electron visits a particular point, the darker the negative becomes. Thus the darkness of a point indicates the probability of finding an electron at that position. This diagram is also known as an *electron density map; electron density* and *electron probability* mean the same thing. When a chemist uses the term *atomic orbital,* he or she is probably picturing an electron density map of this type.

Another way of representing the electron probability distribution for the 1*s* wave function is to calculate the probability at points along a line drawn outward in any direction from the nucleus. The result is shown in Fig. 7.11(b). Note that the probability of finding the electron at a particular position is greatest close to the nucleus and drops off rapidly as the distance from the nucleus increases. We are also interested in knowing the *total* probability of finding the electron in the hydrogen atom at a particular *distance* from the nucleus. Imagine that the space around the hydrogen nucleus is made up of a series of thin spherical shells (rather like layers in an onion), as shown in Fig. 7.12(a). When the total probability of finding the electron in each spherical shell is plotted versus the distance from the nucleus, the plot in Fig. 7.12(b) is obtained. This graph is called the **radial probability distribution.**

اتحاد مسافر البوليتكنك (spu)

The maximum in the curve occurs because of two opposing effects. The probability of finding an electron at a particular position is greatest near the nucleus, but the volume

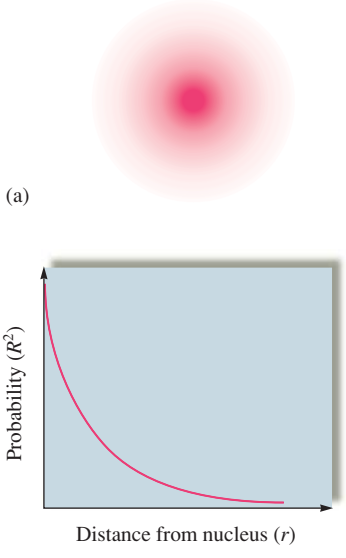



**FIGURE 7.11**
(a) The probability distribution for the hydrogen 1*s* orbital in three-dimensional space. (b) The probability of finding the electron at points along a line drawn from the nucleus outward in any direction for the hydrogen 1*s* orbital.

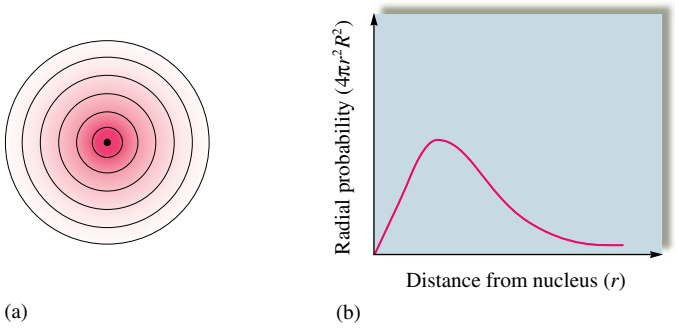



**FIGURE 7.12**
(a) Cross section of the hydrogen 1*s* orbital probability distribution divided into successive thin spherical shells. (b) The radial probability distribution. A plot of the total probability of finding the electron in each thin spherical shell as a function of distance from the nucleus.

of the spherical shell increases with distance from the nucleus. Therefore, as we move away from the nucleus, the probability of finding the electron at a given position decreases, but we are summing more positions. Thus the total probability increases to a certain radius and then decreases as the electron probability at each position becomes very small. For the hydrogen 1*s* orbital, the maximum radial probability (the distance at which the electron is most likely to be found) occurs at a distance of $5.29 \times 10^{-2}$ nm or 0.529 Å from the nucleus. Interestingly, this is exactly the radius of the innermost orbit in the Bohr model. Note that in Bohr's model the electron is assumed to have a circular path and so is *always* found at this distance. In the quantum mechanical model, the specific electron motions are unknown, and this is the *most probable* distance at which the electron is found.

1 Å $= 10^{-10}$ m; the angstrom is most often used as the unit for atomic radius because of its convenient size. Another convenient unit is the picometer:

1 pm $= 10^{-12}$ m

One more characteristic of the hydrogen 1*s* orbital that we must consider is its size. As we can see from Fig. 7.11, the size of this orbital cannot be defined precisely, since the probability never becomes zero (although it drops to an extremely small value at large values of $r$). So, in fact, the hydrogen 1*s* orbital has no distinct size. However, it is useful to have a definition of relative orbital size. *The definition most often used by chemists to describe the size of the hydrogen 1s orbital is the radius of the sphere that encloses 90% of the total electron probability.* That is, 90% of the time the electron is inside this sphere.

Visualization: 1*s* Orbital

So far we have described only the lowest-energy wave function in the hydrogen atom, the 1*s* orbital. Hydrogen has many other orbitals, which we will describe in the next section. However, before we proceed, we should summarize what we have said about the meaning of an atomic orbital. An orbital is difficult to define precisely at an introductory level. Technically, an orbital is a wave function. However, it is usually most helpful to picture an orbital as a three-dimensional electron density map. That is, an electron "in" a particular atomic orbital is assumed to exhibit the electron probability indicated by the orbital map.

## 7.6 Quantum Numbers

When we solve the Schrödinger equation for the hydrogen atom, we find many wave functions (orbitals) that satisfy it. Each of these orbitals is characterized by a series of numbers called **quantum numbers,** which describe various properties of the orbital:

The **principal quantum number** ($n$) has integral values: 1, 2, 3, . . . . The principal quantum number is related to the size and energy of the orbital. As $n$ increases, the orbital becomes larger and the electron spends more time farther from the nucleus. An increase in $n$ also means higher energy, because the electron is less tightly bound to the nucleus, and the energy is less negative.

اتحاد سنافر البوليتكنك (spu)

The **angular momentum quantum number** ($\ell$) has integral values from 0 to $n - 1$ for each value of $n$. This quantum number is related to the shape of atomic orbitals. The value of $\ell$ for a particular orbital is commonly assigned a letter: $\ell = 0$ is called *s*;

**TABLE 7.1 The Angular Momentum Quantum Numbers and Corresponding Letters Used to Designate Atomic Orbitals**

| | | | | | |
|---|---|---|---|---|---|
| **Value of $\ell$** | 0 | 1 | 2 | 3 | 4 |
| **Letter Used** | *s* | *p* | *d* | *f* | *g* |

Number of Orbitals per Subshell

$s = 1$
$p = 3$
$d = 5$
$f = 7$
$g = 9$

**TABLE 7.2 Quantum Numbers for the First Four Levels of Orbitals in the Hydrogen Atom**

| *n* | $\ell$ | Orbital Designation | $m_\ell$ | Number of Orbitals |
|---|---|---|---|---|
| 1 | 0 | 1*s* | 0 | 1 |
| 2 | 0 | 2*s* | 0 | 1 |
| | 1 | 2*p* | −1, 0, +1 | 3 |
| 3 | 0 | 3*s* | 0 | 1 |
| | 1 | 3*p* | −1, 0, 1 | 3 |
| | 2 | 3*d* | −2, −1, 0, 1, 2 | 5 |
| 4 | 0 | 4*s* | 0 | 1 |
| | 1 | 4*p* | −1, 0, 1 | 3 |
| | 2 | 4*d* | −2, −1, 0, 1, 2 | 5 |
| | 3 | 4*f* | −3, −2, −1, 0, 1, 2, 3 | 7 |

$\ell = 1$ is called *p*; $\ell = 2$ is called *d*; $\ell = 3$ is called *f*. This system arises from early spectral studies and is summarized in Table 7.1.

The **magnetic quantum number** ($m_\ell$) has integral values between $\ell$ and $-\ell$, including zero. The value of $m_\ell$ is related to the orientation of the orbital in space relative to the other orbitals in the atom.

$n = 1, 2, 3, \ldots$
$\ell = 0, 1, \ldots (n - 1)$
$m_\ell = -\ell, \ldots 0, \ldots +\ell$

The first four levels of orbitals in the hydrogen atom are listed with their quantum numbers in Table 7.2. Note that each set of orbitals with a given value of $\ell$ (sometimes called a **subshell**) is designated by giving the value of *n* and the letter for $\ell$. Thus an orbital where $n = 2$ and $\ell = 1$ is symbolized as 2*p*. There are three 2*p* orbitals, which have different orientations in space. We will describe these orbitals in the next section.

## Sample Exercise 7.6 Electron Subshells

For principal quantum level $n = 5$, determine the number of allowed subshells (different values of $\ell$), and give the designation of each.

### *Solution*

For $n = 5$, the allowed values of $\ell$ run from 0 to 4 ($n - 1 = 5 - 1$). Thus the subshells and their designations are

| $\ell = 0$ | $\ell = 1$ | $\ell = 2$ | $\ell = 3$ | $\ell = 4$ |
|---|---|---|---|---|
| 5*s* | 5*p* | 5*d* | 5*f* | 5*g* |

اتجاد سنافر البوليتكنك (spu)

*See Exercises 7.57 through 7.59.*

# 7.7 Orbital Shapes and Energies

We have seen that the meaning of an orbital is represented most clearly by a probability distribution. Each orbital in the hydrogen atom has a unique probability distribution. We also saw that another means of representing an orbital is by the surface that surrounds 90% of the total electron probability. These two types of representations for the hydrogen 1*s*, 2*s*, and 3*s* orbitals are shown in Fig. 7.13. Note the characteristic spherical shape of each of the *s* orbitals. Note also that the 2*s* and 3*s* orbitals contain areas of high probability separated by areas of zero probability. These latter areas are called **nodal surfaces,** or simply **nodes.** The number of nodes increases as $n$ increases. For *s* orbitals, the number of nodes is given by $n - 1$. For our purposes, however, we will think of *s* orbitals only in terms of their overall spherical shape, which becomes larger as the value of $n$ increases.

Visualization: Orbital Energies

Visualization: $2p_x$, $2p_y$, $2p_z$ Orbitals

$n$ value
↓
$2p_x$ ← orientation in space
↑
$\ell$ value

The two types of representations for the 2*p* orbitals (there are no 1*p* orbitals) are shown in Fig. 7.14. Note that the *p* orbitals are not spherical like *s* orbitals but have two *lobes* separated by a node at the nucleus. The *p* orbitals are labeled according to the axis of the *xyz* coordinate system along which the lobes lie. For example, the 2*p* orbital with lobes centered along the *x* axis is called the $2p_x$ orbital.

At this point it is useful to remember that mathematical functions have signs. For example, a simple sine wave (see Fig. 7.1) oscillates from positive to negative and repeats this pattern. Atomic orbital functions also have signs. The functions for *s* orbitals are positive everywhere in three-dimensional space. That is, when the *s* orbital function is evaluated at any point in space, it results in a positive number. In contrast, the *p* orbital functions have different signs in different regions of space. For example, the $p_z$ orbital has a positive sign in all the regions of space in which $z$ is positive and has a negative sign when $z$ is negative. This behavior is indicated in Fig. 7.14(b) by the positive and negative signs inside their boundary surfaces. It is important to understand that these are mathematical signs, not charges. Just as a sine wave has alternating positive and negative phases, so too *p* orbitals have positive and negative phases. The phases of the $p_x$, $p_y$, and $p_z$ orbitals are indicated in Fig. 7.14(b).

Visualization: $3d_{x^2-y^2}$, $3d_{xy}$, $3d_{xz}$, $3d_{yz}$, $3d_{z^2}$ Orbitals

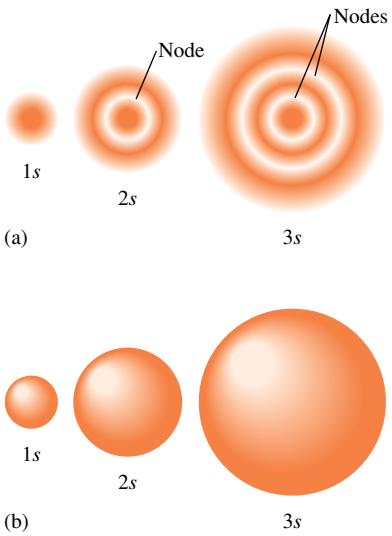



**FIGURE 7.13**
Two representations of the hydrogen 1*s*, 2*s*, and 3*s* orbitals. (a) The electron probability distribution. (b) The surface that contains 90% of the total electron probability (the size of the orbital, by definition).

As you might expect from our discussion of the *s* orbitals, the 3*p* orbitals have a more complex probability distribution than that of the 2*p* orbitals (see Fig. 7.15), but they can still be represented by the same boundary surface shapes. The surfaces just grow larger as the value of $n$ increases.

There are no *d* orbitals that correspond to principal quantum levels $n = 1$ and $n = 2$. The *d* orbitals ($\ell = 2$) first occur in level $n = 3$. The five 3*d* orbitals have the shapes shown in Fig. 7.16. The *d* orbitals have two different fundamental shapes. Four of the orbitals ($d_{xz}$, $d_{yz}$, $d_{xy}$, and $d_{x^2-y^2}$) have four lobes centered in the plane indicated in the orbital label. Note that $d_{xy}$ and $d_{x^2-y^2}$ are both centered in the *xy* plane; however, the lobes of $d_{x^2-y^2}$ lie *along* the *x* and *y* axes, while the lobes of $d_{xy}$ lie *between* the axes. The fifth orbital, $d_{z^2}$, has a unique shape with two lobes along the *z* axis and a belt centered in the *xy* plane. The *d* orbitals for levels $n > 3$ look like the 3*d* orbitals but have larger lobes.

The *f* orbitals first occur in level $n = 4$, and as might be expected, they have shapes even more complex than those of the *d* orbitals. Figure 7.17 shows representations of the 4*f* orbitals ($\ell = 3$) along with their designations. These orbitals are not involved in the bonding in any of the compounds we will consider in this text. Their shapes and labels are simply included for completeness.

So far we have talked about the shapes of the hydrogen atomic orbitals but not about their energies. For the hydrogen atom, the energy of a particular orbital is determined by its size ($n$). All orbitals with the same value of $n$ have the *same energy*—they are said to be **degenerate.** This is shown in Fig. 7.18, where the energies for the orbitals in the first three quantum levels for hydrogen are shown.

اتحاد سنافر البوليتكنك (spu)

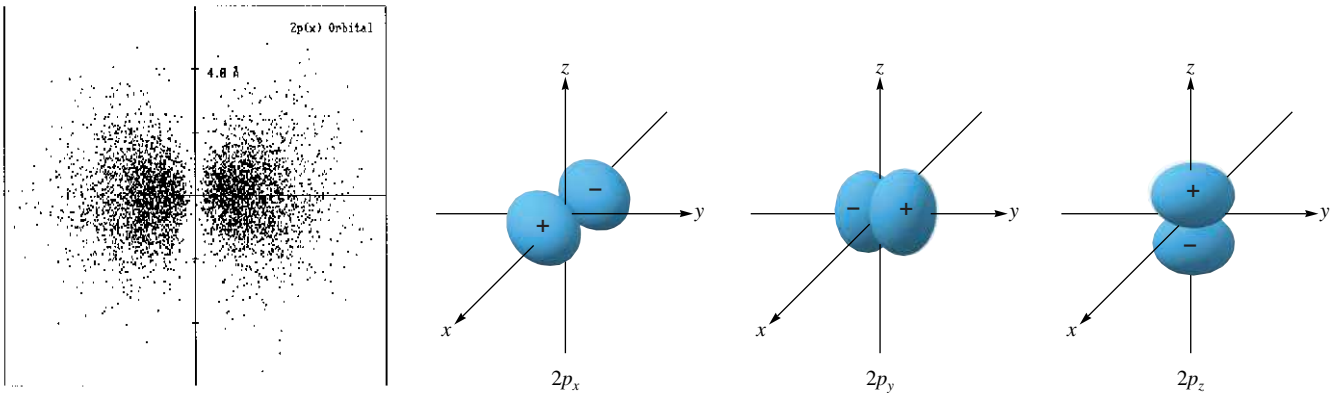



**FIGURE 7.14**
Representation of the 2*p* orbitals. (a) The electron probability distribution for a 2*p* orbital. (Generated from a program by Robert Allendoerfer on Project SERAPHIM disk PC 2402; reprinted with permission.) (b) The boundary surface representations of all three 2*p* orbitals. Note that the signs inside the surface indicate the phases (signs) of the orbital in that region of space.

Hydrogen's single electron can occupy any of its atomic orbitals. However, in the lowest energy state, the *ground state,* the electron resides in the 1*s* orbital. If energy is put into the atom, the electron can be transferred to a higher-energy orbital, producing an *excited state.*

### A Summary of the Hydrogen Atom

- In the quantum (wave) mechanical model, the electron is viewed as a standing wave. This representation leads to a series of wave functions (orbitals) that describe the possible energies and spatial distributions available to the electron.
- In agreement with the Heisenberg uncertainty principle, the model cannot specify the detailed electron motions. Instead, the square of the wave function represents the probability distribution of the electron in that orbital. This allows us to picture orbitals in terms of probability distributions, or electron density maps.
- The size of an orbital is arbitrarily defined as the surface that contains 90% of the total electron probability.
- The hydrogen atom has many types of orbitals. In the ground state, the single electron resides in the 1s orbital. The electron can be excited to higher-energy orbitals if energy is put into the atom.

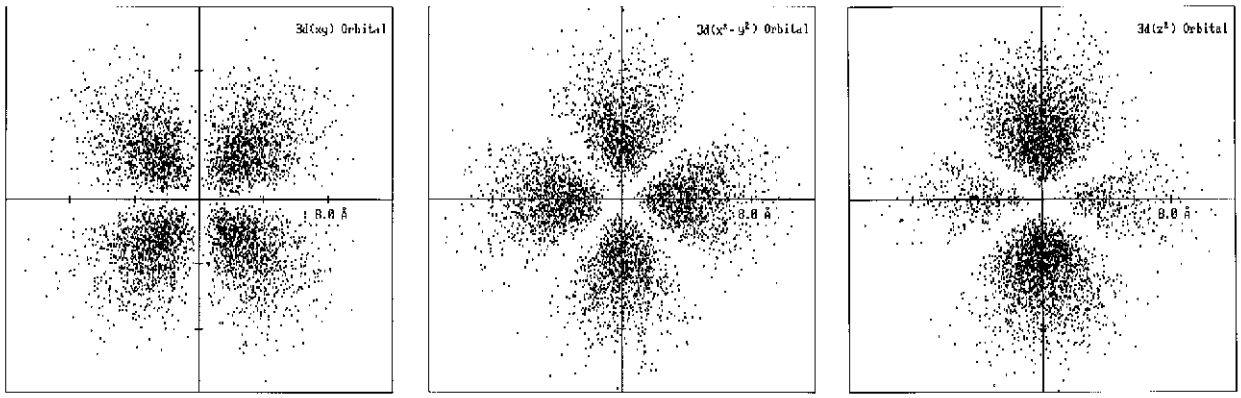


**FIGURE 7.15**
A cross section of the electron probability distribution for a 3*p* orbital.

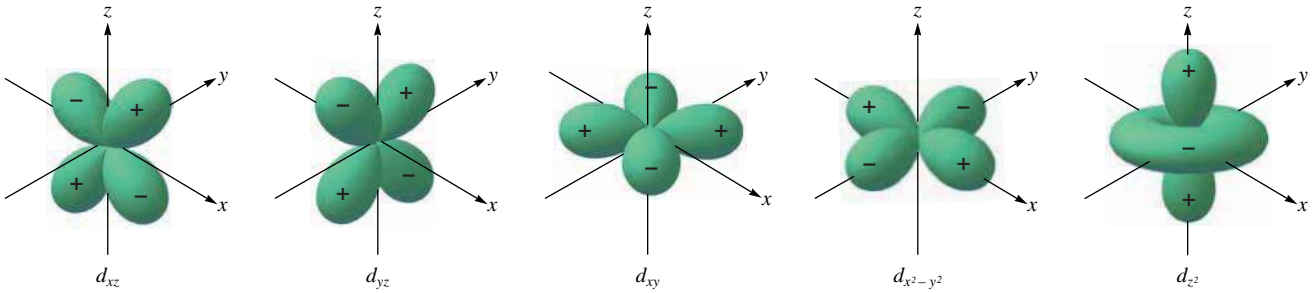
## 7.8 Electron Spin and the Pauli Principle

The concept of **electron spin** was developed by Samuel Goudsmit and George Uhlenbeck while they were graduate students at the University of Leyden in the Netherlands. They found that a fourth quantum number (in addition to $n$, $\ell$, and $m_\ell$) was necessary to account for the details of the emission spectra of atoms. The spectral data indicate that the electron has a magnetic moment with two possible orientations when the atom is placed in an external magnetic field. Since they knew from classical physics that a spinning charge produces a magnetic moment, it seemed reasonable to assume that the electron could have two spin states, thus producing the two oppositely directed magnetic moments

اتحاد مسافر البوليتكنك (spu)

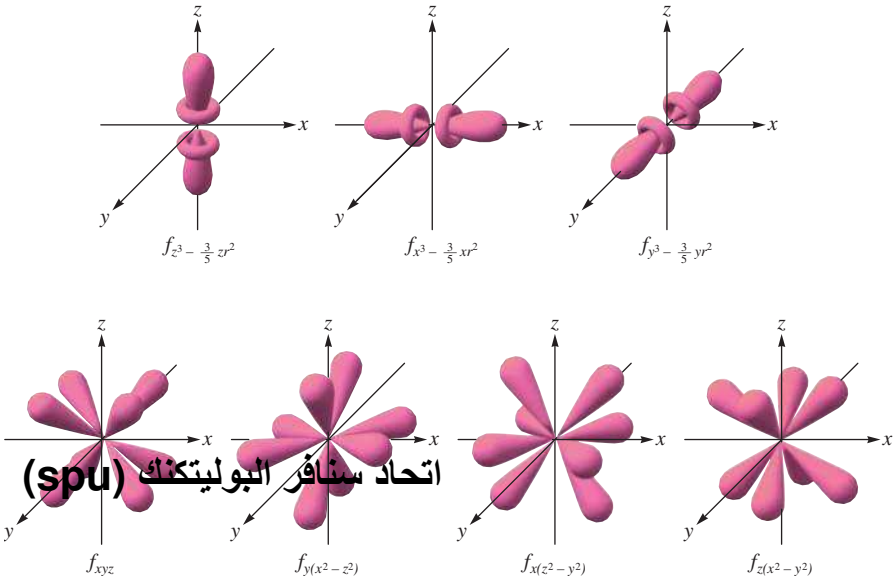



(a)




(b)

**FIGURE 7.16**
Representation of the 3*d* orbitals. (a) Electron density plots of selected 3*d* orbitals. (Generated from a program by Robert Allendoerfer on Project SERAPHIM disk PC 2402; reprinted with permission.) (b) The boundary surfaces of all five 3*d* orbitals, with the signs (phases) indicated.




اتحاد سنافر البوليتكنك (spu)

**FIGURE 7.17**
Representation of the 4*f* orbitals in terms of their boundary surfaces.

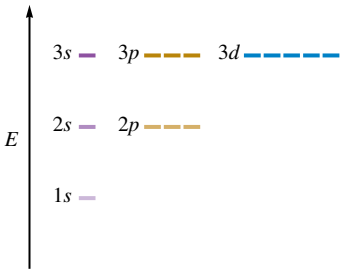



**FIGURE 7.18**
Orbital energy levels for the hydrogen atom.

(see Fig. 7.19). The new quantum number adopted to describe this phenomenon, called the **electron spin quantum number** ($m_s$), can have only one of two values, $+\frac{1}{2}$ and $-\frac{1}{2}$. We can interpret this to mean that the electron can spin in one of two opposite directions, although other interpretations also have been suggested.

For our purposes, the main significance of electron spin is connected with the postulate of Austrian physicist Wolfgang Pauli (1900–1958): *In a given atom no two electrons can have the same set of four quantum numbers ($n$, $\ell$, $m_\ell$, and $m_s$).* This is called the **Pauli exclusion principle.** Since electrons in the same orbital have the same values of $n$, $\ell$, and $m_\ell$, this postulate says that they must have different values of $m_s$. Then, since only two values of $m_s$ are allowed, *an orbital can hold only two electrons, and they must have opposite spins.* This principle will have important consequences as we use the atomic model to account for the electron arrangements of the atoms in the periodic table.

$m_s = +\frac{1}{2}$ or $-\frac{1}{2}$

Each orbital can hold a maximum of two electrons.

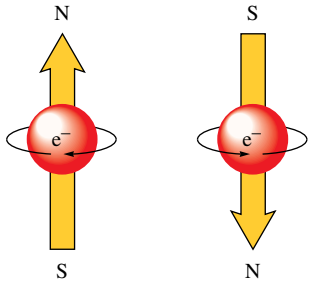



**FIGURE 7.19**
A picture of the spinning electron. Spinning in one direction, the electron produces the magnetic field oriented as shown in (a). Spinning in the opposite direction, it gives a magnetic field of the opposite orientation, as shown in (b).

## 7.9 Polyelectronic Atoms

The quantum mechanical model gives a description of the hydrogen atom that agrees very well with experimental data. However, the model would not be very useful if it did not account for the properties of all the other atoms as well.

To see how the model applies to **polyelectronic atoms,** that is, atoms with more than one electron, let's consider helium, which has two protons in its nucleus and two electrons:

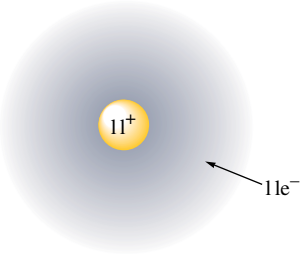



Three energy contributions must be considered in the description of the helium atom: (1) the kinetic energy of the electrons as they move around the nucleus, (2) the potential energy of attraction between the nucleus and the electrons, and (3) the potential energy of repulsion between the two electrons.

Although the helium atom can be readily described in terms of the quantum mechanical model, the Schrödinger equation that results cannot be solved exactly. The difficulty arises in dealing with the repulsions between the electrons. Since the electron pathways are unknown, the electron repulsions cannot be calculated exactly. This is called the *electron correlation problem.*

The electron correlation problem occurs with all polyelectronic atoms. To treat these systems using the quantum mechanical model, we must make approximations. Most commonly, the approximation used is to treat each electron as if it were moving in a *field of charge that is the net result of the nuclear attraction and the average repulsions of all the other electrons.*

For example, consider the sodium atom, which has 11 electrons:




اتحاد سنافر البوليتكنك (spu)

Now let's single out the outermost electron and consider the forces this electron feels. The electron clearly is attracted to the highly charged nucleus. However, the electron also feels the repulsions caused by the other 10 electrons. The net effect is that the electron is not

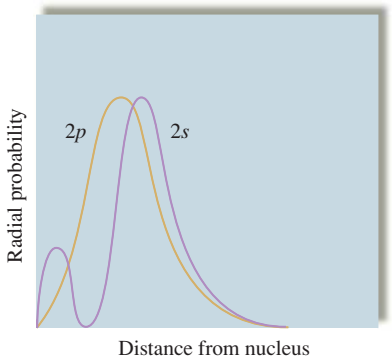
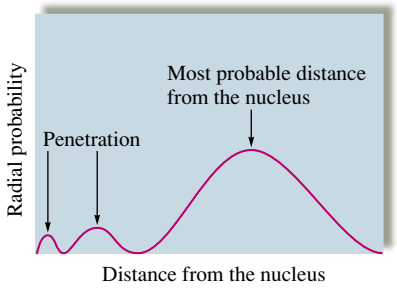
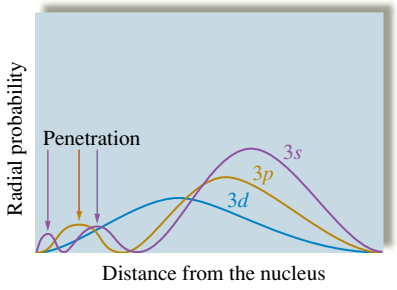



**FIGURE 7.20**
A comparison of the radial probability distributions of the 2*s* and 2*p* orbitals.




(a)




(b)

**FIGURE 7.21**
(a) The radial probability distribution for an electron in a 3*s* orbital. Although a 3*s* electron is mostly found far from the nucleus, there is a small but significant probability (shown by the arrows) of its being found close to the nucleus. The 3*s* electron penetrates the shield of inner electrons. (b) The radial probability distribution for the 3*s*, 3*p*, and 3*d* orbitals. The arrows indicate that the *s* orbital (red arrow) allows greater electron penetration than the *p* orbital (yellow arrow) does; the *d* orbital allows minimal electron penetration.

bound nearly as tightly to the nucleus as it would be if the other electrons were not present. We say that the electron is *screened* or *shielded* from the nuclear charge by the repulsions of the other electrons.

This picture of polyelectronic atoms leads to *hydrogenlike orbitals* for these atoms. They have the same general shapes as the orbitals for hydrogen, but their sizes and energies are different. The differences occur because of the interplay between nuclear attraction and the electron repulsions.

One especially important difference between polyelectronic atoms and the hydrogen atom is that for hydrogen all the orbitals in a given principal quantum level have the same energy (they are said to be *degenerate*). This is not the case for polyelectronic atoms, where we find that for a given principal quantum level the orbitals vary in energy as follows:

$$E_{ns} < E_{np} < E_{nd} < E_{nf}$$

In other words, when electrons are placed in a particular quantum level, they "prefer" the orbitals in the order *s*, *p*, *d*, and then *f*. Why does this happen? Although the concept of orbital energies is a complicated matter, we can qualitatively understand why the 2*s* orbital has a lower energy than the 2*p* orbital in a polyelectronic atom by looking at the probability profiles of these orbitals (see Fig. 7.20). Notice that the 2*p* orbital has its maximum probability closer to the nucleus than for the 2*s*. This might lead us to predict that the 2*p* would be preferable (lower energy) to the 2*s* orbital. However, notice the small hump of electron density that occurs in the 2*s* profile very near the nucleus. This means that although an electron in the 2*s* orbital spends most of its time a little farther from the nucleus than does an electron in the 2*p* orbital, it spends a small but very significant amount of time very near the nucleus. We say that the 2*s* electron *penetrates* to the nucleus more than one in the 2*p* orbital. This *penetration effect* causes an electron in a 2*s* orbital to be attracted to the nucleus more strongly than an electron in a 2*p* orbital. That is, the 2*s* orbital is lower in energy than the 2*p* orbitals in a polyelectronic atom.

The same thing happens in the other principal quantum levels as well. Figure 7.21 shows the radial probability profiles for the 3*s*, 3*p*, and 3*d* orbitals. Note again the hump in the 3*s* profile very near the nucleus. The innermost hump for the 3*p* is farther out, which causes the energy of the 3*s* orbital to be lower than that of the 3*p*. Notice that the 3*d* orbital has its maximum probability closer to the nucleus than either the 3*s* or 3*p* does, but its absence of probability near the nucleus causes it to be highest in energy of the three orbitals. The relative energies of the orbitals for $n = 3$ are

$$E_{3s} < E_{3p} < E_{3d}$$

In general, the more effectively an orbital allows its electron to penetrate the shielding electrons to be close to the nuclear charge, the lower is the energy of that orbital.

A summary diagram of the orders of the orbital energies for polyelectronic atoms is represented in Fig. 7.22. We will use these orbitals in Section 7.11 to show how the electrons are arranged in polyelectronic atoms.

## 7.10 The History of the Periodic Table

The modern periodic table contains a tremendous amount of useful information. In this section we will discuss the origin of this valuable tool; later we will see how the quantum mechanical model for the atom explains the periodicity of chemical properties. Certainly the greatest triumph of the quantum mechanical model is its ability to account for the arrangement of the elements in the periodic table.

اتحاد سنفر البويكنك (spu)

The periodic table was originally constructed to represent the patterns observed in the chemical properties of the elements. As chemistry progressed during the eighteenth and nineteenth centuries, it became evident that the earth is composed of a great many

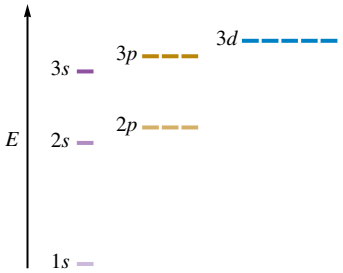



**FIGURE 7.22**
The orders of the energies of the orbitals in the first three levels of polyelectronic atoms.

elements with very different properties. Things are much more complicated than the simple model of earth, air, fire, and water suggested by the ancients. At first, the array of elements and properties was bewildering. Gradually, however, patterns were noticed.

The first chemist to recognize patterns was Johann Dobereiner (1780–1849), who found several groups of three elements that have similar properties, for example, chlorine, bromine, and iodine. However, as Dobereiner attempted to expand this model of *triads* (as he called them) to the rest of the known elements, it became clear that it was severely limited.

The next notable attempt was made by the English chemist John Newlands, who in 1864 suggested that elements should be arranged in *octaves,* based on the idea that certain properties seemed to repeat for every eighth element in a way similar to the musical scale, which repeats for every eighth tone. Even though this model managed to group several elements with similar properties, it was not generally successful.

The present form of the periodic table was conceived independently by two chemists: the German Julius Lothar Meyer (1830–1895) and Dmitri Ivanovich Mendeleev (1834–1907), a Russian (Fig. 7.23). Usually Mendeleev is given most of the credit, because it was he who emphasized how useful the table could be in predicting the existence and properties of still unknown elements. For example, in 1872 when Mendeleev first published his table (see Fig. 7.24), the elements gallium, scandium, and germanium were unknown. Mendeleev correctly predicted the existence and properties of these elements from gaps in his periodic table. The data for germanium (which Mendeleev called "*ekasilicon*") are shown in Table 7.3. Note the excellent agreement between the actual values and Mendeleev's predictions, which were based on the properties of other members in the group of elements similar to germanium.

Using his table, Mendeleev also was able to correct several values for atomic masses. For example, the original atomic mass of 76 for indium was based on the assumption that indium oxide had the formula InO. This atomic mass placed indium, which has metallic properties, among the nonmetals. Mendeleev assumed the atomic mass was probably incorrect and proposed that the formula of indium oxide was really $In_2O_3$. Based on this correct formula, indium has an atomic mass of approximately 113, placing the element among the metals. Mendeleev also corrected the atomic masses of beryllium and uranium.

Because of its obvious usefulness, Mendeleev's periodic table was almost universally adopted, and it remains one of the most valuable tools at the chemist's disposal. For example, it is still used to predict the properties of elements recently discovered, as shown in Table 7.4.

A current version of the periodic table is shown inside the front cover of this book. The only fundamental difference between this table and that of Mendeleev is that it lists the elements in order by atomic number rather than by atomic mass. The reason for this will become clear later in this chapter as we explore the electron arrangements of the atom. Another recent format of the table is discussed in the following section.



**FIGURE 7.23**
Dmitri Ivanovich Mendeleev (1834–1907), born in Siberia as the youngest of 17 children, taught chemistry at the University of St. Petersburg. In 1860 Mendeleev heard the Italian chemist Cannizzaro lecture on a reliable method for determining the correct atomic masses of the elements. This important development paved the way for Mendeleev's own brilliant contribution to chemistry—the periodic table. In 1861 Mendeleev returned to St. Petersburg, where he wrote a book on organic chemistry. Later Mendeleev also wrote a book on inorganic chemistry, and he was struck by the fact that the systematic approach characterizing organic chemistry was lacking in inorganic chemistry. In attempting to systematize inorganic chemistry, he eventually arranged the elements in the form of the periodic table.

Mendeleev was a versatile genius who was interested in many fields of science. He worked on many problems associated with Russia's natural resources, such as coal, salt, and various metals. Being particularly interested in the petroleum industry, he visited the United States in 1876 to study the Pennsylvania oil fields. His interests also included meteorology and hot-air balloons. In 1887 he made an ascent in a balloon to study a total eclipse of the sun.

اتحاد سنافر البوبيكنك (spu)

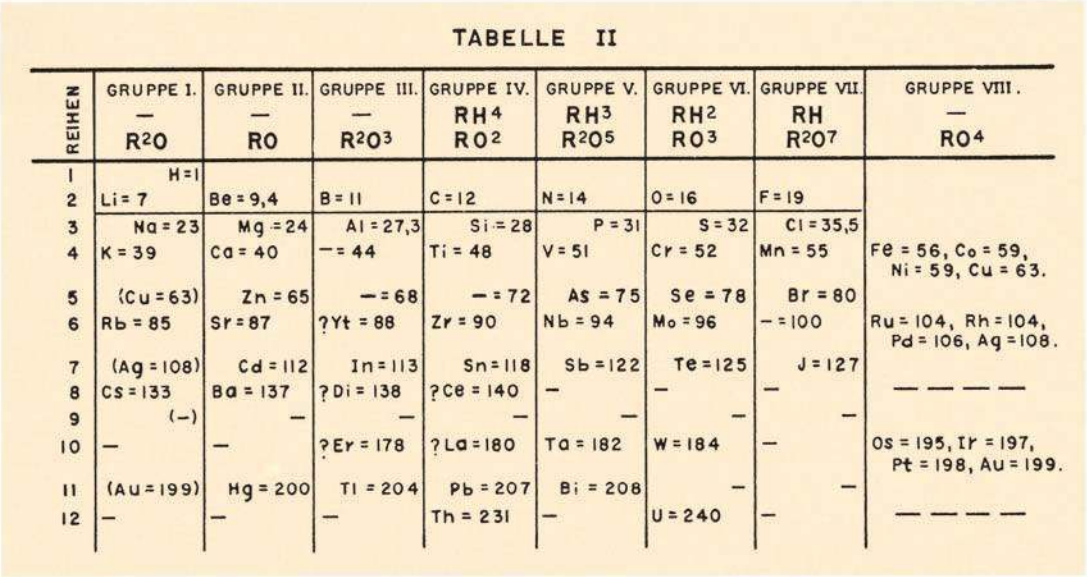


TABELLE II

| REIHEN | GRUPPE I. — $R^2O$ | GRUPPE II. — RO | GRUPPE III. — $R^2O^3$ | GRUPPE IV. $RH^4$ $RO^2$ | GRUPPE V. $RH^3$ $R^2O^5$ | GRUPPE VI. $RH^2$ $RO^3$ | GRUPPE VII. RH $R^2O^7$ | GRUPPE VIII. — $RO^4$ |
|---|---|---|---|---|---|---|---|---|
| 1 | H=1 | | | | | | | |
| 2 | Li=7 | Be=9,4 | B=11 | C=12 | N=14 | O=16 | F=19 | |
| 3 | Na=23 | Mg=24 | Al=27,3 | Si=28 | P=31 | S=32 | Cl=35,5 | |
| 4 | K=39 | Ca=40 | —=44 | Ti=48 | V=51 | Cr=52 | Mn=55 | Fe=56, Co=59, Ni=59, Cu=63. |
| 5 | (Cu=63) | Zn=65 | —=68 | —=72 | As=75 | Se=78 | Br=80 | |
| 6 | Rb=85 | Sr=87 | ?Yt=88 | Zr=90 | Nb=94 | Mo=96 | —=100 | Ru=104, Rh=104, Pd=106, Ag=108. |
| 7 | (Ag=108) | Cd=112 | In=113 | Sn=118 | Sb=122 | Te=125 | J=127 | |
| 8 | Cs=133 | Ba=137 | ?Di=138 | ?Ce=140 | — | — | — | — — — — |
| 9 | (—) | — | — | — | — | — | — | |
| 10 | — | — | ?Er=178 | ?La=180 | Ta=182 | W=184 | — | Os=195, Ir=197, Pt=198, Au=199. |
| 11 | (Au=199) | Hg=200 | Tl=204 | Pb=207 | Bi=208 | — | — | |
| 12 | — | — | — | Th=231 | — | U=240 | — | — — — — |

**FIGURE 7.24**
Mendeleev's early periodic table, published in 1872. Note the spaces left for missing elements with atomic masses 44, 68, 72, and 100.
(From *Annalen der Chemie und Pharmacie*, VIII, Supplementary Volume for 1872, page 511.)

**TABLE 7.3 Comparison of the Properties of Germanium as Predicted by Mendeleev and as Actually Observed**

| Properties of Germanium | Predicted in 1871 | Observed in 1886 |
|---|---|---|
| Atomic weight | 72 | 72.3 |
| Density | 5.5 g/cm$^3$ | 5.47 g/cm$^3$ |
| Specific heat | 0.31 J/(°C · g) | 0.32 J/(°C · g) |
| Melting point | Very high | 960°C |
| Oxide formula | $RO_2$ | $GeO_2$ |
| Oxide density | 4.7 g/cm$^3$ | 4.70 g/cm$^3$ |
| Chloride formula | $RCl_4$ | $GeCl_4$ |
| bp of chloride | 100°C | 86°C |

**TABLE 7.4 Predicted Properties of Elements 113 and 114**

| Property | Element 113 | Element 114 |
|---|---|---|
| Chemically like | Thallium | Lead |
| Atomic mass | 297 | 298 |
| Density | 10 g/mL | 14 g/mL |
| Melting point | 430°C | 70°C |
| Boiling point | 1100°C | 150°C |

(spu) اتحاد سنافر البوليتكنك

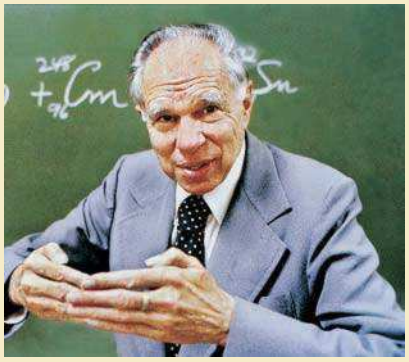


## CHEMICAL IMPACT

### The Growing Periodic Table

The periodic table of the elements has undergone significant changes since Mendeleev published his first version in 1869. In particular, in the past 60 years we have added 20 new elements beyond uranium. These so-called transuranium elements all have been synthesized using particle accelerators.

Edwin M. McMillan and Phillip H. Abelson succeeded in synthesizing the first transuranium element, neptunium (element 93), at the University of California, Berkeley, in 1940. In 1941, Glenn T. Seaborg synthesized and identified element 94 (plutonium), and over the next several years, researchers under his direction at UC Berkeley discovered nine other transuranium elements. In 1945 Seaborg suggested that the elements heavier than element 89 (actinium) were misplaced as transition metals and should be relocated on the periodic table in a series below the transition metals (the actinide series). Seaborg was awarded a Nobel Prize in chemistry in 1951 for his contributions.




Dr. Glenn Seaborg.

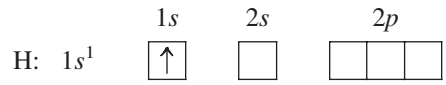
## 7.11 The Aufbau Principle and the Periodic Table

We can use the quantum mechanical model of the atom to show how the electron arrangements in the hydrogenlike atomic orbitals of the various atoms account for the organization of the periodic table. Our main assumption here is that all atoms have the same type of orbitals as have been described for the hydrogen atom. *As protons are added one by one to the nucleus to build up the elements, electrons are similarly added to these hydrogenlike orbitals.* This is called the **aufbau principle.**

***Aufbau*** is German for "building up."

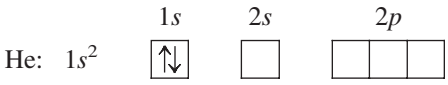
H ($Z = 1$)
He ($Z = 2$)
Li ($Z = 3$)
Be ($Z = 4$)
B ($Z = 5$)
etc.
($Z$ = atomic number)

*Hydrogen* has one electron, which occupies the 1*s* orbital in its ground state. The configuration for hydrogen is written as $1s^1$, which can be represented by the following *orbital diagram:*



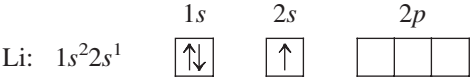

The arrow represents an electron spinning in a particular direction.

The next element, *helium,* has two electrons. Since two electrons with opposite spins can occupy an orbital, according to the Pauli exclusion principle, the electrons for helium are in the 1*s* orbital with opposite spins, producing a $1s^2$ configuration:




*Lithium* has three electrons, two of which can go into the 1*s* orbital before the orbital is filled. Since the 1*s* orbital is the only orbital for $n = 1$, the third electron will occupy the lowest-energy orbital with $n = 2$, or the 2*s* orbital, giving a $1s^22s^1$ configuration:

اتحاد سنافر البوليتكنك (spu)



In recent years, three major research facilities have taken the lead in synthesizing new elements. Along with UC Berkeley, Nuclear Research in Dubna, Russia, and GSI in Darmstadt, Germany, were responsible for synthesizing elements 104–112 by the end of 1996.

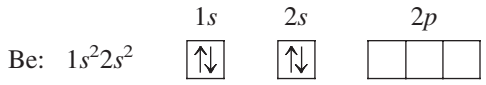
As it turned out, naming the new elements has caused more controversy than anything else connected with their discovery. Traditionally, the discoverer of an element is allowed to name it. However, because there is some dispute among the researchers at Berkeley, Darmstadt, and Dubna about who really discovered the various elements, competing names were submitted. After years of controversy, the International Union of Pure and Applied Chemistry (IUPAC) finally settled on the names listed in the accompanying table.

The name for element 106 in honor of Glenn Seaborg caused special controversy because an element had never before been named for a living person (Dr. Seaborg died in 1999). However, because of Seaborg's commanding stature in the scientific community, the name seaborgium was adopted.

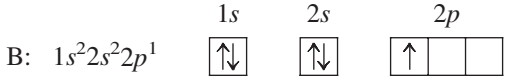
Names for the elements beyond 111 have not been decided, and these elements are represented on many periodic tables with three letters that symbolize their atomic numbers. More traditional names will no doubt be assigned in due time (hopefully with a minimum of controversy).

| Atomic Number | Name | Symbol |
|---|---|---|
| 104 | Rutherfordium | Rf |
| 105 | Dubium | Db |
| 106 | Seaborgium | Sg |
| 107 | Bohrium | Bh |
| 108 | Hassium | Hs |
| 109 | Meitnerium | Mt |
| 110 | Darmstadtium | Ds |
| 111 | Roentgenium | Rg |

The next element, *beryllium,* has four electrons, which occupy the 1*s* and 2*s* orbitals:




*Boron* has five electrons, four of which occupy the 1*s* and 2*s* orbitals. The fifth electron goes into the second type of orbital with $n = 2$, the 2*p* orbitals:



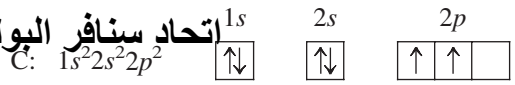

Since all the 2*p* orbitals have the same energy (are degenerate), it does not matter which 2*p* orbital the electron occupies.

*Carbon* is the next element and has six electrons. Two electrons occupy the 1*s* orbital, two occupy the 2*s* orbital, and two occupy 2*p* orbitals. Since there are three 2*p* orbitals with the same energy, the mutually repulsive electrons will occupy *separate* 2*p* orbitals. This behavior is summarized by **Hund's rule** (named for the German physicist F. H. Hund), which states that *the lowest energy configuration for an atom is the one having the maximum number of unpaired electrons allowed by the Pauli principle in a particular set of degenerate orbitals.* By convention, the unpaired electrons are represented as having parallel spins (with spin "up").

For an atom with unfilled subshells, the lowest energy is achieved by electrons occupying separate orbitals with parallel spins, as far as allowed by the Pauli exclusion principle.

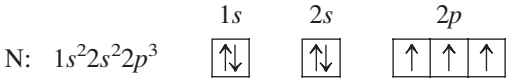
The configuration for carbon could be written $1s^2 2s^2 2p^1 2p^1$ to indicate that the electrons occupy separate 2*p* orbitals. However, the configuration is usually given as $1s^2 2s^2 2p^2$, and it is understood that the electrons are in different 2*p* orbitals. The orbital diagram for carbon is

اتحاد سنافر البوليتكنك (spu)

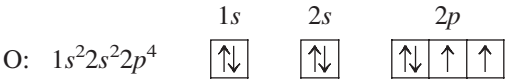



Note that the unpaired electrons in the 2*p* orbitals are shown with parallel spins.

The configuration for *nitrogen,* which has seven electrons, is $1s^2 2s^2 2p^3$. The three electrons in the $2p$ orbitals occupy separate orbitals with parallel spins:

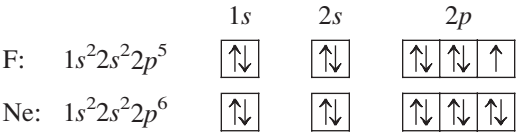



The configuration for *oxygen,* which has eight electrons, is $1s^2 2s^2 2p^4$. One of the $2p$ orbitals is now occupied by a pair of electrons with opposite spins, as required by the Pauli exclusion principle:




[Ne] is shorthand for $1s^2 2s^2 2p^6$.

The orbital diagrams and electron configurations for *fluorine* (nine electrons) and *neon* (ten electrons) are as follows:



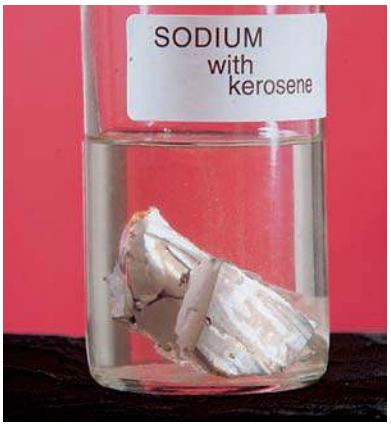

With neon, the orbitals with $n = 1$ and $n = 2$ are now completely filled.

For *sodium,* the first ten electrons occupy the $1s$, $2s$, and $2p$ orbitals, and the eleventh electron must occupy the first orbital with $n = 3$, the $3s$ orbital. The electron configuration for sodium is $1s^2 2s^2 2p^6 3s^1$. To avoid writing the inner-level electrons, this configuration is often abbreviated as $[Ne]3s^1$, where [Ne] represents the electron configuration of neon, $1s^2 2s^2 2p^6$.

The next element, *magnesium,* has the configuration $1s^2 2s^2 2p^6 3s^2$, or $[Ne]3s^2$. Then the next six elements, *aluminum* through *argon,* have configurations obtained by filling the $3p$ orbitals one electron at a time. Figure 7.25 summarizes the electron configurations of the first 18 elements by giving the number of electrons in the type of orbital occupied last.

At this point it is useful to introduce the concept of **valence electrons,** *the electrons in the outermost principal quantum level of an atom.* The valence electrons of the nitrogen atom, for example, are the $2s$ and $2p$ electrons. For the sodium atom, the valence electron is the electron in the $3s$ orbital, and so on. Valence electrons are the most important electrons to chemists because they are involved in bonding, as we will see in the next two chapters. The inner electrons are known as **core electrons.**

Note in Fig. 7.25 that a very important pattern is developing: *The elements in the same group (vertical column of the periodic table) have the same valence electron configuration.* Remember that Mendeleev originally placed the elements in groups based on similarities in chemical properties. Now we understand the reason behind these




Sodium metal is so reactive that it is stored under kerosene to protect it from the oxygen in the air.



A vial containing potassium metal. The sealed vial contains an inert gas to protect the potassium from reacting with oxygen.

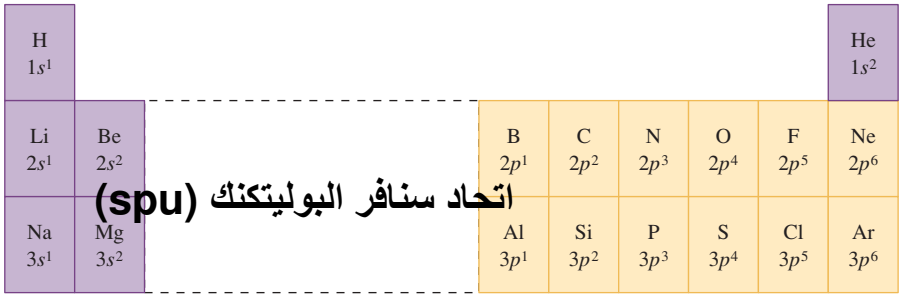



**FIGURE 7.25**
The electron configurations in the type of orbital occupied last for the first 18 elements.

اتحاد سنافر البوليتكنك (spu)

groupings. Elements with the same valence electron configuration show similar chemical behavior.

The element after argon is *potassium*. Since the 3*p* orbitals are fully occupied in argon, we might expect the next electron to go into a 3*d* orbital (recall that for $n = 3$ the orbitals are 3*s*, 3*p*, and 3*d*). However, the chemistry of potassium is clearly very similar to that of lithium and sodium, indicating that the last electron in potassium occupies the 4*s* orbital instead of one of the 3*d* orbitals, a conclusion confirmed by many types of experiments. The electron configuration of potassium is

$$\text{K:}\quad 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 \qquad \text{or} \qquad [\text{Ar}]4s^1$$

The next element is *calcium:*

$$\text{Ca:}\quad [\text{Ar}]4s^2$$



Calcium metal.

The next element, *scandium,* begins a series of 10 elements (scandium through zinc) called the **transition metals,** whose configurations are obtained by adding electrons to the five 3*d* orbitals. The configuration of scandium is

$$\text{Sc:}\quad [\text{Ar}]4s^2 3d^1$$

That of *titanium* is

$$\text{Ti:}\quad [\text{Ar}]4s^2 3d^2$$

And that of *vanadium* is

$$\text{V:}\quad [\text{Ar}]4s^2 3d^3$$

*Chromium* is the next element. The expected configuration is $[\text{Ar}]4s^2 3d^4$. However, the observed configuration is

$$\text{Cr:}\quad [\text{Ar}]4s^1 3d^5$$

The explanation for this configuration of chromium is beyond the scope of this book. In fact, chemists are still disagreeing over the exact cause of this anomaly. Note, however, that the observed configuration has both the 4*s* and 3*d* orbitals half-filled. This is a good way to remember the correct configuration.

The next four elements, *manganese* through *nickel,* have the expected configurations:

$$\text{Mn:}\quad [\text{Ar}]4s^2 3d^5 \qquad \text{Co:}\quad [\text{Ar}]4s^2 3d^7$$
$$\text{Fe:}\quad [\text{Ar}]4s^2 3d^6 \qquad \text{Ni:}\quad [\text{Ar}]4s^2 3d^8$$

The configuration for *copper* is expected to be $[\text{Ar}]4s^2 3d^9$. However, the observed configuration is

$$\text{Cu:}\quad [\text{Ar}]4s^1 3d^{10}$$

In this case, a half-filled 4*s* orbital and a filled set of 3*d* orbitals characterize the actual configuration.

*Zinc* has the expected configuration:

$$\text{Zn:}\quad [\text{Ar}]4s^2 3d^{10}$$

The configurations of the transition metals are shown in Fig. 7.26. After that, the next six elements, *gallium* through *krypton,* have configurations that correspond to filling the 4*p* orbitals (see Fig. 7.26).

The entire periodic table is represented in Fig. 7.27 in terms of which orbitals are being filled. The valence electron configurations are given in Fig. 7.28. From these two figures, note the following additional points:



Chromium is often used to plate bumpers and hood ornaments, such as this statue of Mercury found on a 1929 Buick.

The $(n + 1)s$ orbital fills before the *nd* orbitals.

1. The $(n + 1)s$ orbitals always fill before the *nd* orbitals. For example, the 5*s* orbitals fill in rubidium and strontium before the 4*d* orbitals fill in the second row of transition

(spu) اتحاد سنافر البو لبتكنك

| | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | |
| K $4s^1$ | Ca $4s^2$ | Sc $3d^1$ | Ti $3d^2$ | V $3d^3$ | Cr $4s^1\,3d^5$ | Mn $3d^5$ | Fe $3d^6$ | Co $3d^7$ | Ni $3d^8$ | Cu $4s^1 3d^{10}$ | Zn $3d^{10}$ | Ga $4p^1$ | Ge $4p^2$ | As $4p^3$ | Se $4p^4$ | Br $4p^5$ | Kr $4p^6$ |

**FIGURE 7.26**
Electron configurations for potassium through krypton. The transition metals (scandium through zinc) have the general configuration $[Ar]4s^2 3d^n$, except for chromium and copper.

metals (yttrium through cadmium). This early filling of the *s* orbitals can be explained by the penetration effect. For example, the 4*s* orbital allows for so much more penetration to the vicinity of the nucleus that it becomes lower in energy than the 3*d* orbital. Thus the 4*s* fills before the 3*d*. The same things can be said about the 5*s* and 4*d*, the 6*s* and 5*d*, and the 7*s* and 6*d* orbitals.

Lanthanides are elements in which the 4*f* orbitals are being filled.

2. After lanthanum, which has the configuration $[Xe]6s^2 5d^1$, a group of 14 elements called the **lanthanide series,** or the **lanthanides,** occurs. This series of elements corresponds to the filling of the seven 4*f* orbitals. Note that sometimes an electron occupies a 5*d* orbital instead of a 4*f* orbital. This occurs because the energies of the 4*f* and 5*d* orbitals are very similar.

Actinides are elements in which the 5*f* orbitals are being filled.

3. After actinium, which has the configuration $[Rn]7s^2 6d^1$, a group of 14 elements called the **actinide series,** or the **actinides,** occurs. This series corresponds to the filling of the seven 5*f* orbitals. Note that sometimes one or two electrons occupy the 6*d* orbitals instead of the 5*f* orbitals, because these orbitals have very similar energies.

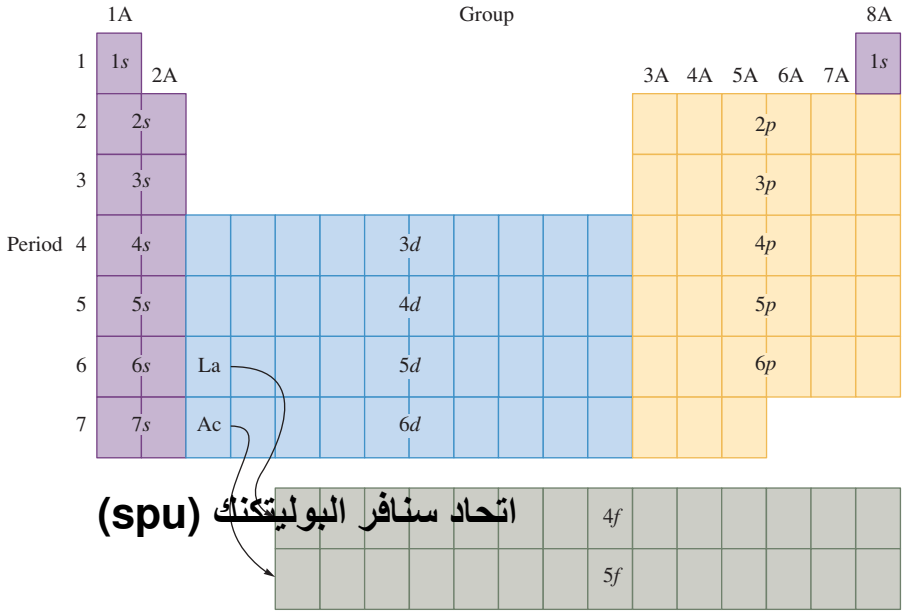


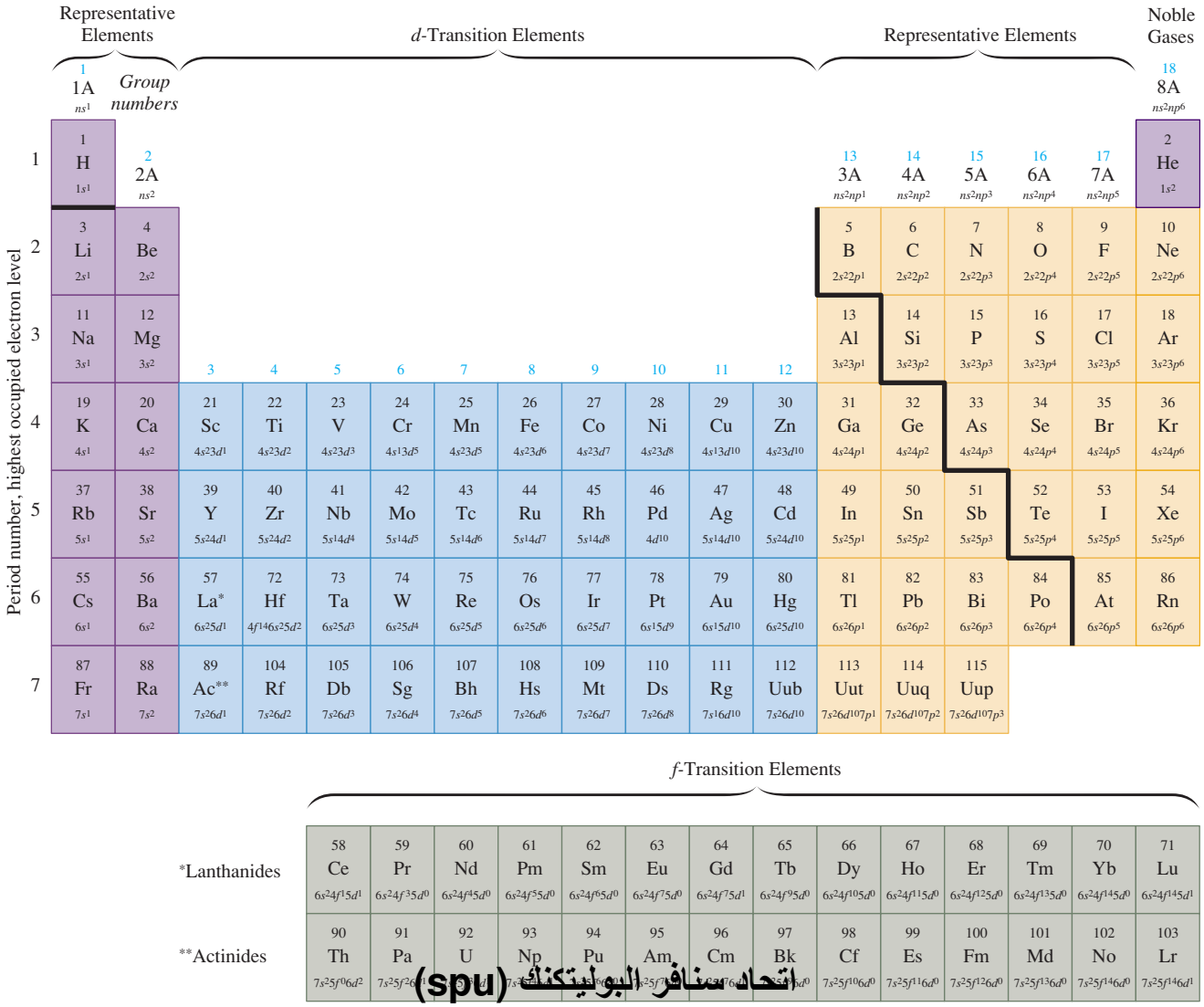

**FIGURE 7.27**
The orbitals being filled for elements in various parts of the periodic table. Note that in going along a horizontal row (a period), the $(n + 1)s$ orbital fills before the *nd* orbital. The group labels indicate the number of valence electrons (*ns* plus *np* electrons) for the elements in each group.

The group label tells the total number of valence electrons for that group.

4. The group labels for Groups 1A, 2A, 3A, 4A, 5A, 6A, 7A, and 8A indicate the *total number* of valence electrons for the atoms in these groups. For example, all the elements in Group 5A have the configuration $ns^2np^3$. (The *d* electrons fill one period late and are usually not counted as valence electrons.) The meaning of the group labels for the transition metals is not as clear as for the Group A elements, and these will not be used in this text.

5. The groups labeled 1A, 2A, 3A, 4A, 5A, 6A, 7A, and 8A are often called the **main-group,** or **representative, elements.** Every member of these groups has the same valence electron configuration.

The International Union of Pure and Applied Chemistry (IUPAC), a body of scientists organized to standardize scientific conventions, has recommended a new form for the periodic table, which the American Chemical Society has adopted (see the blue numbers in Fig. 7.28). In this new version the group number indicates the number of *s*, *p*, and *d* electrons added since the last noble gas. We will not use the new format in this book, but you




اتحاد منافر البوليتكنك (spu)

**FIGURE 7.28**
The periodic table with atomic symbols, atomic numbers, and partial electron configurations.

should be aware that the familiar periodic table may be soon replaced by this or a similar format.

The results considered in this section are very important. We have seen that the quantum mechanical model can be used to explain the arrangement of the elements in the periodic table. This model allows us to understand that the similar chemistry exhibited by the members of a given group arises from the fact that they all have the same valence electron configuration. Only the principal quantum number of the valence orbitals changes in going down a particular group.

It is important to be able to give the electron configuration for each of the main-group elements. This is most easily done by using the periodic table. If you understand how the table is organized, it is not necessary to memorize the order in which the orbitals fill. Review Figs. 7.27 and 7.28 to make sure that you understand the correspondence between the orbitals and the periods and groups.

When an electron configuration is given in this text, the orbitals are listed in the order in which they fill.

Cr: $[Ar]4s^1 3d^5$
Cu: $[Ar]4s^1 3d^{10}$

Predicting the configurations of the transition metals (3*d*, 4*d*, and 5*d* elements), the lanthanides (4*f* elements), and the actinides (5*f* elements) is somewhat more difficult because there are many exceptions of the type encountered in the first-row transition metals (the 3*d* elements). You should memorize the configurations of chromium and copper, the two exceptions in the first-row transition metals, since these elements are often encountered.

## *Sample Exercise 7.7* Electron Configurations

Give the electron configurations for sulfur (S), cadmium (Cd), hafnium (Hf), and radium (Ra) using the periodic table inside the front cover of this book.

### *Solution*

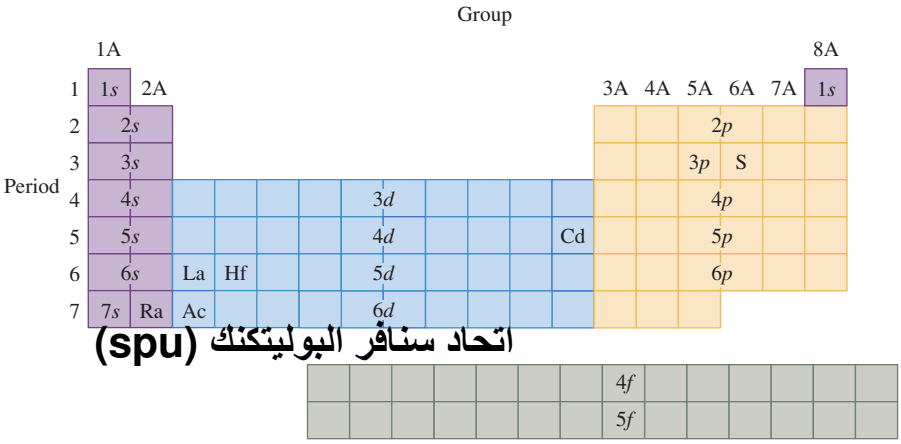
*Sulfur* is element 16 and resides in Period 3, where the 3*p* orbitals are being filled (see Fig. 7.29). Since sulfur is the fourth among the "3*p* elements," it must have four 3*p* electrons. Its configuration is

$$\text{S:}\quad 1s^2 2s^2 2p^6 3s^2 3p^4 \quad \text{or} \quad [\text{Ne}]3s^2 3p^4$$

*Cadmium* is element 48 and is located in Period 5 at the end of the 4*d* transition metals, as shown in Fig. 7.29. It is the tenth element in the series and thus has 10 electrons in the 4*d* orbitals, in addition to the 2 electrons in the 5*s* orbital. The configuration is

$$\text{Cd:}\quad 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} \quad \text{or} \quad [\text{Kr}]5s^2 4d^{10}$$

*Hafnium* is element 72 and is found in Period 6, as shown in Fig. 7.29. Note that it occurs just after the lanthanide series. Thus the 4*f* orbitals are already filled. Hafnium is




اتحاد سنافر البوليتكنك (spu)

**FIGURE 7.29**
The positions of the elements considered in Sample Exercise 7.7.

the second member of the 5*d* transition series and has two 5*d* electrons. The configuration is

$$\text{Hf:}\quad 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^2 \quad \text{or} \quad [\text{Xe}]6s^2 4f^{14} 5d^2$$

*Radium* is element 88 and is in Period 7 (and Group 2A), as shown in Fig. 7.29. Thus radium has two electrons in the 7*s* orbital, and the configuration is

$$\text{Ra:}\quad 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2 4d^{10} 5p^6 6s^2 4f^{14} 5d^{10} 6p^6 7s^2 \quad \text{or} \quad [\text{Rn}]7s^2$$

*See Exercises 7.69 through 7.72.*

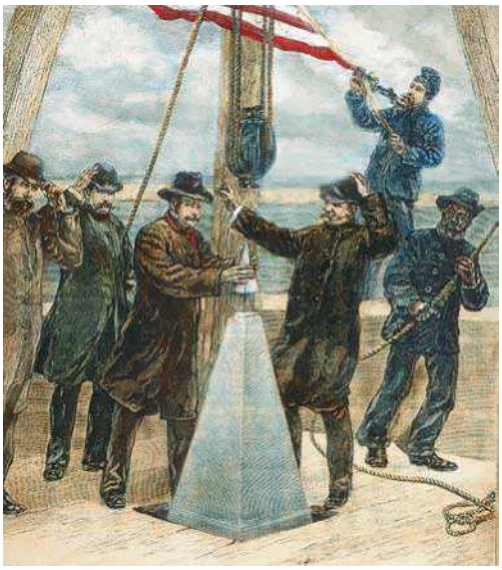
## 7.12 Periodic Trends in Atomic Properties

We have developed a fairly complete picture of polyelectronic atoms. Although the model is rather crude because the nuclear attractions and electron repulsions are simply lumped together, it is very successful in accounting for the periodic table of elements. We will next use the model to account for the observed trends in several important atomic properties: ionization energy, electron affinity, and atomic size.

Visualization: Periodic Table Trends

Ionization energy results in the formation of a **positive** ion.

### Ionization Energy



Setting the aluminum cap on the Washington Monument in 1884. At that time, aluminum was regarded as a precious metal.

*Ionization energy* is the energy required to remove an electron from a gaseous atom or ion:

$$X(g) \longrightarrow X^+(g) + e^-$$

where the atom or ion is assumed to be in its ground state.

To introduce some of the characteristics of ionization energy, we will consider the energy required to remove several electrons in succession from aluminum in the gaseous state. The ionization energies are

$$\begin{aligned}
Al(g) &\longrightarrow Al^+(g) + e^- && I_1 = 580 \text{ kJ/mol}\\
Al^+(g) &\longrightarrow Al^{2+}(g) + e^- && I_2 = 1815 \text{ kJ/mol}\\
Al^{2+}(g) &\longrightarrow Al^{3+}(g) + e^- && I_3 = 2740 \text{ kJ/mol}\\
Al^{3+}(g) &\longrightarrow Al^{4+}(g) + e^- && I_4 = 11{,}600 \text{ kJ/mol}
\end{aligned}$$

Several important points can be illustrated from these results. In a stepwise ionization process, it is always the highest-energy electron (the one bound least tightly) that is removed first. The **first ionization energy** $I_1$ is the energy required to remove the highest-energy electron of an atom. The first electron removed from the aluminum atom comes from the 3*p* orbital (Al has the electron configuration $[\text{Ne}]3s^2 3p^1$). The second electron comes from the 3*s* orbital (since $Al^+$ has the configuration $[\text{Ne}]3s^2$). Note that the value of $I_1$ is considerably smaller than the value of $I_2$, the **second ionization energy.**

This makes sense for several reasons. The primary factor is simply charge. Note that the first electron is removed from a neutral atom (Al), whereas the second electron is removed from a 1+ ion ($Al^+$). The increase in positive charge binds the electrons more firmly, and the ionization energy increases. The same trend shows up in the third ($I_3$) and fourth ($I_4$) ionization energies, where the electron is removed from the $Al^{2+}$ and $Al^{3+}$ ions, respectively.

The increase in successive ionization energies for an atom also can be interpreted using our simple model for polyelectronic atoms. The increase in ionization energy from $I_1$ to $I_2$ makes sense because the first electron is removed from a 3*p* orbital that is higher in energy than the 3*s* orbital from which the second electron is removed. The largest jump in ionization energy by far occurs in going from the third ionization energy ($I_3$) to the fourth ($I_4$). This is so because $I_4$ corresponds to removing a core electron ($Al^{3+}$ has the configuration $1s^2 2s^2 2p^6$), and core electrons are bound much more tightly than valence electrons.

اتحاد ستافر أبوبكر (spu)

**TABLE 7.5 Successive Ionization Energies in Kilojoules per Mole for the Elements in Period 3**

| Element | $I_1$ | $I_2$ | $I_3$ | $I_4$ | $I_5$ | $I_6$ | $I_7$ |
|---|---|---|---|---|---|---|---|
| Na | 495 | 4560 | | | | | |
| Mg | 735 | 1445 | 7730 | Core electrons* | | | |
| Al | 580 | 1815 | 2740 | 11,600 | | | |
| Si | 780 | 1575 | 3220 | 4350 | 16,100 | | |
| P | 1060 | 1890 | 2905 | 4950 | 6270 | 21,200 | |
| S | 1005 | 2260 | 3375 | 4565 | 6950 | 8490 | 27,000 |
| Cl | 1255 | 2295 | 3850 | 5160 | 6560 | 9360 | 11,000 |
| Ar | 1527 | 2665 | 3945 | 5770 | 7230 | 8780 | 12,000 |

*Note the large jump in ionization energy in going from removal of valence electrons to removal of core electrons.

General decrease

General increase

**TABLE 7.6 First Ionization Energies for the Alkali Metals and Noble Gases**

| Atom | $I_1$(kJ/mol) |
|---|---|
| **Group 1A** | |
| Li | 520 |
| Na | 495 |
| K | 419 |
| Rb | 409 |
| Cs | 382 |
| **Group 8A** | |
| He | 2377 |
| Ne | 2088 |
| Ar | 1527 |
| Kr | 1356 |
| Xe | 1176 |
| Rn | 1042 |

First ionization energy increases across a period and decreases down a group.

Table 7.5 gives the values of ionization energies for all the Period 3 elements. Note the large jump in energy in each case in going from removal of valence electrons to removal of core electrons.

The values of the first ionization energies for the elements in the first six periods of the periodic table are graphed in Fig. 7.30. Note that in general as we go *across a period from left to right, the first ionization energy increases.* This is consistent with the idea that electrons added in the same principal quantum level do not completely shield the increasing nuclear charge caused by the added protons. Thus electrons in the same principal quantum level are generally more strongly bound as we move to the right on the periodic table, and there is a general increase in ionization energy values as electrons are added to a given principal quantum level.

On the other hand, *first ionization energy decreases in going down a group.* This can be seen most clearly by focusing on the Group 1A elements (the alkali metals) and the Group 8A elements (the noble gases), as shown in Table 7.6. The main reason for the decrease in ionization energy in going down a group is that the electrons being removed are, on average, farther from the nucleus. As $n$ increases, the size of the orbital increases, and the electron is easier to remove.

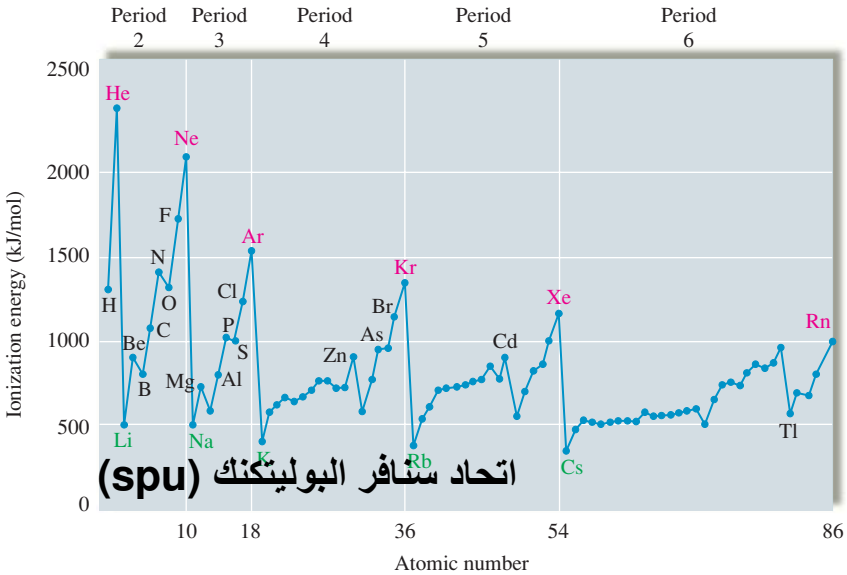



**FIGURE 7.30**
The values of first ionization energy for the elements in the first six periods. In general, ionization energy decreases in going down a group. For example, note the decrease in values for Group 1A and Group 8A. In general, ionization energy increases in going left to right across a period. For example, note the sharp increase going across Period 2 from lithium through neon.

(spu) اتحاد شافر البوليتكنك

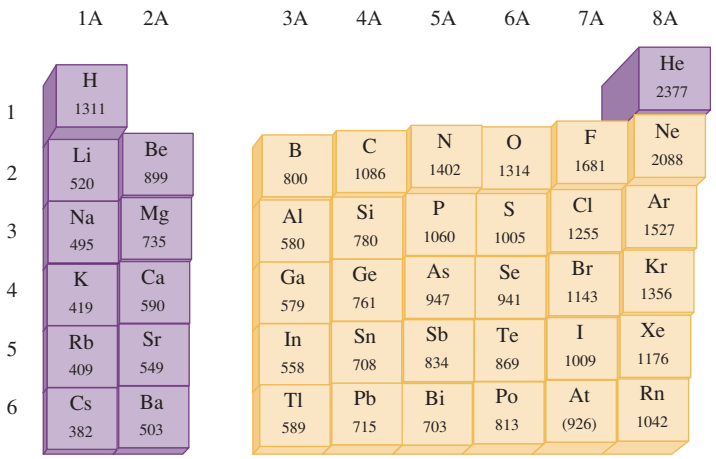



**FIGURE 7.31**
Trends in ionization energies (kJ/mol) for the representative elements.

In Fig. 7.30 we see that there are some discontinuities in ionization energy in going across a period. For example, for Period 2, discontinuities occur in going from beryllium to boron and from nitrogen to oxygen. These exceptions to the normal trend can be explained in terms of electron repulsions. The decrease in ionization energy in going from beryllium to boron reflects the fact that the electrons in the filled 2*s* orbital provide some shielding for electrons in the 2*p* orbital from the nuclear charge. The decrease in ionization energy in going from nitrogen to oxygen reflects the extra electron repulsions in the doubly occupied oxygen 2*p* orbital.

The ionization energies for the representative elements are summarized in Fig. 7.31.

## *Sample Exercise 7.8* Trends in Ionization Energies

The first ionization energy for phosphorus is 1060 kJ/mol, and that for sulfur is 1005 kJ/mol. Why?

### *Solution*

Phosphorus and sulfur are neighboring elements in Period 3 of the periodic table and have the following valence electron configurations: Phosphorus is $3s^2 3p^3$, and sulfur is $3s^2 3p^4$.

Ordinarily, the first ionization energy increases as we go across a period, so we might expect sulfur to have a greater ionization energy than phosphorus. However, in this case the fourth *p* electron in sulfur must be placed in an already occupied orbital. The electron–electron repulsions that result cause this electron to be more easily removed than might be expected.

*See Exercises 7.93 and 7.94.*

## *Sample Exercise 7.9* Ionization Energies

Consider atoms with the following electron configurations:

$$1s^2 2s^2 2p^6$$
$$1s^2 2s^2 2p^6 3s^1$$
$$1s^2 2s^2 2p^6 3s^2$$

اتحاد سنافر البوليتكنك (spu)

Which atom has the largest first ionization energy, and which one has the smallest second ionization energy? Explain your choices.

***Solution***

The atom with the largest value of $I_1$ is the one with the configuration $1s^2 2s^2 2p^6$ (this is the neon atom), because this element is found at the right end of Period 2. Since the $2p$ electrons do not shield each other very effectively, $I_1$ will be relatively large. The other configurations given include $3s$ electrons. These electrons are effectively shielded by the core electrons and are farther from the nucleus than the $2p$ electrons in neon. Thus $I_1$ for these atoms will be smaller than for neon.

The atom with the smallest value of $I_2$ is the one with the configuration $1s^2 2s^2 2p^6 3s^2$ (the magnesium atom). For magnesium, both $I_1$ and $I_2$ involve valence electrons. For the atom with the configuration $1s^2 2s^2 2p^6 3s^1$ (sodium), the second electron lost (corresponding to $I_2$) is a core electron (from a $2p$ orbital).

***See Exercises 7.121 and 7.123.***

## Electron Affinity

Electron affinity is associated with the production of a **negative** ion.

**Electron affinity** *is the energy change associated with the addition of an electron to a gaseous atom:*

$$X(g) + e^- \longrightarrow X^-(g)$$

The sign convention for electron affinity values follows the convention for energy changes used in Chapter 6.

Because two different conventions have been used, there is a good deal of confusion in the chemical literature about the signs for electron affinity values. Electron affinity has been defined in many textbooks as the energy *released* when an electron is added to a gaseous atom. This convention requires that a positive sign be attached to an exothermic addition of an electron to an atom, which opposes normal thermodynamic conventions. Therefore, in this book we define electron affinity as a *change* in energy, which means that if the addition of the electron is exothermic, the corresponding value for electron affinity will carry a negative sign.

Figure 7.32 shows the electron affinity values for the atoms among the first 20 elements that form stable, isolated negative ions—that is, the atoms that undergo the addition of an electron as shown above. As expected, all these elements have negative (exothermic) electron affinities. Note that the *more negative* the energy, the greater the quantity of energy released. Although electron affinities generally become more negative from left to right across a period, there are several exceptions to this rule in each period. The dependence of electron affinity on atomic number can be explained by considering the changes in electron repulsions as a function of electron configurations. For example, the fact that the nitrogen atom does not form a stable, isolated $N^-(g)$ ion, whereas carbon forms $C^-(g)$, reflects the difference in the electron configurations of these atoms. An electron added to nitrogen ($1s^2 2s^2 2p^3$) to form the $N^-(g)$ ion ($1s^2 2s^2 2p^4$) would have to occupy a $2p$ orbital that already contains one electron. The extra repulsion between the electrons in this doubly occupied orbital causes $N^-(g)$ to be unstable. When an electron is added to carbon ($1s^2 2s^2 2p^2$) to form the $C^-(g)$ ion ($1s^2 2s^2 2p^3$), no such extra repulsions occur.

In contrast to the nitrogen atom, the oxygen atom can add one electron to form the stable $O^-(g)$ ion. Presumably oxygen's greater nuclear charge compared with that of nitrogen

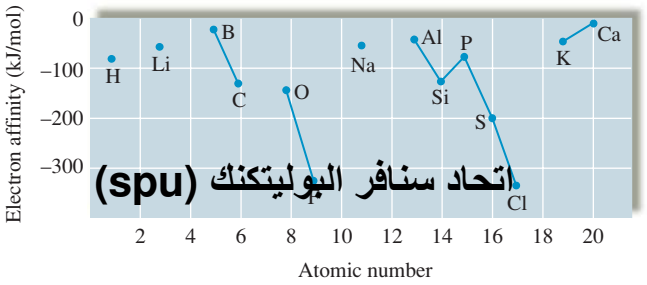



**FIGURE 7.32**
The electron affinity values for atoms among the first 20 elements that form stable, isolated $X^-$ ions. The lines shown connect adjacent elements. The absence of a line indicates missing elements (He, Be, N, Ne, Mg, and Ar) whose atoms do not add an electron exothermically and thus do not form stable, isolated $X^-$ ions.

**TABLE 7.7 Electron Affinities of the Halogens**

| Atom | Electron Affinity (kJ/mol) |
|---|---|
| F | −327.8 |
| Cl | −348.7 |
| Br | −324.5 |
| I | −295.2 |

is sufficient to overcome the repulsion associated with putting a second electron into an already occupied 2*p* orbital. However, it should be noted that a second electron *cannot* be added to an oxygen atom [$O^-(g) + e^- \not\rightarrow O^{2-}(g)$] to form an isolated oxide ion. This outcome seems strange in view of the many stable oxide compounds (MgO, $Fe_2O_3$, and so on) that are known. As we will discuss in detail in Chapter 8, the $O^{2-}$ ion is stabilized in ionic compounds by the large attractions that occur among the positive ions and the oxide ions.

When we go down a group, electron affinity should become more positive (less energy released), since the electron is added at increasing distances from the nucleus. Although this is generally the case, the changes in electron affinity in going down most groups are relatively small, and numerous exceptions occur. This behavior is demonstrated by the electron affinities of the Group 7A elements (the halogens) shown in Table 7.7. Note that the range of values is quite small compared with the changes that typically occur across a period. Also note that although chlorine, bromine, and iodine show the expected trend, the energy released when an electron is added to fluorine is smaller than might be expected. This smaller energy release has been attributed to the small size of the 2*p* orbitals. Because the electrons must be very close together in these orbitals, there are unusually large electron–electron repulsions. In the other halogens with their larger orbitals, the repulsions are not as severe.

## Atomic Radius

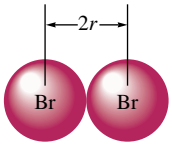



**FIGURE 7.33**
The radius of an atom (*r*) is defined as half the distance between the nuclei in a molecule consisting of identical atoms.

Just as the size of an orbital cannot be specified exactly, neither can the size of an atom. We must make some arbitrary choices to obtain values for **atomic radii.** These values can be obtained by measuring the distances between atoms in chemical compounds. For example, in the bromine molecule, the distance between the two nuclei is known to be 228 pm. The bromine atomic radius is assumed to be half this distance, or 114 pm, as shown in Fig. 7.33. These radii are often called *covalent atomic radii* because of the way they are determined (from the distances between atoms in covalent bonds).

For nonmetallic atoms that do not form diatomic molecules, the atomic radii are estimated from their various covalent compounds. The radii for metal atoms (called *metallic radii*) are obtained from half the distance between metal atoms in solid metal crystals.

The values of the atomic radii for the representative elements are shown in Fig. 7.34. Note that these values are significantly smaller than might be expected from the 90% electron density volumes of isolated atoms, because when atoms form bonds, their electron "clouds" interpenetrate. However, these values form a self-consistent data set that can be used to discuss the trends in atomic radii.

Note from Fig. 7.34 that the atomic radii decrease in going from left to right across a period. This decrease can be explained in terms of the increasing effective nuclear charge (decreasing shielding) in going from left to right. This means that the valence electrons are drawn closer to the nucleus, decreasing the size of the atom.

Atomic radius increases down a group, because of the increases in the orbital sizes in successive principal quantum levels.

Visualization: Determining the Atomic Radius of a Nonmetal (Chlorine)

Visualization: Determining the Atomic Radius of a Nonmetal (Carbon)

***Sample Exercise 7.10***

## Trends in Radii

Predict the trend in radius for the following ions: $Be^{2+}$, $Mg^{2+}$, $Ca^{2+}$, and $Sr^{2+}$.

***Solution***

All these ions are formed by removing two electrons from an atom of a Group 2A element. In going from beryllium to strontium, we are going down the group, so the sizes increase:

$$Be^{2+} < Mg^{2+} < Ca^{2+} < Sr^{2+}$$

Smallest radius ↑ Largest radius

(spu) اتحاد سنافر البوليتكنك

***See Exercises 7.85, 7.86, and 7.89.***

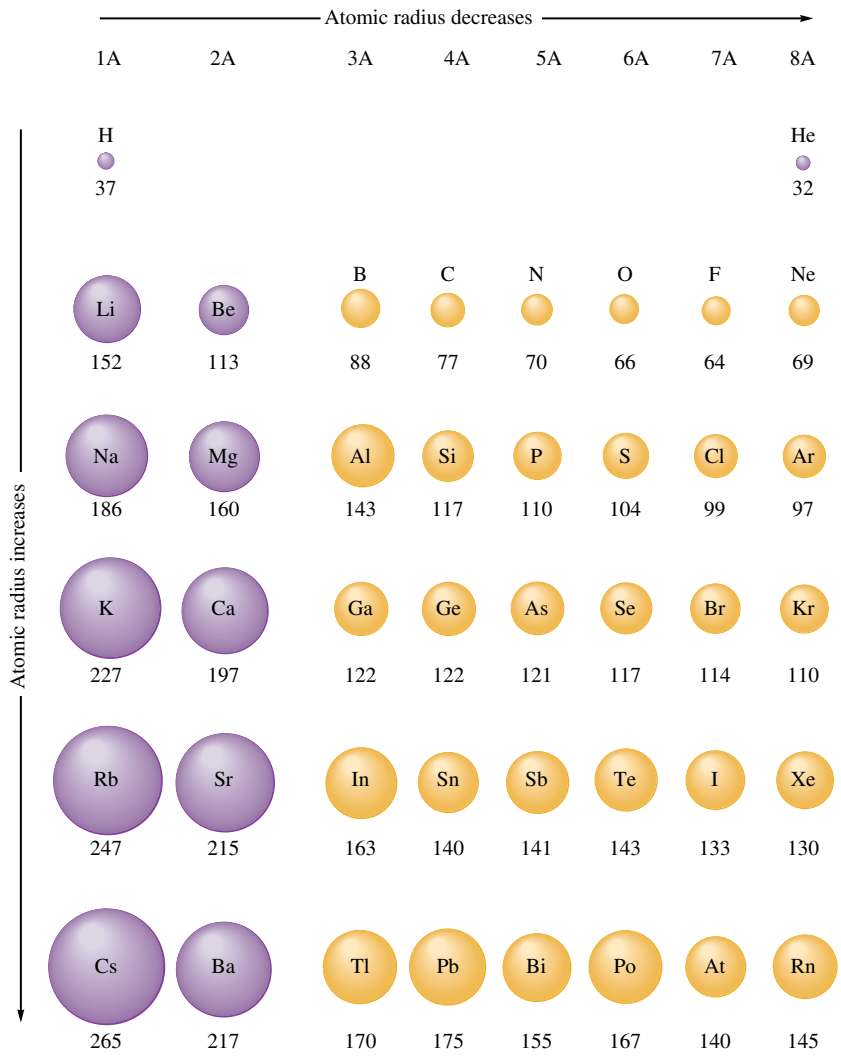



**FIGURE 7.34**
Atomic radii (in picometers) for selected atoms. Note that atomic radius decreases going across a period and increases going down a group. The values for the noble gases are estimated, because data from bonded atoms are lacking.

## 7.13 The Properties of a Group: The Alkali Metals

We have seen that the periodic table originated as a way to portray the systematic properties of the elements. Mendeleev was primarily responsible for first showing its usefulness in correlating and predicting the elemental properties. In this section we will summarize much of the information available from the table. We also will illustrate the usefulness of the table by discussing the properties of a representative group, the alkali metals.

### Information Contained in the Periodic Table

1. The essence of the periodic table is that the groups of representative elements exhibit similar chemical properties that change in a regular way. The quantum mechanical model of the atom has allowed us to understand the basis for the similarity of properties in a group—that each group member has the same valence electron configuration. *It is the number and type of valence electrons that primarily determine an atom's chemistry.*

اتحاد سنافر البولتيكنك (spu)

2. One of the most valuable types of information available from the periodic table is the electron configuration of any representative element. If you understand the organization

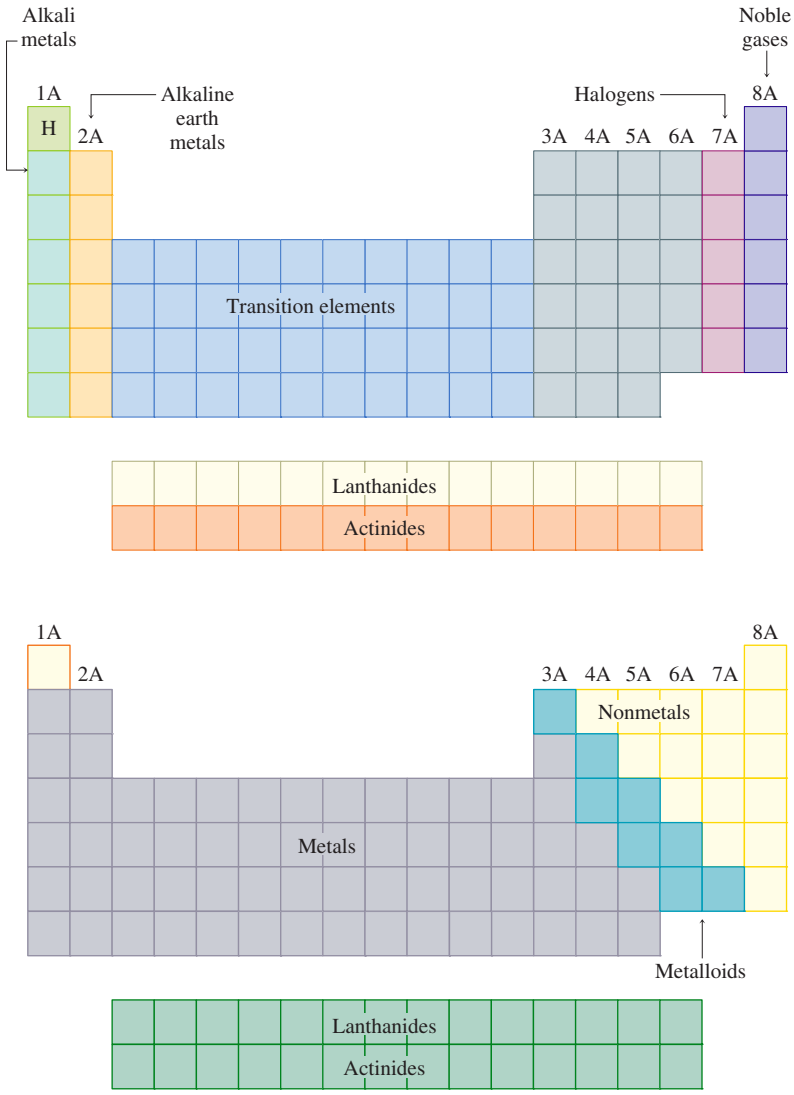



**FIGURE 7.35**
Special names for groups in the periodic table.

of the table, you will not need to memorize electron configurations for these elements. Although the predicted electron configurations for transition metals are sometimes incorrect, this is not a serious problem. You should, however, memorize the configurations of two exceptions, chromium and copper, since these 3*d* transition elements are found in many important compounds.

3. As we mentioned in Chapter 2, certain groups in the periodic table have special names. These are summarized in Fig. 7.35. Groups are often referred to by these names, so you should learn them.

Metals and nonmetals were first discussed in Chapter 2.

4. The most basic division of the elements in the periodic table is into metals and nonmetals. The most important chemical property of a metal atom is the tendency to give up one or more electrons to form a positive ion; metals tend to have low ionization energies. The metallic elements are found on the left side of the table, as shown in Fig. 7.35. The most chemically reactive metals are found on the lower left-hand portion of the table, where the ionization energies are smallest. The most distinctive chemical property of a nonmetal atom is the ability to gain one or more electrons to form an anion when reacting with a metal. Thus nonmetals are elements

اتحاد ستافر البوليكنك (spu)

with large ionization energies and the most negative electron affinities. The nonmetals are found on the right side of the table, with the most reactive ones in the upper right-hand corner, except for the noble gas elements, which are quite unreactive. The division into metals and nonmetals shown in Fig. 7.35 is only approximate. Many elements along the division line exhibit both metallic and nonmetallic properties under certain circumstances. These elements are often called **metalloids,** or sometimes **semimetals.**

## The Alkali Metals

The metals of Group 1A, the alkali metals, illustrate very well the relationships among the properties of the elements in a group. Lithium, sodium, potassium, rubidium, cesium, and francium are the most chemically reactive of the metals. We will not discuss francium here because it occurs in nature in only very small quantities. Although hydrogen is found in Group 1A of the periodic table, it behaves as a nonmetal, in contrast to the other members of that group. The fundamental reason for hydrogen's nonmetallic character is its very small size (see Fig. 7.34). The electron in the small 1*s* orbital is bound tightly to the nucleus.

Hydrogen will be discussed further in Chapter 19.

Some important properties of the first five alkali metals are shown in Table 7.8. The data in Table 7.8 show that in going down the group, the first ionization energy decreases and the atomic radius increases. This agrees with the general trends discussed in Section 7.12.

Other groups will be discussed in Chapters 19 and 20.

The overall increase in density in going down Group 1A is typical of all groups. This occurs because atomic mass generally increases more rapidly than atomic size. Thus there is more mass per unit volume for each succeeding element.

The smooth decrease in melting point and boiling point in going down Group 1A is not typical; in most other groups more complicated behavior occurs. Note that the melting point of cesium is only 29°C. Cesium can be melted readily using only the heat from your hand. This is very unusual—metals typically have rather high melting points. For example, tungsten melts at 3410°C. The only other metals with low melting points are mercury (mp −38°C) and gallium (mp 30°C).

Oxidation–reduction reactions were discussed in Chapter 4.

The chemical property most characteristic of a metal is the ability to lose its valence electrons. The Group 1A elements are very reactive. They have low ionization energies and react with nonmetals to form ionic solids. A typical example involves the reaction of sodium with chlorine to form sodium chloride:

$$2\text{Na}(s) + \text{Cl}_2(g) \longrightarrow 2\text{NaCl}(s)$$

where sodium chloride contains $Na^+$ and $Cl^-$ ions. This is an oxidation–reduction reaction in which chlorine oxidizes sodium. In the reactions between metals and nonmetals,

**TABLE 7.8 Properties of Five Alkali Metals**

| Element | Valence Electron Configuration | Density at 25°C ($g/cm^3$) | mp (°C) | bp (°C) | First Ionization Energy (kJ/mol) | Atomic (covalent) Radius (pm) | Ionic ($M^+$) Radius (pm) |
|---|---|---|---|---|---|---|---|
| Li | $2s^1$ | 0.53 | 180 | 1330 | 520 | 152 | 60 |
| Na | $3s^1$ | 0.97 | 98 | 892 | 495 | 186 | 95 |
| K | $4s^1$ | 0.86 | 64 | 760 | 419 | 227 | 133 |
| Rb | $5s^1$ | 1.53 | 39 | 668 | 409 | 247 | 148 |
| Cs | $6s^1$ | 1.87 | 29 | 690 | 382 | 265 | 169 |

اتحاد سنافر الجوليتكنك (spu)



## CHEMICAL IMPACT

### Potassium—Too Much of a Good Thing Can Kill You

Potassium is widely recognized as an essential element. In fact, our daily requirement for potassium is more than twice that for sodium. Because most foods contain potassium, serious deficiency of this element in humans is rare. However, potassium deficiency can be caused by kidney malfunction or by the use of certain diuretics. Potassium deficiency leads to muscle weakness, irregular heartbeat, and depression.

Potassium is found in the fluids of the body as the $K^+$ ion, and its presence is essential to the operation of our nervous system. The passage of impulses along the nerves requires the flow of $K^+$ (and $Na^+$) through channels in the membranes of the nerve cells. Failure of this ion flow prevents nerve transmissions and results in death. For example, the black mamba snake kills its victims by injecting a venom that blocks the potassium channels in the nerve cells.



The black mamba snake's venom kills by blocking the potassium channels in the nerve cells of victims.

Although a steady intake of potassium is essential to preserve life, ironically, too much potassium can be lethal. In fact, the deadly ingredient in the drug mixture used for executing criminals is potassium chloride. Injection of a large amount of a potassium chloride solution produces an excess of $K^+$ ion in the fluids surrounding the cells and prevents the essential flow of $K^+$ out the cells to allow nerve impulses to occur. This causes the heart to stop beating. Unlike other forms of execution, death by lethal injection of potassium chloride does not harm the organs of the body. Thus condemned criminals who are executed in this manner could potentially donate their organs for transplants. However, this idea is very controversial.



Potassium reacts violently with water.

it is typical for the nonmetal to behave as the oxidizing agent and the metal to behave as the reducing agent, as shown by the following reactions:

$$2Na(s) + S(s) \longrightarrow Na_2S(s)$$
Contains $Na^+$ and $S^{2-}$ ions

$$6Li(s) + N_2(g) \longrightarrow 2Li_3N(s)$$
Contains $Li^+$ and $N^{3-}$ ions

$$2Na(s) + O_2(g) \longrightarrow Na_2O_2(s)$$
Contains $Na^+$ and $O_2^{2-}$ ions

For reactions of the types just shown, the relative reducing powers of the alkali metals can be predicted from the first ionization energies listed in Table 7.8. Since it is much easier to remove an electron from a cesium atom than from a lithium atom, cesium should be the better reducing agent. The expected trend in reducing ability is

$$Cs > Rb > K > Na > Li$$

This order is observed experimentally for direct reactions between the solid alkali metals and nonmetals. However, this is not the order for reducing ability found when the alkali

اتجاد سنافر البوليتكنك (spu)

**TABLE 7.9 Hydration Energies for $Li^+$, $Na^+$, and $K^+$ Ions**

| Ion | Hydration Energy (kJ/mol) |
|---|---|
| $Li^+$ | −510 |
| $Na^+$ | −402 |
| $K^+$ | −314 |

metals react in aqueous solution. For example, the reduction of water by an alkali metal is very vigorous and exothermic:

$$2M(s) + 2H_2O(l) \longrightarrow H_2(g) + 2M^+(aq) + 2OH^-(aq) + \text{energy}$$

The order of reducing abilities observed for this reaction for the first three group members is

$$Li > K > Na$$

In the gas phase potassium loses an electron more easily than sodium, and sodium more easily than lithium. Thus it is surprising that lithium is the best reducing agent toward water.

This reversal occurs because the formation of the $M^+$ ions in aqueous solution is strongly influenced by the hydration of these ions by the polar water molecules. The *hydration energy* of an ion represents the change in energy that occurs when water molecules attach to the $M^+$ ion. The hydration energies for the $Li^+$, $Na^+$, and $K^+$ ions (shown in Table 7.9) indicate that the process is exothermic in each case. However, nearly twice as much energy is released by the hydration of the $Li^+$ ion as for the $K^+$ ion. This difference is caused by size effects; the $Li^+$ ion is much smaller than the $K^+$ ion, and thus its *charge density* (charge per unit volume) is also much greater. This means that the polar water molecules are more strongly attracted to the small $Li^+$ ion. Because the $Li^+$ ion is so strongly hydrated, its formation from the lithium atom occurs more readily than the formation of the $K^+$ ion from the potassium atom. Although a potassium atom in the gas phase loses its valence electron more readily than a lithium atom in the gas phase, the opposite is true in aqueous solution. This anomaly is an example of the importance of the polarity of the water molecule in aqueous reactions.

There is one more surprise involving the highly exothermic reactions of the alkali metals with water. Experiments show that in water lithium is the best reducing agent, so we might expect that lithium should react the most violently with water. However, this is not true. Sodium and potassium react much more vigorously. Why is this so? The answer lies in the relatively high melting point of lithium. When sodium and potassium react with water, the heat evolved causes them to melt, giving a larger area of contact with water. Lithium, on the other hand, does not melt under these conditions and reacts more slowly. This illustrates the important principle (which we will discuss in detail in Chapter 12) that the energy change for a reaction and the rate at which it occurs are not necessarily related.

In this section we have seen that the trends in atomic properties summarized by the periodic table can be a great help in understanding the chemical behavior of the elements. This fact will be emphasized over and over as we proceed in our study of chemistry.

## Key Terms

***Section 7.1***
electromagnetic radiation
wavelength
frequency

***Section 7.2***
Planck's constant
quantization
photon
photoelectric effect
$E = mc^2$
dual nature of light
diffraction
diffraction pattern

***Section 7.3***
continuous spectrum
line spectrum

## For Review

**Electromagnetic radiation**

- Characterized by its wavelength ($\lambda$), frequency ($\nu$), and speed ($c = 2.9979 \times 10^8$ m/s)

$$\lambda\nu = c$$

- Can be viewed as a stream of "particles" called photons, each with energy $h\nu$, where $h$ is Planck's constant ($6.626 \times 10^{-34}$ J · s)

**Photoelectric effect**

- When light strikes a metal surface, electrons are emitted
- Analysis of the kinetic energy and numbers of the emitted electrons led Einstein to suggest that electromagnetic radiation can be viewed as a stream of photons

اتحاد سنافر البوليتكنك (spu)

**Hydrogen spectrum**

- The emission spectrum of hydrogen shows discrete wavelengths
- Indicates that hydrogen has discrete energy levels

***Section 7.4***
quantum model
ground state

***Section 7.5***
standing wave
wave function
orbital
quantum (wave) mechanical model
Heisenberg uncertainty principle
probability distribution
radial probability distribution

***Section 7.6***
quantum numbers
principal quantum number ($n$)
angular momentum quantum number ($\ell$)
magnetic quantum number ($m_\ell$)
subshell

***Section 7.7***
nodal surface
node
degenerate orbital

***Section 7.8***
electron spin
electron spin quantum number
Pauli exclusion principle

***Section 7.9***
polyelectronic atoms

***Section 7.11***
aufbau principle
Hund's rule
valence electrons
core electrons
transition metals
lanthanide series
actinide series
main-group elements (representative elements)

***Section 7.12***
first ionization energy
second ionization energy
electron affinity
atomic radii

***Section 7.13***
metalloids (semimetals)

**Bohr model of the hydrogen atom**
- Using the data from the hydrogen spectrum and assuming angular momentum to be quantized, Bohr devised a model in which the electron traveled in circular orbits
- Although an important pioneering effort, this model proved to be entirely incorrect

**Wave (quantum) mechanical model**
- An electron is described as a standing wave
- The square of the wave function (often called an orbital) gives a probability distribution for the electron position
- The exact position of the electron is never known, which is consistent with the Heisenberg uncertainty principle: it is impossible to know accurately both the position and the momentum of a particle simultaneously
- Probability maps are used to define orbital shapes
- Orbitals are characterized by the quantum numbers $n$, $\ell$, and $m_\ell$

**Electron spin**
- Described by the spin quantum number $m_s$ which can have values of $\pm\frac{1}{2}$
- Pauli exclusion principle: no two electrons in a given atom can have the same set of quantum numbers $n$, $\ell$, $m_\ell$, and $m_s$
- Only two electrons with opposite spins can occupy a given orbital

**Periodic table**
- By populating the orbitals from the wave mechanical model (the aufbau principle), the form of the periodic table can be explained
- According to the wave mechanical model, atoms in a given group have the same valence (outer) electron configuration
- The trends in properties such as ionization energies and atomic radii can be explained in terms of the concepts of nuclear attraction, electron repulsions, shielding, and penetration

## REVIEW QUESTIONS

1. Four types of electromagnetic radiation (EMR) are ultraviolet, microwaves, gamma rays, and visible. All of these types of EMR can be characterized by wavelength, frequency, photon energy, and speed of travel. Define these terms and rank the four types of electromagnetic radiation in order of increasing wavelength, frequency, photon energy, and speed.
2. Characterize the Bohr model of the atom. In the Bohr model, what do we mean when we say something is quantized? How does the Bohr model of the hydrogen atom explain the hydrogen emission spectrum? Why is the Bohr model fundamentally incorrect?
3. What experimental evidence supports the quantum theory of light? Explain the wave-particle duality of all matter. For what size particles must one consider both the wave and the particle properties?
4. List the most important ideas of the quantum mechanical model of the atom. Include in your discussion the terms or names *wave function, orbital, Heisenberg uncertainty principle, de Broglie, Schrödinger,* and *probability distribution.*
5. What are quantum numbers? What information do we get from the quantum numbers $n$, $\ell$, and $m_\ell$? We define a spin quantum number ($m_s$), but do we know that an electron literally spins?
6. How do 2*s* orbitals differ from each other? How do 2*p* and 3*p* orbitals differ from each other? What is a nodal surface in an atomic orbital? What is wrong with 1*p*, 1*d*, 2*d*, 1*f*, 2*f*, and 3*f* orbitals? Explain what we mean when we say that a 4*s* electron is more penetrating than a 3*d* electron.

(spu) اتحاد سنافر البوليتكنك

7. Four blocks of elements in a periodic table refer to various atomic orbitals being filled. What are the four blocks and the corresponding orbitals? How do you get the energy ordering of the atomic orbitals from the periodic table? What is the aufbau principle? Hund's rule? The Pauli exclusion principle? There are two common exceptions to the ground-state electron configuration for elements 1–36 as predicted by the periodic table. What are they?
8. What is the difference between core electrons and valence electrons? Why do we emphasize the valence electrons in an atom when discussing atomic properties? What is the relationship between valence electrons and elements in the same group of the periodic table?
9. Using the element phosphorus as an example, write the equation for a process in which the energy change will correspond to the ionization energy and to the electron affinity.

    Explain why the first ionization energy tends to increase as one proceeds from left to right across a period. Why is the first ionization energy of aluminum lower than that of magnesium, and the first ionization energy of sulfur lower than that of phosphorus?

    Why do the successive ionization energies of an atom always increase? Note the successive ionization energies for silicon given in Table 7.5. Would you expect to see any large jumps between successive ionization energies of silicon as you removed all the electrons, one by one, beyond those shown in the table?
10. The radius trend and the ionization energy trend are exact opposites. Does this make sense? Define electron affinity. Electron affinity values are both exothermic (negative) and endothermic (positive). However, ionization energy values are always endothermic (positive). Explain.

## Active Learning Questions

These questions are designed to be used by groups of students in class. The questions allow students to explore their understanding of concepts through discussion and peer teaching. The real value of these questions is the learning that occurs while students talk to each other about chemical concepts.

1. What does it mean for something to have wavelike properties? Particulate properties? Electromagnetic radiation can be discussed in terms of both particles and waves. Explain the experimental verification for each of these views.
2. Defend and criticize Bohr's model. Why was it reasonable that such a model was proposed, and what evidence was there that it "works"? Why do we no longer "believe" in it?
3. The first four ionization energies for the elements *X* and *Y* are shown below. The units are not kJ/mol.

| | *X* | *Y* |
|---|---|---|
| First | 170 | 200 |
| Second | 350 | 400 |
| Third | 1800 | 3500 |
| Fourth | 2500 | 5000 |

Identify the elements *X* and *Y*. There may be more than one correct answer, so explain completely.

4. Compare the first ionization energy of helium to its second ionization energy, remembering that both electrons come from the 1*s* orbital. Explain the difference without using actual numbers from the text.
5. Which has the larger second ionization energy, lithium or beryllium? Why?
6. Explain why a graph of ionization energy versus atomic number (across a row) is not linear. Where are the exceptions? Why are there exceptions?
7. Without referring to your text, predict the trend of second ionization energies for the elements sodium through argon. Compare your answer with Table 7.5. Explain any differences.
8. Account for the fact that the line that separates the metals from the nonmetals on the periodic table is diagonal downward to the right instead of horizontal or vertical.
9. Explain *electron* from a quantum mechanical perspective, including a discussion of atomic radii, probabilities, and orbitals.
10. Choose the best response for the following. The ionization energy for the chlorine atom is equal in magnitude to the electron affinity for
    **a.** the Cl atom.
    **b.** the $Cl^-$ ion.
    **c.** the $Cl^+$ ion.
    **d.** the F atom.
    **e.** none of these.

(spu) البوليتكنك اتحاد سنافر

Explain each choice. Justify your choice, and for the choices you did not select, explain what is incorrect about them.

**11.** Consider the following statement: "The ionization energy for the potassium atom is negative, because when K loses an electron to become $K^+$, it achieves a noble gas electron configuration." Indicate everything that is correct in this statement. Indicate everything that is incorrect. Correct the incorrect information and explain.

**12.** In going across a row of the periodic table, electrons are added and ionization energy generally increases. In going down a column of the periodic table, electrons are also being added but ionization energy decreases. Explain.

**13.** How does probability fit into the description of the atom?

**14.** What is meant by an *orbital?*

A blue question or exercise number indicates that the answer to that question or exercise appears at the back of the book and a solution appears in the *Solutions Guide*.

## Questions

**15.** What type of relationship (direct or inverse) exists between wavelength, frequency, and photon energy? What does a photon energy unit of a Joule equal?

**16.** Explain the photoelectric effect.

**17.** How does the wavelength of a fast-pitched baseball compare to the wavelength of an electron traveling at 1/10 the speed of light? What is the significance of this comparison? See Sample Exercise 7.3.

**18.** The Bohr model only works for one electron species. Why do we discuss it in this text (what's good about it)?

**19.** Describe the significance of the radial probability distribution shown in Fig. 7.12(b).

**20.** The periodic table consists of four blocks of elements which correspond to *s*, *p*, *d*, and *f* orbitals being filled. After *f* orbitals come *g* and *h* orbitals. In theory, if a *g* block and an *h* block of elements existed, how long would the rows of *g* and *h* elements be in this theoretical periodic table?

**21.** Many times the claim is made that subshells half-filled with electrons are particularly stable. Can you suggest a possible physical basis for this claim?

**22.** Diagonal relationships in the periodic table exist as well as the vertical relationships. For example, Be and Al are similar in some of their properties, as are B and Si. Rationalize why these diagonal relationships hold for properties such as size, ionization energy, and electron affinity.

**23.** Elements with very large ionization energies also tend to have highly exothermic electron affinities. Explain. Which group of elements would you expect to be an exception to this statement?

**24.** The changes in electron affinity as one goes down a group in the periodic table are not nearly as large as the variations in ionization energies. Why?

**25.** Why is it much harder to explain the line spectra of polyelectronic atoms and ions than it is to explain the line spectra of hydrogen and hydrogenlike ions?

**26.** Scientists use emission spectra to confirm the presence of an element in materials of unknown composition. Why is this possible?

**27.** Does the minimization of electron–electron repulsions correlate with Hund's rule?

**28.** In the hydrogen atom, what is the physical significance of the state for which $n = \infty$ and $E = 0$?

**29.** The work function is the energy required to remove an electron from an atom on the surface of a metal. How does this definition differ from that for ionization energy?

**30.** Many more anhydrous lithium salts are hygroscopic (readily absorb water) than are those of the other alkali metals. Explain.

## Exercises

In this section similar exercises are paired.

### Light and Matter

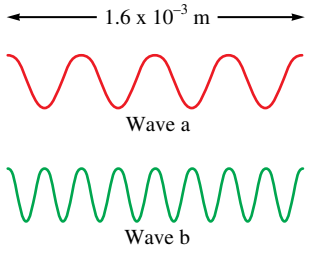
**31.** Photosynthesis uses 660-nm light to convert $CO_2$ and $H_2O$ into glucose and $O_2$. Calculate the frequency of this light.

**32.** An FM radio station broadcasts at 99.5 MHz. Calculate the wavelength of the corresponding radio waves.

**33.** Microwave radiation has a wavelength on the order of 1.0 cm. Calculate the frequency and the energy of a single photon of this radiation. Calculate the energy of an Avogadro's number of photons (called an *einstein*) of this radiation.

**34.** A photon of ultraviolet (UV) light possesses enough energy to mutate a strand of human DNA. What is the energy of a single UV photon and a mole of UV photons having a wavelength of 25 nm?

**35.** Consider the following waves representing electromagnetic radiation:




Which wave has the longer wavelength? Calculate the wavelength. Which wave has the higher frequency and larger photon energy? Calculate these values. Which wave has the faster velocity? What type of electromagnetic radiation are illustrated?

**36.** One type of electromagnetic radiation has a frequency of 107.1 MHz, another type has a wavelength of $2.12 \times 10^{-10}$ m, and another type of electromagnetic radiation has photons with energy equal to $3.97 \times 10^{-19}$ J/photon. Identify each type of electromagnetic radiation and place them in order of increasing photon energy and increasing frequency.

**37.** A carbon sample absorbs energy at a wavelength of 150. nm. The total amount of energy emitted by a carbon sample is $1.98 \times 10^5$ J. Calculate the number of carbon atoms present in the sample, assuming that each atom emits one photon.

اتحاد نافر البوليتكنك (spu)

**38.** A carbon–oxygen double bond in a certain organic molecule absorbs radiation that has a frequency of $6.0 \times 10^{13}\ s^{-1}$.
- **a.** What is the wavelength of this radiation?
- **b.** To what region of the spectrum does this radiation belong?
- **c.** What is the energy of this radiation per photon? Per mole of photons?
- **d.** A carbon–oxygen bond in a different molecule absorbs radiation with frequency equal to $5.4 \times 10^{13}\ s^{-1}$. Is this radiation more or less energetic?

**39.** The work function of an element is the energy required to remove an electron from the surface of the solid element. The work function for lithium is 279.7 kJ/mol (that is, it takes 279.7 kJ of energy to remove one mole of electrons from one mole of Li atoms on the surface of Li metal). What is the maximum wavelength of light that can remove an electron from an atom on the surface of lithium metal?

**40.** It takes 208.4 kJ of energy to remove 1 mole of electrons from an atom on the surface of rubidium metal. How much energy does it take to remove a single electron from an atom on the surface of solid rubidium? What is the maximum wavelength of light capable of doing this?

**41.** Calculate the de Broglie wavelength for each of the following.
- **a.** an electron with a velocity 10.% of the speed of light
- **b.** a tennis ball (55 g) served at 35 m/s (~80 mi/h)

**42.** Neutron diffraction is used in determining the structures of molecules.
- **a.** Calculate the de Broglie wavelength of a neutron moving at 1.00% of the speed of light.
- **b.** Calculate the velocity of a neutron with a wavelength of 75 pm (1 pm = $10^{-12}$ m).

**43.** A particle has a velocity that is 90.% of the speed of light. If the wavelength of the particle is $1.5 \times 10^{-15}$ m, calculate the mass of the particle.

**44.** Calculate the velocities of electrons with de Broglie wavelengths of $1.0 \times 10^{2}$ nm and 1.0 nm, respectively.

## Hydrogen Atom: The Bohr Model

**45.** Calculate the wavelength of light emitted when each of the following transitions occur in the hydrogen atom. What type of electromagnetic radiation is emitted in each transition?
- **a.** $n = 3 \rightarrow n = 2$
- **b.** $n = 4 \rightarrow n = 2$
- **c.** $n = 2 \rightarrow n = 1$

**46.** Calculate the wavelength of light emitted when each of the following transitions occur in the hydrogen atom. What type of electromagnetic radiation is emitted in each transition?
- **a.** $n = 4 \rightarrow n = 3$
- **b.** $n = 5 \rightarrow n = 4$
- **c.** $n = 5 \rightarrow n = 3$

**47.** Using vertical lines, indicate the transitions from Exercise 45 on an energy-level diagram for the hydrogen atom (see Fig. 7.8).

**48.** Using vertical lines, indicate the transitions from Exercise 46 on an energy-level diagram for the hydrogen atom (see Fig. 7.8).

**49.** Does a photon of visible light ($\lambda \approx 400$ to 700 nm) have sufficient energy to excite an electron in a hydrogen atom from the $n = 1$ to the $n = 5$ energy state? from the $n = 2$ to the $n = 6$ energy state?

**50.** An electron is excited from the $n = 1$ ground state to the $n = 3$ state in a hydrogen atom. Which of the following statements are true? Correct the false statements to make them true.
- **a.** It takes more energy to ionize (completely remove) the electron from $n = 3$ than from the ground state.
- **b.** The electron is farther from the nucleus on average in the $n = 3$ state than in the $n = 1$ state.
- **c.** The wavelength of light emitted if the electron drops from $n = 3$ to $n = 2$ will be shorter than the wavelength of light emitted if the electron falls from $n = 3$ to $n = 1$.
- **d.** The wavelength of light emitted when the electron returns to the ground state from $n = 3$ will be the same as the wavelength of light absorbed to go from $n = 1$ to $n = 3$.
- **e.** For $n = 3$, the electron is in the first excited state.

**51.** Calculate the maximum wavelength of light capable of removing an electron for a hydrogen atom from the energy state characterized by $n = 1$. by $n = 2$.

**52.** Consider an electron for a hydrogen atom in an excited state. The maximum wavelength of electromagnetic radiation that can completely remove (ionize) the electron from the H atom is 1460 nm. What is the initial excited state for the electron ($n = ?$)?

**53.** An excited hydrogen atom with an electron in the $n = 5$ state emits light having a frequency of $6.90 \times 10^{14}\ s^{-1}$. Determine the principal quantum level for the final state in this electronic transition.

**54.** An excited hydrogen atom emits light with a wavelength of 397.2 nm to reach the energy level for which $n = 2$. In which principal quantum level did the electron begin?

## Quantum Mechanics, Quantum Numbers, and Orbitals

**55.** Using the Heisenberg uncertainty principle, calculate $\Delta x$ for each of the following.
- **a.** an electron with $\Delta v = 0.100$ m/s
- **b.** a baseball (mass = 145 g) with $\Delta v = 0.100$ m/s
- **c.** How does the answer in part a compare with the size of a hydrogen atom?
- **d.** How does the answer in part b correspond to the size of a baseball?

**56.** The Heisenberg uncertainty principle can be expressed in the form

$$\Delta E \cdot \Delta t \geq \frac{h}{4\pi}$$

where $E$ represents energy and $t$ represents time. Show that the units for this form are the same as the units for the form used in this chapter:

$$\Delta x \cdot \Delta(mv) \geq \frac{h}{4\pi}$$

(sptu) اتحاد سنافر البوليتكنك

**57.** What are the possible values for the quantum numbers $n$, $\ell$, and $m_\ell$?

**58.** Which of the following orbital designations are incorrect: 1*s*, 1*p*, 7*d*, 9*s*, 3*f*, 4*f*, 2*d*?

**59.** Which of the following sets of quantum numbers are not allowed in the hydrogen atom? For the sets of quantum numbers that are incorrect, state what is wrong in each set.
**a.** $n = 3, \ell = 2, m_\ell = 2$
**b.** $n = 4, \ell = 3, m_\ell = 4$
**c.** $n = 0, \ell = 0, m_\ell = 0$
**d.** $n = 2, \ell = -1, m_\ell = 1$

**60.** Which of the following sets of quantum numbers are not allowed? For each incorrect set, state why it is incorrect.
**a.** $n = 3, \ell = 3, m_\ell = 0, m_s = -\frac{1}{2}$
**b.** $n = 4, \ell = 3, m_\ell = 2, m_s = -\frac{1}{2}$
**c.** $n = 4, \ell = 1, m_\ell = 1, m_s = +\frac{1}{2}$
**d.** $n = 2, \ell = 1, m_\ell = -1, m_s = -1$
**e.** $n = 5, \ell = -4, m_\ell = 2, m_s = +\frac{1}{2}$
**f.** $n = 3, \ell = 1, m_\ell = 2, m_s = -\frac{1}{2}$

**61.** What is the physical significance of the value of $\psi^2$ at a particular point in an atomic orbital?

**62.** In defining the sizes of orbitals, why must we use an arbitrary value, such as 90% of the probability of finding an electron in that region?

## Polyelectronic Atoms

**63.** How many orbitals in an atom can have the designation 5*p*, $3d_{z^2}$, 4*d*, $n = 5$, $n = 4$?

**64.** How many electrons in an atom can have the designation 1*p*, $6d_{x^2-y^2}$, 4*f*, $7p_y$, 2*s*, $n = 3$?

**65.** Give the maximum number of electrons in an atom that can have these quantum numbers:
**a.** $n = 4$
**b.** $n = 5, m_\ell = +1$
**c.** $n = 5, m_s = +\frac{1}{2}$
**d.** $n = 3, \ell = 2$
**e.** $n = 2, \ell = 1$

**66.** Give the maximum number of electrons in an atom that can have these quantum numbers:
**a.** $n = 0, \ell = 0, m_\ell = 0$
**b.** $n = 2, \ell = 1, m_\ell = -1, m_s = -\frac{1}{2}$
**c.** $n = 3, m_s = +\frac{1}{2}$
**d.** $n = 2, \ell = 2$
**e.** $n = 1, \ell = 0, m_\ell = 0$

**67.** Draw atomic orbital diagrams representing the ground-state electron configuration for each of the following elements.
**a.** Na
**b.** Co
**c.** Kr
How many unpaired electrons are present in each element?

**68.** For elements 1–36, there are two exceptions to the filling of orbitals as predicted from the periodic table. Draw the atomic orbital diagrams for the two exceptions and indicate how many unpaired electrons are present.

**69.** The elements Si, Ga, As, Ge, Al, Cd, S, and Se are all used in the manufacture of various semiconductor devices. Write the expected electron configuration for these atoms.

**70.** The elements Cu, O, La, Y, Ba, Tl, and Bi are all found in high-temperature ceramic superconductors. Write the expected electron configuration for these atoms.

**71.** Write the expected electron configurations for each of the following atoms: Sc, Fe, P, Cs, Eu, Pt, Xe, Br.

**72.** Write the expected electron configurations for each of the following atoms: Cl, Sb, Sr, W, Pb, Cf.

**73.** Write the expected ground-state electron configuration for the following.
**a.** the element with one unpaired 5*p* electron that forms a covalent with compound fluorine
**b.** the (as yet undiscovered) alkaline earth metal after radium
**c.** the noble gas with electrons occupying 4*f* orbitals
**d.** the first-row transition metal with the most unpaired electrons

**74.** Using only the periodic table inside the front cover of the text, write the expected ground-state electron configurations for
**a.** the third element in Group 5A.
**b.** element number 116.
**c.** an element with three unpaired 5*d* electrons.
**d.** the halogen with electrons in the 6*p* atomic orbitals.

**75.** In the ground state of mercury, Hg,
**a.** how many electrons occupy atomic orbitals with $n = 3$?
**b.** how many electrons occupy *d* atomic orbitals?
**c.** how many electrons occupy $p_z$ atomic orbitals?
**d.** how many electrons have spin "up" ( $m_s = +\frac{1}{2}$ )?

**76.** In the ground state of element 115, Uup,
**a.** how many electrons have $n = 5$ as one of their quantum numbers?
**b.** how many electrons have $\ell = 3$ as one of their quantum numbers?
**c.** how many electrons have $m_\ell = 1$ as one of their quantum numbers?
**d.** how many electrons have $m_s = -\frac{1}{2}$ as one of their quantum numbers?

**77.** Give a possible set of values of the four quantum numbers for all the electrons in a boron atom and a nitrogen atom if each is in the ground state.

**78.** Give a possible set of values of the four quantum numbers for the 4*s* and 3*d* electrons in titanium.

**79.** A certain oxygen atom has the electron configuration $1s^2 2s^2 2p_x^2 2p_y^2$. How many unpaired electrons are present? Is this an excited state of oxygen? In going from this state to the ground state would energy be released or absorbed?

**80.** Which of the following electron configurations correspond to an excited state? Identify the atoms and write the ground-state electron configuration where appropriate.
**a.** $1s^2 2s^2 3p^1$
**b.** $1s^2 2s^2 2p^6$
**c.** $1s^2 2s^2 2p^4 3s^1$
**d.** $[\mathrm{Ar}]4s^2 3d^5 4p^1$
How many unpaired electrons are present in each of these species?

(spu) اتحاد سنافر البوليتكنك

**81.** Which of elements 1–36 have two unpaired electrons in the ground state?

**82.** Which of elements 1–36 have one unpaired electron in the ground state?

**83.** One bit of evidence that the quantum mechanical model is "correct" lies in the magnetic properties of matter. Atoms with unpaired electrons are attracted by magnetic fields and thus are said to exhibit *paramagnetism*. The degree to which this effect is observed is directly related to the number of unpaired electrons present in the atom. Consider the ground-state electron configurations for Li, N, Ni, Te, Ba, and Hg. Which of these atoms would be expected to be paramagnetic, and how many unpaired electrons are present in each paramagnetic atom?

**84.** How many unpaired electrons are present in each of the following in the ground state: O, $O^+$, $O^-$, Os, Zr, S, F, Ar?

## The Periodic Table and Periodic Properties

**85.** Arrange the following groups of atoms in order of increasing size.
- **a.** Te, S, Se
- **b.** K, Br, Ni
- **c.** Ba, Si, F

**86.** Arrange the following groups of atoms in order of increasing size.
- **a.** Rb, Na, Be
- **b.** Sr, Se, Ne
- **c.** Fe, P, O

**87.** Arrange the atoms in Exercise 85 in order of increasing first ionization energy.

**88.** Arrange the atoms in Exercise 86 in order of increasing first ionization energy.

**89.** In each of the following sets, which atom or ion has the smallest radius?
- **a.** H, He
- **b.** Cl, In, Se
- **c.** element 120, element 119, element 117
- **d.** Nb, Zn, Si
- **e.** $Na^-$, Na, $Na^+$

**90.** In each of the following sets, which atom or ion has the smallest ionization energy?
- **a.** Ca, Sr, Ba
- **b.** K, Mn, Ga
- **c.** N, O, F
- **d.** $S^{2-}$, S, $S^{2+}$
- **e.** Cs, Ge, Ar

**91.** Element 106 has been named seaborgium, Sg, in honor of Glenn Seaborg, discoverer of the first transuranium element.
- **a.** Write the expected electron configuration for element 106.
- **b.** What other element would be most like element 106 in its properties?
- **c.** Write the formula for a possible oxide and a possible oxyanion of element 106.

**92.** Predict some of the properties of element 117 (the symbol is Uus, following conventions proposed by the International Union of Pure and Applied Chemistry, or IUPAC).
- **a.** What will be its electron configuration?
- **b.** What element will it most resemble chemically?
- **c.** What will be the formula of the neutral binary compounds it forms with sodium, magnesium, carbon, and oxygen?
- **d.** What oxyanions would you expect Uus to form?

**93.** The first ionization energies of As and Se are 0.947 and 0.941 MJ/mol, respectively. Rationalize these values in terms of electron configurations.

**94.** Rank the elements Be, B, C, N, and O in order of increasing first ionization energy. Explain your reasoning.

**95.** For each of the following pairs of elements

(C and N) (Ar and Br)

pick the atom with
- **a.** more favorable (exothermic) electron affinity.
- **b.** higher ionization energy.
- **c.** larger size.

**96.** For each of the following pairs of elements

(Mg and K) (F and Cl)

pick the atom with
- **a.** more favorable (exothermic) electron affinity.
- **b.** higher ionization energy.
- **c.** larger size.

**97.** The electron affinities of the elements from aluminum to chlorine are −44, −120, −74, −200.4, and −384.7 kJ/mol, respectively. Rationalize the trend in these values.

**98.** The electron affinity for sulfur is more exothermic than that for oxygen. How do you account for this?

**99.** Order each of the following sets from the least exothermic electron affinity to the most exothermic electron affinity.
- **a.** F, Cl, Br, I
- **b.** N, O, F

**100.** Which has the more negative electron affinity, the oxygen atom or the $O^-$ ion? Explain your answer.

**101.** Write equations corresponding to the following.
- **a.** The fourth ionization energy of Se
- **b.** The electron affinity of $S^-$
- **c.** The electron affinity of $Fe^{3+}$
- **d.** The ionization energy of Mg

**102.** Using data from the text, determine the following values (justify your answer):
- **a.** the electron affinity of $Mg^{2+}$
- **b.** the ionization energy of $Cl^-$
- **c.** the electron affinity of $Cl^+$
- **d.** the ionization energy of $Mg^-$ (Electron affinity of Mg = 230 kJ/mol)

## Alkali Metals

**103.** An ionic compound of potassium and oxygen has the empirical formula KO. Would you expect this compound to be potassium(II) oxide or potassium peroxide? Explain.

**104.** Give the name and formula of each of the binary compounds formed from the following elements.
- **a.** Li and N
- **b.** Na and Br
- **c.** K and S

(spu) البوليتكنك اتحاد سنافر

**105.** Cesium was discovered in natural mineral waters in 1860 by R. W. Bunsen and G. R. Kirchhoff using the spectroscope they invented in 1859. The name came from the Latin *caesius* ("sky blue") because of the prominent blue line observed for this element at 455.5 nm. Calculate the frequency and energy of a photon of this light.

**106.** The bright yellow light emitted by a sodium vapor lamp consists of two emission lines at 589.0 and 589.6 nm. What are the frequency and the energy of a photon of light at each of these wavelengths? What are the energies in kJ/mol?

**107.** Does the information on alkali metals in Table 7.8 of the text confirm the general periodic trends in ionization energy and atomic radius? Explain.

**108.** Predict the atomic number of the next alkali metal after francium and give its ground-state electron configuration.

**109.** Complete and balance the equations for the following reactions.
**a.** $Li(s) + N_2(g) \rightarrow$
**b.** $Rb(s) + S(s) \rightarrow$

**110.** Complete and balance the equations for the following reactions.
**a.** $Cs(s) + H_2O(l) \rightarrow$
**b.** $Na(s) + Cl_2(g) \rightarrow$

## Additional Exercises

**111.** Photogray lenses incorporate small amounts of silver chloride in the glass of the lens. When light hits the AgCl particles, the following reaction occurs:

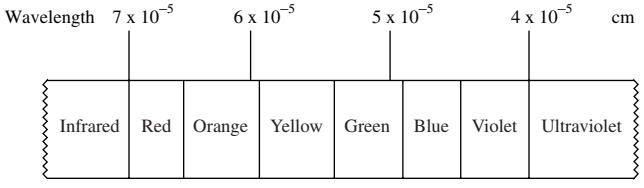
$$AgCl \xrightarrow{h\nu} Ag + Cl$$

The silver metal that is formed causes the lenses to darken. The enthalpy change for this reaction is $3.10 \times 10^2$ kJ/mol. Assuming all this energy must be supplied by light, what is the maximum wavelength of light that can cause this reaction?

**112.** A certain microwave oven delivers 750. watts (joule/s) of power to a coffee cup containing 50.0 g of water at 25.0°C. If the wavelength of microwaves in the oven is 9.75 cm, how long does it take, and how many photons must be absorbed, to make the water boil? The specific heat capacity of water is 4.18 J/°C · g and assume only the water absorbs the energy of the microwaves.

**113.** Mars is roughly 60 million km from earth. How long does it take for a radio signal originating from earth to reach Mars?

**114.** Consider the following approximate visible light spectrum:




Barium emits light in the visible region of the spectrum. If each photon of light emitted from barium has energy $3.29 \times 10^{-19}$ J, what color of visible light is emitted?

**115.** One of the visible lines in the hydrogen emission spectrum corresponds to the $n = 6$ to $n = 2$ electronic transition. What color light is this transition? See Exercise 114.

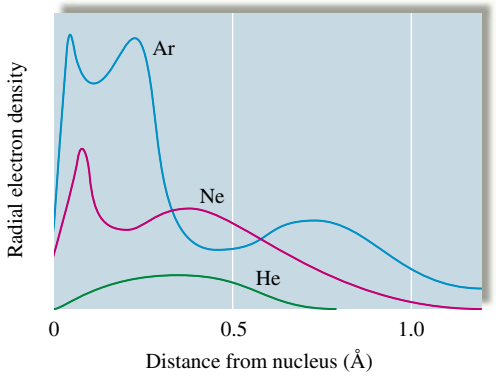
**116.** Using Fig. 7.28, list the elements (ignore the lanthanides and actinides) that have ground-state electron configurations that differ from those we would expect from their positions in the periodic table.

**117.** Are the following statements true for the hydrogen atom only, true for all atoms, or not true for any atoms?
**a.** The principal quantum number completely determines the energy of a given electron.
**b.** The angular momentum quantum number, $\ell$, determines the shapes of the atomic orbitals.
**c.** The magnetic quantum number, $m_\ell$, determines the direction that the atomic orbitals point in space.

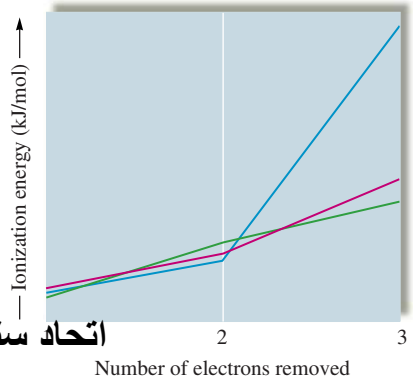
**118.** Although no currently known elements contain electrons in *g* orbitals in the ground state, it is possible that these elements will be found or that electrons in excited states of known elements could be in *g* orbitals. For *g* orbitals, the value of $\ell$ is 4. What is the lowest value of *n* for which *g* orbitals could exist? What are the possible values of $m_\ell$? How many electrons could a set of *g* orbitals hold?

**119.** Consider the representations of the *p* and *d* atomic orbitals in Figs. 7.14 and 7.16. What do the + and − signs indicate?

**120.** Total radial probability distributions for the helium, neon, and argon atoms are shown in the following graph. How can one interpret the shapes of these curves in terms of electron configurations, quantum numbers, and nuclear charges?




**121.** The following graph plots the first, second, and third ionization energies for Mg, Al, and Si.




Without referencing the text, which plot corresponds to which element? In one of the plots, there is a huge jump in energy

(spu) اتحاد سنافر البوليتكنك

between $I_2$ and $I_3$, unlike in the other two plots. Explain this phenomenon.

**122.** An ion having a 4+ charge and a mass of 49.9 amu has 2 electrons with principal quantum number $n = 1$, 8 electrons with $n = 2$, and 10 electrons with $n = 3$. Supply as many of the properties for the ion as possible from the information given. *Hint:* In forming ions for this species, the 4*s* electrons are lost before the 3*d* electrons.
- **a.** the atomic number
- **b.** total number of *s* electrons
- **c.** total number of *p* electrons
- **d.** total number of *d* electrons
- **e.** the number of neutrons in the nucleus
- **f.** the ground-state electron configuration of the neutral atom

**123.** The successive ionization energies for an unknown element are
$I_1 = 896$ kJ/mol
$I_2 = 1752$ kJ/mol
$I_3 = 14{,}807$ kJ/mol
$I_4 = 17{,}948$ kJ/mol
To which family in the periodic table does the unknown element most likely belong?

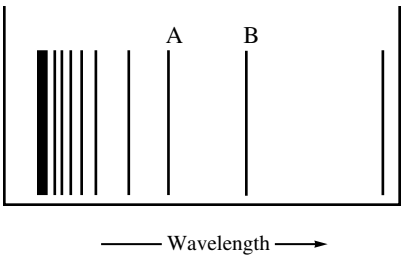
**124.** An unknown element is a nonmetal and has a valence electron configuration of $ns^2np^4$.
- **a.** How many valence electrons does this element have?
- **b.** What are some possible identities for this element?
- **c.** What is the formula of the compound this element would form with potassium?
- **d.** Would this element have a larger or smaller radius than barium?
- **e.** Would this element have a greater or smaller ionization energy than fluorine?

**125.** Using data from this chapter, calculate the change in energy expected for each of the following processes.
- **a.** $Na(g) + Cl(g) \rightarrow Na^+(g) + Cl^-(g)$
- **b.** $Mg(g) + F(g) \rightarrow Mg^+(g) + F^-(g)$
- **c.** $Mg^+(g) + F(g) \rightarrow Mg^{2+}(g) + F^-(g)$
- **d.** $Mg(g) + 2F(g) \rightarrow Mg^{2+}(g) + 2F^-(g)$

## Challenge Problems

**126.** One of the emission spectral lines for $Be^{3+}$ has a wavelength of 253.4 nm for an electronic transition that begins in the state with $n = 5$. What is the principal quantum number of the lower-energy state corresponding to this emission? (*Hint:* The Bohr model can be applied to one-electron ions. Don't forget the $Z$ factor: $Z$ = nuclear charge = atomic number.)

**127.** The figure below represents part of the emission spectrum for a one-electron ion in the gas phase. All the lines result from electronic transitions from excited states to the $n = 3$ state. (See Exercise 126.)




- **a.** What electronic transitions correspond to lines *A* and *B*?
- **b.** If the wavelength of line *B* is 142.5 nm, calculate the wavelength of line *A*.

**128.** When the excited electron in a hydrogen atom falls from $n = 5$ to $n = 2$, a photon of blue light is emitted. if an excited electron in $He^+$ falls from $n = 4$, to which energy level must it fall so that a similar blue light (as with the hydrogen) is emitted? Prove it. (See Exercise 126.)

**129.** The wave function for the $2p_z$ orbital in the hydrogen atom is

$$\psi_{2p_z} = \frac{1}{4\sqrt{2\pi}}\left(\frac{Z}{a_0}\right)^{3/2} \sigma e^{-\sigma/2} \cos\theta$$

where $a_0$ is the value for the radius of the first Bohr orbit in meters ($5.29 \times 10^{-11}$), $\sigma$ is $Z(r/a_0)$, $r$ is the value for the distance from the nucleus in meters, and $\theta$ is an angle. Calculate the value of ${\psi_{2p_z}}^2$ at $r = a_0$ for $\theta = 0$ ($z$ axis) and for $\theta = 90°$ ($xy$ plane).

**130.** Answer the following questions assuming that $m_s$ could have three values rather than two and that the rules for $n$, $\ell$, and $m_\ell$ are the normal ones.
- **a.** How many electrons would an orbital be able to hold?
- **b.** How many elements would the first and second periods in the periodic table contain?
- **c.** How many elements would be contained in the first transition metal series?
- **d.** How many electrons would the set of 4*f* orbitals be able to hold?

**131.** Assume that we are in another universe with different physical laws. Electrons in this universe are described by four quantum numbers with meanings similar to those we use. We will call these quantum numbers $p$, $q$, $r$, and $s$. The rules for these quantum numbers are as follows:

$p = 1, 2, 3, 4, 5, \ldots$.
$q$ takes on positive odd integer and $q \leq p$.
$r$ takes on all even integer values from $-q$ to $+q$. (Zero is considered an even number.)
$s = +\frac{1}{2}$ or $-\frac{1}{2}$

- **a.** Sketch what the first four periods of the periodic table will look like in this universe.
- **b.** What are the atomic numbers of the first four elements you would expect to be least reactive?
- **c.** Give an example, using elements in the first four rows, of ionic compounds with the formulas XY, $XY_2$, $X_2Y$, $XY_3$, and $X_2Y_3$.
- **d.** How many electrons can have $p = 4$, $q = 3$?
- **e.** How many electrons can have $p = 3$, $q = 0$, $r = 0$?
- **f.** How many electrons can have $p = 6$?

**132.** Without looking at data in the text, sketch a qualitative graph of the third ionization energy versus atomic number for the elements Na through Ar, and explain your graph.

**133.** The following numbers are the ratios of second ionization energy to first ionization energy:

Na: 9.9
Mg: 2.0
Al: 3.1
Si: 2.0

اتحاد صنافو البوليتكنك (spu)

P: 1.8
S: 2.3
Cl: 1.8
Ar: 1.8

Explain these relative numbers.

**134.** We expect the atomic radius to increase going down a group in the periodic table. Can you suggest why the atomic radius of hafnium breaks this rule? (See data below.)

| Atomic Radii, in pm | | | |
|---|---|---|---|
| Sc | 157 | Ti | 147.7 |
| Y | 169.3 | Zr | 159.3 |
| La | 191.5 | Hf | 147.6 |

**135.** Consider the following ionization energies for aluminum:

$$Al(g) \longrightarrow Al^{+}(g) + e^{-} \qquad I_1 = 580 \text{ kJ/mol}$$
$$Al^{+}(g) \longrightarrow Al^{2+}(g) + e^{-} \qquad I_2 = 1815 \text{ kJ/mol}$$
$$Al^{2+}(g) \longrightarrow Al^{3+}(g) + e^{-} \qquad I_3 = 2740 \text{ kJ/mol}$$
$$Al^{3+}(g) \longrightarrow Al^{4+}(g) + e^{-} \qquad I_4 = 11{,}600 \text{ kJ/mol}$$

**a.** Account for the trend in the values of the ionization energies.
**b.** Explain the large increase between $I_3$ and $I_4$.
**c.** Which one of the four ions has the greatest electron affinity? Explain.
**d.** List the four aluminum ions given in order of increasing size, and explain your ordering. (*Hint:* Remember that most of the size of an atom or ion is due to its electrons.)

**136.** While Mendeleev predicted the existence of several undiscovered elements, he did not predict the existence of the noble gases, the lanthanides, or the actinides. Propose reasons why Mendeleev was not able to predict the existence of the noble gases.

**137.** An atom of a particular element is traveling at 1.00% of the speed of light. The de Broglie wavelength is found to be $3.31 \times 10^{-3}$ pm. Which element is this? Prove it.

## Integrative Problems

These problems require the integration of multiple concepts to find the solutions.

**138.** As the weapons officer aboard the *Starship Chemistry,* it is your duty to configure a photon torpedo to remove an electron from the outer hull of an enemy vessel. You know that the work function (the binding energy of the electron) of the hull of the enemy ship is $7.52 \times 10^{-19}$ J.
**a.** What wavelength does your photon torpedo need to be to eject an electron?
**b.** You find an extra photon torpedo with a wavelength of 259 nm and fire it at the enemy vessel. Does this photon torpedo do any damage to the ship (does it eject an electron)?
**c.** If the hull of the enemy vessel is made of a metal with an electron configuration of $[Ar]4s^1 3d^{10}$, what metal is this?

**139.** Francium, Fr, is a radioactive element found in some uranium minerals and is formed as a result of the decay of actinium.
**a.** What are the electron configurations of francium and its predicted most common ion?
**b.** It has been estimated that at any one time, there is only one (1.0) ounce of francium on earth. Assuming this is true, what number of francium atoms exist on earth?
**c.** The longest-lived isotope of francium is $^{223}$Fr. What is the total mass in grams of the neutrons in one atom of this isotope?

**140.** Answer the following questions based on the given electron configurations and identify the elements.
**a.** Arrange these atoms in order of increasing size: $[Kr]5s^2 4d^{10} 5p^6$; $[Kr]5s^2 4d^{10} 5p^1$; $[Kr]5s^2 4d^{10} 5p^3$.
**b.** Arrange these atoms in order of decreasing first ionization energy: $[Ne]3s^2 3p^5$; $[Ar]4s^2 3d^{10} 4p^3$; $[Ar]4s^2 3d^{10} 4p^5$.

## Marathon Problem*

This problem is designed to incorporate several concepts and techniques into one situation. Marathon Problems can be used in class by groups of students to help facilitate problem-solving skills.

**141.** From the information below, identify element X.
**a.** The wavelength of the radio waves sent by an FM station broadcasting at 97.1 MHz is 30.0 million ($3.00 \times 10^7$) times greater than the wavelength corresponding to the energy difference between a particular excited state of the hydrogen atom and the ground state.
**b.** Let $V$ represent the principal quantum number for the valence shell of element X. If an electron in the hydrogen atom falls from shell $V$ to the inner shell corresponding to the excited state mentioned above in part a, the wavelength of light emitted is the same as the wavelength of an electron moving at a speed of 570. m/s.
**c.** The number of unpaired electrons for element X in the ground state is the same as the maximum number of electrons in an atom that can have the quantum number designations $n = 2$, $m_\ell = -1$, and $m_s = -\frac{1}{2}$.
**d.** Let $A$ equal the charge of the stable ion that would form when the undiscovered element 120 forms ionic compounds. This value of $A$ also represents the angular momentum quantum number for the subshell containing the unpaired electron(s) for element X.



Get help understanding core concepts and visualizing molecular-level interactions, and practice problem solving, by visiting the Online Study Center at **college.hmco.com/PIC/zumdahl7e.**

*Used with permission from the *Journal of Chemical Education,* Vol. 68, No. 11, 1991, pp. 919–922; copyright © 1991, Division of Chemical Education, Inc.

(spu) البوليتكنك اتحاد سنافر

## Units of Energy

Like we saw with pressure, many different units are used throughout the world for energy.

Joule (J)
calorie (cal)
erg (erg)
electron volts (eV)
British thermal unit (Btu)

SI unit for energy

$$1\,\mathrm{J} = \frac{1\,\mathrm{kg}\cdot\mathrm{m}^2}{\mathrm{s}^2}$$

Amount of energy it takes to raise one gram of water 1ºC

**4.184 Joules (J) = 1 calorie (cal)**

**1 kcal = 1000 calories = 1 Nutritional Calorie (Cal)**

## Energy in Thermochemistry

Heat (q): energy transferred from an object with a higher temperature to an object with a lower temperature

100º C → heat → 50º C

Heat cannot be transferred in the opposite direction!

Work (w): energy transferred against a force

Example: Combustion reactions allow your car to move...performing work



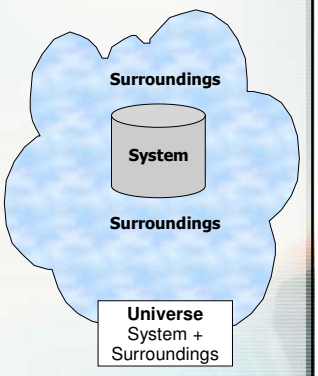

## Thermochemistry

In thermochemistry, the universe is divided into two parts:

The system: The physical process or chemical reaction in which we are interested.

We can define the system anyway we like.

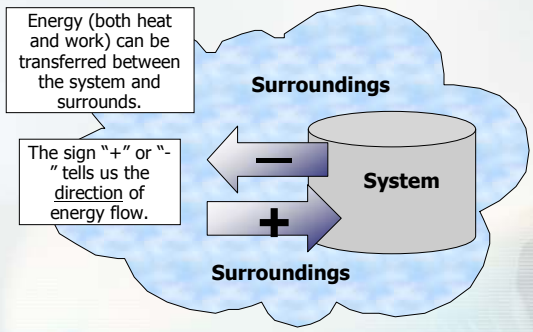
The surroundings: Everything else in the universe.




## Energy flow between system and surroundings

Energy (both heat and work) can be transferred between the system and surrounds.

The sign "+" or "-" tells us the direction of energy flow.




## Specific Heat Capacity

Let's say that I have 5.0 grams of solid aluminum and copper at 25 ºC.

After I add 2.0kJ of heat energy to each sample, will they still be the same temperature?

**5.0 g Al 25°C** ⟷ 2.0 kJ ⟷ **5.0 g Cu 25°C**

Specific Heat: The amount of energy it takes to raise the temperature of 1g of material 1 ºC

Different materials have different specific heats

## Specific Heat

TABLE 8.1 Specific Heats and Molar Heat Capacities for Some Common Substances at 25°C

| Substance | Specific Heat J/(g · °C) | Molar Heat Capacity J/(mol · °C) |
|---|---|---|
| Air (dry) | 1.01 | 29.1 |
| Aluminum | 0.902 | 24.4 |
| Copper | 0.385 | 24.4 |
| Gold | 0.129 | 25.4 |
| Iron | 0.450 | 25.1 |
| Mercury | 0.140 | 28.0 |
| NaCl | 0.864 | 50.5 |
| Water(s) | 2.03 | 36.6 |
| Water(l) | 4.179 | 75.3 |

## Specific heat calculations

$$q = (\text{specific heat})(m)(\Delta T)$$

Heat energy in J

$\frac{J}{g\,^\circ C}$

mass in g

$T_f - T_i$ in °C or K

Example: How much heat energy would it take to raise the temperature of 1000.0g of $H_2O$ from 0.00°C to 100.00°C?

## Calorimetry

**The energy released from the reaction (or whatever is in the calorimeter) is gained by the water. So…**

$$-q_{rxn} = q_{water}$$

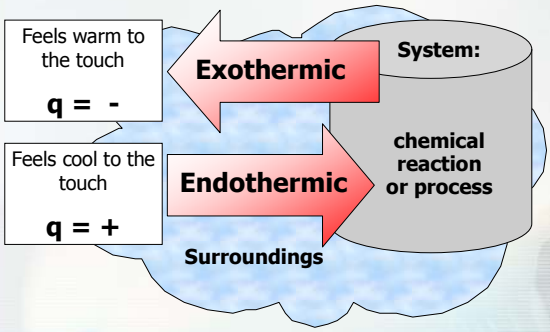
Notice that they have opposite signs, which tells us the direction of transfer.

However, assuming no heat loss, the magnitudes are the same. The amount of heat lost by the reaction is the same amount gained by the water.

## Example

- A 28.2 gram sample of nickel is heated to 99.8°C and placed in a coffee cup calorimeter containing 150.0 grams of water at 23.5°C. After the metal cools, the final temperature of the metal and water is 25.0°C.
- Which substance absorbed heat?
- Which substance released heat?
- Calculate the heat absorbed by the substance you indicated above.

## Exothermic vs. Endothermic processes




## Enthalpies of Reaction

- Determine if the following processes are endothermic or exothermic…
  - Combustion of methane
  - Neutralization of HCl
  - Boiling
  - Melting
  - $CaCO_3$ (s) → CaO (s) + $CO_2$ (g)

## Heat and Enthalpy

So how are q and ΔH related?

At constant pressure:

$$q_p = \Delta H$$

So, using a calorimeter, the ΔH of a chemical reaction can be calculated from the change in heat in the calorimeter.



اتحاد سنافر البوليتكنك (spu)

## Enthalpies of Physical/Chemical Changes

Enthalpy ($\Delta H$) describes the change in heat from a chemical reaction or process.

$\Delta H = H_{products} - H_{reactants}$

Where do these energies come from?

Energy is released if:
$\sum Energy_{reactants} > \sum Energy_{products}$

Energy is absorbed if:
$\sum Energy_{reactants} < \sum Energy_{products}$

## Enthalpies of Phase Changes

- Heat of fusion ($\Delta H_{fus}$): Amount of heat required to melt (solid → liquid)
- Heat of vaporization ($\Delta H_{vap}$): Amount of heat required to evaporate (liquid → gas)
- Heat of sublimation ($\Delta H_{sub}$): Amount of heat required to sublime (solid → gas)

Why are there no values for $\Delta H_{freezing}$, $\Delta H_{condendsation}$, or $\Delta H_{deposition}$?

## Thermodynamic Standard State

$\Delta H°$ — The "°" (knot) denotes enthalpy under standard states

**Standard States**

- A gas at 1 atm
- An aqueous solution (1 M conc.) at a pressure of 1 atm
- Pure liquids and solids
- The most stable form of elements at 1 atm and 25°C (298 K)

## Amount of reactant, and $\Delta H$

In chemistry, we usually specify energy in terms of a specific amount of reactant or product. For example:

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}$ $\Delta H = -241.8$ kJ

Which means if we react 2 mole of $H_2$ with 1 mole of $O_2$, 241.8kJ of heat energy will be released.

But what if we react 4 moles of $H_2$ with 2 moles of $O_2$?

$\Delta H = -483.6$ kJ

This can be read as:
- -241.8 kJ per 2 moles $H_2$
- -241.8 kJ per 1 mole $O_2$
- -241.8 kJ per 2 moles of $H_2O$

## Direction of reaction and $\Delta H$

What about the reverse reaction of:

$2H_{2(g)} + O_{2(g)} \rightarrow 2H_2O_{(g)}$ $\Delta H = -241.8$ kJ

$2H_2O_{(g)} \rightarrow 2H_{2(g)} + O_{2(g)}$ $\Delta H = +241.8$ kJ

All we do is change the sign. Instead of energy being released (as in the first reaction), energy is absorbed when breaking water molecules apart.

## Thermochemical Equations and Stoichiometry

- $2Al\ (s) + Fe_2O_3\ (s) \rightarrow 2Fe\ (s) + Al_2O_3\ (s)$
  $\Delta H° = -852$ kJ
- How much heat is released if 10.0 grams of $Fe_2O_3$ reacts with excess Al?
  - 10.0 g $Fe_2O_3$ → x mol $Fe_2O_3$
  - x mol $Fe_2O_3$ → 852 kJ / 1 mol $Fe_2O_3$

اتحاد سنافر البوليتكنك (spu)

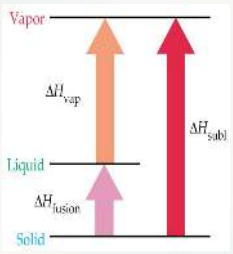
## Hess's Law

- If a compound cannot be directly synthesized from its elements, we must use multiple reactions to calculate the enthalpy of reaction
- *Hess's Law*: change in enthalpy is the same whether the reaction occurs in one step or in a series of steps
- Look at direction of reaction and amount of reactants/products

## Hess's Law

ΔH is a state function…it doesn't matter what route we take to get there. Same change in energy, ΔH.

These two steps (solid → liquid → gas) should have the same change in energy as…




…this one step (solid → gas)



## Hess's Law

- Values of enthalpy change
  - For a reaction in the reverse direction, enthalpy is numerically equal but opposite in sign
    - Reverse direction, heat flow changes; endothermic becomes exothermic (and vice versa)
  - Proportional to the amount of reactant consumed
    - Twice as many moles = twice as much heat
  - Depends on the physical state of the reactants and products (i.e., elemental state)



## Hess's Law

- Does the energy change in a reaction depend on the number of steps in the reaction?
- $H_2(g) + I_2(s) \rightarrow 2HI(g)$ $\Delta H = 53.00$ kJ
  $I_2(s) \rightarrow I_2(g)$ $\Delta H = 62.44$ kJ
- $H_2(g) + I_2(g) \rightarrow 2HI(g)$ $\Delta H = ?$
- How can we add the first two reactions to get the third?
- Hess's Law: $\Delta H_{sum\ of\ steps} = \Delta H_1 + \Delta H_2$



## Application of Hess's Law

- We can use known values of $\Delta H^o$ to calculate unknown values for other reactions
- $P_4(s) + 3\ O_2(g) \rightarrow P_4O_6(s)$ $\Delta H = -1640.1$ kJ
- $P_4(s) + 5\ O_2(g) \rightarrow P_4O_{10}(s)$ $\Delta H = -2940.1$ kJ
- What is $\Delta H^o$ for the following reaction?
  $P_4O_6(s) + 2\ O_2(g) \rightarrow P_4O_{10}(s)$ $\Delta H = ?$



## Hess' Law Problem

- Given:
- $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$ $\Delta H = -802$ kJ
- $H_2O(l) \rightarrow H_2O(g)$ $\Delta H = 88$ kJ
- Find the enthalpy for the following reaction:
- $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ $\Delta H = ?$

اتحاد سنافر البوليتكنك (spu)

## Standard Enthalpies of Formation

- *Standard enthalpy of formation* ($\Delta H^o_f$): heat needed to make *1 mole* of a substance from its stable elements in their *standard states*

  $\Delta H^o_f = 0$ for a stable (naturally occurring) element

- Which of these have $\Delta H^o_f = 0$?
  - CO(g), Cu(s), $Br_2$(l), Cl(g), $O_2$(g), $O_3$(g), $O_2$(s)
- Do the following equations represent standard enthalpies of formation? Why or why not?
  - 2Ag (l) + $Cl_2$ (g) → 2AgCl (s)
  - Ca (s) + $F_2$ (g) → $CaF_2$ (s)

25

## Standard Enthalpies of Formation

- Can use measured enthalpies of formation to determine the enthalpy of a reaction (use Appendix B in back of book)

  $\Delta H^o_{rxn} = \Sigma n \Delta H^o_f$ (products) – $\Sigma n \Delta H^o_f$ (reactants)
  - $\Sigma$ = sum
  - n = number of moles (coefficients)

## Heats of Formation

$\Delta H^o_{rxn} = \Sigma\ n\Delta H^o_{f,products} - \Sigma\ n\Delta H^o_{f,reactants}$

- Calculate values of $\Delta H^o$ for the following rxns:
- 1) $CaCO_3$ (s) → CaO (s) + $CO_2$ (g)
- 2) $2C_6H_6$ (l) + $15O_2$ (g) → $12CO_2$ (g) + $6H_2O$ (l)

| $\Delta H^\circ$ of (in kJ/mol): |
|---|
| CaCO3 (-1207.1) |
| CaO (-635.5) |
| CO2 (-393.5) |
| C6H6 (49.0) |
| CO2 (-393.5) |
| H2O (-285.8) |

27

## Group Work

- Use Standard Heat of Formation values to calculate the enthalpy of reaction for:
- $C_6H_{12}O_6$*(s)* → 2 $C_2H_5OH$*(l)* + 2 $CO_2$*(g)*

  $\Delta H^o_f$ ($C_6H_{12}O_6$*(s)*) = -1260.0 kJ/mol
  $\Delta H^o_f$ ($C_2H_5OH$*(l)*) = -277.7 kJ/mol
  $\Delta H^o_f$ ($CO_2$*(g)*) = -393.5 kJ/mol

## Bond Dissociation Energies

- Bond dissociation energy (BDE). Energy needed to break one 1 mol of a specific bond (eg. H-H, C-H, C=C, C-C, etc.)
- Bond dissociation energy varies somewhat from one molecule to another, or even within one molecule, so in most cases, we have use an *average bond energy (D).*

- *For example:*
- H-OH 502 kJ/mol
- H-O 427 kJ/mol
- H-OOH 431 kJ/mol
- Average = 453.3 kJ/mol for O-H

- Note: there are certain bonds where the BDE is not and average. These are bonds that only exist within diatomic molecules (eg. H-H, H-Br, Cl-Cl, etc.)

## Bond Energy

TABLE 7.1 Average Bond Dissociation Energies, D (kJ/mol)*

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| H—H | 436† | C—H | 410 | N—H | 390 | O—H | 460 | F—F | 159† |
| H—C | 410 | C—C | 300 | N—C | 300 | O—C | 360 | Cl—Cl | 243† |
| H—F | 570† | C—F | 450 | N—F | 270 | O—F | 180 | Br—Br | 193† |
| H—Cl | 432† | C—Cl | 330 | N—Cl | 200 | O—Cl | 200 | I—I | 151† |
| H—Br | 366† | C—Br | 270 | N—Br | 240 | O—Br | 210 | S—F | 310 |
| H—I | 298† | C—I | 240 | N—I | | O—I | 220 | S—Cl | 250 |
| H—N | 390 | C—N | 300 | N—N | 240 | O—N | 200 | S—Br | 210 |
| H—O | 460 | C—O | 350 | N—O | 200 | O—O | 180 | S—S | 225 |
| H—S | 340 | C—S | 260 | N—S | — | O—S | — | | |
| Multiple covalent bonds‡ | | | | | | | | | |
| C—C | 611 | C═C | 835 | C—O | 732 | O—O | 498† | N═N | 945† |

(spd) اتحاد سنافر البوليتكنك

30

## Bond Dissociation Energies

$\Delta H^o_{rxn} = \Sigma BDE \text{ (reactants)} + - \Sigma BDE \text{ (products)}$

endothermic energy input — exothermic energy released

- Or: $\Delta H^o_{rxn} = \Sigma BDE \text{ (reactants)} - \Sigma BDE \text{ (products)}$

  $\Sigma BDE(react) > \Sigma BDE(prod) \rightarrow$ endothermic

  $\Sigma BDE(react) < \Sigma BDE(prod) \rightarrow$ exothermic

- We use only when heats of formation are not available, since bond energies are average values for gaseous molecules

31

## Enthalpy of Reaction

- Use bond energies to calculate the enthalpy change for the following reaction:

| |
|---|
| N≡N = 945 kJ/mol |
| H-H = 436 kJ/mol |
| N-H = 390 kJ/mol |

- $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

  $\Delta H_{rxn} = [BE_{N \equiv N} + 3BE_{H-H}] - [6BE_{N-H}]$
  $\Delta H_{rxn} = [945 + 3(436)] - [6(390)] = -87$ kJ

- measured value = -92.2 kJ

- Why are the calculated and measured values different?

32

## Enthalpy of Reaction

- Use bond energies to calculate the enthalpy change for the decomposition of nitrogen trichloride:
  $NCl_3(g) \rightarrow N_2(g) + Cl_2(g)$ (hint: balance first!!)
  $\Delta H_{rxn}$?

| |
|---|
| N-Cl = 200 kJ/mol |
| N≡N = 945 kJ/mol |
| Cl-Cl = 243 kJ/mol |

33

## Practice Problems

- Which amount of energy is higher?
  - Energy that is released when 55g of Al is cooled from 5˚C to -25 ˚C
  - Energy that is absorbed when 64g of Cu is heated from 298K to 315K?

- 8750 J of heat are applied to a 170 g sample of metal, causing a 56°C increase in its temperature. What is the specific heat of the metal? Which metal is it?

## Practice Problems

- Practice: Worked Example 8.10, page 320….

- Identify how to set up the following problems:
- Calculate the $\Delta H^o$ of reaction for:
  - $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$
    $\Delta H^o_f$ $C_3H_8(g)$: -103.95 kJ/mol; $\Delta H^o_f$ $CO_2(g)$: -393.5 kJ/mol; $\Delta H^o_f$ $H_2O(l)$: -285.8 kJ/mol
- 8750 J of heat are applied to a 170 g sample of metal, causing a 56°C increase in its temperature. What is the specific heat of the metal? Which metal is it?

## Practice Problems

- $C_2H_4(g) + 6F_2(g) \rightarrow 2CF_4(g) + 4HF(g)$ $\Delta H^o$ = ?
  - $H_2(g) + F_2(g) \rightarrow 2HF(g)$ $\Delta H^o$ = -537 kJ
  - $C(s) + 2F_2(g) \rightarrow CF_4(g)$ $\Delta H^o$ = -680 kJ
  - $2C(s) + 2H_2(g) \rightarrow C_2H_4(g)$ $\Delta H^o$ = 52.3 kJ

- Use average bond energies to determine the enthalpy of the following reaction (from Table 7.1).
  - $CH_4(g) + Cl_2(g) \rightarrow CH_3Cl(g) + HCl(g)$
  - ($BE_{C-Cl}$ = 328 kJ/mol)

اتحاد سنافر البوليتكنك (spu)

*Chapter 8*

***Thermochemistry: Chemical Energy***





1

## Energy Changes in Chemical Rxns

- Most reactions give off or absorb energy
- Energy is the capacity to do work or supply heat.
  - ***Heat***: transfer of thermal (kinetic) energy between two systems at different temperatures (from hot to cold)



Metal bar in water



Metal bar drilled

## Types of Energy

- ***Work (w)***: energy transfer when forces are applied to a system
- ***Heat (q)***: energy transferred from a hot object to a cold one
  - Radiant energy→ heat from the sun
  - Thermal energy → associated with motion of particles
  - Potential energy → energy associated with object's position or substance's chemical bonds
  - Kinetic energy → energy associated with object's motion

## Heat versus Temperature

- Describe the difference between the two.
- SI unit of energy: J $1\,\mathrm{J} = \frac{1\,\mathrm{kg}\cdot\mathrm{m}^2}{\mathrm{s}^2}$



Molecular heat transfer

- 1 watt = 1 J/s, so a 100 Watt bulb uses 100 J each second
- We often use the unit of kJ to refer to chemical heat exchanges in a reaction. 1 kJ = 1000 J
- Energy is also reported in calories:
  - Amount of energy needed to raise 1 gram of water by 1°C
  - 1 cal = 4.184 J; 1 Cal = 4184 J
  - Cal (or kcal) is used on food labels

## Energy and Energy Conservation

- ***Heat***: form of energy transferred from object at higher temperature to one at lower temperature (from hot object to cold object)
- ***Thermochemistry***: study of heat changes in chemical reactions, in part to predict whether or not a reaction will occur
- ***Thermodynamics***: study of heat and its transformations
- **First Law of Thermodynamics**: Energy can be converted from one form to another but cannot be created or destroyed

## Conservation of Energy



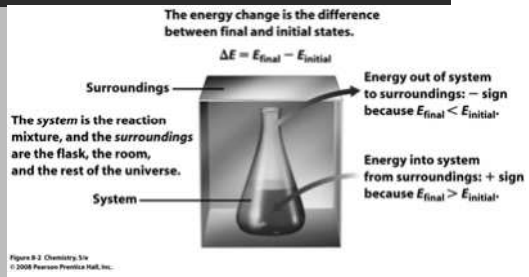






**اتحاد سنافر البوليتكنك (spu)**

## System and Surroundings




- System loses heat (negative); gains heat (positive)

## Endothermic vs Exothermic

- Endothermic reaction: q is positive (q > 0)
  - ♦ Reaction (system) absorbs heat
  - ♦ Surroundings feel cooler
- Exothermic reaction: q is negative (q < 0)
  - ♦ Reaction releases heat
  - ♦ Surroundings feel warmer





## Applications of heat emission/absorption







9

## Specific Heat and Heat Capacity

- ***Specific heat (sp. ht.)***: amount of heat required to raise 1 gram of substance by 1°C
- Use mass, specific heat, and ΔT to calculate the amount of heat gained or lost:
- **$q = ms\Delta T$** → $ms = C$ → **$q = C\Delta T$**
  - ♦ ***Heat capacity (C)***: amount of heat required to raise the temperature of a given quantity of a substance by 1°C; **C = q / ΔT** = J / °C
  - ♦ ***Molar heat capacity ($C_m$)***: amount of heat that can be absorbed by 1 mole of material when temperature increases 1°C; **$q = (C_m)$ x (moles of substance) x (ΔT)** = J / mol • °C

## Specific Heat and Heat Capacity

**TABLE 8.1 Specific Heats and Molar Heat Capacities for Some Common Substances at 25 °C**

| Substance | Specific Heat J/(g · °C) | Molar Heat Capacity J/(mol · °C) |
|---|---|---|
| Air (dry) | 1.01 | 29.1 |
| Aluminum | 0.902 | 24.4 |
| Copper | 0.385 | 24.4 |
| Gold | 0.129 | 25.4 |
| Iron | 0.450 | 25.1 |
| Mercury | 0.140 | 28.0 |
| NaCl | 0.864 | 50.5 |
| Water(*s*)* | 2.03 | 36.6 |
| Water(*l*) | 4.179 | 75.3 |

## Practice Problem

- Calculate the amount of heat transferred when 250 g of $H_2O$ (with a specific heat of 4.184 J/g·°C) is heated from 22°C to 98°C.
- $q = ms\Delta T$
- Is heat being put into the system or given off by the system?
- If a piece of hot metal is placed in cold water, what gains heat and what loses heat? Which one will have a positive q value and which will have a negative q value?

Worked Ex. 8.4, 8.5; Problems 8.10 – 8.12

اتحاد سنافر البوليتكنك (spu)

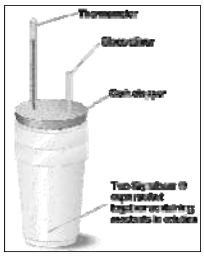
## Practice Problem

- 34.8 g of an unknown metal at 25.2°C is mixed with 60.1 g of $H_2O$ at 96.2 °C (sp. ht. = 4.184 J/g·°C). The final temperature of the system comes to 88.4°C. Identify the unknown metal.
- Specific heats of metals:
  - ♦ Al 0.897 J/g·°C
  - ♦ Fe 0.449 J/g·°C
  - ♦ Cu 0.386 J/g·°C
  - ♦ Sn 0.228J/g·°C

13

## Calorimetry and Heat Capacity

- Heat changes in a reaction can be determined by measuring the heat flow at constant pressure
- Apparatus to do this is called a **calorimeter**.
- Heat evolved by a reaction is absorbed by water; heat capacity of calorimeter is the heat capacity of water.



14

## Example

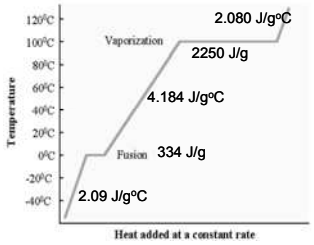
- A 28.2 gram sample of nickel is heated to 99.8°C and placed in a coffee cup calorimeter containing 150.0 grams of water at 23.5°C. After the metal cools, the final temperature of the metal and water is 25.0°C.
- $q_{absorbed} + q_{released} = 0$
- Which substance absorbed heat?
- Which substance released heat?
- Calculate the heat absorbed by the substance you indicated above.

## Enthalpies of Physical/Chemical Changes

- Enthalpy (*H*) describes heat flow into and out of a system under constant pressure
- Enthalpy (a measure of energy) is heat transferred per mole of substance.
- At constant pressure,
  - ♦ $q_p = \Delta H = H_{products} - H_{reactants}$
  - ♦ $\Delta H > 0$ → endothermic (net absorption of energy from environment; products have more internal energy)
  - ♦ $\Delta H < 0$ → exothermic (net loss of energy to environment; reactants have more internal energy)

## Heating a Pure Substance (Water)

- Why does T become constant during melting and evaporating?
- Melting, vaporization, and sublimation are endothermic
- We can calculate total heat needed to convert a 15 gram piece of ice at -20°C to steam at 120°C.




17

## Enthalpies of Phase Changes

- Heat of fusion ($\Delta H_{fus}$): Amount of heat required to melt (solid → liquid)
- Heat of vaporization ($\Delta H_{vap}$): Amount of heat required to evaporate (liquid → gas)
- Heat of sublimation ($\Delta H_{sub}$): Amount of heat required to sublime (solid → gas)
- Why are there no values for $\Delta H_{freezing}$, $\Delta H_{condendsation}$, or $\Delta H_{deposition}$?

اتحاد سنافر البوليتكنك (spu)

## Enthalpies of Reaction

- Determine if the following processes are endothermic or exothermic…
  - ♦ Combustion of methane
  - ♦ Reacting $Ba(OH)_2$ with $NH_4Cl$
  - ♦ Neutralization of HCl
  - ♦ Melting
  - ♦ $CaCO_3$ (s) → CaO (s) + $CO_2$ (g)

19

## Thermochemical Equations

- Shows both mass and enthalpy relationships
- 2Al (s) + $Fe_2O_3$ (s) → 2Fe (s) + $Al_2O_3$ (s) $\Delta H^o$ = -852 kJ
- Amount of heat given off depends on amount of material:
  - ♦ 852 kJ of heat are released for every 2 mol Al, 1 mol $Fe_2O_3$, 2 mol Fe, and 1 mol $Al_2O_3$

## Thermochemical Equations

- 2Al (s) + $Fe_2O_3$ (s) → 2Fe (s) + $Al_2O_3$ (s) $\Delta H^o$ = -852 kJ
- How much heat is released if 10.0 grams of $Fe_2O_3$ reacts with excess Al?
- Worked Ex. 8.3; Problems 8.8, 8.9
- What if we reversed the reaction?
- Heat would have to be put in to make the reaction proceed:
  - ♦ 2Fe (s) + $Al_2O_3$ (s) → 2Al (s) + $Fe_2O_3$ (s) $\Delta H^o$ = +852 kJ

## Hess's Law

- If a compound cannot be directly synthesized from its elements, we can add the enthalpies of multiple reactions to calculate the enthalpy of reaction in question.
- ***Hess's Law***: change in enthalpy is the same whether the reaction occurs in one step or in a series of steps
- Look at direction of reaction and amount of reactants/products

## Hess's Law

- Value changes sign with direction

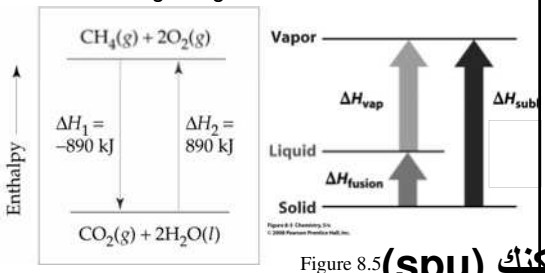



Figure 8.5

23

## Hess's Law

- Values of enthalpy change
  - ♦ For a reaction in the reverse direction, enthalpy is numerically equal but opposite in sign
    - Reverse direction, heat flow changes; endothermic becomes exothermic (and vice versa); sign of $\Delta H$ changes
  - ♦ Proportional to the amount of reactant consumed
    - Twice as many moles = twice as much heat; half as many moles = half as much heat
- $\Delta H_T = \Delta H_1 + \Delta H_2 + \Delta H_3 + \ldots$.

24

اتحاد سنافر البوليتكنك (spu)

## Enthalpy of Chemical Reaction

- Thermochemical equation:
  - ♦ $H_2(g) + I_2(s) \rightarrow 2HI(g)$ $\Delta H = +53.00$ kJ
- Two possible changes:
- Reverse the equation:
  - ♦ $2HI(g) \rightarrow H_2(g) + I_2(s)$ $\Delta H = -53.00$ kJ
- Double the amount of material:
  - ♦ $2H_2(g) + 2I_2(s) \rightarrow 4HI(g)$ $\Delta H = +106.00$ kJ

25

## Hess's Law

- Calculate $\Delta H^o$ for
- $2NO(g) + O_2(g) \rightarrow N_2O_4(g)$ $\Delta H^o = ?$

- $N_2O_4(g) \rightarrow 2NO_2(g)$ $\Delta H^o = 57.2$ kJ
- $NO(g) + \frac{1}{2} O_2(g) \rightarrow NO_2(g)$ $\Delta H^o = -57.0$ kJ

## Hess's Law

- We can use known values of $\Delta H^o$ to calculate unknown values for other reactions
- $P_4(s) + 3\ O_2(g) \rightarrow P_4O_6(s)$ $\Delta H = -1640.1$ kJ
- $P_4(s) + 5\ O_2(g) \rightarrow P_4O_{10}(s)$ $\Delta H = -2940.1$ kJ

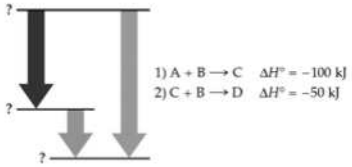
- What is $\Delta H^o$ for the following reaction?
  $P_4O_6(s) + 2\ O_2(g) \rightarrow P_4O_{10}(s)$ $\Delta H = ?$

- Worked Ex. 8.6, 8.7; Problem 8.13, 8.15

27

## Hess's Law

**KEY CONCEPT PROBLEM 8.14** The reaction of A with B to give D proceeds in two steps and can be represented by the following Hess's law diagram.




(a) What is the equation and $\Delta H^\circ$ for the net reaction?
(b) Which arrow on the diagram corresponds to which step, and which arrow corresponds to the net reaction?
(c) The diagram shows three energy levels. The energies of which substances are represented by each?

## Hess's Law

- Given:
- $2NH_3(g) \rightarrow N_2H_4(l) + H_2(g)$ $\Delta H = 54$ kJ
- $\frac{1}{2}N_2(g) + \frac{3}{2}H_2(g) \rightarrow NH_3(g)$ $\Delta H = -69$ kJ
- $CH_4O(l) \rightarrow CH_2O(g) + H_2(g)$ $\Delta H = -195$ kJ

- Find the enthalpy for the following reaction:
- $N_2H_4(l) + CH_4O(l) \rightarrow CH_2O(g) + N_2(g) + 3H_2(g)$
  $\Delta H = ?$ kJ

اتحاد سنافر البوليتكنك (spu)

## Standard Heats of Formation

- ***Standard heat of formation*** ($\Delta H^o_f$): heat needed to make ***1 mole*** of a substance from its stable elements in their ***standard states***
- $\Delta H^o_f = 0$ for a stable (naturally occurring) element
- Which of these have $\Delta H^o_f = 0$?
  - ♦ $CO(g)$, $Cu(s)$, $Br_2(l)$, $Cl(g)$, $O_2(g)$, $O_3(g)$, $O_2(s)$, $P_4(s)$
- Do the following equations represent standard enthalpies of formation? Why or why not?
  - ♦ $2Ag(l) + Cl_2(g) \rightarrow 2AgCl(s)$
  - ♦ $Ca(s) + F_2(g) \rightarrow CaF_2(s)$

30

## Standard Enthalpies of Formation

- Can use measured enthalpies of formation to determine the enthalpy of a reaction (use Appendix B in back of book)
- $\Delta H^o_{rxn} = \Sigma n \Delta H^o_f$ (products) – $\Sigma n \Delta H^o_f$ (reactants)
  - ♦ $\Sigma$ = sum; n = number of moles (coefficients)
- Direct calculation of enthalpy of reaction if the reactants are all in elemental form
  - ♦ Sr (s) + $Cl_2$ (g) → $SrCl_2$ (g)
  - ♦ $\Delta H^o_{rxn} = [\Delta H^o_f (SrCl_2)] – [\Delta H^o_f (Sr) + \Delta H^o_f (Cl_2)]$ = -828.4 kJ/mol

## Standard Enthalpies of Formation

- Table 8.2: Some Common Substances (25°C)

| Substance | Formula | $\Delta H^\circ_f$ (kJ/mol) |
|---|---|---|
| Acetylene | $C_2H_2(g)$ | 226.7 |
| Ammonia | $NH_3(g)$ | –46.1 |
| Carbon dioxide | $CO_2(g)$ | –383.5 |
| Carbon monoxide | $CO(g)$ | –110.5 |
| Ethanol | $C_2H_5OH(l)$ | –277.7 |
| Ethylene | $C_2H_4(g)$ | 52.3 |
| Glucose | $C_2H_{12}O_6(s)$ | –1260 |

| Substance | Formula | $\Delta H^\circ_f$ (kJ/mol) |
|---|---|---|
| Hydrogen chloride | $HCl(g)$ | –92.3 |
| Iron(III) oxide | $Fe_2O_3(s)$ | –824.2 |
| Magnesium carbonate | $MgCO_3(s)$ | –1095.8 |
| Methane | $CH_4(g)$ | –74.8 |
| Nitric oxide | $NO(g)$ | 90.2 |
| Water (g) | $H_2O(g)$ | –241.8 |
| Water (l) | $H_2O(l)$ | –285.8 |

## Heats of Formation

- $\Delta H^o_{rxn} = \Sigma \Delta H^o_{f,products} - \Sigma \Delta H^o_{f,reactants}$
- Calculate values of $\Delta H^o$ for the following rxns:
- 1) $CaCO_3$ (s) → CaO (s) + $CO_2$ (g)
- 2) $2C_6H_6$ (l) + $15O_2$ (g) → $12CO_2$ (g) + $6H_2O$ (l)

- $\Delta H^o_f$ values:
  $CaCO_3$: -1207.1 kJ/mol; CaO: -635.5 kJ/mol; $CO_2$: -393.5 kJ/mol; $C_6H_6$: 49.0 kJ/mol; $H_2O$(l): -285.8 kJ/mol
- Worked Ex. 8.8, 8.9; Problems 8.16, 8.17

33

## Bond Dissociation Energies

- ***Bond Dissociation Energy (or Bond Energy, BE)***: energy required to break a bond in 1 mole of a gaseous molecule
- Reactions generally proceed to form compounds with more stable (stronger) bonds (greater bond energy)



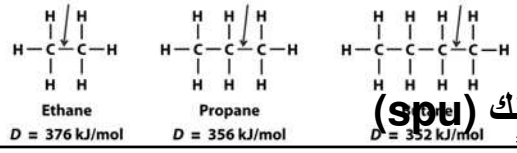




34

## Bond Dissociation Energies

- Bond energies vary somewhat from one molecule to another so we use ***average*** *bond dissociation energy (D)*
- H-OH 502 kJ/mol Avg O-H = 453
- H-O 427 kJ/mol kJ/mol
- H-OOH 431 kJ/mol




35

## Bond Dissociation Energies

TABLE 7.1 Average Bond Dissociation Energies, *D* (kJ/mol)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| H–H | 436[a] | C–H | 410 | N–H | 390 | O–H | 460 | F–F | 159[a] |
| H–C | 410 | C–C | 350 | N–C | 300 | O–C | 350 | Cl–Cl | 243[a] |
| H–F | 570[a] | C–F | 450 | N–F | 270 | O–F | 180 | Br–Br | 193[a] |
| H–Cl | 432[a] | C–Cl | 330 | N–Cl | 200 | O–Cl | 200 | I–I | 151[a] |
| H–Br | 366[a] | C–Br | 270 | N–Br | 240 | O–Br | 210 | S–F | 310 |
| H–I | 298[a] | C–I | 240 | N–I | — | O–I | 220 | S–Cl | 250 |
| H–N | 390 | C–N | 300 | N–N | 240 | O–N | 200 | S–Br | 210 |
| H–O | 460 | C–O | 350 | N–O | 200 | O–O | 180 | S–S | 225 |
| H–S | 340 | C–S | 260 | N–S | — | O–S | — | | |
| Multiple covalent bonds[b] | | | | | | | | | |
| C=C | 611 | C=C | 835 | C=O | 732 | O=O | 498[a] | N=N | 945[a] |

[a] Exact value.

36

اتحاد سنافر البوليتكنك (spu)

## Bond Dissociation Energies

- $\Delta H^o_{rxn} = \Sigma BE$ (reactants) + - $\Sigma BE$ (products)
  endothermic energy input; exothermic energy released
- $\Sigma BE(react) > \Sigma BE(prod) \rightarrow$ endothermic
- $\Sigma BE(react) < \Sigma BE(prod) \rightarrow$ exothermic
- Use only when heats of formation are not available, since bond energies are average values for gaseous molecules

37

## Heats of Reaction

- Use bond energies to calculate the enthalpy change for the following reaction:

$$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$$

- $\Delta H_{rxn} = [BE_{N \equiv N} + 3BE_{H-H}] + [-6BE_{N-H}]$
- $\Delta H_{rxn} = [945 + 3(436)] – [6(390)] = -87$ kJ
- measured value = -92.2 kJ
- Why are the calculated and measured values different?

38

## Heats of Reaction

- Use bond energies to calculate the enthalpy change for the decomposition of nitrogen trichloride:
  $NCl_3$ (g) → $N_2$ (g) + $Cl_2$ (g)
- How many distinct bond types are there in each molecule?
- How many of each bond type do we need to calculate $\Delta H_{rxn}$?
  - $BE_{(N-Cl)} = 200$ kJ/mol
  - $BE_{(N \equiv N)} = 945$ kJ/mol
  - $BE_{(Cl-Cl)} = 243$ kJ/mol

39

## Thermochemistry Calculation Summary

- Use $q = ms\Delta T$ ($s = J/g \cdot {}^oC$)
- If given mass of reactant, convert to moles and multiply by enthalpy to find total heat transferred
- If given multiple equations with enthalpies, use Hess's Law
- If given $\Delta H^o_f$ values: products – reactants
- If given bond energy (BE) values: +reactants + -products

## Practice Problems

- Identify how to set up the following problems:
- Calculate the $\Delta H^o$ of reaction for:
  - $C_3H_8$ (g) + $5O_2$ (g) → $3CO_2$ (g) + $4H_2O$ (l)
  - $\Delta H^o_f$ $C_3H_8$(g): -103.95 kJ/mol; $\Delta H^o_f$ $CO_2$(g): -393.5 kJ/mol; $\Delta H^o_f$ $H_2O$(l): -285.8 kJ/mol
- 8750 J of heat are applied to a 170 g sample of metal, causing a 56°C increase in its temperature. What is the specific heat of the metal? Which metal is it?

## Practice Problems

- $C_2H_4(g) + 6F_2(g) \rightarrow 2CF_4(g) + 4HF(g)$ $\Delta H^o$ = ?
  - $H_2$ (g) + $F_2$ (g) → 2HF (g) $\Delta H^o = -537$ kJ
  - C (s) + $2F_2$ (g) → $CF_4$ (g) $\Delta H^o = -680$ kJ
  - 2C (s) + $2H_2$ (g) → $C_2H_4$ (g) $\Delta H^o = 52.3$ kJ
- Use average bond energies to determine the enthalpy of the following reaction (from Table 7.1).
  - $CH_4$ (g) + $Cl_2$ (g) → $CH_3Cl$ (g) + HCl (g)
    **($BE_{C-Cl} = 328$ kJ/mol)**

**اتحاد سنافر البوليتكنك (spu)**

**Entropy: i**s "measure if degree of randomness."It is denoted by S, it is a state function and extensive property.

$\Delta S = \Delta S_p - \Delta S_r$

In term of energy it is given as:

$\Delta S = (q_{rev}/t)$

**How s & q related?**

When q increases, kinetic energy increase, randomness increases, entropy also increase.

**How S&T related?**

At low t, randomness increases, entropy increases.

Therefore $s = q/t$

unit = $jk^{-1}$

For any reaction to be spontaneous $\Delta s$ = +ve spontaneous

$\triangle s$ = -ve non spontaneous

$\triangle s$ = 0 equilibrium

$\triangle s$ cyclic= 0

Total $s = s_{system} + s_{surrounding}$

$\triangle i_{rreversible} > 0$ or +ve

**Entropy of fusion** = $s = q/t_f$

**Entropy of vaporisation** = $s = q/t_b$

**Entropy of sublimation.** = $s = q/t$

1. Lets calculate the enthalpy change :

$C + 2H_2 \rightarrow CH_4$ $\quad \Delta H = ?$

$C + O_2 \rightarrow 2H_2O$ $\quad \Delta H = 285.8 \times 2$

$C + 2H_2 + 2O_2 \rightarrow CO_2 + 2H_2O$ $\quad -393.5 + (285.8)$

$CH_4 - 2O_2 \rightarrow -CO_2 - 2H_2O$ $\quad -393.5 - 285.8$

$= -679.3$

$C + 2H_2 \rightarrow CH_4$ $\quad 679.3 - (-890.2)$

$-679.3 + 890.2$

$= 210.9 kj/mol$

**Second law of thermodynamics**

To determine spontaneous, total entropy is needed.

**(spu) اتحاد سنافر البوليتكنك**

$\Delta S > 0$ = spontaneous

$\Delta S = 0$ Equilibrium

$\Delta S < 0$ Non-Spontaneous

$S_{total} = S_{system} + S_{Surroundings}$

$\Delta S_{sys}$ depends on the following:-

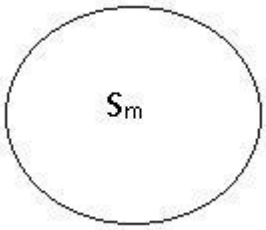
**Absolute entropy depends upon**

- Mass
- Molecular structure
- Temperature &pressure

**Standard molar entropy:**

It is the entropy for 1 mole of substance at 298k & 1 atm.

It is represented as:




Issue with total entropychange:

$\Delta s = \Delta S_{sys} + \Delta S_{urrounding}$ → difficult to calculate

$\Delta S_{um} = -q/T = -\Delta H/T$

$q_p = \Delta H$

$\Delta S_t = \Delta S_{sys} + \Delta S_{urrounding}$

$\Delta S_t = \Delta S_{sys} - \Delta H/T$

$T\Delta S_t = T\Delta S_s - \Delta H$

$\Delta G = -T\Delta S$

$-T\Delta S_t = -T\Delta S + \Delta H$

$Or \Delta G = \Delta H_{sy} - T\Delta S_{sy}$

$\Delta G = -T\Delta s$

**Gibbs free energy**

- It is denoted by G
- It is given by G = H – TS
- It is state function
- Extensive property **(spu) اتحاد سنافر البوليتكنك**
- $\Delta G = \Delta H - T\Delta S$ ( at constant t)

**It is defined as the amount of energy for a system that can be converted into useful work.**

$\Delta G$ = (-ve) spontaneous

$\Delta G$ = 0 Equilibrium

$\Delta G$ = (+ve) Non-Spontaneous

**Spontaneity in terms of free energy change:**

We know

$\Delta G = \Delta H - T\Delta S$

This equation takes into account both the concepts (a) energy factor and other (b) entropy factor

**Depending upon the signs of h and t.s. and their relative magnitudes the following possibilities arise:**

1. When both H and T.S are negative: Energy factor favours and randomness oppose then: if H <T.S(non-spontaneous), G=negative
   - When both H = T.S the process is in equilibrium and G =0
2. when both H and T.S are positive: Energy factor opposes and randomness favours then:
   - If H >T. S process is non spontaneous, G =+ve
   - If H<T.S process is spontaneous, G= -ve
   - If H=T.S the process is in equilibrium, G=0
3. When H = -ve and T.S =+ve, process is spontaneous and G= -ve
4. When H = +ve and T.S =-ve, process is non spontaneous and G= +ve

**Effect of temperature on spontaneity of process**

Endothermic process à high t-> spontaneous

Exothermic process à low t-> spontaneous

**<u>Third law of thermodynamics:</u>**

- At absolute zero, the entropy of perfect crystalline is o.
- Please note there are certain substances which possess certain entropy even at absolute zero. This entropy is also known as residual entropy.

**(spu)** هذه معلومة إضافية البوربوينت

- The origin of residual entropy can be explained on the basis of the disorder which remains at absolute zero in certain crystals composed of ab types of molecules where a and b are similar atoms .
- There is little energy difference between ab—ab—ab and ab –ba—ba –ab and other arrangements so that the molecules adopt the orientation ab and ba at random in solid .this give rise to residual entropy .

# General Chemistry II
# Chapter 13 Lecture Notes
# Chemical Kinetics

**Chemical Kinetics** is concerned with reaction **rates**.
Typical reaction rate units are M/s, the change in concentration of a species per unit time.
There is a huge variation in possible reaction rates.
Increasing reaction rates is the goal of many industrial (and biological) processes.

In a simple reaction, **A → B,** the number of A molecules or [A] decreases with time and the number of B molecules or [B] increases with time.
The reaction rate can be expressed in several useful forms,

Rate = $-\Delta[A]/\Delta t$ or rate = $\Delta[B]/\Delta t$

(Since is it desirable for rate values to be positive, the [A] equation has a negative sign.

Species may be monitored by pressure, pH, spectroscopically or otherwise.

**The Oxidation of Formic Acid by Molecular Bromine**.

$Br_2(aq) + HCOOH(aq) \rightarrow 2Br^{-1}(aq) + 2H^{+1}(aq) + CO_2(g)$

The reaction may be followed by tracking the loss of bromine color spectroscopically using Beer's Law.

The <u>average</u> rate of the reaction can be calculated by measuring the bromine concentration and the time at two different stages of the reaction. The average rate will typically vary with the time interval and the stage of the reaction chosen. Since the reaction rate will decrease as the reactants are used up, a plot of the reaction rate vs. time will be a curve. The *average* rate of the reaction is not necessarily equal to the *instantaneous* rate of the reaction at any given time.

A sketch of $Br_2$(M) vs. time will show steadily decreasing rate.

Rate = $\Delta[Br_2]/\Delta t$

Since $[Br_2]$ is in mol/L and t is in seconds, the rate is expressed with units M/s.

The instantaneous rate at a given point of the reaction is more meaningful, since it is constant for a given reaction at a specified temperature and specified reactant concentrations.

To separate the factors influencing the rate of a reaction, one reactant concentration can be held constant by making it so high that it changes little during the reaction.
For the reaction between bromine and formaldehyde, keeping formaldehyde concentration constant, the rate of disappearance of bromine is proportional to $[Br_2]$.

(spu) اتحاد سنافر البوليتكنك

Therefore, rate $\propto [Br_2]$ or rate = $k[Br_2]$ or $k = \text{rate}/[Br_2]$

k, called the rate constant, therefore equals the reaction rate divided by the bromine concentration *at constant temperature and formaldehyde concentration.*

A sketch of reaction rate (M/s) vs. bromine concentration (M) will be linear. The slope of the line, the rate constant, expressed in $s^{-1}$ therefore remains the same.

The experiment tells nothing about the reaction rate dependence on [HCOOH] or on temperature.

**Decomposition of Hydrogen Peroxide**

Since one product of the reaction is a gas, the reaction rate can be tracked by measuring the pressure exerted by the evolved oxygen (at constant system volume).
$2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$

If an expression involving the molar concentration of oxygen is desired, it can be derived from the measurable gas parameters, P, V, and T. the simplest experimental setup is to keep V and T constant and measure changing P.

From $PV = nRT$, the molar concentration M(mol/L) is equal to $n/V = P/RT$. A change in the concentration of oxygen $\Delta M = \Delta P/RT$. To measure $\Delta M = \Delta[O_2]$ with respect to time,

$\Delta[O_2]/dt = \Delta M/dt = d(\Delta P)dt \bullet 1/RT$

Keep the temperature and volume constant and track the increase in oxygen pressure in a closed system. This can be converted to change in $O_2$ concentration.

**Reaction Rates and Stoichiometry**

How to express reaction rates when the stoichiometry of reactants is not one-to-one? For example:

$2A \rightarrow B$

In this reaction, the rate of disappearance of A is twice the rate of appearance of B.
If rate is written as
$rate = \Delta[B]/\Delta t$,

Then the rate of disappearance of A must be written as $rate = -\frac{1}{2}\Delta[A]/\Delta t$

Note that the coefficient 2 appears in the denominator of the fraction. For 3A, the rate coefficient would have been ⅓, etc.

**The Rate Law**

**اتحاد سنافر البوليتكنك (spu)**

For the general expression of a chemical reaction, $aA + bB \rightarrow cC + dD$, the ***rate law*** is expressed as the rate law ***constant, k,*** times the ***reactant*** concentrations raised to the power of the order of each reactant.

Rate = $k [A]^x [B]^y$

It is tempting to assign x = a, y = b, etc. for the reactant orders, but this is generally not possible. Reaction order often does not correlate to reactant coefficient. ***Rate measurements*** are needed to determine the values of x and y. The reactions is described as being first, second, etc. order in A and first, second etc. order in B. The allover reaction order is the sum of x + y + ···. Reactant orders are typically 0, 1, 2, 3, or sometimes 0.5.

A zero order reactant means that the rate of the reaction is not influenced by the concentration of that particular reactant.

Example: An experimental determination of the order of the reaction between fluorine and chlorine dioxide gas was done by measuring the *initial* reaction rate for different *starting* concentrations of fluorine and chlorine dioxide.

$F_2(g) + 2ClO_2(g) \rightarrow 2FClO_2(g)$

The concentration of one gaseous reactant is held constant while the other is varied to fine the rate order of each reactant. Experimentally it was found that, while keeping $[F_2]$ constant and varying $[ClO_2]$ (doubling, tripling, etc. the concentration), the reaction rate increased exactly in proportion to the change in $[ClO_2]$. This means that the reaction is first order in $[ClO_2]$, or that the reaction rte equals some constant times $[ClO_2]$. Similarly, when $[ClO_2]$ is kept constant and $[F_2]$ is varied, the reaction rte changes in the same ratio. The reaction is also first order in $[F_2]$. The overall reaction is ***second*** order, first order in each of the two reactants. The rate of this reaction is given by

rate = $k[ClO_2] [F_2]$

The units of rate are typically concentration/time, or in most cases, M/s.

For carrying out calculations, it is useful to know the value of k, which is invariant when temperature is constant.

$k = \text{rate}/([ClO_2] [F_2])$

In the case of a second order reaction, the units of k are M/s divided by $M^2 = M^{-1} \cdot s^{-1}$. In a first order reaction, k has units M/s ÷ M = $s^{-1}$. Higher order reactions give more complicated k values.

## Relation Between Reactant Concentration and Time

The mathematical relationships between reactant concentrations and time make possible the prediction of reaction times, per cent completion, half-lives, etc. The algebraic equations are different for ***first*** and for ***second order*** reactions.

اتحاد سنافر البوليتكنك (spu)

### First-Order Reactions

The rate of a first-reaction depends only on the concentration of one reactant, A.

$A \rightarrow$ products

By definition, rate = $- \Delta [A]/\Delta t$ and rate = $k[A]$

Therefore, $- \Delta[A]/\Delta t = k[A]$

$d[A]/dt = -kt$ (Instantaneous rate of change described by calculus.)

$\ln([A]/[A_o]) = - kt$ (Calculus solution to the equation by integration)

$\ln[A] - \ln[A_o] = - kt$ or

$\ln[A] = - kt + \ln[A_o]$

This is a useful equation for graphing since it is in the form $y = mx + b$

Example: For a reaction with $k = 5.50 \times 10^{-3}\ s^{-1}$ at 45.0 ºC, what will be the concentration of A remaining after 12.0 min if the initial concentration of A is 0.200 M?
$\ln[A] = - kt + \ln[A_o] = - 5.50 \times 10^{-3}\ s^{-1}$ (12.0 min x 60 s/1 min) + ln (0.200)
$\ln[A] = - 3.96 + - 1.61 = - 5.57$
$[A] = 3.81 \times 10^{-3}$ M.

The half-life of a reaction is a useful parameter. The half-life of a first-order reaction is characteristic of the reaction and is ***independent*** of the starting concentration of A.

$t = (1/k) \cdot \ln [A_o]/[A]$

$t_{½} = 1/k \cdot \ln (1/0.5) = (1/k) \cdot \ln 2 = (1/k) \cdot 0.693 = 0.693/k$

$t_{½} = 0.693/k$ or $k = 0.693\ t_{½}$

**Second-Order Reactions**

Second order reactions in A can be easily described mathematically. Second order in [A] and [B] and third order reactions will not be covered here.

For second order reactions in A:

$A \rightarrow$ products

اتحاد سنافر البوليتكنك (spu)

rate = $- \Delta [A]/\Delta t$, but now rate = $k[A]^2$

The graphable equation becomes:

$1/[A] = 1/[A_o] + kt$

again, $y = mx + b$

The half-life equation can be obtained by substituting $½[A_o] = [A]$

The half-life, $t_{½}$ then becomes, $t_{½} = (k[A_o])^{-1}$

This makes sense because a large rate constant leads to a faster reaction and short half-life. In second order reactions, half-life does depend on the initial [A] because a higher $[A_o]$ means more frequent collisions between A molecules.

## Activation Energy and Temperature Dependence of Rate Constants

### The Collision Theory of Chemical Kinetics

Common sense dictates that a reaction between two particles can occur only if they collide (usually in a gas or liquid). Therefore expect that a reaction rate is *proportional* to the rate of collision or the number of collisions per second.
The percentage of *successful* collisions in chemical systems varies greatly.
In some systems, almost every collision results in chemical change. (Instantaneous rxn)
In other systems, collisions rarely produce chemical change. (A slow rxn)
Therefore, the reaction rate is not the same as the *collision* rate but is often directly proportional to it.

Chemical reactions occur when the energy of collision is enough to break reactant bands and form product bonds. If there is not enough kinetic energy in colliding species, reactant bonds will not *break* and new, *product* bonds will not form.

It stands to reason that higher temperatures (which increase the frequency of collisions and increase the energy of collisions) will increase all or most reaction rates.

The kinetic energy of the colliding particles must equal or exceed a certain minimum value in order for a reaction to proceed. This minimum energy is called the ***Activation Energy, $E_a$.*** If a collision has enough energy to break some reactant bonds, an ***Activated Complex*** may form. This is an intermediate state, higher in energy than the reactants that may lead to products or may lead back to the original reactants. The products that form may be lower in energy than the reactants (exergonic or exothermic reaction) or may be higher in energy than the reactants (exergonic or exothermic reaction).

Two Sketches of Activated Complex Curves:

### The Arrhenius Equation

**اتحاد سنافر البوليتكنك (spu)**

The Arrhenius Equation relates the rate constant k to the activation energy, R the gas law constant and T the absolute temperature.

$k = A$ times e raised to the power $-E_{act}/RT$

*A* is not the reactant species A, but a dimensionless constant called the collision frequency or frequency factor. $E_{act}$, the reaction activation energy is given in J/mol or kJ/mol, R is equivalent to the gas law constant but is expressed as 8.314 J · (mol · K)$^{-1}$, and T is in kelvins. (Remember that 1/T will be in K$^{-1}$.)

To adapt the equation to graphing, take the natural log of both sides. The equation becomes

$\ln k = \ln A - E_{act}/RT$ or $\ln k = (- E_{act}/R)(1/T) + \ln A$,

again corresponding to $y = mx + b$.

This form of the equation shows that ln k varies as minus $T^{-1}$.
Does this make sense?

For using the Arrhenius Equation algebraically to compare rate constant value at different temperatures, write two equations for different temperatures and therefore different k values.

$\ln k_1 = \ln A - E_{act}/RT_1$ and $\ln k_2 = \ln A - E_{act}/RT_2$

$\ln k_1 - \ln k_2 = - E_{act}/RT_1 + E_{act}/RT_2$

**$\ln (k_1/k_2) = E_{act}/R(1/T_2 - 1/T_2)$** useful for solving for $T_1$ or $T_2$, and

**$\ln (k_1/k_2) = E_{act}/R(T_1 - T_2)(T_1 \cdot T_2)$** useful when both Ts are known.

Example: The rate constant for a first-order reaction is 4.50 x 10$^{-4}$ s$^{-1}$ at 25.0 ºC. What is its rate constant at 50.0 º C if the activation energy is 35.6 kJ/mol? Note that the collision frequency *A*, being the same, has cancelled out of the equation.

$\ln (k_1/k_2) = E_{act}/R(T_1 - T_2)(T_1 \cdot T_2)$ $T_1 = 298.2$ K, $T_2 = 323.2$ K

$\ln k_1 = E_{act}/R(T_1 - T_2)(T_1 \cdot T_2) + \ln k_2$

$\ln (k_2) = \ln k_1 - E_{act}/R(T_1 - T_2)(T_1 \cdot T_2)$
$= -7.71 - 35.6$ kJ/mol/(8.314 J · mol · K)$^{-1}$ · 1000 J/kJ · (-2.589 x 10$^{-4}$ K$^{-1}$)
$= -7.71 + 1.11 = - 6.60$
$k = 1.36\ 10^{-3}$ s$^{-1}$

The Arrhenius equation can also be used to solve algebraically for $E_{act}$, for either k or for either T.

## Reaction Mechanisms

(spu) اتحاد سنافر الدولي بتكنك

Observing the allover course of a reaction and measuring its rate usually tells little about the ***reaction mechanism***. A reaction mechanism is a series of ***elementary reactions*** or ***elementary steps*** that lead from reactants to products. To be reasonable, the sum of the elementary steps must add up to the overall reaction.

Example: the reaction of nitrogen oxide (NO) and oxygen gas does not proceed in one step as shown by the overall equation

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

First, three-body (***termolecular***) collisions are extremely unlikely and rare. Second, $N_2O_2$ (an ***intermediate***) is detected during the reaction.

The two postulated (and plausible) elementary steps are :

$$2NO(g) \rightarrow N_2O_2(g)$$
$$N_2O_2(g) + O_2(g) \rightarrow 2NO_2(g)$$

The add up to the overall observed reaction:

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$

The intermediate $N_2O_2$ is not present in the final product.

Both elementary steps above are bimolecular; they involve the collision of two molecules. Bimolecular and unimolecular reactions (when a single molecule breaks into two or more particles) are by far the most common examples of molecularity.

**Rate Laws and Elementary Steps**

Knowing or postulating the elementary steps can lead to an expression of the rate law of a reaction. For the simple unimolecular reaction where a reactant A changes spontaneously to products,

$A \rightarrow$ products

the elementary step predicts a unimolecular mechanism. That means that A decomposes without regard to anything else present in the system. The rate depends only on the concentration of A (obviously, the more A is present, the faster the reaction proceeds).

Rate = k[A]

For a bimolecular elementary reaction involving two different molecules, A and B:

$A + B \rightarrow$ products and rate $= k[A][B]$

If a reaction proceeds by the collision of two A molecules,
$2A \rightarrow$ products

اتحاد سنافر البوليتكنك (spu)

then, Rate $= k[A][A]$ or Rate $= k[A]^2$

When a chemical reaction proceeds by more than one elementary step and the different elementary steps have different rate laws, the rate law for the overall process is the one for the slowest reaction (the bottleneck), called the ***rate-determining step***.

**Hydrogen Peroxide Decomposition**

The uncatalyzed decomposition of hydrogen peroxide is very slow, especially in the dark. Drugstore 3% hydrogen peroxide lasts for years in brown bottles. Many substances, including the ubiquitous enzyme catalase in the body, hasten (catalyze) the breakdown of hydrogen peroxide to water and oxygen. Iodide ion effectively promotes the reaction. What is the rate law of the allover reaction

$H_2O_2(aq) \rightarrow H_2O_2(l) + O_2(g)$

Experimentally the rate is found to be proportional both to hydrogen peroxide and iodide concentration and first order with respect to both.

Rate $= k[H_2O_2][I^{-1}]$

The experimental observations can be explained by the following two elementary steps:

$k_1$ $\quad H_2O_2 + I^{-1} \rightarrow H_2O + IO^{-1}$

$k_2$ $\quad H_2O_2 + IO^{-1} \rightarrow H_2O + O_2(g) + I^{-1}$

The observed rate law, Rate $= k[H_2O_2][I^{-1}]$, can be explained if the first elementary step is the rate-determining step. Since all the iodide ion is recovered at the end of the reaction, it fits the definition of a catalyst, a substance that speeds up a chemical reaction without itself being consumed in the process.

**The Hydrogen Iodide Reaction**

$H_2(g) + I_2(g) \rightarrow 2HI(g)$

A logical assumption would be that the rate law is first order with respect to both hydrogen gas and iodine vapor. In fact, just that rate law was found experimentally. It was assumed the only elementary step was the simple collision of a $H_2$ molecule with an $I_2$ molecule.

Rate $= k[H_2][I_2]$

The reaction, however, does not take place so simply. First of all, the reaction rate is greatly increases by UV radiation in the reaction vessel. The first elementary step is the UV-induced **fast** dissociation of $I_2$ into two highly reactive I atoms. Then the ***rate-determining*** step is the rare termolecular collision of two I atoms with one $H_2$ molecule to produce two HI molecules. Why does this reaction appear to be second order, first order in both $H_2$ and $I_2$?

**(spu) اتحاد سنافر البوليتكنك**

**Experimental Support for Reaction Mechanisms**

Oxygen-18 labeling of hydrolyzing water to show that the carbonyl C-O single bond breaks and the oxygen atom adjacent to the ester carbonyl group ends up in the parent alcohol, not the carboxylic acid.

Using O-18 labeled water and carbon dioxide, it can be shown that the evolved $O_2$ is derived exclusively from water and not form $CO_2$. The $CO_2$ oxygen atoms are all incorporated into the glucose.

## Catalysis

A catalyst is a substance that increases a reaction rate without itself being consumed in the reaction. In principle, all of a catalyst can be recovered unchanged at the end of a reaction.

In writing a chemical equation, the catalyst is not shown as a reactant or a product, but is written above the arrow.

Example: The decomposition of potassium chlorate when heated is thousands of times more rapid when the catalyst manganese dioxide is present.

$$2KClO_3(s) \xrightarrow{MnO_2} 2KCl(s) + 3O_2(g)$$

Example: The burning of glucose to produce carbon dioxide and water takes place at a much lower temperature in the body in the presence of biological catalysts called *enzymes*. The energy released during air combustion is exactly the same but the reaction takes place at a flame temperature.

$$C_6H_{12}O6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)$$

The energy released or absorbed by a chemical reaction is exactly the same whether a catalyst is present or not.

A catalyst serves to lower the ***Activation Energy*** of a reaction, allowing the bonds to break and reform with much less input of energy.

Three main types of catalysis are Heterogeneous Catalysis, Homogeneous Catalysis and enzyme Catalysis.

## HETEROGENEOUS CATALYSIS

Heterogeneous two-phase) Catalysis is widely used in industrial manufacture of chemicals. The catalyst is usually in the solid phase and the reactants are either gaseous or liquid. The gas or liquid molecules bond temporarily to the atoms at the surface of the catalyst, which allows their internal bonds to break more rapidly or at a lower temperature.

(spu) اتحاد سنافر البوليتكنك

**Catalytic Converter**

In the automobile catalytic converter a Cu or Cr oxide surface catalyzes the *rapid* decomposition of toxic and smog-producing nitrogen oxides in the exhaust to harmless nitrogen and oxygen gas.

$NO_x(g) \rightarrow N_2(g) + O_2(g)$

If the nitrogen oxides were allowed to escape to the urban air, they would persist for days, causing environmental and health effects.

A precious metal catalyst (Pt, Rh, Pd, etc.) converts CO and $O_2$ in the exhaust stream to $CO_2$. In the atmosphere, CO would be slowly oxidized to $CO_2$ anyway but its presence in city air is associated with heart disease and other problems.

**The Haber Synthesis of Ammonia**

Probably the most important industrial process converts nitrogen gas and hydrogen to ammonia for use in fertilizer. The reaction is exothermic (energetically favorable) but it is very slow.

$H_2(g) + N_2(g) \rightarrow NH_3(g)$

$H_2$ molecules adhere readily to the surface of an iron metal catalyst and dissociate readily into H atoms which allows them to react with nitrogen gas. The reaction proceeds much more rapidly at 500 ºC than it would without the catalyst and makes the reaction an economically process.

**Homogeneous Catalysis**

Homogeneous implies that the reaction takes place in solution, with the *soluble* catalyst also in solution.

The classic example is the catalysis of ester hydrolysis by hydrogen ion.

$CH_3COC_2H_5 + H_2O \rightarrow CH_3COOH + C_2H_5OH$

Without $H^{+1}$ the process is very slow and the rate depends only on ester concentration.

Rate $= k [CH_3COC_2H_5]$

With $H^{+1}$, it is much more rapid and the catalyzed reaction is second order, that is first order with respect to both ester and hydrogen ion concentration.

Rate $= k_c [CH_3COC_2H_5] [H^{+1}]$

By the definition of catalysis, $k_c > k$ **(spu) اتحاد سنافر البوليتكنك**

**Enzyme Catalysis**

Enzymes are biological catalysts, always or almost always protein in nature that speed up many biological reactions. Enzymes have an ***active site***, where the ***substrate*** attaches in order for its bonds to be weakened. Life as we know it would not exist without enzymes to speed up otherwise sluggish reactions. Enzymes are very substrate specific, since their active sites will only accept a substrate of the right size, shape and charge distribution.

Low temperatures slow down enzyme-catalyzed reactions because collision rates are lowered. High temperatures also slow sown enzyme-catalyzed reaction rates because high temperatures denature (damage) proteins.

Step by Step: Oxidation Numbers and Balancing Redox reactions.

**Ex. 1) $MnO_4^{-1} + H_2SO_3 \rightarrow Mn^{2+} + HSO_4^{-1}$ (acidic)**

*Rules for Oxidation Numbers (for individual atoms):*

*1) **Any free element (without a charge) is zero.** Ex.) $Zn_{(s)}$, $N_{2(g)}$*

*2) **Monoatomic ions have their charge**. Ex.) $Zn^{2+}_{(aq)}$, $F^{-1}_{(aq)}$*

*3) **Hydrogen is always +1,** unless it is alone with a positive metal, then the H is -1.*
*Ex.) $NaH = Na^{+1} H^{-1}$*

*4) **Oxygen is always -2**, unless it is in a peroxide, then O is -1.*
*Ex.) $H_2O_2 = H^{+1}{}_2O^{-1}{}_2$*

*5) Fluorine is always -1.*

*6) The sum of oxidation numbers in a group of elements equals the charge of the group.*
*Ex.) NaCl so $NaCl^{=0}$, $SO_4^{2-}$ so $SO_4^{=-2}$*

*** Remember charges on Periodic Table Columns, for **<u>single</u>** element ions.*
*($1^{st}$ column = +1, $2^{nd}$ column = +2 etc.)*

**$MnO_4^{-1}$ :** Oxygen is always -2 (Rule 4). If there are 4 oxygens, then 4 X (-2) = -8 for all of the oxygens.

**Mn +** (**-8** for the oxygens) **= -1** (It must all add up to -1, since that is the charge given for the ion.)

Mn must equal +7.

**$(Mn^{+7}O^{-2}{}_4)^{-1}$**
+7 -8 = -1 It works!

**$H_2SO_3$ :** Hydrogen is always +1 (Rule 3). Oxygen is always -2 (Rule 4). If there are 2 hydrogens 2X (+1) = +2. If there are 3 oxygens, 3X(-2) = -6.

(**+2** for the hydrogens) **+ S +** (**-6** for the oxygens) **= 0** (This compound is neutral, no charge written above it.

S must equal +4.

**$H^{+1}{}_2S^{+4}O^{-2}{}_3$**
+2 +4 -6 = 0 It works!

**$Mn^{2+}$** is alone and has a charge, so it must add up to +2 (Rule 2).

(spu) اتحاد سنافي السوليتكنك

$HSO_4^{-1}$ : Hydrogen is always +1 (Rule 3). Oxygen is always -2 (Rule 4). If there are 4 oxygens, 4X(-2) = -8.

(**+1** for the hydrogen) **+ S +** (**-8** for the oxygens) **= -1** (This compound is neutral, no charge written above it.

S must equal +6.

$(H^{+1}S^{+6}O^{-2}{}_4)^{-1}$

+1 +6 -8 = -1 It works!

$(Mn^{+7}O^{-2}{}_4)^{-1} + H^{+1}{}_2S^{+4}O^{-2}{}_3 \rightarrow Mn^{2+} + (H^{+1}S^{+6}O^{-2}{}_4)^{-1}$

****For more practice on oxidation numbers go to the first part of each of the following problems!**

## Steps for balancing redox:

1) *Write the half reactions. (one for oxidation, one for reduction)*

2) *For each half reaction:*
   a) *Balance the electrons, by looking at the oxidation number of the elements that are oxidizing or reducing. (Make sure these elements are first balanced!)*
   b) *Balance all other atoms except H and O. ( If necessary, put in other compounds from the original reaction.)*
   c) *Balance O by adding $H_2O$.*
   d) *Balance H by adding $H^+$.*
   e) *Make sure all elements and charges balance.*

3) *Multiply the half reactions by numbers so that each has the same number of electrons.*
4) *Add half reactions and cancel what you can.*
5) ***If in basic solution**:*
   a) *Notice how many $H^+$ are left. Add the same amount of $OH^-$ as $H^+$, to both sides of the reaction.*
   b) *The $OH^-$ and $H^+$ that are on the same side make $H_2O$.*
   c) *Cancel any $H_2O$ that will cancel.*
6) *Make sure all charges and atoms balance.*

We must balance these in this strange method to get the charges to balance. If we balanced these using the old method, we could get the elements to balance, but the charges would not balance. We have to have the charges balance, since the oxidizing elements are giving electrons to the reducing elements. These numbers of electrons will eventually have to be equal (Step #3). If the two half reactions are physically separated and a wires placed between them, a battery can be made. These are battery reactions, since they transfer electrons and can give off current.

**Look at the charges on the single elements! Not the total charges of all hydrogens or oxygens, only the charge on a single element of that type!**

$(Mn^{+7}O^{-2}{}_4)^{-1} + H^{+1}{}_2S^{+4}O^{-2}{}_3 \rightarrow Mn^{2+} + (H^{+1}S^{+6}O^{-2}{}_4)^{-1}$

**(spu) اتحاد سنافر البوليتكنك**

Find the elements that have changing oxidation numbers going from the left side of the reaction to the right side of the reaction!

chemistrynoteslecture.com © 2011

Sulfur goes from +4 to +6, so it is **oxidizing**; it is getting more **positive**.
Mn goes from +7 to +2, so it is **reducing**; it is getting **smaller (reducing)**.
Write a reaction using only the sulfur compounds (for oxidizing) and write a reaction using only the manganese compounds (for reducing).

**Step 1:**

S Oxidizes
$H_2S^{+4}O_3 \rightarrow HS^{+6}O_4^{-1}$

Mn reduces
$Mn^{+7}O_4^{-1} \rightarrow Mn^{2+}$

Looking at only the +4 and +6 on the sulfurs, add electrons to balance the charge. (Make the charge the same on each side of the reaction.) Always add the electrons to the more positive side.
$+4 \rightarrow +6 + 2e^-$
Now +4 on the left will equal the +6 + (-2 charge from the electrons) on the right.
$+4 = +6 \quad -2$

Do the same for the Mn: only look at the +7 and the +2 on the Mn. Add electrons to the more positive side.
$+7 + 5e^- \rightarrow +2$
$+7 \quad -5 = +2$

**Step 2a): Add negative electrons to balance changing oxidation #'s.**

**+4** → **+6** + $2e^-$
$H_2S^{+4}O_3 \rightarrow HS^{+6}O_4^{-1}$ **+ $2e^-$**

**+7** + $5e^-$ → **2+**
$Mn^{+7}O_4^{-1}$ **+ 5 $e^-$** → $Mn^{2+}$

**Step 2b):** all balanced except H, O

There are no other elements that need balanced other than Oxygen and Hydrogen, so step 2b can be skipped.

Next we will balance Oxygen by adding water ($H_2O$). This will mess up the Hydrogens, but we will fix them in the next step.

We want to add water to whichever side needs Oxygen!
**For Oxidizing:** Looking below we have 3 oxygens on the left side ($H_2SO_3$) and 4 oxygens on the right side ($HSO_4^{-1}$). So we need one more oxygen on the left side to make 4 total oxygens on each side. So we add one $H_2O$ to the left side.

**For Reducing:** Looking below we have 4 oxygens on the left side ($MnO_4^{-1}$) and none on the right side. So we need 4 oxygens on the right side, add 4 $H_2O$ to the right side.

**(spu) اتحاد سنافر البوليتكنك**

chemistrynoteslecture.com © 2011

**Step 2c): Add $H_2O$ to balance O.**

**$H_2O$** + $H_2SO_3$ → $HSO_4^{-1}$ + $2e^-$

$MnO_4^{-1}$ + 5 $e^-$ → $Mn^{2+}$ + **4 $H_2O$**

Next we balance the Hydrogens by adding $H^{+1}$ to whichever side needs more hydrogens.
**For Oxidizing:** There are 4 hydrogens on the left ($H_2O$ + $H_2SO_3$) and 1 hydrogen on the right ($HSO_4^{-1}$). So we need to add 3 hydrogens to the right side to have a total of 4 hydrogens on each side.

**For Reducing:** There are no hydrogens on the left and 8 hydrogens on the right. We need to add 8 hydrogens on the left side.

**Step 2d): Add $H^+$ to balance H.**

$H_2O$ + $H_2SO_3$ → $HSO_4^{-1}$ + $2e^-$ + **3 $H^+$**

**8 $H^+$** + $MnO_4^{-1}$ + 5 $e^-$ → $Mn^{2+}$ + 4 $H_2O$

Next we need to have the same number of electrons on each side, so that we can cancel them in the next step!
**For Oxidizing:** There are 2 electrons.
**For Reducing:** There are 5 electrons. We need to make 10 total (lowest common number). So the whole **oxidizing** reaction is multiplied by 5. (5 times each compound to keep it balanced.) The whole **reducing** reaction is multiplied by 2.

**Step 3: Multiply to get common # of electrons.**

5 ($H_2O$ + $H_2SO_3$ → $HSO_4^{-1}$ + $2e^-$ + 3 $H^+$)

2 ( 8 $H^+$ + $MnO_4^{-1}$ + 5 $e^-$ → $Mn^{2+}$ + 4 $H_2O$)

Next we cancel or reduce whatever we can to make the reaction simpler and then add it up. We can cancel or reduce, if they are on opposite sides of the arrows. If they are on the same side of the arrows, they will add together (see example 3).
The 10 electrons completely cancel from each side (subtract 10 from each side).
15 $H^+$ will subtract from each side, leaving only 1$H^+$ on the left side.
5 $H_2O$ will subtract from each side, leaving only 3 $H_2O$ on the right side.

**اتحاد سنافر البوليتكنك (spu)**

chemistrynoteslecture.com © 2011

**Step 4: Add ½ reactions.**

$5\ H_2O + 5\ H_2SO_3 \rightarrow 5\ HSO_4^{-1} + 10e^- + 15\ H^+)$

$16\ H^+ + 2\ MnO_4^{-1} + 10\ e^- \rightarrow 2\ Mn^{2+} + 8\ H_2O$ cancel $e^-$, reduce $H^+$ & $H_2O$

**$5\ H_2SO_3 + H^+ + 2\ MnO_4^{-1} \rightarrow 5\ HSO_4^{-1} + 2\ Mn^{2+} + 3\ H_2O$**

The order does not matter as long as they are on the correct side of the arrow.
The elements and charges should all be equal on both sides.
See below for the check on the charges:

$5\ H_2SO_3 + H^+ + 2\ MnO_4^{-1} \rightarrow 5\ HSO_4^{-1} + 2\ Mn^{2+} + 3\ H_2O$

$+1 + 2(-1) = -1$ total $\qquad 5(-1) + 2(+2) = -1$ total

It balances chargewise! **You do not look at the individual charges you used to get the electrons, only at the charges on whole ions.** Make sure you did not lose any charges from the original problem!

**Ex. 2) $As_2O_3 + NO_3^{-1} \rightarrow H_3AsO_4 + NO$ (acidic)**

**$As_2O_3$**: Oxygen is -2 (Rule 4). 3 oxygens times -2 equals -6. So the As side must equal +6 (since the compound is neutral), but there are 2 As, so each will equal +3.

$As^{+3}{}_2O^{-2}{}_3$
$+6 \quad -6 = 0$ It works.

**$NO_3^{-1}$**: Oxygen is -2 (Rule 4). 3 oxygens times -2 equals -6. But the whole nitrate ion has to add up to -1. So nitrogen will be +5.

$(N^{+5}O^{-2}{}_3)^{-1}$
$+5 \quad -6 = -1$ It works.

**$H_3AsO_4$:** Hydrogen is +1 (Rule 3). Oxygen is -2 (Rule 4). 3 hydrogens equals +3 and 4 oxygens equals -8. For the compound to be neutral As must be +5.

$H^{+1}{}_3As^{+5}O^{-2}{}_4$
$+3 \quad +5 \quad -8 = 0$ It works.

**NO**: Oxygen is -2 (Rule 4). 1 oxygen times -2 equals -2, so nitrogen must be +2 to make a neutral compound.

$N^{+2}O^{-2}$
$+2 \quad -2 = 0$ It works.

**(spu) اتحاد سنافر البوليتكنك**

$As^{+3}{}_2O^{-2}{}_3 + (N^{+5}O^{-2}{}_3)^{-1} \rightarrow H^{+1}{}_3As^{+5}O^{-2}{}_4 + N^{+2}O^{-2}$

chemistrynoteslecture.com © 2011

As is oxidizing, since it goes from +3 to +5 (getting more positive). (Remember you are looking at the single charges for each element, not the total/overall charge.)
N is reducing, since it goes from +5 to +2 (getting smaller). Now write the half reactions using the compounds containing those elements.

As oxidizes
$As^{+3}{}_2O_3 \rightarrow H_3As^{+5}O_4$

N reduces
$N^{+5}O_3^{-1} \rightarrow N^{+2}O$

**For Oxidizing:** **This time we need to **balance the arsenic (As) before adding the electrons,** since there are 2 As on the left side and only 1 As on the right side.

When we add the 2 in front of the compound containing As, the charge will also be affected.
On the left side of the reaction we have 2 As that have a +3 charge equaling +6.
On the right side of the reaction we have 2 As that have a +5 charge equaling +10.

We will need to add 4 negative electrons to the most positive side, the right side. Now we have a charge of +6 on both sides of the reaction.
$+6 \rightarrow +10 + 4\ e^-$

**For Reducing:** The nitrogens are balanced, so we can go straight to adding electrons. The nitrogen on the left has a +5 charge and the nitrogen on the right has a +2 charge, so we need to add 3 negative electrons to the more positive +5 side. Now we have a charge of +2 on both sides of the reaction.

$3e^- + 5 \rightarrow +2$

**** Balance As before doing electrons,**
**so then multiply charges by 2, since there are 2 Arsenics!!**

**+6** → **+10** +4e-
$As^{+3}{}_2O_3 \rightarrow$ **2** $H_3As^{+5}O_4$ + **4e$^-$**

3e- **+5** → **+2**
**3e$^-$** + $N^{+5}O_3^{-1} \rightarrow N^{+2}O$

Next we add $H_2O$ to balance Oxygen.
**For oxidizing:** we have 3 oxygens on the left and 8 oxygens ( 2 X 4 O) on the right. We need 5 more oxygens on the left side.
**For reducing:** we have 3 oxygens on the left side and 1 oxygen on the right, so we need to add 2 more oxygens to the right side.

**5 $H_2O$** + $As_2O_3 \rightarrow$ 2 $H_3AsO_4$ + $4e^-$
$3e^- + NO_3^{-1} \rightarrow NO +$ **2 $H_2O$**

Next we add $H^{+1}$ to balance the hydrogens.
**For oxidizing:** we have 10 hydrogens on the left and on the right we have 6, so we need 4 more on the right.

(spu) اتحاد سنافر البوليتكنك

chemistrynoteslecture.com © 2011

**For reducing:** we have 0 hydrogens on the left and on the right we have 4, so we need 4 on the right.

$$5\ H_2O + As_2O_3 \rightarrow 2\ H_3AsO_4 + 4e^- + \mathbf{4\ H^+}$$

$$\mathbf{4\ H^+} + 3e^- + NO_3^{-1} \rightarrow NO + 2\ H_2O$$

Next we need to cancel the electrons: we have 4 electrons for oxidizing and 3 for reducing. We can make 12, by multiplying by 3 and by 4.

$$3\ (5\ H_2O + As_2O_3 \rightarrow 2\ H_3AsO_4 + 4e^- + 4\ H^+)$$
$$4\ (4\ H^+ + 3e^- + NO_3^{-1} \rightarrow NO + 2\ H_2O)$$

Then we need to cancel and reduce:
The electrons cancel.
We can cancel (subtract) 8 $H_2O$ from each side.
We can cancel (subtract) 12 $H^+$ from each side.

$$15\ H_2O + 3\ As_2O_3 \rightarrow 6\ H_3AsO_4 + 12e^- + 12\ H^+$$
$$16\ H^+ + 12e^- + 4\ NO_3^{-1} \rightarrow 4\ NO + 8\ H_2O \qquad \text{cancel } e^-\text{, reduce } H_2O \text{ \& } H^+$$
$$\mathbf{7\ H_2O + 3\ As_2O_3 + 4\ H^+ + 4\ NO_3^{-1} \rightarrow 6\ H_3AsO_4 + 4\ NO}$$

The charge balances: +4 -4 =0 zero charge

**Ex.3 ) $MnO_4^{2-} \rightarrow MnO_4^{-1} + MnO_2$ (acidic)**

**$MnO_4^{2-}$:** Oxygen is -2 (Rule 4). 4 oxygens times -2 equals -8. Mn must be +6 for the ion to add up to -2.
$(Mn^{+6}O^{-2}{}_4)^{2-}$
+6 -8 =-2 It works.

**$MnO_4^{-1}$:** Oxygen is -2 (Rule 4). 4 oxygens times -2 equals -8. Mn must be +7 for the ion to add up to -1.
$(Mn^{+7}O^{-2}{}_4)^{-1}$
+7 -8 =-1 It works.

**$MnO_2$**: Oxygen is -2 (Rule 4). 2 oxygens times -2 equals -4. Mn must be +4 for the compound to be neutral.
$Mn^{+4}O^{-2}{}_2$
+4 -4 = 0 It works.

$(Mn^{+6}O^{-2}{}_4)^{2-} \rightarrow (Mn^{+7}O^{-2}{}_4)^{-1}$ + $Mn^{+4}O^{-2}{}_2$

(spu) اتحاد سنافر البوليكنك

Mn is oxidizing (+6 to +7) and Mn is reducing (+6 to +4).

chemistrynoteslecture.com © 2011

Mn oxidizes
$Mn^{+6}O_4^{2-} \rightarrow Mn^{+7}O_4^{-1}$

Mn reduces
$Mn^{+6}O_4^{2-} \rightarrow Mn^{+4}O_2$

Mn is already balanced in each reaction. (There is one on each side.) So we can go straight to balancing the electrons. (Remember electrons are added to the most positive side, so that we have the same charge on both sides.)

$\mathbf{+6} \rightarrow \mathbf{+7} + 1e^-$
$Mn^{+6}O_4^{2-} \rightarrow Mn^{+7}O_4^{-1} + \mathbf{1e^-}$

$2e^- \quad \mathbf{+6} \rightarrow \mathbf{+4}$
$\mathbf{2e^-} + Mn^{+6}O_4^{2-} \rightarrow Mn^{+4}O_2$

**For oxidizing:** this ½ reaction is already balanced and needs no oxygens or hydrogens.

$MnO_4^{2-} \rightarrow MnO_4^{-1} + 1e^-$

**For reducing:** we need to balance oxygens first by adding $H_2O$. We need 2 waters on the right so that we have a total of 4 oxygens on each side.

$2e^- + MnO_4^{2-} \rightarrow MnO_2 + \mathbf{2\ H_2O}$

**For reducing:** we need to balance hydrogen by adding $H^{+1}$. We need 4 $H^+$ on the left, so that we have a total of 4 hydrogens on each side.

$\mathbf{4\ H^+} + 2e^- + MnO_4^{2-} \rightarrow MnO_2 + 2\ H_2O$

Next we need to cancel the electrons. Oxidizing needs multiplied by 2, so that we have 2 electrons in each reaction.

$2( MnO_4^{2-} \rightarrow MnO_4^{-1} + 1e^-)$
$1\ (4\ H^+ + 2e^- + MnO_4^{2-} \rightarrow MnO_2 + 2\ H_2O)$

Then we can cancel electrons.
The $MnO_4$'s do not cancel across (left to right), since the charges are different.
The $MnO_4^{2-}$ that are on the same side add together!

$2\ MnO_4^{2-} \rightarrow 2\ MnO_4^{-1} + 2e^-$
$4\ H^+ + 2e^- + MnO_4^{2-} \rightarrow MnO_2 + 2\ H_2O$
$\mathbf{3\ MnO_4^{2-} + 4\ H^+ \rightarrow 2\ MnO_4^{-1} + MnO_2 + 2\ H_2O}$

Charge balances: -6 +4 = -2 -2

**(spu) اتحاد سنافر البوليتكنك**

chemistrynoteslecture.com © 2011

Basic reactions are the same as acid, except they have one extra step at the end!

**Ex. 4)** $NO_2^{-1} + Al \rightarrow NH_3 + AlO_2^{-1}$ **(basic)**

**$NO_2^{-1}$:** Oxygen is -2 (Rule 4). 2 oxygens times -2 equals -4, but we need the ion to equal -1, so nitrogen will be +3.
$(N^{+3}O^{-2}{}_2)^{-1}$
+3 -4 = -1 It works.

**Al:** aluminum is alone and there is no charge written, so it is zero (Rule 1).

**$NH_3$:** Hydrogen is +1 (Rule 3). 3 hydrogens times +1 equals +3, but we need a neutral compound, so nitrogen will be -3.
$N^{-3}H^{+1}{}_3$
-3 +3 = 0 It works.

**$AlO_2^{-1}$:** Oxygen is -2 (Rule 4). 2 oxygens times -2 equals -4, but we need the ion to equal -1, so aluminum will be +3.
$(Al^{+3}O^{-2}{}_2)^{-1}$
+3 -4 = -1

$$(N^{+3}O^{-2}{}_2)^{-1} + Al^0 \rightarrow N^{-3}H^{+1}{}_3 + (Al^{+3}O^{-2}{}_2)^{-1}$$

Al oxidizes (0 to +3), N reduces (+3 to -3).

Al oxidizes
$Al^0 \rightarrow Al^{+3}O_2^{-1}$

N reduces
$N^{+3}O_2^{-1} \rightarrow N^{-3}H_3$

Al and N are balanced. There is one on each side, so we can do the electrons. Add the electrons to the more positive side, so that we have equal charges on both sides.
**Note on the nitrogens we go from +3 to -3, think of a number line +3, +2, +1, 0, -1, -2,-3 we need to go 6 positions, so we need 6 electrons (-6 + 3 = -3).

**0** → **+3** +3e⁻
$Al^0 \rightarrow Al^{+3}O_2^{-1} + \mathbf{3e^-}$

6e⁻ **+3** → **-3**
$\mathbf{6e^-} + N^{+3}O_2^{-1} \rightarrow N^{-3}H_3$

Next we add $H_2O$ to balance the oxygens.

$$\mathbf{2\ H_2O} + Al \rightarrow AlO_2^{-1} + 3e^-$$

$$\mathbf{2\ H_2O} + NO_2^{-1} \rightarrow NH_3 + \mathbf{2\ H_2O}$$

**(spu)** البوليتكنك ستاذ

Next we add $H^+$ to balance the hydrogens.

chemistrynoteslecture.com © 2011

$2\ H_2O + Al \rightarrow AlO_2^{-1} + 3e^- + \mathbf{4\ H^+}$

$\mathbf{7\ H^+} + 6e^- + NO_2^{-1} \rightarrow NH_3 + 2\ H_2O$

Next we multiply to cancel the electrons.

$2\ (\ 2\ H_2O + Al \rightarrow AlO_2^{-1} + 3e^- + 4\ H^+)$
$1\ (\ 7\ H^+ + 6e^- + NO_2^{-1} \rightarrow NH_3 + 2\ H_2O)$

Next we cancel electrons.
We can cancel (subtract) 2 waters.
We can cancel (subtract) 7 $H^+$.

$4\ H_2O + 2\ Al \rightarrow 2\ AlO_2^{-1} + 6e^- + 8\ H^+$
$7\ H^+ + 6e^- + NO_2^{-1} \rightarrow NH_3 + 2\ H_2O$
$2\ H_2O + 2\ Al + NO_2^{-1} \rightarrow 2\ AlO_2^{-1} + H^+ + NH_3$
$\uparrow \mathbf{1\ H^+}$

For basic we must now get rid of $H^+$, since this signifies acid.
$OH^-$ signifies base, so we will add $OH^-$ to **each** side of the reaction, since the reaction is balanced at this point. **We will add as many $OH^-$ as we have of $H^+$ in the reaction.** This reaction has 1 $H^+$, so we will add 1 $OH^-$ to each side of the reaction. (If we had 3 $H^+$ in the reaction, we would have added 3 $OH^-$ to each side.)

**Step 5: Note how many $H^+$.**
**Add this amount of $OH^-$ to both sides, canceling the $H^+$.**

$2\ H_2O + 2\ Al + NO_2^{-1} \rightarrow 2\ AlO_2^{-1} + \mathbf{H^+} + NH_3$
$\mathbf{+\ 1\ OH^-} \qquad \mathbf{+1\ OH^-}$ ** $H^+ + OH^- \rightarrow H_2O$
$OH^{-1} + 2\ H_2O + 2\ Al + NO_2^{-1} \rightarrow 2\ AlO_2^{-1} + H_2O + NH_3$

The point of adding the $OH^-$ is that when added to $H^+$, it will make water.
($H^+ + OH^- \rightarrow H_2O$) Now on the right we will only have $H_2O$ and not $H^+$.

$2\ H_2O + 2\ Al + NO_2^{-1} \rightarrow 2\ AlO_2^{-1} + \mathbf{H^+} + NH_3$
$\mathbf{+\ 1\ OH^-} \qquad \mathbf{+1\ OH^-}$ ** $H^+ + OH^- \rightarrow H_2O$
$OH^{-1} + 2\ H_2O + 2\ Al + NO_2^{-1} \rightarrow 2\ AlO_2^{-1} + H_2O + NH_3$

If we had 3 $H^+$ and had added 3 $OH^-$, we would have made 3 $H_2O$.
($3H^+ + 3OH^- \rightarrow 3H_2O$) Because they react in a ratio of 1 $H^+$: 1$OH^-$ : 1 $H_2O$.

The last step is to cancel $H_2O$ if they are on opposite sides and add the $H_2O$ together, if they are on the same side. In this case the waters are on opposite sides, so we can cancel (subtract) one of them from each side.

chemistrynoteslecture.com © 2011

$OH^{-1} + 2\ H_2O + 2\ Al + NO_2^{-1} \rightarrow 2\ AlO_2^{-1} + H_2O + NH_3$

Cancel $H_2O$, if necessary.

$$\mathbf{OH^{-1} + 1\ H_2O + 2\ Al + NO_2^{-1} \rightarrow 2\ AlO_2^{-1} + NH_3}$$

Charge balances: -1 -1 = -2 -2

Ex. 5) Do Ex. 3) as if it were BASIC.

$3\ MnO_4^{2-} + 4\ H^+ \rightarrow 2\ MnO_4^{-1} + MnO_2 + 2\ H_2O$

Note the number of $H^+$, and add this amount of $OH^-$ to both sides.

$3\ MnO_4^{2-}$ **+ 4 $H^+$** $\rightarrow 2\ MnO_4^{-1} + MnO_2 + 2\ H_2O$

**+ 4 $OH^-$** **+ 4 $OH^-$**

$3\ MnO_4^{2-}$ **+ 4 $H_2O$** $\rightarrow 2\ MnO_4^{-1} + MnO_2 + 2\ H_2O$ **+ 4 $OH^-$**

** $H^+ + OH^- \rightarrow H_2O$

*(so 4 $H^+$ + 4 $OH^-$ → 4 $H_2O$)*

(ratio is 1:1:1, 4:4:4)

Cancel/Reduce $H_2O$:

$$\mathbf{3\ MnO_4^{2-} + 2\ H_2O \rightarrow 2\ MnO_4^{-1} + MnO_2 + 4\ OH^-}$$

## Redox Reactions with **more than one element oxidizing or reducing**:

Ex. 6) $CrI_3 + Cl_2 \rightarrow CrO_4^{2-} + IO_4^{-1} + Cl^{-1}$ (basic)

$\overset{+3\ -1}{CrI_3} + \overset{0}{Cl_2} \rightarrow \overset{+6}{CrO_4^{2-}} + \overset{+7}{IO_4^{-1}} + \overset{-1}{Cl^{-1}}$ Both Cr and I are oxidizing and Cl is reducing!

Oxidation

$\overset{+3\ -1}{CrI_3} \rightarrow \overset{+6}{CrO_4^{2-}} + 3\ \overset{+7}{IO_4^{-1}}$

Reduction

$\overset{0}{Cl_2} \rightarrow 2\ \overset{-1}{Cl^{-1}}$ ←Balance Iodines and Chlorines.

For oxidation, the number of electrons can be calculated separately for the Cr and I reactions (then totaling the electrons for both reactions) or the Cr and I reactions can be combined together so that the electrons are totaled together in the combined reaction.

(spu) اتحاد سنافر البولينكنف

chemistrynoteslecture.com © 2011

**Oxidation Reactions Done Separately:**

$^{+3}CrI_3 \rightarrow {}^{+6}CrO_4^{2-} + 3\ IO_4^{-1}$

$+3 \rightarrow +6 + \mathbf{3\ e^-}$

Cr Cr

$^{-1}CrI_3 \rightarrow CrO_4^{2-} + 3\ {}^{+7}IO_4^{-1}$

$3 \times -1 \rightarrow 3 \times +7$

I I

$-3 \rightarrow +21 + \mathbf{24\ e^-}$

(-3 = +21 - 24)

**OR**

**Oxidation Combined Together:**

$^{+3\ -1}CrI_3 \rightarrow {}^{+6}CrO_4^{2-} + 3\ {}^{+7}IO_4^{-1}$

$(+3) + (3X\ -1) \rightarrow (+6) + (3X\ +7)$

Cr I Cr I

$+3 + -3 \rightarrow +6 + +21$

$0 \rightarrow +27 + \mathbf{27\ e^-}$

Combine the 3 $e^-$ and 24 $e^-$ electrons, so 27 e- on the right.

Above we also get 27 e-.

Oxidation

$CrI_3 \rightarrow CrO_4^{2-} + 3\ IO_4^{-1} + \mathbf{27\ e^-}$

Reduction

$\mathbf{2\ e^-} + {}^{0}Cl_2 \rightarrow {}^{-1}2\ Cl^{-1}$

Then do your normal balancing with $H_2O$ and $H^+$.

$\mathbf{16\ H_2O} + CrI_3 \rightarrow CrO_4^{2-} + 3\ IO_4^{-1} + 27\ e^- + \mathbf{32\ H^+}$

X2

$2\ e^- + Cl_2 \rightarrow 2\ Cl^{-1}$

X27

$32\ H_2O + 2\ CrI_3 \rightarrow 2\ CrO_4^{2-} + 6\ IO_4^{-1} + \mathbf{54\ e^-} + 64\ H^+$

$\mathbf{54\ e^-} + 27\ Cl_2 \rightarrow 54\ Cl^{-1}$

$32\ H_2O + 2\ CrI_3 + 27\ Cl_2 \rightarrow 2\ CrO_4^{2-} + 6\ IO_4^{-1} + 64\ H^+ + 54\ Cl^{-1}$

$+ 64\ OH^{-1} \qquad + 64\ OH^{-1}$

Basic, so add $OH^-$.

$\mathbf{32\ H_2O} + 2\ CrI_3 + 27\ Cl_2 + \mathbf{64\ OH^{-1}} \rightarrow 2\ CrO_4^{2-} + 6\ IO_4^{-1} + \mathbf{64\ H_2O} + 54\ Cl^{-1}$

Cancel $H_2O$.

**Final Answer:** $2\ CrI_3 + 27\ Cl_2 + 64\ OH^{-1} \rightarrow 2\ CrO_4^{2-} + 6\ IO_4^{-1} + 54\ Cl^{-1} + \mathbf{32\ H_2O}$

Check charge: -64 =? -4 -6 -54 √

Check elements: 2 Cr, 6 I, 54 Cl, 64 O, 64 H =? 2 Cr, 8+24+32 = 64 O, 6 I, 54 Cl, 64 H √

***End of Notes***

**(spu) اتحاد سنافر البوليتكنك**

chemistrynoteslecture.com © 2011